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# The Interactive Stance

*Meaning for Conversation*

Jonathan Ginzburg



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JONATHAN GINZBURG

OXFORD  
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## *List of Acronyms*

|          |                                              |
|----------|----------------------------------------------|
| ASP      | Add Side Participants                        |
| BNC      | British National Corpus                      |
| CA       | Conversational Analysis                      |
| CCUR     | Clarification Context Update Rule            |
| CMT      | Conversational Move Type                     |
| C-Params | Contextual Parameters                        |
| CR       | Clarification Request                        |
| DGB      | Dialogue GameBoard                           |
| DR       | Duplicate Responders                         |
| eset     | empty set                                    |
| FEC      | Focus-Establishing Constituent               |
| GQT      | Generalized Quantifier Theory                |
| HOU      | Higher-Order Unification                     |
| HPSG     | Head Driven Phrase Structure Grammar         |
| ISA      | Indirect Speech Act                          |
| KoS      | not an acronym                               |
| MAG      | Multilogue Adjacency of Grounding/acceptance |
| MAX-QUD  | Maximal element of QUD                       |
| MCI      | Metacommunicative Interaction                |
| MLDSA    | Multilogue Long-Distance Short Answers       |
| neset    | non-empty set                                |
| nelist   | non-empty list                               |
| NI       | Naked Infinitive                             |
| NSA      | Non-Sentential Answer                        |
| NSQ      | Non-Sentential Question                      |
| NSU      | Non-Sentential Utterance                     |
| PTT      | Dutch Telecommunications and Post            |

|       |                                                 |
|-------|-------------------------------------------------|
| QIAC  | Question Introduction Appropriateness Condition |
| QNP   | Quantificational Noun Phrase                    |
| QNUD  | Questions No longer Under Discussion            |
| QPA   | Questions as Propositional Abstracts            |
| QUD   | Questions Under Discussion                      |
| RCH   | Reprise Content Hypothesis                      |
| RF    | Reprise Fragment                                |
| RFC   | Right Frontier Constraint                       |
| SOA   | not an acronym                                  |
| ST    | Situation Theory                                |
| SU+AE | Situational Universe with Abstract Entities     |
| TFS   | Typed Feature Structures                        |
| TRP   | Transition Relevance Point                      |
| TTP   | Turn-Taking Puzzle                              |
| TTR   | Type Theory with Records                        |
| UR    | Update Rule                                     |

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# Interaction, Grammar, and the Behavioural Sciences

A: Knock knock.

B: Who's there?

A: Doctor.

B: Doctor Who?

(Schoolground ditty)

## 1.1 Introduction

Conversation is a form of *interaction*. This book concerns itself with the issue of how to describe certain linguistic features of such interaction, some of which are exemplified elegantly in the following extract from Harold Pinter's play *Betrayal*. Robert confronts his wife Emma after learning of a liaison she has with his best friend:

- (1) a. Emma: We have a flat.
- b. Robert: Ah, I see. (Pause) Nice? (Pause) A flat. It's quite well established then, your ... uh ... affair?
- c. Emma: Yes.
- d. Robert: How long?
- e. Emma: Some time.
- f. Robert: But how long exactly?
- g. Emma: Five years.
- h. Robert: *Five years?*

[Harold Pinter *Betrayal*, (London: Faber and Faber, 1991), 85]

Although this is not a transcript of an actually occurring conversation, it exemplifies certain features of natural dialogue:

- **Coherence:** each conversational move seems to cohere smoothly with its predecessor: questions are followed by answers which, in turn, raise new questions.
- **Conciseness:** conversation is, by comparison with text, a highly efficient medium. Emma's affirmation of the well-established nature of the affair, Robert's



wondering how long the affair has been going on, Emma's informing Robert that it has gone on for five years, and Robert's astonishment at Emma's informing him this, all of this which takes forty odd words of text to convey, takes a dozen words of dialogue.

- **Radical Context Dependence:** Isolated from their occurrence in a dialogue many utterances lose most of their import. None of the utterances (1c–h) could stand on their own in a text. Indeed, some utterances (e.g. (1d), (1h)) resist a univocal sentential paraphrase. At the same time, in context, all these utterances seem readily comprehensible to the conversationalists.

That conversation involves interaction is of course a truism. Equally incontrovertible is that in a conversation the participants use language to engage with each other. What *is* ultimately quite controversial, however, is the claim **that interaction is built into the grammar**, that internalized system underlying our ability to use language. This claim, further sharpened below, is defended in the course of what is the main task undertaken in this book: the development of a theory of meaning for spoken interaction. Now, that task in its full generality is without doubt a grand task worthy of a massive team of scholars. I attempt a restricted version of the task—developing a theory, henceforth referred to as KoS,<sup>1</sup> which can enable a description of the workings of *non-sentential utterances*. Non-sentential utterances are utterances without an overt predicate which in certain settings constitute on some estimates 30 per cent of all utterances (de Waijer 2001).<sup>2</sup> Example (2), an extract from the British National Corpus (BNC) (Burnard 2000), exemplifies their occurrence in a natural setting—John, Sarah, and Sue are attempting to print a file ca. 1990:

- (2) John: Okay which one do you think it is?

Try F1 F1 again and we'll get

Sarah: Shift and F1?

Sue: It's, no.

John: No, just F1 F1.

Sue: It isn't that.

John: F1.

<sup>1</sup> KoS is not an acronym, but the name of an island in the south Sporades group of the Dodecanese. It is loosely connected to *conversation-oriented semantics*.

<sup>2</sup> De Waijer provides figures of 40%, 31%, and 30% respectively for the percentage of *one-word utterances* in the speech exchanged between adults and infant, adult and toddler, and among adults in a single Dutch-speaking family consisting of two adults, one toddler, and one baby across two months. In general, the term 'non-sentential utterance' involves various difficulties in its application, potentially conflating disfluencies with 'canonical' grammatical constructions such as *short answers*. Given this, estimates of the frequency of NSUs are somewhat variable, depending on the classificational criteria applied. For discussion of these and of corpus studies of NSUs see Chapter 7.

Right, and that tells us  
Sue: It's shift F7. (BNC, G4K)<sup>3</sup>

One reaction, given the relative mayhem in evidence in (2), with its miscomprehension, disagreement, and disfluencies, is *despair* and, *inter alia*, the conclusion that spoken language cannot be the input for language acquisition:

- (3) [spoken language constitutes] 'a highly degenerate sample, in the sense that much of it must be excluded as irrelevant and incorrect—thus the child learns rules of grammar that identify much of what he has heard as ill-formed, inaccurate, and inappropriate.' (Chomsky 1972: 171).

Of course few would seriously dispute the importance of studying conversational interaction. And yet the fact of the matter is that, for all the highly insightful and detailed work of many researchers over decades, the dominant paradigms in grammar and semantics have, on the whole, abstracted away from interaction, viewing it as somebody else's problem. This is why a simple conversation, such as (2), still constitutes a significant challenge to formal grammar of just about any theoretical flavour. Even shorn of its disfluencies, (2) is not much better from a classical logician's perspective: what inferences, if any, can be drawn from this transcript given its mutual contradictions? And yet (2) is a perfectly coherent conversation that ultimately leads to a successful practical outcome (a file got printed).

As I will argue shortly, a detailed theory of conversational context is—or at least should be—a vital component for just about all of the disciplines that in some way make up the behavioural sciences. Before that, though, I would like to sharpen and make concrete my claim about interaction being *built in* to the grammar, based on data from non-sentential utterances (NSUs). Why care about NSUs? For the specialist in dialogue the answer is clear: they are a frequently occurring, highly context-dependent class of constructions, which therefore provide various clues to the structure of context. However, NSUs should also be of interest to anyone contending with the issue of *what are the bounds of grammar?*

There are a large variety of views on the nature of grammar. And they play an inevitable role in debates on the nature of how language evolved (see e.g. Pinker and Bloom 1990, Pinker and Jackendoff 2005, Hauser, Chomsky, and Fitch 2002, Chater and Christiansen 2008). On one view, which Chomsky has advocated repeatedly (Chomsky 1986, 2000), grammar is best seen as a disembodied, decontextualized combinatorial system, which interfaces with phonological and semantic modules. Various alternative frameworks have been developed, which, following Montague's lead, give equal weight to a semantic component (Montague 1974b, Pollard and Sag

<sup>3</sup> '(BNC, G4K)' means that this extract is to be found in block G4K of the BNC. Similar notation will be used for other extracts from the BNC. All the examples here were found using SCoRE, Matt Purver's search engine for the BNC (Purver 2001).

1994, Dalrymple 1999, Steedman 1999b, Culicover and Jackendoff 2005) or in some cases have even ‘devalued’ the pivotal role of syntax (Langacker 1987).

The issue of whether to integrate *context* as a separate dimension of grammar was brought to the fore in the work of the generative semanticists in the early 1970s (see e.g. Jerry Morgan’s seminal 1973 paper). A restricted notion of context, enabling some aspects of the meaning of noun phrases to be captured, has been integrated within various grammatical frameworks, following the seminal work of Kamp and Heim (Heim 1982, Kamp 1981, Kamp and Reyle 1993). Indeed, following Montague and Kaplan and much work in Situation Semantics (Montague 1974a, Cresswell 1973, Kaplan 1989, Barwise and Perry 1983, Gawron and Peters 1990), the need for grammar to make reference to an utterance situation in order to accommodate the meaning of indexicals such as ‘I’, ‘You’, ‘Here’, and ‘Now’ is commonly assumed, with much less consensus on the precise mechanism. Similarly, there has been much debate in recent years over where to draw the semantics/pragmatics border. A variety of influential arguments have been put forward arguing that ‘pragmatic reasoning’ can be used in determining ‘what is said’ (Levinson 2000, Recanati 2001, Carston 2002, Asher and Lascarides 2003). Nonetheless, even such maximalist positions concerning pragmatic enrichment of ‘what is said’ are quite consistent with and have not led to challenges to the orthodox assumption about the grammar itself being isolated from interaction.

Still, to the extent we accept that indexicals such as ‘I’, ‘You’, ‘Here’, and ‘Now’ need to be accommodated by the grammar, an assumption that seems reasonably incontrovertible on just about any useful notion of grammar,<sup>4</sup> a similar claim can be made for the NSUs exemplified in (4)—the NSUs themselves are in bold. Examples (4) offer samples from the BNC of ‘yeah’ and ‘yes’, ‘bye’, ‘mhm’, a reprise fragment—a phrase repeated from a previous utterance to seek clarification, and ‘Why?’:

- (4) a. Ann: Can you hear the birds singing? Listen. James: Er (pause) **yeah**. Ann: Can you hear? Bryony: I hear birds singing. Ann: You can hear the birds singing. **Yes**. (BNC, KB8)
- b. Ann: Well put it on the draining board and I’ll wash it and then put it back. (pause) James: Right, I’ll see ya tonight. Ann: **Mhm, mhm**. (pause) James: Tarrah. Ann: mm, **bye** [conversation ends] (BNC, KB8)
- c. Tim: Those pink things that af after we had our lunch. Dorothy: **Pink things?** Tim: Yeah. Er those things in that bottle. Dorothy: Oh I know what you mean. For your throat? (BNC, KBW)
- d. Cherrilyn: Are you still (pause) erm (pause) going to Bristol (pause) on Monday? Fiona: Dunno. Cherrilyn: No? Fiona: I dunno. Doubt it, why? (= *Why do you ask if I’m going to Bristol?*) (BNC, KBL)

<sup>4</sup> It might be possible to come up with a definition of ‘grammar’ which allows it to remain unaffected by context; anyone adopting such a stance will probably not be persuaded by the arguments below.

What sort of meanings do these NSUs have? In section 2.1 of Chapter 2, I will provide some justification for the meaning paraphrased in (5a) for ‘yes’; the meaning of ‘bye’ given in (5b) is justified in Chapter 4—‘bye’ is, in fact, a somewhat tougher nut to crack than might seem at first blush; ‘mmh’, discussed in Chapters 6 and 8, is used to *acknowledge* i.e. signal positive understanding of utterances—it also has a use in acceptances of assertions; the intended content construal of a reprise fragment, paraphrased in (5d), is developed and justified in Chapter 6; in Chapter 8 I argue that ‘Why?’ has a grammaticized class of uses dubbed ‘metacommunicative “Why?”’ as paraphrased in (5e):

- (5) a. A: Yes; meaning of ‘yes’:  $p$ , where  $p$ ? is the current issue under discussion.
- b. A: Bye; meaning of ‘bye’: A seeks to disengage from a conversation with B which has involved at least some discussion.
- c. A: mmh; meaning of ‘mmh’: A acknowledges understanding of B’s latest utterance.
- d. B: Did Jo leave? A: Jo?; intended content meaning of reprise fragment ‘u?’: A asks B what is the intended content of B’s (sub-utterance)  $u$ .
- e. B: Did Jo leave? A: Why?; meaning of metacommunicative ‘Why?’: A asks B of the cause of an utterance by B, by which an issue is raised which remains under discussion.

In all these cases, then, the conventional meaning of a word or a construction involves notions that irreducibly involve reference to interaction—notions such as ‘current issue under discussion’, ‘disengagement from conversation’, ‘acknowledgement of understanding’, and ‘ask intended reference of other’s utterance’. Moreover, as I will argue, this part of meaning does not derive from syntactically driven structure. Thus, in so far as grammar accommodates the conventional meaning of words and constructions, what we have here is the nucleus of an argument to establish that **grammar must link up directly with interaction**. Alternatively put, that **grammar characterizes talk in interaction**.

For every semantic imperialist, who wishes to expand the reach of semantics, there is often a pragmatic sceptic, who would not wish to concede the *conventionality* of all the meanings I will argue for. I should emphasize though that my aim in assimilating to ‘grammar’ certain aspects of linguistic behaviour in dialogue is not to start a potentially unproductive *border dispute*. As we will eventually see, in Chapters 6 through 8, one can certainly offer both linguistic arguments—centred on the diversity of syntactic and phonological parallelism effects associated with NSUs—and arguments based on the very gradual trajectory of their ontogenetic emergence, which suggest that the proper domain for explicating the linguistic properties of NSUs is in the grammar. And yet, my hope is that even the pragmatic sceptic can make use of the formal theory of the fundamental notions of interaction developed here to formulate an alternative. Particularly, given that our analysis of contexts in terms of *total information states*

invites an integration of semantic and pragmatic aspects of meaning. In any case, my claim is that the sceptical pragmaticist will need to work quite hard to develop pragmatic principles of significant detail that will be able to derive the requisite import using pragmatic inference in a way that avoids building in reference to interaction in the semantics.

## 1.2 Why a Theory of Dialogue Context is Important for Behavioural Scientists

Given their interests, analysing the mechanics of conversation should be important business for wide swathes of the behavioural sciences.

- (6) a. **Sharing meanings.** Philosophers, linguists, and psychologists worry about the extent to which contexts and meanings are shared. Among psychologists there has been a long-standing debate with positions ranging from the claim that conversational participants are generally egocentric to the opposite perspective that speakers design their utterances with their addressees in mind (Brown and Dell 1987, Horton and Keysar 1996, Clark 1996, Brennan and Clark 1996, Garrod and Pickering 2004, Barr 2008). As I mentioned earlier, where to draw the line between ‘semantic’ meaning and ‘what is said’, and how context mediates between these is a central area for both linguists and philosophers (e.g. Recanati 2001, Carston 2002, Cappelen and LePore 2005, Borg 2007, Ariel 2008).
- b. **Repair.** Psychologists, spoken dialogue system developers, phoneticians, and sociologists are interested in *repair*, a cover term for the wide range of corrective actions that occur when ‘trouble’ arises in interaction, be it self-directed (as in disfluencies and hesitations) or other-originating (as in clarification requests and corrections) (see e.g. Schegloff, Jefferson, and Sacks 1977a, Levelt 1983, Demetras, Post, and Snow 1986, Shriberg 1994, Heeman and Allen 1999, Brennan and Schober 2001).
- c. **Conversational structure.** The structure of conversation—or the various genres it can embody—concerns literary scholars working on the semiotics of theatre (e.g. Ubersfeld 1999, Elam 2002, De Marinis 1993).
- d. **Argumentation.** Logicians since antiquity and recently argumentation theorists (e.g. Walton, Reed, and Macagno 2010, Prakken 2010) are interested in characterizing good/valid/fallacious arguments. Context is an obvious parameter of any characterization of premises, and—to the extent that the characterization is to be applicable to arguments in real conversations—this must be more or less closely modelled on conversational context.

Nonetheless, the words of Willem Levelt written almost twenty years ago are, unfortunately, still quite apt:

...the literature on speaking is gigantic....Students of conversational analysis, pragmatics, discourse semantics, artificial intelligence, syntax, phonology, speech communication, and phonetics have contributed myriad theoretical insights and empirical findings. The major problem with this huge literature, however, is that it is compartmentalized—phoneticians ignore phonology, conversational analysts ignore discourse semantics and phonetics, students of AI ignore psycholinguistics, and so on. (Levelt 1993: xiii)

Although such a situation is to some extent inevitable in an area that interests multiple disciplines, one plausible underlying cause is the lack of a widely shared framework for describing *dialogue context and interaction*. What is needed is a framework that could capture a variety of linguistic, psychological, and sociological observations in a systematic way, but at the same time be sufficiently *explicit* so as to be usable by formal modellers. Developing KoS to be such a framework, one which allows us to address phenomena and issues such as those in (6), is one of the main aims of this book. To do so requires synthesizing work from a wide range of computational, linguistic, logical, philosophical, psycholinguistic, and sociological approaches.

### 1.3 A Theory of Dialogue Context: Antecedents and Pay-off

Fundamental to KoS is the dynamic strategy to meaning, pioneered by Stalnaker (1978), Lewis (1979), Kamp (1981), Heim (1982), Barwise (1985), Groenendijk and Stokhof (1991), wherein the meaning of a linguistic form is explicated in terms of the effect its use has on existing commonly shared contextual *resources*. This suggests thinking of context as structured by resources which conversational participants keep track of, as demonstrated by linguistic evidence. Working out what these resources are and how to model interaction in their terms will be a focus of Chapters 2, 4, and 6.

Combining the view of context that this leads to with a perspective originating from artificial intelligence work on the design of dialogue systems (e.g. Allen and Perrault 1980) will enable us in Chapter 4 to sketch an explication of the range of interaction conventions specific to a particular domain or genre, and in this way offer formal counterparts akin to Wittgenstein's *language games* (Wittgenstein 1953) and Bakhtin's *speech genres* (Bakhtin 1986).

KoS, then, draws on a long-standing and highly successful tradition in work on meaning, one that goes back at least to Frege. This is a tradition rooted in characterizing communication via its successful episodes, thereby abstracting away from individual differences and from the communicative process. Consequently, I dub this tradition the *communitarian* approach to semantics. We have already seen data that do require one to interleave the communicative process into the semantics—the NSUs in (4). This and a variety of other data first introduced in Chapter 2 will lead me to propose an alternative, far more general perspective to our semantic enterprise—the interactive stance. The interactive stance involves taking seriously the fact that

communication involves multiple agents with distinct beliefs and desires and places importance on explicating the potential for misunderstanding, rejection, and correction, as well as success.

Adopting the interactive stance will allow us to incorporate into KoS insights from two important bodies of work—Conversation Analysis (e.g. Schegloff, Jefferson, and Sacks 1977a, Schegloff 1987b), which lead the way in recognizing the fundamental importance of *repair*, and Clarkian psycholinguistics (e.g. Clark 1996), which has offered detailed documentation of *grounding*—the process by means of which content becomes part of the common ground.

Combining these approaches with the dynamic perspective will also suggest a fundamental desideratum for a theory of meaning. Such a desideratum needs to cover both the range of possibilities associated with successful communication (grounding), as well as with imperfect communication—indeed it has been argued that *miscommunication* is the more general case (see e.g. Healey 2008). Thus, we can suggest that the adequacy of semantic theory involves **the ability to characterize for any utterance type the contextual update that emerges in the aftermath of successful exchange and the range of possible clarification requests otherwise**. This is, arguably, the early twenty-first-century analogue of truth conditions.

In order to fulfil this desideratum we will need a formal framework that can link up the three domains illustrated in Figure 1.1: the external world, grammar, and interaction.

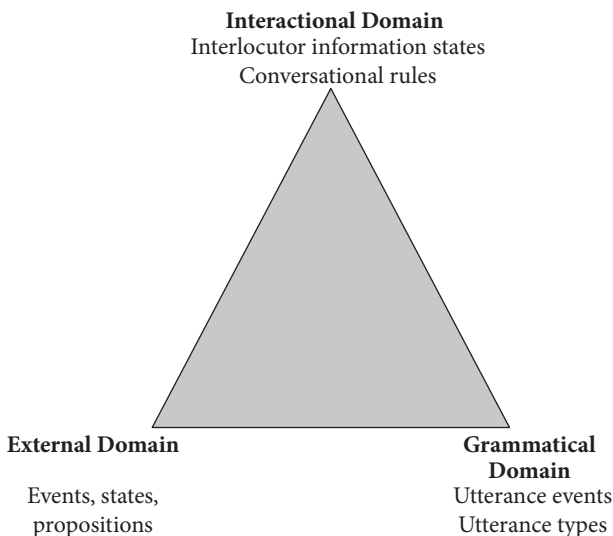


FIGURE 1.1: The Interactional Triangle (with apologies to Morris 1938)

In other words, we want to write grammatical rules that relate utterances to the import they have on the information states of conversational participants. And we also want to write conversational rules that indicate what utterances are possible given the current state of play. In order to do this I employ the formalism of Type Theory with Records (Cooper 2005a, 2011), a formalism that allows one to integrate techniques from a variety of semantic frameworks, including Discourse Representation Theory (Kamp and Reyle 1993), Montague Semantics (Montague 1974b), and Semantic Frame Theory (Fillmore 1985), as well as to build on techniques for the precise description of words and constructions emanating from sign-based grammars (e.g. Sag, Wasow, and Bender 2003) and construction grammar (e.g. Michaelis 2009). What is crucial for current purposes about Type Theory with Records, which distinguishes it from most other existing semantic and grammatical formalisms, is that it provides access to both types and tokens at the object level. Concretely, this enables simultaneous reference to both utterances and utterance types, a key desideratum for modelling metacommunicative interaction.

KoS has descriptive reach from the micro-conversational—e.g. self-repair at the word level—to macro-level phenomena such as multiparty conversation and conversational genres. KoS provides accounts of the opening, middle game, and closing stages of conversation. Corpus studies in Chapters 6 and 7 provide benchmarks for what the theory covers: coverage of the main types of metacommunicative acts in the British National Corpus and of virtually all the types of NSUs in the British National Corpus. These are buttressed by evidence from computer simulations that compare language dynamics with and without metacommunication and from corpus studies of the order of emergence of NSUs in child language, which offer some support to the hierarchy of complexity among NSU constructions that emerges from the analyses I propose. This acquisition data will also help me to argue against the approach of *Underlying Sententialism*, adopted in most generative linguistic work on ellipsis, wherein NSUs are underlyingly canonical sentences that (under conditions of contextual recoverability) involve unpronounced material.

Above and beyond the nitty-gritty of conversation KoS also offers a new perspective on traditional semantic concerns. Adopting the interactive stance will lead us to formulate a new methodological criterion, the Reprise Content Hypothesis—stronger than traditional compositionality—which regulates allowable semantic denotations. This pushes us to eschew higher-order denotations for generalized quantifiers. I also show how to provide an account of pronouns that can allow for their antecedents to occur across self-corrections, across different moves, but also accommodating their ‘bound-variable’ uses.

All in all, by the end what emerges is a highly detailed theory of relevance, taking in the illocutionary, metacommunicative, metadiscursive, genre-based components of this notion. With this in hand, I will also be able to sketch accounts of one class of conversational implicatures and several classes of indirect speech acts.



## 1.4 A Road Map for this Book

Although some background in formal linguistics is needed to follow all the technical developments, the main arguments and phenomena should be comprehensible for others. Chapters 3 and 5 are more technical than the rest, the former on the semantic front, the latter on the grammatical front—the non-linguist reader interested in interaction can skip them and dip back into the summaries of those chapters. The time-pressed reader can skip everything and proceed directly to Chapter 9, where all is revealed—motivation, tools, results, and theoretical implications in non-technical terms: four pictures and fewer than two thousand words. Conversely, the main technical details are summarized in three appendices relating to, respectively, the external world, the interactional domain, and the grammatical domain. There are also more than forty exercises scattered throughout the book; some are routine applications of techniques just exemplified, others explore alternative strategies to ones pursued in the text, and in some cases open questions are presented.<sup>5</sup>

**Chapter 2** The basic structure of the Dialogue GameBoard, the public component of context, is motivated. Arguments are provided for the need to adopt the interactive stance.

**Chapter 3** describes the semantic ontology which is the basis for subsequent representation of dialogue contexts. Type Theory with Records (TTR) is introduced.

**Chapter 4** develops a theory of dialogue moves—including greetings, partings, queries, and assertions—and of dialogue context.

**Chapter 5** A simple grammar is developed as a basis for formulating the theory of metacommunicative interaction and formally describing NSUs. The grammar is based on Head-Driven Phrase Structure Grammar (HPSG), but reformulated in TTR.

**Chapter 6** The theory from Chapter 4 is extended to cover metacommunicative interaction, in particular to account for interaction relating to clarification requests. A corpus-based characterization of the clarification requests that occur in three corpora is provided.

**Chapter 7** The chapter contains a corpus-based characterization of the range of non-sentential utterances that occur in the British National Corpus. Grammatical and contextual analysis of a wide range of NSUs is developed.

**Chapter 8** This chapter considers extensions of previous topics in five directions:

1. How to scale up the theory of two-person interaction developed earlier to multiparty dialogue.
2. How to generalize the theory of metacommunicative interaction to cover self-repair.

<sup>5</sup> Solutions can be obtained by contacting the author.

3. Extending the theory to cover indirect speech acts, more generally the issue of intention recognition in dialogue: how often, diagnostics, etc. An analysis of queries of the form ‘Why?’, in particular those used to request clarification of the goals underlying an utterance.
4. Developmental evidence concerning the emergence of NSUs is presented and an account of the order of emergence is sketched.
5. An extension of the grammar to deal with quantified NPs and anaphora while attempting to satisfy the Reprise Content Hypothesis.

**Chapter 9** Non-technical survey of all key results.

## From Communitarian to Interactive Semantics

### 2.1 Context in Dialogue: an Initial Sketch

The word ‘yes’ is one of the most frequently encountered in conversation, occurring, for instance, 31,960 times in the spoken part of the British National Corpus (approx. 0.3 percent of all word tokens) and 727 times from among the 50,000 word tokens of the London–Lund corpus (Kennedy 1998). It has, nonetheless, not been the object of much formal study.<sup>1</sup> The main difficulty in providing a formal description is pinning down its contextual background. Let us start, then, quite informally. When ‘yes’ stands alone, it typically serves to affirm a proposition, one that has either been asserted or queried:

- (1) a. A: Did Suleyman the Magnificent build the walls of Jerusalem? B: Yes.
- b. A: Bo is finally leaving us tomorrow. B: Yes. It’s a relief.
- c. Robert(1): It’s quite well established then, your . . . uh . . . affair? Emma(2): Yes.
- d. Our affair is quite well established.

In (1c(2)) Emma conveys (1d) to Robert. How does this come about? Essentially, the most obvious formal semantics account one would offer involves the following three components:

- (2) a. **Meaning:** Emma and Robert know the *meaning* of ‘yes’, which can be described tentatively as: *the proposition currently most salient in the context*.
- b. **Context:** Emma and Robert are jointly aware of the evolving context. In particular, they are both aware that as a result of Robert’s query the truth or otherwise of *the affair is well established* became most salient in the context.

<sup>1</sup> For some accounts, which abstract away from its contextual background; see Groenendijk and Stokhof (1984), Hoepelmann (1983). The latter also discusses subtleties concerning the German and French *ja/doch* and *oui/si*.

- c. **Content:** putting (1) and (2) together, the content of Emma's utterance of 'yes' emerges.

Before attempting to describe the meaning of 'yes' more precisely, let me first generalize the principles appealed to in (2), which are more or less uncontroversial reflections of much existing semantic practice. The need to accommodate the dependence of content on context led to the emergence of a distinction between *meaning*, a notion associated with sentences or types of utterances, and *content*, a notion characterizing utterance tokens. This distinction, the importance of which was first brought home to semanticists and logicians by David Kaplan, building also on insights due to Bar-Hillel, Montague, and Cresswell, was further developed by work in Situation Semantics (Barwise and Perry 1983, Gawron and Peters 1990, Cooper and Poesio 1994). The basic task of a semantic analysis, on this view, is to specify the meanings associated with expressions. A meaning of an expression  $E$  on this view is, loosely speaking, a rule which allows a conversationalist to calculate, given a context  $c$ , what content is associated with  $E$  in  $c$ . One apparently subtle point is worth emphasizing, for future reference: expressions are taken to be platonic entities, unlocated in space or time. The alternative perspective, thinking of a meaning as associated with a type of utterance, where an utterance is a spatio-temporally located event involving the sequential enunciation of one or more word(s), has occasionally been proffered (originally in Barwise and Perry 1983 and subsequently in Gawron and Peters 1990, Cooper and Poesio 1994, Poesio 1998), but it has generally aroused scant interest. Let me enshrine this initial general assumption as follows:

- (3) The supra-contextual nature of Semantics: Semantics associates meanings with expressions.

This notion of a meaning as a contextual use rule can be formalized as a function from *contexts* to *contents*. (4) is a simple example of a meaning construed as a function:

- (4) a. I hear you.  
 b.  $f: c \mapsto \text{Hear}(s, a, t)$ , where  $s$  is the speaker in  $c$ ,  $a$  is the addressee, and  $t$  overlaps with the time of  $c$ .

There are two additional principles to mention, both commonly exploited but rarely enunciated. The first, which seems a rather intuitive piece of common sense, concerns the sharing of context:

- (5) Equal Access to Context: as a conversation proceeds, a shared context (the *common ground*) emerges: A has her turn, reaches a transition relevance point

(TRP),<sup>2</sup> then either A proceeds or B takes over **from the common ground point at which A spoke**.

(6) exemplifies why Equal Access seems a plausible assumption: A makes an initial utterance, a query, which either A or B can follow up on:

- (6) a. A(1): Who should we invite to the conference?
- b. A(2): Perhaps Noam, huh?
- c. B(2): Perhaps Noam, huh?

The final principle derives from the work of David Lewis on convention (Lewis 1968). Simplifying somewhat, what Lewis proposed was that a regularity as described in (7a) is what underlies the communicative process involved in conveying 'literal import'. Now (7a) won't work for sentences whose (literal) import varies with context, the lion's share of actually used sentences. However, the intent of the semanticists who brought in the meaning/content distinction was, presumably, to replace (7a) with something along the lines of (7b).<sup>3</sup>

- (7) a. Original Lewisian regularity: Whenever S is uttered, the utterer intends to communicate P and the hearer acquires the belief P.
- b. Contextualized: Whenever A utters S with meaning  $\mu$  in a context  $c$ , A communicates that P, where  $P = \mu[f]$ ,  $f$  is the assignment  $c$  provides for  $\mu$ , and B acquires the belief that P.

Having spelled out some basic principles linking meaning, context, and content, there are two crucial interrelated issues we need to address before we can actually spell out the meaning of 'yes'. Simply put, the issues are:

- (8) a. How is context structured?
- b. How does context evolve?

Contending with the issues in (8) will occupy us for a large part of this book. As mentioned in Chapter 1, my starting point concerning context and its evolution is the *dynamic* approach to meaning (e.g. Stalnaker 1978, Lewis 1979, Kamp 1981, Heim 1982, Barwise 1985, Groenendijk and Stokhof 1991). Stated in its most general form the dynamic approach to meaning proposes that the meaning of a linguistic form is explicated in terms of the effect its use has on existing commonly shared contextual resources. Conversely, this suggests thinking of context as structured by

<sup>2</sup> The term *Transition Relevance Point* comes from the tradition of Conversation Analysis (Schegloff 2007). It refers to a point in a conversation at which control of the turn, in other words who is speaking, could change.

<sup>3</sup> A somewhat refined version of (7b) is provided in the work of David Israel and John Perry, e.g. Israel and Perry (1991). But there is little evidence to suggest that many working semanticists find (7b) objectionable.

resources which conversational participants keep track of, as demonstrated by linguistic evidence. Attention has been focussed on two types of *changing* contextual resources. First, in order to explicate assertion and presupposition, Stalnaker (1978), Lewis (1979), and (somewhat less directly) Clark and Marshall (1981) urged context to be construed as a resource that represents the commonly accepted information at any given point in conversation. A second contextual resource which has received wide attention was introduced in order to explicate nominal anaphoric possibilities available at any stage of processing a text or monologue. This resource is harder to describe in theory-neutral terms, not least because its nature depends crucially on the meaning one decides to assign to pronouns. The commonest approach, based on a view of pronouns as bound variables, has been to view this resource as a set of variable assignments. Views of pronouns as definite descriptions of some kind assume the resource to be a set of properties, situations, or some similar entity with which an individual can be uniquely identified. For the present I concentrate attention on the former resource, the set of commonly accepted assumptions or the (contextual) FACTS(base) as I shall sometimes refer to them.<sup>4</sup>

Consider the dialogue in (9). This provides a simple example of the dynamics of the FACTS and its effect on what can be uttered. Assuming that an audience has accepted (9(1, 3)), this opens the way for an utterance like (9(5)) in which the speaker refers to the woman named Cherie. More generally, when a speaker utters a sentence like (9(5)) she relies on the presence of a fact such as conveyed by (9(3)) in FACTS—or at least maintains this as a pretence. This assumption might be based on knowledge that this fact was explicitly introduced into the facts base at an earlier stage of the conversation or be assumed to be commonly known via some other source. Utterance (9(7)) illustrates how once an assertion has been accepted, the fact associated with it can be used in a presuppositional context, here exemplified by ‘Given this . . .’

- (9) A(1): There is this guy Tony. B(2): mm. A(3): He’s got a wife called Cherie. B(4): Right. A(5): Anyway, one day Cherie was caught without a ticket and was given a fine. B(6): hmm. A(7): Given this, none of us need to buy tube tickets.

For reasons such as those discussed above, Stalnaker chooses to *identify* a context at time *t* with the set of assumptions common to the conversational participants. *Presupposition*, that information which conversationalists can take as given, is intrinsically dynamic and shared. For a conversational setting the account is incomplete in a variety of ways which relate to the fact that interaction is taking place. Let me focus for now on two problems which relate to our current quarry, specifying the meaning of ‘yes’. Both problems relate to the fact that somehow context needs to be *structured* in a way that Stalnaker’s context sets do not countenance.

Given that a conversational participant makes a particular move, say an assertion, we would like context to enable us to characterize the possible follow-ups to that move.

<sup>4</sup> How to integrate anaphora into a dialogue context is discussed in Chapter 8.

In particular, we would like to allow a subsequent utterance such as 'yes' to pick up the proposition which was asserted. But if a particular item of illocutionary information has been added in the normal way to the set of common-ground facts, it has neither more nor less influence on what comes next than any other information in the common ground. In other words, the preconditions for the next move are the totality of what has been accepted hitherto in the common ground. To take an informal example:

- (10) a. Initial common ground: *The date today is 5 January 2011, the conversation is taking place in France, the weather outside is sunny, ...*  
b. A: Bill left yesterday.  
c. New common ground: *The date today is 5 January 2011, The conversation is taking place in France, the weather outside is sunny, A asserts that Bill left yesterday, ...*

What this indicates is, unsurprisingly enough, the need to somehow *isolate*, within the context at each point after a conversational move has been made, information that pertains to that move, most prominently the content of the move. Without worrying for the moment about how this is to be represented formally, we can simply posit that at any given point from among the facts of a context there is a distinguished element which is designated as the LatestMove. Then we can redescribe the evolution of the context in (10) as follows:

- (11) a. Initial common ground: facts: *The date today is 5 January 2011, the conversation is taking place in France, the weather outside is sunny, ...*  
A: Bill left yesterday.  
b. New common ground: facts: *The date today is 5 January 2011, the conversation is taking place in France, the weather outside is sunny, A asserts that Bill left yesterday, ...* LatestMove: A asserts that Bill left yesterday.

In fact, thinking of conversation in such terms is familiar from a number of traditions of discourse and conversation analysis. In such analyses (see e.g. Sinclair and Coulthard 1975, Nofsinger 1991, Carletta et al. 1996) the LatestMove information is either identified as the dialogue context or constitutes one of the basic levels of structure. The idea being that a given action by one participant sets up a restricted set of follow-up actions by the other participant. The simplest notion of move structure proposed in the Conversational Analysis (CA) literature (reviewed most recently in Schegloff 2007) is adjacency. In other words, it is assumed that various moves in dialogue come in move/countermove pairs:

- (12) a. A: Who left? B: Bill (query/reply)  
b. A: Open the window please! B: Sure (command/acceptance)  
c. A: Hi! B: Hiya! (greeting/counter-greeting)

Positing the existence of LatestMove as a distinguished component of the context also has some cognitive justification. The claim in the CA literature, based on substantial examination of corpora, is that any reaction to an initial member of an adjacency pair that is not an appropriate countermove is *marked* in the sense that it will be accompanied by hesitation or hedging. This claim has been refined somewhat to take into account moves that pertain to the communication process such as clarifications and acknowledgements (see e.g. Clark 1996), as we discuss below.

Let us return to the problem of specifying the meaning of ‘yes’. Armed with the initial structure we have imposed on context, we could hypothesize the following as the meaning of ‘yes’:

- (13) *The proposition introduced by LatestMove into the context.*

A cursory examination of any conversational corpus will attest that this description covers a high percentage of the occurrences of ‘yes’. Nonetheless, the description is intrinsically incomplete. This is exemplified in (14a). The proposition introduced by the LatestMove in the context for (14a(4)) is A’s clarification, which conveys essentially (14b). However, (14a(4)) need not affirm (14b), rather it can convey (14c), which is the proposition queried by A in (14(1)). Excerpts (15a, b) from the BNC provide additional examples of non-local resolution of ‘yes’:

- (14) a. A(1): Did Billie show up at all? B(2): Billie? A(3): Billie Whitechapel. B(4):  
Yes.  
b. The person A was asking whether she showed up is Billie Whitechapel.  
c. Billie showed up.
- (15) a. Richard: He started he’s back next weekend.  
Anon 6: Who?  
Richard: Morse.  
Anon 6: No he’s not.  
Richard: Oh yes (BNC, KSV)  
b. Richard: Well we’ll get the motor work done first  
Anon 6: Mm  
Richard: then play table tennis.  
Anon 6: No.  
Richard: Yes. (BNC, KSV)

These data point to an additional component of context we need to bring into the picture. If we tried to refine Stalnaker’s view of assertion using the structure we have so far, we would describe things in this way:

- (16) LATEST-MOVE: A asserts *p*;  
Follow-ups:



- a. option 1: B does not object: add *p* to FACTS.
- b. option 2: B does object: ???

Let us put to one side the question whether option 1 should be *B does not object* or *B actively agrees*. In fact, as we will argue in Chapter 4, with Clark (1996), the latter is probably more apt. In any case, the really problematic aspect within the current view is how to formulate option 2. Stalnaker's account abstracts away from this, but in so doing misses out on an important aspect of assertion, namely the propensity to cause *discussion* to occur. This is exemplified in the dialogue in (17) where B's assertion in (1) is challenged by c in (2); this challenge raises an issue (3) whose discussion lasts over some seven turns, the conclusion of which is a reiteration of the corrected assertion by c:

- (17) B(1): you've obviously started a trend.  
 c(2): no I don't think we're definitely following a trend.  
 B(3): who was in front?  
 c(4): oh all sorts of people—we've definitely bandwaggoned.  
 A(5): mm  
 c(6): well certainly of of many of of Jo's and your friends surely.  
 A(7): yeah, May and Jan, all sorts of people, loads of Bristol people, most of Bristol people in fact, if they knew entirely [4 sylls unclear] they should have  
 c(8): oh yes—I'm getting definitely getting very long in the tooth.  
 A(9): (laughs)  
 B(10): I sympathize.  
 c(11): so uh there's no point in my pretending I'm setting a trend.
- London–Lund Corpus (text S02.11.02, turns 172–82)

This dialogue suggests two conclusions, one specific to assertion, the other of much wider significance. Excerpt (17) brings out the fact that an assertion *that p*, apart from having the potential to increment FACTS with *the fact that p*, can also give rise to discussion which tries to establish *whether p*, which might itself lead to the discussion of other questions before it can be resolved. The more general conclusion concerns the existence of an additional contextual resource conversational participants need to keep track of, one which regulates the coherence of discussion, namely *the issue or question currently under discussion*. In fact, since more than one question can be under discussion at a given point without conversational chaos ensuing, it will turn out to be useful to think of this component of context as a set which carries an ordering relation of some kind. The maximal element of this set, call it QUD (acronym for *questions under discussion*), will constitute the current topic of discussion.

The role of questions in structuring context has been recognized in a variety of works, including Hamblin (1970), Carlson (1983), Van Kuppevelt (1995),

Ginzburg (1994 and 1996), Roberts (1996), and Larsson (2002). Hamblin (1970) contains an illuminating history of 2000 years' work in logic which employs some notion of question/answer structure to analyse rhetorical and logical fallacies. This work also contains a rudimentary propositional-logic-like system for analysing dialogues. Carlson (1983) and van Kuppevelt (1995) offer an analysis of dialogue structure based on a hierarchical ordering of questions. This is applied particularly to explicate notions of topic and focus. Van Kuppevelt (1996) further applies his theory to offer an account of scalar implicature. Roberts (1996) develops an account of context, discussed further in Chapter 4, which resembles the view developed here and earlier in Ginzburg (1994) in certain respects. Her main applications concern intonational focus and adverbial *association with focus*.

How does QUD evolve? On the account I will develop in this book, questions, semantic entities whose nature is discussed in Chapter 3, can get introduced into the context from a number of different sources, the most prototypical source being illocutionary acts such as querying and assertion. As we will see in Chapters 4, 6, and 8, modelling the posing of a question  $q$  as involving incrementation of QUD with  $q$  provides an account that generalizes to a wide number of cases, from the prototypical two-person *ignorant querier poses question to a responder who she has grounds for believing knows the answer*, through exam querying (*knowledgeable querier poses question to a responder who might or might not know the answer*), through self-answering, and querying in multiparty situations. The generality of the KoS account constitutes a significant advantage over intention-based analyses of querying, common in logic since Åqvist (1965) and Hintikka (1976), in pragmatics since Searle (1969), and particularly influential in dialogue system design in light of the work of the Toronto school (e.g. Allen and Perrault 1980, Cohen and Perrault 1979).

Other triggers for the update of QUD, which will emerge below, include locutionary acts and specific conversational genres; the latter will provide us with a means of integrating the insights of the Toronto school into context. In its evolution QUD is intermediate between FACTS, which (at least to a first approximation) grows as a conversation proceeds, and LatestMove, which changes with each move. As illustrated in (17), a question can remain under discussion for a number of turns. I will suggest that the factors that determine how long are essentially 'pragmatic', or to use a term I prefer, agent-dependent. Roughly, a question will remain under discussion as long as there are indications that there is additional *useful* information about that question that can be imparted.

One of the major claims I will attempt to defend, in Chapter 7, is that QUD is a resource on the basis of which resolution of dialogue ellipsis can be achieved. A simple example of this concerns our quarry in this section, 'yes'. We can propose the following as the meaning for 'yes':

- (18)  $p$ , where  $p?$  is maximal in QUD.

This will offer a uniform account of the cases we have seen so far: in those cases where ‘yes’ is used to affirm the positive option in a polar question  $p?$  (e.g. (1a, c), (14), (15)) the polar question has been introduced to QUD as a consequence of the query act. Although other questions might have been introduced in the interim, as in (14), (15), if the issues they raise are downdated from QUD,  $p?$  will return to become QUD-maximal. Similarly, in cases where ‘yes’ is affirming a previously asserted proposition, the resolution builds on the presence of  $p?$  in QUD as a consequence of an assertion.

Let us take stock. My starting point was certain fundamental assumptions about how meaning, context, and content relate to each other, embodied, more or less explicitly, in existing formal semantic work. The next stop was to consider the nature of context and its evolution. How does context evolve? The basic idea, pioneered in work in dynamic semantics, is simple: linguistic actions change resources. I have motivated the need for a *layered* view of context, which is to say that the contextual resources are characterized not merely by the *facts* that the conversational participants view as *common*, components of context that (to a first approximation at least) are permanent components once introduced, but also by two more transient components. The first pertains to conversational move structure: each move by a conversational participant sets up a restricted set of options for follow-up, pertaining either to the illocutionary nature of the move or, as I shall discuss in more detail below, to its locutionary nature. A second component of context, initial motivation for which was provided above, is those questions that are raised for discussion and which condition both what can be said (information pertaining to that question) and how (‘dialogue ellipsis’).

A context, on the view I have sketched so far, will have the structure in (19). I will refer to this component of context as a *dialogue gameboard* (DGB), representing as it does information that arises from public interactions. We will shortly motivate the assumption that each conversational participant has their own DGB, in addition to a *private* component of their information state.

- (19) 
$$\left[ \begin{array}{l} \text{FACTS : set of facts} \\ \text{LatestMove : (illocutionary) fact} \\ \text{QUD : partially ordered set of questions} \end{array} \right]$$

In order to manipulate such an entity, in developing a semantics where context drives content, we will need a theory of such entities. Such a theory will be provided by Type Theory with Records (Cooper 2005a), to be introduced in Chapter 3, where I develop a semantic ontology that will underpin the grammar, the theory of interaction, and the theory of meaning. With these two components in hand, introduced respectively in Chapters 4 and 5, we will be able to formalize the sketch of the meaning of ‘yes’ given in (18).

## 2.2 Communitarian Semantics Bumps into Conversation

In the previous section I articulated certain principles relating meaning, context, and content which describe (what I take to be) the commonly accepted practice of much work in semantics. This general picture, which I dubbed *Communitarian Semantics* in section 1.1, is one in which semantics characterizes successful interaction, abstracting away from individual differences among the interlocutors. In this section I point out certain dialogue phenomena that call this picture into question. In (20) are repeated three principles I discussed above, together with a fourth principle given in (20d). I have dubbed this principle Weak Montogovianism in light of a tendentious comment made by Montague in Montague (1974c). What Montague said was that he failed ‘to see any great interest in syntax except as a preliminary to semantics’. Although this remark occasioned much opprobrium for Montague,<sup>5</sup> the formulation I have put forward seems commonly adhered to. That is, the syntax (or phonology) actually has no role beyond bringing about the composition of the content. With this task achieved, the syntactic information might as well decay instantly, since it is not projected into the context. Apart from some isolated examples in the computational literature (e.g. Kehler 1993, Lappin and Gregory 1997), formal accounts of context change (e.g. those formulated in Discourse Representation Theory (Kamp and Reyle 1993), in Dynamic Predicate Logic and its relatives (e.g. Groenendijk and Stokhof 1991, Dekker 2004) and in Segmented Discourse Representation Theory (Asher and Lascarides 2003)) have assumed that syntactic information has no role to play in a theory of context; one important exception to this is Poesio and Traum’s PTT outgrowth from DRT, on which see below).<sup>6</sup>

### (20) Communitarian Semantics: some assumptions

- a. Contextualized Lewisian Regularity: whenever A utters S with meaning  $\mu$  in  $c$ , A communicates that P, where  $P = \mu[f]$ ,  $f$  is the assignment  $c$  provides for  $\mu$ , and B acquires the belief that P.
- b. Equal Access to Context: As a conversation proceeds, a common ground emerges: A has her turn, reaches a transition relevance point (TRP), then either A proceeds or B takes over **from the common-ground point at which A spoke**.
- c. The supra-contextual nature of Semantics: Semantics associates meanings with sentences and constituents of sentences.
- d. Weak Montogovianism: only the *content* of utterances (*not* their syntactic or phonological properties) contributes information that persists in the context.

<sup>5</sup> Folklore has it that this comment was, in fact, tongue very much in cheek.

<sup>6</sup> Possibly this assumption was reinforced by the slogan ‘meaning/content is preserved, structure decays’, prominent following the famous experimental results of Sachs (1967), which apparently showed just that. Later psycholinguistic results which showed a more complex picture (e.g. Kintsch and Bates 1977, Keenan, MacWhinney, and Mayhew 1977) received less prominence.

I start by considering Equal Access to Context and the Contextualized Lewisian Regularity. My basic claim is that these principles build in an altogether untenable symmetry between speaker and addressee, which abstracts away from the complexity of the communicative process, the existence of which dialogue provides many overt indications.

Consider the conversation in (21): this is primarily a *metacommunicative* conversation—Damion tries to figure out who is the person from whom Terry snagged a drink, referred to initially with a pronoun, then with the proper noun ‘Sam’, and when even that doesn’t work (there being multiple Sams in Terry and Damion’s acquaintance), as ‘the other one [not with Kevin]’.

- (21) Terry: Yeah I but think he gave me all his drink.  
Damion: Who?  
Terry: Sam.  
Damion: He gave it to you?  
Terry: No, no, I was (laughing) drinking all his drinks.  
Damion: Which Sam?  
Terry: Sam, Sam, the one  
Damion: The one  
Terry: who was totally pissed.  
Damion: Oh [last or full name] or the other one?  
Terry: No have you got, don’t you know the other one?  
Damion: No there’s two we know.  
Terry: Yeah.  
Damion: The one with Kevin  
Terry: The one with the longish  
Damion: or the other one.  
Terry: it’s the other one.  
Damion: Oh right.  
(BNC, KR2)

(21) despite being a perfectly normal and unremarkable conversation points to some very significant issues, highlighting intrinsic shortcomings of the lion’s share of contemporary semantic frameworks. For a start, the conversation points to a difficulty with the meaning/content relation as standardly implemented (see section 2.1):<sup>7</sup> Damion spends just about the entire conversation without a referent associated with ‘he’, and ‘Sam’, though eventually the matter is put right. This is hard to square with the Contextualized Lewisian Regularity: where a function gets fed its value (a contextual tuple of some kind), yielding a content as value, or no value whatever if something goes wrong. This technical quibble is a symptom of a far deeper malaise: with some notable

<sup>7</sup> Thanks to Robin Cooper for enlightening conversations on this issue.

exceptions to be discussed below, linguistic semantics still operates under the simplifying assumption that **perfect communication** obtains—nothing does go wrong and interpretation leads to an identical update on Terry's and Damion's information states. But this means giving up on trying to make sense of conversations such as (21) and most concretely the analysis of the meaning of NSUs there such as 'Who?', 'Which Sam', and 'Oh [last or full name] or the other one?' that express metacommunicative queries.

For all its plausibility, there is reason to doubt a simplistic version of Equal Access. The data in (22) and (23), involving the resolution of NSUs, are incompatible with (the above version of) Equal Access. Example (22) illustrates that the contextual possibilities for resolving the fragment 'Bo?' are distinct for the original speaker A and the original addressee B. Whereas there is one common possibility, the short answer reading, only B has the two clarification request readings, whereas only A has a self-correction reading, albeit one that probably requires an additional elaboratory follow-up:

- (22) a. A: Who does Bo admire? B: Bo?  
         Reading 1 (**short answer**): Does Bo admire Bo?  
         Reading 2 (**clausal confirmation**): Are you asking who Bo (of all people) admires?  
         Reading 2 (**intended content**): Who do you mean 'Bo'?  
       b. A: Who does Bo admire? Bo?  
         Reading 1 (**short answer**): Does Bo admire Bo?  
         Reading 2 (**self-correction**): Did I say 'Bo'?

As I will discuss in Chapter 6, fragments of this kind, henceforth *reprise fragments* (RFs), are frequent: the **clausal confirmation** and **intended content** readings constituting approximately 30 per cent of all clarification requests (CRs).

Example (23) is an even more striking illustration of the phenomenon I have dubbed the Turn-Taking Puzzle (see Ginzburg 1997): here the resolution accorded to the bare 'why' changes according to who keeps or takes over the turn. The resolution that can be associated with 'Why?' if A keeps the turn is unavailable to B were s/he to have taken over **and** vice versa. (23c) shows that these facts cannot be reduced to coherence or plausibility—the resolution unavailable to A in (23a) yields a coherent follow-up to A's initial query if it is expressed by means of a non-elliptical form:

- (23) a. A: Which members of this audience own a parakeet? Why? (= Why own a parakeet?)  
       b. A: Which members of this audience own a parakeet? B: Why? (= Why are you asking which members of this audience own a parakeet?)  
       c. A: Which members of this audience own a parakeet? Why am I asking this question?

The Turn-Taking Puzzle vitiates Equal Access, since it demonstrates in a rather direct way that, at least some of the time, the contextual options available to one participant are distinct from those available to the other(s) participant(s). At the same time, the Turn-Taking Puzzle is not an argument for solipsism, for instance of a type advocated in Relevance Theory (Sperber and Wilson 1986), the only influential approach which explicitly avoids assuming Equal Access. For a start, the Turn-Taking Puzzle reflects an asymmetry of production, not understanding: it seems clear that both participants would understand the potential contributions of the other.<sup>8</sup> Second, merely adopting a solipsistic approach does not offer an account of the particular asymmetries displayed and readings available in the Turn-Taking Puzzle.

Let us now re-examine the two other assumptions I have called the Supra-contextual Nature of Semantics and Weak Montogovianism. In standard semantic frameworks the various basic semantic notions (truth or update conditions) are defined for *sentences*. Thus, despite the fact that it is by now universally recognized that it is only *relative to a context* that truth is to be evaluated or an update performed, the distinction between a sentence *S* (whose effect is relativized to a context) and an utterance *u*, a spatio-temporally located event where words get uttered in succession, remains more often than not unheeded. Indeed there have been arguments that the sentence-based approach is preferable, most prominently by Kaplan (1989).<sup>9</sup>

On the issue whether there might be a role for platonic entities such as sentences in grounding inference, as suggested in Kaplan (1989), I will remain silent. The data I point to, which requires adopting an utterance-based perspective, concerns the evolution of context in dialogue. Example (24) illustrates that if *A* makes the utterance in (24a) a variety of facts about the utterance (in boldface in (24b–e)) potentially enter into the common ground. This is evinced by the possibility of embedding them under a factive-presupposition predicate such as ‘interesting.’ Example (24) exemplifies two classes of facts about the utterance that become presupposable, facts about the content of sub-utterances (24b–d) and also facts that concern solely the phonology and word order of the utterance.

- (24) a. *A*: Did Mark send you a love letter?  
 b. *B*: No, though it’s interesting **that you refer to Mark/my brother/our friend**  
 c. *B*: No, though it’s interesting **that you bring up the sending of love letters**  
 d. *B*: No, though it’s interesting **that you ask about Mark’s epistolary habits**  
 e. *B*: No, though it’s interesting **that the final two words you just uttered start with ‘I’**

The cases in (25) are all naturally occurring examples, respectively spontaneous speech, text, and email: (25a) illustrates reference to the fact that a sub-utterance was

<sup>8</sup> I owe this point to Mandy Simons, in conversation.

<sup>9</sup> And see also Richard (1993) and Partee (1985) for similar sentiments; for some arguments contra, along with a good review of Kaplan’s and related approaches, see Israel and Perry (1996).

a misreference; (25b), analogous to the constructed (24e), illustrates reference to the phonological properties of a sub-utterance; (25c) illustrates reference to the word used in a sub-utterance:

- (25) a. This is work done with Marilyn Kutich, no Mary Ellen Kutich. Now that's an interesting speech error (Herb Clark in a talk, 26 January 2006).  
 b. Tomorrow night will also see Milan—just a point behind Inter—face yet another make-or-break game away to Livorno. Quite apart from their horror at facing any side whose name begins with those three letters,<sup>10</sup> there's plenty to worry Milan here. (*The Guardian*, 20 December 2005)  
 c. 'Struth,' says Jon Mann, clearly not trying to hide his nationality. 'You Poms fluke one Test match (by just two runs), now you think you can win the series! Not a chance mate—I reckon you guys have had one Fosters too many. And even rain won't save you this time, Bruce, because you need to win one of the remaining matches. Sorry guys I mentioned "win" there, you Poms might need to look that word up.' (Test match over-by-over coverage, *The Guardian*, 25 August 2005)

The fact that non-semantic information such as phonology or word order associated with an utterance needs to filter into the context and stay there for a while does not of course mean that all such information is equally *accessible*. The lion's share of these facts cannot actually be picked up by means of ellipsis/anaphora. Thus, consider (26) as follow-ups to (24a):

- (26) a. B: (No, though) why? (= *Why are you asking whether Mark sent me a love letter*; cannot mean: *Why do you refer to Mark/my brother/our friend, Why do you bring up the sending of love letters*, etc.).  
 b. B: (No, though) Don't you think that's a bit over-inquisitive? ('that' = *your asking me whether Mark sent me a love letter*)  
 c. B: (No, though) that is amazing—Jill asked me the same question yesterday. ('that' = *your asking me whether Mark sent me a love letter*)

I will refer to those facts that are easily accessible as *topical*. How to model topicality will be discussed in Chapter 8.

A further argument for this utterance-based perspective is that the sub-utterance tokens figure in the content of CRs. The examples in (27) all involve an intrinsic anaphoricity: in these examples the issue is not *what do you mean by leaving or who is Bo/we in general*, but *what do you mean by leaving or who is Bo/we in the sub-utterance needing clarification*.

- (27) a. A: Max is leaving. B: leaving?  
 b. A: Did Bo leave? B: Who is Bo?  
 c. A: We're fed up. B: Who is we?

<sup>10</sup> Milan had lost the 2005 Champions League to *Liverpool*, after surrendering a three-goal lead.



We have already seen some data that go against Weak Montogovianism: the pre-suppositions about phonology and word order that emerge in the aftermath of an utterance. An additional set of phenomena, discussed in detail in Chapters 6 and 7, are phonological and syntactic parallelism phenomena in CRs and NSUs. Example (28) illustrates that clausal confirmation readings require partial syntactic parallelism: a phrase used to clarify that an antecedent sub-utterance  $u_1$  must match  $u_1$  categorially:

- (28) a. A: I phoned him. B: Him? / #He?<sup>10a</sup>  
b. A: Did he phone you? B: He? / #Him?  
c. A: Did he adore the book. B: Adore? / #Adored?  
d. A: Were you cycling yesterday? B: Cycling?/Biking?/#Biked?

An even stronger parallelism requirement applies to the intended content reading of a reprise fragment: this involves (segmental) phonological identity with the source:

- (29) A: Did Bo leave? B: Max? (cannot mean: intended content reading: **Who are you referring to?** or **Who do you mean?**)

Of course the persistence of non-semantic information in context is very much an empirical question, but discarding Weak Montogovianism seems a necessary first step in accounting for a growing body of work in psycholinguistics that demonstrates the existence of non-semantically-based priming (see e.g. Branigan, Pickering, and Cleland 2000, Garrod and Pickering 2004).

### 2.3 Integrating Semantics and Metacommunication

In the previous section I presented data that call into question certain fundamental assumptions of a large body of semantic theory, in linguistics, the philosophy of language, and computational work, a view I have dubbed Communitarian Semantics. I state the positive conclusions we can draw on the basis of this data, the nucleus of the interactive stance, the stance which will guide the development of KoS.

- (30) a. **Meaning, content, and clarification potential:** knowledge of meaning includes knowledge of clarification interaction possibilities. In other words, an important benchmark for any semantic theory of dialogue is to accommodate as coherent and characterize the range of clarificatory potential of utterances.  
b. **Speaker/addressee asymmetries:** the contexts available to the conversationalists in the aftermath of an utterance are NOT identical, as illustrated by various turn-taking puzzles. This entails a cognitive architecture in which there is no single common ground, but distinct yet coupled DGBs, one per conversationalist.

<sup>10a</sup> Here and below, the symbol # marks an utterance as (possibly grammatical but) infelicitous.

- c. **Rich but graded update:** information pertaining to syntactic and phonological aspects of an utterance, not merely the utterance's illocutionary *content*—on whatever theoretical explication of this concept—becomes presupposed following a grounded utterance. This point has also been argued for extensively by Massimo Poesio; see e.g. Poesio and Traum (1997), Poesio (1998), Poesio and Rieser (2010). This requires a means of keeping track of non-semantic information associated with an utterance in context after its successful processing. In turn, this necessitates a grammatical ontology that furnishes all this information in parallel—a *sign-based grammar* of some kind, as explained in Chapter 5.

The bottom line is the need for an enrichment of the semantic perspective by integrating metacommunicative and illocutionary aspects of interaction and by taking a more holistic view of the contribution of utterances to context. As we discuss in more detail in Chapter 6, there is in fact evidence from computational simulations (see among others Macura 2007) that a metacommunicative component in interaction is not some incidental add-on but rather plays a vital role in maintaining a language from irretrievably diverging, Tower of Babel-style, across its speakers. Apart from its importance in accounting for phenomena of a kind we have just seen, clarification potential becomes an important tool for various other purposes: it offers the possibility of defining a strengthened notion of *compositionality* (see the *Reprise Content Hypothesis* introduced in Purver 2004a, discussed in Chapter 5). It also enables us to shed some new light on the crucial issue of the extent of integration of 'mind reading', viz. unpublicized intention recognition involved in interpretation and direct/indirect content (see Chapter 8).

A theory that can accommodate the requirements in (30) has not existed to date, but KoS draws on a number of existing approaches, outwith the communitarian semantics mainstream, which far from shunting aside metacommunication, have focussed significant attention on it. One research tradition which *has* emphasized the robustness of the communicative process despite the ever-present possibilities of *trouble* is Conversation Analysis (CA) (Schegloff, Jefferson, and Sacks 1977b, Jefferson 1983, Schegloff 1992, 2007). The basic CA notion of coherence is the *adjacency pair*, as mentioned in section 2.1. Any turn that deviates from the structure imposed by adjacency pair is viewed as *repair*:

By 'repair' we refer to efforts to deal with trouble in speaking, hearing, or understanding talk in interaction. 'Trouble' includes such occurrences as misarticulations, malapropisms, use of a 'wrong' word, unavailability of a word when needed, failure to hear or to be heard, trouble on the part of the recipient in understanding, incorrect understandings by recipients, and various others. Because anything in talk can be a source of trouble, everything in conversation is, in principle, 'repairable.'

(Schegloff 1987b: 210)

Repair, then, is a very general cover term for a wide range of moves, including corrections, clarification requests, and clarifications. CA offers an essentially structurally-based taxonomy of repair: *self-initiated within utterance correction*, (other-initiated) *next-turn repair initiations* (NTRIs),<sup>11</sup> *third-position* (by the original utterer), and *fourth-position* (by the original addressee) corrections. These are illustrated, respectively, in (31):<sup>12</sup>

- (31) a. A: Did Bo ... Bro leave?  
b. A: Did Bro leave? B: Bro?  
c. A: Did Bro leave? B: There he is. A: I meant Bro Eliasson.  
d. A: Did Bro leave? B: There he is. A: Oh dear. B: He's not called Bro in any case.

Many significant observations have originated with CA, including the following two: first, addressees tend to avoid repairing speakers in mid utterance ('the preference for self-initiated repair'), second, other-initiated repair is almost invariably in the next turn.<sup>13</sup> CA emphasizes the need for relating though not conflating self-repair and other-repair, a preliminary sketch of which is provided in Chapter 8.

On the other hand, largely by design, CA provides no formal analysis as such, either on the level of grammar (the structure and meaning of repair utterances), or on the level of contextual development or cognitive modelling. In particular, no theory is provided for what might constitute 'repairable trouble'. Indeed CA has, again for methodological reasons, eschewed (quantative) empirical characterization of repair phenomena, for instance the distribution of actually occurring CRs, both formally and functionally. The above quote from Schegloff suggests that 'anything in talk can be a source of trouble'. If true, this would be a bit of a disaster for the prospects of a computational theory of repair. In fact, as we will see in Chapter 6, corpus studies of CRs demonstrate the existence of various systematic restrictions on the aspects of an utterance for which conversational participants seek clarification. The lion's share of CRs concern reference resolution and mishearing. The range of actual and possible CRs, in turn, feeds in as a minimal adequacy criterion for the theory of utterance representation, yielding clues as to what are potential sources of uncertainty for conversational participants.

Probably the most influential cognitively-oriented perspective which builds into interaction a layer of metacommunication derives from the work of Herb Clark and his associates (Clark and Wilkes-Gibbs 1986, Clark and Schaefer 1989, Clark 1996), on *grounding* an utterance. The main claim that originates with Clark and Schaefer

<sup>11</sup> In Schegloff (2000), based on a variety of data, Schegloff points out the weakness of this term since adjacency is built too strongly into clarification interaction.

<sup>12</sup> For whatever it's worth, fourth-position repair is, nonetheless, very rare indeed, as Pat Healey (p.c.) has pointed out to me.

<sup>13</sup> But see discussion in Schegloff (2000). See also discussion of clarification potential concerning unpublishized intention in Chapter 8, where locality is very naturally avoided.

(1989) is that any dialogue move  $m_1$  made by A must be grounded by the other conversational participant B before it enters the common ground.<sup>14</sup> B does this only if she understands  $m_1$  relative to her own purposes. Otherwise, B must issue a clarification request. As far as grounding a completed utterance is concerned, Clark suggests that the basic mechanism is by providing a second-part adjacency pair response, though of course there is also the option of an explicit acknowledgement move.

Clark's work on grounding is driven by corpus and experimental data, but is not formalized. However, it has inspired a significant body of computational and formal work, starting with Traum (1994). Traum offers a computationally explicit account, formulated as a finite state machine, of the possible range of move *types* (e.g. 'acknowledgement', 'correction', 'repair', etc.) that follow up on an utterance, and also provides a semantics for such moves, in terms of the background and effect on an agent mental state. Subsequently, Traum's account of grounding has been fused with a version of Discourse Representation Theory developed by Massimo Poesio (see e.g. Poesio and Traum 1997, 1998, Poesio and Rieser 2010) in a framework called PTT.<sup>15</sup> In addition to integrating grounding interaction into the general semantic interpretation process, the tools of DRT provide for an account of anaphoric uses to previous utterances, important in an analysis of grounding acts. One of the important insights that PTT develops, tying in with the realist perspective on utterances argued for above, is that of utterances as complex events constituted from *micro conversational events*, the events associated with the utterance of individual words and phrases. This yields an elegant account of the acknowledgement of utterances midstream and the potential for collaborative utterances.

A crucial ingredient in Clark's notion of grounding is the notion of *understanding an utterance relative to an interlocutor's current purpose*. Clark does not formalize or reductively analyse this notion and, for the most part, neither has PTT work.<sup>16</sup> This reflects, as with earlier work in the Communitarian Semantics tradition, an almost exclusive concentration on (perceived) successful communication, where grounding actually takes place.

A related problem for Clark's model concerns collaboration: Clark's assumption is that grounding by the other conversational participant is a necessary condition for utterances to enter *the* common ground. Allwood (1995), an additional important source on integration of metacommunication into interaction, argues that one-sided communicative acts are possible, citing examples such as (32).

- (32) a. I was referring to Bertrand Russell but she did not hear me. (Example (11) in Allwood 1995.)  
 b. I warned him but he did not hear me. (Example (7) in Allwood 1995.)

<sup>14</sup> Traum (1994) argues persuasively for weakening this assumption and allowing for certain moves—primarily acknowledgements—to go unacknowledged.

<sup>15</sup> Like KoS, PTT is not an acronym, but relates in some fashion to its progenitors' initials.

<sup>16</sup> For discussion, see Traum (1999).

Allwood suggests that ‘Contributions in the form of “acknowledging feedback”, (a notion from Allwood 1976—J.G.) are not needed to constitute speech acts but rather to inform the interlocutor of the extent to which his communicative objectives are met’ (Allwood 1995: 8). In line with Allwood’s perspective, in KoS an utterance will have an asymmetrical effect on context. It will be immediately integrated by its speaker into her DGB, but it will be integrated by the addressee only if she believes she has a means of instantiating the contextual parameters in a way consistent with the speaker.

## A Semantic Ontology for Dialogue

### 3.1 Introduction

Khalil wonders whether Jerusalem is calm. It's a reasonable question and has been for a millenium or two. How are we to describe Khalil's information state? It depends of course on what we're aiming to achieve. As I explained in Chapter 2, one of our aims is to come up with an account of dialogue context that can predict, for instance, the range of possible clarification requests to an utterance. To do that one needs a quite fine-grained view of the make-up of information states. For instance, as a speaker of Arabic, Khalil knows Jerusalem primarily by the name 'ElQuds', though when he speaks English he refers to it as 'Jerusalem'. Khalil's Greek friend Cybeli knows the latter name, but not 'ElQuds'. A fine-grained view of the make-up of information states will need to distinguish between the update effected by the Arabic (1a), as opposed to its English gloss (1b). This, given that e.g. (1c), but not its English gloss (1d), can be used to request a clarification of (1a), but not of (1b):

- (1) a. ElQuds hadi'a elyom?
- b. Is Jerusalem calm today?
- c. Eyš bi'ani ElQuds?
- d. What do you mean 'Jerusalem'?

And yet, semantics classically concerned itself with classifying the import of utterances in terms of their 'external import'. In other words, focussing on truth conditions or descriptive content or similar notions—tying an utterance to a range of situations it accurately describes (or analogous notions for non-declaratives). And such a concern is of course still warranted for dialogue, not least because we want to capture the commonality in import between Khalil's asking (1a) and Cybeli's asking (1b) more or less simultaneously. Our task in this chapter is to develop an ontology that is up to both tasks—provide entities adequate for describing the fine-grained updates caused by utterances, as well as the somewhat more coarse-grained 'external reality'.

Concretely, a semantic ontology for dialogue needs to provide the following:

- (2) a. **A theory of events/situations:** a natural starting point for the ontology is some notion of an event or situation, a spatio-temporally located part of the

world. These are the entities we appeal to in order to explicate reference, speech events, and more generally the external reality with which agents interact.

- b. **A theory of abstract entities:** propositions and questions play an important role in structuring context, as discussed in Chapter 2. The ontology needs to offer an explication of propositions, questions, and a number of related entities.
- c. **Grounding/clarification interaction conditions:** the ontology needs to provide entities off which one can read grounding/clarification interaction conditions associated with utterance types. This also involves supplying the semantic means for a meaning/context relation that can accommodate the existence of partial utterance comprehension.

As reviewed in Ginzburg (2010b), situation semantics (e.g. Barwise and Perry 1983, Barwise and Etchemendy 1987, Cooper and Poesio 1994, Seligman and Moss 1997, Ginzburg and Sag 2000) provides an ontology that fulfils the first two of these three desiderata. The most recent version of this ontology, originating in Ginzburg and Sag (2000), was dubbed a *Situational Universe with Abstract Entities* (SU+AE). Its development was in line with the original situation-theoretic strategy, embodied in the following quote:

- (3) The primitives of our theory are all real things: individuals, properties, relations, and space-time locations. Out of these and objects available from the set theory we construct a universe of abstract objects. (Barwise and Perry 1983: 178)

*Situational Universes with Abstract Entities* underpin a wide range of linguistic applications, including:

- 1. perception complements, event anaphora, definite NP reference, based on a theory of situations;<sup>1</sup>
- 2. a fine-grained theory of propositions that offers a solution to the liar paradox;
- 3. a theory of questions as propositional abstracts that provides a detailed account of answerhood.

However, as a new millennium dawned, the theory was hamstrung by a number of foundational problems. The logical underpinnings for the theory in terms of non-well-founded set theory, the most comprehensive of which is Seligman and Moss (1997), were rather complex.<sup>2</sup> Concretely, simultaneous  $\lambda$ -abstraction with

<sup>1</sup> For computational work on definites using situations, integrated also with visual information, see Poesio (1993). For experimental work on the resolution of definites in conversation taking a closely related perspective see Brown-Schmidt and Tanenhaus (2008).

<sup>2</sup> Much of the motivation for the theory is in Barwise (1989a). See also the series of volumes *Situation Theory and its Applications*.

restrictions is a tool with a variety of uses, including quantification, questions, and the specification of attitudinal states and meanings. Its complex set-theoretic characterization made it difficult to use. Concomitantly, the theory in this form required an auxiliary coding into a distinct formalism (e.g. typed feature structures) for grammatical and computational applications. Neither of these versions of the theory provides an adequate notion of role dependency, whereas this has become standard in recent treatments of anaphora and quantification, on which much semantic work has been invested in frameworks such as Discourse Representation Theory and Dynamic Semantics.<sup>3</sup>

Motivated to some extent by such concerns, the situation-theoretic outlook has been redeveloped using tools from Type Theory with Records (TTR), a framework initiated by Robin Cooper—see Cooper (2011) for the most detailed and extensive presentation of TTR. Ever since Sundholm and Ranta's pioneering work (Sundholm 1986, Ranta 1994), there has been interest in using constructive type theory (CTT) (often referred to as Martin-Löf Type Theory (MLTT)) as a framework for semantics (see e.g. Fernando 2001, Krahmer and Piwek 1999). TTR is a model-theoretic outgrowth of CTT. Its provision of entities at both levels of tokens and types distinguishes TTR from (standard) Discourse Representation Theory,<sup>4</sup> for instance, where the witnesses are at a model-theoretic level, distinct from the level of discourse representations. Providing entities at both levels of tokens and types allows one to combine aspects of the typed feature structures world and the set-theoretic world, enabling its use as a computational grammatical formalism.

As we will see, TTR enables one to build an ontology for dialogue that satisfies all the desiderata I posited in (2). In particular, by combining the situation-theoretic conception of propositions with the fine-grained notions of utterance types TTR provides, we will be able to construct entities which encode grounding/clarification interaction conditions.

## 3.2 The Type-Theoretic World: Basic Notions

### 3.2.1 Judgements

The most fundamental notion of TTR is the typing *judgement*  $a : T$  classifying an object  $a$  as being of type  $T$ . Examples are given in (4). Those in (4a, b) involve *basic* 'atomic' types IND(ividual) and TIME. In (4c)  $run(b, t)$  is a *complex type*, that arises by assigning the entities  $b, t$ , respectively to the argument roles of *run*;  $arg1_{IND}$  requires

<sup>3</sup> See Gawron and Peters (1990) for a detailed theory of anaphora and quantification in situation semantics, though one that is not dynamic.

<sup>4</sup> There are versions of DRT that do allow for the presence of witnesses in the logical representation, e.g. Compositional DRT (Muskens 1996), employed to underpin the PTT dialogue framework (Poesio and Rieser 2010).



its fillers to be : IND, whereas  $arg2_{TIME}$  requires its fillers to be : TIME. Ranta (1994) suggested that elements such as  $s$  in (4c) be viewed as events or situations.

- (4) a.  $b : IND$   
 b.  $t : TIME$   
 c.  $s : run(arg1_{IND} : b, arg2_{TIME} : t)$

### 3.2.2 Records and Record Types

A useful innovation in TTR relative to earlier version of type theory is records and record types. A record is a set of fields assigning entities to labels, partially ordered by a notion of *dependence* between fields. Its general form is as in (5):

$$(5) \begin{bmatrix} l_1 = val_1 \\ l_2 = val_2 \\ \dots \\ l_n = val_n \end{bmatrix}$$

Together with records come record *types*. Technically, a record type is simply a record where each field represents a judgement rather than an assignment, as in (6).

$$(6) \begin{bmatrix} l_1 : T_1 \\ l_2 : T_2 \\ \dots \\ l_n : T_n \end{bmatrix}$$

The basic relationship between the two is that a record  $r$  is of type  $RT$  if each value in  $r$  assigned to a given label  $l_i$  satisfies the typing constraints imposed by  $RT$  on  $l_i$ . More precisely,

$$(7) \text{ The record } \begin{bmatrix} l_1 = a_1 \\ l_2 = a_2 \\ \dots \\ l_n = a_n \end{bmatrix} \text{ is of type: } \begin{bmatrix} l_1 : T_1 \\ l_2 : T_2 \\ \dots \\ l_n : T_n \end{bmatrix} \\ \text{iff } a_1 : T_1, a_2 : T_2, \dots, a_n : T_n$$

Not all the fields in  $r$  need to be ‘disciplined’ by  $RT$ . Thus, assuming the judgements in (8a) hold, then the record in (8b) is of all the types (8c–f). Indeed, all records are of the empty type (8f), the type that imposes no constraints.

- (8) a. bo : IND, 2pm, Dec 20 : TIME, batumi : LOC  
 b.  $\left[ \begin{array}{ll} \text{runner} = & \text{bo} \\ \text{time} = & \text{2pm, Dec 20} \\ \text{place} = & \text{batumi} \end{array} \right]$   
 c.  $\left[ \begin{array}{ll} \text{runner} & : \text{IND} \\ \text{time} & : \text{TIME} \\ \text{place} & : \text{LOC} \end{array} \right]$  d.  $\left[ \begin{array}{ll} \text{runner} & : \text{IND} \\ \text{time} & : \text{TIME} \end{array} \right]$  e.  $\left[ \text{runner} : \text{IND} \right]$   
 f.  $\left[ \right]$

### 3.2.3 Situations and Frames

Cooper (2005a) proposes that events/situations be modelled as records.<sup>5</sup> The type of a situation with a woman riding a bicycle would then be the one in (9a). A record of this type (a *witness* for this type) would be as in (9b), where the required corresponding typing judgements are given in (9c):

- (9) a.  $\left[ \begin{array}{ll} x & : \text{IND} \\ c1 & : \text{woman}(x) \\ y & : \text{IND} \\ c2 & : \text{bicycle}(y) \\ \text{time} & : \text{TIME} \\ \text{loc} & : \text{LOC} \\ c3 & : \text{ride}(x, y, \text{time}, \text{loc}) \end{array} \right]$  b.  $\left[ \begin{array}{ll} \dots & \\ x & = a \\ c1 & = p1 \\ y & = b \\ c2 & = p2 \\ \text{time} & = t0 \\ \text{loc} & = l0 \\ c3 & = p3 \\ \dots & \end{array} \right]$   
 c. a : IND; p1 : woman(a); b : IND; p2 : bicycle(b); t0 : TIME; l0 : LOC; p3 : ride(a, b, t0, l0);

As another illustration of similar notions, consider the *frames* introduced by Fillmore to explicate lexical meaning.<sup>6</sup> Cooper (2011) shows how frames can be modelled as record types. He illustrates this with the frame *Ambient temperature*, as defined in the Berkeley FrameNet2 (10a), motivated by examples such as (10b, c):<sup>7</sup>

<sup>5</sup> In a series of recent papers (e.g. Fernando 2007a, b), Tim Fernando has provided a type-theoretic account of the *internal* make-up of situations. Events and situations are represented by strings of temporally ordered observations, on the basis of which the events and situations are recognized. This allows a number of important temporal constructions to be derived, including Allen's basic interval relations (Allen 1983) and Kamp's event structures (Kamp 1979). Observations are generalized to temporal propositions, leading to event types that classify event instances.

<sup>6</sup> For an early reference see e.g. (Fillmore 1985), with the current Framenet database and references accessible from <<http://framenet.icsi.berkeley.edu/>>.

<sup>7</sup> <<http://framenet.icsi.berkeley.edu/>> consulted on 16 February 2011.

- (10) a. The Temperature in a certain environment, determined by Time and Place, is specified.  
 b. It's too hot to do anything today.  
 c. Nome is pretty cold this time of year.

Simplifying somewhat, this leads one to posit that this frame has core elements including:

- **Attribute:** The temperature feature of the weather
- **Place:** The Place where it is a certain Temperature
- **Temperature:** A quantity or other characterization of the Temperature of the environment.
- **Time:** The Time during which an ambient environment has a particular Temperature

This frame can be modelled by the record type in (11a); an instance of such a frame would be a record as in (11b), satisfying the constraints in (11c):

- (11) a.  $\text{AmbTemp} =_{\text{def}} \left[ \begin{array}{ll} x & : \text{IND} \\ \text{e-time} & : \text{TIME} \\ \text{e-loc} & : \text{LOC} \\ c_{\text{temp-at-in}} & : \text{temp\_at\_in}(\text{e-time}, \text{e-location}, x) \end{array} \right]$
- b.  $\left[ \begin{array}{ll} x & = -18 \\ \text{e-time} & = 3:45\text{AM}, 17 \text{ Feb } 2011 \\ \text{e-loc} & = \text{Nome} \\ c_{\text{temp-at-in}} & = \text{o1} \end{array} \right]$
- c.  $-18 : \text{IND}; 3:45\text{AM}, 17 \text{ Feb } 2011 : \text{TIME}; \text{Nome} : \text{LOC}; \text{o1} : \text{temp\_at\_in}(3:45\text{AM}, 17 \text{ Feb } 2011, \text{Nome}, -18)$

**Exercise 3.2.1** Formalize the frame Communication, defined in (12):

- (12) A Communicator conveys a Message to an Addressee; the Topic—the entity that the proposition or propositions conveyed relate to—and Medium of the communication—the physical or abstract setting in which the Message is conveyed—also may be expressed.

### 3.2.4 Propositions

Propositional-like entities, more intensional than events/situations, are a necessary ingredient for accounts of illocutionary acts, as well as of attitude reports. *Sets* of situations, although somewhat more fine-grained than sets of worlds, also succumb to sophisticated variants of logical omniscience (see e.g. Soames's puzzle: Soames 1985).

Building on a conception articulated thirty years earlier by Austin (Austin 1961), Barwise and Etchemendy (1987) developed a theory of propositions in which a proposition is a structured object  $prop(s, \sigma)$ , individuated in terms of a situation  $s$  and a situation type  $\sigma$ . Given the ‘:’ relation between situations and their types there is a straightforward notion of truth and falsity:

- (13) a.  $prop(s, \sigma)$  is true iff  $s : \sigma$  ( $s$  is of type  $\sigma$ ).  
 b.  $prop(s, \sigma)$  is false iff  $s \not:\sigma$  ( $s$  is not of type  $\sigma$ ).

There are two ways to maintain the insights of an Austinian approach in TTR, implicitly Austinian or explicitly so. Cooper (2005a) develops the former, in which a proposition  $p$  is taken to be a record type. A witness for this type is a situation, as e.g. (9b). On this strategy, a witness is not directly included in the semantic representation. Indeed, record types *are* competitive in such a role: they are sufficiently fine-grained to distinguish identity statements that involve distinct constituents. As will become clear once we illustrate how contents emerge from meanings in this set-up in section 3.3.2, (14a) would correspond to the record type in (14c), whereas (14b) would correspond to the record type in (14d). Moreover, in this set-up substitutivity of co-referentials (14e) and cross-linguistic equivalents ((14f) is the Hebrew equivalent of (14a)) can be enforced:

- (14) a. Enescu is identical with himself.  
 b. Poulenc is identical with himself.  
 c.  $\left[ c : \text{Identical}(\text{enesco}, \text{enesco}) \right]$   
 d.  $\left[ c : \text{Identical}(\text{poulenc}, \text{poulenc}) \right]$   
 e. He is identical with himself.  
 f. Enesku zehe leacmo.

A situational witness for the record type could also be deduced to explicate cases of event anaphora, as in (15); indeed, a similar strategy will be invoked when we sketch an analysis of nominal anaphora in Chapter 8:

- (15) a. A: Jo and Mo got married yesterday. It was a wonderful occasion.  
 b. A: Jo’s arriving next week. B: No, that’s happening in about a month.

Nonetheless, here we develop an explicitly Austinian approach, where the situational witness is directly included in the semantic representation. The original Austinian conception was that  $s$  is a situation deictically indicated by a speaker making an assertion<sup>8</sup>—teasing out the semantic difference between implicit and explicit witnesses is

<sup>8</sup> One could also construe  $s$  as *evidence* (a body of knowledge, a database) which provides support (or otherwise) for the type  $\sigma$ . The potential fruitfulness of this construal was pointed out to me by Nicholas Asher (p.c.); Barwise was possibly hinting in this direction in talking about (Sherlock) Holmesian propositions in Barwise (1989b).

a difficult semantic task.<sup>9</sup> However, propositions can also play a role in characterizing the communicative process: in Chapter 6 I will show that *locutionary propositions* individuated in terms of an utterance event  $u_0$  as well as to their grammatical type  $T_{u_0}$  allow one to simultaneously define update and clarification potential for utterances. In this case, there are potentially many instances of distinct locutionary propositions, which need to be differentiated on the basis of the utterance token—minimally any two utterances classified as being of the same type by the grammar.

Assuming we adopt an explicitly Austinian approach, then the type of propositions is the record type (16a). The correspondence with the situation-semantics conception is quite direct. We can define truth conditions as in (16b).

$$(16) \quad \begin{array}{l} \text{a. Prop} =_{\text{def}} \left[ \begin{array}{ll} \text{sit} & : \text{Record} \\ \text{sit-type} & : \text{RecType} \end{array} \right] \\ \text{b. A proposition } p = \left[ \begin{array}{ll} \text{sit} & = s_0 \\ \text{sit-type} & = ST_0 \end{array} \right] \text{ is true iff } s_0 : ST_0 \end{array}$$

### 3.3 Constructing Entities in the Type-Theoretic World

#### 3.3.1 *Connecting to the Real World: Situations and their Types*

The preceding section gave a taste of the basic entities of TTR. In what follows I show how the Type-Theoretic World can provide us with the remaining notions we need to develop our theory of interaction. What then is the Type-Theoretic World? Cooper (2005a, 2011) offers essentially the following answer:<sup>10</sup> the Type-Theoretic World consists of entities classified by types. The entities and types are built recursively from basic entities and types by means of:

1. complex types: predicates whose typed argument roles get assigned entities from the world (technically, a given *model*)
2. record and record types defined with respect to a set of objects used as labels
3. type construction operations: function types, list/set types, singleton types, dual types, and conjunction/disjunction types

The type construction operations I assume here, given in (17), are with two exceptions by and large familiar from various past semantic work. Function types are familiar from Montague Semantics, though as we will see, a TTR setting provides a rich notion of abstraction; they will be used to model questions, but also grammatical and conversational rules; list types are commonly used in constraint-based grammars; conjunction/disjunction types are familiar logical notions. The less familiar constructors are dual and singleton types. Dual types correspond to negative situation types

<sup>9</sup> An additional consideration is that this allows us to maintain a conceptual distinction between questions and property denotations.

<sup>10</sup> This is a simplified description of Cooper's proposal, which the careful reader would do well to consult.

in situation semantics and are used to explicate the effect of negation. Singleton types are an innovation in TTR (going back to Coquand, Pollack, and Takeyama 2003), enabling cross-label dependencies to be captured within record types:

- (17) a. **function types:** if  $T_1$  and  $T_2$  are types, then so is  $(T_1)T_2$ , the type of functions from elements of type  $T_1$  to elements of type  $T_2$ .  $f : (T_1)T_2$  iff  $\{a | a : T_1\}$  is the domain of  $f$  and the range of  $f$  is a subset of  $\{a | a : T_2\}$ .
- b. **List/finite set types:** if  $T$  is a type, then  $List(T)$  ( $FiniteSet(T)$ ), the type of lists (finite sets) each of whose members is of type  $T$ , is also a type. The list  $[a_1, \dots, a_n] : List(T)$  (finite set  $\{a_1, \dots, a_n\}$ ) iff for all  $i$ ,  $a_i : T$ .
- c. **Meet/Join types:** if  $T_1$  and  $T_2$  are types, then so are  $T_1 \wedge T_2$  and  $T_1 \vee T_2$ :  $a : T_1 \wedge T_2$  ( $T_1 \vee T_2$ ) iff  $a : T_1$  and (or)  $a : T_2$ .
- d. **Dual types:** if  $T$  is a type, then so is  $\neg T$ .  $a : \neg T$  iff for some type  $S$ , where  $S$  and  $T$  are incompatible,  $a : S$ .
- e. **Singleton types:** if  $T$  is a type and  $x : T$ , then  $T_x$  is a type.  $a : T_x$  iff  $a = x$ .

Since types and entities constructed from types are in the world, we want to classify them by means of types as well. This has the potential of leading to paradoxes, similar to Russell's paradox. As Cooper (2011) explains, one way of avoiding this is by means of *stratification*: this means building the universe in strata: classifying elements of strata  $1, \dots, n$  by means of types that emanate from (at the highest) stratum  $n + 1$ . In light of this, we can summarize our view of what the type-theoretic world consists of:

- (18) **Type-Theoretic World (Cooper 2011, simplified)**  
 TYPEDWORLD =  $\langle Type^n, BasicType, ComplexType^n, RecType^n, \langle A, F^n \rangle \rangle$
- a.  $Type^n$  is the set of types of order  $n$ , built up recursively using type construction operations.
- b.  $BasicType$ : IND, TIME, LOC, ...
- c.  $ComplexType^n$  ('interface with external reality'): tuples consisting of entities [from the model] and predicates.
- d.  $RecType^n$ : set of records, record types defined with respect to a set of objects used as labels.
- e.  $\langle A, F^n \rangle$  is a model ( $A^n$  assigning entities to  $BasicType$ , and  $F^n$  assigns tuples to  $ComplexType^n$ ).

I now show how meanings, substitution, outcomes, compound and negative propositions, questions, and answerhood are all characterizable within the Type-Theoretic World.

### 3.3.2 Abstraction

Function types allow one to model abstraction. As Cooper (2005a) points out, although abstraction in TTR works in a deceptively familiar 'type-theoretic' way, the existence of record typing yields a rich notion of abstraction. It is simultaneous and

restricted, i.e. it allows for multiple entities to be abstracted over simultaneously while encoding restrictions.

Concretely, (a simplified) analysis of the *meaning* (in the Montague–Kaplan sense from Chapter 2) of the sentence ‘I see Bo’ would be the function  $M$  given in (19a): this is a function from records with fields  $x, t, y, s$  for the speaker, utterance time, referent of ‘Bo’, and described situation, plus two fields  $p1, p2$  encoding relationships among or properties of the elements assigned to these fields; the value this function gets is a record type with one field *cont* whose value is restricted to be the proposition

$$\left[ \begin{array}{lcl} \text{sit} & = & M.s \\ \text{sit-type} & = & [c : \text{see}(M.x, M.t, M.y)] \end{array} \right], \text{ here the notation ‘} M.x, M.t, M.y, M.s \text{’ signifies}$$

the value the labels  $x, t, y, s$  have in any record appropriate to be an argument of  $M$ . (19b) exemplifies such a record, which would thus be assigned the record type in (19c) as its value for  $M$ . More technically, since it maps records of the type  $\text{Context}_{I \text{ see } Bo}$  into record types, the function  $M$  is of type  $(\text{Context}_{I \text{ see } Bo})\text{RecType}$ .

$$(19) \quad \begin{array}{l} \text{a. } M : \text{Context}_{I \text{ see } Bo} = \left[ \begin{array}{lcl} x & : & \text{IND} \\ t & : & \text{TIME} \\ p1 & : & \text{speak}(x, t) \\ y & : & \text{IND} \\ p2 & : & \text{named}('Bo', y) \\ s & : & \text{Rec} \end{array} \right] \mapsto \\ \\ \text{b. } \left[ \begin{array}{lcl} \text{cont} = & \left[ \begin{array}{lcl} \text{sit} = M.s \\ \text{sit-type} = [c : \text{see}(M.x, M.t, M.y)] \end{array} \right] & : \text{Prop} \\ x = x_0 \\ t = t_0 \\ p1 = c1 \\ y = y_0 \\ p2 = c2 \\ s = s_0 \end{array} \right], \text{ where } x_0 : \text{IND}, t_0 : \text{TIME}, c1 : \text{speak}(x_0, t_0), y_0 : \text{IND}, c2 : \\ \text{named}(Bo, y_0) \\ \\ \text{c. } \left[ \begin{array}{lcl} \text{cont} = & \left[ \begin{array}{lcl} \text{sit} & = & s_0 \\ \text{sit-type} & = & [c : \text{see}(x_0, t_0, y_0)] \end{array} \right] & : \text{Prop} \end{array} \right] \end{array}$$

Meanings viewed in this way are somewhat brittle. As we discussed in Chapter 2, with reference to example (21), in practice one can cope fine without fully instantiating a meaning, at least until clarification interaction, which can be extended over many turns, is concluded. However, as we will see in Chapter 5, where we integrate meanings

into a grammar, it is straightforward to reconfigure meaning functions like (19) in a way that allows them to be robust and accommodate partial comprehension.

**Exercise 3.3.1** Sketch the meaning of the sentence ‘I am speaking to you now’. Does the analysis capture the fact that this sentence cannot be used to make false claims?

Record types also bring with them familiar notions of substitution. For instance, (20) exemplifies substituting one label  $z$  for another one  $x$  within a record type:

$$(20) \left[ \begin{array}{ll} x & : \text{IND} \\ t & : \text{TIME} \\ p1 & : \text{say}(x,t) \\ y & : \text{IND} \\ p2 & : \text{named}(\text{Bo},y) \end{array} \right] = (x \rightsquigarrow z)$$

$$\left[ \begin{array}{ll} z & : \text{IND} \\ t & : \text{TIME} \\ p1 & : \text{say}(z,t) \\ y & : \text{IND} \\ p2 & : \text{named}(\text{Bo},y) \end{array} \right]$$

This operation, characterized in somewhat more detail in Cooper (2005a), will be of some use in dealing with the meaning of expressions that effect abstraction such as *wh*-phrases.

### 3.3.3 Meets, Joins, and Compound Propositions

Meet and join types are useful for various applications. I defined these above in the traditional Boolean way:

- (21) a. **Meet/Join types:** if  $T_1$  and  $T_2$  are types, then so are  $T_1 \wedge T_2$  and  $T_1 \vee T_2$ .  
 b.  $a : T_1 \wedge T_2$  ( $T_1 \vee T_2$ ) iff  $a : T_1$  and (or)  $a : T_2$ .

The ability to combine record types by means of meets will enable us to abbreviate the grammatical and conversational rules we define in subsequent chapters. Now given two record types  $RT_1$  and  $RT_2$ ,  $RT_1 \wedge RT_2$  will not in general be a record type (a set of atomic judgements). However, we will exploit the fact demonstrated by Cooper (2011) that there exists a record type we will denote as  $RT_1 \wedge_{\text{merge}} RT_2$  which is *equivalent* to  $RT_1 \wedge RT_2$  in the sense that  $a : RT_1 \wedge RT_2$  iff  $a : RT_1 \wedge_{\text{merge}} RT_2$ .

We illustrate some examples of  $\wedge_{\text{merge}}$ :

$$(22) \text{ a. } \left[ f : T_1 \right] \wedge_{\text{merge}} \left[ g : T_2 \right] = \left[ \begin{array}{l} f : T_1 \\ g : T_2 \end{array} \right]$$

$$\text{ b. } \left[ f : T_1 \right] \wedge_{\text{merge}} \left[ f : T_2 \right] = \left[ f : T_1 \wedge T_2 \right]$$



$$c. \begin{bmatrix} \text{spkr} & : & \text{IND} \\ c & : & \text{woman}(\text{spkr}) \\ \text{addr} & : & \text{IND} \end{bmatrix} \wedge_{\text{merge}} \begin{bmatrix} \text{addr} & : & \text{IND} \\ c1 & : & \text{woman}(\text{addr}) \end{bmatrix} = \begin{bmatrix} \text{spkr} & : & \text{IND} \\ c & : & \text{woman}(\text{spkr}) \\ \text{addr} & : & \text{IND} \\ c1 & : & \text{woman}(\text{addr}) \end{bmatrix}$$

One very minor use of  $\wedge$  is that it enables us to define the class of true propositions, which is occasionally useful. In (23) after having introduced *sit-type* as a field for a record type, it occurs as one of the types the field *sit* must satisfy, which is preceisely our definition of truth for propositions.

$$(23) \text{ TrueProp} =_{\text{def}} \begin{bmatrix} \text{sit-type} & : & \text{RecType} \\ \text{sit} & : & \text{Record} \wedge \text{sit-type} \end{bmatrix}$$

Finally, we can define compound propositions, crucial in defining answerhood to questions, based on the meet/join operations. Let us start with disjunctive propositions. We define first a disjunctive witness—one of the situational witnesses, depending on which of these is indeed a witness:

$$(24) \text{ Given } s_1, s_2 : \text{Rec} \text{ and } T_1, T_2 : \text{RecType}, \\ \text{Disjrec}((s_1, T_1), (s_2, T_2)) =_{\text{def}} \begin{cases} s_1, & \text{if } s_1 : T_1 \\ s_2, & \text{otherwise, if } s_2 : T_2 \\ [] & \text{(the empty record), otherwise} \end{cases}$$

Given this, we can then define the disjoined proposition (the operation  $\vee_{\text{prop}}$ ) as follows:

$$(25) \begin{bmatrix} \text{sit} & = & s_1 \\ \text{sit-type} & = & T_1 \end{bmatrix} \vee_{\text{prop}} \begin{bmatrix} \text{sit} & = & s_2 \\ \text{sit-type} & = & T_2 \end{bmatrix} =_{\text{def}} \begin{bmatrix} \text{sit} & = & \text{Disjrec}((s_1, T_1), (s_2, T_2)) \\ \text{sit-type} & = & T_1 \vee T_2 \end{bmatrix}$$

It is fairly straightforward to show that the disjunctive proposition is true iff at least one of its disjuncts is true.<sup>11</sup>

One might feel a bit anxious due to the somewhat contingent identity conditions of this disjunctive proposition (or rather its situational component). However, one can

<sup>11</sup> Assume without loss of generality that  $\begin{bmatrix} \text{sit} & = & s_1 \\ \text{sit-type} & = & T_1 \end{bmatrix}$  is true. Then  $s_1 : T_1$ , so  $\text{Disjrec}((s_1, T_1), (s_2, T_2)) = s_1$ . Consequently,  $\begin{bmatrix} \text{sit} & = & s_1 \\ \text{sit-type} & = & T_1 \end{bmatrix} \vee_{\text{prop}} \begin{bmatrix} \text{sit} & = & s_2 \\ \text{sit-type} & = & T_2 \end{bmatrix} = \begin{bmatrix} \text{sit} & = & s_1 \\ \text{sit-type} & = & T_1 \vee T_2 \end{bmatrix}$ . And this latter proposition is true as  $s_1 : T_1 \vee T_2$  (Note continues opposite).

motivate this choice based on data such as (26). Here the situational anaphor ‘that’—or ‘whatever happened’—seems able to refer to this split eventuality:

- (26) A: Bill left town or his bicycle has been stolen.  
 B: Yeah, that must have happened more than five hours ago.

With conjunction there are a number of options. Since records are sets of assignments to labels, one can define an operation  $\cup$  which combines two records into one. Such an operation is well defined if the two records have disjoint labels and is exemplified in (27):

$$(27) \quad r1 = \begin{bmatrix} x1 = a \\ y1 = b \end{bmatrix}, r2 = \begin{bmatrix} x2 = c \\ y2 = d \end{bmatrix}, \text{ then } r1 \cup r2 = \begin{bmatrix} x1 = a \\ y1 = b \\ x2 = c \\ y2 = d \end{bmatrix}$$

Given this, one could define the conjoined proposition (the operation  $\vee_{prop}$ ) as in (28a). This operation would satisfy (28b).<sup>12</sup> However, because  $\cup$  merges the two records without leaving a trace of the original constituents, only a weak form of conjunction reduction is maintained, given in (28c):

- (28) a.  $\begin{bmatrix} \text{sit} = s_1 \\ \text{sit-type} = T_1 \end{bmatrix} \wedge_{prop} \begin{bmatrix} \text{sit} = s_2 \\ \text{sit-type} = T_2 \end{bmatrix} =_{def} \begin{bmatrix} \text{sit} = s_1 \cup s_2 \\ \text{sit-type} = T_1 \wedge T_2 \end{bmatrix}$   
 b. If  $\begin{bmatrix} \text{sit} = s_1 \\ \text{sit-type} = T_1 \end{bmatrix}$  and  $\begin{bmatrix} \text{sit} = s_2 \\ \text{sit-type} = T_2 \end{bmatrix}$  are true, then so is  $\begin{bmatrix} \text{sit} = s_1 \\ \text{sit-type} = T_1 \end{bmatrix} \wedge_{prop} \begin{bmatrix} \text{sit} = s_2 \\ \text{sit-type} = T_2 \end{bmatrix}$ .  
 c. If  $\begin{bmatrix} \text{sit} = s_1 \\ \text{sit-type} = T_1 \end{bmatrix} \wedge_{prop} \begin{bmatrix} \text{sit} = s_2 \\ \text{sit-type} = T_2 \end{bmatrix}$  is true, then there exist records  $s'_1$  and  $s'_2$  such that  $s'_1 : T_1$  and  $s'_2 : T_2$

If one wishes to have a stronger version of conjunction reduction (viz. the converse of (28b)), one can use the standard constructive type theory  $\wedge_{ctt}$ , whose witnessing condition is given in (29a) in terms of an ordered pair of entities. In that case, propositional conjunction could be defined as (29b):

Conversely, assume  $\begin{bmatrix} \text{sit} = \text{Disjrec}((s_1, T_1), (s_2, T_2)) \\ \text{sit-type} = T_1 \vee T_2 \end{bmatrix}$  is true. Again assume without loss of generality that  $\text{Disjrec}((s_1, T_1), (s_2, T_2)) = s_1$ . But this means that  $s_1 : T_1$ , and therefore one of the disjuncts is true.

<sup>12</sup> If  $\begin{bmatrix} \text{sit} = s_1 \\ \text{sit-type} = T_1 \end{bmatrix}$  and  $\begin{bmatrix} \text{sit} = s_2 \\ \text{sit-type} = T_2 \end{bmatrix}$  are true, then  $s_1 : T_1$ , so also  $s_1 \cup s_2 : T_1$  (the subrecord  $s_1$  of  $s_1 \cup s_2$  already satisfies the constraints imposed by  $T_1$ ). By similar reasoning  $s_1 \cup s_2 : T_2$ . Hence,  $s_1 \cup s_2 : T_1 \wedge T_2$ .

$$(29) \quad \begin{array}{l} \text{a. } \langle s_1, s_2 \rangle : T_1 \wedge_{ctt} T_2 \text{ if } s_1 : T_1 \text{ and } s_2 : T_2. \\ \text{b. } \left[ \begin{array}{cc} \text{sit} & = s_1 \\ \text{sit-type} & = T_1 \end{array} \right] \wedge_{prop} \left[ \begin{array}{cc} \text{sit} & = s_2 \\ \text{sit-type} & = T_2 \end{array} \right] =_{def} \left[ \begin{array}{cc} \text{sit} & = \langle s_1, s_2 \rangle \\ \text{sit-type} & = T_1 \wedge_{ctt} T_2 \end{array} \right] \end{array}$$

### 3.3.4 Outcomes

The class of outcomes was introduced in Ginzburg and Sag (2000), following Portner (1997), to describe goals and serve as the denotata of imperatives, subjunctives, and (certain uses of) infinitives, as in (30):

- (30) a. Sit!  
 b. Bo demands that Mo sit.  
 c. Mo's goal is to leave.

Outcomes are closely related to propositions, with the main difference being temporal—outcomes are intrinsically future, but with a temporal dimension which is typically unanchored (at speech time), which makes them useful theoretical entities for reasoning about future action. (31) illustrates the futurity and the fact that an outcome can remain unfulfilled:

- (31) a. Go home. You can *then* take a nap.  
 b. Mo requested Bo to resign. He never did.

Truth is not applicable to such entities; what is applicable is the notion of being *fulfilled*. We can explicate this in an Austinian fashion—with fields for a situation and a situation type abstract, of which a temporal argument has been abstracted away. We define the type Irrealis—temporal abstracts over the class of record types:

$$(32) \quad \text{Irrealis} =_{def} ([t : \text{Time}])\text{RType}$$

An outcome will be a record exemplified in (33a)—corresponding to the outcome *that Bo leave*, the type Outcome is given in (33b):<sup>13</sup>

$$(33) \quad \begin{array}{l} \text{a. } \left[ \begin{array}{cc} \text{sit} & = s \\ \text{irr-sit-type} & = (r : [t : \text{Time}]) [c : \text{Leave}(b, r.t)] \end{array} \right] \\ \text{b. } \text{Outcome} =_{def} \left[ \begin{array}{cc} \text{sit} & : \text{Record} \\ \text{irr-sit-type} & : \text{Irrealis} \end{array} \right] \end{array}$$

<sup>13</sup> Strictly speaking, the irr-sit-type in (33a) should be as in (i), with a label introducing *b*:

$$(i) \quad (r : [t : \text{Time}]) \left[ \begin{array}{cc} b : \text{IND} \\ c : \text{Leave}(b, r.t) \end{array} \right]$$

However, the referents of proper names will usually be introduced as values for contextual parameters, as in (19). So for simplicity, this notational/conceptual abuse is tolerated.

The fulfilledness conditions of an outcome  $\left[ \begin{array}{l} \text{sit} = s_0 \\ \text{irr-sit-type} = p_0 \end{array} \right]$  involve the existence of a situation  $s_1$  which is situated temporally after  $s_0$  such that  $s_1$  witnesses an instantiation of  $p_0$ . This is the sense in which outcomes are ‘future’. In other words, (33a) will be fulfilled iff there exists a time  $t_1$  within the situation  $s_1$  such that  $t_1$  is subsequent to  $s_0$  and such that the proposition (34a) is true, or equivalently, such that (34b):

$$(34) \quad \begin{array}{l} \text{a. } \left[ \begin{array}{l} \text{sit} = s_1 \\ \text{sit-type} = [c : \text{Leave}(b, t_1)] \end{array} \right] \\ \text{b. } s_1 : [c : \text{Leave}(b, t_1)] \end{array}$$

Thus, the witnesses for a given outcome’s fulfilledness can be characterized in terms of the following mapping:<sup>14</sup>

$$(35) \quad \text{For an outcome } \left[ \begin{array}{l} \text{sit} = s_0 \\ \text{irr-sit-type} = p_0 \end{array} \right] \text{ Fulfillers } =_{\text{def}} \left[ \begin{array}{l} s_1 : \text{Record} \\ \text{fulfil-time} : \text{Time} \\ c1 : \text{anterior}(s_0, s_1) \\ p = \left[ \begin{array}{l} \text{sit} = s_1 \\ \text{sit-type} = p_0(\text{fulfil-time}) \end{array} \right] : \text{TrueProp} \end{array} \right]$$

**Exercise 3.3.2** Sketch the meaning of an imperative sentence such as (36):

(36) Leave!

### 3.3.5 Questions as Propositional Abstracts in Type Theory with Records

**3.3.5.1 Questions as Propositional Abstracts** The theory of questions I adopt here is one which views a question as a propositional abstract. Ginzburg and Sag (2000) develop a situation-theoretic version of this theory in some detail, whereas

<sup>14</sup> The notation in (i) is shorthand for (ii). This exemplifies the appeal to the singleton type constructor introduced in (17b), which requires (iii):

- (i)  $p = a : T$
- (ii)  $p : T_a$
- (iii)  $p : T$  and  $p = a$

Krifka (2001) argues for a variant of this view in a structured meanings framework. For current dialogical purposes, one can offer two main arguments for this view:<sup>15</sup>

- Any theory of questions needs to associate an abstract-like object with interrogatives to explicate the resolution of short answers. As (37) illustrates, the nature of the question asked strongly influences the semantic type, and to some extent, the form of the short answer. A unary ‘who’-interrogative gives rise to NP short answers, a unary ‘when’-interrogative gives rise to temporal phrase short answers, a polar question gives rise to adverbial short answers, etc. Examples (37) provide indications that following an interrogative use a semantic object become accessible in the context, one that enables a short answer to express a propositional content. In other words, this object is a function from a class of objects determined by the question (persons, times, reasons respectively in (37a–c)) into the class of propositions.
- (37) a. A: Who attended the meeting?  
       B: Mo./No students./A friend of Jo’s.
- b. A: When did Bo leave?  
       B: Yesterday./At two.
- c. A: Why did Maire cross the road?  
       B: Because she thought no cars were passing.
- Explicating answerhood involves a characterization that utilizes in some form an abstract-like object—either explicitly or in terms of an open sentence or similar.

If questions simply *are* abstracts, then the fact that abstracts need to be associated with interrogative utterances follows automatically.

**3.3.5.2 Some Examples** In light of my earlier comments about abstraction, a (basic, non-compound) question will be a function from records into propositions. As such, questions are automatically part of the type-theoretic ontology. Let us start by considering some very simple examples of interrogatives and their TTR representations. (38) exemplifies the denotations we can assign to a unary *wh*, and a binary *wh*-interrogative. The unary question ranges over instantiations by persons of the proposition *run*(*x*), the binary question ranges over instantiations by pairs of persons *x* and things *y* of the proposition *touch*(*x*, *y*):<sup>16</sup>

<sup>15</sup> For alternative perspectives see Groenendijk and Stokhof (1997) and the papers in Aloni, Butler, and Dekker (2006); for a recent review of work on questions see Ginzburg (2010).

<sup>16</sup> I will occasionally use notation such as ?P(*x*...) for polar questions or  $\lambda x R(x \dots)$  etc. as abbreviations for these TTR notations.

(38) a. who ran

$$\begin{aligned} &\text{TTR representation—}(r : \begin{bmatrix} x & : & \text{IND} \\ \text{rest} & : & \text{person}(x) \end{bmatrix}) \\ &\begin{bmatrix} \text{sit} & = & r_1 \\ \text{sit-type} & = & \begin{bmatrix} c : \text{run}(r.x) \end{bmatrix} \end{bmatrix} \quad \text{That is, a function that maps records} \\ &r : T_{\text{who}} = \begin{bmatrix} x & : & \text{IND} \\ \text{rest} & : & \text{person}(x) \end{bmatrix} \quad \text{into propositions of the form} \\ &\begin{bmatrix} \text{sit} & = & r_1 \\ \text{sit-type} & = & \begin{bmatrix} c : \text{run}(r.x) \end{bmatrix} \end{bmatrix} \end{aligned}$$

b. who touched what

$$\begin{aligned} &\text{TTR representation—}(r : \begin{bmatrix} x & : & \text{IND} \\ \text{rest1} & : & \text{person}(x) \\ y & : & \text{IND} \\ \text{rest2} & : & \text{thing}(y) \end{bmatrix}) \\ &\begin{bmatrix} \text{sit} & = & r_1 \\ \text{sit-type} & = & \begin{bmatrix} c : \text{touch}(r.x, r.y) \end{bmatrix} \end{bmatrix} \quad \text{That is, a function that maps} \\ &\text{records } r : T_{\text{who,what}} = \begin{bmatrix} x & : & \text{IND} \\ \text{rest1} & : & \text{person}(x) \\ y & : & \text{IND} \\ \text{rest2} & : & \text{thing}(y) \end{bmatrix} \\ &\text{into propositions of the form } \begin{bmatrix} \text{sit} & = & r_1 \\ \text{sit-type} & = & \begin{bmatrix} c : \text{touch}(r.x, r.y) \end{bmatrix} \end{bmatrix} \end{aligned}$$

What of polar questions? Ginzburg and Sag (2000) proposed that these are 0-ary abstracts, though the technical apparatus involved in explicating this notion in their framework based on non-well-founded set theory was quite complex. TTR, however, offers a simple way to explicate 0-ary abstraction. If we think of a unary abstract as involving a domain type with one field which directly influences the value of the function, a binary abstract as one whose domain type contains two such fields, etc., then by analogy the domain type of a 0-ary type would simply be the empty type  $[]$ . This makes a 0-ary abstract a constant function from the universe of all records (since every record is of the type  $[]$ ). (39) exemplifies this:

(39) a. Did Bo run

TTR representation—(  $r : \left[ \begin{array}{l} \text{sit} \\ \text{sit-type} \end{array} \right]$   $\left[ \begin{array}{l} = r_1 \\ = [c : \text{run}(b)] \end{array} \right]$  ). That is,  
 the function that maps records  $r : T_0 = \left[ \begin{array}{l} \text{sit} \\ \text{sit-type} \end{array} \right]$  (i.e. *any* record)  
 into the proposition  $\left[ \begin{array}{l} \text{sit} = r_1 \\ \text{sit-type} = [c : \text{run}(b)] \end{array} \right]$

The fact that questions individually are part of the type-theoretic world is not the end of the story. For various linguistic tasks (e.g. specifying the selectional requirements of verbs like ‘ask’, ‘wonder’, and ‘investigate’), and for various dialogical tasks (e.g. the formulation of various conversational rules), one needs to appeal to a type Question. This means that we need to have a characterization of this type within TTR. One such characterization is given in Ginzburg (2005); an alternative, more refined characterization is given in Appendix A. These characterizations are based on the idea that one can build a series of increasingly inclusive function types that for any question of order  $n$   $q_n$  eventually include  $q_n$ .<sup>17</sup>

### 3.3.6 Dual Types, Negative Propositions, and Negative Questions

Negation is a crucial notion for questions and for explicating answerhood. One of the fundamental assumptions of situation semantics has been that situation types come in positive/negative pairs, so that in current terms every complex type comes in two flavours, a positive and a negative. Assuming the positive ones to be (notationally) unmarked, one can notate the corresponding negative with the standard  $\neg$  symbol, as in (40a). Because situations are partial, there is a difference between a situation failing to be correctly classified by  $T$  and being correctly classified by  $\neg T$ . For any situation  $s$  and situation type  $T$ , (40b) holds, but (40c) generally fails. The intuition, originally motivated by data such as (40d),<sup>18</sup> is that classifying  $s$  with  $\neg T$  means that  $s$  actually

<sup>17</sup> The characterization in Ginzburg (2005) relies on the fact that within a version of TTR (i) is a theorem, being a subcase of Theorem 4 of Constable (2002):

- (i) Function type subsumption  
 For any types  $A, A', B$  if  $A \sqsubseteq A'$ , then  $(A)'B \sqsubseteq (A')B$ .

This entails, for instance, that given the fact that (ii) holds, i.e. that the domain of binary who-what-questions is a subtype of the domain type of unary who-questions, which is, in turn, a subtype of the domain type of polar questions, it follows that (iii), i.e. that the type of binary who-what-questions is a supertype of the type of unary who-questions, which is, in turn, a subtype of the domain type of polar questions:

- (ii)  $T_{\text{who,what}} \sqsubseteq T_{\text{who}} \sqsubseteq T_0$   
 (iii)  $(T_0)Prop \sqsubseteq (T_{\text{who}})Prop \sqsubseteq (T_{\text{who,what}})Prop$

<sup>18</sup> See Cooper (1998a) for discussion, for postulates that attempt to pin down the nature of  $\neg T$  in situation semantics, and for arguments that the requirements on the perceived scene satisfying (i) go beyond Davidsonian accounts (e.g. Higginbotham 1983, 1994), which are limited to the far weaker (ii):

possesses information which rules out  $T$  (e.g. Mary avoiding contact with Bill), rather than simply lacking concrete evidence for  $T$  (e.g. Ralph shutting his eyes).

- (40) a.  $s : \neg T$   
b. Either  $s : T$  or  $s : \neg T$   
c. Either  $s : T$  or  $s : \neg T$   
d. Ralph saw Mary not serve Bill.

We can enshrine this ‘constructive’ view of negation in TTR—assuming a notion of type *incompatibility* such as (41a)<sup>19</sup>—in the definition given in (41b). In order to be fully viable *incompatibility* needs to be further restricted using some notion of *alternativehood* (Cohen 1999). In some cases what the alternatives amount to is fairly straightforward and even lexicalized—classifying the table as *not black* requires evidence that it is green or brown or blue, say. But in general, figuring out the alternatives, as Cohen illustrates, is of course itself context-dependent, relating *inter alia* to issues currently under discussion. An obvious consequence of (41b) is (41c).<sup>20</sup> This allows us to impose (41d), which is useful for some purposes as we will shortly see.<sup>21</sup>

- (41) a. The types  $S$  and  $T$  are incompatible  $=_{def}$  it is never the case that  $s : S$  and  $s : T$ .  
b. **Dual types:** if  $T$  is a type, then so is  $\neg T$ .  $a : \neg T$  iff for some type  $S$ , where  $S$  and  $T$  are incompatible  $a : S$   
c. If  $s : T$ , then  $s : \neg \neg T$ ; if  $s : \neg T$ , then  $s : \neg \neg \neg T$ .  
d.  $\neg \neg T = T$

Given this view of dual types, the truth of negative propositions involves the situation  $s$  witnessing a negative situation type  $\neg T$ . This, in turn, following (41b), involves  $s$  supporting the factuality of a positive situation type  $S$  alternative to  $T$ :

- (42)  $\left[ \begin{array}{l} \text{sit} = s \\ \text{sit-type} = \neg T \end{array} \right]$  is true iff  $s : \neg T$  iff there exists a (positive alternative)  $S$  such that  $s : S$

This view of negation jibes well with recent psycholinguistic findings, as summarized in a review of current work on working memory representations involved in the

- (i)  $s : \neg \text{Serve}(m, b)$   
(ii)  $s : \neg \text{Serve}(m, b)$

<sup>19</sup> Arguably, we need a stronger notion than (41a) since incompatibility has a modal character. A more detailed account of negation for dialogue is developed in Cooper and Ginzburg (2011), where one possible approach concerning *incompatibility* is proposed.

<sup>20</sup> If  $s : T$  and also  $s : \neg T$ , then there exists  $S$  incompatible with  $T$  such that  $s : S$ , which contradicts the assumption that  $s : T$ . If  $s : \neg T$ , then as before there exists  $S$  incompatible with  $T$  such that  $s : S$ , which rules out  $s : T$ .

<sup>21</sup> Cooper and Ginzburg (2011) argue that (41d) can and should be weakened to  $s : T$  iff  $s : \neg \neg T$ .



comprehension of negation in texts (Kaup 2006). Kaup argues that the approach that accords best with current evidence is an experiential-simulations view of comprehension. On this view, comprehenders construct mental simulations—grounded in perception and action—of the states of affairs described in a text. Kaup offers experimental evidence that comprehending a negative sentence (e.g. Sam is not wearing a hat) involves simulating a scene consistent with the negated sentence. She suggests that indeed initially subjects simulate an ‘unnegated’ scene (e.g. involving Sam wearing a hat). Tian, Breheny, and Ferguson (2010) offer additional evidence supporting the simulationist perspective. However, they argue against the ‘two-step’ view of negation (viz. unnegated and then negated), in favour of a view driven by dialogical coherence, based on QUD, which we further discuss in section 4.4.3.

The constructive view of negation also has consequences for polar questions. The fact that polar questions are 0-ary abstracts over propositions constructed from records and record types has a consequence that, in contrast to approaches where questions are characterized in terms of exhaustive answerhood conditions (see Groenendijk and Stokhof 1997), positive and negative polar interrogatives are assigned distinct denotations. For instance, (43a) and (43b), due to Hoepelmann (1983), would be assigned the 0-ary abstracts in (43c) and (43d) respectively:

- (43) a. Is 2 an even number?  
 b. Isn't 2 an even number?  
 c.  $\left[ \begin{array}{l} \text{sit} \\ \text{sit-type} \end{array} \right] = \left[ \begin{array}{l} s \\ c : \text{EvenNumber}(2) \end{array} \right]$   
 d.  $\left[ \begin{array}{l} \text{sit} \\ \text{sit-type} \end{array} \right] = \left[ \begin{array}{l} s \\ c : \neg \text{EvenNumber}(2) \end{array} \right]$

This means we can explicate the distinct presuppositional backgrounds associated with positive and negative polar interrogatives. For instance, Hoepelmann, in arguing for this distinction, suggests that a question like (43a) is likely to be asked by a person recently introduced to the odd/even distinction, whereas (43b) is appropriate in a context where, say, the opaque remarks of a mathematician sow doubt on the previously well-established belief that *two is even*. The latter can be tied to the factuality conditions of negative situation types.

These contextual differences gives rise in some languages including French and Georgian to distinct words to affirm a positive polar question (*oui, xo*) and a negative polar question (*si, diax*); the relevant QUD presupposition is discussed in section 7.5. Nonetheless, as we will now see, given the definitions of answerhood available in this system, then, assuming (41c), positive and negative interrogatives specify identical answerhood relations. Hence, the identity of truth conditions of sentences like (44) can be captured:

- (44) a. Kim knows whether Bo left.  
b. Kim knows whether Bo did not leave.

### 3.3.7 Answerhood

Answerhood is one of the essential testing grounds for a theory of questions. Abstracts can be used to *underspecify* answerhood. This is important given that NL requires a variety of answerhood notions, not merely exhaustive answerhood or notions straightforwardly definable from it. Moreover, as with questions, answerhood needs to be explicable type-theory-internally. This is because answerhood figures as a constituent relation of the lexical entries of resolutive verbs<sup>22</sup> and in rules regulating felicitous responses in dialogue management (see section 4.4.3). For current purposes this means that we need to be able to define notions of answerhood as types.

There are a number of notions of answerhood that are of importance to dialogue. One relates to coherence: any speaker of a given language can recognize, independently of domain knowledge and of the goals underlying an interaction, that certain propositions are *about* or *directly concern* a given question.

How can this notion of answerhood be characterized? The simplest notion of answerhood we can define on the basis of an abstract is one I will call, following Ginzburg and Sag (2000), *simple answerhood*:

- (45) a.  $p$  is a simple answer to  $q$  iff  $p$  is an instantiation of  $q$  or a negation of such an instantiation  
b. For a polar question:  $\{r \mid \text{SimpleAns}(r, \lambda\{ \}p)\} = \{p, \neg p\}$   
c. For a unary *wh*-question:  $\{r \mid \text{SimpleAns}(r, \lambda\{b\}p(b))\} = \{p(a_1), \dots, p(a_n), \neg p(a_1), \dots, \neg p(a_n)\}$

In terms of TTR, we can characterize simple answerhood as follows. We first define an auxiliary notion of *atomic answerhood*, specified in terms of the mapping in (46a). Here the label *shortans* is a field for an entity whose type is the domain of the question,<sup>23</sup> whereas the label *propans* is a field for a proposition that constitutes an atomic answer. Negative atomic answers are negations of atomic answers. Joining these two types yields simple answerhood:

- (46) Given a question  $q : (A)B$ :
- $\text{AtomAns}(q) =_{\text{def}} \left[ \begin{array}{ll} \text{shortans} & : A \\ \text{propans} = q(\text{shortans}) & : \text{Prop} \end{array} \right]$
  - $\text{NegAtomAns}(q) =_{\text{def}} \left[ \begin{array}{ll} \text{shortans} & : A \\ \text{propans} = \neg q(\text{shortans}) & : \text{Prop} \end{array} \right]$
  - $\text{SimpleAns}(q) =_{\text{def}} \text{AtomAns}(q) \vee \text{NegAtomAns}(q)$

<sup>22</sup> For more detailed discussion see Ginzburg and Sag (2000: ch. 3, section 3.2 and ch. 8, section 8.3).

<sup>23</sup> This is a mnemonic for the fact that short answers involve a term of this type.

To exemplify this, the domain type of (47a) is (47b), and therefore for a proposition to be appropriate for the field *propans* in (46a), it needs to be as in (47c), whereas to be appropriate for the field *propans* in (46b), it needs to be as in (47d). A polar question such as (47e) is a constant function, so that the only atomic answer of (47e) is the single value of that function, namely (47f); consequently the only negative atomic answer of (47e) is (47g). Finally, (47g) would be the only atomic answer of the negative polar question in (47h), whereas given our assumption that  $\neg\neg T = T$ , discussed in section 3.3.6, (47f) would be the only negative atomic answer of (47h). This is what ensures that positive and negative polar answers have the same simple answerhood relation. And, since the other answerhood relations we define are built above this notion of answerhood, this applies also to them.

$$\begin{aligned}
 (47) \quad & \text{a. } q = (r : \left[ \begin{array}{ll} x & : \text{IND} \\ \text{rest} & : \text{person}(x) \end{array} \right]) \left[ \begin{array}{ll} \text{sit} & = r_1 \\ \text{sit-type} & = \left[ c : \text{run}(r.x) \right] \end{array} \right] \\
 & \text{b. } \left[ \begin{array}{ll} x & : \text{Ind} \\ \text{rest} & : \text{person}(x) \end{array} \right] \\
 & \text{c. If } r = \left[ \begin{array}{ll} x & = a \\ \text{rest} & = \text{pl} \end{array} \right], \text{ then } q(r) = \left[ \begin{array}{ll} \text{sit} & = r_1 \\ \text{sit-type} & = \left[ c : \text{run}(a) \right] \end{array} \right] \\
 & \text{d. If } r = \left[ \begin{array}{ll} x & = a \\ \text{rest} & = \text{pl} \end{array} \right], \text{ then } \neg q(r) = \left[ \begin{array}{ll} \text{sit} & = r_1 \\ \text{sit-type} & = \left[ c : \neg \text{run}(a) \right] \end{array} \right] \\
 & \text{e. } (r : \left[ \right]) \left[ \begin{array}{ll} \text{sit} & = r_1 \\ \text{sit-type} & = \left[ c : \text{run}(b) \right] \end{array} \right] \\
 & \text{f. } \left[ \begin{array}{ll} \text{sit} & = r_1 \\ \text{sit-type} & = \left[ c : \text{run}(b) \right] \end{array} \right] \\
 & \text{g. } \left[ \begin{array}{ll} \text{sit} & = r_1 \\ \text{sit-type} & = \left[ c : \neg \text{run}(b) \right] \end{array} \right] \\
 & \text{h. } (r : \left[ \right]) \left[ \begin{array}{ll} \text{sit} & = r_1 \\ \text{sit-type} & = \left[ c : \neg \text{run}(b) \right] \end{array} \right]
 \end{aligned}$$

**Exercise 3.3.3** Characterize the answerhood relations associated with *Did Jo see Brendan*, *Who saw Brendan*, and *Who saw who*; assume that the queried situation is the same for all questions. What relationship do the answerhood relations to the questions stand to each other?

Simple answerhood covers a fair amount of ground. But it clearly underdetermines aboutness. On the polar front, it leaves out the whole gamut of answers to polar questions that are weaker than  $p$  or  $\neg p$  such as conditional answers ‘If  $r$ , then  $p$ ’ (e.g. 48a) or weakly modalized answers ‘probably/possibly/maybe/possibly not  $p$ ’ (e.g. (48b)). As far as *wh*-questions go, it leaves out quantificational answers (48c–g), as well as disjunctive answers. These missing class of propositions are pervasive in actual linguistic use:

- (48) a. Christopher: Can I have some ice-cream then?  
           Dorothy: you can do if there is any. (BNC, KBW)  
       b. Anon: Are you voting for Tory?  
           Denise: I might. (BNC, KBU, slightly modified)  
       c. Dorothy: What did grandma have to catch?  
           Christopher: A bus. (BNC, KBW, slightly modified)  
       d. Rhiannon: How much tape have you used up?  
           Chris: About half of one side. (BNC, KBM)  
       e. Dorothy: What do you want on this?  
           Andrew: I would like some yogurt please. (BNC, KBW, slightly modified)  
       f. Elinor: Where are you going to hide it?  
           Tim: Somewhere you can’t have it. (BNC, KBW)  
       g. Christopher: Where is the box?  
           Dorothy: Near the window. (BNC, KBW)

One straightforward way to enrich simple answerhood is to consider the relation that emerges by closing simple answerhood under disjunction.

- (49)  $p$  is About  $q$  iff  $p$  entails a disjunction of simple answers.

Ginzburg (1995) and Ginzburg and Sag (2000) show that aboutness as defined in (49) seems to encompass the various classes of propositions exemplified in (48). Thus:

- By defining *possibly*( $p$ ) as involving (a) the truth of  $p \vee r$ , where  $p, r$  are incompatible and (b)  $\neg p$  not being proven, one can show *possibly*( $p$ ) is About  $p?$ . By monotonicity, all modalities stronger than ‘possibly’ are also about  $p?$ .
- Existence-entailing generalized quantifiers such as existentials satisfy aboutness in virtue of their entailing the truth of a positive instantiation of a given *wh*-question.
- Generalized quantifiers such as ‘At most one  $N$ ’ or ‘Few  $N$ ’ satisfy aboutness in virtue of their entailing the truth of a positive or negative instantiation of a given *wh*-question.<sup>24</sup>

<sup>24</sup> Note that in frameworks like Situation Semantics or TTR, a proposition such as ‘At most one student left’ can be made to entail—within a suitable extension to cover generalized quantifiers, for some individual ‘Kim’, the truth of ‘Either Kim left or Kim did not leave’. But it will not entail, for instance, the truth of ‘Either Kim is tall or Kim is not tall’.

We can capture this in terms of the mapping in (50); here  $p1$  and  $p2$  are fields for two simple answers<sup>25</sup> and  $propans$  is a proposition that entails their disjunction:<sup>26</sup>

(50) Given a question  $q : (A)B$ :

$$a. \text{Aboutness}(q) =_{\text{def}} \left[ \begin{array}{ll} r1 & : \text{SimpleAns}(q) \\ p1 = r1.\text{propans} & : \text{Prop} \\ r2 & : \text{SimpleAns}(q) \\ p2 = r2.\text{propans} & : \text{Prop} \\ \text{propans} & : \text{Prop} \\ c & : \text{Entails}(\text{propans}, p1 \vee_{\text{prop}} p2) \end{array} \right]$$

Answerhood in the aboutness sense is clearly distinct from a highly restricted notion of answerhood, that of being a proposition that *resolves* or constitutes *exhaustive information* about a question. This latter sense of answerhood, which is restricted to true propositions, has been explored in great detail in the formal semantics literature, since it is a key ingredient in explicating the behaviour of interrogatives embedded by resolutive predicates such as ‘know’, ‘tell’, and ‘discover’. This notion of answerhood relates to the intuitions of an agent possessing information about a question which forecloses discussion thereof:

- (51) a. I wondered about who/when/why Bo left, so I asked. Mo told me ..., so now I know who/when/why Bo left.  
 b. An important issue is what we should do next. We’ve been discussing this in some detail. But the issue hasn’t been resolved as yet.

For now I mention two relevant notions, one purely semantic, the other pragmatically relativized. The former, strong exhaustiveness, due originally to Groenendijk and Stokhof (1984), approximates resolvedness in cases where interaction is relatively transparent. As its name implies, this is just about the most restrictive notion of resolvedness that could be entertained:

- (52)  $p$  is a strongly exhaustive answer to  $q$  iff  $p$  is true and entails every  $p_i$  which is a true simple answer to  $q$ .

The strongly exhaustive answer can be constructed as in (53). The functional restriction is the standard TTR way of encoding a universal quantification, akin to the classical discourse representation theoretic technique (Kamp 1981):

<sup>25</sup> If  $r : \text{SimpleAns}(q)$ , then since  $r : \text{AtomAns}(q)$  or  $r : \text{NegAtomAns}(q)$ ,  $r$  is a record of the form in (i),  $p1$  then is the value  $r$  gets on the  $\text{propans}$  field:

(i)  $r = \begin{bmatrix} \text{shortans} & = a \\ \text{propans} & = p \end{bmatrix}$

<sup>26</sup> This definition presupposes the existence of an entailment relation between propositions:  $\text{Entails}(p, r)$  iff it cannot be the case that  $p$  is true and  $r$  is false.

$$(53) \quad \text{StrongExhAns}(q) =_{\text{def}} \left[ \begin{array}{l} \text{exhans} : \text{Prop} \\ \text{c0} : \left( \begin{array}{l} \text{r} : \text{SimpleAns}(q) \\ \text{p} = \text{r.propans} : \text{TrueProp} \end{array} \right) \\ \left[ \text{c1} : \text{Entails}(\text{exhans}, \text{p}) \right] \end{array} \right]$$

The characterization in (53) seems to straightforwardly capture the resolvedness conditions for a polar question. In other words, a resolving answer needs to entail whichever of  $p$  or  $\neg p$  is factual, assuming the question is decided:

- (54) Bo knows whether  $p$  (whether Mo is asleep). Hence, either Jo knows  $p$  (that Mo is asleep) or Jo knows  $\neg p$  (that Mo is not asleep).

The situation with *wh*-questions is far more complicated. As discussed in *inter alia* Boër and Lycan (1985), Ginzburg (1995), Asher and Lascarides (1998), Lahiri (2002) and van Rooy (2003), there are various pragmatic factors which seem to come into the picture when evaluating whether a proposition resolves a question, in a given context. These include the goals associated with the interaction and the knowledge states of the conversational participants. How to integrate them, however, is very much still an unresolved issue. To illustrate the complexity—whether the inference pattern illustrated in the scenario in (55) is valid depends on precise details concerning Anders’s knowledge:

- (55) Anders: I wanted to get to José’s house on the Tube. So I asked Werner where José’s house is. Werner told me that the house is near Pimlico station. So now I know where José’s house is.

An analysis of resolvedness needs to integrate semantic constraints that derive directly from the question with agent-relative information. (56a) defines the relation of being a *potentially resolving answer* (originally formulated in Ginzburg 1995), being either an atomic answer or the negative universal answer.<sup>27,28</sup> In (56b), the semantic constraints involve a resolving answer entailing a potentially resolving answer, whereas the agent-relative condition is cashed out in terms of a relation (not further analysed here) *epistemically sufficient*, that holds between a proposition, an agent, and an outcome. This relation is intended to represent a proposition providing information to an agent that is sufficient for an outcome to be fulfilled.

- (56) a. Given a question  $q : (A)B$ ,  $\text{PotResolvAns}(q) =_{\text{def}} \text{AtomAns}(q) \vee \left[ \text{propans} = \forall x : A \neg q(x) : \text{Prop} \right]$

<sup>27</sup> For simplicity I have written the latter here in a hybrid first-order, type-theoretic notation.

<sup>28</sup> For polar questions this reduces to the characterization in (53), given that a polar question ( $[ ]$ )  $p$  has the sole instantiation  $p$  and so the corresponding negative universal is  $\neg p$ .

$\text{ResolvingAnswers}(q) =_{\text{def}} (\text{p resolves } q \text{ relative to B's desired outcome } o)$

|                               |                          |
|-------------------------------|--------------------------|
| p                             | : TrueProp               |
| B                             | : IND                    |
| o                             | : outcome                |
| p1 = PotResolveAns(q).propans | : Prop                   |
| c2                            | : Entails(p, p1)         |
| c3                            | : want(B, o)             |
| c4                            | : ep-sufficient(p, B, o) |

**Exercise 3.3.4** Is there a way to reduce resolvedness to exhaustiveness by fixing certain of the agent-dependent parameters?

### 3.3.8 *Dependence and Influence*

So far we have discussed answerhood, a relation between propositions and questions, which plays an important role in regulating which responses can be provided to queries. Any inspection of corpora, nonetheless, reveals the underdiscussed fact that many queries are responded to with a query.<sup>29</sup> A large proportion of these, exemplified in (57), are clarification requests pertaining to the literal content of the previous utterance, which I discuss in Chapters 6 and 7.

- (57) a. Anon 1: Did, did, did, did they have the union in there at all or (pause)?  
       Anon 2: Have what dear? (BNC, K62)  
       b. Anon 2: Should there be er a change in the rules?  
       Anon 3: What rules? (BNC, K6A)

But in addition to these, there are a variety of query responses whose content more directly addresses the question posed, as exemplified in the constructed (58) and the corpus-based (59):

- (58) a. A: Who murdered Smith? B: Who was in town?  
       b. A: Who is going to win the race? B: Who is going to participate?
- (59) a. Carol: Right, what do you want for your dinner?  
       Chris: What do you (pause) suggest? (BNC, KBJ)  
       b. Chris: Where's mummy?  
       Emma: Mm?  
       Chris: Mummy?  
       Emma: What do you want her for? (BNC, KBJ)

One class of these query responses can be characterized via the semantic relations of *dependence*, or its converse *influence*, as in (60); this view was first articulated by

<sup>29</sup> Thanks to Staffan Larsson, Jon Shaheen, and David Traum for discussions of this issue.

Carlson (1983), see also extensive logical analysis of question/question relations and their use in inquiry by Andrzej Wiśniewski, e.g. Wiśniewski (2001, 2003).

- (60) a. Who killed Smith depends on who was in town at the time.
- b. What Chris wants for dinner depends on what Carol will suggest.

One possible definition of these notions is in (61b). Its intuitive rationale is this: discussion of  $q_2$  will necessarily bring about the provision of information about  $q_1$ :

- (61) a.  $q_2$  can be used to respond to  $q_1$  if  $q_2$  influences  $q_1$  ( $q_1$  depends on  $q_2$ )
- b.  $q_2$  influences  $q_1$  iff any proposition  $p$  such that  $p$  Resolves  $q_2$ , also satisfies  $p$  entails  $r$  such that  $r$  is About  $q_1$ .

But there are question responses that don't fit in this characterization. One such case is (62a), which is metacommunicative, not at the level of content but of underlying goals or similar. I discuss this class of queries in section 8.3.6. One could, arguably, subsume (62b), repeated from (59) above, and, more indirectly, (62c) to this class. Here there is no dependence as such between the question asked and the question replied, since e.g. (62d) is false. If anything there is dependence of the question replied, but on a 'metaquestion' such as in (62e):

- (62) a. Cherrilyn (1): You got a pound? Jessica (2): Why? (= Why does Cherrilyn ask if Jessica has got a pound?) (BNC, KBL)
- b. Chris: Where's mummy? Emma: Mm? Chris: Mummy? Emma: What do you want her for? (BNC, KBJ)
- c. A: Who's coming tomorrow? B: Which people do you know in the department?
- d. Who is coming tomorrow depends on which people you know in the department.
- e. Which answer to 'who is coming tomorrow' will be useful depends on which people you know in the department.

One could try and unify (62b, c) and the cases subsumed above by dependence/influence via the discourse relation of *question elaboration* proposed by Asher and Lascarides (2003):

- (63) If  $Q\text{-Elab}(\alpha, \beta)$  holds between an utterance  $\alpha$  uttered by A, where  $g$  is a goal associated by convention with utterances of the type  $\alpha$ , and the question  $\beta$  uttered by B, then any answer to  $\beta$  must elaborate a plan to achieve  $g$  (simplified version of Asher and Lascarides 2003: 323)

The relation of Q-Elab, motivated by interaction in cooperative settings, is vulnerable to examples such as those in (64). (64a) has one understanding that could even be characterized using dependence (*What I like depends on what you like*), but in its original context is used simply as a coherent retort. (64b) could possibly be used in



political debate without it necessarily involving an attempt to discover an answer to the first question asked; similarly for (64c):<sup>30</sup>

- (64) a. A: What do you like? B: What do you like? (based on examples from Tom Stoppard, *Rosencrantz and Guildenstern Are Dead*, Faber and Faber, 1967)  
 b. A: What is Sarkozy going to do about it? B: What is Aubry?  
 c. Patient: Do you think I'm insane now. Doctor: Do you think so? (From Scheflen 1961, via Schegloff 2007)

Given this, I retain (61a) as one source underlying the coherence of question replies, returning to this general issue when discussing the status of the ordering on QUD in section 4.3.3 and later on the issue of metadiscussion in section 8.3.1.

### 3.4 Summary

In this section, I have sketched an ontology which underpins KoS. I have motivated this ontology by a combination of semantic, logical, and cognitive considerations.

As the underlying logical framework, we use Type Theory with Records (TTR), a model-theoretic descendant of Martin–Löf Type Theory. One reason for this is that, in contrast to Type Feature Structure-based frameworks, whose notation is quite similar, TTR provides us with a token/type distinction at the object level. This feature, which we illustrate in detail in Chapters 4, 5, and 6, is crucial for dealing with the grounding/clarification potential of utterances.

The most fundamental notions of TTR are:

- **The typing judgement:**  $a : T$  classifying an object  $a$  as being of type  $T$ .
- **Records:** A record is a set of fields assigning entities to labels of the form (65a), partially ordered by a notion of *dependence* between the fields—dependent fields must follow fields on which their values depend. A concrete instance is exemplified in (65b). Records are used here to model events, including utterances, and dialogue gameboards.

- (65) a. 
$$\left[ \begin{array}{l} l_1 = val_1 \\ l_2 = val_2 \\ \dots \\ l_n = val_n \end{array} \right]$$
- b. 
$$\left[ \begin{array}{ll} x & = -28 \\ e\text{-time} & = 2\text{AM, 17 Feb 2011} \\ e\text{-loc} & = \text{Nome} \\ c_{temp-at-in} & = \text{ol} \end{array} \right]$$

<sup>30</sup> Although in its original context, which continued—*Patient: No, of course not. Doctor: But I think you are—perhaps this was going on.*

- **Record Types:** a record type is simply a record where each field represents a judgement rather than an assignment, as in (66a). (66b) is a possible type for (65b), assuming the conditions in (66c) hold. As explained in Chapter 5, records types are used to model utterance types (aka *signs*); in Chapters 4 and 6, we will use record types to express rules of conversational interaction.

- (66) a. 
$$\begin{bmatrix} l_1 : T_1 \\ l_2 : T_2 \\ \dots \\ l_n : T_n \end{bmatrix}$$
- b. 
$$\begin{bmatrix} x & : & \text{IND} \\ \text{e-time} & : & \text{TIME} \\ \text{e-loc} & : & \text{LOC} \\ c_{temp-at-in} & : & \text{temp\_at\_in}(\text{e-time}, \text{e-location}, x) \end{bmatrix}$$
- c.  $-28 : \text{IND}; 3:45\text{AM}, 17 \text{ Feb } 2011 : \text{TIME}; \text{Nome} : \text{LOC}; o1 : \text{temp\_at\_in}(3:45\text{AM}, 17 \text{ Feb } 2011, \text{Nome}, -28)$

TTR offers a straightforward way for us to model propositions and questions using records and functions:

- **Propositions:** A proposition is a record of the form in (67a). The type of propositions is the record type (67b) and truth can be defined as in (67c):

- (67) a. 
$$\begin{bmatrix} \text{sit} & = & r_0 \\ \text{sit-type} & = & p_0 \end{bmatrix}$$
- b. 
$$\begin{bmatrix} \text{sit} & : & \text{Record} \\ \text{sit-type} & : & \text{RecType} \end{bmatrix}$$
- c. A proposition  $\begin{bmatrix} \text{sit} & = & r_0 \\ \text{sit-type} & = & p_0 \end{bmatrix}$  is true iff  $r_0 : p_0$

I have shown how to define negation, disjunction, and conjunction operations for this view of propositions.

- **Questions:** A question can be identified as a propositional abstract—a function from records into propositions:

- (68) a. who ran
- b. TTR representation—
- $$(r : \begin{bmatrix} x & : & \text{IND} \\ \text{rest} & : & \text{person}(x) \end{bmatrix}) \mapsto \begin{bmatrix} \text{sit} & = & r_1 \\ \text{sit-type} & = & [c : \text{run}(r.x)] \end{bmatrix}$$

That is, a function that maps records  $r : T_{who} =_{def}$

$$\left[ \begin{array}{ll} x & : \text{IND} \\ \text{rest} & : \text{person}(x) \end{array} \right] \text{ into propositions of the form}$$

$$\left[ \begin{array}{ll} \text{sit} & = r_1 \\ \text{sit-type} & = [c : \text{run}(r.x)] \end{array} \right]$$

I have shown how to define a number of notions of answerhood for questions, along with the relations of question dependence and influence. All these will be of importance in developing a theory of relevance for dialogue.

The ontology satisfies the following requirements:

1. It is *intentional*: providing entities such as propositions and questions which can be shared across participants and languages and enabling us to develop a theory of event anaphora.
2. It is fine-grained: it does not run into logical omniscience or Soames's puzzle. It is not too fine-grained, so it allows for identity of content across translation or synonymous reformulation, leaving an account of, for instance, clarification potential to a theory of locutionary propositions (see Chapter 6).
3. It allows for distinctions to be made between various kinds of 'informational entities', including events, situation types, propositions, questions, and outcomes. Given the structured means by which the latter three are constructed, it is possible to spell out a semantic commonality, needed to explicate adverbial modification and nominal quantification, the latter exemplified in (69a). On the account provided here, we associate the descriptive contents—i.e. modulo illocutionary force—in (69b–d) with the corresponding sentences uttered in a single context:

- (69) a. Everyone vacate the building! Did everyone vacate the building? Everyone vacated the building.
- b. Bo will leave  $\mapsto \left[ \begin{array}{ll} \text{sit} & = s \\ \text{sit-type} & = [c : \text{leave}(b,t)] \end{array} \right]$
- c. Will Bo leave?  $\mapsto \left( \left[ \begin{array}{ll} \text{sit} & = s \\ \text{sit-type} & = [c : \text{leave}(b,t)] \end{array} \right] \right)$
- d. Leave, Bo!  $\mapsto \left[ \begin{array}{ll} \text{sit} & = s \\ \text{irr-sit-type} & = ([t : \text{Time}]) [c : \text{leave}(b,t)] \end{array} \right]$

What is common across the three utterances is the situation type  $[c : \text{leave}(b,t)]$ . Where indicative and subjunctive/imperative verb forms differ is that in the latter, an abstraction occurs over the temporal argument.

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## Basic Interaction in Dialogue

### 4.1 Introduction

Conversation is often likened to a game. This is quite a useful metaphor: conversation as a joint, rule-governed activity, whose progress is relative to a constantly shifting context. As in most games the beginning and end are reasonably constrained—they involve an exchange of greetings and partings respectively of greater or lesser extent. What will happen in the middle is typically less predictable, as in a football game after set pieces:<sup>1</sup> once an initial move is made, what happens subsequently (or rather what is expected to happen) is constrained by certain conventions but this still leaves things quite open-ended. The task of the dialogue analyst is, arguably, not significantly different from that of the syntactician or semanticist: whereas the latter describe conventionally acceptable sequences of words and the effects they give rise to among conversation participants in terms of syntactic categories and semantic entities, the dialogue analyst describes conventionally acceptable moves and the effects they give rise to among conversation participants in terms of information states. In fact, towards the end of this chapter I will suggest a somewhat more global perspective, in which the analysis is at the level of types of conversation stretches, or as I will call them *genres*, a notion related to the activity types of Allwood (1995) and also akin if not quite coextensive with the CA notion of *speech exchange systems* (see e.g. Schegloff 1987a, 2007).<sup>2</sup>

<sup>1</sup> I mean here *Association* football or soccer; both rugby and American football are of course far more predictable.

<sup>2</sup> This view of the dialogue analyst as not dissimilar to the syntactician or semanticist is potentially provocative to those who view conversational action as being much less rule-bound and more open to variation. I have in mind critiques such as Levinson's dismissal of discourse analysis (see Levinson 1983: 287–94).

Two main points can be made in this regard. First, in the current set-up, as will become clear, by positing conversational rules, we do not put conversational participants in a straitjacket—we are merely making hypotheses about norms or expectations, particularly in reactive situations. Second, one of the insights gained from the recent 'statistical turn' in computational linguistics is that obtaining wide empirical coverage of syntax using discrete rule systems is difficult. Given this, there is no *a priori* reason to think that wide coverage of dialogue coherence can be achieved using *simple* rules. However, just as with sentence-level syntax, this need not lead to nihilism—one can certainly strive to produce wide coverage by combining rule-based and statistical techniques.

One methodological constraint I adopt is that the account I develop for two person dialogue (duologue) should both scale down to monologue and up to multilogue. Take as an example querying. Here we have the following possible set-ups:

- (1) a. Monologue: self-answering (A: *I met an... what's the word? Ichthyologist today.*)
- b. Duologue: querier/responder (A: *Who should we invite? B: Perhaps Noam.*)
- c. Multilogue: multiple discussants (A: *Who should we invite? B: Perhaps Noam. C: Martinu. D: Bedrich. ...*)

A basic aim is that the commonality across these distinct interaction types can be captured by the account developed here. This contrasts with a view of querying derived from Speech Act Theory (e.g. Searle 1969), still widely assumed see e.g. Asher and Lascarides (2003), where there is a very tight link to intentional categories of two-person dialogue ('... Speaker wants Hearer to provide an answer ... Speaker does not know the answer ...').

As with sentence-level syntax and semantics the basic strategy for achieving cross-contextual generality is to assume some type of *compositionality*. I will decompose interaction protocols using a technique pioneered by Larsson (2002) and Cooper (2006), namely using Type Theory with Records (TTR) to describe conversational update rules. This chapter builds on the semantic ontology developed in Chapter 3, but abstracts, for the most part, from the communicative process. This means adopting the simplifying assumption that the content of a given utterance is transmitted flawlessly from speaker to addressee. This, in turn, means that the internal structure of utterances will not be at issue. I will abandon the assumption of perfect communication in Chapter 6, after having developed an account of dialogue-oriented grammar in Chapter 5. Larsson's conversational rules, which underpin a number of implemented systems, emphasize the public/private interface. Generally they describe the modification of the total information state (TIS) which encapsulates both public and private aspects. They are explicitly intended for duologue (two-person dialogue), in which the interlocutors make utterances on alternate turns.<sup>3</sup> The rules I write below attempt to be somewhat more general and emphasize the public aspect of information states in so far as possible. There will be appeal to private, agent-relative parameters—the hidden variables of interaction—at those junctures underdetermined by public input.

The public/private dichotomy in conversational participant representation is the first topic I discuss. Following that I discuss the components of the public part of ISs that will figure in the account of interaction, which constitutes the bulk of this chapter. In particular, I offer rules for greeting, parting, querying, and assertion in dialogue.

<sup>3</sup> Indeed where one of the interlocutors is a (computational) system and the other a user, though this does not influence the set-up very much.

I briefly discuss some applications of the framework to monologue and conclude by considering how genre-specific aspects of conversation fit in to the emerging picture of interaction.

## 4.2 Public vs Private Context in Interaction

In psycholinguistics there has been a long-standing controversy concerning whether conversational participants are egocentric or monitor their interlocutor's information state (or multiply, in case of multilogue). In fact, views are more subtle than this dichotomy suggests. On the one hand are proposals that audience monitoring is almost entirely emergent, as a late, delayed inference (e.g. Brown and Dell 1987), retrospectively via self-monitoring and adjustment (e.g. Horton and Keysar 1996), or most radically via priming (Garrod and Pickering 2004). Conversely, there are the views that speakers design their utterances with their addressees in mind—*audience design* (Clark 1996)—or implicitly come to agreements as to how objects should be referred to—*conceptual pacts* (Brennan and Clark 1996).<sup>4</sup> In deciding which architecture to adopt it is worth considering the reasons analyses of conversational use have made reference both to a public aspect (the current state of the conversation, conversational scoreboard, context etc.), as well as to the non-public, private parts of a conversationalist. The former is the domain much previous work in semantics has focussed on and conditions linguistic behaviour such as:

- **proper name usage:** the successful use of a proper name, say 'Prashant', to refer to an individual A requires that 'Prashant' be publicly known as a name for A.<sup>5</sup>
- **anaphora and ellipsis resolution:** the successful use of a pronoun, say 'she', requires the common knowledge—or related notion—of a previous utterance bearing feminine gender that fixes its reference (or commonly perceptible entity referable as a female). Similar conditions *mutatis mutandis* apply to pro-verbs such as bare English auxiliaries and other elliptical constructions we discuss below.

The role of the non-public part has been emphasized in AI and related computational work for such notions as:

- **Sincerity:** does public behaviour by A (assertion that *p*; query *q*) match A's unpublicized state (belief that *p*; interest in *q*)?
- **Cooperativity:** does A's response to B show that A is taking into account B's (potentially non-public) goals?

<sup>4</sup> A wide range of views on this controversy can be found in Behavioural and Brain Sciences (2004).

<sup>5</sup> The phrase 'successful use' is of course not unproblematic in a dialogue setting: *successful for whom?* is the first up; second is: what means does an addressee have in case of partial understanding (accommodation, CRs etc.). The issues which surround it—grounding and clarification interaction—will be addressed in Chapter 6. For the moment, I take a speaker-centric point of view.

- **Intonation:** in recent work Beyssade and Marandin (2007a) have suggested that certain intonational contours in French used in conveying content *p* involve the speaker attributing to the addressee non-public beliefs concerning *p*:

Falling contours are used when the speaker presents the context of her utterance as a non-defective context (*a context in which the speaker assumes consistency of his assumptions about the current issue with the addressee's assumptions*—J.G.). On the other hand, non-falling contours are used when the speaker presents the context of her utterance as a defective context.

(Beyssade and Marandin 2007a: 10).

In light of this, we can enumerate a number of possible models of the information states of conversational participants. Model (2a) eschews positing a public domain—all updates are simply relative to a given participant; (2b) goes to the other extreme, eschewing any reference to unpublicized information, all information is public and shared. Model (2c) assumes that conversational participants keep track of their own and of their interlocutor's non-public information; (2d) partitions the information state into a private and a public part, which is shared among the participants; finally, (2e), like (2d), partitions the IS into public/private, but relativizes both to the individual conversational participants:

- (2) a. A: ⟨A.private⟩, B: ⟨B.private⟩  
 b. A: ⟨public⟩, B: ⟨public⟩  
 c. A: ⟨A.private, A.B.private⟩, B: ⟨B.private, B.A.private⟩  
 d. A: ⟨public, A.private⟩, B: ⟨public, B.private⟩  
 e. A: ⟨A.public, A.private⟩, B: ⟨B.public, B.private⟩

Model (2a) is conceptually the most parsimonious position: it recognizes that in modelling conversational agents we need access to their non-public information states. However, it misses out on shared/unshared information distinctions crucial for explicating various conditions underlying linguistic use exemplified above. In the discussion of their Interactive Alignment approach to dialogue, Garrod and Pickering (2004) seem to adopt a variant of this architecture—dialogue on their view proceeds primarily as a sequence of the participants priming each other as a consequence of their utterances. I use the word 'variant' since Pickering and Garrod do not acknowledge any phenomena that involve appealing to non-public information.

Model (2b) is the standard formal semantics approach. It underpins linguistic use that relies on shared information. Roberts (e.g. Roberts 1996) has explicitly argued for the primacy of this architecture as a notion of context, on methodological grounds. The idea being that the common ground constitutes a normative (albeit idealized) entity with which conversationalists strive to identify their information states. In so far as one is interested in cognitive modelling, however, such an architecture gives up on any sort of other-modelling. It can only handle mismatches across participants exemplified by the Turn-Taking Puzzle (see (23), Chapter 2) by considering the individual information states. This means that actually a certain redundancy emerges, as we

have three entities: the individual information states and the disembodied common ground.

Model (2c) has been advocated in recent work by Beyssade and Marandin (2007a) and defended also by Schober in Behavioural and Brain Sciences (2004). The main problem with such an architecture is that it is either cognitively expensive or static. Thus, on its most faithful implementation it is cognitively expensive and involves redundancy, since it requires each conversational participant to reflect after each utterance who is committed to what. Indeed this problem is exacerbated, as pointed out by Pickering and Garrod (Behavioural and Brain Sciences 2004), in multilogue, suggesting as it does that conversation is like simultaneous chess, harder and harder the more participants are around. Experience tells us that, in some sense, this is not the case: conversations with three or six persons *can* be had with reasonable ease. Nonetheless, this argument cannot be taken too far—it is clear (and backed by corpus evidence, e.g. Eshghi and Healey 2006) that a multilogue *is* a qualitatively different experience from a duologue. Before a detailed account of duologue/multilogue differences emerges—some initial discussion is provided in Chapter 8—we should be careful not to make strong claims about duologue/multilogue scalability.

An alternative construal of this architecture is to minimize the cognitive load involved by forgoing continual commitment separation. However, this means that the conversational participants cannot be said to be *keeping track* of each other's unpublicized states. All they are doing is making very occasional hypotheses about these states. In other words, such states constitute contextual parameters that need to be instantiated on an *ad hoc* basis. Susan Brennan and her collaborators (see Galati and Brennan 2006, Brennan and Hanna 2009) have proposed to call such an approach a *1-bit model*. They argue that various partner-oriented adaptations such as *lexical entrainment* can all be explicated within such an approach, without positing an elaborate, computationally costly 'partner model'.

Model (2d) represents the null hypothesis of Lewis's scoreboard metaphor for context (Lewis 1979). In this approach some of the action takes place in the shared component and some originates in the private component. Each conversational participant does not in general keep track of the other's private component, but given that her own is explicitly modelled she can reason about the other's, most simply, but not exclusively of course, by assuming it is similar to her own and the shared part.

Model (2e) is the architecture assumed in KoS since its earliest days (Ginzburg 1994). It is also essentially what Poesio, Rieser, and Traum assume in their PTT framework (e.g. Poesio and Traum 1997, Poesio and Rieser 2010).<sup>6</sup> Rather than viewing context as an entity abstracted from the conversational participants, each conversa-

<sup>6</sup> Poesio and Traum (1997) distinguish between the following components of each conversational participant's information state: DU (ungrounded (sub-)utterances), G (grounded (sub-)utterances), and the unpublicized state. In principle A.DU and B.DU can be mismatched. See Chapter 6 for more on this and for exposition of how the DU/G dichotomy gets handled in KoS.



tional participant is assigned their own gameboard, where they record their version of the public conversational action and relative to which they compute their possible reactions. Much of the time the conversational participants' DGBs are identical, but differences can arise given the intrinsic speaker/addressee differences: a speaker is committed to knowing what his utterance means,<sup>7</sup> whereas an addressee has no such commitment. This is what I take to underlie the TTP in its various versions discussed in Chapter 2—semantic options across participants need not be identical.

Such differences become even more marked in multilogue, where, as we discuss in Chapter 8, following Goffman (1981), distinct types of non-speaker (direct addressee, side participant, overhearer etc.) with distinct semantic options need to be accommodated.

### 4.3 Public Context in Interaction

The discussion in section 2.1 led to the characterization of a conversational participant's dialogue gameboard as a data structure whose primary attributes are the following:<sup>8</sup>

- FACTS: set of commonly agreed-upon facts.
- QUD ('questions under discussion') : a partially ordered set that specifies the currently discussed questions.
- LATEST-MOVE: content of *latest move* made.

Before offering a formal TTR characterization of DGBs and their role in conversational updates, I would like to discuss briefly some assumptions about the different components of the DGB.

#### 4.3.1 FACTS

FACTS represents the shared knowledge that conversational participants utilize during a conversation. More operationally, this amounts to information that a conversational participant can use embedded under presuppositional operators:<sup>9</sup>

<sup>7</sup> Although disfluencies, discussed in Chapter 8, requires this assumption to be made more subtle incrementally.

<sup>8</sup> I do not develop a treatment of instructions/requests/commands. See Beyssade and Marandin (2007b) for an account of such moves within an extended and distinctive version of KoS, building on a proposal in Portner (2004). In order to develop their account Beyssade and Marandin, following Portner, posit an extra field in the DGB, which they dub the *To-Do-List*. The *To-Do-List* is of type *set(outcome)*.

<sup>9</sup> Of course, if we wish we could reduce all context to FACTS. Thus, if A has just said *p*, B can factualize this as 'Given that you just claimed that *p*'. Similarly with QUD: if A and B are discussing who is the greatest living Kyrgyz writer, A can say 'Given that we're discussing who is the greatest living Kyrgyz writer.' This sort of 'factive reentrancy' is an important property of the public context. On the other hand we get a more perspicuous theory of context by recognizing its distinct facets rather than by clumping everything together.

- (3) a. ...Given that
- b. ...Since we know that

There are a number of significant details that need to be worked out concerning FACTS: the most important being its ontological make-up and its structure. For most purposes we need in this work, we can stay fairly agnostic about these, nonetheless. Thus, whether the elements of FACTS are propositions or facts (a distinct class of entities from *true* propositions, see Asher 1996, Peterson 1997, Ginzburg and Sag 2000 for some recent discussion) or some other entity is an interesting problem for natural language metaphysics, but one which we can steer clear of here. For simplicity, I will assume that the witnesses of FACTS are propositions.<sup>10</sup>

Similarly, there are various interesting and important issues concerning the structure of FACTS: should we assume it to be closed under various operations (conjunction/disjunction (Asher 1993), agent-relative consequence, logical consequence etc.)?<sup>11</sup> It is clear that if FACTS is to underpin presuppositional behaviour, it cannot allow for anything but very weak closure conditions. Thus, if A accepts *p*, following B's assertion of *p*, if FACTS is closed under some operation OP, we would need to assume that all *p'* that are in the OP-closure of *p* are also in FACTS. This means that such *p'* also count as presupposed. This is not unreasonable with weak closure conditions such as conjunction/disjunction, but is steadily less plausible as the closure condition becomes more cognitively complex. The resolution of this issue will not be of much direct import to us here generally. The two places where it will emerge again are in Chapter 6, where we appeal to a form of conjunction reduction to explain certain utterance-related presuppositional behaviour, and in Chapter 8, when we discuss the possible antecedents of bare 'Why'-phrases—utterances of the form 'Why?' The antecedents for these need to be presupposed, but moreover somehow germane to the current question under discussion. This will suggest the need for some additional structure within FACTS.

#### 4.3.2 MOVES

In Chapter 2 I motivated the need for a contextual attribute *LatestMove*, a distinguished fact that characterizes the content of the most recent move made. The main motivation was to segregate from the entire presuppositional mush information on the basis of which coherent reactions could be computed. For subsequent development we will need two generalizations of this attribute. One generalization, of relevance already in this chapter, and brought to the fore in multilogue in Chapter 8,

<sup>10</sup> If, as much evidence suggests, non-propositional elements should actually be used, a distinction that might prove of practical importance in generation, the required modifications to the update rules we posit will be minimal.

<sup>11</sup> One other issue is whether to treat FACTS as a set of propositions or as a single proposition that corresponds as it were to the conjunction of all the members of FACTS. I assume the former since it is more flexible than the latter. But for current purposes this will not make much difference.

is not to restrict attention solely to the *latest* move made. Hence we will keep track of previously made utterances using an attribute Moves: this will be list-valued and its first element will be LatestMove. A further generalization, which will be implemented in Chapter 6 and onwards, concerns the *type* of witnesses of MOVES. Although in most situations the most significant aspect of an utterance is its content, I will nonetheless take the type of LatestMove to be a *locutionary proposition*, a proposition that characterizes the utterance effecting the move and the linguistic sign associated with the utterance. The motivation for this, already sketched in Chapter 2, relates directly to the need to provide grounding/clarification interaction conditions.

It is an interesting theoretical question as well as an empirical and practical one: what extent ('window size') should Moves have? I will not have any definitive proposal with respect to this issue, but will consider some data on this as we proceed.

### 4.3.3 QUD

The property of being in QUD that a question can bear is primarily taken to correlate with the fact that that question constitutes a 'live issue': a question that has been *introduced for discussion* at a given point in the conversation and not yet been *down-dated*. How exactly a question gets introduced into and downdated from the DGB is a complex issue which will be discussed in detail here, in particular in section 4.4.3; additional, indirect ways for questions to get added into QUD will emerge in section 4.6 (via domain-based accommodation, based on Larsson 2002) and in Chapter 6 (via clarification context accommodation, which allows clarification questions to emerge from partially understood utterances). In order to say anything more substantive about QUD, we need to address the issue of the ordering of its elements: although more than one question can be under discussion at a given point, locally (i.e. at the point a given utterance is made) one question will tend to predominate. Thus, we need to assume that QUD is ordered in some way, by means of an ordering denoted as in (4a). When we use list notation for QUD, we will denote this as (4b):

- (4) a.  $q_1 \succ_{A.DGB.QUD} q_2$   
       b.  $\langle q_1, q_2 \rangle$

This order is construed as 'takes conversational precedence'.<sup>12</sup> In other words,  $q_1 \succ_{A.DGB.QUD} q_2$  requires A to believe that  $q_1$  deserves precedence—for a number of possible reasons I discuss below—over  $q_2$  at this stage of the conversation. The maximal element of QUD I will often refer to simply as QUD-maximal or MaxQUD. It plays the role often dubbed in previous work as *discourse topic*. The most important consequence of being MaxQUD is that it circumscribes what can be said: it introduces an expectation for utterances that are *specific* to  $q$ , that is are either partial answers

<sup>12</sup> I will usually omit the subscript from  $\succ$ , unless it is important to distinguish whose ordering is involved.

or sub-questions of  $q$ . More technically, they are either propositions  $p$  such that  $\text{About}(p, q)$  or questions  $q_1$  such that  $\text{Influence}(q_1, q)$ . We formulate this, informally for now, as follows:

- (5) a.  $q$ -specificity: If  $C$  has the turn and  $q$  is MaxQUD, then  $C$  can make an utterance specific to  $q$ .
- b. An utterance  $u$  is specific to a question  $q$  iff  $\text{content}(u)$  is either (a) a proposition  $p$  such that  $\text{About}(p, q)$  or (b) a question  $q_1$  such that  $\text{Influence}(q_1, q)$ .

As much past literature has observed, QUD-maximality—or related notions pertaining to ‘information structure’—has various linguistic reflexes, licensing the use of certain syntactic structures and intonational tunes. The relation between QUD-maximality and the potential for non-sentential utterances is discussed in Chapter 7. The commonest propositional NSUs are *short answers* (where the antecedent provides a unary *wh*-question) and *Affirmative Answer particles* (e.g. ‘yes’). I assume that the possibility of uttering a short answer/affirmative answer particle appropriate to  $q$  is a necessary condition for  $q$  being the MaxQUD. This provides us with a concrete test for QUD-maximality. The structure of QUD is exploited in particular in the resolution of long-distance NSUs such as the constructed (6a) and the naturally occurring (6b):<sup>13</sup>

- (6) a. A(1): Who is coming to the barbecue?  
 B(2): The barbecue on Sunday?  
 A(3): The 29th yes.  
 B(4): Sunday is the 28th.  
 A(5): Oh right, yes the 28th.  
 B(6): The one Sam’s organizing?  
 A(7): Yes.  
 B(8): Will it be on even if it snows?  
 A(9): Sam hasn’t said anything.  
 B(10): Right. Anyway, I’d guess Sue and Pat for sure, maybe Alex too.
- b. Allan(1): How much do you think?  
 Cynthia(2): Three hundred pounds.  
 Sue(3): More.  
 Cynthia(4): A thousand pounds.  
 Allan(5): More.  
 Unknown(6): (unclear utterance)  
 Allan(7): Eleven hundred quid apparently. [BNC, G4X]

<sup>13</sup> This is not an uncontroversial assumption, disputed *inter alia* by Larsson (2002), who essentially doubts the existence of long-distance NSUs. Given this, he effects a bifurcation in the function of QUD: a field *ISSUES* tracks questions that have been introduced and not been resolved, whereas QUD is a very *local* field structured as an *open stack*, tracking only very recently introduced questions. I return to this issue in Chapter 7 and in particular in Chapter 8.

At any stage of a simple dialogue conducted in a cooperative spirit, a *stack* will emerge on QUD: the question added in last becomes the maximal element. The basic argument is this: assume A poses a question  $q_1$  which elicits as response from B a question  $q_2$ . Since B can be assumed to attempt to provide information about  $q_1$  were she able to do so, the fact that she has not done so must mean that B cannot do so without first (attempting to) resolve  $q_2$ , that is he believes that  $\text{Influence}(q_2, q_1)$ . Now if A also believes that  $\text{Influence}(q_2, q_1)$ , then it will be rational for both A and B to adopt:  $q_2 \succ q_1$ .

The assumption that QUD is a stack, often made in past literature,<sup>14</sup> although it has very good coverage, is ultimately a consequence of the fact that the order according to which issues are presented in conversation is not arbitrary but typically reflects issue dependence (see Chapter 3, section 3.3.8). When this does not obtain, the simple stack structure can be overridden. Examples of this arise in multilogue (see Chapter 8), when a speaker makes multiple successive moves (see examples in section 4.5), and indeed when conversational participants do not share beliefs about issue dependence.<sup>15</sup> It is perfectly *coherent*, if potentially impolite, for a conversational participant to respond to a question with another question which *she* views as taking precedence. This can lead to deadlocked Buridan's Ass-like situations where one conversational participant needs to give way but it's not clear which one will:

- (7) a. A: Who did Bill invite?  
       B: Which of his friends do you know?  
       A: Before I can answer this, you really need to answer my question.  
       B: But I cannot answer it before you answer mine.
- b. Guildenstern: What's your name?  
       Rosencrantz: What's yours?  
       Guildenstern: I asked first. (Tom Stoppard, *Rosencrantz and Guildenstern Are Dead*, p. 33)

Indeed in extreme cases an *effect* of an utterance can simply be to reverse the ordering on QUD. In (8(1)) A starts by posing a question to which B responds. In (8(3)) A suggests that B was wrong in thinking that B's question takes precedence. In (8(4)) B accepts A's correction. Consequently, A's question become maximal in QUD, taking precedence over B's question, and indeed even an elliptical response is licit as (8(4))

<sup>14</sup> See e.g. Roberts (1996).

<sup>15</sup> A fourth case for non-stack behaviour is *cross-talk*, cases where two distinct conversations occur involving conversational participants who participate in both conversations:

- (i) A (to B): What did they have in that shop? C (to D, E, and overheard by A): Where have these workshops taken place?  
       D: First one was in Kabul.  
       B (to A): Great stuff from Miyake.  
       A (to C, D, E): Second one was in Peshawar, no? (modelled on an example due to Justine Cassell (p.c.))

shows. A then accepts B's assertion about his question, downdates the question from QUD, and then addresses B's question, possibly using ellipsis, as in (8(5b)).

- (8) A(1): Where can I use my BTT preferred customer credit card?  
 B(2): Which malls in town do you plan visiting?  
 A(3): Well, as far as I can tell, we need to get to the bottom of my question first.  
 B(4): OK, sir. Let me talk to my supervisor [phone call]. As far as she knows, Harrods, M&S, Tesco, and William Hill.  
 A(5a): Interesting. So I'm off to Gateshead and the Metrocentre.  
 A(5b): Right, so Gateshead and the Metrocentre!

Note that one could get an equally acceptable dialogue in which QUD got precisely the reverse ordering:

- (9) A(1): Where can I use my BTT preferred customer credit card?  
 B(2): Which malls in town do you plan visiting?  
 A(5b): Right. Gateshead and the Metrocentre.  
 B(4): OK, sir. Let me talk to my supervisor [phone call]. As far as she knows, Harrods, M&S, Tesco, and William Hill.

The difference between the two dialogues might still be attributed to distinct underlying domain plans:<sup>16</sup> in (8) where A shops seems to be determined by where he can pay with his BTT card, whereas in (9) there is no such implication. However, it seems that no such difference can be detected in the following two dialogues:

- (10) A(1): Who's on the Mafia's hit-list at the moment?  
 B(2): Which hit-men are currently in town?  
 A(3): Well, I really think that we need to get to the bottom of my question first.  
 B(4b): OK. Let me talk to Enzo [phone call]. As far as he knows, Maldini and Chiesa.  
 A(5b): Interesting. So, Masserotti and Baggio, ha?
- (11) A(1): Who's on the Mafia's hit-list at the moment?  
 B(2): Which hit-men are currently in town?  
 B(4b): Masserotti and Baggio.  
 A(5b): Interesting. So, Maldini and Chiesa, ha?

The most direct conclusion we draw is that conversational structure is in part *constructed* by the participants. This is not in any way to downplay the importance of the structure or the influence it has on how the conversation is carried out. But it is to say that the structure must be analysed in a way that makes it intrinsic to the conversational participants and affectable by them. In particular, then, we conclude

<sup>16</sup> Following the view of Grosz and Sidner (1986) that intentional structure drives discourse structure; see Asher and Lascarides (1998) for critical discussion in the context of anaphora resolution.

that the information about the ordering on QUD is internal to the DGB and moves should be able to affect it to some extent, as exemplified in (8) and (10) above. This issue is directly tied up with the issue of *agreement to discuss an issue*, to which I return, albeit briefly, in Chapter 8.

## 4.4 Basic Interaction: the Evolution of the DGB

### 4.4.1 Conversational Rules

Having informally described the components of the DGB, we can now proceed to describe patterns of conversational interaction in terms of the evolution of the DGB. The first thing we need to determine is a TTR description of the DGB since conversational rules will involve specifying how one DGB configuration can be modified into another.

DGBs are records all of which are characterized by the type *DGBType*. Given our discussion above, the *DGBType* will have fields for Facts, Moves, and QUD. In addition it will also have fields which track who is holding the turn, as well as a time:<sup>17</sup>

$$(12) \quad \text{DGBType (initial definition)} =_{\text{def}} \left[ \begin{array}{ll} \text{spkr} & : \text{IND} \\ \text{addr} & : \text{IND} \\ \text{utt-time} & : \text{Time} \\ \text{c-utt} & : \text{addressing}(\text{spkr}, \text{addr}, \text{utt-time}) \\ \text{Facts} & : \text{Set}(\text{Prop}) \\ \text{Moves} & : \text{list}(\text{IllocProp}) \\ \text{QUD} & : \text{poset}(\text{Question}) \end{array} \right]$$

We call a mapping that indicates how one DGB can be modified by conversationally related action a *conversational rule*, and the types specifying its domain and its range respectively the *preconditions* and the *effects*.

Typically, such rules will involve a preconditions type which is more specific (i.e. a subtype of) the DGB type, similarly with the effects; in a small number of cases, the rules will also involve reference to non-public components of the information state. Notation-wise, a conversational rule will, therefore, be specified as in (13a). We will often notate such a mapping as in (13b):

<sup>17</sup> As I indicated in Chapter 3, in TTR the labels in a type are essentially arbitrary. Hence, what restricts the witnesses for the *spkr*, *addr* fields in (12) to be the speaker and addressee of an interaction is the complex type *addressing* (*spkr*, *addr*, *utt-time*). Nonetheless, for the most part, we will omit this specification and that of the utterance time in the interest of clutter elimination. It should always be understood as being present when the *spkr*, *addr* fields are present.

- (13) a.  $\text{preconds} : \text{PRECONDS} \sqsubseteq \text{DGB} \mapsto \text{effects} : \text{EFFECTS} \sqsubseteq \text{DGB}$   
 b. 
$$\left[ \begin{array}{l} \text{pre(conds)} \\ \text{effects} \end{array} : \left[ \begin{array}{l} \dots \\ \dots \end{array} \right] \right]$$

Just as the grammar specifies ways to construct larger units by composing smaller units, here we can recognize one fundamental compositional principle:

- (14) Composition of conversational rules: given two conversational rules *part1*, *part2* that satisfy  $\text{preconds}(\text{part2}) \sqsubseteq \text{effects}(\text{part1})$  they can be composed yielding a new conversational rule whose  $\text{preconds} = \text{preconds}(\text{part1})$  and whose  $\text{effects} = \text{effects}(\text{part2})$

Rule (14) will be the basic tool we use in decomposing protocols and establishing a coherence principle that provides the basis for explicating the coherence of a sequence of moves. I formulate this principle, in section 4.4.5, after discussing some examples of update rules in the upcoming sections.

*Adjacency pairs*, exemplified in (12) of Chapter 2, involve the addressee of *part1* becoming the speaker of *part2*, and the successive move types related by a coherence relation. One of our main tasks is to elaborate this relation, but assuming it as given, we could specify adjacency pairs, or somewhat more specifically *two person turn exchange pairs* (2-PTEP) as in (15). This specifies an input DGB, one in which the LatestMove is  $P1(\text{spkr}, \text{addr})$ , for some illocutionary relation  $P1$ , *spkr* the speaker and *addr* the addressee. This can be coherently followed up by a DGB whose LatestMove is  $P2(\text{addr}, \text{spkr})$ , for some illocutionary relation  $P2$ , where the previous addressee *addr* is now the speaker and the previous speaker *spkr* is now the addressee.

- (15) **two-person turn exchange pairs (2-PTEP)**

$$\left[ \begin{array}{l} \text{pre} \\ \text{effects} \end{array} : \left[ \begin{array}{l} \text{spkr} : \text{Ind} \\ \text{addr} : \text{Ind} \\ P1 : \text{IllocRel} \\ \text{LatestMove} = P1(\text{spkr}, \text{addr}) : \text{IllocProp} \\ \text{qud} : \text{poset}(\text{Question}) \\ \text{facts} : \text{Set}(\text{Prop}) \\ \text{spkr} = \text{pre.addr} : \text{Ind} \\ \text{addr} = \text{pre.spkr} : \text{Ind} \\ P2 : \text{IllocRel} \\ \text{Moves} = \left\langle P2(\text{spkr}, \text{addr}), \right. \\ \quad \left. \text{pre.LatestMove}, \dots \right\rangle : \text{list}(\text{IllocProp}) \\ \text{cl} : \text{AdjPair}(P1, P2) \\ \text{qud} = \text{pre.qud} : \text{poset}(\text{Question}) \\ \text{facts} = \text{pre.facts} : \text{Set}(\text{Prop}) \end{array} \right] \right]$$



(15) contains one notational abbreviation adopted subsequently throughout the book, one that relates to *illocutionary propositions*. These are the content of conversational moves. An example of such a proposition is in (16a), where in addition to the two standard fields for a proposition, fields exist for speaker, addressee, and descriptive content. (16b) is the general characterization of the type *IllocProp*. For notational economy I will typically abbreviate (16a) as (16c), omitting the fields, retaining only the illocutionary complex type component:

- (16) a. 
$$\left[ \begin{array}{ll} \text{sit} & = \text{ul} \\ \text{spkr} & = \text{x} \\ \text{addr} & = \text{y} \\ \text{a} & = \text{p} \\ \text{R} & = \text{Assert} \\ \text{sit-type} & = [\text{cl} : \text{R}(\text{spkr}, \text{addr}, \text{a})] \end{array} \right]$$
- b. 
$$\text{IllocProp} =_{\text{def}} \left[ \begin{array}{ll} \text{sit} & : \text{Record} \\ \text{spkr} & : \text{Ind} \\ \text{addr} & : \text{Ind} \\ \text{a} & : \text{AbstSemObj} \\ \text{R} & : \text{IllocRel} \\ \text{sit-type} = [\text{cl} : \text{R}(\text{spkr}, \text{addr}, \text{a})] & : \text{RecType} \end{array} \right]$$
- c.  $\text{R}(\text{spkr}, \text{addr}, \text{p})$

Let us now consider some much more specific types that relate to particular kinds of interactions.

#### 4.4.2 *Greeting and Parting*

I start by considering simple conversational rules, those for greeting and parting. An initiating greeting typically occurs dialogue-initially. We can capture this by postulating a conversational rule for greeting that has as a precondition the requirement that Moves is the empty list. The primary contextual effect of such a greeting, as far as we are concerned here, is simply providing the addressee with the possibility of reciprocating with a counter-greeting.<sup>18</sup> A greeting has of course some additional import involving non-hostility, perhaps even friendliness, an import that plausibly has

<sup>18</sup> I say *possibility* because conversations initiated with greetings have perfectly coherent continuations without a counter-greeting being uttered:

- (i) A: Hi Mo. How are you?  
B: OK. Where are you heading?

Subsequently we will see how to build in the apparent norm that conversations among acquaintances typically involve an exchange concerning each other's well-being.

correlates in similar interactions in the animal world.<sup>19</sup> This import could be encoded within a suitable logic for conversational moves, though I will not do so here. The conversational rule associated with greeting is given in (17). The preconditions state that both Moves and QUD need to be empty; the sole effect is to initialize Moves with the illocutionary proposition *greet*(*A*, *B*), *A* the speaker, *B* the addressee. This will constitute the precondition for a counter-greeting.

$$(17) \left[ \begin{array}{l} \text{pre} \\ \text{effects} \end{array} : \left[ \begin{array}{ll} \text{spkr} & : \text{Ind} \\ \text{addr} & : \text{Ind} \\ \text{moves} = \langle \rangle & : \text{list}(\text{IllocProp}) \\ \text{qud} = \{ \} & : \text{poset}(\text{Question}) \\ \text{facts} & : \text{Prop} \end{array} \right] \right]$$

$$\left[ \begin{array}{ll} \text{spkr} = \text{pre.spkr} & : \text{Ind} \\ \text{addr} = \text{pre.addr} & : \text{Ind} \\ \text{LatestMove} = \text{Greet}(\text{spkr}, \text{addr}) & : \text{IllocProp} \\ \text{qud} = \text{pre.qud} & : \text{poset}(\text{Question}) \\ \text{facts} = \text{pre.facts} & : \text{Prop} \end{array} \right]$$

In what follows we adopt a more economical notation: the preconditions can be written as a *merge* of two record types  $DGBType^- \wedge_{\text{merge}} PreCondSpec$ , one of which  $DGBType^-$  is a supertype of  $DGBType$  and therefore represents predictable information common to all conversational rules;  $PreCondSpec$  represents information specific to the preconditions of this particular interaction type. Concretely for (17):

$$(18) \text{ a. } DGBType^- = \left[ \begin{array}{ll} \text{spkr} & : \text{Ind} \\ \text{addr} & : \text{Ind} \\ \text{facts} & : \text{Prop} \end{array} \right]$$

$$\text{ b. } PreCondSpec = \left[ \begin{array}{ll} \text{moves} = \langle \rangle & : \text{list}(\text{IllocProp}) \\ \text{qud} = \{ \} & : \text{poset}(\text{Question}) \end{array} \right]$$

Similarly, the effects can be written as a merge of two record types  $DGBType^0 \wedge_{\text{merge}} ChangePrecondSpec$ , where  $DGBType^0$  is a supertype of the preconditions and

<sup>19</sup> On the latter, see Lorenz (1963) for the classic description and Hauser (1995) for a more recent perspective.

*ChangePreCondSpec* represents those aspects of the preconditions that have changed. So for (17):

$$(19) \quad \begin{array}{l} \text{a. } \text{DGBType}^0 = \left[ \begin{array}{ll} \text{spkr} = \text{pre.spkr} & : \text{Ind} \\ \text{addr} = \text{pre.addr} & : \text{Ind} \\ \text{qud} = \text{pre.qud} & : \text{poset(Question)} \\ \text{facts} = \text{pre.facts} & : \text{Prop} \end{array} \right] \\ \text{b. } \text{ChangePreCondSpec} = \left[ \text{LatestMove} = \text{Greet}(\text{spkr}, \text{addr}) : \text{IllocProp} \right] \end{array}$$

Given this, the information that characterizes a given conversational rule as far as the preconditions go is *PreCondSpec*, whereas as far as the effects go it is *ChangePreCondSpec*, so we can *abbreviate* conversational rules as in (20a), which for the rule for greeting yields (20b).

$$(20) \quad \begin{array}{l} \text{a. } \left[ \begin{array}{ll} \text{pre} & : \text{PreCondSpec} \\ \text{effects} & : \text{ChangePreCondSpec} \end{array} \right] \\ \text{b. } \text{Greeting} =_{\text{def}} \left[ \begin{array}{ll} \text{pre} & : \left[ \begin{array}{ll} \text{moves} = \langle \rangle & : \text{list(IllocProp)} \\ \text{qud} = \{ \} & : \text{poset(Question)} \end{array} \right] \\ \text{effects} & : \left[ \text{LatestMove} = \text{Greet}(\text{spkr}, \text{addr}) : \text{IllocProp} \right] \end{array} \right] \end{array}$$

It should be emphasized that this is merely *notational abbreviation*—in particular, the labels in a record type need to be introduced before they can be the constituents of a complex type. This is particularly pertinent with respect to the labels associated with speaker and addressee. Note that there are three basic possibilities in two-person interaction: the turn remains the same (21a), the turn changes (21b), or the turn is underspecified—it is up for grabs for either conversationalist, as in (21c), which picks the speaker from among the conversationalists and the addressee as the distinct other element in this set. No turn change gets abbreviated away in the notational convention introduced in (20). The other two cases will get notated explicitly—turn change by specifying the effects as in (21b), but turn underspecification by means of a merge-type conjunction, as in (21d), and exemplified in section 4.4.3 and many other cases:

$$(21) \quad \text{a. } \text{No-Turn-Change} =_{\text{def}} \left[ \begin{array}{ll} \text{pre} & : \left[ \begin{array}{ll} \text{spkr} & : \text{Ind} \\ \text{addr} & : \text{Ind} \end{array} \right] \\ \text{effects} & : \left[ \begin{array}{ll} \text{spkr} = \text{pre.spkr} & : \text{Ind} \\ \text{addr} = \text{pre.addr} & : \text{Ind} \end{array} \right] \end{array} \right]$$

$$\begin{aligned}
\text{b. Turn-Change} &=_{\text{def}} \left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{spkr} : \text{Ind} \\ \text{addr} : \text{Ind} \end{array} \right] \\ \text{effects} : \left[ \begin{array}{l} \text{spkr} = \text{pre.addr} : \text{Ind} \\ \text{addr} = \text{pre.spkr} : \text{Ind} \end{array} \right] \end{array} \right] \\
\text{c. Turnholder-Underspecified} &=_{\text{def}} \left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{spkr} : \text{Ind} \\ \text{addr} : \text{Ind} \end{array} \right] \\ \text{effects} : \left[ \begin{array}{l} \text{PrevAud} = \{ \text{pre.spkr}, \text{pre.addr} \} : \text{Set}(\text{Ind}) \\ \text{spkr} : \text{Ind} \\ \text{c1} : \text{member}(\text{spkr}, \text{PrevAud}) \\ \text{addr} : \text{Ind} \\ \text{c2} : \text{member}(\text{addr}, \text{PrevAud}) \\ \quad \wedge \text{addr} \neq \text{spkr} \end{array} \right] \end{array} \right] \\
\text{d. effects} : \text{TurnUnderspec} \wedge_{\text{merge}} [\dots]
\end{aligned}$$

We can exemplify turn change in the specification of counter-greeting: ignoring again the non-hostility import, a counter-greeting simply grounds the original greeting, requires no response, nor has other contextual effects. Indeed the conversation that starts with a greeting and ends with a counter-greeting e.g. *A: Hi, B: Hi* is surely a not uncommon one.<sup>20,21</sup> The effect it has is to modify the value of LatestMove with the illocutionary proposition CounterGreet(B, A). This rule using the abbreviatory convention introduced in (20) and (21) is given in (22):

$$(22) \text{ Countergreeting} =_{\text{def}} \left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{LatestMove} = \text{Greet}(\text{spkr}, \text{addr}) : \text{IllocProp} \\ \text{qud} = \{ \} : \text{poset}(\text{Question}) \end{array} \right] \\ \text{effects} : \left[ \begin{array}{l} \text{spkr} = \text{pre.addr} : \text{Ind} \\ \text{addr} = \text{pre.spkr} : \text{Ind} \\ \text{LatestMove} = \text{CounterGreet}(\text{spkr}, \text{addr}) : \text{IllocProp} \end{array} \right] \end{array} \right]$$

Parting is in some sense the mirror image of greeting. In this case, the second-part action is the more substantive. The main issue with parting is explicating what it means for a conversation to be at a stage that allows it to be terminated. This is

<sup>20</sup> I happen to have had two such conversations during the week in which this passage was written.

<sup>21</sup> I am introducing here a distinction between greeting and *counter-greeting*, motivated by the existence in some languages of forms usable only as *responses* to greetings (e.g. Arabic 'marhabteyn', 'sabax elnur', Hebrew 'boker mecuyan', etc.), intonational differences (e.g. in English initiating greetings involve fall, responsive greetings involve rise). The alternative is to characterize the background for greetings disjunctively, an instance of the initiating/reactive distinction further discussed below.

partly captured by the requirement that QUD be empty, for instance because all issues previously raised have indeed been discussed sufficiently for current purposes. However, mere emptiness of QUD does not seem quite strong enough—apparently parting cannot take place without a minimum of interaction, as (23a, b, c) suggest; though, as (23d) indicates, this interaction can be very threadbare indeed:

- (23) a. (Conversation-initially) A: Hi.  
           B: #Bye.  
       b. (Conversation-initially) A: #Bye.  
       c. (Conversation initially) A: Hi.  
           B: Hi.  
           A: #Bye.  
           B: #Bye.  
       d. (Conversation-initially)  
           A: Hi.  
           B: I'm off.  
           A: OK.  
           B: Bye.  
       e. (Conversation-initially)  
           A: Hi.  
           B: They've bombed the Via Dolorosa.  
           A: Oh my God.  
           B: #Bye.  
       f. A: Hi.  
           B: They've bombed the Via Dolorosa.  
           A: Oh my God.  
           B: Anyway, I'm off.  
           A: OK.  
           B: Bye.

So what does 'minimal interaction having taken place' amount to? One way of cashing this out is in terms of discussion of a certain issue having taken place. For instance, in section 4.6, I will suggest that a way of characterizing a conversation between reasonably intimate conversational participants is that such a conversation includes *inter alia* some bare reference to each of the conversational participants' personal situation. That such discussion took place can be captured by requiring that FACTS contain facts resolving the two issues,  $\lambda P.P(A)$  ('How is A'),  $\lambda P.P(B)$  ('How is B'). This will probably work in most cases, but does not cover cases like (24) suggested to me by Emanuel Schegloff (p.c.):

- (24) a. A: Hi. B: Hi. (B's infant on the floor starts crying, B attends to the child, the child does not relent, A retreats: ) A: Bye.

- b. A: Hi. B: Hi. (C, a close colleague of A, arrives and starts a technical discussion to which B cannot contribute; after a minute B departs, whispering: )  
B: Bye.

I will, therefore, include in the preconditions for parting a fact  $MinInteraction(\{A, B\})$  holding of the group of interlocutors. This can be reductively analysed as  $Resolved(\{\lambda P.P(A), \lambda P.P(B)\})$ , but leaves open the possibility of an alternative analysis. Section 4.6 will open up the possibility of an analysis that takes a more global view of the conversation. In light of this, the basic description of parting is in (25)—given the precondition of an empty QUD and  $MinInteraction(\{spkr, addr\})$  holding of the conversational participants, one can increment MOVES so that  $Part(spkr, addr)$  becomes LatestMove.

$$(25) \text{ Parting} =_{def} \left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{ll} \text{qud} = \{ \} & : \text{poset(Question)} \\ \text{facts} & : \text{Set(Prop)} \\ \text{f} = \text{MinInteraction}(\{ \text{spkr}, \text{addr} \}) & : \text{Prop} \\ \text{cl} & : \text{member(f, facts)} \end{array} \right] \\ \text{effects} : \left[ \text{LatestMove} = \text{Part(spkr, addr)} : \text{IllocProp} \right] \end{array} \right]$$

Counter-parting, specified in (26a), is the exact analogue of counter-greeting: given A's parting from B, B is provided with the possibility of reciprocating with a counter-parting. This, in turn, as indicated in (26b) enables the conversation to be terminated. That is, the effect of the counter-parting is a state in which the conversational participants are disengaging:<sup>22</sup>

$$(26) \text{ a. Counterparting} =_{def} \left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{ll} \text{LatestMove} = \text{Part(spkr, addr)} & : \text{IllocProp} \\ \text{qud} = \{ \} & : \text{poset(Question)} \end{array} \right] \\ \text{effects} : \left[ \begin{array}{ll} \text{spkr} = \text{pre.addr} & : \text{Ind} \\ \text{addr} = \text{pre.spkr} & : \text{Ind} \\ \text{LatestMove} = \text{CounterPart(spkr, addr)} & : \text{IllocProp} \end{array} \right] \end{array} \right]$$

<sup>22</sup> What disengaging amounts to is not something about which I will say anything here. In Macura (2007), for instance, the import is that the two agents walk away from each other and do not resume talking for a certain period. Arguably, disengaging should not be classified as a *move*, but as a fact of the common ground. Nothing much rides on that here though.

b. Disengaging = *def*

$$\left[ \begin{array}{l} \text{pre} \\ \text{effects} \end{array} : \left[ \begin{array}{l} \text{LatestMove} = \text{CounterPart}(\text{spkr}, \text{addr}) : \text{IllocProp} \\ \text{qud} = \{ \} : \text{poset}(\text{Question}) \\ \text{LatestMove} = \text{Disengaged}(\{ \text{spkr}, \text{addr} \}) : \text{IllocProp} \end{array} \right] \right]$$

In the conversational rules I provided for greeting and parting, DGB dynamics is limited to LatestMove. Asking and asserting and their associated countermoves, answering and accepting, involve changes in QUD and FACTS as well. Let us consider these now.

**Exercise 4.4.1** Offer an analysis of the move that involves placing the word ‘toe’ in Scrabble. Consider carefully what fields are required for the Scrabble DGB.

#### 4.4.3 Asking, Asserting, Answering, and Accepting

Broadly speaking, queries and assertions are either *issue-initiating*—they introduce an issue unrelated to those currently under discussion—or they are *reactive*—they involve a reaction to a previously raised issue. As we will shortly see, accounting for the latter using DGB-based conversational rules is simple. These can also be used to explicate the *effects* issue-initiating moves have. What will have to wait, until section 4.6, is discussion of the background of such moves, which is intrinsically tied in with the activity-specific aspects of interaction.

To get going, I will make an initial tentative assumption, an initialization principle dubbed *Free Speech*, formulated informally in (27a), and formalized in (27b). The formal version takes as its preconditions a DGB with empty QUD, and allows subsequently for either conversationalist to utter *anything*.

(27) a. Free Speech: in the absence of an issue in QUD any issue may be raised.

$$\left[ \begin{array}{l} \text{pre} \\ \text{effects} \end{array} : \left[ \begin{array}{l} \text{qud} = \langle \text{poset}(\text{question}) \rangle \\ \text{TurnUnderspec} \wedge_{\text{merge}} \left[ \begin{array}{l} r : \text{AbSemObj} \\ R : \text{IllocRel} \\ \text{LatestMove} = R(\text{spkr}, \text{addr}, r) : \text{IllocProp} \end{array} \right] \end{array} \right] \right]$$

*Free Speech* has some descriptive accuracy, witness those well-known lulls in conversation, particularly one with multiple participants, where people are looking for *something*, *anything* to talk about.

Let me start by considering how to describe the primary effect a query has on context. The most prototypical query exchange involves *q* getting posed by A, B adopting *q* and providing a response. This can be described by the protocol in (28), which is to be read timewise like musical notation—time proceeding from left to right (top to bottom represents temporal parallelism):

## (28) cooperative query exchange

LatestMove.Cont = Ask(A,q): IllocProp

1. A: push q onto QUD; release turn

2. B: push q onto QUD; take turn; make q-specific-utterance

Two aspects of this protocol are not query-specific: first, the protocol is—in common with the ones we have seen for greeting and parting—a two-person turn exchange pair (2-PTEP), as we defined in (15) above. Second, the specification make q-specific-utterance is an instance of a constraint we associated in section 4.3.3 informally with QUD, namely QSPEC. QSPEC is what characterizes the contextual background of reactive queries and assertions. We can describe it by means of the conversational rule in (29). This says that if  $q$  is QUD-maximal, then subsequent to this either conversational participant may make a move constrained to be  $q$ -specific (i.e. either About or Influencing  $q$ ).

(29) QSPEC =<sub>def</sub>

$$\left[ \begin{array}{l} \text{pre} : \left[ \text{qud} = \langle q, Q \rangle \text{ poset}(\text{Question}) \right] \\ \text{effects} : \text{TurnUnderspec} \wedge_{\text{merge}} \left[ \begin{array}{ll} \begin{array}{l} r \\ R \\ \text{LatestMove} = R(\text{spkr}, \text{addr}, r) \\ cl \end{array} & \begin{array}{l} : \text{AbSemObj} \\ : \text{IllocRel} \\ : \text{IllocProp} \\ : \text{Qspecific}(r, q) \end{array} \end{array} \right] \end{array} \right]$$

Thus, the only *query-particular* aspect of the protocol is the incrementation of QUD with  $q$ , which can be described simply as in (30): given a question  $q$  and ASK(A, B, q) being the LatestMove, one can update QUD with  $q$  as QUD-maximal:

(30) Ask QUD-incrementation =<sub>def</sub>

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{ll} q & : \text{Question} \\ \text{LatestMove} = \text{Ask}(\text{spkr}, \text{addr}, q) & : \text{IllocProp} \end{array} \right] \\ \text{effects} : \left[ \text{qud} = \langle q, \text{pre.qud} \rangle : \text{poset}(\text{Question}) \right] \end{array} \right]$$

This completes my initial explication of a query  $q$  as the incrementation of QUD by  $q$  combined with QUD-determined answerhood and turn exchange. Let us turn to consider assertion. In the aftermath of an assertion  $p$  by A to B, there seem to be two basic alternative developments: either B accepts the assertion or she chooses to discuss  $p$ ?. This is illustrated in the following constructed dialogue: at the third turn two options are illustrated—acceptance (3''), and discussion (3'). (4') continues discussion of the issue *whether Helen will agree to come*. (5') exemplifies the case where the discussion terminates with the original assertion being accepted, whereas (6) exemplifies the case where the negation of the original assertion wins out:



- (31) A(1): Who will agree to come?  
 B(2): Helen.  
 A(3'): I doubt Helen will want to come after last time. A(3''): I see.  
 B(4'): Nah, I think she's forgiven and forgotten.  
 A(5'): OK. A(5''): No, she definitely has not.  
 B(6): OK.

Accepting *p* involves explicitly indicating that B assents to *p* and hence that FACTS, which represents information which is taken to be shared, should be incremented with *p*. B's explicit signal serves as an important cue which allows A's DGB to remain aligned with B's, indicating to A that he should add *p* to FACTS.<sup>23</sup>

What of the alternative option—starting a discussion of *p*? or related issues? The simplest way to do so is to communicate *p*?. This can be uttered either by the asserter, as a request for confirmation, often called a *check* move (see Carletta et al. 1996), or by the responder, as an expression of surprise, doubt or disbelief—see (32). In addition the context seems to allow the full range of utterances that are available as responses to *p*?, in other words *p*?-specific utterances, as exemplified in (33).<sup>24</sup>

- (32) a. A: Bo is in town. B: Is Bo in town?  
 b. A: Bo is in town, right?  
 c. A: Bo is in town, is he?  
 d. A: Bo is in town. B: Is he?
- (33) a. A: Bo is in town. B: He can't be.  
 b. A: Bo is in town. B: Has his bicycle been sighted?  
 c. A: Bo is in town. B: Why would he bother coming?  
 d. A: Is Bo in town? B: He can't be.  
 e. A: Is Bo in town? B: Has his bicycle been sighted?

<sup>23</sup> One might also wonder whether all assertions in dialogue need to be explicitly accepted: this is a rather subtle issue, which is not straightforward to resolve. One problem is that the precise import of many affirmative utterances is not clear: they can either be construed as indicating *comprehension* of the previous (let us assume assertoric) utterance or *acceptance* of the assertion or both. Thus, for a given utterance determining whether the responder implicitly indicates comprehension and explicitly accepts the assertion or merely acknowledges the utterance is difficult. A second issue is that, at least for conversations where the conversational participants are visible to each other, gesture (head nodding, eye contact etc.) also provides an option by means of which affirmative moves can be made (see Nakano et al. 2003). It is certainly much more correct to assume (with e.g. Clark 1996) that acceptance requires in general what one might call *affirmative action* than to assume that *lack of objection* (as in Stalnaker 1978) is sufficient. This issue receives a significant new twist in the context of multilogue, where implicit acceptance is a far more significant option, as will be discussed further in Chapter 7.

<sup>24</sup> Note though that whereas *p*? is a possible response to an assertion of *p*, it is not a possible response to asking *p*?. Hence the branch corresponding to 'Discuss *p*?' as a post-assertion utterance is not identical to the class of reactions to a query *p*?, a point we take up further shortly.

Of course utterances like (i(2)) are felicitous, but they function as clarification requests of (i(1)), with readings as in (ii) and (iii), as is explained in detail in Chapter 6:

- (i) A(1): Did Bo leave? B(2): Did Bo leave?  
 (ii) Are you asking whether Bo left?  
 (iii) What do you mean by asking 'Did Bo leave?'?

f. A: Is Bo in town? B: Why would he bother coming?

This protocol for assertion can be summarized as follows:

- (34) cooperative assertion exchange (initial version)
1. LatestMove.Cont = Assert(A, p) : IllocProp
  2. A: release turn
  3. B: take turn; Option 1: Discuss p?  
Option 2: Accept p
- (35)
1. LatestMove.Cont: Accept(B, p)
  2. B: increment FACTS with p;
  3. A: increment FACTS with p;

Comparing this protocol for assertion with the protocol for querying reveals one significant asymmetry. In the assertion protocol, QUD does not get updated as a result of the assertion. QUD will only get updated if B chooses the discussion option. This asymmetry is, arguably, somewhat inelegant—if QUD records the issues under discussion in a conversation, then it seems, intuitively, that there is an issue on the table as a consequence of an assertion, namely the issue of *whether p*. This intuition has concrete ramifications: in the aftermath of an assertion *p*, it is possible for either speaker to utter a wide range of elliptical propositional modifiers, which are equally felicitous in the aftermath of a query *p?*:

- (36)
- a. A: Bo is in town. B: Yes/No/Perhaps/Definitely.
  - b. A: Is Bo in town? B: Yes/No/Perhaps/Definitely.
  - c. A: Bo is dangerous. Yes?/No?

Given the protocol above, there is no commonality between a post-query context and a post-assertion context, necessary to describe the contextual background of propositional modifiers, as in (36). There is a simple modification to the protocol in (34), which will allow us to capture this contextual commonality, as well as the intuition that *p?* is on the table. This is simply to assume that an effect of assertion is to update QUD with the question *p?* as its topmost element.<sup>25</sup> Let us assume, in line with the assumptions adopted so far concerning querying, a relatively ‘cooperative’ contextual development—in other words, there is no possibility of B declining to address A’s assertion. In line with this, B’s initial move on recognizing that A asserted *p* is to adopt *p?* as QUD-maximal.

- (37) cooperative assertion exchange with topic incrementation
1. LatestMove.Cont = Assert(A, p): IllocProp

<sup>25</sup> This has been taken to imply that a post-assertion context is *identical* to a post-polar query context (see Beyssade and Marandin 2007b). This apparent implication is not correct, however: since although these two types of context are identical with respect to QUD, they have a distinct LatestMove value. Hence, a post-assertion context *will* license acceptance and check responses, in contrast to a post-polar question context.

2. A: push  $p?$  onto QUD, release turn
3. B: push  $p?$  onto QUD, take turn; Option 1: Discuss  $p?$ ,  
Option 2: Accept  $p$

- (38) 1. LatestMove.Cont = Accept(B,p) : IllocProp  
 2. B: increment FACTS with  $p$ ; pop  $p?$  from QUD;  
 3. A: increment FACTS with  $p$ ; pop  $p?$  from QUD;

What are the components of this protocol? Not specific to assertion is the fact that it is a 2-PTEP; similarly, the possibility of discussing issues relating to  $p?$  is a consequence of QSPEC. This leaves three novel assertion-specific components: QUD incrementation with  $p?$ , posing  $p?$ , and acceptance. Of these three, the first is a straightforward analogue of the conversational rule we posited for asking, (30) above: given a proposition  $p$  and ASSERT(A, B,  $p$ ) being the LatestMove, one can update QUD with  $p?$  as QUD-maximal:

- (39) Assert QUD-incrementation =<sub>def</sub>
- $$\left[ \begin{array}{lcl} \text{pre} & : & \left[ \begin{array}{ll} p & : \text{Prop} \\ \text{LatestMove} = \text{Assert}(\text{spkr}, \text{addr}, p) & : \text{IllocProp} \end{array} \right] \\ \text{effects} & : & \left[ \text{qud} = \langle p?, \text{pre.qud} \rangle : \text{poset}(\text{Question}) \right] \end{array} \right]$$

Assertion checking licenses an asking of  $p?$  as a follow-up of asserting  $p$ , the checker can be either the original asserter or the original addressee. An asserter's demand for confirmation does not introduce a new issue but it does change the context, forcing a response about the issue from B, as in (40a, b), rather than allowing her to accept in an understated fashion, as in (40c):

- (40) a. A: Bo is leaving, right? B: Right / #I see / #Aha.  
 b. A: Bo is leaving, isn't he? B: Right / #I see / #Aha.  
 c. A: Bo is leaving. B: I see / Aha.

I distinguish a Check-move from an Ask-move because the former will typically be followed by a confirmation of  $p$ , NOT a fresh assertion of  $p$ . The latter, however, would be coherent as a follow-up to asking  $p?$ . (41) licences a Check move of whether  $p$  by either A or B in the aftermath of A's assertion of  $p$ :

- (41) Assertion checking =<sub>def</sub>
- $$\left[ \begin{array}{lcl} \text{pre} & : & \left[ \begin{array}{ll} p & : \text{Prop} \\ \text{LatestMove} = \text{Assert}(\text{spkr}, \text{addr}, p) & : \text{IllocProp} \\ \text{qud} = \langle p?, Q1 \rangle & : \text{poset}(\text{Question}) \end{array} \right] \\ \text{effects} & : & \text{TurnUnderspec} \wedge_{\text{merge}} \left[ \text{LatestMove} = \text{Check}(\text{spkr}, \text{addr}, p?): \text{IllocProp} \right] \end{array} \right]$$

Acceptance is a somewhat more involved matter because a lot of the action is not directly perceptible. The labour can be divided here in two: on the one hand is the action brought about by an acceptance utterance (e.g. ‘mmh’, ‘I see’). The background for an acceptance by B is an assertion by A and the effect is to modify LatestMove:

(42) Accept move =<sub>def</sub>

$$\left[ \begin{array}{l} \text{pre} \\ \text{effects} \end{array} : \left[ \begin{array}{ll} \text{spkr} & : \text{Ind} \\ \text{addr} & : \text{Ind} \\ \text{p} & : \text{Prop} \\ \text{LatestMove} = \text{Assert}(\text{spkr}, \text{addr}, \text{p}) & : \text{IllocProp} \\ \text{qud} = \langle \text{p?}, \text{pre.qud} \rangle & : \text{poset}(\text{Question}) \end{array} \right] \right]$$

$$\left[ \begin{array}{l} \text{effects} \end{array} : \left[ \begin{array}{ll} \text{spkr} = \text{pre.addr} & : \text{Ind} \\ \text{addr} = \text{pre.spkr} & : \text{Ind} \\ \text{LatestMove} = \text{Accept}(\text{spkr}, \text{addr}, \text{p}) & : \text{IllocProp} \end{array} \right] \right]$$

In so far as we distinguish Check moves from Ask moves, we also need to distinguish acceptance from confirmation: Checks cannot be followed up by acceptance moves, as we saw above in (40a, b). Beyond their subtle preconditional difference—assertoric vs check preceding move—I do not propose a difference between acceptances and confirmations:

(43) Confirm move =<sub>def</sub>

$$\left[ \begin{array}{l} \text{pre} \\ \text{effects} \end{array} : \left[ \begin{array}{ll} \text{p} & : \text{Prop} \\ \text{LatestMove} = \text{Check}(\text{spkr}, \text{addr}, \text{p?}) & : \text{IllocProp} \\ \text{qud} = \langle \text{p?}, \text{pre.qud} \rangle & : \text{poset}(\text{Question}) \end{array} \right] \right]$$

$$\left[ \begin{array}{l} \text{effects} \end{array} : \left[ \begin{array}{ll} \text{spkr} = \text{pre.addr} & : \text{Ind} \\ \text{addr} = \text{pre.spkr} & : \text{Ind} \\ \text{LatestMove} = \text{Confirm}(\text{spkr}, \text{addr}, \text{p}) & : \text{IllocProp} \end{array} \right] \right]$$

The second component of acceptance and confirmation is the incrementation of FACTS by  $p$ . This is not quite as straightforward as it might seem: as I will argue shortly, a question  $q$  needs to be removed from QUD once FACTS contains an element that resolves  $q$ . In particular, one consequence of accepting  $p$  is the need to ensure that  $p?$  gets downdated from QUD. More generally, we need to verify for all existing elements of QUD that they are not resolved by the new value of FACTS. Hence, accepting  $p$  involves both an update of FACTS and a downdate of QUD—minimally just removing  $p?$ , potentially removing other questions as well. This joint process of FACTS update / QUD-downdate is formulated in (44): given an acceptance or confirmation of  $p$  by B,  $p$  can be unioned into FACTS, whereas QUD is modified by

the function *NonResolve*. *NonResolve* is a function that maps a partially ordered set of questions *poset*(*q*) and a set of propositions *P* to a partially ordered set of questions *poset'*(*q*) which is identical to *poset*(*q*) modulo those questions in *poset*(*q*) resolved by members of *P*.

(44) Fact Update/ QUD-Downdate =<sub>def</sub>

$$\left[ \begin{array}{l} \text{pre} \\ \text{effects} \end{array} : \left[ \begin{array}{ll} \begin{array}{l} p \\ \text{LatestMove} = \text{Accept}(\text{spkr}, \text{addr}, p) \vee \\ \text{Confirm}(\text{spkr}, \text{addr}, p) \\ \text{qud} = \langle p?, \text{pre.qud} \rangle \end{array} & \begin{array}{l} : \text{Prop} \\ \\ : \text{IllocProp} \\ : \text{poset}(\text{Question}) \end{array} \end{array} \right] \right]$$

$$\left[ \begin{array}{l} \text{effects} \\ \text{qud} \end{array} : \left[ \begin{array}{ll} \begin{array}{l} \text{facts} = \text{pre.facts} \cup \{ p \} \\ \text{qud} = \text{NonResolve}(\text{pre.qud}, \text{facts}) \end{array} & \begin{array}{l} : \text{Set}(\text{Prop}) \\ : \text{poset}(\text{Question}) \end{array} \end{array} \right] \right]$$

One aspect of (44) bears refinement: this concerns the conditions governing the downdating of questions from QUD. This is *not* a deterministic process, the primary reason for that being the fact that resolvedness is, as discussed earlier in Chapter 3, not purely semantic—viz. a two-place relation between facts and questions, as assumed in (44). Rather, resolvedness is to a certain extent an *agent-relative* notion. This is exemplified in (45), originally provided as (17) in Chapter 2, where the question raised in move (45(3)) is discussed in moves (45(4)), (45(6)), and (45(7)), before a return to the original issue ensues. Note that discussion of (45(3)) could have terminated after any of the answers provided:

- (45) B(1): you've obviously started a trend  
 c(2): no, I don't think. (2a) we're definitely following a trend  
 B(3): who was in front  
 c(4): oh all sorts of people—we've definitely bandwaggoned  
 A(5): mm  
 c(6): well certainly of of many of of Jo's and your friends surely  
 A(7): yeah, May and Jan, all sorts of people, loads of Bristol people, most of Bristol people in fact [...]  
 c(8): oh yes—I'm getting definitely getting very long in the tooth  
 A(9): (laughs)  
 B(10): I sympathize  
 c(11): so uh there's no point in my pretending I'm setting a trend

The rudiments of how to handle the necessary parametrization will be provided in section 4.6 and are the subject of Exercise 4.6.1. In section 8.3.1 a further enrichment of downdating possibilities will be provided, the option being to downdate when it becomes clear no further information about a given issue will be provided.

#### 4.4.4 How QUD and FACTS relate: preconditions for querying and assertion

So far I have considered FACTS, Moves, and QUD as three distinct but *independent* fields. However, there is one rather clear relation that holds between Facts and QUD: a question *q* can be in QUD only if there is no fact *f* in FACTS such that *f* resolves *q*. The evidence for this, a significant restriction on possible DGBs, comes both from queries and assertions.

Consider first queries: the presence, in FACTS, of a resolving answer to the question vitiates the possibility of a successful interrogatory use. This is illustrated in (46):

- (46) a. A: John and Bill visited me yesterday. B: Interesting . . . # Who visited you yesterday?  
 b. A: Every major journalist was at the press conference. B: Right. . . . # Who was at the press conference?<sup>26</sup>  
 c. A: Bill came/did not come to the party.  
 B: Right. . . . # but did Bill come to the party?

Note that it is not mere salience of the information that matters but whether it has been accepted. When the latter condition fails, a question *can* be successfully reposed despite information about it being salient:<sup>27</sup>

- (47) a. B: Who will solve that problem?  
 A: John and Bill might.  
 B: No I don't think so.  
 b. A: No? So, *who* will solve that problem?
- (48) A: I think Jill came to the party.  
 B: I doubt it.  
 A: Why?  
 B: I didn't see her bike parked outside.  
 A: That's your evidence.  
 B: Mmh.  
 A: Jack says he thinks he saw her there.  
 C: Well, so, *did* she or didn't she come to the party?

Why state such a condition in terms of the DGB? Early work on queries (e.g. Hintikka 1976, Åqvist 1965, Searle 1969) took an overly epistemic view, in particular assuming that one of either the querier or the responder was required to *lack* information that,

<sup>26</sup> Note that B can follow up A's assertion by saying 'Who?' or 'Which journalists?', but these 'sluiced' questions, further discussed in Chapter 7, have quite distinct contents, meaning roughly 'Who/which journalists are [the] journalist(s) who were at the press conference.'

<sup>27</sup> Note that the second time around, the *form* used to pose the question is different: in (47b) the 'who' is accented, which is not the case in (47a); similarly in (48b) the auxiliary is accented.

in current terms, resolves the question posed.<sup>28</sup> In fact, it is quite easy to come up with examples that show that, in principle, ignorance at any level of iterated belief between speaker or hearer allows for the posing of queries. For instance, consider (49):

- (49) A(1): When I think of women politicians, I don't think I can name a single compassionate one: Indira Gandhi, Golda Meir, ...  
 B(2): Well, but who's your role model?  
 A(3): Petra Kelly.  
 B(4): That proves my point, doesn't it?

A plausible scenario underlying (49) is one in which B knows who A's role model is, knows that A knows who A's own role model is, and assumes that A is not aware that he knows who her role model is.<sup>29</sup> This suggests that rather than considering *individual conversational participant* knowledge we need to consider some more public repository, which needs to *lack* the requisite information. I will assume, following Hamblin (1970) and Carlson (1983), that the DGB is indeed the right level to state this, though other subtly different candidates could also be considered.<sup>30</sup>

The following informal formulation suggests itself:

- (50) The Unresolvedness Condition on Querying:  
 A question  $q_1$  can be posed by A only if there does not exist a fact  $\tau$  such that  $\tau \in \text{FACTS}$  and  $\tau$  resolves  $q_1$ .

Not surprisingly, (50) brings up the same public/private issues that we discussed earlier with respect to QUD downdating: in order to pose a question  $q_1$  it must be the case that the DGB lacks a fact that resolves  $q$ . But resolvedness after all involves two (typically) non-public parameters, the goal, and the inferential capabilities relative to which the resolving fact entails the goal. And in this case too it seems like these parameters need not derive from the public domain. Thus, in (51) after the initial two turns the dialogue could continue either as (51b) or (51c): in the former case A adjudges the information he received as insufficient for his goal, one component of which ('not getting any refusals') he decides to make explicit. Consequently, he allows himself to repose the question. In (51c), where apparently A's goal is less specific, the

<sup>28</sup> Searle says: There are two types of questions [queries in current terminology—J.G.]: (a) real questions, (b) exam questions. In real questions S wants to know (find out) the answer; in exam questions, S wants to know if H knows. (Searle 1969: 66).

<sup>29</sup> As discussed further below, if A is aware that B knows who her role model is, she will typically not respond in (3) with an answer, but rather will treat B's utterance in (2) more or less as an assertion. A might say 'Right, good point' or 'hmm, but that is an exception.'

<sup>30</sup> For instance, (one-sided) common knowledge *simpliciter*. Cocktail pleasantries and barber/client interaction (Stalnaker 1974) might be viewed as problematic for this latter possibility; but perhaps these involve a pretence that common knowledge fails to exist. The main difference between the DGB and (one-sided) common knowledge seem to lie in whether it is crucial that the information has been explicitly provided. As soon as we allow for inference within the DGB, however, it becomes difficult to tease apart the difference.

information counts as resolving, so it would be strange to repose the question unless a different goal was adopted (e.g. one that involves identifying the set of candidate invitees):

- (51) a. A: Who can we invite?  
       B: Some guys from the department and a couple of students.  
       b. A: I need you to be more specific; I'm not going to cope with a lot of polite refusals. Come on tell me: who exactly can we invite?  
       c. A: Right, so whoever I'll bump into tomorrow during coffee I'll ask over.

One can generalize (50), viewing it as a constraint about the introduction of questions into QUD: whether directly by querying or as a side effect of other dialogue acts. Thus, we have already seen how assertion *that*  $p$  can be analysed as introducing a question *whether*  $p$  into QUD, and we shall see various further examples of this type of analysis in what follows. We can generalize (50) to serve as a constraint that precludes a question from being introduced into QUD if information that resolves it is already established in the DGB. For the case of assertion, as is simple to check, what the generalized QIAC would enforce is the infelicity of asserting a proposition which *contradicts* a common-ground fact, as well as that of a proposition which is already in the common ground:

- (52) Question Introduction Appropriateness Condition (QIAC):  
 A question  $q_1$  can be introduced into QUD by A only if there does not exist a fact  $\tau$  such that  $\tau \in \text{FACTS}$  and  $\tau$  resolves  $q_1$

The QIAC can be incorporated straightforwardly into our theory by adding a further restriction into the definition of the DGBType as follows:<sup>31</sup>

- (53) DGBType (interim definition with QIAC) = *def*
- |                  |                                                                     |
|------------------|---------------------------------------------------------------------|
| spkr             | : Ind                                                               |
| addr             | : Ind                                                               |
| utt-time         | : Time                                                              |
| c-utt            | : addressing(spkr, addr, utt-time)                                  |
| Facts            | : Set(Prop)                                                         |
| Moves            | : list(IllocProp)                                                   |
| QUD              | : poset(Question)                                                   |
| non-resolve-cond | : $\forall q \in \text{QUD} [\neg \text{Resolve}(\text{FACTS}, q)]$ |

What happens with interrogative uses that occur in contexts where the QIAC would be violated? In Chapter 8, I discuss two constructions—often conflated as *rhetorical questions*—whose contextual background involves using a resolved question.

<sup>31</sup> The same techniques that can be used to refine (44) to take care of the agent relativity of resolvedness could be applied to the type definition in (53).



4.4.5 *Move Coherence and Some Worked Examples*

I consider now some example dialogues that illustrate the various rules discussed in previous sections. With this in hand, we will be able to formulate a coherence principle characterizing when a move or a sequence of moves is coherent.

The focus here will be on contextual development at the level of moves, hence I will abbreviate descriptive contents with the obvious choices— $q_i$  for questions,  $p_i$  for propositions, ignoring the internal structure of contents and issues of contextual resolution of meaning.

Consider first the following exchange, taking place conversation-initially:

- (54) A: Hi.  
 B: Hi.  
 A: Who's coming tomorrow?  
 B: Several colleagues of mine (are coming).  
 A: I see.  
 B: Mike (is coming) too.

In order for a DGB to satisfy the preconditions for a greet move, it needs to be of the type in (55a), so a record as in (55b):<sup>32</sup>

- (55) a.  $\left[ \begin{array}{ll} \text{spkr} & : \text{Ind} \\ \text{addr} & : \text{Ind} \\ \text{Moves} = \langle \rangle & : \text{list}(\text{IllocProp}) \\ \text{qud} = \{ \} & : \text{poset}(\text{Question}) \\ \text{facts} = \text{commonground1} & : \text{Set}(\text{Prop}) \end{array} \right]$
- b.  $\left[ \begin{array}{ll} \text{spkr} & = \text{A} \\ \text{addr} & = \text{B} \\ \text{Moves} & = \langle \rangle \\ \text{qud} & = \langle \rangle \\ \text{facts} & = \text{cgl} \end{array} \right]$

Taking (55b) as input and applying the greet rule (17a), yields:

<sup>32</sup> In this section I use the spelled-out version of conversational rules, where all types are as specific as the DGB type.

$$(56) \left[ \begin{array}{ll} \text{spkr} & = A \\ \text{addr} & = B \\ \text{Moves} & = \langle \text{Greet}(A, B) \rangle \\ \text{qud} & = \langle \rangle \\ \text{facts} & = \text{cgl} \end{array} \right]$$

Now the DGB in (56) is of the type (57a), which means it satisfies the preconditions of a counter-greeting. Applying the counter-greeting rule, (22), to (56) yields (57b):

$$(57) \begin{array}{ll} \text{a.} & \left[ \begin{array}{ll} \text{spkr} & : \text{Ind} \\ \text{addr} & : \text{Ind} \\ \text{LatestMove} = \text{Greet}(\text{pre.spkr}, \text{pre.addr}) & : \text{IllocProp} \\ \text{qud} = \{ \} & : \text{poset}(\text{Question}) \\ \text{facts} = \text{commonground1} & : \text{Set}(\text{Prop}) \end{array} \right] \\ \text{b.} & \left[ \begin{array}{ll} \text{spkr} & = B \\ \text{addr} & = A \\ \text{Moves} & = \langle \text{CounterGreet}(B, A), \text{Greet}(A, B) \rangle \\ \text{qud} & = \langle \rangle \\ \text{facts} & = \text{cgl} \end{array} \right] \end{array}$$

The greetings stage has passed. QUD is empty—Free Speech can kick in—either conversational participant can raise any issue they wish. A raises the issue of *who is coming tomorrow*, which we will denote as  $q_0$ . Applying Free Speech to (57b) yields (58a); this record satisfies the preconditions of the type (58b), in other words is the appropriate context for Ask QUD-incrementation. The output of that rule is (58c):

$$(58) \text{ a.} \left[ \begin{array}{ll} \text{spkr} & = A \\ \text{c1} & = p1 \\ \text{addr} & = B \\ \text{c2} & = p2 \\ \text{r} & = q0 \\ \text{Moves} & = \langle \text{Ask}(A, B, q), \text{CounterGreet}(B, A), \text{Greet}(A, B) \rangle \\ \text{qud} & = \langle \rangle \\ \text{facts} & = \text{cgl} \end{array} \right]$$

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| b. | $\left[ \begin{array}{ll} \text{pre} & : \left[ \begin{array}{ll} \text{spkr} & : \text{Ind} \\ \text{addr} & : \text{Ind} \\ \text{r} & : \text{Question} \\ \text{LatestMove} = \text{Ask}(\text{spkr}, \text{addr}, \text{q}) & : \text{IllocProp} \\ \text{qud} & : \text{poset}(\text{Question}) \\ \text{facts} = \text{commonground1} & : \text{Set}(\text{Prop}) \end{array} \right] \\ \text{effects} & : \left[ \begin{array}{ll} \text{spkr} = \text{pre.spkr} & : \text{Ind} \\ \text{addr} = \text{pre.addr} & : \text{Ind} \\ \text{LatestMove} = \text{pre.LatestMove} & : \text{IllocProp} \\ \text{qud} = \langle \text{r}, \text{pre.qud} \rangle & : \text{poset}(\text{Question}) \\ \text{facts} = \text{pre.facts} & : \text{Set}(\text{Prop}) \end{array} \right] \end{array} \right]$ |
| c. | $\left[ \begin{array}{ll} \text{spkr} & = \text{A} \\ \text{addr} & = \text{B} \\ \text{r} & = \text{q0} \\ \text{Moves} & = \langle \text{Ask}(\text{A}, \text{B}, \text{q0}), \text{CounterGreet}(\text{B}, \text{A}), \text{Greet}(\text{A}, \text{B}) \rangle \\ \text{qud} & = \langle \text{q0} \rangle \\ \text{facts} & = \text{cg1} \end{array} \right]$                                                                                                                                                                                                                                                                                                                                                                                                                                             |

Since QUD is now non-empty, QSPEC is applicable: an utterance specific to  $q0$  is required, an instance of which is the assertion by B that *Several colleagues of B's are coming*, which we denote with  $p1$ . Applying this rule leads to the DGB in (59a), which can in turn, entirely analogously to (58a) undergo Assertion QUD-incrementation, yielding (59b):

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (59) a. | $\left[ \begin{array}{ll} \text{spkr} & = \text{B} \\ \text{addr} & = \text{A} \\ \text{r} & = \text{p1} \\ \text{Moves} & = \langle \text{Assert}(\text{B}, \text{A}, \text{p1}), \text{Ask}(\text{A}, \text{B}, \text{q0}), \text{CounterGreet}(\text{B}, \text{A}), \text{Greet}(\text{A}, \text{B}) \rangle \\ \text{cl} & = \text{p2} \\ \text{qud} & = \langle \text{q0} \rangle \\ \text{facts} & = \text{cg1} \end{array} \right]$ |
| b.      | $\left[ \begin{array}{ll} \text{spkr} & = \text{B} \\ \text{addr} & = \text{A} \\ \text{Moves} & = \langle \text{Assert}(\text{B}, \text{A}, \text{p1}), \text{Ask}(\text{A}, \text{B}, \text{q0}), \text{CounterGreet}(\text{B}, \text{A}), \text{Greet}(\text{A}, \text{B}) \rangle \\ \text{qud} & = \langle \text{p1?}, \text{q0} \rangle \\ \text{facts} & = \text{cg1} \end{array} \right]$                                          |

One contextual possibility at this point, which is realized in (54), is for A to accept B's assertion. This yields the DGB in (60a), which can in turn lead to the incrementation of FACTS with  $p1$  and the corresponding downdating of  $p1?$ , hence the DGB in (60b):

$$\begin{array}{ll}
 (60) \text{ a. } & \left[ \begin{array}{l} \text{spkr} = A \\ \text{addr} = B \\ \text{Moves} = \left\langle \begin{array}{l} \text{Accept}(A, B, p1), \text{Assert}(B, A, p1), \text{Ask}(A, B, q0) \\ \text{CounterGreet}(B, A), \text{Greet}(A, B) \end{array} \right\rangle \\ \text{qud} = \langle p1?, q0 \rangle \\ \text{facts} = \text{cgl} \end{array} \right] \\
 \text{b. } & \left[ \begin{array}{l} \text{spkr} = A \\ \text{addr} = B \\ \text{Moves} = \left\langle \begin{array}{l} \text{Accept}(A, B, p1), \text{Assert}(B, A, p1), \text{Ask}(A, B, q0) \\ \text{CounterGreet}(B, A), \text{Greet}(A, B) \end{array} \right\rangle \\ \text{qud} = \langle q0 \rangle \\ \text{facts} = \text{cgl} \cup \{ p1 \} \end{array} \right]
 \end{array}$$

Finally, given that  $q0$  is again QUD-maximal in (60b), QSPEC is applicable again—a  $q0$ -specific utterance is in order, for instance B's assertion *Mike is coming too*, denoted here as  $p2$ . The contextual development here mirrors (59):

$$\begin{array}{ll}
 (61) \text{ a. } & \left[ \begin{array}{l} \text{spkr} = B \\ \text{addr} = A \\ \text{Moves} = \left\langle \begin{array}{l} \text{Assert}(B, A, p2), \text{Accept}(A, B, p1), \\ \text{Assert}(B, A, p1), \text{Ask}(A, B, q0) \\ \text{CounterGreet}(B, A), \text{Greet}(A, B) \end{array} \right\rangle \\ \text{qud} = \langle q0 \rangle \\ \text{facts} = \text{cgl} \end{array} \right] \\
 \text{b. } & \left[ \begin{array}{l} \text{spkr} = B \\ \text{addr} = A \\ \text{Moves} = \left\langle \begin{array}{l} \text{Assert}(B, A, p2), \text{Accept}(A, B, p1), \\ \text{Assert}(B, A, p1), \text{Ask}(A, B, q0) \\ \text{CounterGreet}(B, A), \text{Greet}(A, B) \end{array} \right\rangle \\ \text{qud} = \langle p2?, q0 \rangle \\ \text{facts} = \text{cgl} \cup \{ p1 \} \end{array} \right]
 \end{array}$$

We can summarize the contextual development that has occurred as follows:

| (62) | Utt.    | DGB Update<br>(Conditions)                                                                               | Rule                                  |
|------|---------|----------------------------------------------------------------------------------------------------------|---------------------------------------|
|      | initial | MOVES = $\langle \rangle$<br>QUD = $\langle \rangle$<br>FACTS = $\text{cgl}$                             |                                       |
|      | 1       | LatestMove := Greet(A, B)                                                                                | greeting                              |
|      | 2       | LatestMove := CounterGreet(B, A)                                                                         | counter-greeting                      |
|      | 3       | LatestMove := Ask(A, B, $q_0$ )<br>QUD := $\langle q_0 \rangle$                                          | Free Speech<br>Ask QUD-incrementation |
|      | 4       | LatestMove := Assert(B, A, $p_1$ )<br>(About( $p_1$ , $q_0$ ))<br>QUD := $\langle p_1?, q_0 \rangle$     | QSPEC<br>Assert QUD-incrementation    |
|      | 5       | LatestMove := Accept(A, B, $p_1$ )<br>QUD := $\langle q_0 \rangle$<br>FACTS := $\text{cgl} \cup \{p_1\}$ | Accept<br>Fact update/QUD-downdate    |
|      | 6       | LatestMove := Assert(B, A, $p_2$ )<br>(About( $p_2$ , $q_0$ ))<br>QUD := $\langle p_2?, q_0 \rangle$     | QSPEC<br>Assert QUD-incrementation    |

With equal ease we can deal with a case where the assertion leads to discussion:<sup>33</sup>

- (63) A(1): Who will agree to come?  
 B(2): Helen and Jelle.  
 A(3): I doubt Helen will want to come after last time.  
 B(4): I think she's forgiven and forgotten.  
 A(5): OK.

| (64) | Utt.    | DGB Update<br>(Conditions)                                                                                         | Rule                                  |
|------|---------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------|
|      | initial | MOVES = $\langle \rangle$<br>QUD = $\langle \rangle$<br>FACTS = $\text{cgl}$                                       |                                       |
|      | 1       | LatestMove := Ask(A, B, $q_0$ )<br>QUD := $\langle q_0 \rangle$                                                    | Free Speech<br>Ask QUD-incrementation |
|      | 2       | LatestMove := Assert(B, A, $p_1$ )<br>(About( $p_1$ , $q_0$ ))<br>QUD := $\langle p_1?, q_0 \rangle$               | QSPEC<br>Assert QUD-incrementation    |
|      | 3       | LatestMove := Assert(A, B, $p_2$ )<br>(About( $p_2$ , $p_1?$ ))<br>QUD := $\langle p_2?, p_1?, q_0 \rangle$        | QSPEC<br>Assert QUD-incrementation    |
|      | 4       | LatestMove := Assert(B, A, $p_3$ )<br>(About( $p_3$ , $p_2?$ ))<br>QUD := $\langle p_3?, p_2?, p_1?, q_0? \rangle$ | QSPEC<br>Assert QUD-incrementation    |
|      | 5       | LatestMove := Accept(A, B, $p_3$ )<br>QUD := $\langle p_1?, q_0 \rangle$<br>FACTS := $\text{cgl} \cup \{p_3\}$     | Accept<br>Fact update/QUD-downdate    |

<sup>33</sup> Note that on the analysis provided here after A's acceptance in (63(5)), the original issue raised in (63(2)) remains potentially unresolved. This seems reasonable, given that A could still raise the issue of whether Jelle is or is not going to agree to come.

In Chapter 3 we noted that one possible response to a given query use of  $q$  is by means of a question  $q_1$ , as long as  $q_1$  is a question which *influences*  $q$ . Thus, within the set-up as it stands we can straightforwardly analyse dialogues such as the following:

- (65) A(1): Who should we invite for tomorrow?  
 B(2): Who will agree to come?  
 A(3): Helen and Jelle and Fran and maybe Sunil.  
 B(4): (a) I see. (b) So, Jelle I think.  
 A(5): OK.

(66)

| Utt.    | DGB Update<br>(Conditions)                                                                                    | Rule                                  |
|---------|---------------------------------------------------------------------------------------------------------------|---------------------------------------|
| initial | MOVES = $\langle \rangle$<br>QUD = $\langle \rangle$<br>FACTS = cgl                                           |                                       |
| 1       | LatestMove := Ask(A, B, $q_0$ )<br>QUD := $\langle q_0 \rangle$                                               | Free Speech<br>Ask QUD-incrementation |
| 2       | LatestMove := Ask(B, A, $q_1$ )<br>Influence( $q_1, q_0$ )<br>QUD := $\langle q_1, q_0 \rangle$               | QSPEC<br>Assert QUD-incrementation    |
| 3       | LatestMove := Assert(A, B, $p_1$ )<br>(About( $p_1, q_1$ ))<br>QUD := $\langle p_1?, q_1, q_0 \rangle$        | QSPEC<br>Assert QUD-incrementation    |
| 4a      | LatestMove := Accept(B, A, $p_1$ )<br>FACTS := $\text{cgl} \cup \{p_1\}$<br>QUD := $\langle q_0 \rangle$      | Accept<br>Fact update/QUD downdate    |
| 4b      | LatestMove := Assert(B, A, $p_2$ )<br>(About( $p_2, q_0$ ))<br>QUD := $\langle p_2?, q_0 \rangle$             | QSPEC<br>Assert QUD-incrementation    |
| 5       | LatestMove := Accept(A, B, $p_2$ )<br>FACTS := $\text{cgl} \cup \{p_1, p_2\}$<br>QUD := $\langle q_0 \rangle$ | Accept<br>Fact update/QUD-downdate    |

In (67) and (68) we have an illustration of the interaction involved in Check moves:

- (67) A(1): (a) Bo is now in Essen, (b) is he?  
 B(2): Yes.

(68)

| Utt. | DGB Update<br>(Conditions)                                                                              | Rule                                     |
|------|---------------------------------------------------------------------------------------------------------|------------------------------------------|
| 1a   | LatestMove := Assert(A, B, $p_1$ )<br>QUD := $\langle p_1? \rangle$                                     | Free Speech<br>Assert QUD-incrementation |
| 1b   | LatestMove := Check(A, B, $p_1?$ )<br>QUD := $\langle p_1? \rangle$                                     | Check                                    |
| 2    | LatestMove := Confirm(B, A, $p_1$ )<br>FACTS := $\text{cgl} \wedge p_1$<br>QUD := $\langle q_0 \rangle$ | Confirm<br>Fact update/QUD-downdate      |

**Exercise 4.4.2** Provide a detailed trace through the dialogues in (69):

- (69) a. Anon 6: Who is coming back next weekend? Richard: Morse. Anon 6: No he's not. Richard: Oh yes.  
 b. B: you've obviously started a trend. c: no I don't think we're definitely following a trend. B: who was in front? c: oh all sorts of people—we've definitely bandwaggoned. A: mm. c: well certainly of of many of of Jo's and your friends surely. A: yeah May and Jan, all sorts of people, loads of Bristol people, most of Bristol people in fact. c: oh yes—I'm getting definitely getting very long in the tooth. A: (laughs) B: I sympathize. c: so uh there's no point in my pretending I'm setting a trend.

Having considered a number of examples, we can now formulate a notion that allows the coherence of a move to be captured—and by extension that of a sequence of moves.<sup>34</sup> Pre-theoretically, *coherence* relates a move  $m_1$  to a DGB  $dgb_0$  just in case there is a way to successfully update  $dgb_0$  with  $m_1$ . The basic concept introduced here is contextual *m(ove)-coherence* defined in (70a) as applying to  $m_1$  and  $dgb_0$  just in case there is a conversational rule  $c_1$  which maps  $dgb_0$  to  $dgb_1$  and such that  $dgb_1$ 's LatestMove value is  $m_1$ . **Pairwise M(ove)-Coherence**, defined in (70b), applies to a pair of moves  $m_1, m_2$  if  $m_1$  is **M-Coherent** relative to some DGB  $dgb_0$  and there is a sequence of updates leading from LatestMove being  $m_1$  to LatestMove being  $m_2$ . Finally, **Sequential M(ove)-Coherence**, defined in (70c), applies to a sequence of moves  $m_1, \dots, m_n$  just in case each successive pair of moves are **Pairwise M-Coherent**:

- (70) a. **M(ove)-Coherence**: Given a set of conversational rules  $\mathcal{C}$  and a dialogue gameboard  $dgb_0 : DGBType$ , a move  $m_1 : IllocProp$  is **m(ove) $_{\mathcal{C}}$  $^{dgb_0}$ -coherent** iff there exists  $dgb_1 : DGBType, c_1 \in \mathcal{C}$  such that  $c_1(dgb_0) = dgb_1$  and  $dgb_1.LatestMove = m_1$ .  
 b. **Pairwise M(ove)-Coherence**: Given a set of conversational rules  $\mathcal{C}$ , two moves  $m_1, m_2$  are **m(ove) $_{\mathcal{C}}$ -pairwise-coherent** iff there exists  $dgb_0 : DGBType$  such that  $m_1$  is **m(ove) $_{\mathcal{C}}$  $^{dgb_0}$ -coherent** and there exist  $dgb_i, c_i, (1 \leq i \leq k-1, dgb_i : DGBType, c_i \in \mathcal{C})$  such that  $c_{i+1}(dgb_i) = dgb_{i+1}$  and  $dgb_i.LatestMove = m_1$ , whereas  $dgb_k.LatestMove = m_2$ .  
 c. **Sequential M(ove)-Coherence**: A sequence of moves  $m_1, \dots, m_n$  is **m $_{\mathcal{C}}$ -coherent** iff, for any  $1 \leq i, m_i, m_{i+1}$  are **m $_{\mathcal{C}}$ -pairwise-coherent**.

## 4.5 Scaling Down to Monologue

In this work I concentrate on dialogues where turns change after one move: one speaker poses a question or makes an assertion, the other speaker reacts to this,

<sup>34</sup> I say 'moves'—illocutionary propositions—rather than 'utterances' because for the moment the focus is entirely on the content of utterances, ignoring their form and the process that leads to the content getting recognized.

and so forth. Turns in which more than a single illocutionary move is made involve many additional issues—though presumably at least a good number of these (e.g. those concerning tense and aspect) can be analysed using techniques developed for text/monologue. In recent years there has been, following Mann and Thompson (1987), much work on characterizing the nature of DISCOURSE RELATIONS, coherence relations that can hold between adjacent clauses in text/monologue. For extensive formal semantic development of such notions see in particular work in SDRT and formal pragmatics, e.g. Asher (1993), Asher and Lascarides (2003), Jasinskaja (2006).

It is not my intention here to develop an extension of the framework to multiple-move turns, although this particular domain as far as dialogue goes has not been much explored to date, though see Asher and Lascarides (2003) and Prévot (2003). I merely want to show that with the existing tools we can at the very least provide a rudimentary sketch to accommodate some simple instances, including cases like self-answering and successive querying/assertion. This will be necessary in order for us to provide an account for a number of key dialogue phenomena including the TURN-TAKING PUZZLE.

Consider first self-answering: asking a question does not require *an addressee* to provide a response, the speaker can address the issue herself, as exemplified in (71); note that in (71c) the self-answer isn't even discussed. Queries like (72), by no means 'rhetorical', frequently do not even involve the expectation that the other conversational participant might offer a response:

- (71) a. Vicki: When is, when is Easter? March, April? (BNC, KC2)  
 b. Unknown: When was that? That was last week wasn't it? (BNC, KC2)  
 c. Frederick: When's it taking place? (pause)  
 Joan: Erm it's the last Saturday of half term so I should think it's about erm (pause) when's half term? Eighteenth on the Monday I think (pause) so it'll be twenty (pause) erm it's the fifth is it when I, no it won't (pause) it's not as late as that is it (pause) eighteenth, nineteenth, twentieth, twenty one (pause) about the twenty second, something like that. (BNC, KC2)
- (72) a. Carol 133 Well it's (pause) it's (pause) er (pause) what's his name? Bernard Matthews' turkey roast. (BNC, KBJ)  
 b. Brian: you could encourage, what's his name? Neil. (BNC, KSR)

Self-answering is directly accommodated by QSPEC given that it licenses MaxQuD-specific utterances regardless of who the speaker of LatestMove is.

Another consequence of QSPEC is the possibility it provides for the posing of two successive questions by a single speaker, where the second question influences the first. This means that the second query becomes QUD-maximal. This is indeed a commonly occurring pattern in dialogue:



- (73) a. Ann: What are your shifts next week? Can you remember offhand?  
James: Yes. I'm early Monday and Tuesday (pause) and Wednesday (pause)  
a day off Thursday (pause) Friday (pause) late (4968–71)  
b. Ann: Anyway, talking of over the road, where is she? Is she home?  
Betty: No. She's in the Cottage. (5121–24)

A straightforward way to deal with cases such as following, where a *wh*-question maintains its maximality subsequent to a polar answer, will emerge once we refine the ordering on QUD in section 8.1:

- (74) a. Arthur: How old is she? Forty?  
Evelyn: Forty one! (456–8)  
b. Evelyn: what time is it? is it eight thirty five yet?  
Arthur: twenty past (6527–8)

QSPEC seems to be able to handle the commonest case of successive questions. Nonetheless, not all cases of successive questions do involve a second question which is a sub-question of the first:

- (75) a. A(1): Who will Max be inviting? When will these guests be arriving?  
B(2): Mary and Bill.  
A(3): Aha.  
B(4): Tomorrow or the day after most likely.  
b. A(1): Is Bill ill? Should we send the doctor along?  
B(2): Bill is just fine.  
A(3): Aha.  
B(4): So, no need to call the doctor.

Notice that here it is the first question that gets posed that takes precedence and only after it gets answered does the second question get addressed, and indeed it would be somewhat strange to do otherwise. Indeed, in many cases, a dialogue like (75b) would terminate after B's initial contribution. Notice also that, in contrast to cases covered by QSPEC, these cases are strange if the second question is posed by the addressee of the first question—one gets the feeling that the original question was ignored:

- (76) a. A(1): Who will Max be inviting?  
B(2): When will these guests be arriving?  
b. A(1): Is Bill ill?  
B(2): Should we send an ambulance?

In order to accommodate such cases we could postulate the conversational rule in (77): this allows a speaker to follow up an initial question with a non-influencing question, where the initial question remains QUD-maximal:

(77)  $QCoord =_{def}$ 

|         |   |                                                                                                                                                                                                                                                                                                               |
|---------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pre     | : | $\left[ \begin{array}{ll} q & : \text{Question} \\ \text{LatestMove} = \text{Ask}(\text{spkr}, \text{addr}, q) & : \text{IllocProp} \\ \text{qud} = \langle q, Q \rangle & : \text{poset}(\text{Question}) \end{array} \right]$                                                                               |
|         |   |                                                                                                                                                                                                                                                                                                               |
| effects | : | $\left[ \begin{array}{ll} q1 & : \text{Question} \\ c1 & : \neg \text{Influence}(q1, \text{pre}.q) \\ \text{LatestMove} = \text{Ask}(\text{spkr}, \text{addr}, q1) & : \text{IllocProp} \\ \text{qud} = \langle \text{pre}.q, q1, \text{pre}.Q \rangle & : \text{poset}(\text{Question}) \end{array} \right]$ |
|         |   |                                                                                                                                                                                                                                                                                                               |
|         |   |                                                                                                                                                                                                                                                                                                               |
|         |   |                                                                                                                                                                                                                                                                                                               |

A dialogue illustrating the workings of QCoord is (78), a trace through which is provided in (79):

- (78) a. A: (1a) Who will Max be inviting? (1b) When will these guests be arriving?  
 B(2): Mary and Bill.  
 A(3): Aha.  
 B(4): Tomorrow or the day after most likely.

(79)

| Utt.    | DGB Update<br>(Conditions)                                                                           | Rule                                  |
|---------|------------------------------------------------------------------------------------------------------|---------------------------------------|
| initial | MOVES = $\langle \rangle$<br>QUD = $\langle \rangle$<br>FACTS = $\text{cgl}$                         |                                       |
| 1a      | LatestMove := Ask(A, B, $q0$ )<br>QUD := $\langle q0 \rangle$                                        | Free Speech<br>Ask QUD-incrementation |
| 1b      | LatestMove := Ask(A, B, $q1$ )<br>$\neg \text{Influence}(q1, q0)$<br>QUD := $\langle q0, q1 \rangle$ | QCoord<br>QSPEC                       |
| 2       | LatestMove := Assert(B, A, $p0$ )<br>(About( $p0, q0$ ))<br>QUD := $\langle p1?, q0, q1 \rangle$     | Assert QUD-incrementation             |
| 3       | LatestMove := Accept(A, B, $p0$ )<br>FACTS := $\text{cgl} \wedge p0$<br>QUD := $\langle q1 \rangle$  | Accept<br>Fact update/QUD-downdate    |
| 4b      | LatestMove := Assert(B, A, $p1$ )<br>(About( $p1, q1$ ))<br>QUD := $\langle p1?, q1 \rangle$         | QSPEC<br>Assert QUD-incrementation    |

One coarse aspect of (77) is the relationship it places between  $q$  and  $q_1$ —*non-influence*. I have assumed this to ensure complementarity with QSPEC, but this presumably is not precise enough. Note though that when the relationship between the two questions seems indeterminate, this licenses clarification, as in the following example (originally discussed in Ginzburg (1994)):

- (80) A(1): I'm going to ask you a couple of questions.  
 B(2): Shoot.  
 A(3): When did you start studying French? Who was the biggest influence you had at school?  
 B(4): In which order should I take these?

Let us turn finally—and briefly—to compound moves that involve assertions. QSPEC allows for successive assertion query pairs  $p_1, q_1$ , where  $q_1$  is a sub-question of  $p$ ?

QSPEC also allows for successive assertions  $p_1, p_2$ , where  $p_2$  is About  $p_1$ ?, as exemplified in (81):

- (81) A(1a): Several people showed up. (1b) Bill did.  
 B(2): Aha.  
 A(3): Max did.  
 B(4): I see.

From this emerges an implicit mechanism for accepting  $p_1$ : when the later assertion  $p_2$  is accepted, the issue associated with the earlier assertion  $p_1$  will be downdated iff FACTS (incremented with  $p_2$ ) resolves  $p_1$ ?:

(82)

| Utt. | DGB Update<br>(Conditions)                                                                                                                           | Rule                                     |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 1a   | LatestMove := Assert(A, B, $p_1$ )<br>QUD := $\langle p_1? \rangle$                                                                                  | Free Speech<br>Assert QUD-incrementation |
| 1b   | LatestMove := Assert(A, B, $p_2?$ )<br>QUD := $\langle p_2?, p_1? \rangle$<br>Influence( $p_2?, p_1?$ )                                              | QSPEC<br>Assert QUD-incrementation       |
| 2    | LatestMove := Accept(B, A, $p_2$ )<br>FACTS := $\text{cgl} \wedge p_2$<br>QUD := $\langle p_1? \rangle$                                              | Accept<br><br>Fact update/QUD-downdate   |
| 3    | LatestMove := Assert(A, B, $p_3$ )<br>QUD := $\langle p_3?, p_1? \rangle$<br>Influence( $p_3?, p_1?$ )                                               | QSPEC<br>Assert QUD-incrementation       |
| 4    | LatestMove := Accept(B, A, $p_3$ )<br>FACTS := $\text{cgl} \wedge p_2 \wedge p_3$<br>QUD := $\langle \rangle$<br>$\rightarrow (p_2 \wedge p_3, p_1)$ | Accept<br><br>Fact update/QUD downdate   |

## 4.6 Conversational Genres

I have so far discussed queries and assertions that arise *reactively*. This is not accidental—the conventions regulating the *initiating* of such moves, conversation-initially and periodically during extended interactions, seem far less domain-independent, far more activity-dependent, strongly conditioned by factors such as politeness, prior acquaintance between conversationalists, and so forth. The most challenging problems here relate to what one might call *applied pragmatics*—how to model the interaction that characterizes a given activity type. Some dialogue system designers would say,<sup>35</sup> inspired by Wittgenstein (1953), that building a dialogue system essentially starts and ends with this and such a view could be adopted by the working semanticist. Nonetheless, it seems a defensible strategy to try and isolate some domain-independent conversational possibilities (e.g. with respect to how questions are asked and responded to or positive/negative feedback is provided), while acknowledging the possibility that any given domain might involve moves that are specialized in some way. For now I sketch some general principles concerning activity types, which allow us to say something about initiating queries and assertions that goes a step beyond the principle of free speech, which I introduced earlier.

Free Speech postulates that when QUD is empty one can say whatever one likes. This is true when one stands on the proverbial soap box or, to a somewhat lesser extent, is sitting in a restaurant with friends, but this is, obviously enough, not sufficiently restrictive in general. Intuitively we would like to say that an issue can be introduced, by a query or an assertion, if it *relates to the current activity*. What does it mean to *relate to the current activity*? In some cases the activity is very clearly defined and tightly constrains what can be said, as in examples (83a–e). In other cases the activity is far less restrictive on what can be said, as in examples (83f–h):

- (83) a. Buying a train ticket: *client* wants a train ticket: *client* needs to indicate where to, when leaving, if return, when returning, which class, *salesperson* needs to indicate how much needs to be paid.
- b. Planning a trip with a travel agent: issues to be discussed include—where to? When? Which class? How? How long for? Are there visa restrictions? Is travel insurance needed? How much needs to be paid?
- c. Buying in a bakery: *client* needs to indicate what baked goods are desired, *baker* needs to indicate how much needs to be paid.
- d. Buying goods in a minimarket located in a petrol station: *client* needs to show what she bought, *salesperson* needs to check if *client* bought petrol and to tell *client* how much needs to be paid.

<sup>35</sup> Rudnicky (2004) expresses such views.

- e. Departmental meeting: first: the agenda is ... what can be said about agenda item 1? what can be said about agenda item 2? ..., what can be said about agenda item  $n$ ?
- f. Chatting among friends: first: how are conversational participants and their near ones?
- g. Chatting with a young child: first: how are conversational participants and their near ones?
- h. Buying in a bakery from a long-standing acquaintance: combination of (c) and (f).

Trying to operationalize activity relevance presupposes that we can classify conversations into various *genres*; I use the term *genre* following the seminal paper (Bakhtin 1986) to denote a particular type of interactional domain. I shall not offer a proposal for a taxonomy, nor undertake the even harder task of actually performing such a classification of dialogues.<sup>36</sup> These are both important tasks for future work, though open-ended like the construction of a dictionary. I will use the resources of TTR to sketch a means of genre classification, inspired most directly by a proposal of Larsson (2002). I use the common intuition in much linguistic classification (lexicons, constructions) that some genres involve very specific conditions, whereas others can be specified in very general terms.

How to classify a conversation into a genre? One way is by providing a description of an information state of a conversational participant who has successfully completed such a conversation. This is a reasonable way of doing things as long as the final state does not lose 'significant' information concerning what took place during the conversation. On the view of contextual evolution described here the final state of a conversational participant will be a DGB of the type in (84):

$$(84) \left[ \begin{array}{ll} \text{facts} & : \text{Set(Prop)} \\ \text{qud} = \{ \} & : \text{poset(question)} \\ \text{moves} & : \text{list(IllocProp)} \end{array} \right]$$

At the end of a conversation QUD is empty, but the issues that have been discussed during the conversation can be recovered by examining FACTS: we can introduce a simple refinement of FACTS update/QUD-downdate such that every acceptance leads to the recording not solely of the facts added but also which questions were resolved by these facts and downdated. One can implement this in two ways: either by introducing facts of the form **Resolve**( $p$ ,  $q$ ) into FACTS; alternatively, one can introduce a field dubbed QNUD ('questions no longer under discussion') to keep track of such questions. The former strategy has the advantage of preserving the

<sup>36</sup> Such taxonomies are implicit in a number of large-scale corpora such as the BNC and the Swedish Spoken Language Corpus at Göteborg University (Allwood 1999).

information of which fact resolves which question. Such facts have semantic importance: they underwrite resolvedness presuppositions (see Chapter 3) and are exploited in rhetorical question constructions (see Chapter 8). For current purposes I will resort to the latter, more direct strategy. Technically, then, (85) differs from (44) in introducing a field *qnud* in the preconditions and updating it with the questions downdated from QUD.

(85) Fact update/QUD-downdate (version with QNUD update) = *def*

$$\left[ \begin{array}{lcl} \text{pre} & : & \left[ \begin{array}{lcl} p & : & \text{Prop} \\ \text{LatestMove} = \text{Accept}(\text{spkr}, \text{addr}, p) \vee \\ \text{Confirm}(\text{spkr}, \text{addr}, p) & : & \text{IllocProp} \\ \text{qud} = \langle p?, \text{pre.qud} \rangle & : & \text{poset}(\text{Question}) \\ \text{qnud} & : & \text{set}(\text{question}) \\ \text{facts} = \text{commonground1} & : & \text{Set}(\text{Prop}) \end{array} \right] \\ \text{effects} & : & \left[ \begin{array}{lcl} \text{facts} = \text{pre.facts} \cup \{ p \} & : & \text{Set}(\text{Prop}) \\ \text{qud} = \text{NonResolve}(\text{pre.qud}, \text{facts}) & : & \text{poset}(\text{Question}) \\ \text{qnud} = \text{pre.qnud} \cup (\text{pre.qud} \setminus \text{qud}) \end{array} \right] \end{array} \right]$$

**Exercise 4.6.1** Formulate Fact update/QUD downdate so that facts of the form **Resolve**(*p*, *q*) get introduced when *q* gets downdated as a consequence of *p*'s acceptance.

Final states of a conversation will then be records of type *T*, for *T* a subtype of *DGBType<sub>fin</sub>*. I rename this latter to *GenreType* since it may be identified as the general type of all conversational genres:

(86) *DGBType<sub>fin</sub>* = *def* *GenreType* = *def*

$$\left[ \begin{array}{lcl} \text{facts} & : & \text{Set}(\text{Prop}) \\ \text{qnud} & : & \text{set}(\text{question}) \\ \text{moves} & : & \text{list}(\text{IllocProp}) \end{array} \right]$$

Note that we can, if we so wish, distinguish a genre in which only a restricted set of issues *I* can be discussed from a genre in which *I* includes the issues to be discussed, by using manifest fields. The former case will be as in (87a), whereas the latter will be as in (87b):

- (87) a.  $\left[ \text{qnud} = \mathcal{I} : \text{list}(\text{question}) \right]$   
 b.  $\left[ \begin{array}{l} \text{qnud} : \text{list}(\text{question}) \\ \text{c: } \mathcal{I} \subset \text{qnud} \end{array} \right]$

Different effects—whether questions needs to be discussed in a certain order—can also be achieved depending on whether we assume QNUD to be an unordered set, partially ordered, or a list. One can also introduce conditions about the composition of MOVES, though I do not do so here; see Exercise 4.6.3 for one example along these lines.

Let us consider some highly simplified examples: Casual conversation among acquaintances seems to be governed by a convention that an initial greeting optionally raises as MaxQUD an issue glossable as  $\lambda P.P(A)$  ('How is A'),  $\lambda P.P(B)$  ('How is B'), A and B being the conversational participants. QSPEC then licenses assertions such as 'You look well/as young as ever/pale etc.' This is captured in terms of the type CasualChat given in (88). At the other extreme of this very minimally specific type is PETROLMARKET, mentioned above in (83e). BakeryChat is somewhere between these in restrictiveness:

(88) a. CasualChat =<sub>def</sub>

|          |                                                            |
|----------|------------------------------------------------------------|
| A        | : Ind                                                      |
| B        | : Ind                                                      |
| utt-time | : TimeInterval                                             |
| c1       | : Speak(A, t) $\vee$ Speak(B, t)                           |
| facts    | : Set(Prop)                                                |
| qnud     | : list(question)                                           |
| c2       | : $\{\lambda P.P(A), \lambda P.P(B)\} \subset \text{qnud}$ |
| moves    | : list(IllocProp)                                          |

b. Petrolmarket =<sub>def</sub>

|          |                                                                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| A        | : Ind                                                                                                                                           |
| B        | : Ind                                                                                                                                           |
| utt-time | : TimeInterval                                                                                                                                  |
| c1       | : Speak(A, t) $\vee$ Speak(B, t)                                                                                                                |
| facts    | : Set(Prop)                                                                                                                                     |
| qnud     | : list(question)                                                                                                                                |
| c2       | : $\left\{ \lambda x.\text{InShopBuy}(A, x), \right.$<br>$\left. ?\text{BuyPetrol}(A), \lambda x.\text{Pay}(A, x) \right\} \subset \text{qnud}$ |
| moves    | : list(IllocProp)                                                                                                                               |

c. BakeryChat =<sub>def</sub>

|          |                                                                                                                                        |
|----------|----------------------------------------------------------------------------------------------------------------------------------------|
| A        | : Ind                                                                                                                                  |
| B        | : Ind                                                                                                                                  |
| utt-time | : TimeInterval                                                                                                                         |
| c1       | : Speak(A, t) $\vee$ Speak(B, t)                                                                                                       |
| facts    | : Set(Prop)                                                                                                                            |
| qnud     | : list(question)                                                                                                                       |
| c2       | : $\left\{ \lambda P.P(A), \lambda P.P(B), \lambda x.InShopBuy(A, x), \right. \left. \lambda x.Pay(A, x) \right\} \subset \text{qnud}$ |
| moves    | : list(IllocProp)                                                                                                                      |

**Exercise 4.6.2** Construct instances of each of the above three genres.

**Exercise 4.6.3** Specify a genre type for interaction in court. This should involve roles for participants in such interaction and, for instance, the characteristic first and last moves ('This session is now over').

Before we introduce *activity relevance* let me first introduce an auxiliary notion, that of the *outcome defined by a DGB and a genre*. Recall that we introduced an outcome in Chapter 3 as an entity used to represent the future classification of a situation. Thus, an outcome is a record with two fields, one of type record, the other of type *irrealis* (viz. a temporally abstracted record type). Given this, a DGB *dgb0* and a genre type *G0* define an outcome, given in (89), which will be fulfilled iff *dgb0* represents a conversation potentially classifiable by *G0*:

$$(89) \quad \text{outcome}(dgb0, G0) = \left[ \begin{array}{ll} \text{sit} & = dgb0 \\ \text{irr-sit-type} & = (r : [\text{utt-time} : \text{TimeInterval}]) \ G0 \ (\text{utt-time} \rightsquigarrow r.\text{utt-time}) \end{array} \right]$$

We can now pin down the basic intuition concerning activity relevance. In current terms—one can make an initiating move *m0* relative to the current DGB *dgb0*, and *G0*, the genre one assumes the conversation to be, if and only if one believes that updating *dgb0* with *m0* results in an outcome(*dgb0*  $\oplus_{\text{moves}}$  *m0*, *G0*) that can be fulfilled. This means that making move *m0* given what has happened so far (represented in *dgb0*) can be *anticipated* to conclude as final state *dgb1* which is a conversation of type *G0*:

(90) *m0* is relevant to *G0* in *dgb0* for *A* (**GenreRelevant**(*m*, *G0*, *dgb0*)) iff *A* believes that outcome(*dgb0*  $\oplus_{\text{moves}}$  *m0*, *G0*) will be fulfilled. That is, iff there exists *dgb1* such that *dgb0*  $\oplus_{\text{moves}}$  *m0*  $\sqsubset$  *dgb1* and such that *dgb1* : *G0*



A slight generalization to this arises from the following consideration: an initiating move  $m0$  might in itself carry QUD or FACTS presuppositions, in other words involve some form of accommodation. I will restrict myself here to QUD accommodation, following proposals of Larsson (2002). In order to make this tractable, one needs to ensure a very tight fit between the QUD accommodated entity  $q(m0)$  and the content of  $m0$ . I will assume that the appropriate relation is *co-propositionality*.<sup>37, 38</sup>

- (91)  $m0$  is relevant to  $G0$  in  $dgb0$  QUD-presupposing  $q(m0)$   
 ( $\text{GenreRelevant}^{qudpresupp}(m, q(m0), dgb0, G0)$ ) iff A believes that the outcome  
 $\text{outcome}(dgb \cup \left[ \begin{array}{l} dgb.moves = \langle m0, dgb.moves \rangle \\ dgb.qud = \langle q0, dgb.qud \rangle \\ cl: \text{Copropositional}(qud-contrib(m0.cont), q(m0)) \end{array} \right],$   
 $G0$ ) will be fulfilled.

Beyond the issue of the nature of conversation genre, there are a number of other primarily representational issues that arise: first, it seems clear that formulating the rule for move initiation necessarily involves interfacing with *non-public* information. This is because an initiating query/assertion can be uttered when virtually no explicit interaction between the participants has taken place. Even if a certain amount of interaction has taken place, it is rare for the conversational participants to negotiate in detail about the characteristics of the activity in which they are engaged. Thus, Initiating Move will require us to make recourse to *non-public* or *private* beliefs, and hence to make explicit some notion of the non-public part of an information state. Given framework congruity I will build on the proposal of Larsson (2002), which is inspired by much previous work in the BDI tradition (for a review see Bratman 1987, Georgeff and Lansky 1987) and intended primarily as an architecture for a dialogue system. In Larsson's account the private part of the information state can be partitioned in two: there is a dynamic part, which can be modified during a conversation and a static part, which cannot. The dynamic part consists of *BEL*, a set of beliefs, *PLAN*, a stack of plans, and *AGENDA*, a stack of actions. The static part consists of a lexicon, a database, and *DOMAIN*, the locus for information about the domain for which the system is built. To get an idea of how this fits together I informally describe (slightly simplified and modified versions of) three of Larsson's rules, *findPlan*, *selectFromPlan*, and *SelectAsk*:

<sup>37</sup> Two questions  $q_0$  and  $q_1$  are co-propositional if there exists a record  $r$  such that  $q_0(r) = q_1(r)$ . This means that, modulo their domain, the questions involve similar answers. For instance 'Whether Bo left', 'Who left', and 'Which student left' (assuming Bo is a student).

<sup>38</sup>

- (i)  $qud-contrib(m0.cont)$  is  $m0.cont$  if  $m0.cont$  : Question  
 (ii)  $qud-contrib(m0.cont)$  is ? $m0.cont$  if  $m0.cont$  : Prop

- (92) a. findPlan: given the presence of the action *respond*(*q*) maximal in AGENDA, then if DOMAIN contains a plan *P* with the component *respond*(*q*):  
 (a) remove *respond*(*q*) from AGENDA, (b) push *P* onto PLAN.  
 b. selectFromPlan: given that action *A* is maximal in plan *P*, which is maximal in PLAN and that AGENDA is empty: push *A* onto AGENDA.  
 c. selectAsk: given that *ask*(*q*) is maximal in AGENDA:  
 choose *ask*(*q*) as one of the next utterances to make.

Of these three findPlan is the most interesting from our current perspective, the other two are provided for completeness. findPlan is stated from the perspective of an addressee (in Larsson's case the system) that encounters a question *q*. In essence what it amounts to is simply this: a query *q* is felicitous iff there exists a plan *p* such that *q* features in *p*. More precisely, a query is associated with a search for a plan *P* in DOMAIN which contains *respond*(*q*) as a constituent—our action relevance is inspired by and is a generalization of this latter. Given this, *P* is loaded into PLAN. This latter is a crucial additional point: once a conversational participant has identified the genre of the conversation, s/he needs to maintain that identification for the duration of the conversation (or conversational stretch).<sup>39</sup>

It is useful to abstract away from the complex, implementationally motivated, internal structure of the private part of an IS in Larsson's system and consider what is clearly necessary for our current, somewhat more general purposes. The static components are ones whose existence we presuppose—in particular a grammar, though we do not assume a single domain is involved. As for the dynamic components, clearly private beliefs are a necessary private counterpart to the public FACTS, whereas AGENDA is a corresponding counterpart to Moves. PLAN, on the other hand, is a type of information which does not have a public counterpart, but plays an important role. I will presuppose its existence, though rename it GENRE, as in (93a). This leads naturally enough to the total information state structure in (93b):

- (93)
- a.  $PRTYPE =_{def} \begin{bmatrix} \text{genre} & : & \text{GenreType} \\ \text{beliefs} & : & \text{Prop} \\ \text{agenda} & : & \text{list}(\text{IllocProp}) \end{bmatrix}$
- b.  $\text{TotalInformationState (TIS)} =_{def} \begin{bmatrix} \text{dgb} & : & \text{DGBType} \\ \text{private} & : & \text{PRTYPE} \end{bmatrix}$

<sup>39</sup> Of course, as with most other conversational practices the possibility of error exists, which requires some sort of backtracking. I will not consider this here.

**Exercise 4.6.4** By making reference to the private part of an information state, refine Fact Update/QUD-Downdate to allow for agent relativity of question resolvedness.

We can finally turn to formulate Initiating move, a refinement of Free Speech, as follows: an initiating move *ip0* can be made by either conversational participant relative to A's information state given that: (a) QUD is empty and (b) given that the current genre is *G0*, A believes that *ip0* uttered relative to *q0* is relevant to *G0* in *dgb0*. In such a case, *ip0* becomes the LatestMove and *q0* becomes MaxQUD, with *ip0* constrained to be co-propositional with *q0*.

(94) Initiating Move =<sub>def</sub>

$$\left[ \begin{array}{l} \text{dgb} : \left[ \text{qud} = \langle \rangle \text{ poset}(\text{Question}) \right] \wedge \text{DGB} \\ \text{pre} : \left[ \begin{array}{l} \text{genre} : \text{GenreType} \\ \text{beliefs} : \text{Prop} \\ \text{agenda} : \text{list}(\text{IllocProp}) \\ \text{private} = \left[ \begin{array}{l} \text{ip0} : \text{IllocProp} \\ \text{q0} : \text{Question} \\ \text{c1} : \rightarrow (\text{beliefs}, \\ \text{GenreRelevant}^{\text{qudpresupp}}(\text{ip0}, \text{q0}, \text{dgb}, \text{genre}) ) \end{array} \right] : \text{PRTYPE} \end{array} \right] \\ \text{effects} : \text{TurnUnderspec} \wedge_{\text{merge}} \left[ \begin{array}{l} \text{LatestMove} = \text{pre.private.ip0} : \text{IllocProp} \\ \text{qud} = \langle \text{pre.private.q0} \rangle : \text{list}(\text{question}) \\ \text{c3} : \text{Copropositional}(\text{qud-contrib}(\text{pre.private.ip0}), \text{pre.private.q0}) \end{array} \right] \end{array} \right]$$

To conclude this section, let us exemplify the working of Initiating Move. Consider first (95). There are two noteworthy moves here, B's utterances (95(2)) and (95(4)). The former can be analysed as B believing that the conversational genre is CasualChat (see (88a) above). This introduces the need to discuss the issues  $\lambda P.P(A)$ ,  $\lambda P.P(B)$ , the second of which (95(2)) is About. A accepts B's assertion, after which B decides—somewhat rudely or at least egocentrically—that the issue  $\lambda P.P(A)$  is resolved—the fact that A is there standing with him is sufficient information about A's current state.<sup>40</sup> This then licenses a parting utterance:

- (95) A(1): Hi.  
 B(2): I'm off.  
 A(3): OK.  
 B(4): Bye.

<sup>40</sup> Note that the conversation has a perfectly coherent alternative continuation in which B's state issue is discussed:

B: (4') You?  
 A(5'): I'm around for now.  
 B(6'): OK. Bye.

In more detail: as B is about to make his move (95(2)), assume his information state is as in (96):

$$(96) \left[ \begin{array}{lcl} & \left[ \begin{array}{lcl} \text{Spkr} & = & A \\ \text{addr} & = & B \\ \text{MOVES} & = & \langle \text{Greet}(A,B) \rangle \\ \text{QUD} & = & \langle \rangle \\ \text{FACTS} & = & \text{cgl} \end{array} \right] \\ \text{dgb} & = & \\ & & \\ & & \\ & & \\ & & \\ & & \\ \text{private} & = & \left[ \begin{array}{lcl} \text{genre} & = & \text{CasualChat} \\ & & \\ \text{beliefs} & = & \left\{ \begin{array}{l} \dots, \\ \text{Fulfillable}(\text{outcome}(\text{dgb0} \mapsto \text{dgb} \cup \\ \left[ \begin{array}{lcl} \text{Moves} & = & \langle \text{Assert}(B, A, \text{Off}(B)), \text{dgb.moves} \rangle \\ \text{qud} & = & \langle \lambda P.P(B), \text{dgb.qud} \rangle \end{array} \right] \rangle), \\ \dots \end{array} \right\} \\ \text{agenda} & = & \text{a1} \\ \text{cl} & = & \text{pr1} \end{array} \right] \end{array} \right]$$

This will underwrite the application of Initiating Move, leading to the following DGB update—the issue  $\lambda P.P(B)$  is accommodated, serving as background to the utterance:

$$(97) \left[ \begin{array}{lcl} \text{Spkr} & = & B \\ \text{addr} & = & A \\ \text{MOVES} & = & \langle \text{Assert}(B, A, \text{Off}(B)), \text{Greet}(A, B) \rangle \\ \text{QUD} & = & \langle \lambda P.P(B) \rangle \\ \text{FACTS} & = & \text{cgl} \end{array} \right]$$

A accepts B's assertion, which B takes as a cue for his parting move.

**Exercise 4.6.5** Trace in detail through (95).

**Exercise 4.6.6** Provide an alternative account of parting on the basis of Initiating Move. Specify that all conversations should have parting interaction as constituent moves.

As our final example, let us consider (98). This is an example of how our approach can fairly straightforwardly deal with a mixture of genres, thereby proceeding beyond approaches that handle only a single domain at a time. Initially there is a greeting and information exchanged about the state of the participants. Then there is interaction concerning the purchase of baked goods. Following that there is an exchange concerning what A will be doing over the weekend. Finally, the issue of what B needs to pay is resolved:

- (98) A(1): Hi! How are you?  
 B(2): Fine. And you?  
 A(3): Not bad. What do you want to buy?  
 B(4): I want two croissants.  
 A(5): OK.  
 B(6): And a Finnbrot.  
 A(7): Sure.  
 B(8): Doing anything interesting this weekend?  
 A(9): Hertha are at home.  
 B(10): Nice.  
 A(11): Is that all?  
 B(12): Yes.  
 A(13): Two euros sixty please.

In order to see how this example can be analysed, I trace through the issues as they get resolved; for brevity I omit polar questions introduced by assertions:

(99)

| Utt. | QNUD                                                                                                                                                                                                   |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2    | $\text{QNUD} := \langle \lambda P.P(B) \rangle$                                                                                                                                                        |
| 3    | $\text{QNUD} := \langle \lambda P.P(B), \lambda P.P(A) \rangle$                                                                                                                                        |
| 7    | $\text{QNUD} := \langle \lambda P.P(A), \lambda P.P(B),$<br>$\lambda x.\text{InShopBuy}(B, x) \rangle$                                                                                                 |
| 12   | $\text{QNUD} := \langle \lambda P.P(A), \lambda P.P(B),$<br>$\lambda x.\text{InShopBuy}(B, x) ? \exists x.\text{Doing}(A, x : \text{interesting}(x)) \rangle$                                          |
| 13   | $\text{QNUD} := \langle \lambda P.P(A), \lambda P.P(B), \rangle$<br>$\lambda x.\text{InShopBuy}(B, x), ? \exists x.\text{Doing}(A, x : \text{interesting}(x))$<br>$\lambda x.\text{Pay}(B, x) \rangle$ |

This gives an indication that at all points the conversation is of type BakeryChat (see above (88c)).

## 4.7 Summary

In this chapter I have exemplified dialogue analysis in KoS using two key notions:

- **Dialogue Gameboards (DGBs):** a DGB is a conversational participant's own version of the public context, structured by fields for the turn holder, her interlocutor, the commonly accepted assumptions (FACTS), the moves that have taken place (MOVES), and the questions that have been introduced and remain unresolved (QUD). Concretely, the type of the DGB is given in (100):

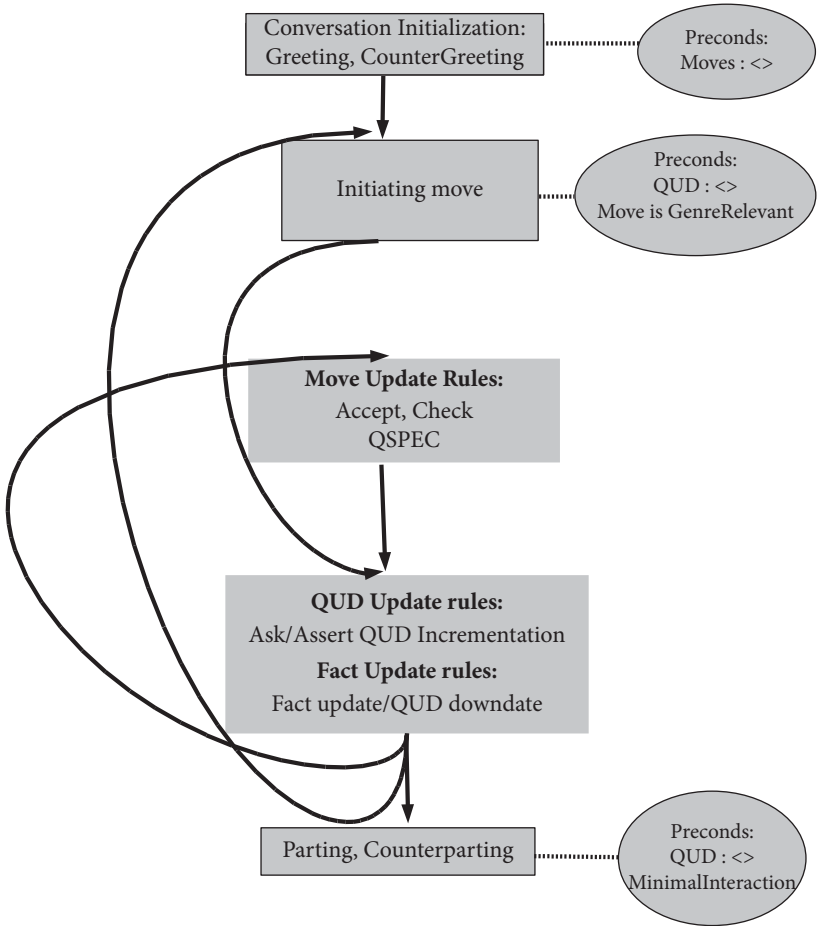


FIGURE 4.1: Dynamics of context underlying illocutionary interaction

(100)  $DGBType =_{def}$ 

|                  |                                               |
|------------------|-----------------------------------------------|
| spkr             | : Ind                                         |
| addr             | : Ind                                         |
| utt-time         | : Time                                        |
| c-utt            | : addressing(spkr, addr, utt-time)            |
| Facts            | : Set(Prop)                                   |
| Moves            | : list(IllocProp)                             |
| QUD              | : poset(Question)                             |
| non-resolve-cond | : $\forall q \in QUD[\neg Resolve(FACTS, q)]$ |

- **Conversational rules:** We call a mapping that indicates how one DGB can be modified by conversationally related action a *conversational rule*, and the types specifying its domain and its range respectively the *preconditions* and the *effects*. In a small number of cases conversational rules actually make reference to the entire information state.

Using these notions I have then offered:

- a formal description of greeting and parting: these are interaction types where the dynamics is primarily on the level of MOVES
- a formal description of querying and assertion and related moves (accepting, checking, confirming etc.)

I showed that this account scales down to monologue to cover *inter alia* self-querying. I also illustrated the need for certain specifically monological conversational rules, given that certain sequences of well-formed monologues do not allow for coherent variants that involve turn exchange.

The final contribution of this chapter is an attempt to to explicate *initiating* moves, moves that do not occur as a reaction to a prior issue. This explication relies on a classification of conversations into types or *genres*, which can be done straightforwardly on the basis of our theory of DGBs.

The basic picture of conversational structure that emerges from this chapter is summarized in Figure 4.1: a conversation is initialized by greeting interaction. This leads to an initiating move and a sequence of reactive moves building on that move. When that sequence is terminated, with QUD becoming empty once more, one returns again to an initiating move. This continues until one final emptying of QUD which presages parting interaction.

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# A Grammar for Dialogue

## 5.1 Introduction

In Chapter 4 I offered a view of dialogue interaction that ignores the internal structure of utterances, abstracting away as it does from the communicative process. In order to give up on this latter simplifying assumption, a task undertaken in Chapter 6, we will need to have the means of taking into account the internal structure of utterances. The current chapter is devoted to outlining the basic features of a grammatical framework for dialogue. Apart from providing a vital component in KoS's account of grounding/clarification potential, this grammar will underpin a formal description of the workings of non-sentential utterances. My aim will not be to develop a new grammatical framework, but merely to modify certain aspects of existing approaches to make them dialogue-friendly. In particular, I propose a dialogically-oriented methodological principle, the Reprise Content Hypothesis, intrinsically more restrictive than compositionality, for regulating allowable semantic contents.

My starting point will be the Head-Driven Phrase Structure Grammar (HPSG)-formulated grammar expounded in Ginzburg and Sag (2000). That work developed a grammar of interrogatives, declaratives, and other clause types (including imperatives, subjunctives, and infinitives). This grammar draws on two convergent approaches, *Construction Grammar* and *Sign-Based Grammar*. It draws on Construction Grammar (Fillmore 1999, Michaelis 2009) for its potential for providing highly detailed descriptions of phrasal constructions in such a way that not *all* semantic aspects of a construction need to derive from its constituents. This will be of significance when we tackle non-sentential utterances. Sign-based grammars, whether in HPSG, in Categorical Grammar (see e.g. Calder, Klein, and Zeevat 1988, Moortgat 1997), or in versions of Lexical Functional Grammar (see e.g. Muskens 2001), have some important architectural features as far as dialogue is concerned, already briefly motivated in Chapter 2. Their architecture satisfies the property of incremental correspondence (Johnson and Lappin 1999): utterance representations encode phonological, syntactic, semantic, and contextual information *fractally*.<sup>1</sup> That

<sup>1</sup> Johnson and Lappin distinguish grammars built on an incremental correspondence architecture from grammars based on an *Interface Model* architecture. My use of 'fractal' in this context follows Pollard (1997).



is, the requisite representation format contains heterogeneous (viz. phonological, syntactic, semantic, and contextual) information and, moreover, this applies uniformly as the parts get smaller and smaller. As I argue in Chapter 6, utterance types of this kind are a good approximation for (one component of) the entity which conversationalists share during the process of grounding a speech event  $se_0$ . This entity (a) encodes medium-term presuppositions about  $se_0$  and (b) (in case the addressee does not fully comprehend) is used to specify context for a clarification request concerning any sub-utterance down to the word level. Thus, an adequacy requirement for any grammatical formalism applied to dialogue is the ability to provide a theory of such entities. This will rule out, for instance, derivational theories in the transformational tradition because they intrinsically fail to satisfy *Incremental Correspondence*.

The first task I undertake here is to argue that, in so far as we want use our utterance types as the basis for a theory of communicative interaction, then using a grammar based on records and record types is superior to a grammar formulated in terms of typed feature structures (TFSs). Specifically, I will propose abandoning the use of type feature structures and of unification as a semantic glue for TFSs. The alternative is straightforward: utterance types can be modelled as record types, whereas actual utterance tokens—speech events—can be modelled as records, as indeed events in general are modelled in Chapter 3. I will call the resulting grammatical framework  $HPSG_{TTR}$ , contrasting it with  $HPSG_{TFS}$ , the type feature structure-based version.<sup>2</sup>

My discussion will relate to TFS-formulated grammar, though a number of the arguments—centred around the Reprise Content Hypothesis—apply to other grammatical formalisms, assuming one intends to use them for dialogue. Following this, I develop a small fragment of English in  $HPSG_{TTR}$  with which one can analyse some simple interrogatives and declaratives. I conclude the chapter with a discussion of how *Conversational Move Type* (aka ‘illocutionary force’) should be integrated in the grammar.

## 5.2 TTR-Based Grammar: Motivation

### 5.2.1 Unification and the Reprise Content Hypothesis

Unification has been used as the glue for semantic composition in various frameworks, in particular HPSG. Thus, in Pollard and Sag (1994) (non-quantificational) content is inherited from head daughter to mother (notated here using daughter  $\nearrow$  mother) in at least the following cases:

- (1) a.  $V \nearrow VP \nearrow S$
- b.  $CN \nearrow NP$

<sup>2</sup> The alternative I pursue here was initiated in Cooper and Ginzburg (2002) and Cooper (2005b); a number of the driving motivations are pursued in ongoing work by Carl Pollard (e.g. Pollard 2004).

In Ginzburg and Sag (2000)  $VP \nearrow S$  was no longer a tenable possibility since this latter work distinguished the denotata of interrogative, declarative, and imperative clauses that were potentially built up from identical VPs. Nonetheless,  $V \nearrow VP$ ,  $CN \nearrow NP$  inheritance was still maintained.

As pointed out in Ginzburg (1999a) and Purver and Ginzburg (2004), postulating semantic identity between daughter and mother is intuitively unsatisfying *if sub-utterances are taken as having meaningful contents*.<sup>3</sup> We can convert this intuition into concrete data using reprise fragments (RFs), the construction first mentioned in Chapter 1.1. Recall that RF probes the content of sub-utterances:

- (2) a. A: Did Bo finagle a raise? B: (i) Bo? (ii) finagle?
- b. **Clausal confirmation reading:** Are you asking if BO (of all people) finagled a raise/Bo FINAGLED a raise (of all actions)
- c. **Intended content reading:** Who is Bo?/What does it mean to finagle?

Inspired by RF and other reprise constructions, Purver and Ginzburg (2004) postulate the Reprise Content Hypothesis (RCH) as a methodological principle and offer evidence for its consistency with many currently assumed semantic denotations. The strong version of the RCH is given in (3):<sup>4</sup>

- (3) Reprise Content Hypothesis (strong version)  
*A fragment reprise question queries exactly the standard semantic content of the fragment being reprised.* (Purver and Ginzburg 2004)

The RCH is intuitively attractive, as it provides us with a version of Occam's Razor: do not postulate any part of a semantic representation that cannot be observed via a reprise question. The RCH supports, but is stronger than, the usual criterion of compositionality. Compositionality, as standardly formulated, leaves underdetermined the question of which part contributes what. Admitting the RCH means responsibility for the content assigned not only to the complete utterance but also to each component (or at least each reprisable and semantically potent component).

Assuming the RCH, let us consider the common unification-based approach to constructing the meanings of certain referential NPs (see e.g. Sag and Wasow 1999, Ginzburg and Sag 2000). CN content is identified with that of the NP mother, and thus taken to be a parameter whose referent is an individual (the NP referent). In contrast to this view, in Montogovian semantics CNs have typically been analysed as properties. This view seems to be supported by the RCH: paraphrases which concern

<sup>3</sup> This is not a foregone conclusion. See Johnson and Lappin (1999) for discussion of grammatical frameworks that deny this.

<sup>4</sup> The weak version differs from the strong one in allowing that the reprise question queries a *part* of the content, leaving wiggle room for positing content undetectable by such queries.

an intended referent of the NP containing the CN (e.g. the ‘Which X ...’ paraphrases) do not appear to be available, even when the NP might appear to be referential (see (5)).<sup>5</sup>

- (4) a. Emma: Got a comb anywhere?  
Helena: Comb?  
Emma: Even if it's one of those <pause> tremmy [sic] pretend combs you get with a Barbie doll, oh this'll do! <pause> Don't know what it is, but it'll do!
- b. Is it a *comb* that you are asking if I've got?  
# Which comb are you are asking if I've got?  
(BNC, KCE)
- (5) a. Carol: We'll get the turkey out of the oven.  
Emma: Turkey?  
Carol: Well it's <pause> it's <pause> er <pause> what's his name? Bernard Matthews' turkey roast.  
Emma: Oh it's looks horrible!
- b. Are you saying the thing we'll get out is a *turkey*?  
What concept/property do you mean by 'turkey'?  
# Which turkey are you saying we'll get out?  
# Is it this/that turkey you're saying we'll get out? (BNC, KBJ)

Similarly, the evidence from RFs argues against the unification (1a), wherein the content of a verb is identified with the content of a VP, *a fortiori* an S. Verbs are better analysed as properties or relations rather than as propositional entities:

- (6) a. Joyce: He had some stuff nicked, a ski jacket which cost me seventy five quid it were half, the rest it should of been a hundred and fifty  
Ann: Nicked?  
Joyce: Nicked  
Alec: Mm  
Joyce: Pinched  
Ann: Aargh
- b. Are you saying it was *nicking* that happened to his stuff?  
What property do you mean by 'nicked'?  
(BNC, KB2)

<sup>5</sup> Similar problems apply to approaches such as Minimal Recursion Semantics (Copestake, Flickinger, Pollard, and Sag 2005) in which the content of an NP mother is constructed by set union (*amalgamation*) over the content of its daughters (sets of *elementary predications*, simple pieces of propositional information). This again results in CN content including the individual referent of the mother NP.

The TTR formulation will allow us to use Montague Semantics-style denotations for functors (verbs, common nouns). This is one way—historically speaking the most conservative way—of achieving a division of semantic content across different expressions consistent with the RCH; one could equally achieve this using a different approach, for instance the glue language approach (see e.g. Dalrymple 1999). As we will see in Chapter 8, the broadly Montogovian approach to semantic composition has also an important role in explicating the complexity of NSU constructions, an explication that jibes well with evidence about children's acquisition of NSUs. Nonetheless, in section 8.5 we will see that with regards to NPs, data relating to the RCH points to individual or set denotations, NOT the generalized quantifier denotations, routinely assumed in much of Montogovian-inspired semantics.

### 5.2.2 The Semantic Undesirability of Typed Feature Structures

In the grammars developed in Pollard and Sag (1987) and (1994) typed feature structures (TFSs) were used as a data structure that encodes all grammatically pertinent information, including semantic information. For the latter this meant encoding situation-theoretic entities, primarily situation types—usually called SOAs in the HPSG literature. A similar strategy was employed in Ginzburg and Sag (2000), though the status of TFSs in the semantic domain was somewhat undermined: given the existence of an explicit TFS  $\rightarrow$  Situation Theory interpretation,<sup>6</sup> the TFSs were viewed as *dispensable* logical forms. However, if one takes this line, it remains unresolved how the semantic component interfaces with the rest of the grammar.

What is the problem with using TFSs to encode semantic entities? The basic point is that *HPSG<sub>TTR</sub> directly provides semantic entities, whereas HPSG<sub>TFS</sub> simulates them*. Apart from the undesirability of intermediary coding,<sup>7</sup> this turns out to have practical consequences if one wishes to use signs as utterance representations in dialogue—HPSG<sub>TFS</sub> is insufficiently expressive for the task.

It is perhaps easiest to grasp this by means of an example that compares a TFS-based account and one couched in TTR.

The grammar fragment developed in section 5.3 below posits the sound/syntax/meaning constraint in (7a) as a rule of English; PHON encodes requirements on the phonological type of the utterance, CAT requirements on the syntactic category, DGB-PARAMS encodes the domain of the meaning function (see section 3.3.2), and CONT(ENT) encodes the value the meaning function gets when instantiated. For a speech event *se0*, (7b), to be classified as being of this type, the requirements in (7c) will need to be met:<sup>8</sup>

<sup>6</sup> See Appendix B, pp. 388–9.

<sup>7</sup> Properties of the intermediate 'stratum' take on a life of their own.

<sup>8</sup> A convention I employ here to distinguish phonological tokens and types is to refer to the latter with English words and the former with a mock representation of their pronunciation.

- (7) a. 
$$\left[ \begin{array}{l} \text{phon : did jo leave} \\ \text{cat} = V[+\text{fin}] : \text{syncat} \\ \text{dgb-params : } \left[ \begin{array}{l} \text{s0: SIT} \\ \text{t0: TIME} \\ \text{j: IND} \\ \text{c3: Named(j, jo)} \end{array} \right] \\ \text{cont} = ? \left[ \begin{array}{ll} \text{sit} & = \text{s0} \\ \text{sit-type} & = [c: \text{Leave(j, t0)}] \end{array} \right] : \text{Question} \end{array} \right]$$
- b. 
$$\left[ \begin{array}{ll} \text{phon} & = \text{di jo liv} \\ \text{cat} & = V[+\text{fin}] \\ \text{dgb-params} & = \left[ \begin{array}{ll} \text{s0} & = \text{sit0} \\ \text{t0} & = \text{time0} \\ \text{j} & = \text{j0} \\ \text{c3} & = \text{c30} \end{array} \right] \\ \text{cont} & = ? \left[ \begin{array}{ll} \text{sit} & = \text{s0} \\ \text{sit-type} & = [c: \text{Leave(j, t0)}] \end{array} \right] \end{array} \right]$$
- c. di jow liv : did jo leave;  
 sit0 : SIT, time0 : TIME, j0 : IND, c30 : Named(j0, jo)  
 cont0 = ? 
$$\left[ \begin{array}{ll} \text{sit} & = \text{sit0} \\ \text{sit-type} & = \text{Leave(j0, time0)} \end{array} \right] : \text{Question}$$

Specifically: a witness for the type (7a) includes a phonetic token, contextual parameters—a situation, a time, an individual named Jo—and the question entity  $? \left[ \begin{array}{ll} \text{sit} & = \text{sit0} \\ \text{sit-type} & = \text{Leave(j0, time0)} \end{array} \right]$ , a 0-ary function into propositions. Thus, the fact

that DGB-PARAMS represents the type of entities needed to instantiate a meaning is a direct consequence of what it means to be a witness of this type. In addition, the values of the CONT field *are* already the semantic entities. Thus, the function in (8a) is, as discussed in Chapter 3, of the type in (8b), which is a subtype of the type in (8c). This latter is the type of a question such as (8d). This yields a natural notion of dependence among questions, answerhood etc.:

- (8) a.  $r : T_0 = \left[ \begin{array}{l} \end{array} \right]$   
 $\mapsto \left[ \begin{array}{ll} \text{sit} & = r_1 \\ \text{sit-type} & = c: \text{leave(j, t)} \end{array} \right]$

- b.  $(T_0 \rightarrow Prop)$   
 c.  $(T_{who} (= [ \begin{array}{ll} x & : \text{Ind} \\ \text{rest} & : \text{person}(x) \end{array} ] ) \rightarrow Prop)$   
 d.  $r : T_{who} \mapsto \left[ \begin{array}{ll} \text{sit} & = r_1 \\ \text{sit-type} & = c: \text{leave}(r.x, t) \end{array} \right]$

This explanatory state of affairs contrasts with the HPSG<sub>TFs</sub> account of this example, given in (9). This AVM *looks* very much like the type (7a), but appearances can be and in this case actually are deceiving.

$$(9) \left[ \begin{array}{l} \text{PHON did jo leave} \\ \text{CAT S} \\ \text{DGB-PARAMS } \left\{ \begin{array}{l} \left[ \begin{array}{ll} \text{INDEX} & b \\ \text{RESTR} & \{\text{named}(\text{Jo})(b)\} \end{array} \right] \\ \left[ \begin{array}{ll} \text{INDEX} & t \\ \text{RESTR} & \{\text{precedes}(t, k)\} \end{array} \right] \\ \left[ \begin{array}{ll} \text{INDEX} & s \\ \text{RESTR} & \{ \} \end{array} \right] \end{array} \right\} \\ \text{CONT } \left[ \begin{array}{l} \text{question} \\ \text{PARAMS } \{ \} \\ \text{PROP } \left[ \begin{array}{ll} \text{SIT} & s \\ \text{SOA} & \text{leave}(j, t) \end{array} \right] \end{array} \right] \end{array} \right]$$

In (9) CONT is *intended* as representation of the situation theoretic abstract  $\lambda\{ \} \text{leave}(j, t)$ . But, as Penn (2000) puts it (in discussing a related set of issues), ‘At this point, feature structures are not being used as a formal device to represent knowledge, but as a formal device to represent data structures that encode formal devices to represent knowledge.’<sup>9</sup> Similarly, DGB-PARAMS is *intended* as a representation of the contextual parameters that need to be instantiated, but there is no explicit way of modelling this.

This latter point can be amplified. As mentioned earlier, the interaction over grounding of a speaker A’s utterance  $u$  addressed to B typically leads to two outcomes: either (a) B acknowledges  $u$  (directly, gesturally or implicitly) and then (b) responds to the content of  $u$ . Alternatively, B utters a clarification question about some unclear aspect of  $u$ . As we will see in Chapter 6, this interaction can be explicated as an attempt to find a type  $T_u$  that uniquely classifies  $u$ . This involves *inter alia* recognizing

<sup>9</sup> Penn (2000: 63).

the words used and instantiating the contextual parameters specified in  $T_u$ . Seeking clarification involves utilizing a partially instantiated content and posing a question constructed from  $u$  and  $T_u$ . The standard view of the meaning/content relationship as a function from contexts to contents chokes in the face of a *partially* specified context. Reconstructing the meaning/content relationship in terms of two fields, DGB-PARAMS and CONTENT, originating in standard HPSG, allows in the current setting for the possibility of partially instantiating a content and maintaining this as the semantic representation of an utterance for the duration of clarification interaction until a more detailed instantiation is available (see the discussion of example (21) in Chapter 2). Here a partial witness for DGB-PARAMS field  $T_u.dgb-params$  is a record  $r_0$  that is extendible to  $r_1$  such that  $r_1 : T_u.dgb-params$ . This is exemplified in (10), where  $r_0$  lacks fields for  $j, c3$ :

$$(10) \quad \begin{array}{l} \text{a. } T_u.dgb - params = \left[ \begin{array}{ll} s0 & : \text{SIT} \\ t0 & : \text{TIME} \\ j & : \text{IND} \\ c3 & : \text{Named}(j, jo) \end{array} \right] \\ \\ \text{b. } r_0 = \left[ \begin{array}{ll} \text{phon} & = \text{di jo liv} \\ \text{cat} & = \text{V}[+fin] \\ \text{dgb-params} & = \left[ \begin{array}{ll} s0 & = \text{sit0} \\ t0 & = \text{time0} \end{array} \right] \end{array} \right] \\ \\ \text{c. } r_1 = \left[ \begin{array}{ll} \text{phon} & = \text{di jo liv} \\ \text{cat} & = \text{V}[+fin] \\ \text{dgb-params} & = \left[ \begin{array}{ll} s0 & = \text{sit0} \\ t0 & = \text{time0} \\ j & = j0 \\ c3 & = c30 \end{array} \right] \end{array} \right] \end{array}$$

Another ingredient in clarification interaction is building the clarification question itself. This can be modelled by means of a function which maps the partially instantiated content into a context, constructed from  $u$  and  $T_u$  using one of a handful of *Clarification Context Update Rules*. One crucial ingredient in this modelling is the ability to build functions from utterance tokens and utterance types into types of contexts, characterized in terms of various semantic objects such as propositions and questions. This is straightforward in HPSG<sub>TTR</sub> given the fact that it enables the construction of function types. On the other hand, HPSG<sub>TFs</sub> does not provide such tools.

A third advantage of HPSG<sub>TTR</sub> is that both utterance tokens (records) and signs (record types) become available simultaneously in a natural way. Having both around

will be crucial in providing a semantics for metacommunicative utterances, where as we saw in Chapter 2, reference is made to utterance tokens on the basis of the type of utterance they appear to manifest. For instance, (11) has a reading which means roughly *which individual named Bo were you referring to in the previous utterance*, NOT *who does Bo refer to in general*.<sup>10</sup>

(11) A: Did Bo leave? B: Bo?

The types and tokens corresponding to an utterance and its parts get integrated in an information state.

Previous approaches to formalizing (HPSG) grammars were, with one exception I am aware of, formulated either in terms of tokens OR, more recently, types. Thus, originally HPSG viewed a grammar as something emergent from the tokens of NL—Pollard and Sag (1987) suggested thinking about feature structures as ‘partial descriptions of signs (or sign tokens) and other linguistic objects which occur as part of signs’.<sup>11</sup> This was violently repudiated in Pollard and Sag (1994): ‘One thing that [language] certainly does not consist of is individual linguistic events or utterance tokens, for knowledge of these is not what is shared among the members of a linguistic community.’<sup>12</sup> Paul King, on the other hand, in his formalizations of HPSG (see e.g. (King 1989, 1996), does view a grammar as a characterization of the class of well-formed utterance *tokens*. In King (1996), he identifies a token as a pair  $\langle u, I \rangle$  of an entity  $u$  and an interpretation  $I$  (in a technical sense that King develops). From this, he constructs types as equivalence classes of indiscernable tokens. Whether King’s theory could serve to underpin linguistic description of metacommunicative utterances is not straightforward to discern since it is not clear that his formalism allows for types as first class entities.

## 5.3 A TTR-Based Fragment

### 5.3.1 Preliminaries

As mentioned previously, the approach taken here is to view linguistic ontology as an instance of the general ontology. What this means is that utterance events, like other events, are a kind of record, whereas lexical entries and phrasal rules are explicated as types. The types, modulo certain refinements on the semantic/contextual front, are for concreteness’ sake identical to (or simplified versions of) the types emanating from Ginzburg and Sag (2000). I present the types in a very underspecified form,

<sup>10</sup> This is not to say that CRs cannot address type information as well, particularly in interaction between linguistic unequals (parents/children, native/local).

<sup>11</sup> Pollard and Sag (1987 : 28).

<sup>12</sup> Pollard and Sag (1994 : 14).



omitting most details apart from those directly relevant to building up the context and content.<sup>13</sup>

All words and phrases are *signs*, i.e. they are of type (12). In other words, the field PHON has as its type a list from elements of the type *phonform*; the field CAT is a record type with a field HEAD whose values are taken from the type P(art)o(f)Sp(eech) (e.g. N(oun), V(erb), D(eterminer), P(reposition));<sup>14</sup> the field CONT has as its type the basic type SemObj—the range of possible denotations; the field GAPS is used to encode long-distance dependencies, and the fields DGB-PARAMS and QUEST-DOM, whose use is explained below, have as their type the type RecType. Non-referential elements such as whPs and quantificational NPs are handled, respectively, via QUEST-DOM and the field Q-PARAMS introduced in section 8.5 of Chapter 8.<sup>15</sup> It is common in many versions of HPSG to posit a type *synsem* which consists, essentially, of the fields of the sign, apart from PHON, in other words as in (12b). The type *synsem* also has its own intricate feature geometry, intended to explicate *inter alia* which information is relevant for complement selection. For reasons of notational simplicity, I will abstract away from such details, though this omission has no theoretical import.

(12) a.  $\text{Sign} =_{\text{def}}$

$$\left[ \begin{array}{ll} \text{phon} & : \text{list}(\text{phonform}) \\ \text{cat} & : [\text{head} : \text{PoS}] \\ \text{gaps} & : \text{list} \left( \begin{array}{ll} \text{cat} & : [\text{head} : \text{PoS}] \\ \text{cont} & : [\text{x} : \text{Ind}] \end{array} \right) \\ \text{dgb-params} & : \text{RecType} \\ \text{cont} & : \text{SemObj} \\ \text{quest-dom} & : \text{list}(\text{RecType}) \end{array} \right]$$

b.  $\text{Sign} = \text{SynSem} \wedge_{\text{merge}} [\text{phon} : \text{list}(\text{phonform})]$

The DGB represents publicized context for us, albeit an individually relativized version thereof. It interfaces with grammar via the field DGB-PARAMS.<sup>16</sup> This field represents the constraints on the DGB associated with a given utterance type. The minimal

<sup>13</sup> It is, nonetheless, relatively straightforward to incorporate the requisite syntactic information in the types. For some exemplification see Cooper (2008).

<sup>14</sup> Notation such as  $V[+\text{fin}]$  is shorthand for the value  $\begin{bmatrix} \text{head} & = & V \\ \text{fin} & = & + \end{bmatrix}$ . Hence,  $\text{cat.head} = V[+\text{fin}]$  :

POS is an alternative notation for  $\text{cat.head} : \text{POS}_{V[+\text{fin}]}$ , where  $\text{POS}_{V[+\text{fin}]}$  is the subtype of POS whose single member is  $V[+\text{fin}]$ . In such cases we will omit the rest of the cat specification—on grounds of expository simplification.

<sup>15</sup> The fragment developed here does not handle scope ambiguities due to sentence embedding. Hence, there is no STORE-like mechanism or alternative means for emulating scope underspecification.

<sup>16</sup> This way of thinking of contextual parameters is inspired by Purver (2004a).

possible constraint involves [DGB-PARAMS : [ ]]. In general, contextual information will be of one of the following types:

- (13) a. **Indexical parameters:** the DGB-PARAMS of a root clause, discussed in section 5.5, will introduce speaker, addressee, utterance time parameters. A witness for such a type will be a sub-record of the current DGB, specifically the fields in the DGB that pertain to the turn holder, addressee, and utterance time.
- b. **Presuppositions:** content elements that involve presuppositions to be grounded in FACTS, for instance, naming restrictions, will have the form  $\left[ \begin{array}{l} x : \text{Ind} \\ \text{rest}^{\text{FACTS}} : p(x) \end{array} \right]$ . A witness for such a type has the form  $\left[ \begin{array}{l} x = a \\ \text{rest}^{\text{FACTS}} = p0 \end{array} \right]$ , with the constraints  $a : \text{Ind}$ ,  $p0 : p(a)$ , with the latter to be construed as the requirement that the proposition  $\left[ \begin{array}{l} \text{sit} = p0 \\ \text{sit-type} = p(a) \end{array} \right]$  be an element of the current DGB.FACTS
- c. **Direct reference to DGB elements:** the grammar can refer to elements of the current DGB. Thus, when analysing NSUs a frequently utilized parameter will be the maximal element of QUD, MaxQuD, as will be the maximal element of PENDING, MaxPending. DGB-PARAMS types specified as  $\left[ \text{max-qud} : \text{Question} \right]$  are shorthand for  $\left[ \begin{array}{l} q : \text{Question} \\ \text{dgb} : \text{DGBType} \\ \text{rest} : \text{Maximal}(q, \text{dgb.qud}) \end{array} \right]$ . A witness for such a type provides the MaxQuD of the current DGB. Similarly, *mutatis mutandis* for reference to other DGB elements.

### 5.3.2 A Simple Lexicon

For our initial illustrative purposes, I include solely two kinds of NPs—proper names and *wh*-phrases. Quantificational NPs and pronouns, whose contextual specification is quite intricate, are discussed in Chapter 8.

Proper names are handled as in (14)—this provides a lexical entry for the name ‘Jo’. The value for CAT is self-explanatory. Of more interest are the phon, DGB-PARAMS, and CONT fields. A proper name generally varies with context (and is potentially ambiguous or unknown to the addressee). Hence, it projects a contextual parameter, which restricts this potential referent to bear the name which is the type of the phon field. Not that the identity is with the type, NOT the value, illustrating the need for a formalism that can accommodate such dependencies and have types as arguments of predicates. For this reason the type itself is introduced as a field in the sign, to enable

it to appear as a value subsequently in the record type. This introduction of types as fields will recur when we need to use the types as values. The content is typed, as will be the case with all NPs, with the record type for an individual entity. For a referential NP, this is restricted to be identical with the contextual parameter.

$$(14) \left[ \begin{array}{ll} \text{jo} & : \text{Phontype} \\ \text{phon} & : \text{jo} \\ \text{cat.head} = \text{N} & : \text{PoS} \\ \text{cont} & : \left[ \text{x} : \text{IND} \right] \\ \text{dgb-params} & : \left[ \begin{array}{ll} \text{y} = \text{cont.x} & : \text{IND} \\ \text{rest}^{\text{facts}} & : \text{Named}(\text{y}, \text{jo}) \end{array} \right] \end{array} \right]$$

An argument-filling *wh*-phrase like *who* gets assigned a minimally different lexical entry, as in (15). It has a content type similar to a proper name, but does not project a contextual parameter. The fact that the argument role it is associated with will get abstracted over, i.e. will specify the domain of a question, is captured by assigning it a non-trivial value for the field  $\text{QUEST}(\text{ION})\text{-DOM}(\text{AIN})$ .<sup>17</sup> The sole member of the list constitutes the domain (or a supertype thereof) of the function denoted by the interrogative clause in which ‘*who*’ is a constituent.  $\text{QUEST-DOM}$  is list-typed rather than a straight record type because we need to accommodate cases where a single sentence contains multiple occurrences of *wh*Ps, which end up belonging to distinct domains—this occurs most prototypically when one *wh*P is used directly and the other as a reprise, as discussed in Chapters 6 and 7.

$$(15) \left[ \begin{array}{ll} \text{phon} & : \text{who} \\ \text{cat.head} = \text{N} & : \text{PoS} \\ \text{cont} & : \left[ \text{x} : \text{IND} \right] \\ \text{quest-dom} = \left\langle \left[ \begin{array}{ll} \text{y} = \text{cont.x} & : \text{IND} \\ \text{rest1} & : \text{person}(\text{index}) \end{array} \right] \right\rangle & : \text{list}(\text{RecType}) \end{array} \right]$$

The lexical entries for verbs here are straightforward—essentially TTR counterparts to (lower-order) Montogovian contents. I introduce a number of Montague-style abbreviations for the types of verbal contents, as in (16a–b). Also, to avoid overly long path names, I introduce the abbreviations in (16d, e):

<sup>17</sup> Previously known as *STORE*. This latter name, whose origin is steeped in assembly language terminology, obscures the semantic role this information plays.

- (16) a.  $IV =_{def} ([subj.cont.x : Ind])RecType$   
 b.  $TV =_{def} ([comp.cont.x : Ind])([subj.cont.x : Ind])RecType$   
 c.  $x^s =_{def} subj.cont.x$   
 d.  $x^c =_{def} comp.cont.x$

The symbol *runs* is specified as a function that maps a record assigning a value to its subject to a type that holds of (an event) if that individual runs.

- (17) 
$$\left[ \begin{array}{ll} \text{phon} & : \text{runs} \\ \text{cat} & : \left[ \begin{array}{ll} \text{head} = V & : \text{PoS} \\ \text{subj} & : \langle \text{synsem} \rangle \end{array} \right] \\ \text{cont} = (r : [x^s : Ind]) [c : \text{run}(r.x^s)] & : IV \end{array} \right]$$

The symbol *visits* is similar *mutatis mutandis*: I include here two variants, one for its ‘canonical’ use, the other for its use in object extraction cases, where no object is projected, but rather this argument is associated with the GAPS valency field.

- (18) a. 
$$\left[ \begin{array}{l} \text{phon} : \text{visit} \\ \text{cat} : \left[ \begin{array}{ll} \text{head} = V & : \text{PoS} \\ \text{subj} & : \langle \text{synsem} \rangle \\ \text{comps} & : \langle \text{synsem} \rangle \end{array} \right] \\ \text{cont} = (r1 : [x^c : Ind]) (r2 : [x^s : Ind]) [c : \text{visit}(r2.x^s, r1.x^c)] : TV \end{array} \right]$$

- b. The gaps version to be used in object extraction cases:<sup>18</sup>

$$\left[ \begin{array}{l} \text{phon} : \text{visit} \\ \text{cat} : [\text{head} = V : \text{PoS}] \\ \text{gaps} : \text{nelist} \left( \left[ \begin{array}{ll} \text{cat} : [\text{head} : \text{PoS}] \\ \text{cont} : [x : \text{Ind}] \end{array} \right] \right) \\ \text{cont} = (r : [x^s : Ind]) [c : \text{visit}(r.x, \text{gaps.cont}.x)] : IV \end{array} \right]$$

<sup>18</sup> In most cases dealt with here when there is reference to non-empty GAPS, it can be assumed to be a list of length one. Consequently, I will write e.g. *gaps.cont*, as opposed to the more correct *gaps.1.cont*, where *gaps.1* denotes the first element of the list.

The symbol *did* is analysed as denoting an identity function that maps denotations of type *RecType* into denotations of type *RecType*:

$$(19) \quad \left[ \begin{array}{ll} \text{phon} & : \text{did} \\ \text{cat} & : \left[ \begin{array}{ll} \text{head} = V & : \text{PoS} \\ \text{subj} & : \langle \text{synsem} \rangle \end{array} \right] \\ \text{cont} = \text{ID}_{(\text{RecType})\text{RecType}} & : (\text{RecType}) \text{RecType} \end{array} \right]$$

### 5.3.3 A Simple Grammar

I now turn to illustrate a number of phrasal types, including a basic head complement rule, a declarative rule, a polar interrogative rule, and a *wh*-interrogative rule. I offer one relatively detailed specification and then proceed highly schematically.

Equation (20) is a specification for the type *hd-comp-ph*, which combines a lexical head with a complement. This specification is different from the lexical ones we saw above in that a field for *DTRS*, the field by which immediate dominance is usually represented in HPSG, is provided. Equation (20) is simplified in a number of respects—most importantly, the standard specification allows for *n* complements, whereas here only one is provided for.<sup>19</sup> Nonetheless, it illustrates the basic aspects of phrasal composition appealed to—usually implicitly—subsequently:

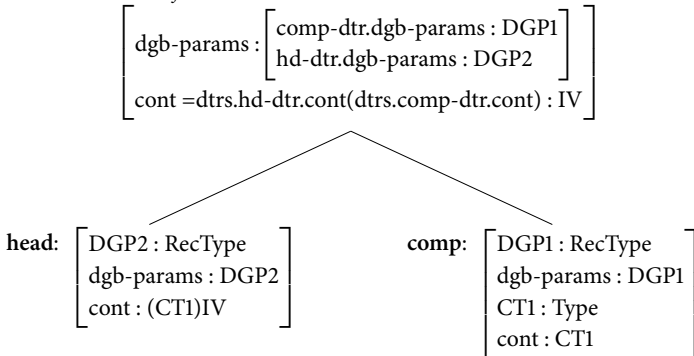
- **syntax:** information ‘percolates’ from mother to head daughter, using dependencies between labels, cashed out using singleton types. Thus, the mother’s *CAT.HEAD* is identified with the head daughter’s value for this field, as are their values for the valence field *CAT.SUBJECT*. In contrast, whereas the mother’s value for the valence field *COMPS* is empty, the head daughter’s value for this is a list consisting of the complement daughter’s value for the field *synsem*.
- **dgb-params:** the *DGB-PARAMS* value of the mother is a record type which combines—each via their own label—the *DGB-PARAMS* types of the daughters.
- **content:** the content of the phrase arises by functional application—the head daughter’s content taking its complement’s content as its argument.

<sup>19</sup> This simplification could be eliminated by defining an *n*-ary composition operation.

(20)  $hd-comp-ph =_{def}$ 

$$\left[ \begin{array}{l}
 \left[ \begin{array}{l}
 \left[ \begin{array}{l}
 CT1 : \text{Type} \\
 DGP1 : \text{RecType}
 \end{array} \right] \\
 \text{comp-dtr} : \left[ \begin{array}{l}
 \text{cat} : \text{syncat} \\
 \text{synsem} : \left[ \begin{array}{l}
 \text{dgb-params} : DGP1 \\
 \text{cont} : CT1
 \end{array} \right]
 \end{array} \right] \\
 \text{dtrs} : \left[ \begin{array}{l}
 \left[ \begin{array}{l}
 \text{head} : \text{PoS} \\
 \text{cat} : \left[ \begin{array}{l}
 \text{comps} = \langle \text{comp-dtr.synsem} \rangle : \text{list}(\text{synsem}) \\
 \text{subj} : \text{list}(\text{synsem})
 \end{array} \right] \\
 DGP2 : \text{RecType} \\
 \text{dgb-params} : DGP2 \\
 \text{cont} : (CT1)IV
 \end{array} \right] \wedge \text{word} \\
 \text{hd-dtr} : \left[ \begin{array}{l}
 \text{head} = \text{hd.dtr.head} : \text{PoS} \\
 \text{cat} : \left[ \begin{array}{l}
 \text{comps} = \langle \rangle : \text{list}(\text{synsem}) \\
 \text{subj} = \text{hd.dtr.subj} : \text{list}(\text{synsem})
 \end{array} \right] \\
 \text{dgb-params} : \left[ \begin{array}{l}
 \text{comp-dtr.dgb-params} : DGP1 \\
 \text{hd-dtr.dgb-params} : DGP2
 \end{array} \right] \\
 \text{cont} = \text{dtrs.hd-dtr.cont}(\text{dtrs.comp-dtr.cont}) : IV
 \end{array} \right]
 \end{array} \right]
 \end{array} \right]
 \end{array}$$

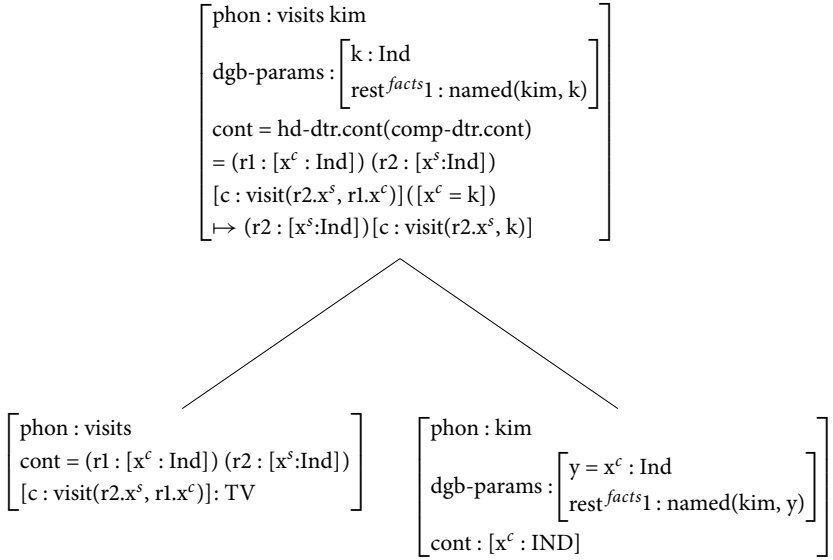
Subsequent phrasal specifications will omit most of the information made explicit in (20), in particular omitting the lion's share of syntactic information. In addition, the phrasal constituency will be displayed in graphical form using tree notation. The pared-down specification is illustrated in (21):

(21)  $hd-comp-ph =_{def}$ 

As an example consider (22), where I use the following notational convention: daughters are identified by the value of their phonology and semantics, and their dgb-params if this is non-empty. The content arises by applying the function denoted by the hd-dtr to the cont field of the comp-dtr, a record of the form  $[x^c = k]$ . Since the

head daughter introduces no DGB-PARAMS information, then via path simplification and label renaming, exemplified in (23), the mother's DGB-PARAMS is equivalent to the complement daughter's.<sup>20</sup>

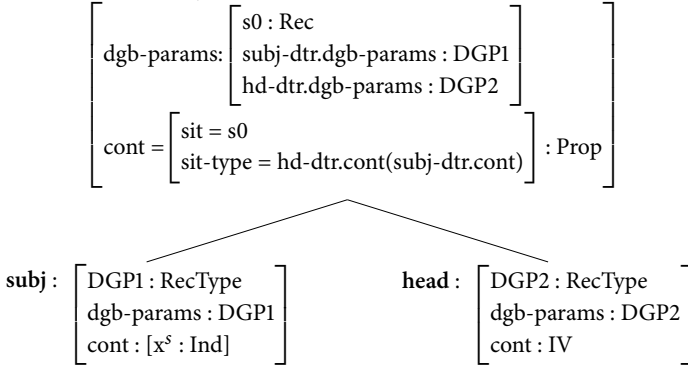
(22) visits Kim



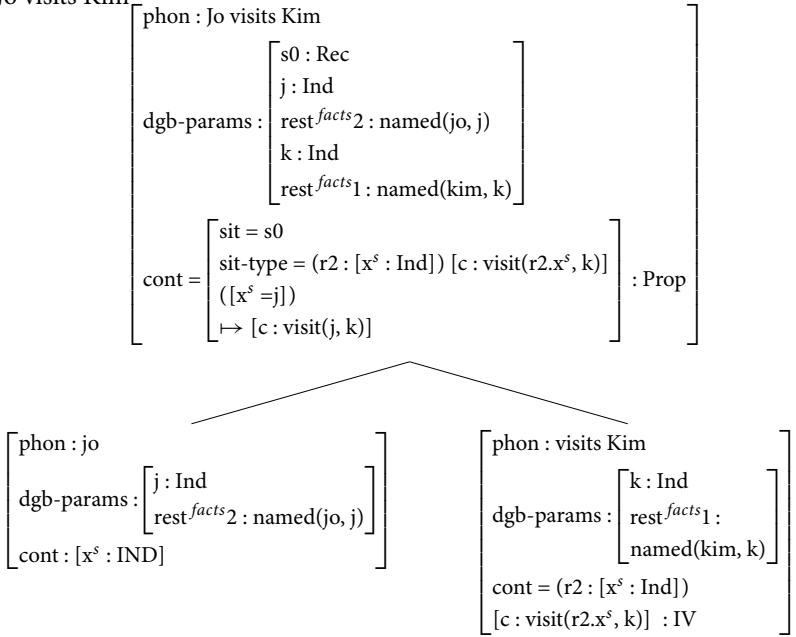
- (23) dgb-params :  $\left[ \begin{array}{l} \text{comp-dtr.dgb-params : } \left[ \begin{array}{l} y = x^c : \text{Ind} \\ \text{rest}^{facts}1 : \text{named}(\text{kim}, y) \end{array} \right] \\ \text{head-dtr.dgb-params : } [ ] \end{array} \right]$
- $\mapsto$  dgb-params :  $\left[ \begin{array}{l} \text{comp-dtr.dgb-params.y} = x^c : \text{Ind} \\ \text{rest}^{facts}1 : \text{named}(\text{kim}, \text{comp-dtr.dgb-params.y}) \end{array} \right]$
- $\mapsto$  (renaming comp-dtr.dgb-params.y to z)
- dgb-params :  $\left[ \begin{array}{l} z = x^c : \text{Ind} \\ \text{rest}^{facts}1 : \text{named}(\text{kim}, z) \end{array} \right]$
- $\mapsto$  (since the value of  $x^c = k$ ):
- dgb-params :  $\left[ \begin{array}{l} k : \text{Ind} \\ \text{rest}^{facts}1 : \text{named}(\text{kim}, k) \end{array} \right]$

The type *decl-hd-subj-cl* in (24a) is the analogue of the standard 'S  $\rightarrow$  NP VP' rule. It builds a proposition whose situation field is contextually provided and whose sit-type field arises by applying the head-dtr's content to the subj-dtr's content. An example is provided in (24b): the DGB-PARAMS amalgamation involves a reduction entirely analogous to (23).

<sup>20</sup> For detailed discussion of such path simplification and renaming operations see Cooper (2011).

(24) a. *decl-hd-subj-cl* =<sub>def</sub>

b. Jo visits Kim

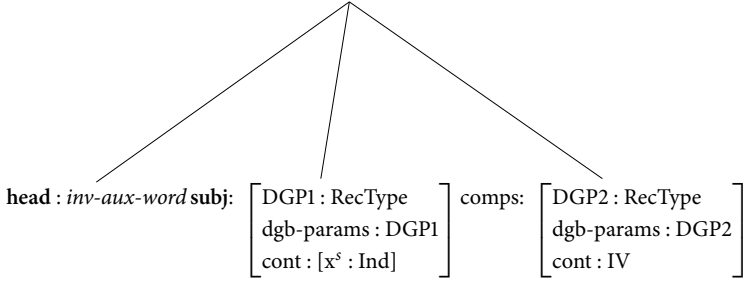


Let us consider two subject-auxiliary rules. The first is a phrasal type for polar interrogatives. Its content is a 0-ary abstract, abstracted from a proposition, whose situation field is contextually provided and whose sit-type field arises by applying the head-dtr's content to the type formed by applying the complement-dtr to the subj-dtr's content. DGB-PARAMS amalgamation is akin to (24a).



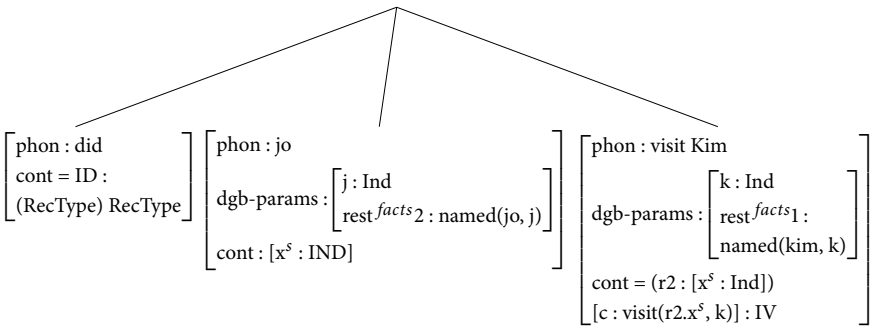
(25) a. *pol-int-cl* =<sub>def</sub>

$$\left[ \begin{array}{l} \text{dgb-params:} \left[ \begin{array}{l} s0 : \text{Rec} \\ \text{subj-dtr.dgb-params} : \text{DGP1} \\ \text{comp-dtr.dgb-params} : \text{DGP2} \end{array} \right] \\ \text{cont} = ? \left[ \begin{array}{l} \text{sit} = s0 \\ \text{sit-type} = \text{hd-dtr.cont}(\text{comp-dtr.cont}(\text{subj-dtr.cont})) \end{array} \right] : \text{Question} \end{array} \right]$$



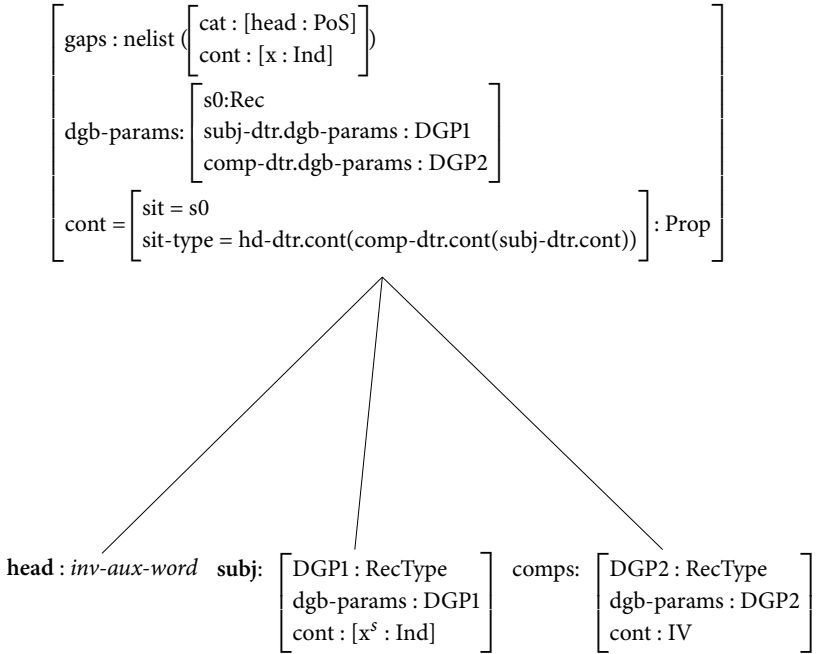
b. Did Jo visit Kim?

$$\left[ \begin{array}{l} \text{phon} : \text{did Jo visit Kim} \\ \text{dgb-params:} \left[ \begin{array}{l} s0 : \text{Rec} \\ j : \text{Ind} \\ \text{rest}^{facts2} : \text{named}(jo, j) \\ k : \text{Ind} \\ \text{rest}^{facts1} : \text{named}(kim, k) \end{array} \right] \\ \text{cont} = ? \left[ \begin{array}{l} \text{sit} = s0 \\ \text{sit-type} = \text{ID}_{(\text{RecType})\text{RecType}} \\ (r2 : [x^s : \text{Ind}]) [c : \text{visit}(r2.x^s, k)] \\ ([x^s = j]) \\ \mapsto [c : \text{visit}(j, k)] \end{array} \right] : \text{Question} \end{array} \right]$$

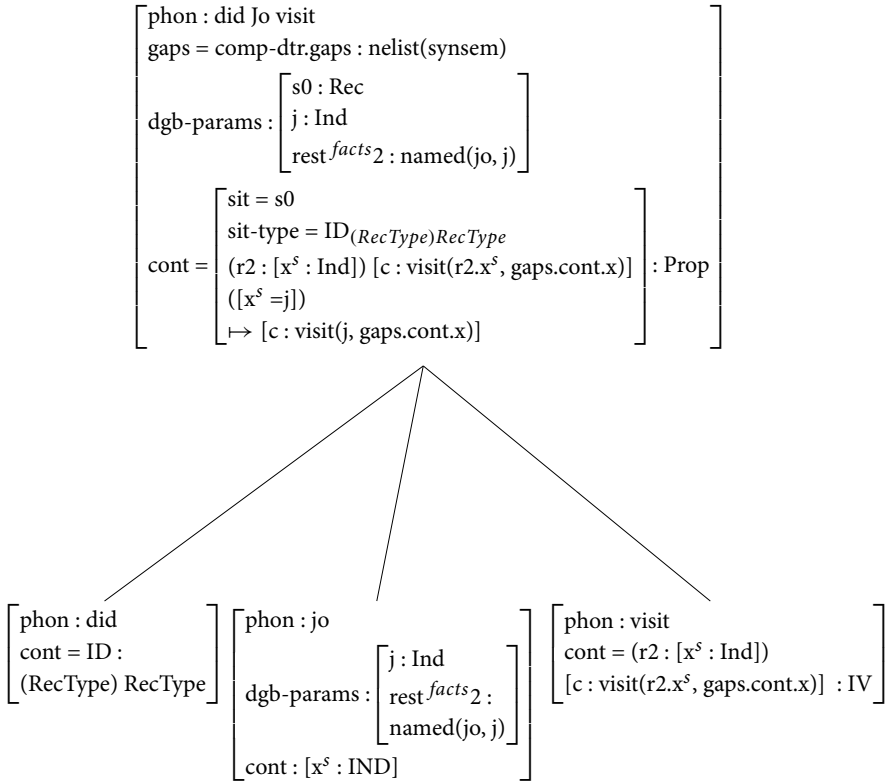


A second subject-auxiliary phrasal type we will need in building filler-gap dependencies differs minimally from *pol-int-cl*. Syntactically, it involves a non-empty GAPS value. Its content is a proposition, whose situation field is contextually provided and whose sit-type field arises by applying the head-dtr's content to the type formed by applying the complement-dtr to the subj-dtr's content:

(26) a. *decl-inv-cl* =<sub>def</sub>

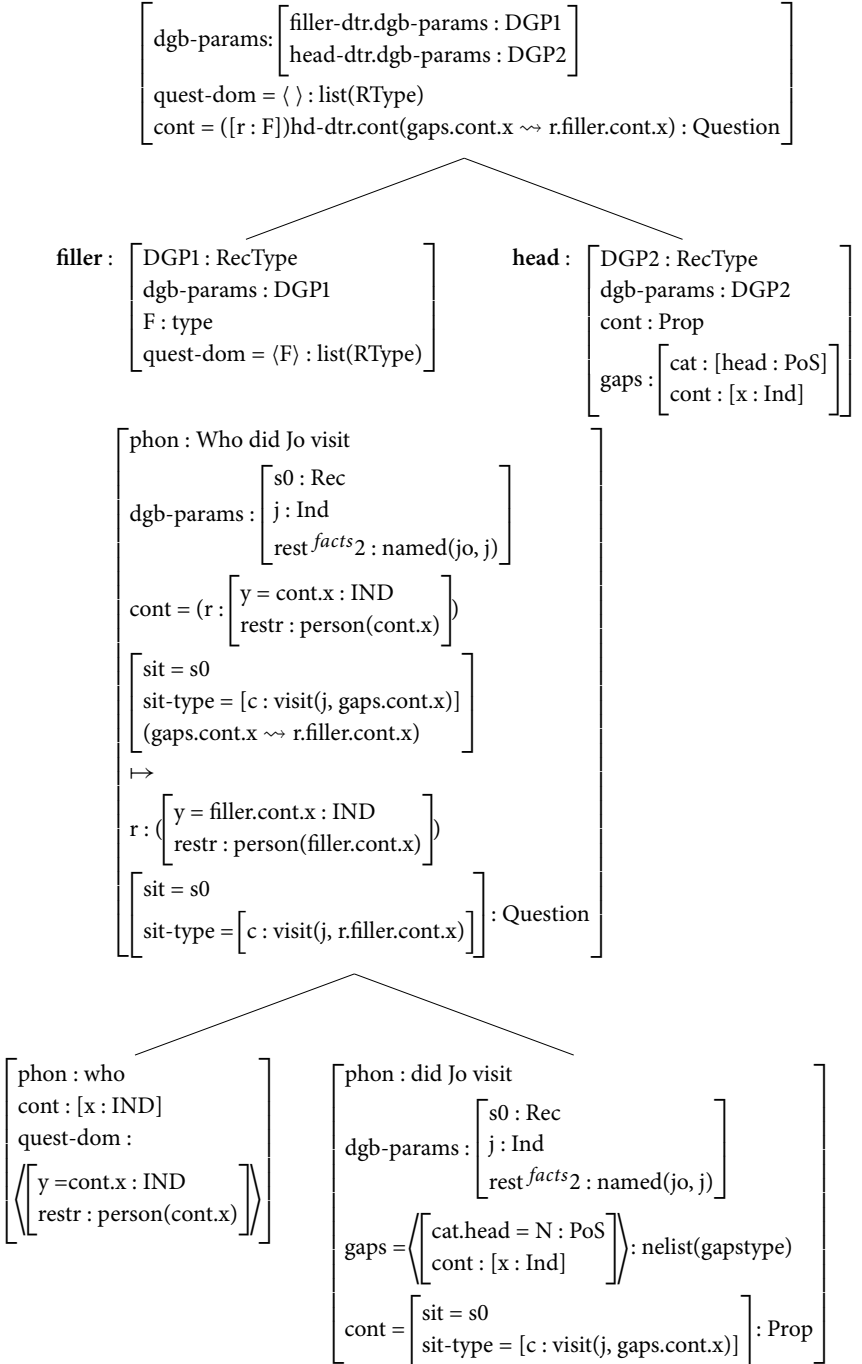


## b. Did Jo visit ?



Finally we turn to two interrogative types. The first is for a ‘canonical’ *wh*-interrogative that involves a filler gap construction, the filler being a dislocated *wh*-phrase. Here the content arises by forming a function, call it  $q$ . The domain of  $q$  is the type that constitutes the filler’s quest-dom value. The range of  $q$  is the proposition that constitutes the content of the hd-dtr, but where the index corresponding to gaps.cont.x is substituted with the index of the filler’s content.<sup>21</sup> DGB-PARAMS amalgamation is similar to previous phrasal rules.

<sup>21</sup> This constraint is simplified in a number of ways. On the semantic front the simplifications include two in particular: first, the constraint only allows for unary *wh*-questions; second, and relatedly, the clause’s quest-dom is stipulated to be empty. For greater generality, with some attendant complexity, see Ginzburg and Sag (2000).

(27) a. *wh-ns-int-cl* =<sub>def</sub>

**Exercise 5.3.1** Refine (27) so that it can handle multiple *wh*-interrogatives.

The last type I mention is a type that underlies non-reprise *in situ wh*-interrogatives such as (28a), intonation queries such as (28b), and various NSUs we will encounter in Chapter 7:

- (28) a. A: Mary's bringing a moussaka. B: And you're bringing what?  
b. Bo runs?

(28b) can be constructed using the type in (29a), as exemplified in (29b): the type involves building a polar question from a proposition; DGB-PARAMS is identical to that of the head daughter.

$$(29) \quad a. \text{polarization} =_{\text{def}} \left[ \begin{array}{l} \text{dgb-params} : \text{DGPI} \\ \text{cont} = ?[\text{hd-dtr.cont}] : \text{Question} \end{array} \right]$$

$$\begin{array}{c} | \\ \text{hd-dtr} : \left[ \begin{array}{l} \text{DGPI} : \text{RecType} \\ \text{dgb-params} : \text{DGPI} \\ \text{cont} : \text{Prop} \\ \text{quest-dom} = \langle \rangle : \text{RType} \end{array} \right] \end{array}$$

b. Bo runs?

$$\left[ \begin{array}{l} \text{phon} : \text{Bo runs} \\ \text{dgb-params} : \left[ \begin{array}{l} \text{s0} : \text{Rec} \\ \text{b} : \text{Ind} \\ \text{rest}^{facts_1} : \text{named}(\text{bo}, \text{b}) \end{array} \right] \\ \text{cont} = ? \left[ \begin{array}{l} \text{sit} = \text{s0} \\ \text{sit-type} = [\text{c} : \text{run}(\text{b})] \end{array} \right] : \text{Question} \end{array} \right]$$

$$\begin{array}{c} | \\ \text{hd-dtr} : \left[ \begin{array}{l} \left[ \begin{array}{l} \text{s0} : \text{Rec} \\ \text{b} : \text{Ind} \\ \text{rest}^{facts_1} : \text{named}(\text{bo}, \text{b}) \end{array} \right] \\ \text{cont} = \left[ \begin{array}{l} \text{sit} = \text{s0} \\ \text{sit-type} = [\text{c} : \text{run}(\text{b})] \end{array} \right] : \text{Prop} \\ \text{quest-dom} = \langle \rangle : \text{RType} \end{array} \right] \end{array}$$

(28a) can be constructed using the type in (30a), as exemplified in (30b):

(30) a. *unary-dir-is-cl* =<sub>def</sub>

$$\left[ \begin{array}{l} \text{dgb-params} : \text{DGPI} \\ F : \text{RecType} \\ G = [x : \text{Ind}] \wedge_{\text{merge}} F \\ \text{cont} = (r : G) \text{hd-dtr.} \text{cont}(\text{quest-dom.l.x} \rightsquigarrow \text{r.x}) : \text{Question} \end{array} \right]$$

$$\text{hd-dtr:} \left[ \left[ \begin{array}{l} \text{DGPI} : \text{RecType} \\ \text{dgb-params} : \text{DGPI} \\ \text{cont:Prop} \\ \text{quest-dom} : \langle G \rangle \end{array} \right] \right]$$

b. Bo phoned who?

$$\left[ \begin{array}{l} \text{phon} : \text{Bo phoned who} \\ \text{dgb-params} : \left[ \begin{array}{l} s0 : \text{Rec} \\ b : \text{Ind} \\ \text{rest}^{facts_1} : \text{named}(\text{bo}, b) \end{array} \right] \\ \text{cont} = (r : \left[ \begin{array}{l} x : \text{IND} \\ \text{restr: person}(x) \end{array} \right]) \left[ \begin{array}{l} \text{sit} = s0 \\ \text{sit-type} = [c : \text{phone}(b, \text{r.x})] \end{array} \right] : \text{Question} \end{array} \right]$$

$$\text{hd-dtr:} \left[ \begin{array}{l} \text{dgb-params} : \left[ \begin{array}{l} s0 : \text{Rec} \\ b : \text{Ind} \\ \text{rest}^{facts_1} : \text{named}(\text{bo}, b) \end{array} \right] \\ \text{cont} = \left[ \begin{array}{l} \text{sit} = s0 \\ \text{sit-type} = [c : \text{phone}(b, x)] \end{array} \right] : \text{Prop} \\ \text{quest-dom} : \left\langle \begin{array}{l} [x : \text{IND}] \\ \text{restr: person}(x) \end{array} \right\rangle \end{array} \right]$$

The types in (29a) and (30a) can be generalized into a single type, as they were in Ginzburg and Sag (2000); for an example of such a general schema in an analogous case see (87) in section 6.6.3.

## 5.4 Keeping Track of Constituents

I need to mention one relatively minor enhancement we need to make to the standard sign-based ontology. This enhancement plays a key role in capturing cross-utterance parallelism, agreement, and scope and anaphoric antecedency, though its primary motivation is to enable us to describe certain rules that regulate grounding and clarification interaction.

Various grammatical formalisms employ means of encoding *immediate* phrasal constituency in utterance types. In HPSG this information is handled in terms of the feature *DTRS*. Although the value of *DTRS* is limited to immediate constituents, by recursion we eventually have access to more remote constituents. Concretely, a token of a sentential type will have values for all sub-utterances in paths emanating from the highest level *DTR*. Access to these sub-utterance tokens is a key desideratum of various conversational rules, above all Clarification Context Update Rules (CCURs): given that a contextual parameter emanating from a deeply embedding constituent is as clarifiable as an immediate constituent—recall that in intended content readings—(see section 2.2 for initial discussion of these)—the to-be-clarified sub-utterance  $u_i$  is actually a component of the clarification question ‘what did you mean by uttering  $u_i$ ?’ Here we can adopt one of two possible strategies: the first is to enable access to non-immediate constituents by expressive means that in some way effect a recursion, an approach implemented by Gregory and Lappin (1999).<sup>22</sup> The other strategy, one I adopt here is to enhance the representation itself so it keeps track of *all* constituents, not merely the immediate ones. This is done by positing an additional, set-valued field in the type definition of signs dubbed *CONSTIT(UE)NTS*, as in Ginzburg and Cooper (2004). This leads to two notational decisions. First, *CONSTITS* values are depicted using the words that make them up. Second, *DTRS* is omitted from utterance type representations (except for cases where we state general constraints).<sup>23</sup> Given this, a typical utterance type will have the form in (31):

<sup>22</sup> As part of a search procedure utilized by a post-parsing module which effects ellipsis resolution, Gregory and Lappin define a procedure whose essence is to calculate all constituents, direct or otherwise, for a given sign. Given that signs represent information about utterances which constitute part of a conversational participant’s linguistic competence, then in so far as information about (not solely immediate) constituents is manipulated as part of this competence, it seems preferable to encode this information in the utterance representations, rather than appeal to a post-parsing module whose theoretical status is unclear.

<sup>23</sup> In fact, having posited *CONSTITS* one could eliminate *DTRS*: this by making the value of *CONSTITS* be a set of sets whose first-level elements are the immediate constituents. Such a move could have interesting theoretical consequences for the grammar, though I am not going to explore these here.

- (31) 
$$\left[ \begin{array}{ll} \text{phon} & : \text{is georges here} \\ \text{cat} = V[+\text{fin}] & : \text{PoS} \\ \text{constits} = \{\text{is, georges, here, is georges here}\} : \text{set}(\text{sign}) \\ \text{dgb-params} : \left[ \begin{array}{ll} \text{spkr} & : \text{IND} \\ \text{addr} & : \text{IND} \\ \text{c1} & : \text{address}(s, a) \\ \text{s0} & : \text{SIT} \\ \text{l} & : \text{LOC} \\ \text{g} & : \text{IND} \\ \text{c3} & : \text{Named}(g, \text{'georges'}) \end{array} \right] \\ \text{cont} = \text{Ask}(\text{spkr}, \text{addr}, ? \left[ \begin{array}{l} \text{sit} = \text{s0} \\ \text{sit-type} = \text{In}(l, g) \end{array} \right]) & : \text{IllocProp} \end{array} \right]$$

The value of CONSTITS is specified by the type in (32), which all phrasal types are assumed to satisfy; this specification is analogous, if not identical, to the HPSG *Non-local Amalgamation Constraint*.<sup>24</sup>

- (32) CONSTITS amalgamation:
- $$\left[ \begin{array}{l} \text{dtrs} = d0 : \text{list}(\text{sign}) \\ \text{constits} : \text{set}(\text{sign}) \\ \text{c1} : \forall r (\text{member}(r, \text{constits}) \leftrightarrow \text{member}(r, d0) \vee \text{Constitsmember}(r, d0)) \end{array} \right]$$

*Constitsmember*( $r, d0$ ) holds iff  $r$  is a member of one of  $d0$ 's constits' sets. In other words: if there exist  $c, d$  such that  $\text{member}(d, d0)$ ,  $c = d.\text{constits}$ , and  $\text{member}(r, c)$ .

## 5.5 Root Utterances

### 5.5.1 Integrating Move Types into Grammatical Analysis

Traditionally, grammars have been written in a way that abstracts away from *use*. 'S's were given denotations as propositions and that, more or less, was that. If the grammar is intended for a text, the implicit assumption was that, ultimately, the force of the text is an assertion by the author of the conjunction of all sentences. A similar sort of simplifying assumption could be taken to apply to a sequence of sentences construed as a monologue.

This strategy has been abandoned for text processing in work that assumes sentences to be related by *rhetorical relations*, perhaps most explicitly in SDRT (Asher and Lascarides 2003). This strategy is even less tenable for dialogue. Categorizing utterances in terms of a notion of illocutionary force or conversational move type (CMT) is common in corpus-based work (for some recently proposed CMT taxonomies,

<sup>24</sup> Signs of type *word* are assumed to have the empty set as value for CONSTITS.



see Carletta et al. 1996, Core and Allen 1997). Indeed any analysis of dialogue that incorporates the insights of speech act theory presupposes that an utterance ultimately gets associated with a CMT. Nonetheless, there exist few attempts to integrate such notions into contemporary formal grammatical work. In part, this is due to the fact that most grammatical formalisms to date have been designed with monologue or text in mind, where this issue is easier to put aside than in conversational settings. A more principled reason for this lacuna is perhaps the phenomenon of *multifunctionality* (see e.g. Allwood 1978): it is often the case that a given utterance serves more than one purpose—an assertion can function also as an offer, a query as a suggestion etc. This has often led to the feeling that issues pertaining to CMT belong entirely to the realm of pragmatics. Although no worked-out pragmatic theory as to how CMTs get assigned to utterances has emerged to date, the one influential series of attempts to subsume CMT into the grammar, based on the Performative Hypothesis (PH), is generally viewed to have been a resounding failure (see Levinson 1983: 247–63).<sup>25</sup>

Here I argue that CMT can and should be integrated in the semantic analyses provided by the grammar. That is, CMT is a parameter of meaning conventionally associated with certain words and classes of phrases. For instance, in hearing an utterance by A of a sentence such as (33a), I claim that a competent interlocutor B knows that its meaning is the template schematically given as (33b), *not* simply the proposition (33c). That is, B knows that in order to ground A's utterance she must try to instantiate the parameters *A*, *t*, *l*, *P* within the template given in (33b) in such a way as to satisfy the constraints provided by the grammar (e.g. *A* must be the speaker, *t* must be a time the day after utterance time, *P* ranges over a set that includes {assert, threaten, promise, . . .}, but not over, for instance, {ask, exclaim, apologize, . . .}).<sup>26</sup>

<sup>25</sup> There has been some recent work reviving the PH in terms of *functional projections*, see e.g. Speas and Tenny (2003). Speas and Tenny set out to account for the 'particular and tiny inventory of grammaticized speech acts [arguing] that they are constrained by the operation of familiar syntactic processes and principles on syntactic configurational structure. The four types of grammaticized speech acts [assertion, querying, imperative, and subjunctive—J.G.] exhaust the possible operations on the structure of the Speech Act Phrase' (Speas and Tenny, p. 336). I reserve judgement here on whether these forces really exhaust the causally available forces—indeed, Speas and Tenny do not discuss moves such as greeting, parting, thanking, and apologies discussed below in the body of the text.

Ross was quite explicit in making claims for synonymy (deep structural identity) between performatives and sentence types lacking explicit illocutionary force indicators, claims thoroughly refuted in past literature. Speas and Tenny (2003) avoid any such claims, and thereby also the problematic consequence of recursion of illocutionary force predicates. They evade this problem by underspecifying completely the illocutionary force predicate: 'Our claim, on the other hand, is that every sentence has one and only one Speech Act Projection, with an abstract structure that constrains what can be coindexed with the seat of knowledge, but gives no other specific information about whether the speech act is a telling, a warning, a report' (Speas and Tenny, p. 338). This conflicts with the claim of Ginzburg and Sag (2000), discussed in the text below, that declaratives do actually have a primary force of asserting, queries have a primary force of asking, etc.

<sup>26</sup> How any of these values get instantiated, if indeed B manages to do so, can involve highly complex reasoning (involving e.g. domain-specific knowledge, reasoning about intentions etc.) with which of course the grammar as such provides no assistance. However, the use of such reasoning to resolve the value of a constituent of content also affects constituents of content (e.g. tense and anaphora) that lie uncontroversially

- (33) a. A: I will leave tomorrow.  
 b.  $P(A, B, \text{leave}(\text{leaver} : A, \text{time} : t, \text{location} : l))$   
 c.  $\text{leave}(A, \text{time} : t, \text{location} : l))$

Our first argument concerns the existence of words that actually carry their CMT on their sleeve. Examples of such words are given in (34):

- (34) a. [Context: A sees B as she enters a building] A: Hi.  
 b. [Context: A enters train carriage, sees B leave] A: Bye.  
 c. [Context: in a bus queue A slips and unintentionally pushes B] A: Sorry.  
 d. [Context: B is a bus conductor who gives A a ticket.] A: Thanks.

A competent speaker of English might paraphrase each of these utterances as in (35):

- (35) a. A *greeted* B.  
 b. A *bid farewell* to B.  
 c. A *apologized* to B (*for having pushed her*).  
 d. A *thanked* B (*for giving her a ticket*).

This can be used as evidence that these words are associated with meanings schematized as in (36). In these representations, the main predicate constitutes the CMT associated with the utterance, whereas *m(es)s(a)g(e)-arg* indicates the semantic type of the propositional/descriptive content selected by the CMT. Note a contrast illustrated in (35): whereas both (the relations denoted by) *apologize* and *thank* are specified for arguments constituting the descriptive content, there is no such selection by *greet* and *bid farewell*. This provides some of the motivation for assuming that these latter should not be specified for a *msg-arg*, in other words that such speech acts have no descriptive content.

- (36) a. Hi: **greet**(*speaker, addressee, msg-arg: none*)  
 b. Bye: **bid farewell**(*speaker, addressee, msg-arg: none*)  
 c. Sorry: **apologize**(*speaker, addressee, msg-arg: event*)  
 d. Thanks: **thank**(*speaker, addressee, msg-arg: event*)

If we assumed the existence of a ‘post-semantic module’ which associates CMTs with the (descriptive) contents provided by the grammar, we would run into significant problems. To get the right result for *hi*, we would need to assume that a null descriptive content however represented somehow gets associated with the CMT **greet**. But this would result in a problem with *bye*, utterances of which equally lack a descriptive content. Assuming underspecification—e.g. null descriptive content associates with, say, **greet**  $\vee$  **bid farewell**—would lead to the unintuitive expectation that *hi* and *bye* potentially allow for multiple CMTs. Note that this argument is *independent* of the

within the realm of semantics. Hence, this cannot be used as an argument against integrating CMT within grammatical analysis.

assumption that *hi* and *bye* lack descriptive content; it applies equally to pairs such as *sorry* and *thanks*, which clearly do possess a descriptive content. Assuming that eventive descriptive contents are associated with the CMT of **apologize** or alternatively with **thank** or are underspecified between, say, **apology** and **thank**, would lead to similar problems *mutatis mutandis*. Thus, in their representation in the lexicon such words must have a CMT associated with them.

A second argument concerns reprise utterances. As mentioned already in Chapter 2 and discussed in detail in Chapter 6, utterances such as B's in (37a, b) can be understood (on the 'clausal confirmation' reading, where the addressee verifies she has understood the content of the utterance correctly) as in the respective parenthesized paraphrases; whereas B's utterance in (37c) unambiguously involves the adjacent parenthesized content:

- (37) a. A: Who left? B: Who left? (clausal reading: *Are you asking who left?*)  
b. A: Go home Billie. B: Go home? (clausal reading: *Are you ordering me, of all things, to go home?*)  
c. A: Did Belula resign? B: Did WHO resign? (unambiguously: *Who<sub>i</sub> are you asking whether i resigned?*)

If such paraphrases are the correct basis for an analysis of such utterances, this indicates that, in reprise utterances at least, CMT (the CMT of the preceding utterance, to be precise) can become a constituent of the *descriptive* content of an utterance. In other words, CMT becomes a constituent of the content the grammar incontrovertibly needs to build up.

Ginzburg and Sag (2000) suggest that reprise utterances provide a probe that allows one to filter away the indirect force of an utterance and establish a single direct CMT with a given utterance.<sup>27</sup> They argue this on the basis of examples like (38), uttered outside a West End theatre currently showing a best-selling musical:

- (38) (1) Stina: I have a ticket for tonight's performance.  
(2) Padraig: You have a ticket for tonight's performance?  
(3) Stina: Yes.
- (39) a. I'm offering to sell a ticket for tonight's performance.  
b. Are you claiming that you have a ticket for tonight's performance?  
c. Are you saying that you wish to sell a ticket for tonight's performance?  
d. I'm claiming that I have a ticket for tonight's performance.  
e. I'm offering to sell a ticket for tonight's performance.

<sup>27</sup> Using reprises as such a probe was first suggested to us by Richmond Thomason in an oral discussion that followed presentation of Ginzburg and Sag (1999).

Stina's utterance (38(1)) could naturally be understood to convey (39a). However, Pádraig's reprise—(38(2))—merely requests clarification of the claim Stina made; it can be understood solely as (39b), not as (39c). This can be further demonstrated by noting that *yes* in (38(3)) conveys (39d) in this context, but cannot convey (39e), despite the salience of the offer.<sup>28</sup>

A final point worth making is that, far from casting doubt on the assumption that grammatically associated CMTs exist, it seems that the phenomenon of multifunctionality *strengthens* the need for the assumption. In order to deal with indirectly conveyed messages such as (39a), one will need to state domain axioms whose antecedents will often involve a content with a grammatically associated CMT. For instance, If A states to B that she has a ticket, she might wish to sell it to B, rather than simply If agent A has a ticket, she might wish to sell it to B. Programming a robot with the latter axiom is a recipe for disaster, as the robot will hassle any approaching theatre-goer, rather than solely loudly declaiming touts.

### 5.5.2 Root Clauses

Assigning CMT can occur in the lexicon, as with the interjections that were discussed above, or on the phrasal level. I discuss the latter here and deal with words in Appendix C and some examples in subsequent chapters. The constraint for a *generic root clause*, which I will refine shortly, is given in (40).<sup>29</sup> The constraint simply states that a root clause's content is a proposition that ascribes to situation  $u_0$  the type  $R(s, a, t, hd - dtr.cont)$ , where  $R$  is an illocutionary relation that depends on the descriptive content being embedded, and  $s, a, t$  are the speaker, addressee, utterance time. This proposition reflects the belief a competent speaker would form concerning the utterance she has just made, shared by the addressee if communication is successful.

<sup>28</sup> Actually, as already discussed in Chapter 1, reprises exemplify an additional reading dubbed the *intended content reading*, which involves a request for reformulation of the import of the reprised (sub)-utterance. Thus, for an referential NP utterance, as in (i), this will be understood as a request for reference resolution:

- (i) Andie: Did Jo leave?  
     Bo: Jo?  
     Andie: Your cousin.

Given this, reprises such as (38(2)) will also yield readings paraphrasable as (ii), where the inferred component of content is *not* necessarily filtered away:

- (ii) Shi: What do you mean by saying *you have a ticket for tonight's performance*?

*yes*, however, is an inappropriate response to this reading.

<sup>29</sup> The attribute 'generic' serves to contrast it with a number of less 'canonical' root clause types, which we encounter in Chapter 8.

(40) *generic-root-cl* =<sub>def</sub>

$$\left[ \begin{array}{l} \text{dgb-params : } \left[ \begin{array}{l} \text{u0 : Rec} \\ \text{spkr: IND} \\ \text{addr: IND} \\ \text{utt-time: TIME} \\ \text{c1 : addressing(spkr, addr, utt-time)} \\ \text{head-dtr.dgb-params : DGP1} \end{array} \right] \\ \\ \text{cont} = \left[ \begin{array}{l} \text{sit} = \text{u0} \\ \text{sit-type} = \left[ \begin{array}{l} \text{R = IllocRel(hd-dtr.cont) : IllocReln} \\ \text{c1 : R(spkr, addr, utt-time, hd-dtr.cont)} \end{array} \right] \end{array} \right] : \text{Prop} \end{array} \right]$$
  

$$\text{head-dtr : } \left[ \begin{array}{l} \text{DGP1 : RecType} \\ \text{dgb-params : DGP1} \\ \text{cont : AbstSemObj} \end{array} \right]$$

(40) assigns the labour to fixing the illocutionary relation associated with a given root clause directly to the semantics, making it dependent on the content of the embedded message. This is in line with the approach developed in Ginzburg and Sag (2000). They assumed the following minimal theory relating semantic types to CMTs:

- (41) a.  $m : \text{SemObj} \mapsto \text{IllocRel}(m)$   
 b.  $\text{IllocRel}(m) = \text{Assert}$ , if  $m : \text{Prop}$   
 c.  $\text{IllocRel}(m) = \text{Ask}$ , if  $m : \text{Question}$   
 d.  $\text{IllocRel}(m) = \text{Order}$ , if  $m : \text{Outcome}$   
 e.  $\text{IllocRel}(m) = \text{Exclaim}$ , if  $m : \text{Fact}$

The assumption of (41) is that semantic typehood is sufficient to determine the illocutionary relation of a given utterance. It is of course simple enough to formulate a weaker theory. In such a theory, propositional messages would be associated with a type *PropMove* that consists of, for instance, asserting, threatening, and promising, whereas interrogative messages are associated with a type *QuestMove* that consists of,

for instance, asking and reasserting (the move type associated with ‘rhetorical questions’) and so forth. The illocutionary relation associated with a given utterance would then be semantically underspecified as a contextual parameter. Indeed in Chapter 8 I propose additional types of root clauses whose semantics is somewhat ‘non-canonical’, covering certain classes of conventionalized indirect speech acts; see also Beyssade and Marandin (2006) for a significantly more sophisticated analysis of CMTs.

**Exercise 5.5.1** Reformulate (40) so that CMTs are underspecified contextual parameters.

**Exercise 5.5.2** Write lexical entries for ‘Hi’, ‘Sorry’, ‘Bye’ and use the *interj-cl* schema from Appendix C to offer analyses of utterances of such words.

(40) is insufficiently constrained in a number of ways. As it stands, nothing stops one from using a root clause as ‘input’ to the constraint and building a root clause in which, say, an asking is asserted. That is, root clauses should not be re-entrant. To block this, I reuse a brute force method from Ginzburg and Sag (2000). Whereas all signs that cannot be complements of an embedding predicate bear the specification  $I(NDEPENDENT)C(LAUSE) : +$ , Ginzburg and Sag (2000) introduce a further partition among such signs, depending as to whether or not they can play a role in recursive operations of the grammar. Those that cannot are designated as  $ROOT : +$ . In other words, utterance types classified as  $ROOT : +$  are grammatically non-re-entrant. In addition, we want to block the possibility of illocutionary uses of certain message-denoting expressions. These include NPs (42a, b) and CPs (42c, d):

- (42) a. The question of who Mo attacked  
       b. Bo’s claim  
       c. whether Bo left  
       d. that the revolution failed.

One syntactic way of achieving this is to specify root clauses in English to be  $V[+fin]$ . In addition to this, root clauses need to be specified as the ‘termination point’ for unbounded dependencies such as QUEST-DOM and GAPS. Putting this together yields the following constraint:

$$\begin{array}{l}
 (43) \quad \text{generic-root-cl} =_{def} \\
 \left[ \begin{array}{l}
 \text{cat} = \left[ \begin{array}{l} \text{head} = V[+\text{fin}] : \text{PoS} \\ \text{IC} = + : \text{Boolean} \\ \text{ROOT} = + : \text{Boolean} \end{array} \right] : \text{syncat} \\
 \\
 \text{dgb-params:} \left[ \begin{array}{l} \text{u0} : \text{Rec} \\ \text{spkr} : \text{IND} \\ \text{addr} : \text{IND} \\ \text{utt-time} : \text{TIME} \\ \text{cl} : \text{addressing}(s, a, t) \\ \text{head-dtr.dgb-params} : \text{DGP1} \end{array} \right] \\
 \\
 \text{cont} = \left[ \begin{array}{l} \text{sit} = \text{u0} \\ \text{sit-type} = \left[ \begin{array}{l} \text{R} = \text{IllocRel}(\text{hd-dtr.cont}) : \text{IllocReln} \\ \text{cl} : \text{R spkr, addr, utt-time, hd-dtr.cont} \end{array} \right] \end{array} \right] : \text{Prop}
 \end{array} \right] \\
 \\
 \text{head-dtr:} \left[ \begin{array}{l}
 \text{cat} = \left[ \begin{array}{l} \text{head} = V[+\text{fin}] : \text{PoS} \\ \text{IC} = + : \text{Boolean} \\ \text{ROOT} = - : \text{Boolean} \end{array} \right] : \text{syncat} \\
 \\
 \text{DGP1} : \text{RecType} \\
 \text{dgb-params} : \text{DGP1} \\
 \text{cont} : \text{AbstSemObj} \\
 \text{quest-dom} : \langle \rangle \\
 \text{gaps} : \langle \rangle
 \end{array} \right]
 \end{array}$$

**Exercise 5.5.3** Provide analyses of (44) as root clauses:

- (44) a. Who did Bo visit?  
 b. Bo sees Kim.

**Exercise 5.5.4** What rules out an analysis of ‘Did Bo see who?’ as a ‘canonical’ (non-reprise) question in the current system?

## 5.6 Summary

In this chapter I have outlined  $\text{HPSG}_{TTR}$ , a variant of HPSG which uses TTR as its formal underpinning. I have offered some arguments as to why for dialogue this approach constitutes an improvement over a typed feature structure-based approach to grammar, in particular by arguing against the use of typed feature structures

and the use of unification in semantic composition. In their place, I have proposed the use of records and record types, and  $\lambda$ -calculus tools for semantic composition. The main linguistic motivation for abandoning unification has been adherence to the Reprise Content Hypothesis, a methodological principle about the interface between syntax and semantics that is strictly stronger than compositionality. The arguments against typed feature structures derive from the indirectness they impose on semantic representation and the weakness of the tools they provide for the semantic arsenal. The notions introduced in this chapter, fundamental in constructing KoS's theory of communicative interaction, are:

- **Utterances:** these are treated as records, in line with our representation of events as records.
- **Lexical and Phrasal types:** these are treated as record types. The type which characterizes all words and phrases is given in (45):

(45) Sign =

|            |                                                                                                                      |     |                |      |             |
|------------|----------------------------------------------------------------------------------------------------------------------|-----|----------------|------|-------------|
| phon       | : list(phonform)                                                                                                     |     |                |      |             |
| cat        | : [head : PoS]                                                                                                       |     |                |      |             |
| gaps       | : list( <table> <tr> <td>cat</td><td>: [head : PoS]</td></tr> <tr> <td>cont</td><td>: [x : Ind]</td></tr> </table> ) | cat | : [head : PoS] | cont | : [x : Ind] |
| cat        | : [head : PoS]                                                                                                       |     |                |      |             |
| cont       | : [x : Ind]                                                                                                          |     |                |      |             |
| dgb-params | : RecType                                                                                                            |     |                |      |             |
| cont       | : SemObj                                                                                                             |     |                |      |             |
| quest-dom  | : list(RecType)                                                                                                      |     |                |      |             |

- **Meaning in grammar:** as far as semantics goes, utterance types represent *meanings*, a representation which is broken down into two essential components, DGB-PARAMS, which represents those context-dependent aspects of a meaning that get instantiated during grounding, and CONTENT, which represents the content an utterance gets by combining information from DGB-PARAMS and content-independent information.

The chapter contains a small fragment covering declarative clauses, polar interrogatives, and *wh*-interrogatives, a fragment that will be put to use in subsequent chapters, as we develop grammatical analyses of reprise clauses and various kinds of non-sentential utterances. The final topic discussed, crucial for the interfacing of grammar and interaction, is the integration of conversational move types (aka *illocutionary force*) in the grammar. This will play an important role in my account of certain cases of indirect speech acts.



## Grounding and CRification

Jeremy Paxman: You don't drink privately?

Charles Kennedy: What do you mean, 'privately'?

Jeremy Paxman: By yourself, a bottle of whisky late at night?

[George] Galloway [MP] is recorded reassuring his Excellency [Uday Hussein] that 'I'd like you to know we are with you' 'til the end.' Who did he mean by 'we'? Who did he mean by 'you'? And what 'end' did he have in mind? He hasn't said.

(From a report in the *Cambridge Varsity* by Jon Swaine, 17 February 2006)

Is The War Salvageable? That depends on what we mean by 'the war' and what we mean by 'salvage'.

(Andrew Sullivan's Blog *The Daily Dish*, September 2007)

### 6.1 Introduction

Metacommunicative interaction is continually taking place in conversation: its commonest manifestation are acknowledgement acts performed *sotto voce* and/or gesturally (gaze, head nods etc.). There is some evidence, discussed below, that such acts can also be performed by non-human primates. This same evidence, very tentative at this stage no doubt, suggests that what non-human primates cannot do is engage in discussion of a partially comprehended utterance. Humans, on the other hand, participate in such interaction, the posing of Clarification Requests (CRs) or simply *CRification*, on average once every twenty turns.<sup>1</sup>

CRs have been studied in a variety of settings: with and among normally developing children, children with hearing, pragmatic, or SL impairments, autistic and Downs

<sup>1</sup> Some readers absolutely detest the neologism 'CRification'. I apologize to them in advance, but in this chapter it is an important ink-saving measure. In its defence, I also think that the 'else case' of grounding deserves a (more or less) monomorphemic term.

syndrome children, the deaf, elderly people, and indeed among ‘normal’ adults.<sup>2</sup> An unambiguous finding from this large body of work is the unfailing ability to use and comprehend CRs, documented from the age of earlier than 1 and until later than 97.

KoS has hitherto incorporated the simplifying assumption that **perfect communication** obtains. Operatively speaking, this means that as soon as Ariadne makes a conversational move, it is assumed to effect an identical update on Ariadne’s and Barabbas’s information states. This simplifying assumption is built in to the view of language as an entity shared by a homogeneous speech community, assumed in all but a smattering of work in generative grammar and formal semantics to date.

In Chapter 2, I presented data that demonstrates the need to integrate illocutionary and metacommunicative interaction: this included the Turn-Taking Puzzle, which shows that with respect to resolving NSU content there are simultaneously both metacommunicative and illocutionary possibilities. We also saw the potential emergence after each utterance of various utterance-oriented presuppositions, concerning the utterance’s form and the speaker’s intentions about its intended content. I then surveyed work which offers the starting point for such an integration—work on repair by Sacks, Schegloff, Jefferson, and others in the CA tradition; work on grounding by Clark and his collaborators, subsequently implemented and formalized in the PTT framework by Poesio and Traum; work by Allwood on feedback.

The basic desideratum to which the data and previous work leads us is essentially this: a protocol in which an utterance event triggers either an update of the DGB if conditions enable grounding to occur, otherwise CRification takes place. This means we need to characterize what are the conditions that enable grounding to occur and how does it affect the DGB; in addition, what kinds of CRification might ensue when grounding is precluded.

The chapter is structured as follows: I start with an empirical study of CRification, given that communicative mismatches give very good clues about what needs to be *matched* for communication to succeed. The data offers striking evidence that the range of trouble is highly constrained, one of the reasons for the tractability of CRification. The corpus studies of CRs, allied with the data we saw in Chapter 2, constitute the empirical basis driving the theory subsequently developed. In section 6.3 two simple interaction systems are sketched. These systems illustrate the bare essentials required for grounding and CRification. Although highly simplified, the systems underpin computational simulations that demonstrate the pivotal contribution MCI makes in language acquisition and maintenance. Sections 6.5, 6.6, and 6.7 spell out the informal model sketched out in section 6.4, dealing respectively with successful grounding, semantically oriented CRification, and how to integrate the two in a single protocol. Section 6.8 extends the account to a further class of CRs—‘repetition CRs’. Their study is important, pushing us away from an overly semantic perspective on CRification.

<sup>2</sup> See Saxton et al. (2005) for extensive references to this work.

## 6.2 Clarification Requests: Form and Content

There is a wide range of CRs in conversation. A characterization of this range is important because it gives an indication of possible routes down which CRification can go. More generally, it is one of the most explicit pieces of evidence we have of the distribution of sources of *trouble* in interaction and the techniques for resolving such trouble. In this section I summarize the results of several corpus and experimental studies of CRs, as well as briefly surveying the linguistic properties of one of the most pervasive CR types, so-called Reprise Fragments (RF).<sup>3</sup>

### 6.2.1 CRs: A Bottom-Up Taxonomy

The first study of CRs I discuss—the most extensive to date as far as I know—is work reported in Purver, Ginzburg, and Healey (2001) and Purver (2004a), where all the motivating data is to be found. This is based on a random sampling of the ten million word dialogue sub-corpus of the BNC, containing about 150,000 words. To maintain a spread across dialogue domain, region, speaker age, etc., this sub-portion was created by taking a 200-speaker-turn section from fifty-nine transcripts.

Eight form categories were identified, examples from each are given in (1), which (apart from (1i)) are all assumed uttered as CRs to (1a). The category *Wot* includes words typically used to request (more or less verbatim) repetitions; *Explicit* comprises phrases which constitute CRs context-independently, in virtue of their ‘compositional meaning’; *Literal Reprise* are CRs which repeat verbatim the troubled utterance;<sup>4</sup> *Wh-substituted Reprise* are like *Literal Reprise* except for the replacement of a constituent by a whP; *Reprise Sluices* are bare whPs used as CRs; *Reprise Fragments* are bare phrases used as CRs; *Gaps* are utterances where a CR is obtained by omitting the targetted constituent from a sentence; finally, *Fillers* are CRs whose antecedent is an unfinished sentence and where the CR contains a guess as to the intended utterance:<sup>5</sup>

- (1) a. A: Did Bo leave?
- b. **Wot**: B: Eh? / What? / Pardon?
- c. **Explicit (exp)**: B: What did you say? / Did you say ‘Bo’? / What do you mean ‘leave’?
- d. **Literal Reprise (lit)**: B: Did BO leave? / Did Bo LEAVE?
- e. **Wh-substituted Reprise (sub)**: B: Did WHO leave? / Did Bo WHAT?
- f. **Reprise Sluice (slu)**: B: Who? / What? / Where?
- g. **Reprise Fragments (RF)**: B: Bo? / Leave?

<sup>3</sup> Also known in past work by the term *Clarification Ellipsis* (CE).

<sup>4</sup> ‘Verbatim’ should be taken with the caveat that interlocuter-oriented indexicals (‘I’, ‘You’, etc.) in the source utterance will typically be substituted with the appropriate correlate—‘I’ with ‘you’ and vice versa.

<sup>5</sup> Schegloff (1987b) reports, based on observations due to Besnier, that among Samoans this class of CRs does not exist, but is subsumed by a *Wh-substituted Reprise*—this arising, allegedly, because of a strong reluctance to make guesses about another interlocutor’s non-public intentions.

- h. **Gap:** B: Did Bo ...?
- i. **Filler (fil):** A: Did Bo ... B: Win? (Table I from Purver 2006)

Four classes of contents were identified: they can be exemplified in the form of Explicit CRs:<sup>6,7</sup>

- (2) a. **Repetition:** What did you say? Did you say 'Bo'?
- b. **Clausal Confirmation:** Are you asking if Bo left? You're asking if who left?
- c. **Intended Content:** What do you mean? Who is 'Bo'?
- d. **Correction:** Did you mean to say 'Bro'?

In practice, though, most CRs are not of the Explicit category. Many CR utterances are multiply ambiguous. The most extreme case is RF, which seems able to exhibit all four readings, though as we will see below in practice 99 per cent of cases found in Purver, Ginzburg, and Healey (2001) were either **Clausal Confirmation** or **Intended Content**. The distinction between the Clausal Confirmation and Intended Content readings is intuitively clear: Clausal Confirmation readings constitute a kind of confirmation question, specifically with respect to the semantic contribution of a particular constituent within the entire clausal content. Intended Content readings, in contrast, query the content a speaker intends to associate with a given (sub-)utterance, with no commitment to having resolved the remaining content of the clause. Examples of **Clausal Confirmation** and **Intended Content** readings from the BNC, with a proposed paraphrase, are given in (3) and (4) respectively; (4) also illustrates three form types uttered in close proximity and Dorothy's move (4(8)) gives direct evidence for the proposed paraphrase:

- (3) a. Marsha: yeah that's it, this, she's got three rottweilers now and  
      Sarah: three?  
      Marsha: yeah, one died so only got three now. (BNC, KP2 )  
      Are you saying she's got THREE rottweilers now?
- b. Unknown: Will you meet me in the drama studio?  
      Caroline: Drama studio?  
      Unknown: Yes I've got an an audition. (BNC, KP3)  
      Are you asking if I should meet you in (of all places) the DRAMA STUDIO?

<sup>6</sup> In previous work what I call here **Intended content** has been called a *constituent* reading, what I call here **Clausal confirmation** has been called a *clausal* reading, and what I call here **Repetition** has been referred to as either a *lexical* or a *form identification* reading.

<sup>7</sup> As we will see below, correction readings are rare in everyday conversation and I will not say much about them in what follows. Purver (2006) points out that they are far commoner in certain types of human-computer conversation, particularly ones in which errors abound due to less than 100 per cent reliable automatic speech recognition. Moreover, as Gruenstein (2002) points out, the commoner reading in such cases is (iii), not (ii):

- (i) A: Fly to the base. B: OK, I'm flying to the tower. A: No, the base.
- (ii) You mean to say that you're flying to the base.
- (iii) You should be flying to the base.

- (4) a. Tim (1): Could I have one of those (unclear)? Dorothy (2): Can you have what? Tim (3): Have one of those things. Dorothy (4): What things? Tim (5): Those pink things that af after we had our lunch. Dorothy (6): Pink things? Tim (7): Yeah. Er those things in that bottle. Dorothy (8): Oh **I know what you mean.** For your throat?  
b. What do you mean 'pink things'?

The Repetition reading is expressed most commonly by the Wot lexical class when the target is the entire utterance:

- (5) a. Gary: It's, it's okay I mean (pause) when he's away (pause, clears throat, pause) things go, actually things go a lot smoother.  
Clare: Sorry?  
Gary: Things go a lot smoother when he's not here. (KSR)  
b. S. Bust: Great memorial I think really isn't it?  
E. Bust: Beg pardon?  
S. Bust: Be a good appropriate memorial if we can afford it. (KM8)  
c. Unknown: Does she have to get permission to do it?  
Catherine: Pardon?  
Unknown: Does she have to have permission to do it?  
d. Kitty: How's Nat?  
Anon 4 Fine.  
Kitty Oh oh!  
Anon 4 Gone to the cinema tonight or summat.  
Kitty Eh?  
Anon 4 Gone to the cinema.

In such a case the only presupposition required is that an utterance has been produced by the other conversational participant; the CR poser can in principle have failed to recognize any of the words that constituted the utterance. Indeed, unless she is a stenographer, she is for all intents and purposes in an equivalent situation to someone who heard correctly but failed to understand the intended content, the background for **Intended Content** readings of RF or **Explicit** questions such as 'what do you mean'. Nonetheless, for the CR responder there *is* a difference: whereas an **Intended Content** reading requires the responder to reformulate in some way, a Repetition reading permits the responder to repeat her utterance verbatim (though as the examples in (5) above illustrate, minor reformulations are common).

It seems that RF can also be used simply to elicit (sub-clausal) repetition, as in (6), though such uses turn out to be quite rare. (6b(4)) illustrates a point that is of some importance particularly with respect to self-corrections discussed in Chapter 8—that material which is corrected by a subsequent utterance is not erased from the public

record. This goes against the widespread computational and psycholinguistic strategy to treat corrections as a low-level, pre-semantic process.<sup>8</sup>

- (6) a. A: Have a laugh and joke with Dick./ B: Dick?/A: Have a laugh and joke with Dick. (BNC, KB7 )  
 b. Catherine(1): I use Safeway's Special Formula Doing One or something. Cassie(2): Doing One? Catherine(3): Two in One. Cassie(4): You said Doing One. (BNC, KP4)

Ginzburg and Cooper (2004) argue in detail that the multiplicity of understandings of RF (clausal confirmation/intended content/repetition) is an ambiguity, rather than a consequence of a radically underspecified meaning.<sup>9</sup> Indeed the BNC provides numerous examples of misunderstandings concerning RF interpretation, where a speaker intends one interpretation, is misunderstood, and clarifies his original interpretation:

- (7) a. George ... you always had er er say every foot he had with a piece of spun yarn in the wire/Anon1: Spun yarn?/George: Spun yarn, yes/ Anon1: What's spun yarn?  
 George: Well that's like er tarred rope (BNC, H5G)  
 b. A: Have a laugh and joke with Dick./ B: Dick?/A: Have a laugh and joke with Dick./B: Who's Dick? (BNC, KB7)  
 c. Andy: Bloody lucky though./Monica: pikey! Typical! / Andy: Pikey? /Nick: Pikey! /Andy: What's pikey? What does pikey mean? / Monica: I dunno. Crusty. (BNC, KP6)

I will not rehearse the arguments beyond restating (and slightly correcting) one argument, one which has significant import for how we need to conceive of context. This argument derives from the fact that RF displays parallelism on a syntactic and phonological level with its source. Revealingly, clausal confirmation readings, on the one hand, and intended content and repetition readings, on the other, involve distinct parallelism conditions, suggesting that different linguistic mechanisms underlie the distinct understandings. Indeed the existence of syntactic and phonological parallelism in CRs across utterances is further evidence to that provided earlier in Chapter 2 that context in grounding/CRification needs to track non-semantic sign information, not merely content. I make this latter argument in more detail in section 6.4.

<sup>8</sup> I would like to thank David Schlangen for alerting me to this point.

<sup>9</sup> Actually the arguments there primarily concerned the clausal confirmation/intended content distinction. So, in principle, even accepting those arguments, we need to argue for the autonomy of Clausal Confirmation from Repetition and Intended Content from Repetition. The argument about distinct parallelism conditions provided below separates Clausal Confirmation from Repetition. As for the Intended Content vs Repetition distinction: the latter queries what the former presupposes ('what did you mean when saying ...'), hence an obvious difficulty in collapsing the two.

Clausal Confirmation readings do not require phonological identity between target and source:

- (8) a. A: Did Bo leave? B: My cousin? (Are you asking if BO, my cousin, left?)  
 b. A: Did she annoy Bo? B: Sue? (Are you asking if SUE annoyed Bo?)

Nonetheless, they require partial syntactic parallelism: an XP used to clarify an antecedent sub-utterance  $u_1$  must match  $u_1$  categorially:

- (9) a. A: I phoned him. B: Him? / #He?  
 b. A: Did he phone you? B: He? / #Him?  
 c. A: Did he adore the book. B: Adore? / #Adored?  
 d. A: Were you cycling yesterday? B: Cycling?/Biking?/#Biked?

That Repetition readings of RF involve (segmental) phonological identity with their source follows from their very nature ('Did you say ...'). And this requirement also applies to Intended Content readings of RF:<sup>10</sup>

- (10) A: Did Bo leave? B: Max? (cannot mean: Intended Content reading: 'Who are you referring to?' or 'Who do you mean?')

At the level of pre-theoretic intuitions we can make a couple more comments on form/content correlations. Of the other form types, Gap and Filler are of necessity confined to a Repetition reading.<sup>11</sup> Reprise Sluices are a tricky case: they are primarily used for CRs on a semantic level, but whether the associated reading is Clausal Confirmation or Intended Content is usually hard to tease out; either reading seems plausible. Given the commoner presuppositional background of Clausal Confirmation readings one could argue, as has Purver (2004a), that it should be given preference as a default.

We can now turn to the results of the BNC corpus study described in Purver (2004a), a slightly modified version of Purver, Ginzburg, and Healey (2001). The basic procedure followed was this: in the (common) case of ambiguity of reading, the response(s) of other dialogue participants were examined to determine which reading was chosen by them. The ensuing reaction of the CR initiator was then used to judge whether this interpretation was acceptable. If the CR initiator gave no reaction, the reading was assumed to have been acceptable.

The results are given in Table 6.1 as cumulative percentages from a total of 418 CRs that were found in 11,800 turns.

<sup>10</sup> RF does seem to exhibit readings, which are halfway between a clausal confirmation and an intended content reading, which one could refer to as 'polar intended content' readings. These do not require phonological parallelism, but are also consistent with incomplete understanding of the rest of the utterance:

(i) A: Did she annoy Bo? B: My cousin? (= are you referring to my cousin?) A: Yes. B: When?

<sup>11</sup> Though in the case of fillers the term 'repetition' is a bit misleading.

TABLE 6.1 CR form and type as percentage of CRs – BNC all domains

|                      | exp  | lit | sub | slu  | rf   | gap | fil | wot  | oth | Total |
|----------------------|------|-----|-----|------|------|-----|-----|------|-----|-------|
| Clausal-Confirmation | 2.6  | 6.2 | 1.4 | 11.5 | 24.9 | 0   | 0   | 0    | 0.5 | 47.1  |
| Intended-Content     | 7.7  | 0   | 0   | 0    | 1.7  | 0   | 0   | 5.0  | 0   | 14.4  |
| Repetition           | 0.7  | 0   | 2.2 | 1.4  | 0.2  | 0.5 | 4.1 | 25.6 | 0   | 34.7  |
| Correction           | 0.7  | 0.5 | 0   | 0    | 1.2  | 0   | 0   | 0    | 0   | 2.4   |
| other                | 0    | 0   | 0   | 0    | 1.0  | 0   | 0   | 0.5  | 0   | 1.5   |
| Total                | 11.7 | 6.7 | 3.6 | 12.9 | 29.0 | 0.5 | 4.1 | 31.1 | 0.5 | 100.0 |

Some observations about these results:

- CRs were found to make up around 3.5 per cent of turns (418/11,800) when calculated over all domains.
- Forms: The most common forms of CR can be seen to be the wot and RF forms, with each making up over 25 percent of CRs. Explicit CRs and Reprise Sluices are also common, each contributing over 10 per cent of CRs. Other forms are all around 5 per cent or less.
- Readings: nearly 50 per cent of CRs can be successfully interpreted as having a Clausal Confirmation reading, although both the Repetition (about 35 per cent) and Intended Content (about 15 per cent) readings also make up a significant proportion. Note though that these figures would be altered somewhat if Reprise Sluices were taken to have Intended Content rather than Clausal Confirmation readings—this would reduce the share of Clausal Confirmation to 35 per cent and increase the share of Intended Content to 25 per cent.
- Coverage: the coverage of the corpus by the forms and readings listed here is good, with only 0.5 per cent of CR readings (viz. two sentences) and about 1.5 per cent of CR forms (viz. six sentences) being classified as *other*.

I have dubbed the taxonomy of Purver, Ginzburg, and Healey (2001) ‘bottom up’ since it originates by grouping together certain actually found constructions on the basis of their core syntactic and semantic properties. Consequently, the taxonomy has very good coverage, but is susceptible to sparse-data lacunas. I mentioned earlier that one of the interests in studying CRs is that it enables us to study the distribution of communicative trouble. Table 6.1 gave us one indication with respect to this issue (approx. 50 per cent Clausal Confirmation, 35 per cent Repetition, 15 per cent Intended Content). Now, as (11a) indicates, *a priori* ANY sub-utterance is clarifiable, including function words like ‘the’, as in (11b):

- (11) a. Who rearranged the plug behind the table?  
 b. Who? / rearranged?/ the plug? / behind? / the table?  
 c. A: Is that the shark? B: The? B: Well OK, A. (based on an example in the film *Jaws*.)



The consequences this has for utterance representation is that we need to ensure that for a given utterance each *sub*-utterance is accessible as an antecedent, a need for which motivated our field *CONSTITTS*; see section 5.4. While the potential for repetition-oriented CRification clearly applies to all utterances and their parts, it is much more of an open question whether this is true for semantically/pragmatically-oriented CRification. I mention briefly several studies that have explored this issue.

Healey et al. (2003) and Purver et al. (2003) describe an experimental technique and corresponding set of experiments concerning responses to CRs, in particular, how the likelihood and nature of answers to CRs is affected by the nature of the source. This was achieved by use of a specially designed text-based chat tool which allowed CRs of particular types to be artificially introduced into natural dialogues in a controlled and synchronous manner, and unknown to the dialogue participants. Their (spontaneous) responses were then classified, and correlations with the nature of the source examined. Among other things, this enabled one to check the hypothesis that CRs whose source is a content word (e.g. noun or verb) would be more likely to cause an Intended Content reading than a CR of a function word (e.g. preposition or determiner); the assumption being that dialogue participants would normally assume that the meaning of function words is well known in a particular linguistic community and that, as a result, a reprise of a function word is more likely to signal need for confirmation or repetition.

Healey et al. (2003) showed that the content/function distinction has a large effect on likelihood of response to an RF query: 65 of 143 content words (45 per cent) were explicitly answered (responded to in a way other than refusal or querying the question), but only 6 of 41 function words (15 per cent). The effect on the attributed form and reading is also strong. No function word RFs seemed to be interpreted as fragments with Clausal Confirmation or Intended Content readings: all six which received explicit answers were classified as gaps or non-clarificational questions, or in one case an RF with a Repetition reading. In contrast, 60 of 65 (92 per cent) of answered content word RFs were interpreted as Clausal Confirmation or Intended Content Reprise Fragments, with only one being interpreted as a gap. In other words, function word repeats are very unlikely to be interpreted as CRs with the function word as the primary source.

This basic result, the difference in clarification potential of open/closed-class words, was confirmed by corpus studies reported in Purver (2004a) and Purver, Ginzburg, and Healey (2006). These provide a more fine-grained view of trouble by considering the distribution of CRs relative to the syntactic category of the antecedent (the trouble source). The materials used were the same as the original study reported in Purver, Ginzburg, and Healey (2001). The main result is displayed in Tables 6.2 and 6.3, split for display purposes. In these tables *unc1* denotes that the source was marked 'unclear' in the BNC transcription, *unkn* that the source cannot be identified, *sent* that the

TABLE 6.2 CR reading vs source category

|                 | uncl | sent | utt | unkn | np | pn | pro | cn | adj |
|-----------------|------|------|-----|------|----|----|-----|----|-----|
| <i>cla-conf</i> | 12   | 27   | 5   | 1    | 37 | 25 | 33  | 23 | 4   |
| <i>int-cont</i> | 3    | 2    | 20  | 0    | 5  | 9  | 5   | 13 | 0   |
| <i>repet</i>    | 45   | 0    | 68  | 5    | 4  | 2  | 0   | 9  | 4   |
| <i>cor</i>      | 0    | 2    | 0   | 0    | 4  | 0  | 1   | 1  | 0   |
| <i>oth</i>      | 0    | 0    | 2   | 0    | 1  | 0  | 0   | 0  | 0   |
| <i>Total</i>    | 60   | 31   | 95  | 6    | 51 | 36 | 39  | 46 | 8   |

TABLE 6.3 CR reading vs source category (cont.)

|                 | adv | prep | conj | det | mod | v | vp | wp | <i>Total</i> |
|-----------------|-----|------|------|-----|-----|---|----|----|--------------|
| <i>cla-conf</i> | 1   | 0    | 0    | 8   | 15  | 0 | 6  | 0  | 197          |
| <i>int-cont</i> | 0   | 1    | 0    | 0   | 1   | 0 | 1  | 0  | 60           |
| <i>repet</i>    | 0   | 0    | 0    | 0   | 0   | 3 | 2  | 3  | 145          |
| <i>cor</i>      | 0   | 0    | 0    | 1   | 0   | 0 | 0  | 1  | 10           |
| <i>oth</i>      | 1   | 0    | 0    | 1   | 1   | 0 | 0  | 0  | 6            |
| <i>Total</i>    | 2   | 1    | 0    | 10  | 17  | 3 | 9  | 4  | 418          |

source was a full sentence (though possibly a sub-utterance), *utt* that the source was the complete utterance, and the rest coincide with standard abbreviations for syntactic categories.

Purver, Ginzburg, and Healey (2006) provides a statistical analysis of some of the patterns indicated in Tables 6.2 and 6.3. The most perplexing of these perhaps, which does not seem amenable to a straightforward explanation, is that nominal CRs outweigh verbal ones by more than 40:1, a figure that cannot be explained in terms of token, type, average rarity, or information content ratios between these categories.

For a discussion of these and additional factors involved in disambiguating CRs see Purver (2004a) and Purver (2006), which explain how these are incorporated in the dialogue system CLARIE. CLARIE, an information-state-based dialogue system,<sup>12</sup> interprets and generates the full range of CRs discussed above, using principles akin to those discussed below.

### 6.2.2 Unavailable CRs: An Action Ladder-inspired CR-Taxonomy

So far we have been concerned with the range of CRs actually found in a general-purpose corpus (the BNC), coupled with discussion of the properties of one of the

<sup>12</sup> The paradigm of *information-state-based* dialogue systems is motivated in Larsson and Traum (2003).

commonest forms for posing a CR, namely reprising a fragment of the utterance. We have seen evidence for the asymmetries in the distribution of CRs, relative to the lexical category of the trouble source. One can also ask a more general question: what CRs could (in principle) be asked but (in practice) fail to?

Some answers about this question come via a number of recent studies of CRs that take a more 'top-down' approach to CR classification than we have above, starting from a space of possible processing problems and attempting to map these to actually occurring CRs. The space of possible grounding problems is derived from *action ladders* proposed in Allwood (1995) and Clark (1996). These ladders are intended to represent successful communication as a hierarchy of levels such that grounding at a higher level presupposes grounding at all previous levels:

- (12) 4. intention (proposal)
- 3. meaning and understanding
- 2. presentation / identification
- 1. execution / attention

This strategy allows for a more fine-grained classification of the functions of CRs than the bottom-up strategy. It is particularly well suited for generation. For instance, Brennan and Hulstén (1995) use a refined set of levels in (12) to generate various different types of feedback for an application-independent spoken language system. Some examples of this are given in (13a, b). Larsson (2002), basing himself more directly on (12), generates various different types of negative feedback for the dialogue system GODIS, an instance of which is in (13c):

- (13) a. (*System detects a sound but cannot recognize an utterance*):  
      User: Call Susan.  
      System: What was that again?
- b. (*System has heard, parsed, and mapped onto an interpretation but cannot map the interpretation onto a command it can perform*):  
      User: Find Chris.  
      System: Sorry, I don't know how to find Chris.
- c. S: What city do you want to travel from? U: I want to travel from Kuala Lumpur. S: Sorry, Kuala Lumpur is not in the database. So, what city do you want to travel from?

On the other hand, this strategy runs a risk and raises an interesting issue: the risk is shoehorning actually expressed content to predefined functions; the issue it raises: does each rung really correspond to and allow one to characterize a class of CRs?

Schlangen (2004) and Rodriguez and Schlangen (2004) offer (14) as potential CRs corresponding to the different levels:

(14)

| Level of action                  | Kind of problem                        | Example                               |
|----------------------------------|----------------------------------------|---------------------------------------|
| 1: Execution / Attention         | Channel                                | huh?                                  |
| 2: Presentation / Identification | Acoustic problem                       | Pardon?                               |
| 3: Signal / Recognition          | Lexical problem                        | What's a double torx?                 |
|                                  | Parsing problems                       | Did you have a telescope, or the man? |
|                                  | Reference resolution problem:          |                                       |
|                                  | NP-reference                           | Which square?                         |
| 4: Proposal / Consideration      | Deictic-reference                      | Where is 'there'?                     |
|                                  | Action-reference                       | What's 'to kowtow'?                   |
|                                  | Problem with recognizing the intention | Why? You want me to give you this?    |

Rodriguez and Schlangen's form inventory, given in (15), is somewhat richer than that of Purver, Ginzburg, and Healey (2001): particularly useful is the fact that it explicitly tracks the relation between the antecedent and the CR (15c) and intonation (15d):

- (15) Rodriguez and Schlangen (2004) form features:
- Mood:** decl, polar-q, wh-q, alt-q, imp, other
  - Completeness:** wot, partial, complete
  - Relation to antecedent:** repetition, addition, reformulation, independent
  - Intonation:** rising, falling

Rodriguez and Schlangen (2004) applied the taxonomy just described to the Bielefeld Corpus of German task-oriented human-human dialogue (SFB-360, 2000) (model aeroplane construction instructions), which consists of twenty-two dialogues, with 3,962 dialogue turns and 35,813 words. 230 CRs were found in the 3,962 turns examined. This constitutes 5.8 per cent of turns. Ninety-five per cent of CRs were adjacent to their source, in contrast to Purver, Ginzburg, and Healey (2001) for whom the corresponding figure is 85 per cent.<sup>13</sup>

The results of Rodriguez and Schlangen (2004) are summarized in (16):

- (16)
- Form: 76.5% (non-sentential) fragments, 62.6% of these were reformulations of previous content, 24.8% were repetitions
  - Form: only 9.6% of CRs were independently formulated (i.e. do not contain material from the clarification target)
  - Function: 50% reference resolution, 14% ambiguous, 22% intention evaluation, 11% acoustic
  - Answers: 45% yes/no, 20% reformulation, 13% elaboration, 12% repetition, 8% no reaction

<sup>13</sup> The difference is presumably to be explained in terms of two factors: (a) all the Bielefeld dialogues were two-person, whereas there are many multilogues in the BNC, and (b) the Bielefeld dialogues are task-oriented, where invariably only one participant (the model aeroplane *constructor*) asks CRs.

These results are broadly in line with the results of Purver, Ginzburg, and Healey (2001) discussed earlier: the majoritarian form is fragmentary, indeed to a larger extent than in Purver, Ginzburg, and Healey (2001). As for function: there are three main functions—the majoritarian being reference resolution, the other two are acoustic CRs and intention evaluation CRs. Of these three the only one which is not fully consistent with the PGH taxonomy is intention evaluation. If viewed as *intended reference* it corresponds well to the intended content category of PGH. However, Rodríguez and Schlangen seem to view such CRs as relating to the highest level in the Allwood–Clark hierarchy. In Chapter 8, where the issue of (non-public) intention recognition is discussed, I suggest that reanalysis of their data suggests that only a small percentage of these CRs warrant such an analysis.

It is also worth pointing out what CRs do NOT seem to exist: two of the most highly researched areas in formal and computational grammar are syntactic ambiguity (e.g. prepositional attachment) and scope ambiguity. However, **not a single CR concerned with syntactic or scope ambiguity has been found**, suggesting that these are not domains that involve much uncertainty for interlocutors in human conversation.<sup>14</sup>

As a final piece of evidence on CRs I mention a corpus study of CRs in an English task-oriented corpus described in Rieser and Moore (2005a), whose main results are summarized in (17). This study used the human–human travel reservation dialogues available as part of the Carnegie-Mellon Communicator Corpus, and annotated them using Rodríguez and Schlangen's taxonomy. The corpus then annotated consists of 31 dialogues of transcribed telephone speech, with 2,098 dialogue turns and 19,395 words. In this case, 98 CRs were found in 2,098 dialogue turns (4.6%).<sup>15</sup>

(17)

| Source              | Freq. (%) |
|---------------------|-----------|
| NP-reference        | 40        |
| acoustic            | 31        |
| intention           | 8         |
| belief confirmation | 6         |
| ambiguity           | 4         |
| contact             | 4         |
| others              | 3         |
| relevance           | 2         |
| several             | 2         |

<sup>14</sup> Of course, their non-occurrence, while of some note, does not mean that *in principle* such CRs cannot occur. I discuss some constructed examples of such CRs and their implications in section 8.5.

<sup>15</sup> In this case too, as discussed in Chapter 8, there was probably a significant *overuse* of the category *intention* (in the sense of level 4 of the Allwood–Clark ladder)—no more than a couple of cases seem to require this categorization. The rest are all either Clausal Confirmation or Intended Content CRs. The category *belief confirmation* corresponds to the category *clausal confirmation* of Purver, Ginzburg, and Healey (2001).

### 6.2.3 Empirical Conclusions

In section 6.1 I mentioned Schegloff's claim that 'Because anything in talk can be a source of trouble, everything in conversation is, in principle, "repairable".' In this section I have attempted to temper this claim somewhat by considering the evidence provided in the main by corpus studies and also experimental studies of CRs, as well as discussing certain properties of one of the most common constructions used in CRs, the Reprise Fragment.

The three corpus studies I surveyed here differ in their backgrounds—two concern English conversations, one German; two involve task-oriented conversations (though of a quite different nature), the most comprehensive involves a wide range of primarily free unrestricted conversation types. And yet, important commonalities emerge:

- **Restricted range of contents:** the function of CRs seems, to a very large extent, to consist of either (a) confirming or querying intended content or (b) requesting repetition of a misheard (sub-)utterance.
- **Syntactic and phonological parallelism:** CRs frequently exhibit (segmental) phonological parallelism with their source, indeed for certain form/content combinations this is a grammatical requirement; this requirement is weakened to syntactic parallelisms for other constructions.
- **No evidence for structural or scope underdetermination:** CRs provide us with some evidence concerning what interlocutors do not appear to be uncertain about: although syntactic and scope ambiguity are a big issue for computational systems, CRs provide no evidence for uncertainty on this score among human users.

The empirical survey of actually occurring CRs brings with it some obvious tasks, which are discharged in this chapter:

- Characterize the contextual background which makes the various classes of CRs coherent; do so in a way that allows for the exhibited syntactic and phonological parallelism to be captured.
- Provide a grammatical analysis of explicit CRs—these are in fact already accommodated by our existing grammar. Extend the grammar to cover reprises (literal and substitution) and Wot CRs.

A grammatical analysis for NSU CRs are provided in Chapters 7 and 8, once the general background for NSU resolution is in place.

## 6.3 Grounding and CRification: Two Very Simple Systems

The MCI apparatus available to an adult natural-language speaker is very powerful indeed, as we saw above, allowing her to request clarification of sub-utterances with

pinpoint accuracy. However, before considering a detailed characterization of this, in order to understand the essence of MCI, one can restrict attention to two much simpler systems which bypass syntax. Using computational simulations,<sup>16</sup> Ginzburg and Macura (2006) have shown that a population of agents possessing the capability to play the second of these language games, the **ack-huh game**, is intrinsically better equipped to maintain coordination of their individual beliefs about meanings than a population of agents whose metalinguistic development is based solely on introspection. Indeed, in a multigenerational setting Macura and Ginzburg (2006) show that whereas the language of the CR-using community remains relatively stable across generations, the language of a community without CRs collapses eventually to a single multiply ambiguous word. An even stronger result emerges from Macura (2007): he introduces resource competition in a foraging scenario and shows that in an increasingly complex language the ability to interact metacommunicatively is overwhelmingly adaptive. There is also some suggestive evidence, discussed shortly, that **ack-huh** constitutes a biologically relevant crossing point with respect to cognitive complexity.

### 6.3.1 *The Utt(erance) Ack(nowledgement) Game.*

The first game I mention is one I will call the *utt(erance) ack(nowledgement) game*. Its description is very simple:

- (18) The *utt(erance) ack(nowledgement) game*
- a. Given an utterance  $u_0$  consisting of a string ( $\text{word}_1, \dots, \text{word}_i, \dots, \text{word}_n$ ) by the master, the novice may respond with the utterance  $u_1 = \text{word}_i$ .
  - b.  $u_1$  is assigned content: *novice acknowledges that an utterance including the word  $\text{word}_i$  happened*. This fact now becomes part of the novice's and master's common ground.

Notice that in this game the novice's utterance does not commit him to understanding the words he utters. The reward for playing this game is shared interaction with the master. Playing the *utt-ack* game requires from the novice two basic capabilities:

- A phonological imitation and segmentation module<sup>17</sup>
- The ability to form (one-sided) mutual beliefs.

Who can play this game? Arguably, the initial stage of speech for human neonates consists in various variants of this game:<sup>18</sup>

<sup>16</sup> The environment is modelled loosely after the Sugarscape environment (Epstein and Axtell 1996), in that it is a spatial grid containing different plants. Plants can be perceived and disambiguated by the agents. Agents walk randomly in the environment and when proximate to one another engage in a brief conversational interaction concerning visible plants. An agent's field of vision consists of a grid of fixed size originating from his location. Hence proximate agents have overlapping but not identical fields of vision.

<sup>17</sup> Hence the game can be played in one-word mode, i.e. does not require the novice to have syntactically complex generation capabilities.

<sup>18</sup> Indeed a common variant on this is one where the infant plays the role of the master, uttering sounds which the adult in the role of novice reprises without knowledge of their meaning or content.

- (19) a. Mother: you go in the room and kill a fly.  
           Eve: fly.  
       b. Mother: that'll keep you busy.  
           Eve: busy. (From Eve (Brown 1973) at age 1;6)

According to Greenfield and Rumbaugh (1993) both chimps and bonobos can also play this game, as in the following attested examples that they provide:<sup>19</sup>

- (20) a. Kelly (human carer): We are going to see the GIBBONS.  
           Kanzi (bonobo): GIBBONS. (As per Kanzi's earlier request.)  
       b. (Kanzi points to refrigerator and after Rose has opened it points to a bowl of raspberries.)  
           Rose (human carer): We will call these FOOD. (*There being no lexigram for raspberries on the keyboard.*)  
           Kanzi: FOOD. (Kanzi indicates no desire to eat the raspberries, goes over and looks out of the window.)  
       c. Kelly: Let's see what's on TELEVISION  
           Mulika (bonobo): TELEVISION. (Walks over to video deck and gestures to it, awaiting input of tape by Kelly.)

### 6.3.2 A Rudimentary Game with CRs: The *ack-huh?* Game.

A more advanced game is the *ack-huh?* game:

- (21) The *ack-huh?* game.  
       Given an utterance  $u_0$  by the master, the novice may either (a) acknowledge  $u_0$  or (b) query the content of  $u_0$ .

For instance:

- (22) a. Master: grass  
           Novice: huh?  
           Master: [points at the object] That thing.  
       b. Master: This is a banana.  
           Novice: banana?  
           Master: Sweet.

Who can play this game? Human neonates certainly can (from about 30 months) (see e.g. Ninio and Snow 1996, Ginzburg and Kolliakou 2010). However, according to Greenfield and Rumbaugh (1993), neither chimps nor bonobos ever use repetition to pose CRs, indeed they do not pose CRs.

What additional capabilities does playing *ack-huh?* involve? The key feature is awareness of synonymy between utterances, i.e. awareness of the possibility to reformulate in a way that preserves content. In the absence of that any content-oriented

<sup>19</sup> Words given in small capitals are ones uttered via a lexigram 'typewriter'.



*discussion* will be circular—A will utter *u*, B will query it, A will repeat *u*, etc.<sup>20</sup> Example (23) illustrates this in a dialogue between a mother and her child, who has not yet understood the need for reformulation in content-oriented CRification. The cycle is finally relieved by the child's father:

- (23) Tim: Who gave you that? Dorothy: Who gave me what? Tim: That? Dorothy: What? Tim: That? Andrew: The cassette. Dorothy: A man. (BNC, KBW)

Awareness of synonymy in turn presupposes that utterances are entities that the language user recognizes as part of her domain. In particular, that they can be arguments of predicates (see below, for instance, the protocol (24)).

Thus, the simplest agent with the ability to engage in (content-oriented) CRification is an agent who can (a) communicate a content along the lines of 'I don't understand *that utterance* (*u1*)' and (b) recognize a reformulation of *u1* as such. A natural way of capturing both (a) and (b) is to say that, given an utterance *u* uttered by A, the agent must be able to discuss the issue *what did A mean by uttering u*. In terms of compositionality, given an agent who can reflect and form questions about entities in the domain, this means that once the predicate 'mean' is in the language, then basic clarification requests can be expressed. But the latter is not necessary for *conveying* CRs, as these could arise in non-decomposable way (e.g. 'eh', 'pardon', and the like).

A protocol for the **ack-huh?** game is in (24):

- (24) a. Master utters *u0*  
 b. Option 1: Grounding  
     Novice utters *u1* such that  $\text{cont}(u1) = \text{Understand}(B, u0)$   
 c. Option 2: CRification  
     Novice utters *u1* such that  $\text{cont}(u1) = \lambda x \text{Mean}(A, u, x)$   
     Master answers  $\lambda x \text{Mean}(A, u, x)$ , typically by a deictic act.

In most settings, CRs are about utterance tokens (i.e. they resolve the content of a given use, not the meaning associated with the utterance).<sup>21</sup> Nonetheless, from **ack-huh** it is a short step to lexical acquisition and dynamics—one simply needs to formulate rules that enable the lexicon to be updated as a consequence of answers to CRs (see e.g. Purver 2004a, Macura 2007; see also Saxton, Houston-Price, and Dawson 2005 for experimental evidence that CRification constitutes a form of corrective input for young children's grammatical errors).

However, staying on topic—the absolute basics of MCI—if we want to formulate lexical entries for words or phrases that a novice could use in **ack-huh?**, we need to have a means of referring to the property an utterance has while it is under grounding.

<sup>20</sup> There is a crucial difference between this and a variant of this game where mere repetition of utterance is desired, as in the repetition uses of the CR category *Wot*.

<sup>21</sup> After all, explaining a rule underlying a *meaning* is quite a bit trickier than a one-off *content*.

I will call such an utterance **pending**—also subsequently the name for the field for ungrounded utterances in a slightly enriched DGB.<sup>22</sup>

Given this, we can posit the lexical entries for ‘mmh’ and ‘huh?’ provided in (25) and (26) respectively; (26) is based on an entry for ‘pardon’ in Purver (2004a).<sup>23</sup> In terms of contextual processing, using these words requires from an interlocutor the ability to resolve who is speaking, who is being addressed, and the pending utterance. Given that, (25a) enables the speaker to acknowledge the pending utterance, whereas (25b) to ask for its content:

$$(25) \left[ \begin{array}{l} \text{phon : mmh} \\ \text{cat.head = } \textit{interjection} : \textit{syncat} \\ \\ \text{dgb-params : } \left[ \begin{array}{l} \text{spkr : IND} \\ \text{addr : IND} \\ \text{pending : utt} \\ \text{c2 : address(addr, spkr, pending)} \end{array} \right] \\ \text{cont = Understand(spkr, addr, pending) : IllocProp} \end{array} \right]$$

$$(26) \left[ \begin{array}{l} \text{phon : huh} \\ \text{cat.head = } \textit{interjection} : \textit{PoS} \\ \\ \text{dgb-params : } \left[ \begin{array}{l} \text{spkr : IND} \\ \text{addr : IND} \\ \text{pending : utt} \\ \text{c2 : address(addr, spkr, pending)} \end{array} \right] \\ \text{cont = Ask(spkr, addr,} \\ \quad \lambda x \text{ Mean(addr, pending, x)) : IllocProp} \end{array} \right]$$

## 6.4 Interacting over Grammatically Governed Content: Informal Picture

In describing utterance integration we need to describe a DGB update which will simultaneously allow us to explain either of the two main options that occur in the aftermath of an utterance:

<sup>22</sup> In the PTT framework (Poesio and Traum 1997) this is captured by means of a list-valued discourse referent named UDUS (Ungrounded Dialogue UnitS).

<sup>23</sup> Acknowledgement via reprise requires one additional technical device, the ability to reprise words or constituents that comprise an utterance. This is a very significant feature of both child and adult MCI system, as we discuss below, but it not a *necessary* component of an MCI system. See Chapter 7 for a description of such uses.

1. the utterance is understood
  - in which case LatestMove will be updated with *u*
2. the utterance fails to be completely understood
  - in which case CRification concerning *u* will ensue

In this section I discuss things informally, at a basic conceptual level moving to a more detailed formal account in the following three sections.<sup>24</sup>

Consider the following two utterances Ariadne might address to Barabbas:

- (27) a. Ariadne: Bye  
b. Ariadne: Did Jo leave?

What is Ariadne trying to convey to Barabbas? In (27a), Ariadne is parting from Barabbas, whereas in (27b) she is asking him a question. What question? Not knowing Ariadne, her acquaintances, or the circumstances under which she asked (27b), essentially all we can say—a bit pedantically—is that she is attempting to convey to Barabbas her question whether the property she has referred to with her utterance of *leave* holds of the person she has referred to with the name *Jo*. In the best of circumstances, Barabbas will cope fine: he knows *Jo* refers to, say, Barabbas's cousin Jo Sredovic (as opposed to the Joes Cocker, Strummer, or Cole, say) and that at issue is whether, say, Jo is no longer staying at Barabbas's place (as opposed to having left for a trip or resigned her job or left for the day ...). Nonetheless, the contrast between (27a) and (27b) brings out a familiar but crucial feature of natural language, the existence of utterances whose content is context-dependent. The task of understanding an utterance involves finding values for those contextually varying aspects of meaning, as encoded already in the dependence of CONTENT on DGB-PARAMS, introduced in Chapter 5.

Modelling this condition for successful grounding provides one obvious constraint on the representation of utterance types: as we already pointed out in Chapter 2, such a representation must involve in some way a means of varying a set of certain parameters (the *contextual parameters*) to contents. This much is familiar already from early work on context dependence by Montague (1974a) and Kaplan (1989) to more recent work in situation semantics (Gawron and Peters 1990, Cooper and Poesio 1994). In the earlier works in particular, context dependence was viewed as, more or less, restricted to a small class of indexical expressions (along the lines of 'I', 'You', 'now', 'here'). However, as we discussed in section 6.2, once one takes clarificational potential seriously, even the utterly mundane (27b) hints that the affected class of expressions is far larger, including *minimally* all open-class lexical items. Whereas in the case of

<sup>24</sup> The model informally sketched here draws substantially on the one developed in Ginzburg and Cooper (2004). One difference is in the ontological category used to represent utterances. In the latter paper, formalized using HPSG<sub>TFS</sub>, this was taken to be a *sign*. Here, given the availability of types and tokens in the underlying formalism, TTR, we use *locutionary propositions*. Differences concerning formalization are discussed briefly in the summary section.

indexicals the semantic values are fixed to a certain extent directly by the external context,<sup>25</sup> in general these values are fixed with reference to<sup>26</sup> the speaker's intentions. In a room containing a number of people named 'John', (28a) is a possible utterance, even without gestural acts that pick a single referent; similar comments apply to the pronominal use in (28b) and both the verb 'leave' and temporal modifier 'very soon' in (28c). Successful reference is not guaranteed, of course.

- (28) a. John should be given the prize.  
 b. Millie met with Melissa and after that she was very satisfied.  
 c. Maxine will leave the institute very soon.

Indeed, one can assimilate to this the so-called literal/metaphorical bifurcation:<sup>27</sup> the potential for metaphorical uses constitutes an additional pervasive case of contextual variability.

What of the *failure* to ground an utterance? There are many possible reasons, but as we saw in section 6.2 one can divide these in three main classes:

1. content-specific: uncertainty about resolving the content of *u*
2. generalized problems:
  - (a) form recognition
  - (b) utterance plan recognition.

This partition derives from whether or not the cause of failure relates to the semantic complexity of *u*. Even an apparently straightforward utterance such as (27a) can give rise to CRs relating to its form or to its utterance plan.<sup>28</sup>

I will return to the generalized problem cases later—repetition in section 6.8, utterance plan recognition in Chapter 8. What do problems with resolving the content of *u* amount to? There are two main cases to consider here, the first of which is, to a certain extent, an instance of the second. The first case is lexical ignorance—Barabbas

<sup>25</sup> *Direct reference* was a big theme in the philosophy of language of the decade and a bit starting around 1977 (e.g. Perry 1977). This means that, for instance, 'I' denotes whoever is speaking, regardless of the identity the speaker associates with him or herself.

<sup>26</sup> I use this hedged collocation rather than the more direct 'by' to allow for cases such as those discussed by Clark (1996) or (i) below, where the speaker's initial underspecified reference is fixed by the addressee.

(i) Maggie: Yeah. He'll probably get (pause) what's his name?

Kathleen: Who?

Maggie: Er (pause) tt (pause) him!

Kathleen: Which him?

Him that lived on caravan site?

Maggie: That's him! (BNC, KCX)

<sup>27</sup> See Stone (2008) for the relationship between metaphorical meaning and dialogue understanding.

<sup>28</sup> Of course even the *content* of 'Bye' is contextually dependent—an overhearer of Ariadne's utterance of 'Bye' needs to recognize Ariadne and that she is addressing Barabbas. But in anything but utterly bizarre circumstances (for human as opposed to robotic agents), by the point 'Bye' has been uttered Barabbas will have recognized Ariadne. The only contextual dependence of the utterance can lie in an unusual associated utterance plan or it being unexpected that Ariadne is terminating the conversation at that point.

simply has not come across one of the words or idioms used by Ariadne. Of course such cases are more common in interaction among linguistic unequals (adult/child, local/foreigner, scholarly/illiterate, sports-mad/sports-abhorrent etc.), but in increasingly complex societies it can occur even among 'well-matched' individuals who could hardly be argued to be speaking distinct dialects. Example (29a) is a headline from the financial pages of a daily newspaper: its understanding at the most basic level intended involves awareness of what (or who) is QinetiQ and what constitutes opening up a flotation, facts which once eluded the author of this book (reading over a neighbour's shoulder in a train); (29b) is from an email exchange between a departmental administrator and a (highly experienced) professor of that department:

- (29) a. Ariadne: QinetiQ is finally opening up their flotation  
 b. Administrator: The School Office would like to know whether we are holding a PG Programme board prior to the School Board on 1st March.  
 Prof. C: What is a 'PG Programme board' please? (attested example in a UK university)

The second and more general problem case for Barabbas is that, there is (at least) one word or phrase  $u_i$  where, despite his being familiar with  $u_i$ , he is uncertain of the intended use  $u_i.cont$  in the current context. This is the more general case for a number of reasons: for a start, unless the need is explicitly indicated by Barabbas, the response to a CR provides an answer indicating the intended reference in the current context, not the meaning. Even if lexical ignorance is clearly indicated by Barabbas, Ariadne is unlikely to provide an exhaustive explication of meaning, but will elucidate briefly the relevant intended sense. Moreover, at least with regards to adult native language speech, we can assume that lexical ignorance is restricted to open-class items and that morphology guides Barabbas at least to recognizing the general semantic class the word (or even phrasal idiom) belongs to. Hence, an answer about reference rather than meaning will be of some utility, paving the way for a modification of his permanent lexicon.

Given inability to ground Ariadne's utterance, Barabbas can decide to let things lie, in other words simply to *existentially quantify* the problematic semantic element and work with this somewhat weakened content. This is exemplified in (30), due to David Schlangen (p.c.), where Max could respond as shown despite being ignorant of who Belulah is:

- (30) Babette: Did Belulah call? Max: No one called today.

See Exercise 53 below, following Israel and Perry (1991), Cooper (1998b), for the operation of existential quantification of contextual parameters.<sup>29</sup>

<sup>29</sup> This is also the strategy a reader of a book engages in when out of laziness s/he does not look up a word whose sense is not obvious.

Alternatively, Barabbas can decide to engage in discussion concerning the problematic content,  $u_i.cont$ , posing one of a number of possible questions concerning  $u_i.cont$ . A clarification question does not relate to an existing question in QUD—it is introduced to discuss a side issue, hence clarification interaction involves a form of QUD accommodation. In this sense there will be a commonality with our earlier account of initiating moves. As we saw in section 6.2, there is a range of coherent clarification questions by Barabbas concerning  $u$  which the original speaker, Ariadne, can understand, and find perfectly coherent, most prototypically RF:

(31) Ariadne: Did Jo leave?

Barabbas: Jo?

What underpins the asking of content-oriented CRs? I assume the existence of (for now) two Clarification Context Update Rules (CCURs).<sup>30</sup> The status of the CCURs as components of conversational competence is entirely akin to that of other conversational rules such as those introduced in Chapter 4. Reprise Fragments give us some indication concerning both the contents of possible clarification questions as well as the requisite QUD accommodation. One type of clarification interaction, which I will refer to as Parameter Identification, is a generalization of the clarification interaction we saw in the ack-huh game: it involves conveying a question paraphrasable as *what is the intended reference of  $u_i$ ?* (for (31) *who are you referring to as 'Jo'?*). The partially updated context in which such a clarification takes place is such that simply repeating the segmental phonology of  $u_i$  using a distinctive intonation (e.g. focus-associated rise with spreading high tone or focus-associated fall with spreading low tone) enables that question to be expressed. Parameter Identification concerns itself exclusively with a particular sub-utterance, which is unresolvable, with no commitment to understanding the rest of the utterance. The other type of clarification interaction, one which I will refer to as Parameter Focussing, enables Barabbas to seek *confirmation* concerning  $u_i.cont$  (for (31)—*Was Ariadne asking if JO SREDOVIC (of all people) left?*). In order to affect such a focus, Barabbas accommodates into QUD a question that arises by instantiating all contextual parameters except for  $u_i.cont$  and abstracting over  $u_i.cont$ . The confirmation question can be conveyed in such a context by uttering with intonational patterns mentioned before any apparently co-referential phrase whose syntactic category is identical to  $u_i$ 's (for (31) an NP such as 'My cousin', 'That woman we met last week', 'Jo Sredovic' etc.).

<sup>30</sup> Called in previous work *coercion operations*, a term originating in theoretical computer science with respect to treatment of functions that receive input of the wrong type. The analogy will hold—CRification is triggered by incomplete contextual parameter assignments. But more seems to be happening here than merely the reshaping of a function—an *interaction* towards fixing the incomplete context.

So far I have described informally the two main branches that can ensue following an utterance: grounding vs CRification. An important issue raised by clarification interaction concerning an utterance *u* is: what is the contextual input of CCURs? Before addressing this issue directly, let me mention an enhancement we will need to make to the structure of the DGB to process CRs, an enhancement briefly mentioned in section 6.3. We posit a list-valued field *PENDING* which keeps track of ungrounded utterances.<sup>31</sup> These remain on *PENDING* until information resolving the requisite CR is forthcoming. The question about the input to CCURs can be reformulated as: what type is associated with the field *PENDING*?

Consider first the perspective of the speaker of a given utterance. As soon as she has completed her utterance, we *need* to assume that she updates her information state with the content of her utterance. She can after all keep the turn and follow up on the initial utterance using material from the initial utterance:

(32) Ariadne: (1) Did Jo leave? (2) (If so,) When?

However, whereas Ariadne is, at least to a first approximation, omniscient about her own utterance,<sup>32</sup> there is of course no guarantee that Barabbas, the addressee, will manage to combine the information he obtained in processing the utterance with his existing linguistic and background assumptions to yield a coherent content. As I stressed above, Ariadne would find (a range of) requests for clarification from Barabbas to be entirely coherent and interpretable. In light of this, we need to assume that Ariadne's information state contains some representation which enables her to interpret and recognize the coherence of a class of possible clarification queries that Barabbas might make. Conversely, in so far as possible, one would like this representation to be 'independently motivated', i.e. a representation that Ariadne might plausibly be expected to have associated with her information state for her own processing needs. What entities are plausible candidates?

The most parsimonious candidate is of course the content of the utterance, since we know that this has to be in Ariadne's information state regardless. The problem is, however, that on most conceptions of content, be they relatively coarse-grained (e.g. possible-worlds-based) or more fine-grained (e.g. as explicated in situation

<sup>31</sup> Note that the assumption about the ordering on *PENDING* is of a quite different nature from the corresponding one involving *QUD*, discussed in section 4.3.3, where I argued that the ordering on *QUD* ultimately derives from the conversational participants' not necessarily public intentions. The assumption about the last-in-first-out nature of *PENDING* presumably derives from 'lower level' constraints on memory.

<sup>32</sup> Malapropic speakers undermine this assumption. Their behaviour is, nonetheless, viewed as comical at best or even bizarre. In Chapter 8, I also make allowance for self-correction.

semantics), the content is simply insufficiently structured to enable the requisite ‘back-tracking’ to be defined. Thus (uttered in a single context) the contents expressed by the sentences in (33a–c) are all taken to be identical, a singular proposition involving the individual *j* and the property *President*, as in (33d):

- (33) a. Ariadne: Jill is the president.  
 b. Ariadne: [Pointing at Jill] She is the president.  
 c. Ariadne: That tall woman over there is the president.  
 d.  $\langle \text{President}(j) \rangle$

Clearly, (33a–c) can lead to different and pairwise inappropriate clarification queries:

- (34) a. Barabbas: Who is Jill?  
 b. Ariadne: She?  
 c. Barabbas: Which tall woman?

Yet more significantly—Barabbas will not have the content as appropriate input—commonly a CR is posed precisely by an interlocutor who cannot fully resolve an utterance’s content.

A much more promising candidate than the content of an utterance is its meaning, suitably conceived. As we mentioned already in Chapter 2, meanings in something like the intended sense were originally introduced by Montague and Kaplan to help explicate the logical properties of indexicals. Chapter 3 provides a TTR implementation of meanings as functions from records, which provide values for certain parameters (the *contextual parameters*), to contents. Meanings provide a useful notion for conceptualizing grounding/clarification potential. This is because the range of contextual parameters offers a possible characterization of the contextually variable and hence potentially problematic constituents of utterance content. It is important to stress, in line with our observations above about words such as *leave*, *flotation*, and the like, that if we conceive of meanings as entities which characterize potential sources of misunderstanding, then predicates associated with verbs, common nouns, and adjectives will also need to be assumed to project parameters which require instantiation in context. So something like (35b) is required rather than the ‘classical’ (35a). I discussed experimental evidence about which lexical categories are viewed to be clarifiable in section 6.2.

- (35) a. I hear you.  
 $\lambda s, a, t \text{Hear}(s, a, t)$ , where *s* is addressing *a* at *t*.  
 b. QinetiQ opened up their flotation.  
 $\lambda s, a, q, f, t \text{Assert}(s, a, O(q, f, t))$ , where *s* is addressing *a* at *t*, *q* is named ‘Qinetiq’, *O* refers to a concept of ‘opening up’, *f* has the property referred to as a ‘flotation’, ...



Nonetheless although meanings in the generalized sense we have introduced here are, in some form or other, a key component in characterizing clarification potential, they are not quite sufficient, even when we limit ourselves to content-oriented CRs. One problem is the familiar one of grain. In terms of the concept or property that they represent, one would be hard-pressed to distinguish the meanings of words such as *attorney*, *lawyer*, and *advocate*.<sup>33</sup> And yet, since knowledge of language is not uniform, it is clear that the clarification potential of the sentences in (36) is not identical. One can be acquainted with the word *lawyer* but not with *attorney*. Moreover, since the words are distinct phonologically, which word was used initially makes a difference as to how the clarification can be formulated:

- (36) a. Ariadne: Jo is a lawyer. Barabbas: A lawyer?/What do you mean a lawyer?/#What do you mean an advocate?/#What do you mean an attorney?  
 b. Ariadne: Jo is an advocate. Barabbas: #What do you mean a lawyer?/An advocate?/What do you mean an advocate?/#What do you mean an attorney?  
 c. Ariadne: Jo is an attorney. Barabbas: #What do you mean a lawyer?/#What do you mean an advocate?/What do you mean an attorney?/An attorney?

A related point arises from considering the following punful exchange:

- (37) Ariadne: It rained horribly yesterday.  
 Barabbas: It?  
 Ariadne (laughs): Oh the sky, I suppose.

Although Barabbas's clarification query is sheer smart-aleckry, it is interpretable, despite the fact that the expletive does not, on most accounts, contribute an argument filler. This interpretability depends on reference to the utterance pronounced *it*, not simply on whether a meaning parameter is projected.

These data, together with our earlier data on syntactic and phonological parallelism exhibited by RF, suggest that a credible candidate for the type of PENDING—though even this candidate will need some fine-tuning—is the utterance *type* associated by the grammar with the clarification target.<sup>34</sup>

<sup>33</sup> As Eve Clark (p.c.) has reminded me, these words, popular figures in philosophical puzzles, are operative in distinct English-speaking countries, so on the whole do not really coexist in most speakers' lexicons.

<sup>34</sup> The use of the definite article with respect to the grammatically associated type is potentially problematic, both conceptually, given considerations of ambiguity for the addressee, and formally, given that we are using a polymorphic-type system. I will address both points below, but will continue to use the definite for simplicity.

It is not difficult to see that using an utterance's grammatically associated type as input to CCURs eliminates the problem of grain exemplified above in (36). Similarly with respect to allowing parallelism phenomena associated with RF to be explicated—the details of this are discussed in Chapter 7. Just a little less obvious perhaps is why the strategy of preserving the utterance type in the DGB is not merely a worst-case one in preparation for CRs by the addressee, but one that has independent utility. Here we can point to the fact that even in case of successful grounding, as we observed already in section 2.2, various presuppositions get introduced into the DGB whose source can only derive from the utterance type, not merely its content or meaning.

In talking about *the* utterance type grammatically associated with an utterance some care must be employed, however. Using the definite article is somewhat dangerous. Uniqueness is normally correct as far as the speaker goes, since the grammar allows us to build such a type, e.g. (38):

$$(38) \quad \text{IGH} =_{\text{def}} \left[ \begin{array}{l} \text{phon : is georges here} \\ \text{cat = V[+fin,+root] : syncat} \\ \text{constits = \{is, georges, here, is georges here\} : set(sign)} \\ \\ \text{dgb-params : } \left[ \begin{array}{l} \text{spkr: IND} \\ \text{addr: IND} \\ \text{c1 : addressing(spkr, addr)} \\ \text{s0: SIT} \\ \text{l: LOC} \\ \text{g: IND} \\ \text{c3: Named(g, 'georges')} \end{array} \right] \\ \\ \text{cont= Ask(spkr, addr, ?} \left[ \begin{array}{l} \text{sit = s0} \\ \text{sit-type = [c : In(l, g)]} \end{array} \right] \text{) : IllocProp} \end{array} \right]$$

On the other hand, we cannot assume that the addressee will always be able to zero in on such a type from his speech percept, for a variety of reasons—ambiguity, lexical ignorance, noisy environment. Since these are common causes for clarification, we need to ensure that we do not define clarification potential in terms of an entity we cannot ensure is available to the addressee.

The upshot of this is that on occasion the addressee will need to use a somewhat underspecified type as input to the CCUR, which TTR offers straightforwardly,<sup>35</sup>

<sup>35</sup> See Cooper (2008).

though I offer here no theory of ‘partial parsing’ that will explain the emergence of such types. Mishearing (38) could lead to a type such as (39), which in addition to IllocProp we introduced in Chapter 4, presupposes very general types, *EnglishString* and 3-wordset that characterize English speech events of length three:

$$(39) \quad T_{3\text{word-polar}} = \left[ \begin{array}{l} \text{phon} : \text{EnglishString} \\ \text{cat} = V[+\text{fin}, +\text{root}] : \text{syncat} \\ \text{constits} : 3\text{-wordset} \\ \text{dgb-params} : \left[ \begin{array}{l} \text{spkr} : \text{IND} \\ \text{addr} : \text{IND} \\ \text{cl} : \text{address}(s, a) \end{array} \right] \\ \text{cont} : \text{IllocProp} \end{array} \right]$$

Although the utterance type associated with an utterance is a necessary component in the input to grounding/CCURs, it is not sufficient. The most obvious components missing are those context-dependent components of content—the contextual parameters together with the entire utterance’s content which, depending on the information available to the interlocutor, the contextual parameters partially or fully *instantiate*. More generally, what are missing are the sub-utterance tokens, the constituents of the utterance. These figure in the content of CRs. The examples in (40), repeated from Chapter 2, all involve an intrinsic anaphoricity:

- (40) a. A: Max is leaving. B: leaving?  
 b. A: Did Bo leave? B: Who is Bo?  
 c. A: We’re fed up. B: Who is we?

In these examples the issue is not *what do you mean by leaving* or *who is Bo/we in general*, but *what do you mean by leaving* or *who is Bo/we in the sub-utterance needing clarification*.

The conclusion this leads us about the type of PENDING is that it combines tokens of the utterance and constituents, the elements making up the content of the difference, and the utterance type. An entity that fits this specification is something we will call a *locutionary proposition*, a (n Austinian) proposition (in the sense of section 3.2.4) whose SIT field is instantiated by (a record characterizing) the utterance and whose SIT-TYPE field is filled by the utterance type that emerges as a consequence of parsing. An example of a locutionary proposition is (41), corresponding to an utterance of the sentence ‘Is George here?’:

$$\begin{aligned}
 (41) \quad & \left[ \begin{array}{l} \text{sit} = \left[ \begin{array}{l} \text{phon} = \text{izjorjhiya} \\ \text{cat} = V[+\text{fin}, +\text{root}] \\ \text{constits} = \left\{ \begin{array}{l} \text{u1(iz), u2(jorj),} \\ \text{u3(hiya), u4(izjorjhiya)} \end{array} \right\} \\ \\ \text{dgb-params} = \left[ \begin{array}{l} \text{spkr} = A \\ \text{addr} = B \\ \text{s0} = \text{sit1} \\ l = l0 \\ g = g0 \\ c3 = \text{pr1} \end{array} \right] \\ \\ \text{cont} = \text{Ask}(A, B, ? \left[ \begin{array}{l} \text{sit} = \text{sit1} \\ \text{sit-type} = [c : \text{In}(l0, g0)] \end{array} \right] ) \end{array} \right] \\ \\ \text{sit-type} = \left[ \begin{array}{l} \text{phon} : \text{is george here} \\ \text{cat} = V[+\text{fin}, +\text{root}] : \text{syncat} \\ \text{constits} = \{\text{is, george, here, is george here}\} : \text{set}(\text{sign}) \\ \\ \text{dgb-params} : \left[ \begin{array}{l} \text{spkr: IND} \\ \text{addr: IND} \\ \text{c1 : address(s, a)} \\ \text{s0: SIT} \\ l: \text{LOC} \\ g: \text{IND} \\ \text{c3: Named(g, 'george')} \end{array} \right] \\ \\ \text{cont} = \text{Ask}(\text{spkr}, \text{addr}, ? \left[ \begin{array}{l} \text{sit} = \text{s0} \\ \text{sit-type} = [c : \text{In}(l, g)] \end{array} \right] ) : \text{IllocProp} \end{array} \right] \end{array}
 \end{aligned}$$

Now (41) is a rather unwieldy representation, from a graphical point of view. To the extent it is necessary to display both type and token simultaneously, I will notate this as a single type, where the token values are represented as manifest fields;<sup>36</sup> usually more radical abbreviations will be employed. In the single type notation, (41) becomes (42):

$$(42) \left[ \begin{array}{l} \text{phon} = \text{izjorjhiya} : \text{is george here} \\ \text{cat} = V[+\text{fin}, +\text{root}] : \text{syncat} \\ \text{constits} = \left\{ \begin{array}{l} \text{u1(iz), u2(jorj),} \\ \text{u3(hiya), u4( izjorjhiya)} \end{array} \right\} : \{\text{is, george, here, is george here}\} \\ \text{dgb-params} = \left[ \begin{array}{l} \text{spkr} = A \\ \text{addr} = B \\ \text{s0} = \text{sitl} \\ \text{l} = \text{l0} \\ \text{g} = \text{g0} \\ \text{c3} = \text{prl} \end{array} \right] : \left[ \begin{array}{l} \text{spkr: IND} \\ \text{addr: IND} \\ \text{c1 : address(s,a)} \\ \text{s0: SIT} \\ \text{l: LOC} \\ \text{g: IND} \\ \text{c3: Named(g, 'george')} \end{array} \right] \\ \text{cont} = \text{Ask}(\text{spkr}, \text{addr}, ? \left[ \begin{array}{l} \text{sit} = \text{s0} \\ \text{sit-type} = \left[ \text{c : In(l, g)} \right] \end{array} \right]) : \text{IllocProp} \end{array} \right]$$

Let us summarize this section: we started by informally describing the grounding process in which an utterance gets integrated in an IS or spawns a clarification. One conclusion to emerge from this discussion is that conversational participants possess as part of their dialogic competence knowledge of rules integrating clarification interaction into contexts: CCURs are operations by means of which the partial understanding of an utterance *u* can effect an update of an IS in which a clarification question about *u* can be posed. Consideration of the range of the required inputs and outputs of such operations, as evinced by the phenomenon of RF, leads to the conclusion that **updates resulting from utterances cannot be defined in purely semantic terms**. We considered what representational entity could serve as the input for CCURs, as well as being entities that both speaker and addressee have reason to preserve in their DGBs. After considering various candidates (contents, meanings, signs, ...), I ended up plumping for *locutionary propositions*—entities that combine utterance and sub-utterance tokens with grammatically associated utterance types. This leads to a modification in our definition of the DGBType so that it contains a field Pending that keeps track of ungrounded utterances, as in (43):

<sup>36</sup> This notation is inspired by a comment of Robin Cooper's. In fact, if we wished to avoid using Austinian propositions, we could use instantiated types such as (42) as an alternative.

(43) DGBType (final version) =<sub>def</sub>

|                  |                                                      |
|------------------|------------------------------------------------------|
| spkr             | : IND                                                |
| addr             | : IND                                                |
| utt-time         | : Time                                               |
| c-utt            | : addressing(spkr, addr, utt-time)                   |
| Pending          | : list(LocProp)                                      |
| Moves            | : list(LocProp)                                      |
| Facts            | : Set(Prop)                                          |
| QUD              | : poset(Question)                                    |
| non-resolve-cond | : $\forall q \in QUD[\neg \text{Resolve}(FACTS, q)]$ |

An important issue I will discuss though not fully resolve in section 6.5 is the type of Moves: LatestMove will be required to be henceforth of type *LocProp*, for reasons similar to those motivating the type of Pending. But what of earlier moves? Since the view that all moves have the same type in the DGB is the simpler one I will assume that in (43), though both options will be considered in our analyses.

## 6.5 Successful Grounding

In this and the next two sections I develop a relatively detailed model for integrating utterances into the DGB. The model spells out the informal picture sketched in previous sections, which can be summarized in terms of the protocol in (44):

(44) **Utterance-processing protocol**

For an agent A with IS *I*: if a locutionary proposition  $p_u = \begin{bmatrix} \text{sit} & = & u \\ \text{sit-type} & = & T_u \end{bmatrix}$

is Maximal in PENDING:

(a) Try to:

1. Construct a witness for the contextual parameters of  $T_u$
2. Integrate  $p_u$  in A.DGB using a Moves update rule
3. If (1)–(2) are successful,  $u$  is removed from Pending

(b) Else: seek a witness for the contextual parameters of  $T_u$  by asking a CR

I make no claims as such about the cognitive reality of this model, in particular for the way it is broken down into successive stages. These are posited solely due to semantic necessity.

### 6.5.1 Starting Point

There are two presuppositions that form a necessary starting point of utterance integration. One is the assumption that an utterance has occurred, perceived by both

participants; the other that the utterance—a speech event  $u$ —can be classified using a grammatical type  $T_u$ , the output of a parser. Given this, the locutionary proposition in which the speech event fills the *SIT* field and the grammatical type fills the *SIT-TYPE* field is integrated into the DGB as (the maximal) element of *PENDING*. For reasons that will become clear shortly we separate this initial stage from what could in principle be viewed as occurring simultaneously—the instantiation of the meaning with (non-core speech event) contextual parameters. This initial step is assumed to occur with any just-perceived utterance, whether it will be grounded or will trigger CRification. Hence we cannot in general assume that a conversational participant who is updating *PENDING* believes that  $u : T_u$  since, for instance,  $u$  might lack witnesses for contextual parameters of  $T_u$ . What we can assume in either case is something weaker: the belief that there is a record  $u1$  which is an extension of  $u$  such that  $u1 : T_u$ . This is the semantics we associate with the predicate ‘Classifies’, relating utterance tokens and grammatical types.<sup>37</sup>

The basic rule for updating *PENDING* can, thus, be stated concisely as in (45): this states that given a DGB  $d$  and a speech event  $u$ , presupposed to be the most recent utterance,<sup>38,39</sup> and classifiable by the grammatical type  $T_u$ , the locutionary proposition comprising  $u$  and  $T_u$  becomes the maximal element in *PENDING* and the contextual assignment in  $u$  determines the values for *SPKR* and *ADDR*:

(45) Pending Update =<sub>def</sub>

$$\left[ \begin{array}{lcl} \text{pre} & : & \left[ \begin{array}{l} T_u : \text{GramType} \\ u : \text{Rec} \\ d : \text{dgb} \\ c1 : \text{member}(\text{MostRecentSpeechEvent}(u), d.\text{FACTS}) \\ c2 : \text{member}(\text{Classifies}(T_u, u), d.\text{FACTS}) \end{array} \right] \\ \text{effects} & : & \left[ \begin{array}{l} \text{spkr} = u.\text{dgb-params.spkr} : \text{IND} \\ \text{addr} = u.\text{dgb-params.addr} : \text{IND} \\ \text{pending} = \left\langle \left[ \begin{array}{ll} \text{sit} & = u \\ \text{sit-type} & = T_u \end{array} \right], \text{pre.pending} \right\rangle : \text{list}(\text{locProp}) \end{array} \right] \end{array} \right]$$

<sup>37</sup> Strictly speaking, ‘Classifies’ should be relative to a grammar  $\mathcal{G}$ , of which  $T_u$  is an element. But I will ignore this, for the most part.

<sup>38</sup> The notation ‘MostRecentSpeechEvent( $u$ )’ is intuitive but flawed, given the context dependence of the predicate ‘MostRecent’. In a treatment that incorporated tense and/or event structure, a satisfactory treatment could be developed.

<sup>39</sup> This presupposition’s emergence is a prime candidate for explication in terms of the *copresence heuristics* of Clark and Marshall (1981).

A fussier version of (45) is (46), where I have made the structure of  $u$  more explicit, as of the presuppositions  $p1$ , that  $u$  is the most recent utterance and  $p2$ , that  $u$  is classifiable by the grammatical type  $T_u$ . This, in turn, involves the existence of situations *audio* and *parse* in which  $u$  was perceived and parsed. I will on the whole be a bit sloppy and prefer formulations like (45) over the more official but less readable (46):

(46) Pending Update =<sub>def</sub>

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} T_u : \text{GramType} \\ \\ u = \left[ \begin{array}{ll} \text{phon} & = p \\ \text{cat} & = \text{syncat1} \\ \text{constits} & = \text{con1} \\ \text{dgb-params} & = \left[ \begin{array}{ll} \text{spkr} & = s \\ \text{addr} & = a0 \\ \text{utt-time} & = t0 \end{array} \right] \\ \text{cont} & = \text{cont1} \end{array} \right] : \text{sign} \\ \\ \text{audio} : \text{Rec} \\ \text{parse} : \text{Rec} \\ \text{d} : \text{dgb} \\ p1 = \left[ \begin{array}{ll} \text{sit} & = \text{audio} \\ \text{sit-type} & = \text{MostRecentSpeechEvent}(u) \end{array} \right] : \text{Prop} \\ p2 = \left[ \begin{array}{ll} \text{sit} & = \text{parse} \\ \text{sit-type} & = \text{Classifies}(T_u, u) \end{array} \right] : \text{Prop} \\ c1 : \text{member}(p1, \text{FACTS}) \\ c2 : \text{member}(p2, \text{FACTS}) \end{array} \right] \\ \\ \text{effects} : \left[ \begin{array}{l} \text{spkr} = u.\text{dgb-params}.\text{spkr} : \text{IND} \\ \text{addr} = u.\text{dgb-params}.\text{addr} : \text{IND} \\ \text{pending} = \left\langle \left[ \begin{array}{ll} \text{sit} & = u \\ \text{sit-type} & = T_u \end{array} \right], \text{pre.pending} \right\rangle : \text{list}(\text{locProp}) \end{array} \right] \end{array} \right]$$

Semantically, the most significant aspect of utterance integration is anchoring the contextual parameters of the utterance's type relative to the context provided by the DGB. Formally speaking, I conceptualize this as the search within the DGB for the *witness intended by the speaker for the DGB-PARAMS label from  $T_u$* . This is for the moment our version of the *grounding criterion* (Clark and Schaefer 1989). Contextual instantiation can result in a record  $u \uplus \text{dgb-param-witness}$  such that



$u \uplus \text{dgb-paramwitness} : T_u$ —the utterance can be grounded. Alternatively, the most one can do is construct a witness which is a sub-record of  $T_u.\text{dgb} - \text{params}$ , in which case CRification can ensue. Contextual instantiation will occur as soon as an utterance has taken place, but it can also take place subsequently, as when more information is provided as a consequence of CRification—hence our separation of this process from Pending Update.

In order to formulate Contextual Instantiation we need to introduce the relation of *ContextualExtension* that holds between two utterance records:

- (47)  $\text{ContextualExtension}(u, v)$  iff  $u, v : \text{sign}$  and  $u.f = v.f$  on all sign fields except DGB-PARAMS, and  $u.\text{dgb-params} \sqsubseteq v.\text{dgb-params}$

In light of this, Contextual Instantiation can be stated as follows: given the Max-Pending locutionary proposition  $p = \begin{bmatrix} \text{sit} & = u \\ \text{sit-type} & = T_u \end{bmatrix}$  and a record  $w$  such that (i)  $w$  contextually extends  $u$  and such that (ii)  $w.\text{dgb} - \text{params}$  is a sub-record of the dgb-param anchoring intended by  $u$ 's speaker, integrate  $w$  into  $p$ :<sup>40</sup>

- (48)  $\text{Contextual Instantiation} =_{\text{def}}$

$$\left[ \begin{array}{lcl} \text{pre} & : & \begin{bmatrix} T_u : \text{GramType} \\ u : \text{Rec} \\ \text{spkr} : \text{IND} \\ \text{MaxPending} = \begin{bmatrix} \text{sit} & = u \\ \text{sit-type} & = T_u \end{bmatrix} : \text{LocProp} \\ w : \text{Rec} \\ c1 : \text{ContextualExtension}(u, w) \\ p1 = \text{IntendClassify}(\text{spkr}, w.\text{dgb-params}, T_u.\text{dgb-params}) : \text{Prop} \\ c2 : \text{member}(p1, \text{private.beliefs}) \end{bmatrix} \\ \text{effects} & : & \begin{bmatrix} \text{MaxPending} = \begin{bmatrix} \text{sit} & = w \\ \text{sit-type} & = T_u \end{bmatrix} : \text{LocProp} \end{bmatrix} \end{array} \right]$$

In line with our comments in Chapter 5 about representing meanings using the separate fields DGB-PARAMS and CONTENT, when we instantiate a content, we preserve this separation. Thus, a witness for (49a) will be (49b):<sup>41</sup>

<sup>40</sup> Henceforth, I adopt the convention that utterance records that contain contextual witnesses are designated with  $w$ :  $w0, w1, w2, \dots$

<sup>41</sup> I would like to thank Robin Cooper for his suggestions, which led me to a significant rethink here.

- (49) a. 
$$\left[ \begin{array}{l} \text{dgb-params : } \left[ \begin{array}{l} \text{spkr : IND} \\ \text{addr : IND} \\ \text{utt-time : Time} \\ \text{c1 : address(spkr, addr)} \\ \text{s0 : SIT} \\ \text{l : LOC} \\ \text{g : IND} \\ \text{c3 : Named(g, 'georges')} \end{array} \right] \\ \text{cont= Ask(spkr, addr, ?} \left[ \begin{array}{l} \text{sit} = \text{s0} \\ \text{sit-type} = [\text{c : In(l, g)}] \end{array} \right] \text{) : IllocProp} \end{array} \right]$$
- b. 
$$\left[ \begin{array}{l} \text{dgb-params = } \left[ \begin{array}{l} \text{spkr} = \text{A} \\ \text{addr} = \text{B} \\ \text{c1} = \text{utt1} \\ \text{s0} = \text{sit1} \\ \text{l} = \text{l0} \\ \text{g} = \text{g0} \\ \text{c3} = \text{pr1} \end{array} \right] \\ \text{cont} = \text{Ask(spkr, addr, ?} \left[ \begin{array}{l} \text{sit} = \text{s0} \\ \text{sit-type} = [\text{c : In(l, g)}] \end{array} \right] \text{) : IllocProp} \end{array} \right]$$

Apart from the fact that this allows for instantiation to be an incremental process, crucial for CRification, it also enables revisions to be made straightforwardly. Corrections seem to have a potentially unbounded range, as illustrated in (50a, b). In both cases a correction to what seems like a grounded contextual referent occurs and is integrated seamlessly. Note that this is possible not merely with nominal referents (50a), but also with a verbal sub-utterance (50b):

- (50) a. A: Is Bo here? B: He left this morning. A: Oh really? B: Yes, his flight left at 4 am. A: What day did he arrive last week? B: Tuesday. A: Wait, I meant Bo Smith. B: Oh, OK, he's in Paris at present.  
 b. A: When did Bo leave? B: Late last night. A: Oh really? B: Yes, a truck came for his stuff at 4 am. He was helping them load it up. A: Wait, I meant resign. B: Oh, that happened last month.

The obvious conclusion from (50) is that content needs to allow the values of contextual parameters to be modifiable. Maintaining DGB-PARAMS and CONT as separate fields, with the instantiation of content emerging indirectly allows for this possibility.

Actually, there are grounds to think we need a somewhat stronger principle than Contextual Instantiation, one which involves not merely extending  $u$  contextually, but also  $T_u$  on other levels as well. Consider the (only mildly fanciful) constructed dialogue in (51): here B could be posing his CR unsure whether ‘off’ is being used as an adjective (synonymously with ‘spoilt’) or as an adverb (synonymously with ‘sent away’):

(51) A: Are the strawberries off? B: Off? A: Spoilt/On the way to the depot.

B’s initial representation of the sub-utterance ‘off’, before CRification, will need to involve underspecification of its syntactic category (e.g. via disjunction). Once she receives an answer to her CR, she will then resolve this underspecification in her representation of A’s original utterance. A more prototypical case arises when a repetition CR occurs. In such a case the desired information is, arguably, for the phonological type *sit-type.phon* of the target utterance. Accepting the CR will involve an update of that field and, as a consequence, of other fields as well.

Hence we posit Pending Extension, which generalizes Contextual instantiation. We define a notion of subsumption between locutionary propositions:<sup>42</sup>

$$(52) \text{ LocPropExtension}(p1 = \begin{bmatrix} \text{sit} & = & u \\ \text{sit-type} & = & T_u \end{bmatrix}, p2 = \begin{bmatrix} \text{sit} & = & v \\ \text{sit-type} & = & T_v \end{bmatrix}) \text{ iff } p1, p2:$$

LocProp and (a) for all sign fields either  $u.f = v.f$  or  $u.f \sqsubset v.f$  and (b)  $T_v \sqsubset T_u$

Given this, we can define Pending Extension as in (53):

$$(53) \text{ Pending Extension} =_{\text{def}} \left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} T_u : \text{GramType} \\ u : \text{Rec} \\ \text{spkr} : \text{IND} \\ \text{MaxPending} = \begin{bmatrix} \text{sit} & = & u \\ \text{sit-type} & = & T_u \end{bmatrix} : \text{LocProp} \\ p1 : \text{LocProp} \\ \text{cl} : \text{Extension}(p, p1) \\ p1 = \text{IntendClassify}(\text{spkr}, u, p1.\text{sit-type}) : \text{Prop} \\ \text{c2} : \text{member}(p1, \text{private.beliefs}) \end{array} \right] \\ \text{effects} : \left[ \text{MaxPending} = p1 : \text{locProp} \right] \end{array} \right]$$

**Exercise 6.5.1** Formulate a rule that updates *MaxPending* such that a subset of DGB-PARAMS becomes existentially quantified in the content.

<sup>42</sup> Whereas a record with more fields subsumes a record with fewer fields, the reverse is true for record types. A record with fewer fields imposes fewer constraints and, therefore, other things being equal will have more witnesses. Hence, in (52) the difference in  $\sqsubset$  at the record and the record-type levels.

**Exercise 6.5.2** In what ways would Pending Extension need to be modified to deal with corrections?

### 6.5.2 Updating Move Update Rules

In Chapter 4, a theory of coherence for moves was developed. This includes rules like QSPEC ('if  $q$  is MaxQuD, A/B can make a  $q$ -specific move'), QCoord ('if  $q$  is MaxQuD, the current speaker can make a  $q$ -independent move'), Accept Move ('if A has asserted  $p$ , B can accept  $p$ '), and Initiating Move ('if QUD is empty, A/B can make a move relevant to the current conversational type'). We need to refine these rules to take into account the fact that new moves 'originate' in PENDING. This is fairly straightforward: it simply involves adding to the preconditions of each such rule the condition that MaxPending is *true*,<sup>43</sup> which in particular requires that the content is properly instantiated and there is an assumption of the utterance being grounded. Technically it will also have the effect of shifting the conditions characterizing the new move to the preconditions.<sup>44</sup> For instance, QSPEC and Accept move become respectively the rules in (54a, b) respectively (new fields are in bold, conditions that used to be in *effects* are italicized).<sup>45</sup> (54a) indicates that, given  $q$  being MaxQUD,  $l$ , the MaxPending utterance, which is true, and  $R(a, b, r)$  being the content of  $l$ , which is  $q$ -specific, then  $l$  can become LatestMove and be removed from Pending; (54b) indicates that, given that the content of the LatestMove is *Assert*( $A, B, p$ ),  $p?$  being MaxQUD,  $l$ , the MaxPending utterance, which is true, and *Accept*( $B, A, p$ ) being the content of  $l$ , turn change having occurred, then  $l$  can become LatestMove and be removed from Pending:

$$(54) \quad \text{a. QSPEC} =_{\text{def}} \left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{qud} = \langle q, \dots \rangle : \text{poset}(\text{Question}) \\ r : \text{AbSemObj} \\ R : \text{IllocRel} \\ \text{I}^{\text{content}} = R(\text{spkr}, \text{addr}, r) : \text{IllocProp} \wedge \text{TrueProp} \\ \text{cl} : \text{Qspecific}(r, q) \\ \text{pending} = \langle l, \text{rest} \rangle : \text{list}(\text{LocProp}) \end{array} \right] \\ \text{effects} : \left[ \begin{array}{l} \text{LatestMove} = l : \text{LocProp} \\ \text{pending} = \text{pre.pending.rest} : \text{list}(\text{LocProp}) \end{array} \right] \end{array} \right]$$

<sup>43</sup> Using the truth of MaxPending, arguably, makes for a crisper formulation, but is not necessary. One can simply declare  $u : \text{Rec} \wedge T_u$ .

<sup>44</sup> This is merely because we introduce MaxPending in the preconditions. It does not change the substance of the rule.

<sup>45</sup> I abbreviate all cases of instantiated content, that is fields of the form ' $X.\text{sit.content}$ ' as ' $X^{\text{content}}$ '.

b. Accept move =<sub>def</sub>

|         |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pre     | : | $\begin{aligned} &\text{spkr: IND} \\ &\text{addr: IND} \\ &p : \text{Prop} \\ &\text{LatestMove}^{\text{content}} = \text{Assert}(\text{addr}, \text{spkr}, p) : \text{IllocProp} \\ &\text{qud} = [p?, \text{pre.qud}] : \text{poset}(\text{Question}) \\ &\text{I}^{\text{content}} = \text{Accept}(\text{spkr}, \text{addr}, p) : \text{IllocProp} \wedge \text{TrueProp} \\ &\text{pending} = \langle l, \text{rest} \rangle : \text{list}(\text{LocProp}) \end{aligned}$ |
| effects | : | $\begin{aligned} &\text{LatestMove} = l : \text{LocProp} \\ &\text{pending} = \text{pre.pending.rest} : \text{list}(\text{LocProp}) \end{aligned}$                                                                                                                                                                                                                                                                                                                             |

In refining these rules from Chapter 4 one should note that I am assuming that the value of *LatestMove* is a locutionary proposition. The most important reason for this, subsequently explained in section 6.7, is to enable speakers to coherently integrate CRs to their own utterances into the DGB. But more generally, as far as Moves goes, there is a very significant issue about the representation of linguistic information with which we need to contend. There are two essential choices: either (a) project away the non-semantic information from the utterance information once it is no longer *LatestMove*, in which case all that gets maintained in the long term is the content, or (b) maintain the utterance information intact; (b) is implemented in (54). The choice between these strategies is partly an empirical issue and partly dependent on the area of intended application. The cognitive psychology literature on memory for discourse provides some evidence for the rapid decay of purely structural information in many but by no means all circumstances.<sup>46</sup> Adding to the mixed picture is the existence of a growing body of work that demonstrates the existence of non-semantically-based priming (see e.g. Branigan, Pickering, and Cleland 2000, Garrod and Pickering 2004). Recall that Purver, Ginzburg, and Healey (2001) report that approximately 80 per cent of the clarification requests found in a random sample of the BNC concern the most recent utterance, whereas 96 per cent concern one of the four most recent utterances. The latter two facts suggest that non-semantic information associated with an utterance *u* has decreasing utility for conversationalists the further time passes from *u*.

It seems clear that a strategy which preserves structure has the potential for explaining a wide variety of data and, coupled with a theory of activation, could underpin a mechanistic theory of processing.<sup>47</sup> The two possible formulations of the relevant condition, one structure preserving, the other semantics-only preserving, are given in (55a, b). Formulation (55a) merely involving incrementing Moves with *MaxPending*,

<sup>46</sup> See Fletcher (1994) for a review.

<sup>47</sup> The simplest—and unquestionably most simplistic—theory of activation is temporally driven. For instance, Purver (2004a) imposes a bound on how many utterances are maintained in MOVES—he adopts  $n = 4$  given that 96% of CRs fall in this window.

whereas (55b) involves incrementing Moves with MaxPending but also projecting from the previous LatestMove everything but its content *and* the *dgb*-params which instantiates it, in line with our earlier remarks. Formulation (55b) would actually require us to modify the type of Moves to *rec*, as the projected value is no longer a LocProp, not even a proposition.

$$\begin{aligned}
 (55) \quad & \text{a. Utterance-based version:} \quad \left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{MaxPending} : \text{LocProp} \wedge \text{TrueProp} \\ \text{Moves} = \langle m, \text{rest} \rangle : \text{List}(\text{LocProp}) \end{array} \right] \\ \text{effects} : [\text{Moves} = \langle \text{pre.MaxPending}, \text{pre.Moves} \rangle] \end{array} \right] \\
 & \text{b. Semantics-only version:} \quad \left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{MaxPending} : \text{LocProp} \wedge \text{TrueProp} \\ m : \text{LocProp} \\ \text{Moves} = \langle m, \text{rest} \rangle : \text{List}(\text{Rec}) \end{array} \right] \\ \text{effects} : \left[ \text{Moves} = \left\langle \begin{array}{l} \text{pre.MaxPending}, \\ \text{dgb-params} = \text{pre.MaxPending.m.sit.dgb-params} \\ \text{cont} = \text{pre.MaxPending.m}^{\text{content}} \end{array} \right\rangle, \text{rest} \right] \end{array} \right]
 \end{aligned}$$

**Exercise 6.5.3** Reformulate all existing Move update rules, some in an utterance-based, others in a content-only version.

Let us consider (56) as an illustration of how Pending update, contextual instantiation, and LatestMove update operate.

- (56) A(1): Is George here?  
 B(2): George isn't here.

(57) repeats from (38) a minimally detailed sketch of the type our grammar might associate with (56(1)). I will refer to this type as 'IGH', a mnemonic practice I will use continually henceforth.

$$(57) \quad \text{IGH} =_{\text{def}} \left[ \begin{array}{l} \text{phon} : \text{is george here} \\ \text{cat} = \text{V}[+\text{fin}, +\text{root}] : \text{syncat} \\ \text{constits} = \{\text{is}, \text{george}, \text{here}, \text{is george here}\} : \text{set}(\text{sign}) \\ \text{dgb-params} : \left[ \begin{array}{l} \text{spkr} : \text{IND} \\ \text{addr} : \text{IND} \\ \text{cl} : \text{address}(\text{s}, \text{a}) \\ \text{s0} : \text{SIT} \\ \text{l} : \text{LOC} \\ \text{g} : \text{IND} \\ \text{c3} : \text{Named}(\text{g}, \text{'george'}) \end{array} \right] \\ \text{cont} = \text{Ask}(\text{spkr}, \text{addr}, ? \left[ \begin{array}{l} \text{sit} = \text{s0} \\ \text{sit-type} = [\text{c} : \text{In}(\text{l}, \text{g})] \end{array} \right]) : \text{IllocProp} \end{array} \right]$$

A witness appropriate to serve as the precondition for pending update for (56(1)) is given in (58). The main point to note is the presence in FACTS of appropriate witnesses for the contextual parameters 'I' and 'g', corresponding to the sub-utterances 'here' and 'George':

$$(58) \left[ \begin{array}{l} T_u = IGH \\ \\ u = u0 = \left[ \begin{array}{l} \text{phon} = \text{izjorjhiya} \\ \text{cat} = V[+\text{fin}, +\text{root}] \\ \text{constits} = \left\{ \begin{array}{l} u1(\text{iz}), u2(\text{jorj}), \\ u3(\text{hiya}), u4(\text{izjorjhiya}) \end{array} \right\} \\ \text{dgb-params} = \left[ \begin{array}{ll} \text{spkr} & = A \\ \text{addr} & = B \\ \text{utt-time} & = t0 \end{array} \right] \\ \text{cont} = \text{Ask}(\text{spkr}, \text{addr}, ? \left[ \begin{array}{ll} \text{sit} & = s0 \\ \text{sit-type} & = [c : \text{In}(l, g)] \end{array} \right] ) \\ \\ d = \text{dgb0} = \left[ \begin{array}{ll} \text{spkr} & = A \\ \text{addr} & = B \\ \text{Pending} & = \langle \rangle \\ \text{Moves} & = \langle \rangle \\ \text{qud} & = \langle \rangle \\ \text{facts} & = \left\{ \begin{array}{l} \text{In}(l, \{A, B\}), \text{Named}('George', g), \\ \text{MostRecentSpeechEvent}(u0), \\ \text{Classify}(?IGH, u0) \dots \end{array} \right\} \end{array} \right] \end{array} \right]$$

Applying pending update to (58) will yield a new DGB *dgb1*, as in (59a). Given this DGB a witness exists for DGB-PARAMS that will yield a completely instantiated content, say (59b). Hence, applying Contextual Instantiation, this witness is integrated into PENDING, as in (59c). This, in turn, satisfies the preconditions of (the pendified) Free Speech, which allows us to update to *dgb3*, (59c).<sup>48</sup> Finally, this serves as appropriate input to Ask QUD-incrementation, hence *dgb4*, (59d):

<sup>48</sup> I am appealing to Free Speech rather than the official Initiating move for simplicity, to avoid intricacies of conversational genre.

$$(59) \text{ a. } \text{dgb1} = \left[ \begin{array}{ll} \text{spkr} & = A \\ \text{addr} & = B \\ \text{pending} & = \left\langle \left[ \begin{array}{ll} \text{sit} & = u0 \\ \text{sit-type} & = \text{IGH} \end{array} \right] \right\rangle \\ \text{qud} & = \text{dgb0.qud} \\ \text{facts} & = \text{dgb0.facts} \\ \text{moves} & = \text{dgb0.moves} \end{array} \right]$$

$$\text{b. } \left[ \begin{array}{ll} \text{spkr} & = A \\ \text{addr} & = B \\ \text{utt-time} & = t0 \\ s0 & = \text{sit1} \\ l & = l0 \\ g & = g0 \\ c3 & = \text{pr1} \end{array} \right]$$

$$\text{c. } \text{dgb2} = \left[ \begin{array}{ll} \text{spkr} & = A \\ \text{addr} & = B \\ \text{pending} & = \left\langle \left[ \begin{array}{ll} \text{sit} & = w0 \\ \text{sit-type} & = \text{IGH} \end{array} \right] \right\rangle \\ \text{qud} & = \text{dgb0.qud} \\ \text{facts} & = \text{dgb0.facts} \\ \text{moves} & = \langle \rangle \end{array} \right]$$

$$\text{d. } \text{dgb3} = \left[ \begin{array}{ll} \text{spkr} & = A \\ \text{addr} & = B \\ \text{pending} & = \langle \rangle \\ \text{qud} & = \text{dgb0.qud} \\ \text{facts} & = \text{dgb0.facts} \\ \text{moves} & = \left\langle \left[ \begin{array}{ll} \text{sit} & = w0 \\ \text{sit-type} & = \text{IGH} \end{array} \right] \right\rangle \end{array} \right]$$



$$e. \text{dgb4} = \left[ \begin{array}{ll} \text{spkr} & = A \\ \text{addr} & = B \\ \text{pending} & = \langle \rangle \\ \text{qud} & = \langle p1? \rangle \\ \text{facts} & = \text{dgb0.facts} \\ \text{moves} & = \left\langle \left[ \begin{array}{ll} \text{sit} & = w0 \\ \text{sit-type} & = \text{IGH} \end{array} \right] \right\rangle \end{array} \right]$$

Now for B's response (56(2)). Formula (60a) is a minimally detailed sketch of the type our grammar could associate with B's utterance. A witness appropriate to serve as the precondition for pending update for this utterance is given in (60b)—it incorporates dgb5, the DGB after the (presumed common) perception of *u1*. Formula (60c) is the result of applying pending update to that record:

$$(60) \quad a. \text{NGIH} =_{\text{def}} \left[ \begin{array}{l} \text{phon : george isn't here} \\ \text{cat} = V[+\text{fin}, +\text{root}] : \text{syncat} \\ \text{constits} = \{\text{george, isn't, here, george isn't here}\} : \text{set}(\text{sign}) \\ \text{dgb-params : } \left[ \begin{array}{l} \text{spkr: IND} \\ \text{addr: IND} \\ \text{c1 : address(s, a)} \\ \text{s0: SIT} \\ \text{l: LOC} \\ \text{g: IND} \\ \text{c3: Named(g, 'george')} \end{array} \right] \\ \text{cont} = \text{Assert}(\text{spkr}, \text{addr}, \left[ \begin{array}{l} \text{sit} = s0 \\ \text{sit-type} = [c : \neg \text{In}(l, g)] \end{array} \right]) : \text{IllocProp} \end{array} \right]$$

$$b. \left[ \begin{array}{ll} T_u & = \text{NGIH} \\ u & = u1 \\ d & = \text{dgb5} = \left[ \begin{array}{l} \text{spkr} = A \\ \text{addr} = B \\ \text{pending} = \langle \rangle \\ \text{qud} = \text{dgb4.qud} \\ \text{facts} = \text{dgb0.facts} \cup \left\{ \begin{array}{l} \text{MostRecentSpeechEvent}(u1), \\ \text{Classify}(\text{NGIH}, u1) \end{array} \right\} \\ \text{moves} = \left\langle \left[ \begin{array}{ll} \text{sit} & = w0 \\ \text{sit-type} & = \text{IGH} \end{array} \right] \right\rangle \end{array} \right] \end{array} \right]$$

$$c. \text{ dgb6} = \left[ \begin{array}{ll} \text{spkr} & = B \\ \text{addr} & = A \\ \text{pending} & = \left\langle \left[ \begin{array}{ll} \text{sit} & = u1 \\ \text{sit-type} & = \text{NGIH} \end{array} \right] \right\rangle \\ \text{qud} & = \text{dg4.qud} \\ \text{facts} & = \text{dgb5.facts} \\ \text{moves} & = \left\langle \left[ \begin{array}{ll} \text{sit} & = w0 \\ \text{sit-type} & = \text{IGH} \end{array} \right] \right\rangle \end{array} \right]$$

The context again enables contextual instantiation and (the pending) QSPEC (see (54a)) to proceed. Composing these two steps yields dgb7. Here there are two possible variants, depending on whether we adopt the utterance-based or the semantics-only versions:

(61) a. Utterance-based QSPEC update:

$$\text{dgb7} = \left[ \begin{array}{ll} \text{spkr} & = B \\ \text{addr} & = A \\ \text{pending} & = \langle \rangle \\ \text{qud} & = \text{dg4.qud} \\ \text{facts} & = \text{dgb5.facts} \\ \text{moves} & = \left\langle \left[ \begin{array}{ll} \text{sit} & = w1 \\ \text{sit-type} & = \text{NGIH} \end{array} \right], \left[ \begin{array}{ll} \text{sit} & = w0 \\ \text{sit-type} & = \text{IGH} \end{array} \right] \right\rangle \end{array} \right]$$

b. Semantics-only QSPEC update:

$$\text{dgb7} = \left[ \begin{array}{ll} \text{spkr} & = B \\ \text{addr} & = A \\ \text{pending} & = \langle \rangle \\ \text{qud} & = \text{dg4.qud} \\ \text{facts} & = \text{dgb5.facts} \\ \text{moves} & = \left\langle \left[ \begin{array}{ll} \text{sit} & = w1 \\ \text{sit-type} & = \text{NIGH} \end{array} \right], \left[ \begin{array}{ll} \text{dgb-params} & = w0.\text{dgb-params} \\ \text{cont} & = w0.\text{cont} = \text{Ask}(\text{spkr}, \text{addr}, ?\text{In}(l, g)) \end{array} \right] \right\rangle \end{array} \right]$$

Finally, we can apply Assert QUD-incrementation and obtain *dgb8*:

(62) a. Utterance-based update:

$$dgb8 = \left[ \begin{array}{ll} \text{spkr} & = B \\ \text{addr} & = A \\ \text{pending} & = \langle \rangle \\ \text{qud} & = \langle \neg p1?, p1? \rangle \\ \text{facts} & = dgb5.\text{facts} \\ \text{moves} & = \left\langle \left[ \begin{array}{ll} \text{sit} & = w1 \\ \text{sit-type} & = \text{NGIH} \end{array} \right], \left[ \begin{array}{ll} \text{sit} & = w0 \\ \text{sit-type} & = \text{IGH} \end{array} \right] \right\rangle \end{array} \right]$$

b. Semantics-only update:

$$dgb8 = \left[ \begin{array}{ll} \text{spkr} & = B \\ \text{addr} & = A \\ \text{pending} & = \langle \rangle \\ \text{qud} & = \langle \neg p1?, p1? \rangle \\ \text{facts} & = dgb5.\text{facts} \\ \text{moves} & = \left\langle \left[ \begin{array}{ll} \text{sit} & = w1 \\ \text{sit-type} & = \text{NGIH} \end{array} \right], \left[ \begin{array}{ll} \text{dgb-params} & = w0.\text{dgb-params} \\ \text{cont} & = \text{Ask}(\text{spkr}, \text{addr}, ?\text{In}(l, g)) \end{array} \right] \right\rangle \end{array} \right]$$

Consider now (63), a variant on the dialogue in (56). This example demonstrates that the fact that the utterance took place (as classified by its content) becomes presupposed after grounding, which inspection of *dgb8* in (62) indicates is indeed the case:

(63) A: (1) Is George here? B: (2) George isn't here, (3) but it's interesting **that you ask if he is here.**

One significant remaining issue is precisely what an utterance being *presupposed* amounts to exactly. Using the semantics-only update, this is straightforward: what gets added to Moves is the instantiated content (64a). End of story. With the utterance-based update what gets added to Moves is the entire locutionary proposition (64b). This is, essentially, a conjunctive proposition renderable in English as in (64c):

- (64) a.  $\left[ \begin{array}{ll} \text{dgb-params} & = \text{w0.dgb-params} \\ \text{cont} & = \text{Ask(spkr, addr, ?In(l, g))} \end{array} \right]$   
 b.  $\left[ \begin{array}{ll} \text{sit} & = \text{w0} \\ \text{sit-type} & = \text{IGH} \end{array} \right]$   
 c. the acoustic signal *izjorhiya* is a sequence of words ‘is George here’; the syntactic category is  $V[+\text{root}]$ ; the content is identical to the proposition  $\text{Ask}(A, B, ?\text{In}(l_0, g_0)); \dots$

Hence what remains is to define conjunction reduction within a suitable logic. In other words, whereas in section 4.3.1 we suggested that FACTS be closed under conjunction, here we need to assume the converse: closure under conjunction reduction.

With respect to (63), it is clear that the semantically-oriented update is simpler. However, it has nothing to offer for the variety of facts discussed in Chapter 2, section 2.2, simplified examples of which are (65): whereas these can be inferred from the relevant locutionary proposition, in a fairly direct manner (65a) or somewhat more intricately (65b).

- (65) a. A: You need to win. Sorry, I used the word ‘win’.  
 b. A: I like Lucien. B: The last two words you uttered start with ‘I’.

As we will shortly see, some inferences the need for which we have seen here will be directly underwritten by the CCURs posited to underwrite clarification interaction.

**Exercise 6.5.4** (66) is another variant on the dialogue in (56). How would the contextual evolution of this dialogue differ from the DGB evolution following (56)?

- (66) A: Hi  
 B: Is George here?  
 A: George is not here.

## 6.6 Clarification Context Update Rules

I now turn to formalizing the two types of semantically-based clarification interaction discussed above in section 6.4. This will serve as the basis for characterizing the clarification potential of utterances. Essentially we need to find a means for ensuring the coherence of CRs—as the Conversation Analysts have already recognized—given that CRs involve a suspension of the standard adjacency coherence patterns. Rather, coherence is based not on relations between adjacent moves, but arises between a pending utterance  $u_0$  and a clarification move  $cr_0$  which it triggers. More specifically, coherence is mediated via an accommodated question defined on the basis of  $u_0$ . With

respect to the possibility of making an utterance whose coherence derives from an accommodated question, there is a commonality that goes beyond clarification interaction and applies equally to the accommodation associated with initiating moves, discussed in section 4.6.

The rule in (67) allows LatestMove to be updated with a locutionary proposition  $p1$  pertaining to the most recent utterance  $u$ —viz. its situational component is  $u$ —constrained to be co-propositional with a question accommodated as MaxQUD:

$$(67) \text{ Utterance Interpolation with Question Accommodation} =_{def} \left[ \begin{array}{ll} \text{pre} & : \text{DGBType} \\ & \left[ \begin{array}{l} T_u : \text{GramType} \\ u : \text{Rec} \\ p1 = \left[ \begin{array}{ll} \text{sit} & = u \\ \text{sit-type} & = T_u \end{array} \right] : \text{Prop} \end{array} \right] \\ \text{effects} & : \left[ \begin{array}{l} c2: \text{True}(p1) \\ q1 : \text{Question} \\ \text{qud} = \langle q1, \text{pre.qud} \rangle : \text{list}(\text{question}) \\ c3: \text{Copropositional}(p1^{\text{content}}, q1) \\ \text{facts} = \text{pre.facts} \cup \{\text{MostRecentSpeechEvent}(u)\} \\ \text{moves} = \langle p1, \text{pre.moves} \rangle \end{array} \right] \end{array} \right]$$

As such, though, this rule says nothing about clarification—it is indeed applicable to move initiation, as demonstrated in Exercise 6.6.1. The clarification-specific aspect will be provided by specifying the accommodated question to be about the utterance whose clarification it targets.

**Exercise 6.6.1** Consider the rule provided in Chapter 4 for initiating moves. Modify it in line with our earlier discussion of move update rules and then derive it from a combination of Utterance Interpolation with Question Accommodation and an initiating move-specific specification.

### 6.6.1 Parameter Identification

The first CCUR we discussed previously is Parameter Identification: for a given sub-utterance  $u1$  its output is a DGB in which MAX-QUD queries the content the previous speaker intended to convey with  $u1$ .

The formulation of this rule presupposes the existence of a relation *Mean* that holds between the speaker of an utterance, the utterance, and the intended content.

In general, we can identify this latter with the value instantiated by *dgb-params* for that utterance:<sup>49</sup>

(68)  $\text{Mean}(A, u, c)$  iff  $u.\text{dgb-param.spkr} = A$  and  $u.\text{dgb-param.x} = c$

This definition will enforce as a precondition for Parameter Identification that it can only pertain to sub-utterances that have a value for *DGB-PARAMS*. This is motivated by the assumption that what gets queried in intended content CRs is the intended instantiation of contextual parameters, exemplified in (69):

- (69) a. Lynne: Why shouldn't I have decent clothes and shoes?  
       Chris: really good clothes, what do you mean by really good clothes?  
       Lynne: Expensive well reasonably expensive (BNC, KBM)
- b. Unknown: Yes so we orbit  
       Mr H. A. Redfern: You turn around every once every twenty four hour.  
       What do you mean by we?  
       Unknown: Earth. (BNC, JAA)

Purver and Ginzburg (2004) note that all examples of sub-utterance (content-oriented) CRs found in their paper (and similarly in the study reported in Purver 2004a) related to instantiation in context of direct content. With CRs of sentential utterances, things are different—an intended content CR *can* query something beyond the direct content—after all, assuming the content of the parts has been resolved, there is not much to query with regard to the complete *direct* content. Such CRs can query how the direct content fits in context, more indirect meanings, and the like. I return to discuss this issue and more generally the issue of indirect meanings in section 8.3.

We can turn to actually formulating Parameter Identification. Its essence is given in (70)—given *MaxPending*, *A*'s utterance whose situational component is  $v_0$ , and  $u_1$ , a sub-utterance of  $v_0$ —a member of  $v_0$ 's *CONSTITUTES* set—then turn change can be effected and the issue 'What did *A* mean by uttering  $u_1$ ' be accommodated as *MaxQUD*:

<sup>49</sup> CRs frequently pertain to sub-utterances, not merely root utterances. Given this, we will need to assume that the relation *Mean* itself pertains also to sub-utterances. This in turn presupposes that the *DGB-PARAMS* associated with sub-utterances is defined for *SPKR*. This is obvious once one considers intra-utterance dynamics, allows for collaborative utterances and the like, which I have on the whole not done. For whatever it's worth, *DGB-PARAMS* clearly does need to be defined for *ADDR* and *UTT-TIME* down to the word level regardless, given examples such as (i) and (ii), based on examples in Levinson (1983), Pollard and Sag (1994).

(i) I want you, you, and you to come up on to the stage (based on an example from Pollard and Sag 1994)  
 (ii) Clap your hands now and now

(70) Parameter Identification =<sub>def</sub>

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{MaxPending} = \left[ \begin{array}{l} \text{sit} = v0 \\ \text{sit-type} = T_{v0} \end{array} \right] : \text{LocProp} \\ p1 = \left[ \begin{array}{l} \text{sit} = u1 \\ \text{sit-type} = T_{u1} \end{array} \right] : \text{LocProp} \\ \text{cl: member}(u1, v0.\text{constits}) \end{array} \right] \\ \text{effects} : \left[ \begin{array}{l} \text{spkr} = \text{pre.addr} : \text{IND} \\ \text{addr} = \text{pre.spkr} : \text{IND} \\ \text{MaxQUD} = \lambda x \text{ Mean}(\text{pre.spkr}, \text{pre.u1}, x) : \text{Question} \end{array} \right] \end{array} \right]$$

Combining Parameter Identification together with Utterance Interpolation yields (71):

(71) Parameter Identification  $\wedge_{\text{merge}}$  Utterance Interpolation =

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{MaxPending} = \left[ \begin{array}{l} \text{sit} = v0 \\ \text{sit-type} = T_{v0} \end{array} \right] : \text{LocProp} \\ p0 = \left[ \begin{array}{l} \text{sit} = u1 \\ \text{sit-type} = T_{u1} \end{array} \right] : \text{LocProp} \\ \text{cl: member}(u1, v0.\text{constits}) \end{array} \right] \\ \text{effects} : \left[ \begin{array}{l} \text{spkr} = \text{pre.addr} : \text{IND} \\ \text{addr} = \text{pre.spkr} : \text{IND} \\ u : \text{sign} \\ T_u : \text{GramType} \\ q1 = \lambda x \text{ Mean}(\text{pre.spkr}, \text{pre.u1}, x) : \text{Question} \\ p1 = \left[ \begin{array}{l} \text{sit} = u \\ \text{sit-type} = T_u \end{array} \right] : \text{Prop} \\ \text{c2: True}(p1) \\ \text{qud} = \langle q1, \text{pre.qud} \rangle : \text{list}(\text{question}) \\ \text{c3: Copropositional}(p1^{\text{content}}, q1) \\ \text{facts} = \text{pre.facts} \cup \{\text{MostRecentSpeechEvent}(u)\} \\ \text{moves} = \langle p1, \text{pre.moves} \rangle \end{array} \right] \end{array} \right]$$

Let us now consider some examples illustrating how parameter identification works in practice. In order to do this, we are faced for the first time with a truly *dialogic* dilemma: from whose perspective to describe the context? In all these cases, as is characteristic of CRification, updates differ depending whether done from the perspective of the original speaker, who *can* successfully integrate her utterance in her IS, or from the perspective of the person posing the CR, who cannot. This, as we discussed earlier and will see in detail below, is what underlies the Turn-Taking Puzzle. Since we are interested in considering how CRs work we adopt for now the CR querier's perspective.

Let us reconsider (72a), which we discussed earlier in the context of successful grounding, which can give rise to CRs such as (72b, c):

- (72) a. A(1): Is George here?  
 b. B(2a): Who do you mean 'George'?  
 c. B(2b): Do you mean George Sand?

Our starting point, as with successful grounding, is the speech event and its aftermath—pending update. The sole difference I introduce from the earlier DGB (dgb0 in (58)) is the absence of an appropriate naming presupposition for 'Georges' and consequently the lack of a witness for the dgb-param  $g$ :<sup>50</sup>

$$(73) \left[ \begin{array}{ll} T_u & = \text{IGH} \\ u & = u0 \\ & \left[ \begin{array}{ll} \text{spkr} & = A \\ \text{addr} & = B \\ \text{Pending} & = \langle \rangle \\ \text{Moves} & = \langle \rangle \\ \text{qud} & = \langle \rangle \\ \text{facts} & = \left\{ \begin{array}{l} \dots \text{In}(l, \{A, B\}), \\ \text{MostRecentSpeechEvent}(u0), \\ \text{Classify}(\text{IGH}, u0), \dots \end{array} \right\} \end{array} \right] \end{array} \right]$$

Applying pending update and contextual instantiation leads to a DGB identical to dgb2 in (59b), save for the discrepancy in FACTS and the lack of witness in  $w0'$ , which in turn prevents the content from being fully instantiated:

<sup>50</sup> Note that this discrepancy is introduced for exemplification's sake only. Nothing technically depends on the absence of the presupposition or the absence of a dgb-param witness. This is desirable, as CRs are possible also in cases where witnesses exist but there is some uncertainty about them, indeed even in cases where no such uncertainty exists (see below for discussion of such cases with respect to acknowledgement). At the same time, we do have the ability to encode uncertainty and/or wondering about the relationship between an utterance and its classifying type if we so wished.



$$(74) \quad \text{dgb2}' = \left[ \begin{array}{ll} \text{spkr} & = A \\ \text{addr} & = B \\ \\ \text{pending} & = \left\langle \left[ \begin{array}{l} \text{sit} = w0' = \\ \left[ \begin{array}{l} \text{phon} = \text{izjorjhiya} \\ \text{cat} = V[+\text{fin}, +\text{root}] \\ \text{constits} = \left\{ \begin{array}{l} v1(\text{iz}), v2(\text{jorj}), \\ v3(\text{hiya}), v4(\text{izjorjhiya}) \end{array} \right\} \end{array} \right] \\ \text{dgb-params} = \left[ \begin{array}{ll} \text{spkr} & = A \\ \text{addr} & = B \\ \text{utt-time} & = t0 \\ s0 & = \text{sit1} \\ l & = l0 \\ c3 & = \text{pr1} \end{array} \right] \\ \text{cont} = \text{Ask}(\text{spkr}, \text{addr}, ? \left[ \begin{array}{ll} \text{sit} & = s0 \\ \text{sit-type} & = [c : \text{In}(l, g)] \end{array} \right]) \\ \text{sit-type} = \text{IGH} \end{array} \right] \right\rangle \\ \text{qud} & = \text{dgb0.qud} \\ \text{facts} & = \text{dgb0'.facts} \\ \text{moves} & = \langle \rangle \end{array} \right]$$

In the pending utterance in (74) there are four constituents, each of which is clarifiable. If 'jorj' is chosen to be the clarification target using Parameter Identification, the resulting context can be specified as in (75). MaxQUD is the issue glossable as 'Who do you mean, "George"'. This context is specified to allow as the clarifying utterance (whose type—Clarification Request Grammatical Type—is abbreviated CRGT) any such that is co-propositional with MaxQUD:

$$(75) \quad \left[ \begin{array}{ll} \text{spkr} & = B \\ \text{addr} & = A \\ T_u & = \text{CRGT} \\ u & = u1 \\ \\ \text{pending} & = \left\langle \left[ \begin{array}{ll} \text{sit} & = w0' \\ \text{sit-type} & = \text{IGH} \end{array} \right] \right\rangle \\ \\ c3 & = \text{pr3} \\ q1 & = \lambda x \text{ Mean}(\text{pre.spkr}, \text{pre.v2}, x) : \text{Question} \\ \text{qud} & = \langle q1 \rangle \\ \text{facts} & = \text{facts1}' = \text{dgb0.facts} \cup \{\text{MostRecentSpeechEvent}(u1)\} \\ \\ \text{moves} & = \left\langle \left[ \begin{array}{ll} \text{sit} & = w1 \\ \text{sit-type} & = \text{CRGT} \end{array} \right] \right\rangle \end{array} \right]$$

Both (72b) and (72c) are co-propositional with MaxQUD. This renders either of the CRs felicitous in this context. Let us assume for concreteness that (72c) was uttered, then B could apply Ask QUD-incrementation and reach the following DGB:

$$(76) \text{ dgb3}' = \left[ \begin{array}{ll} \text{spkr} & = B \\ \text{addr} & = A \\ \text{pending} & = \left\langle \left[ \begin{array}{ll} \text{sit} & = w0' \\ \text{sit-type} & = \text{IGH} \end{array} \right] \right\rangle \\ \text{qud} & = \langle ?\text{Mean}(A, v2, \text{gs}), \lambda x \text{Mean}(A, v2, x) \rangle \\ \text{facts} & = \text{facts1}' \\ \text{moves} & = \langle \text{Ask}(B, A, ?\text{Mean}(A, u2, \text{gs})) \rangle \end{array} \right]$$

We can also deal with corrections, as in (77). B's corrective utterance is co-propositional with  $\lambda x \text{Mean}(A, u2, x)$ , and hence allowed in by the specification in (75):

- (77) a. A: Is George here?  
b. B: You mean Jo.

**Exercise 6.6.2** Calculate the Parameter Identification specification as a follow up to (72a) if (a) 'here' and (b) 'Is George here' are chosen as clarifiable constituents.

**Exercise 6.6.3** How could you use the apparatus developed above to analyse Cassie's utterance in (78)? What difficulties do you encounter?

- (78) Bonnie: (laugh)  
Cassie: What are you laughing at?  
Bonnie: (laugh)

In Chapter 7 I will show how NSUs such as (79) give rise to similar contents:

- (79) a. A: Is George here  
B: George? (= Who do you mean, 'George?')  
b. 1st agent: The red wooden disc!  
2nd agent: This one? (= Do you mean this one?)

### 6.6.2 Parameter Focussing

A second kind of clarification interaction we dubbed earlier *Parameter Focussing*: the essence of the operation, formulated in (80), is the accommodation of a question in which a contextual parameter  $u1.dgb\text{-param}$  gets abstracted away from the entire MaxPending utterance  $v0$ 's content in which— $u1.dgb\text{-param}$  apart—the contextual parameters have been *instantiated*. In other words, there is a certain assumption that the rest of the utterance has been understood.

(80) Parameter Focussing =<sub>def</sub>

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{MaxPending} = \left[ \begin{array}{l} \text{sit} = v0 \\ \text{sit-type} = T_{v0} \end{array} \right] : \text{LocProp} \\ p0 = \left[ \begin{array}{l} \text{sit} = u1 \\ \text{sit-type} = T_{u1} \end{array} \right] : \text{LocProp} \\ \text{cl} : \text{member}(u1, v0.\text{constits}) \end{array} \right] \\ \text{effects} : \left[ \begin{array}{l} \text{spkr} = \text{pre.addr} : \text{Ind} \\ \text{addr} = \text{pre.spkr} : \text{Ind} \\ \text{MaxQUd} = \lambda x v0.\text{cont}(u1.\text{dgb-param} \rightsquigarrow x) : \text{Question} \end{array} \right] \end{array} \right]$$

Integrating Parameter Focussing with utterance interpolation we obtain (81):

(81) Parameter Focussing  $\wedge_{\text{merge}}$  utterance interpolation:

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{MaxPending} = \left[ \begin{array}{l} \text{sit} = v0 \\ \text{sit-type} = T_{v0} \end{array} \right] : \text{LocProp} \\ p0 = \left[ \begin{array}{l} \text{sit} = u1 \\ \text{sit-type} = T_{u1} \end{array} \right] : \text{LocProp} \\ \text{cl} : \text{member}(u1, v0.\text{constits}) \end{array} \right] \\ \text{effects} : \left[ \begin{array}{l} \text{spkr} = \text{pre.addr} : \text{Ind} \\ \text{addr} = \text{pre.spkr} : \text{Ind} \\ T_{u2} : \text{GramType} \\ u2 : \text{Rec} \\ p1 = \left[ \begin{array}{l} \text{sit} = u2 \\ \text{sit-type} = T_{u2} \end{array} \right] \\ c2 : \text{True}(p1) \\ q1 = \lambda x v0.\text{cont}(\text{pre}.u1.\text{dgb-param} \rightsquigarrow x) : \text{Question} \\ \text{qud} = \langle q1, \text{pre.qud} \rangle : \text{list}(\text{question}) \\ c3 : \text{Copropositional}(u1.\text{cont}, \text{maxqud}) \\ \text{facts} = \text{dgb0.facts} \cup \{\text{MostRecentSpeechEvent}(u2)\} \\ \text{moves} = \langle p1, \text{pre.moves} \rangle \end{array} \right] \end{array} \right]$$

We can exemplify the effect of Parameter Focussing with respect to clarifying both complete and parts of utterances. Consider first (82): B's CR can be analysed by taking the clarification target to be the entire utterance (the head daughter of the root

utterance). The specification for such a CR obtained by Parameter Focussing is given in (83): an appropriate CR needs to be co-propositional with MaxQUD glossable as *what did you claim/say?*

- (82) a. A(1): We are screwed.  
b. B(2): Are you saying that we're in trouble?

- (83) a. 
$$\left[ \begin{array}{l} \text{spkr} = A \\ \text{addr} = B \\ \\ \text{Pending} = \left\langle \begin{array}{l} \text{sit} = w0 = \left[ \begin{array}{l} \text{phon} = \text{wirskrud} \\ \text{cat} = V[+\text{fin}, +\text{root}] \\ \text{constits} = \{ u1(\text{wir}), u2(\text{skrud}), u3(\text{wirskrud}) \} \\ \\ \text{dgb-params} = \left[ \begin{array}{l} \text{spkr} = A \\ \text{addr} = B \\ \text{utt-time} = t0 \\ s0 = \text{sitl} \end{array} \right] \\ \text{content} = \text{Assert}(\text{spkr}, \text{addr}, \\ \text{TroubleFor}(\{\text{spkr}, \text{addr}\}, \text{utt-time})) \end{array} \right. \\ \text{sit-type} = \text{WAS} \end{array} \right\rangle \\ \\ \text{Moves} = m1 \\ \text{qud} = \langle \rangle \\ \text{facts} = \text{cgl} \end{array} \right]$$
- b. 
$$\left[ \begin{array}{l} \text{spkr} = B \\ \text{addr} = A \\ T_u = \text{CRGT} \\ u = w1 \\ \\ \text{pending} = \left\langle \left[ \begin{array}{l} \text{sit} = w0 \\ \text{sit-type} = \text{WAS} \end{array} \right] \right\rangle \\ \\ q1 = \lambda x \text{Assert}(A, B, x) \\ \text{qud} = \langle q1 \rangle \\ c3 = \text{pr3} \\ \text{facts} = \text{dgb0.facts} \cup \{\text{MostRecentSpeechEvent}(w1)\} \\ \\ \text{moves} = \left\langle \left[ \begin{array}{l} \text{sit} = w1 \\ \text{sit-type} = \text{CRGT} \end{array} \right] \right\rangle \end{array} \right]$$

Let us now consider some sub-utterance CRs arising with parameter focussing. Utterances (84b–d) are possible CRs to (84a):

- (84) a. A(1): Is George here?  
 b. B(2a): Are you asking if George SAND is here?  
 c. B(2b): Is George here?  
 d. B(2c): Is WHO here?

The CR spec of (84a) using Parameter Focussing with clarification target ‘George’ is:

$$(85) \left[ \begin{array}{l} \text{spkr} = B \\ \text{addr} = A \\ T_u = \text{CRGT} \\ u = w1 \\ \text{pending} = \left\langle \left[ \begin{array}{ll} \text{sit} & = w0' \\ \text{sit-type} & = \text{IGH} \end{array} \right] \right\rangle \\ c3 = \text{pr3} \\ q1 = \lambda x \text{ Ask}(A, \text{In}(l0, x)) : \text{Question} \\ \text{qud} = \langle q1 \rangle \\ \text{facts} = \text{facts}1' = \text{dgb0.facts} \cup \{\text{MostRecentSpeechEvent}(w1)\} \\ \text{moves} = \left\langle \left[ \begin{array}{ll} \text{sit} & = w1 \\ \text{sit-type} & = \text{CRGT} \end{array} \right], \text{pre.moves} \right\rangle \end{array} \right]$$

Given this and the fact that (the content of) ‘Are you asking if George SAND is here’ is co-propositional with MaxQUD in (85) the felicity of (84b) follows immediately. To deal with (84c,d) I need to provide an analysis of reprise or echo clauses, to which I turn immediately in the next subsection.

In Chapter 7, I will show how to handle NSUs such as (86), whose contextual appropriateness is licensed by Parameter Focussing:

- (86) a. BO? (= Are you asking if BO left?)  
 b. WHO? (WHO are you asking if they left?)

### 6.6.3 Reprise clauses

Reprise clauses are a significant clause type: they constitute a type of clause which in various cases, including (84d) is a string whose *sole analysis* is as a CR. They provide a particularly cast-iron argument for the need to have a conception of grammar which has communicative interaction at its core.

Reprise clauses are in a fairly direct sense a grammaticized counterpart of Parameter Focussing: in the analysis of Ginzburg and Sag (2000), which I adopt here, they are a radically anaphoric clause, constrained to be co-propositional with the maximally pending utterance ( $u1$  in (87)), whereas its head daughter—arising via ‘standard compositional rules’ (e.g. the ones we saw in Chapter 5)—is constrained to be identical to that utterance’s descriptive content ( $r$  in (87)). Given this, they satisfy the contextual specification of Parameter Focussing and are licensed as CRs of such utterances.

The sole grammatical condition that remains to be specified concerns the domain of the question. Here an almost entirely analogous situation obtains as with the type *dir-is-int-cl*, discussed in Chapter 5: (in practice) there are two subcases, a polar question [where the domain is null] and a unary question [involving substitution over a single label], though here I state it in general terms. In a more grammatically-oriented presentation, this aspect would not need to be stated in the construction’s specification, but would follow by inheritance.<sup>51</sup>

(87) *repr-int-cl*:

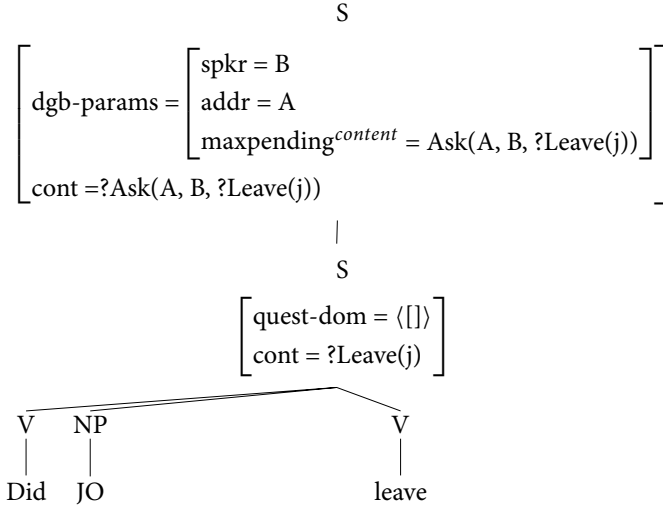
$$\left[ \begin{array}{l} \text{cat.head} = V[+\text{fin}] : \text{PoS} \\ \\ \text{dgb-params} : \left[ \begin{array}{l} \text{spkr} : \text{IND} \\ \text{addr} : \text{IND} \\ \text{cl} : \text{Addr}(\text{spkr}, \text{addr}) \\ \text{P} : \text{IllocRel} \\ \text{r} : \text{AbstSemObj} \\ \text{maxpending}^{\text{content}} = \text{P}(\text{addr}, \text{spkr}, \text{p}) : \text{IllocProp} \end{array} \right] \\ \\ \text{hd-dtr} : \left[ \begin{array}{l} \text{F} : \text{Type} \\ \text{quest-dom} = \langle \text{F} \rangle : \text{list}(\text{RType}) \\ \text{cont} = \text{p} : \text{IllocProp} \end{array} \right] \\ \text{cont} = ([\text{r} : \text{F}])\text{MaxPending}^{\text{content}}(\langle l_i \rightsquigarrow \text{r.l}_i \rangle_{i \in \text{labels}(\text{F})}) : \text{Question} \\ \text{quest-dom} = \langle \rangle : \text{list}(\text{RType}) \end{array} \right]$$

**Exercise 6.6.4** Show that *repr-int-cl* could alternatively be specified using MaxQUD, without using MaxPending.

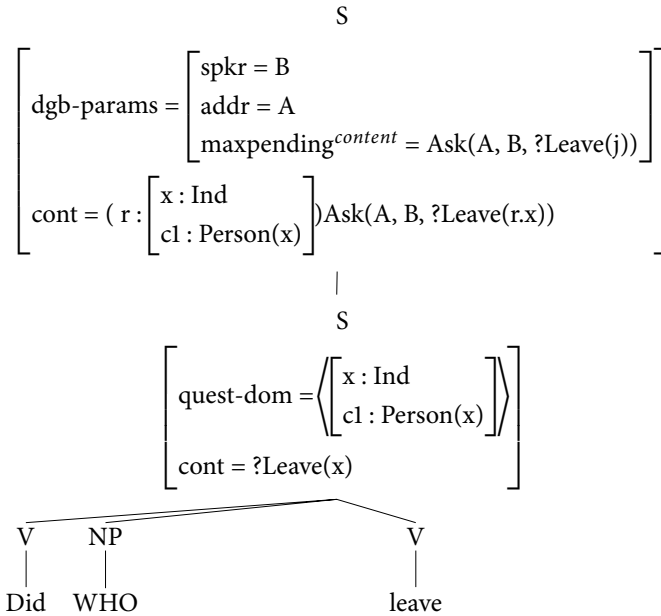
Two instances of *repr-int-cl* are given in (88)—a *literal reprise* and a *wh-subst reprise*:

<sup>51</sup> See Ginzburg and Sag (2000) for extensive exemplification.

- (88) A: Did Jo leave? B: Did Jo leave? (=Is A asking if Jo left)



- (89) A: Did Jo leave? B: Did WHO leave? (=Who<sub>i</sub> are you asking if *i* left?)



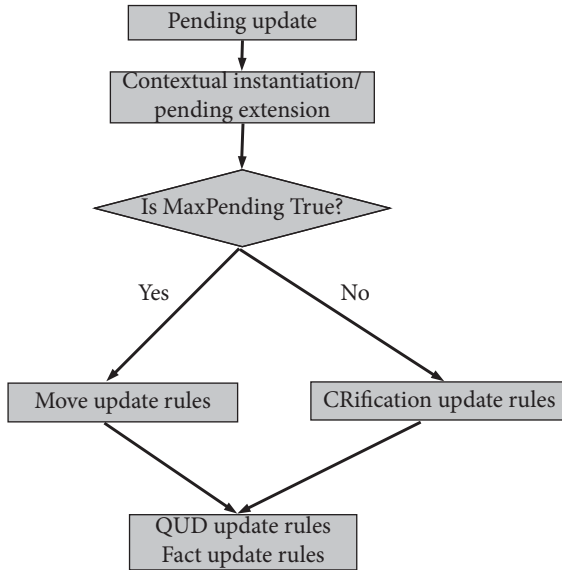


FIGURE 6.1: Utterance Integration Process (interim view)

## 6.7 A Protocol Combining Grounding and CRification

Given what we did in the last two sections, coming up with a general utterance integration protocol involves *essentially* sewing together the successful grounding option (Moves update rules such as QSPEC and Accept move) with the CRification option (Utterance interpolation with question accommodation, parameter identification/focussing), and the various QUD and FACTS updates. The utterance integration process as I have described it so far can be summarized using the flow diagram in Fig. 6.1. This is almost a complete picture. To see one missing piece let us try to analyse an exchange that contains CRification.

- (90) A(1): Is George here?  
 B(2): Is WHO here?  
 A(3): George Sand.  
 B(4): Ah, (5) no.

A utters (90(1)), which B cannot ground because, say, she is unsure about the intended reference of *George*. We assume she applies Parameter Focussing to initiate CRification. Given our protocol, after B's utterance (90(2)) A and B would have the following respective DGBs:



(91) a. A: Is George here? B: Is WHO here?

$$b. A.dgb7 = \left[ \begin{array}{l} \text{spkr} = B \\ \text{addr} = A \\ \text{pending} = \left\langle \left[ \begin{array}{ll} \text{sit} & = w1 \\ \text{sit-type} & = IWH \end{array} \right] \right\rangle \\ \text{qud} = \langle p1? \rangle \\ \text{facts} = \left\{ \begin{array}{l} \text{In}(l, \{A, B\}) \\ \text{Named}(\text{'George'}, g), \\ \text{2ndMostRecentSpeechEvent}(u0), \\ \text{Classify}(IGH, u0) \\ \text{MostRecentSpeechEvent}(u1), \\ \text{Classify}(IWH, u1) \end{array} \right\} \\ \text{moves} = \left\langle \left[ \begin{array}{ll} \text{sit} & = w0 \\ \text{sit-type} & = IGH \end{array} \right] \right\rangle \end{array} \right]$$

$$c. B.dgb7' = \left[ \begin{array}{l} \text{spkr} = B \\ \text{addr} = A \\ \text{pending} = \left\langle \left[ \begin{array}{ll} \text{sit} & = w0' \\ \text{sit-type} & = IGH \end{array} \right] \right\rangle \\ \text{qud} = \langle \lambda x \text{ Ask}(A, B, ?\text{In}(l, x)) \rangle \\ \text{facts} = \left\{ \begin{array}{l} \text{In}(l, \{A, B\}) \\ \text{Named}(\text{'George'}, g), \\ \text{2ndMostRecentSpeechEvent}(u0), \\ \text{Classify}(IGH, u0) \\ \text{MostRecentSpeechEvent}(u1), \\ \text{Classify}(IWH, u1) \end{array} \right\} \\ \text{moves} = \left\langle \left[ \begin{array}{ll} \text{sit} & = w1 \\ \text{sit-type} & = IWH \end{array} \right] \right\rangle \end{array} \right]$$

Something interesting has happened here—well actually it's mundane for a dialogic perspective on language, but alien to a communitarian perspective. A and B have processed the same utterances and yet we find discrepancies in their representation of public context—their DGBs: A has the question of *whether George is here* as the sole

member of QUD, whereas the utterance ‘Is WHO here?’ remains pending: it remains pending because there is no Moves update rule that can apply to it. In contrast, B’s QUD consists of the question *who are you asking if they’re here*, whereas the utterance ‘Is George here’ is pending. Mismatches of this type can be used to explicate the Turn-Taking Puzzle (see examples (22) in Chapter 2) once one has a theory of resolution of NSUs, as we shall in Chapter 7. However, as we noted earlier, Turn-Taking Puzzle-type mismatches are exhibited intrinsically on the level of production, but need not arise at the level of comprehension. That is, in the current example—and more generally—it is rarely the case that the author of an utterance  $u$  fails to understand CRs relating to  $u$ , let alone to recognize their coherence.

In order to enable this possibility, we offer a rule that *inter alia* allows the integration of CRs concerning  $u$  by the author of  $u$ . As we have seen above, a CCUR (for the moment either Parameter Identification or Parameter Focussing) and a sub-utterance  $u1$  give rise to a question, the question accommodated as MaxQUD when  $u1$  is the input to the CCUR. Call that question  $\text{CCUR.qud}(u1)$ . The rule in (92) says that if the speaker of LatestMove is the current addressee,  $p$  is pending, and  $u1$  is a constituent of LatestMove, one can update moves with  $p$  and QUD with  $\text{CCUR.qud}(u1)$ , so long as  $p$  is co-propositional with  $\text{CCUR.qud}(u1)$ . Notice that the formulation of this rule depends crucially on LatestMove being a locutionary proposition. If, contrariwise, it consisted merely of content-related information, the QUD accommodation, for instance, would not be definable.

(92) CR Accommodation =<sub>def</sub>

$$\left[ \begin{array}{l} \text{pre} \\ \text{effects} \end{array} : \left[ \begin{array}{l} \text{MaxPending} = \left[ \begin{array}{l} \text{sit} = v1 \\ \text{sit-type} = T_{v1} \end{array} \right] : \text{LocProp} \\ \text{LatestMove} = \left[ \begin{array}{l} \text{sit} = v0 \\ \text{sit-type} = T_{v0} \end{array} \right] : \text{LocProp} \\ \text{addr} = v0.\text{dgb-params.spkr} : \text{Ind} \\ u1 : \text{Sign} \\ c1 : \text{member}(u1, v0.\text{constits}) \\ q1 = \text{CCUR.qud}(u1) \\ c2 : \text{CoPropositional}(q1, \text{MaxPending}) \\ \text{LatestMove} = \text{Pre.MaxPending} \\ \text{MaxQUD} = \text{pre.q1} \end{array} \right] \right]$$

A couple of comments are in order about CR Accommodation: in practice it can be utilized only by the speaker of LatestMove: he being the current addressee means that another utterance  $v1$  has ensued and its speaker will have already updated her

LatestMove with v1 (since speakers always integrate their own utterances immediately into their DGB). A more technical point is that the rule in (92) appears to make reference to a CCUR in the preconditions, namely the specification  $\text{CCUR.qud}(u1)$ . This is merely convenient shorthand: in so far as we have posited two CCURs, Parameter Identification and Parameter Focussing, we could have instead used the specification in (93); and this disjunction will increase once a couple of additional CCURs are posited (one in the text, one as an exercise). Arguably, this is the source of the not infrequently attested confusion between clarification readings, examples of which we saw in section 6.2.

$$(93) \left[ \begin{array}{l} \text{CCUR.QUD}(u1) = \left\{ \begin{array}{l} \lambda x \text{ v0.cont}(u1.\text{dgb-param} \rightsquigarrow x), \\ \lambda x \text{ Mean}(\text{pre.spkr}, \text{pre}.u1, x) \end{array} \right\} : \text{set}(\text{Question}) \\ \text{q1} : \text{Question} \\ \text{c1} : \text{member}(\text{q1}, \text{CCUR.QUD}(u1)) \end{array} \right]$$

Incorporating CR accommodation changes slightly the picture we had previously of the utterance integration process. The new picture can be described as follows:

(94) **Utterance-processing protocol**

For an agent A with IS  $I$ : if a locutionary proposition  $p_u = \left[ \begin{array}{l} \text{sit} = u \\ \text{sit-type} = T_u \end{array} \right]$

is Maximal in PENDING:

- (a) if  $p_u$  is true, try to integrate  $p_u$  in A.DGB using a Moves update rule
- (b) otherwise: try to accommodate as a CR to LatestMove
- (c) else: seek a witness for  $T_u$  by asking a CR

Or graphically, as in Figure 6.2 (opposite page).

The most direct effect CR accommodation has is that it enables speakers to integrate CRs to their own utterances into the DGB. Thus, returning to our example (91), if A applies CR accommodation, choosing Parameter Focussing as the accommodated question, his DGB becomes (95). This is better aligned with if not identical to B's  $\text{dgb7}'$ :

$$(95) \text{ A.dgb8} = \left[ \begin{array}{l} \text{spkr} = B \\ \text{addr} = A \\ \text{pending} = \langle \rangle \\ \text{qud} = \langle \lambda x \text{ Ask}(A, B, ?\text{In}(l, x)), p1? \rangle \\ \text{facts} = \text{dgb7.facts} \\ \text{moves} = \left\langle \left[ \begin{array}{ll} \text{sit} & = \text{w1} \\ \text{sit-type} & = \text{IWH} \end{array} \right], \left[ \begin{array}{ll} \text{sit} & = \text{w0} \\ \text{sit-type} & = \text{IGH} \end{array} \right] \right\rangle \end{array} \right]$$

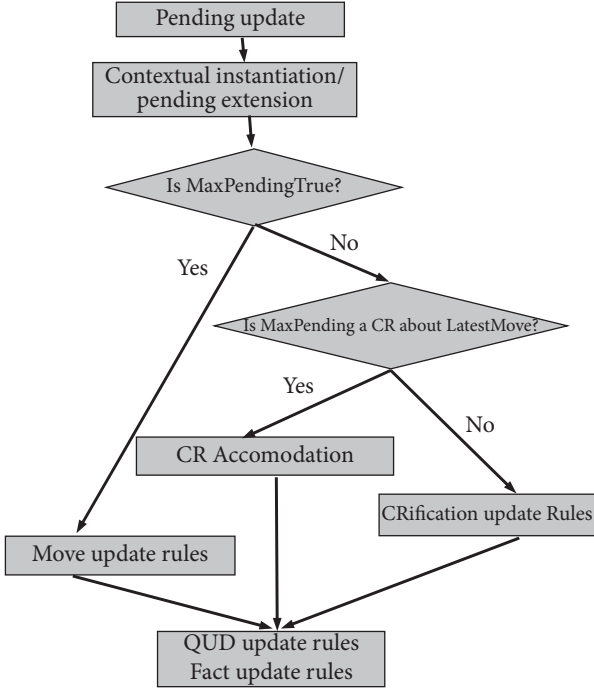


FIGURE 6.2: Utterance Integration Process

We can now proceed to the second part of the CRification dialogue, repeated as (96a): this is straightforward with one exception—the *dependification* of ‘Is George here’. In more detail: in (96a(3)) A addresses MaxQUD, asserting p13. Taking B’s perspective, this leads to *dgb8’*:

(96) a. A(3): (I’m asking about) George Sand. B:(4) Ah, (5) no.

$$\begin{aligned}
 \text{b. B.dgb8'} = & \left[ \begin{array}{l} \text{spkr} = A \\ \text{addr} = B \\ \text{pending} = \left\langle \left[ \begin{array}{ll} \text{sit} & = w0' \\ \text{sit-type} & = \text{IGH} \end{array} \right] \right\rangle \\ \text{qud} = \left\langle ?\text{Ask}(A, B, ?\text{In}(lo, gs)), \lambda x \text{ Ask}(A, B, ?\text{In}(lo, x)) \right\rangle \\ \text{facts} = \text{dgb7'}. \text{facts} \\ \text{moves} = \left\langle \left[ \begin{array}{ll} \text{sit} & = w3 \\ \text{sit-type} & = \text{GS} \end{array} \right], \left[ \begin{array}{ll} \text{sit} & = w1 \\ \text{sit-type} & = \text{IWH} \end{array} \right] \right\rangle \end{array} \right]
 \end{aligned}$$

B accepts this assertion in (96a(4))—leading to  $\text{dgb9}'$ , in which the clarification question has been downdated from QUD; B can now perform contextual instantiation to  $\text{MaxPending}$  ( $\text{dgb10}'$ ), then update  $\text{LatestMove}$  with ‘Is George here’ ( $\text{dgb11}'$ ), which becomes in turn  $\text{MaxQUD}$  ( $\text{dgb12}'$ ). (96(5)) can then be uttered as an answer to  $\text{MaxQUD}$ :

$$(97) \quad \text{B.dgb9}' = \left[ \begin{array}{l} \text{spkr} = \text{B} \\ \text{addr} = \text{A} \\ \text{pending} = \left\langle \left[ \begin{array}{ll} \text{sit} & = \text{w0}' \\ \text{sit-type} & = \text{IGH} \end{array} \right] \right\rangle \\ \text{qud} = \langle \rangle \\ \text{facts} = \{ \text{Ask}(\text{A}, \text{B}, ?\text{In}(\text{lo}, \text{gs})) \} \cup \text{dgb7}'.\text{facts} \\ \text{moves} = \left\langle \left[ \begin{array}{ll} \text{sit} & = \text{w4} \\ \text{sit-type} & = \text{Ah} \end{array} \right], \right. \\ \quad \left. \left[ \begin{array}{ll} \text{sit} & = \text{w3} \\ \text{sit-type} & = \text{GS} \end{array} \right], \right. \\ \quad \left. \left[ \begin{array}{ll} \text{sit} & = \text{w1} \\ \text{sit-type} & = \text{IWH} \end{array} \right] \right\rangle \end{array} \right]$$

$$(98) \quad \text{B.dgb10}' = \left[ \begin{array}{l} \text{spkr} = \text{B} \\ \text{addr} = \text{A} \\ \text{pending} = \left\langle \left[ \begin{array}{ll} \text{sit} = \text{w0} = & \left[ \begin{array}{ll} \text{spkr} & = \text{A} \\ \text{addr} & = \text{B} \\ \text{utt-time} & = \text{t0} \\ \text{dgb-params} = \begin{array}{ll} \text{s0} & = \text{sit1} \\ \text{l} & = \text{lo} \\ \text{g} & = \text{gs} \\ \text{c3} & = \text{pr1} \end{array} \end{array} \right] \\ \text{cont} = \text{Ask}(\text{spkr}, \text{addr}, ? \left[ \begin{array}{ll} \text{sit} & = \text{s0} \\ \text{sit-type} & = [\text{c} : \text{In}(\text{l}, \text{g})] \end{array} \right] ) \\ \text{sit-type} = \text{IGH} \end{array} \right] \right\rangle \\ \text{qud} = \text{dgb9}'.\text{qud} \\ \text{facts} = \text{dgb9}'.\text{facts} \\ \text{moves} = \text{dgb9}'.\text{moves} \end{array} \right]$$

- a. B.dgb11' = 
$$\left[ \begin{array}{l} \text{spkr} = B \\ \text{addr} = A \\ \text{pending} = \langle \rangle \\ \text{qud} = \text{dgb9'}. \text{qud} \\ \text{facts} = \text{dgb9'}. \text{facts} \\ \text{moves} = \left\langle \left[ \begin{array}{l} \text{sit} = w0 \\ \text{sit-type} = \text{IGH} \end{array} \right], \text{dgb10'}. \text{moves} \right\rangle \end{array} \right]$$
- b. B.dgb12' = 
$$\left[ \begin{array}{l} \text{spkr} = B \\ \text{addr} = A \\ \text{pending} = \langle \rangle \\ \text{qud} = \langle ?\text{In}(\text{lo}, \text{gs}) \rangle \\ \text{facts} = \text{dgb9'}. \text{facts} \\ \text{moves} = \text{dgb11'}. \text{moves} \end{array} \right]$$

One class of phenomena that the analysis developed here does not fully extend to are long-distance repairs, the possibility for which we noted earlier in section 6.5.1. In CA there is the notion of *third position repairs*, exemplified in (99), where A understands from B's answer that B misidentified which George A had in mind (Sand instead of Perec, say):

- (99) A(1): Is Georges here? B(2): George left yesterday. A(3): I meant George Perec.

After (99(2)), A could use CR-accommodation to instantiate as MaxQUD the issue 'What did A mean by Georges'. This would allow her to coherently integrate the utterance (99(3)) in her DGB. However, in the current set-up there is no principle enabling B to integrate (99(3)) in his DGB since he grounded (99(1)) and his own utterance (99(2)). In any case though, the term *third position repairs* is misleading since, as we noted earlier, such repairs can occur quite a bit further down the line. In this there is a clear contrast with CRs. (100) illustrates that whereas long-distance corrections are available, this is not the case for most types of CRs, the only exceptions being *explicit* CRs:

- (100) a. A(1): Is Bo here? B(2): He left this morning. A(3): Oh really? B(4): Yes, his flight left at 4am. A(5): What day did he arrive last week? B(6): Tuesday. A(7): Wait, I meant Bo Smith.  
 b. A(7') Wait, Which Bo did you mean ? / #Bo? / #Is WHO here? / #WHO?  
 c. A(1): When did Bo leave? B(2): Late last night. A(3): Oh really? B(4): Yes, a truck came for his stuff at 4am. He was helping them load it up. A(5): Wait, I meant resign.  
 d. A(5'): wait, what did you mean by 'leave'? / #leave? / #When did Bo what?

Explicit CRs and corrections involve anaphoricity, but this clearly has a different source from that which underwrites the resolution of CRs in general. Long-distance corrections also bring with them complicated issues of backtracking—how much of what was said during the misunderstanding should remain in the DGB. I leave this unresolved in the current work.

### 6.7.1 *Coherence Revisited*

In Chapter 4 I introduced several notions of *coherence*, one of which allows the coherence of a sequence of *moves* to be explicated. Formulating this at the level of moves as opposed to *utterances* was an interim measure, in line with the fact that I was abstracting away from metacommunicative issues. Now though, I can remedy the problem. One additional modification is to define transitions on (total) information states, as opposed to merely DGBs, in light of the fact that certain of the conversational rules make reference to the more inclusive entity. This notion is important conceptually, but is also of practical significance, as we will see in Chapter 8, where it plays a crucial role in defining *irrelevance*, the trigger for certain implicatures.

Here I will restrict myself to generalizing the move-based notion of *M(ove)-Coherence*; the modification of the other notions is analogous *mutatis mutandis*.<sup>52</sup> It involves saying that an utterance  $u_1$  is *U(tterance)-Coherent* iff  $u_1$  can be incorporated in an information state  $I_1$  as the situational component of a locutionary proposition  $prop(u_1, T_{u_1})$  a contextual extension of which,  $prop(w_1, T_{u_1})$ , becomes LatestMove. The definition is now relative to both the set of conversational rules and to a grammar from which the types  $T_u$  originate.

- (101) **U(tterance)-Coherence:** Given a set of conversational rules  $\mathcal{C}$ , a grammar  $\mathcal{G}$  and an information state  $I_0 : TIS$ , an utterance  $u_1$  is **U(tterance) $_{\mathcal{C}, \mathcal{G}}^{I_0}$ -coherent** iff there exist  $c_1, \dots, c_{k+1} \in \mathcal{C}$ ,  $T_{u_1} \in \mathcal{G}$ ,  $k \geq 0$ , such that  $c_1(I_0) = I_1, \dots, c_{k+1}(I_k) = I_{k+1}$  such that  $I_{k+1}.DGB.LatestMove = prop(w_1, T_{u_1})$ , where  $prop(w_1, T_{u_1})$  contextually extends  $prop(u_1, T_{u_1})$  and for every  $0 < i \leq k$ ,  $I_i.DGB.MaxPending = prop(w'_i, T_{u_1})$ , where  $prop(w'_i, T_{u_1})$  contextually extends  $prop(u_1, T_{u_1})$ .<sup>53</sup>

Thus, exemplifying this with the input information state being one in which a query was posed, this would amount to the following. An utterance  $u$  is **U(tterance) $_{\mathcal{C}, \mathcal{G}}^{I_0}$ -relevant** iff there exist  $c_1, \dots, c_{k+1} \in \mathcal{C}$ ,  $T_u \in \mathcal{G}$ ,  $k \geq 0$ , such that  $c_1(I_0) = I_1, \dots, c_{k+1}(I_k) = I_{k+1}$ , where  $\mathcal{C}$ 's information state  $I_0$  satisfies (102a); where by means of a sequence of updates the locutionary proposition  $p_u = prop(u, T_u)$  becomes the value of LatestMove (condition (102b)); and the final element of the

<sup>52</sup> The reformulation can be found in Appendix B.

<sup>53</sup> 'Every  $0 < i \leq k \dots$ ' is to be read logically, so that in case  $k = 0$ , the rest of the clause is not required to be true.

sequence of updates  $I_{k+1}$  is such that one of the conditions in (102c–f) is satisfied— $u$  is either  $q$ -specific, an appropriate CR, relates to the issue of willingness to discuss  $q$ , or is genre-relevant:

- (102) a.  $I_0.DGB.LatestMove = v$ ;  $v.content = Ask(A, q)$   
 b.  $I_{k+1}.DGB.LatestMove = p_u$   
 c.  $p_u.content$  is  $q$ -specific relative to  $I.DGB$ , Or  
 d.  $p_u.content$  is CoPropositional with some question  $q_0$  that satisfies  $q_0 = CCUR1.effects.maxqud(I_0.DGB.MaxPending)$  for some CCUR CCUR1, Or  
 e. One of  $C$ 's beliefs in  $I_0$  is that: for some  $G_0$  there exists  $dgb1$  such that  $(I_0.DGB \oplus p_u) \sqsubset dgb1$ , and such that  $dgb1 : G_0$

## 6.8 Repetition Clarification Requests

A common CR discussed in section 6.2 are repetition CRs such as (5), two examples of which are repeated here:

- (103) a. Gary: It's, it's okay I mean (pause) when he's away (pause, clears throat, pause) things go, actually things go a lot smoother.  
 Clare: Sorry?  
 Gary: Things go a lot smoother when he's not here. (BNC, KSR)  
 b. S.Bust: Great memorial I think really isn't it?  
 E.Bust: Beg pardon?  
 S.Bust: Be a good appropriate memorial if we can afford it. (BNC, KM8)

There seem to be two alternative approaches one could take to describing repetition CRs. The first would be very much in line with the strategy we have taken to Parameter Identification and parameter focussing: simply posit an analogous rule. This assumes that the interaction involved is analogous to other CR interactions—an utterance is partially understood, this leads to a question being asked, it gets answered, which leads to the utterance being fully understood.<sup>54</sup> The alternative, call it RequestRepeat, would be to assume that repetition CRs involve a more degenerate, lower-level type of interaction: an utterance is very, very partially understood, this leads to a request for it to be repeated, which it is. End of story.<sup>55</sup> The main expense

<sup>54</sup> Or at least better understood, as in:

(i) A: He's just gone to the cinema.  
 B: Pardon?  
 A: He went to the cinema just now.  
 B: Who?

<sup>55</sup> Such an approach is hinted at by Larsson and Zaenen (2000).



from our current perspective would be the need to posit a rule for responding to such moves—in any case we will need to posit a repetition UR and modification, albeit trivial, to CR accommodation (viz. adding a disjunct which refers to the CCUR-produced MaxQUD). The advantage of the more general approach is that it captures the fact that repetition CRs share the same intonation contours as other queries, commonly involve question words (though other words such as ‘sorry’, ‘pardon’, and ‘eh’ do this work as well), and indeed can be achieved by using sentential interrogatives (e.g. ‘What did you say?’). Moreover, these facts seem to carry over to other languages.<sup>56</sup> A potential advantage of the RequestRepeat approach would be demonstration that repetition CRs involve very limited variation, i.e. invariably involve (more or less verbatim) repetition. I will not try to resolve the choice here: I will formulate the more general approach, leaving the RequestRepeat approach as an exercise.

As already mentioned, to develop the so called general approach to repetition CRs we need to postulate a rule quite analogous to Parameter Identification. The difference can be captured by associating with repetition CRs the MaxQUD in (104a), which also allows for CRs such as (104b):

- (104) a. What did you say?  
b. Did you say ‘five’?

Let us assume a type *Utter* that holds between the speaker of an utterance, the utterance, and a phonological type that classifies the utterance:<sup>57</sup>

- (105)  $Utter(A, u, T)$  iff  $u.dgb-param.spkr = A$  and  $u.phon : T$

We can then posit the following UR, whose essence is this: given a just occurred (sub-)utterance, MaxQUD can become the issue of what did the previous speaker say:<sup>58</sup>

<sup>56</sup> They carry over *inter alia* to Hebrew and to Modern Greek (Dimitra Kolliakou—p.c.).

<sup>57</sup> Repetition CRs seem to be able pertain to sub-utterances, not merely root utterances, as examples like (i) and (ii) show:

- (i) Stanley: How much do you want? Enid: It’s one twenty nine please. Stanley: One (pause) twenty did you say? (BNC, KE2)  
(ii) Terence: if they were built in sixty six, and I think they were, they are now, how old are they now? (pause) Lucy: Sixty six did you say? (BNC, KE2)

Hence our earlier comment about the need for DGB-PARAMS being defined for basic indexical features down to the word level applies here too.

<sup>58</sup> It is a bit troubling that given the semantics we have given to ‘Utter’, answers that do not repeat verbatim the initial utterance are strictly speaking *false*. This could be blamed on short-term memory or could suggest we need a more refined predicate for MaxQUD. The difficulty is that we do want to maintain a distinction from ‘What did you mean’ questions, so do not want to allow for too much reformulatory leeway.

(106) Repetition

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{MaxPending} = \left[ \begin{array}{l} \text{sit} = v0 \\ \text{sit-type} = T_{v0} \end{array} \right] : \text{LocProp} \\ p0 = \left[ \begin{array}{l} \text{sit} = u1 \\ \text{sit-type} = T_{u1} \end{array} \right] : \text{LocProp} \\ \text{cl: member}(u1, v0.\text{constits}) \end{array} \right] \\ \text{effects} : \left[ \begin{array}{l} \text{Spkr} = \text{pre.addr} \\ \text{Addr} = \text{pre.spkr} \\ \text{MaxQUD} = \lambda x \text{ Utter}(\text{pre.spkr}, \text{pre.u1}, x) : \text{Question} \end{array} \right] \end{array} \right]$$

**Exercise 6.8.1** Provide an update rule to explicate *gap*-type CRs (Purver, 2004(b)), exemplified in (107):

(107) A: Did Bo leave?  
B: Did Bo ...?

As an illustration, consider the example in (108):

(108) A(1): Is George here? B(2): What did you say? A(3): Is George here? B(4): mmh.

Let us consider A and B's DGBs after B's utterance (108(2)). As far as A goes there is nothing new to say. For B let us assume that after hearing (108(1)), she associated with it the type in (109b). Her DGB at this point is (109c), where  $u2'$  is the head daughter of the complete utterance  $u0'$ :

$$(109) \text{ a. } A.\text{dgb7} = \left[ \begin{array}{l} \text{spkr} = B \\ \text{addr} = A \\ \text{pending} = \left\langle \left[ \begin{array}{l} \text{sit} = w1 \\ \text{sit-type} = \text{WDYS} \end{array} \right] \right\rangle \\ \text{qud} = \langle p1? \rangle \\ \text{facts} = \left\{ \begin{array}{l} \text{In}(l, \{A, B\}), \\ \text{Named}('George', g), \\ \text{2ndMostRecentSpeechEvent}(u0), \\ \text{Classify}(\text{IGH}, u0), \\ \text{MostRecentSpeechEvent}(u1), \\ \text{Classify}(\text{IWH}, u1) \end{array} \right\} \\ \text{moves} = \left\langle \left[ \begin{array}{l} \text{sit} = w0 \\ \text{sit-type} = \text{IGH} \end{array} \right] \right\rangle \end{array} \right]$$

$$\begin{aligned}
 \text{b. } T_{3\text{word-polar}} &= \left[ \begin{array}{l} \text{phon : EnglishString} \\ \text{cat} = V[+\text{fin}, +\text{root}] : \text{syncat} \\ \text{constits} = \left\{ \begin{array}{l} \text{word1, word2, word3,} \\ \text{word1word2word3} \end{array} \right\} \\ \text{dgb-params} : \left[ \begin{array}{l} \text{spkr: IND} \\ \text{addr: IND} \\ \text{cl : address(s, a)} \end{array} \right] \\ \text{cont : IllocProp} \end{array} \right] \\
 \text{c. } B.\text{dgb7}' &= \left[ \begin{array}{l} \text{spkr} = B \\ \text{addr} = A \\ \text{pending} = \left\langle \left[ \begin{array}{ll} \text{sit} & = u0' \\ \text{sit-type} & = T_{3\text{word-polar}} \end{array} \right] \right\rangle \\ \text{qud} = \langle \lambda x \text{ Utter}(A, u2', x) \rangle \\ \text{facts} = \left\{ \begin{array}{l} \text{In}(l, \{A, B\}), \\ \text{Named}(\text{'George', } g), \\ \text{2ndMostRecentSpeechEvent}(u0'), \\ \text{Classify}(\text{IGH}, u0'), \\ \text{Mostrecentsspeechevent}(u1), \\ \text{Classify}(\text{IWH}, u1) \end{array} \right\} \\ \text{moves} = \left\langle \left[ \begin{array}{ll} \text{sit} & = w1 \\ \text{sit-type} & = \text{WDYS} \end{array} \right] \right\rangle \end{array} \right]
 \end{aligned}$$

Using CR accommodation and Ask QUD incrementation, A updates to the following DGB:

$$(110) \quad A.\text{dgb8} = \left[ \begin{array}{l} \text{spkr} = B \\ \text{addr} = A \\ \text{pending} = \langle \rangle \\ \text{qud} = \langle \lambda x \text{ Utter}(A, u2, x), p1? \rangle \\ \text{facts} = \text{dgb7.facts} \\ \text{moves} = \left\langle \left[ \begin{array}{ll} \text{sit} & = w1 \\ \text{sit-type} & = \text{WDYS} \end{array} \right], \left[ \begin{array}{ll} \text{sit} & = w0 \\ \text{sit-type} & = \text{IGH} \end{array} \right] \right\rangle \end{array} \right]$$

A utters (108(3)): I denote the type associated with (108(3)) as ‘Utt-A-IGH’, which is distinct from ‘IGH’ because on this analysis (108(3)) is a short answer. Its more prolix version would be ‘I said “Is George here?”’.<sup>59</sup> A’s DGB is now—after composing several updates—(111a); B accepts A’s assertion in (108(4)) and she can update to (111b).

$$\begin{aligned}
 (111) \quad a. \quad A.dgb9 = & \left[ \begin{array}{l} \text{spkr} = A \\ \text{addr} = B \\ \text{pending} = \langle \rangle \\ \text{qud} = \langle ?\text{Utter}(A, u0, \text{IGH}), \lambda x \text{ Utter}(A, u0, x), \text{pl?} \rangle \\ \text{facts} = \text{dgb7.facts} \\ \text{moves} = \left\langle \left[ \begin{array}{ll} \text{sit} & = w2 \\ \text{sit-type} & = \text{Utt-A-IGH} \end{array} \right], \left[ \begin{array}{ll} \text{sit} & = w1 \\ \text{sit-type} & = \text{WDYS} \end{array} \right], \left[ \begin{array}{ll} \text{sit} & = w0 \\ \text{sit-type} & = \text{IGH} \end{array} \right] \right\rangle \end{array} \right] \\
 b. \quad B.dgb9 = & \left[ \begin{array}{l} \text{spkr} = A \\ \text{Addr} = B \\ \text{pending} = \left\langle \left[ \begin{array}{ll} \text{sit} & = u0 \\ \text{sit-type} & = T_{3\text{word-polar}} \end{array} \right] \right\rangle \\ \text{qud} = \langle \rangle \\ \text{facts} = \text{dgb7'.facts} \cup \{ \text{Utter}(A, u0, \text{IGH}) \} \\ \text{moves} = \left\langle \left[ \begin{array}{ll} \text{sit} & = w3 \\ \text{sit-type} & = \text{mmh} \end{array} \right], \left[ \begin{array}{ll} \text{sit} & = w2 \\ \text{sit-type} & = \text{Utt-A-IGH} \end{array} \right], \left[ \begin{array}{ll} \text{sit} & = w1 \\ \text{sit-type} & = \text{WDYS} \end{array} \right] \right\rangle \end{array} \right]
 \end{aligned}$$

Given this, she can now perform pending extension on MaxPending, resulting in (112a). This is ripe for application of free speech and Ask QUD incrementation, resulting in (112b):

<sup>59</sup> I take it that on a RequestRepeat analysis (108(3)) *should* be analysed as ‘IGH’ because the utterance is simply being reperformed.

$$(112) \quad a. \quad B.dgb10 = \left[ \begin{array}{l} \text{spkr} = B \\ \text{Addr} = A \\ \text{pending} = \left\langle \left[ \begin{array}{ll} \text{sit} & = w0 \\ \text{sit-type} & = T_{IGH} \end{array} \right] \right\rangle \\ \text{qud} = \langle ?\text{Utter}(A, u0, IGH), \lambda x \text{ Utter}(A, u0, x) \rangle \\ \text{facts} = dgb9'.\text{facts} \\ \text{moves} = dgb9'.\text{moves} \end{array} \right]$$

$$b. \quad B.dgb11 = \left[ \begin{array}{l} \text{spkr} = A \\ \text{Addr} = B \\ \text{pending} = \langle \rangle \\ \text{qud} = \langle pl? \rangle \\ \text{facts} = dgb9.\text{facts} \\ \text{moves} = \left\langle \left[ \begin{array}{ll} \text{sit} & = w0 \\ \text{sit-type} & = T_{IGH} \end{array} \right], \right. \\ \quad \left[ \begin{array}{ll} \text{sit} = w3 \\ \text{sit-type} = mmh \end{array} \right], \\ \quad \left[ \begin{array}{ll} \text{sit} & = w2 \\ \text{sit-type} & = \text{Utt-A-IGH} \end{array} \right], \\ \quad \left. \left[ \begin{array}{ll} \text{sit} & = w1 \\ \text{sit-type} & = \text{WDYS} \end{array} \right] \right\rangle \end{array} \right]$$

Note that A's DGB at this point will—all going well—be (113), which is similar but not quite identical with B's:

$$(113) \quad A.dgb10 = \left[ \begin{array}{l} \text{spkr} = A \\ \text{Addr} = B \\ \text{pending} = \langle \rangle \\ \text{qud} = \langle pl? \rangle \\ \text{facts} = dgb9'.\text{facts} \\ \text{moves} = \left\langle \left[ \begin{array}{ll} \text{sit} & = w3 \\ \text{sit-type} & = mmh \end{array} \right], \right. \\ \quad \left[ \begin{array}{ll} \text{sit} & = w2 \\ \text{sit-type} & = \text{Utt-A-IGH} \end{array} \right], \\ \quad \left[ \begin{array}{ll} \text{sit} & = w1 \\ \text{sit-type} & = \text{WDYS} \end{array} \right], \\ \quad \left. \left[ \begin{array}{ll} \text{sit} & = w0 \\ \text{sit-type} & = T_{IGH} \end{array} \right] \right\rangle \end{array} \right]$$

**Exercise 6.8.2** Offer an account of Repetition CRs in terms of a RequestRepeat act. Use this to analyse (108).

**Exercise 6.8.3** Write a lexical entry for ‘eh?’

## 6.9 Summary

In this chapter I have discussed how to integrate metacommunicative with illocutionary interaction. In fact, the set-up within KoS is such that this integration is straightforward. In light of our previous recognition that there is no single *context* in conversation but rather *coupled, but potentially mismatched* dialogue gameboards, solely one modification is required to the structure of the DGB, the postulation of a field:

- **Pending**, whose members are ungrounded utterances. **Pending** is a list of *locutionary propositions*, propositions consisting of an utterance record and a (grammatical) type which classifies it.

This means that the DGB can now be described in the following terms:

$$(113) \text{ DGBType (final version) } =_{\text{def}} \left[ \begin{array}{ll} \text{spkr} & : \text{IND} \\ \text{addr} & : \text{IND} \\ \text{utt-time} & : \text{TIME} \\ \text{c-utt} & : \text{addressing}(\text{spkr}, \text{addr}, \text{utt-time}) \\ \text{Pending} & : \text{list}(\text{LocProp}) \\ \text{Moves} & : \text{list}(\text{LocProp}) \\ \text{Facts} & : \text{Set}(\text{Prop}) \\ \text{QUD} & : \text{poset}(\text{Question}) \\ \text{non-resolve-cond} & : \forall q \in \text{QUD} [\neg \text{Resolve}(\text{FACTS}, q)] \end{array} \right]$$

I surveyed detailed corpus studies of Clarification Requests (CRs) that demonstrate the striking conclusion that communicative trouble in conversation is strongly circumscribed. Thus, the primary function of CRs are:

1. **Content-oriented**: confirming or querying intended content,
2. **Form-oriented** requesting repetition of a misheard (sub)-utterance.

This is what makes the formulation of a theory of CRification tractable. In the current set-up, grounding and CRification can be described using update rules entirely akin to those used for illocutionary interaction in Chapter 4:

- **Grounding:** ensues when the utterance type fully classifies the utterance token. Associating locutionary propositions with dialogue moves, rather than solely their illocutionary content, enables us to account for the vast range of utterance-related presuppositions that potentially accumulate post-utterance and also for priming phenomena.
- **CRification:** ensues when the utterance type calculated is weak (e.g. incomplete word recognition); or when further information is needed to spell out a token (e.g. incomplete contextual resolution).

Using the taxonomy of Purver, Ginzburg, and Healey (2001) as a measure of coverage:

- **Contextual coherence:** an analysis has been provided for all four categories.
- **Grammatical analysis:** an analysis has been provided for the sentential and lexical form categories. Chapter 7 provides grammatical analysis for the two major non-sentential form categories, Reprise Fragments and sluicing.

Indeed although the approach described here is inspired in certain ways by Clark's collaborative model, our maintaining of individual conversational participants' autonomy means that we are not susceptible to the weaknesses Allwood (1995) points out for Clark's model, for instance inability to accommodate one-sided dialogue moves.

Finally, some remarks on formalization: the account of grounding/CRification given here is formalized in TTR, following an initial sketch in Cooper and Ginzburg (2002), in contrast to earlier analyses such as Ginzburg and Cooper (2001, 2004), which employ typed feature structures. The present account represents an advance on the latter in that most of the principles postulated here cannot be satisfactorily formalized using typed feature structures.<sup>60</sup> This is primarily because the principles make simultaneous reference to entities both at the level of tokens and types, a duality which eludes a TFS-based formalism.

<sup>60</sup> See also earlier discussion of these issues in Chapter 5, section 5.2.

## Non-Sentential Utterances in Conversation

‘When I saw the ball in the net I said, “Me with my head? Absurd”’, Di Natale told reporters.  
(*The Guardian*, 13 September 2007)

### 7.1 Introduction

This chapter integrates material from earlier chapters in an attempt to provide a comprehensive treatment of the meaning and use underlying NSUs, primarily in English, but with occasional cross-linguistic glances. Section 7.2 provides the empirical basis for the rest of the chapter: a taxonomy of NSUs is described and results from a corpus study of the BNC using this taxonomy are provided. The rest of the chapter provides analyses of prominent members of the taxonomy: ‘Yes’, ‘No’, short answers, sluicing (and its various subclasses), and Reprise Fragments. How to analyse NSUs? In section 7.3 I argue against a number of existing *unitary* approaches to NSUs, approaches in which NSU resolution is associated with a single, typically extra-grammatical mechanism. In so doing I will use both fairly standard linguistic argumentation—for instance, pointing to various syntactic and semantic divergences between NSUs and their putative sentential correlates—as well as issues of scalability. I will also rely on developmental evidence, though the bulk of this will be presented in Chapter 8, in section 8.4.

In line with earlier discussions about grammar in Chapter 5, the basic analysis developed for incorporating NSUs in the grammar is constructionist, constructions where context rather than syntax drives interpretation. From a semantic point of view, the approach developed here is one that treats NSUs essentially as indexicals—the primary contextual parameter being the maximal element of QUD, buttressed in some cases by an antecedent sub-utterance, dubbed the Focus Establishing Constituent



(FEC), whose role is explained below.<sup>1</sup> The basic claim embodied in this view is that, given a detailed, precise, and, to use the stock phrase, independently motivated theory of dialogue context, the theory of NSUs is actually relatively simple.

This chapter restricts attention to two person dialogue. We will see in Chapter 8 that there are important differences concerning NSU resolution in the multiparty case, primarily with respect to the possible distance between antecedent and NSU.

## 7.2 Data and Basic Theoretical Picture

### 7.2.1 NSUs: a Taxonomy

NSUs are, in principle, highly ambiguous utterances: (1a) will get radically different understandings depending on whether it is uttered after one or other of (1b, c, d):

- (1) a. B: On the table?
- b. A: Where to put it?
- c. A: He left his leg on the table.
- d. A: I must have put it um ...

Indeed there have been claims in the literature (see e.g. Allen and Perrault 1980 and in particular Carberry 1991) that NSUs can in general only be disambiguated and resolved on the basis of domain-specific heuristics that involve plan recognition. While this might be true of isolated examples, on the whole this appears to be an overly strong conclusion. In this section I introduce a semantically-based taxonomy of NSUs, first proposed in Fernández and Ginzburg (2002).<sup>2</sup> This taxonomy has very good coverage of NSUs as they appear in the BNC and can be reliably coded by human annotators. Moreover, this taxonomy has been shown to be machine-learnable by Fernández, Ginzburg, and Lappin (2007) using a variety of supervised algorithms, on the basis of a small number of morphosyntactic, semantic, and contextual features. Indeed, in the sections of the chapter following this one, I will seek to demonstrate the following: one can explicate the various categories of NSUs by employing domain-independent methods of constructing a context and making use of a dialogue-oriented grammar.

Tables 7.1 and 7.2 provide a basic description and exemplification of the taxonomy, along with a crude clustering of the classes according to function.<sup>3</sup>

<sup>1</sup> One could argue that NSUs should rather be viewed as *anaphoric*, as Alex Lascarides (p.c.) has suggested to me. ‘Indexical’ is intended to emphasize that NSUs make essential reference to one of the basic contextual parameters, which on the analysis presented here is MAX-QUD.

<sup>2</sup> For a discussion of other taxonomies, including in particular the recent taxonomy in Schlangen (2003), see Fernández (2006).

<sup>3</sup> The labelling of the entire corpus of NSUs was done by one expert annotator. As reported in Fernández (2006), a subsequent pilot study with two additional, non-expert annotators was carried out to assess the reliability of the taxonomy. These annotated a total of fifty randomly selected instances (containing a minimum of two instances of each NSU class as labelled by the expert annotator) with the classes in the taxonomy. The agreement obtained by the three annotators was reasonably good, yielding a kappa score of

TABLE 7.1 A Taxonomy for NSUs

| NSU class                 | Description                                                           | BNC Example                                                            |
|---------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------|
|                           | Positive Feedback                                                     |                                                                        |
| Plain Acknowledgement     | signal understanding or acceptance                                    | A: I know that they enjoy debating these issues.<br>B: <i>Mhm.</i>     |
| Repeated Acknowledgement  | Acknowledgements containing verbatim repetition of previous utterance | A: I'm at a little place called Ellenthorpe.<br>B: <i>Ellenthorpe.</i> |
| Metacommunicative queries |                                                                       |                                                                        |
| R(eprise) F(ragments)     | fragments used as CRs                                                 | A: There's only two people in the class B: <i>Two people?</i>          |
| Check Question            | requests explicit feedback                                            | A: So <pause> I'm allowed to record you. <i>Okay?</i>                  |
| Sluice                    | <i>wh</i> -question NSU                                               | A: Only wanted a couple weeks. B: <i>What?</i>                         |
| Filler                    | completes unfinished utterance                                        | A: [...] twenty two per cent is er <pause> B: <i>Maxwell?</i>          |

These classes are not equally fine-grained. In certain cases a single class encodes an NSU type that is, in the final analysis, ambiguous, though these ambiguities are difficult to resolve even in context and probably rarely are. Thus, the class *Plain Acknowledgement* contains *affirmative lexemes* such as 'mmh' and 'yeah', which can be used to acknowledge understanding, as well as to accept an assertion. These are, in principle, distinct acts, but it is usually quite hard to tease them apart.<sup>4</sup> The same applies to *Check Questions*, which can be used to check acceptance of an assertion or to verify understanding.

In contrast, some classes contain ambiguous NSU types that do need to be resolved, even if occasionally they remain ambiguous in context. This applies to Reprise Fragments whose intrinsic ambiguity between a *clausal confirmation* and an *intended content* reading was discussed in Chapter 6. It applies with even greater vengeance to the class *Sluice*. This class is discussed in greater detail in section 7.8. For the moment it suffices to mention that *wh*-NSUs have more than one understanding, the commonest a metacommunicative question as illustrated in Table 7.1, but also a reading as a straightforward 'information query' (A: *Someone left.* B: *Who?*). These

0.76. The non-expert annotators were also asked to identify the antecedent sentence of each NSU. Using the expert annotation as a gold standard, they achieved 96% and 92% accuracy in this task.

<sup>4</sup> See Schlangen (2005) on the difficulties of machine-learning this distinction.

TABLE 7.2 A Taxonomy for NSUs (cont.)

| Answers                        |                                                  |                                                                                               |
|--------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Short Answer                   | answer-conveying<br>NSUs to <i>wh</i> -questions | A: Who's that?<br>B: <i>My Auntie Peggy.</i>                                                  |
| Affirmative Answer             | positive answer to<br>polar query                | A: Do I put a cross?<br>B: <i>Yes.</i>                                                        |
| Rejection                      | negative answer to<br>polar query, assertion     | A: I think I left<br>it too long. B: <i>No no.</i>                                            |
| Repeated Affirmative<br>Answer | Affirmative response<br>with repetition          | A: Did you shout very loud?<br>B: <i>Very loud</i> , yes.                                     |
| Helpful Rejection              | negative answers,<br>with alternative            | A: Is that Mrs. John?<br>B: <i>No, Mrs. Billy.</i>                                            |
| Propositional Modifier         | propositional adverbs                            | A: I wonder if that would be<br>worth getting?<br>B: <i>Probably not.</i>                     |
| Extension Moves                |                                                  |                                                                                               |
| Factive Modifier               | Factive adjectives                               | A: There's your keys.<br>B: <i>Oh great!</i>                                                  |
| Bare Modifier                  | Adjuncts                                         | A: [...] they got men and<br>women in the same dormitory!<br>B: <i>With the same showers!</i> |
| Conjunction + frag             | NSUs introduced<br>by conjunctions               | A: Alistair erm he's, he's made<br>himself coordinator.<br>B: <i>And section engineer.</i>    |

ambiguities, which are reliably distinguishable<sup>5</sup> and have various linguistic reflexes, justify introducing subclasses into the taxonomy. I will do so when offering distinct analyses of their meanings and contextual backgrounds. I avoid doing so here simply for reasons of expository economy.

### 7.2.2 A corpus study of NSUs

The taxonomy described in the previous subsection was used in a number of corpus studies of the BNC (see e.g. Fernández and Ginzburg 2002, Fernández 2006). As reported in Fernández (2006), this taxonomy was used to manually classify a randomly selected section of 200 speaker-turns from 54 BNC files. The examined sub-corpus contains 14,315 sentences. A total of 1,299 NSUs were found. Of these, 1,283 (98.9 per cent) were successfully labelled according to the taxonomy, as illustrated in

<sup>5</sup> For RFs see Purver, Ginzburg, and Healey (2001); for sluicing see Fernández, Ginzburg, and Lappin (2004).

TABLE 7.3 NSUs in a sub-corpus of the BNC: ordered by frequency

| NSU class                   | Example                                    | Total       |
|-----------------------------|--------------------------------------------|-------------|
| Plain Acknowledgement       | A: ... B: <i>mmh</i>                       | 599         |
| Short Answer                | A: <i>Who left?</i> B: <i>Bo</i>           | 188         |
| Affirmative Answer          | A: <i>Did Bo leave?</i> B: <i>Yes</i>      | 105         |
| Repeated Acknowledgement    | A: <i>Did Bo leave?</i> B: <i>Bo, hmm.</i> | 86          |
| C(larification) E(llipsis)  | A: <i>Did Bo leave?</i> B: <i>Bo?</i>      | 79          |
| Rejection                   | A: <i>Did Bo leave?</i> B: <i>No.</i>      | 49          |
| Factive Modifier            | A: <i>Bo left.</i> B: <i>Great!</i>        | 27          |
| Repeated Affirmative Answer | A: <i>Did Bo leave?</i> B: <i>Bo, yes.</i> | 26          |
| Helpful Rejection           | A: <i>Did Bo leave?</i> B: <i>No, Max.</i> | 24          |
| Sluice                      | A: <i>Someone left.</i> B: <i>Who?</i>     | 24          |
| Check Question              | A: <i>Bo isn't here.</i> B: <i>Okay?</i>   | 22          |
| Filler                      | A: <i>Did Bo ...</i> B: <i>leave?</i>      | 18          |
| Bare Modifier Phrase        | A: <i>Max left.</i> B: <i>Yesterday.</i>   | 15          |
| Propositional Modifier      | A: <i>Did Bo leave?</i> B: <i>Maybe.</i>   | 11          |
| Conjunction + fragment      | A: <i>Bo left.</i> B: <i>And Max.</i>      | 10          |
| <b>Total dataset</b>        |                                            | <b>1283</b> |

Table 7.3, the remaining sixteen instances did not fall in any of the categories of the taxonomy and were labelled 'Other'. Table 7.3 shows the distribution of NSU categories in the aforementioned sub-corpus of the BNC, whereas Table 7.4 presents this data by functional groupings.

What's in the corpus and what isn't? The majority class by far is acknowledgement, followed fairly closely by various kinds of answers. Metacommunicative queries constitute about ten per cent of all NSUs. The actually occurring NSUs are, then, predominantly the products of directly produced illocutionary and metacommunicative dialogue acts. What is missing from this line-up, interestingly, are NSUs which require a heavy dose of genre-based reasoning. Of course such NSUs do occur, as we discuss in section 7.10, where we show how to integrate them into our analysis. This does, nonetheless, point to a view of NSUs as above all 'indexical'—drawing on publicly, domain-independent context.

### 7.3 Analysing NSUs: Some Existing Strategies

How are we to develop a grammatical theory of NSUs? The somewhat crude functional partition embodied in the supercategories of the NSU taxonomy (see Tables 7.1 and 7.3) give a rough idea of NSU characteristics but is insufficiently refined for linguistic analysis.

**TABLE 7.4 NSUs in a sub-corpus of the BNC: classified by function**

| NSU class                             | Total |
|---------------------------------------|-------|
| <b>Positive Feedback: 685</b>         |       |
| Plain Acknowledgement                 | 599   |
| Repeated Acknowledgement              | 86    |
| <b>Answers: 413</b>                   |       |
| Short Answer                          | 188   |
| Affirmative Answer                    | 105   |
| Rejection                             | 49    |
| Repeated Affirmative Answer           | 26    |
| Helpful Rejection                     | 24    |
| Propositional Modifier                | 11    |
| <b>Metacommunicative queries: 132</b> |       |
| C(larification) E(llipsis)            | 79    |
| Check Question                        | 22    |
| Reprise Sluice                        | 13    |
| Filler                                | 18    |
| <b>Extension Moves: 63</b>            |       |
| Factive Modifier                      | 27    |
| Bare Modifier                         | 15    |
| Direct Sluice                         | 11    |
| Conjunction + fragment                | 10    |

Before actually developing an analysis, it is worth considering a number of possible strategies. A small number of NSU types have been analysed in past work, primarily direct sluicing and short answers. One class of approaches subsumes NSU constructions into another grammatical phenomenon such as phonological reduction or anaphora—ellipsis resolution is associated with a single, typically extra-grammatical mechanism. I will dub such approaches *unitarian* theories. An alternative strategy views NSU constructions as actually being in some significant way *sui generis*. Theories that follow the latter route are *constructionist* theories in which NSUs are incorporated in the grammar as distinct constructions, which specify among other things the contextual characteristics which govern their use.

Developmental evidence about NSUs is detailed in section 8.4. One of the main findings I discuss there is that (adult-like) NSUs do not appear more or less simultaneously, but in fact emerge over a period of one-and-a-half to two years. This poses difficulty for theories in which NSU resolution is associated with a single mechanism,

since once the child has acquired one type of NSU, s/he has by necessity acquired *the* mechanism for generating NSUs. In order to explain the absence of a delayed NSU type, one will have to ensure that the input to this mechanism which gives rise to the delayed type is absent. Particularly when an NSU type with a similar pragmatic background to the delayed one is already present (e.g. as we will see with sluicing in the case of RFs), this condition will be unlikely to occur. Indeed I will show there that this condition fails for the two unitary strategies discussed here.

Within the unitarian approaches is included the commonest generative approach, which I dub (*Underlying NSU*) *sententialism*. On this view NSUs such as sluicing and short answers are analysed as underlyingly *sentential* constructions (for sluicing see e.g. Ross 1969, Chung, Ladusaw, and McCloskey 1995, Merchant 2002; for short answers see Morgan 1973, Merchant 2004.) On this view, which has attracted a variety of implementations, the claim is that grammar does not need to reify NSUs as such—no new rules or principles need to be posited for individual NSU constructions. A non-grammatical module serves—under appropriate conditions—to recover material that is phonologically redundant and remains unpronounced. One of the major motivations for this type of approach is to account for parallelism phenomena between source and target that are exhibited by NSUs using grammatical principles that apply to non-elliptical sentences. We have seen examples of such parallelism with RFs in Chapter 6, and this is further exemplified by the case parallelism between source and target pointed out by Ross for sluicing (the German examples provided here are minor variants on Ross's original ones), and by Morgan for short answers (the Hebrew examples provided here are from Ginzburg and Sag 2000):

- (2) a. Er will jemandem schmeicheln, aber sie wissen nicht  
 He wants someone-dat flatter, but they know not  
 wem/#wen.  
 who-dat/#who-acc.  
 'He wants to flatter someone, but they don't know whom.'
- b. Er will jemanden loben, aber sie wissen nicht wen/#wem.  
 He wants someone-acc praise, but they know not who-acc/#who-dat.  
 'He wants to praise someone, but they don't know whom.'
- (3) a. A: lemi hixmeta? B: #moti/lemoti.  
 To-who flattered-2nd-sg? #Moti/to-Moti  
 A: Who did you flatter? B: Moti.
- b. A: et mi šibax? B: et moti/#lemoti.  
 def-acc who praised-2nd-sg? def-acc Moti/#to-Moti  
 A: Who did you praise? B: Moti.

Accounting for (categorical/phonological) parallelism conditions is, without doubt, important and their existence provides clues as to the architecture of context, as I

have already emphasized in Chapter 6. The problem in trying to explain these by reducing NSUs to ‘canonical’ sentential constructions are the syntactic and semantic divergences that exist between NSUs and their putative sentential correlates. Examples of these abound. We consider in turn short answers, sluicing, and exclamative phrases.

• **Short answers:**

There does not seem to be a single sentential form whose well-formedness correlates with the well-formedness of short answers: some authors (e.g. von Stechow 1991, Rooth 1993) assume that the reconstruction correlate is a sentential form in which the fragment is inserted in the *in situ* position from which they assume the *wh*-phrase in the original question was fronted. Others (e.g. Merchant 2004) assume the correlate is a sentential form in which the short answer is fronted from the sentential nucleus. Other possibilities are *it*-clefts and pseudo-clefts. But none of these is a good predictor of the felicity of short answers. (4)—variants of data pointed out by Morgan (1973)—illustrate cases where neither *in situ* nor fronted correlate are well formed while the short answer is felicitous; (4c–e) illustrate similar data from Hebrew:

- (4) a. A: Who silenced Malan? B: Not Bill.  
 b. \*Not Bill silenced Malan.  
 c. A: *ma moti axal?* B: *klum.*  
     What Moti ate-3rd-sg? nothing  
     A: What did Moti eat? B: Nothing.  
 d. \**moti axal klum.*  
 e. \**klum moti axal.*

(5) is a case where the *in situ* correlate is OK, while the fronted, cleft, and pseudo-cleft are bad and the short answer is felicitous:<sup>6</sup>

- (5) a. A: What did Jill say? B: Nothing/everything.  
 b. Jill said nothing/everything.  
 c. # Nothing/Everything, Jill said.  
 d. # It was nothing/everything that Jill said.  
 e. # What Jill said was nothing/everything.

<sup>6</sup> Admittedly the version of a cleft headed by a ‘there’ phrase does not run into either of these semantic problems:

- (i) There was nothing that Jill said.  
 (ii) There were few Frenchmen that Jill met in Paris.

But, as is well known, this construction is infelicitous with a whole range of quantifiers, whereas these are perfectly felicitous as answers:

- (iiia) A: Who showed up to the press conference? B: Every journalist I know.  
 (iiib) # There were/was every journalist I know that showed up to the press conference.

See Morgan (1973) for detailed argumentation illustrating the problematic nature of pseudo-clefts and *it*-clefts as ‘reconstruction correlates’ for short answers.

Merchant (2004) is aware of cases like (5)—he cites data similar to (5c) due to Postal (1993). His view of such cases, in common with *inter alia* Lasnik (2001), is that the short answer is a way of *saving* the overtly unrealizable sentential form. As Stainton (2006) points out, this strategy drastically weakens the refutability of a sententialist strategy since it means all miscorrelations between the fragment and sentential form can be explained away.

Morgan's original claim was that the well-formedness of fragments involves checking whether reconstructed forms satisfy binding theory (BT). Morgan suggests that a Condition C violation (in contemporary terminology) in the reconstructed form makes the short answer infelicitous on the basis of examples such as the following:

- (6) a. A: What does John<sub>i</sub> think?  
       B: # That the bastard<sub>i</sub> is spying on him.  
       b. # John<sub>i</sub> thinks that the bastard<sub>i</sub> is spying on him.

Similarly, he suggests cases like (7) should be explained as violations of principle B:

- (7) a. A: Who does Bill<sub>i</sub> like? B: himself<sub>i</sub>/# him<sub>i</sub>  
       b. Bill likes himself<sub>i</sub> /# him<sub>i</sub>

(6) seems equally explicable in terms that refer to point of view. Epithets are typically linked to the point of view of the speaker and are, therefore, difficult to link with the point of view of another agent. Thus, consider (8): it is difficult to get the epithet to be co-referential with 'John' despite the fact that there is no structural conflict:

- (8) John<sub>i</sub> informed me that various interesting facts were discovered last week. ? The most interesting one was the fact that the bastard<sub>i</sub> will go shopping next week.

Nonetheless, epithet-containing short answers are felicitous in certain cases where the reconstructed form is not. This then is another instance of the miscorrelation between reconstruction correlates and short answers:

- (9) a. A: Who likes that bastard Bill? B: Only the bastard himself.  
       b. # Only [the bastard himself]<sub>i</sub> likes that bastard Bill<sub>i</sub>

Moving on to the pronoun/anaphora cases, I note for a start an example where the short answer is badly out of synch with its reconstruction correlates (either fronted or dislocated versions), both of which are infelicitous (on the intended indexing). In (10) binding theory (assuming the *in situ* correlate) predicts that the reflexive short answer should be infelicitous, whereas in fact it is good, in contrast to the pronominal version, predicted to be good by binding theory. The fronted reconstruction correlate predicts both short answers to be bad:

- (10) a. A: Who does Jill think Bill desires? B: Herself/#Her.  
       b. # Jill thinks Bill desires herself.  
       c. Jill thinks Bill desires her.  
       d. # Herself/Her Jill thinks Bill desires.



An alternative explanation of such cases is provided by two independently motivated assumptions: (a) a short answer is focal and hence must be accented (see e.g. Taglicht 1984),<sup>7</sup> (b) accented pronouns in English must be contrastive. A consequence of this is to rule out co-reference in examples such as (11):

- (11) a. A: Who does Bill<sub>i</sub> like? B: Himself<sub>i</sub>/# Him<sub>i</sub>  
 b. A: Who does Jill think Bill desires? B: Herself/#Her.

One additional problem for reconstruction, in this case for the fronted version, derives from an argument due to Stainton (2006)—short answers from *in situ wh*-interrogatives where the *wh*-phrase in the question is in an island configuration. The ungrammaticality of the (fronted) reconstruction correlate, contrasting with the felicity of the short answer, is illustrated in (12) for a *wh*-island violation. Similar cases can be constructed for other island configurations:

- (12) a. A: We don't know whether a solution will be provided by which approach?  
 B: The functional approach.  
 b. \*The functional approach we don't know whether a solution will be provided by.

In addition to syntactic miscorrelations between short answers and their reconstruction correlates, there are also semantic divergences. Here is one instance, based on observations by Corblin (1994) for French, adapted for Hebrew data in Ginzburg (1999b)—the Hebrew data makes the point somewhat more strongly. The observations are these:

- the short answer (13a) has only a  $\forall\exists$  reading (for (13a) this amounts to a reading where everyone met at least one visitor.)
- the reconstructed version only has the reading in which no meetings/no inspections took place (in French this is the *preferred* reading;  $\forall\exists$  also available)

- (13) a. A: mi lo pagash af exad miamevakrim? B2: af exad.  
 A: who not met no one of the visitors? B2: no one.  
 'Who met none of the visitors?' 'No one.'  
 b. af exad lo pagash af exad miamevakrim.  
 'No one met any of the visitors.'

- **Direct Sluicing:** the fact that (direct) sluicing diverges from its putative reconstruction correlate—an interrogative clause involving a fronted *wh*-phrase—has been known since Ross (1969), who first described the construction. Whereas sluicing is possible with antecedents within islands, *wh*-fronting is of course not. In a variety of derivational accounts this was explained by assuming that sluicing

<sup>7</sup> For the relevant notion of 'focus' in the current framework, see Ginzburg (1999b), Engdahl, Larsson, and Ericsson (2000). In any case, though, on most analyses short answers are focal, in the various explanations the term has undergone.

was derived at a level of representation which was not constrained by subadjacency (see e.g. Ross 1969, Levin 1982, Chung, Ladusaw, and McCloskey 1995). A more recent strategy, once multiple strata were no longer available, has been to explain sluicing as a *repair strategy*. As I mentioned with respect to short answers, such a strategy is problematic from a methodological point of view if sententialism is motivated by the assumption that the NSUs *are* canonical sentential structures. In any case, there is additional data pointing to divergence of sluicing and fronted *wh*-interrogative clauses. One property which seems to distinguish fronted *wh*-phrases from those that are *in situ* is the possibility of licensing modification by intensifiers such as *the hell/heck/fuck/blazes/on earth*, as discussed in some detail by Ginzburg and Sag (2000):

- (14) a. What on earth will you do with it?  
 b. You will do what (# on earth) with it?  
 c. Who on earth saw Mo?  
 d. Who on earth saw who?  
 e. #Who saw who on earth?

Ginzburg and Sag (2000) point out that sluicing does NOT license intensive modification of this kind:

- (15) A: Somebody stood outside the room. B: Who? / #Who the hell? / Who the hell stood outside the room?

- **Exclamative sluicing (exclams):**

Finally, in this regard, I mention a variant of sluicing—bare exclamative phrases ('exclams'), exemplified in (16):

- (16) A: Hercules cleaned the Augean stables.  
 B1: What a hero (Hercules was).  
 B2: What a terrible task (to clean the Augean stables).  
 B3: What a horrible place (the Augean stables were).

I am not aware of a sententialist analysis of exclams.<sup>8</sup> Example (17) illustrates that whereas exclamative clauses are embeddable, exclams are not:

- (17) A: What a shot! / \*It's amazing what a shot. / It's amazing what a shot she made.

A significantly distinct unitarian approach to the description of ellipsis constructions is the higher-order unification (HOU) approach pioneered in Dalrymple et al. (1991) for VP ellipsis, and extended to short answers in Pulman (1997) and to corrections in Gardent, Kohlhase, and van Leusen (1996). On this view an ellipsis construction involves the recovery of a property from context such that when it is applied to

<sup>8</sup> A proper treatment of exclams presupposes an analysis of quantified NPs, a sketch of which is provided in Chapter 8. See Exercise 8.5.2 in Chapter 8 for further discussion of exclams.

the content of the ellipsis target, it resolves the interpretation of the elliptical construction. The property is recovered from context by using higher-order unification to solve an equation that relates the target with its antecedent. In distinction from *Underlying NSU sententialism*, HOU approaches make no appeal to syntactic spelling-out of linguistic material. However, in common with *Underlying NSU sententialism*, NSUs are viewed as arising from the application of a single generalized extra-grammatical mechanism. The HOU-based account offers a single mechanism that uniformly accounts for VP ellipsis, focus constructions, and certain NSU constructions such as short answers and corrections. An important insight in HOU is the need to track parallelism between target and source, and while this is not encoded dynamically in HOU approaches, I will show below how it can be implemented. The most intrinsic difficulties for HOU involve scalability. First, HOU is not possible when the semantic type of the target is distinct from that of the source, as when the target denotes a question and the source a proposition or vice versa. Such cases abound for NSUs (short answers, sluicing, certain cases of RF, 'Yes'/'No' etc.). Pulman (1997) attempts to bypass this problem by associating propositional contents with interrogatives. However, such a move is semantically dubious (Groenendijk and Stokhof 1997, Ginzburg and Sag 2000). Second, there are various types of NSUs which cannot possibly arise as solutions of HOU—these include intended content RFs and exclams:<sup>9</sup>

- (18) a. A: Did Bo leave? B: Bo? (*Who are you referring to as 'Bo'?* or similar paraphrase)  
 b. A: Hercules cleaned the Augean stables.  
 B1: What a hero (Hercules was).  
 B2: What a terrible task (to clean the Augean stables).

One additional approach, which does not fit in either of the above categories is that of Stainton (2006). Stainton offers a variety of insightful arguments against existing approaches to NSU constructions, primarily against sententialist ones, some of which I have noted above. He argues in favour of a *pragmatic* approach, in which NSUs are also bare *semantically* (i.e. non-proposition or question-denoting), whereas their content arises via pragmatic inference at the level of conversationalist information states. Given the emphasis here on resolution relative to the DGB there is clearly some commonality in approach. One difference between Stainton's approach and the current one is that Stainton does not offer an explicit theory of information states, nor of the resolution process. Indeed, a pragmatic approach that does not reify NSUs as distinct constructions faces difficulties with explicating the distinct kinds of syntactic/phonological parallelism conditions which distinct kinds of NSU constructions exhibit and their contextual dynamics. Moreover, his account is susceptible to similar

<sup>9</sup> Ginzburg and Cooper (2004) do show that HOU can yield solutions for the clausal confirmation RFs.

arguments on the developmental effects as the HOU approach: once pragmatics can kick in for short answers and *wh*-phrases are available to the child, s/he should be able to produce sluices. Admittedly, there is relatively little detail about the pragmatic resolution of sluices and MCI-oriented NSUs are not discussed, so it is a bit difficult to make strong negative claims in this respect.

To summarize, we have seen the difficulties faced by two influential *unitarian* approaches to NSUs—approaches in which NSUs are explicated in terms of a single, extra-grammatical mechanism, namely sententialism and HOU-based resolution. Sententialism is undermined by the divergent properties fragments exhibit from their putative canonical sentential correlates. HOU scales up badly. My evaluation has been at a strategic level, abstracting for the most part from specifics of proposals:<sup>10</sup> neither of the two approaches I considered has been formulated in such a way as to interface explicitly with dialogue context—issues of how to project relevant non-semantic information into the context have not been addressed. Nonetheless, in what follows I will draw on insights deriving from both approaches. Sententialism emphasizes the importance of capturing (non-semantically-based) parallelism conditions, whereas HOU underlines the need to track parallelism between the fragment and (an element of) the source.

## 7.4 Towards a Theory of NSU Constructions

In the previous section, we considered some of the difficulties faced by two influential *unitarian* approaches to NSUs—approaches in which NSUs are explicated in terms of a single, extra-grammatical mechanism.

The alternative adopted here is the strategy of **dialogue-oriented constructionism** (DOC). This embodies the claim that the context-oriented aspect of NSU words and constructions is as conventional as, for instance, ‘traditional’ indexicals (‘I’, ‘here’, ‘now’)—there being no reasonable criterion which will leave the former outside the purview of grammar and the latter inside. Adopting DOC permits the use of precise methods of linguistic description for such words and expressions, rather than relegating this to, say, domain-specific knowledge.<sup>11</sup> Moreover, we will show how DOC can offer a particularly explicit account of parallelism phenomena on the basis of the dynamics of context.

<sup>10</sup> For a detailed consideration of recent sententialist approaches to sluicing, see Sag and Nykiel (2011). Sag and Nykiel marshal extensive cross-linguistic evidence, allied with psycholinguistic and historical data, against a number of putative generalizations that have been taken to support a sententialist position. In particular, they dispute the generalization, due to Merchant (2001), that a language will allow preposition stranding under sluicing if and only if it allows preposition stranding under regular *wh*-movement.

<sup>11</sup> In saying this, there is no claim that domain-specific knowledge has no role to play in NSU resolution, just as it has in the resolution of standard indexicals, see e.g. recent work on the subtleties involved in resolving ‘you’ in conversation (Frampton et al. 2009).

**TABLE 7.5 NSUs sorted according to Direct Context (DC) and Partial Parallelism (ParPar)**

|         | –DC                                            | +DC                      |
|---------|------------------------------------------------|--------------------------|
| –ParPar | <i>Why<sub>meta</sub></i><br>Repetition sluice | Affirmative Answer       |
|         |                                                | Rejection                |
|         |                                                | Factual Modifier         |
|         |                                                | Propositional Modifier   |
|         |                                                | Plain Acknowledgement    |
| +ParPar | Reprise Fragment                               | Short Answer             |
|         | Reprise Sluice                                 | Repeated Aff. Answer     |
|         | Filler                                         | Direct Sluice            |
|         | Genre-sensitive                                | Helpful Rejection        |
|         | Short Answer/Sluice                            | Bare Modifier Phrase     |
|         |                                                | Conjunction + fragment   |
|         |                                                | Repeated Acknowledgement |

Concretely, a linguistic analysis of NSUs involves three dimensions: (a) the contextual parameters involved, (b) the inference (or lack thereof) involved in constructing the context, and (c) combinatory operations relating the fragment phrase and the contextual parameters, constrained by parallelism conditions. Items (a) and (b) will turn out to be useful features in terms of which to categorize NSUs. We will analyse the complexity of NSUs in terms of (a)–(c) in section 8.4. Table 7.5 partitions the NSUs using the features +/- Direct Context (DC) and +/- Partial Parallelism (ParPar): +DC means that the NSU type does not involve any accommodation; +ParPar means that the NSU type requires access to a constituent of the utterance that gives rise to MaxQUD.

Where do we stand now in terms of accounting for this body of data?

1. **–ParPar + DC NSUs:** We have already given an analysis of plain acknowledgements in Chapter 6. That analysis has the drawback that it does not allow for acknowledgements of *increments* of utterances. This shortcoming can be eliminated within an incremental perspective on PENDING discussed in Chapter 8. This move is also the main ingredient necessary for an analysis of *filler* NSUs (Purver 2004a). I sketched an informal analysis of ‘Yes’ in Chapter 2, which, given the apparatus developed since, is relatively straightforward to make concrete and extend to other +DC, –PP NSU types. This is discussed in section 7.5.
2. **–ParPar –DC NSUs:** One class of uses of repetition sluices, and their ilk (‘Pardon’, ‘Sorry’), to pose intended content CRs, has already been discussed in section 6.3.2—a lexical entry is posited there as (26); the contextual background for the other class of uses is discussed in section 6.8. *Why<sub>meta</sub>*, uses of ‘Why?’ in CRs, are discussed in section 8.3.

3. **+ParPar +DC NSUs:** Analysing the class of +ParPar NSUs will require us to enrich context slightly to represent the (pool of candidate) sub-utterance antecedents. The enrichment is discussed in section 7.6 and analyses of short answers and direct sluicing follow in sections 7.7 and 7.8.
4. **+ParPar –DC NSUs:** Given this and the theory of MCI developed in Chapter 6, the analysis of –DC +ParPar NSUs such as Reprise Fragments and reprise sluicing, discussed in section 7.9, will be fairly straightforward. To accommodate genre-sensitive NSUs will also require little further work, as demonstrated in section 7.10.

## 7.5 Propositional Lexemes

The propositional lexemes include words such as ‘yes’, ‘no’, and adverbials such as ‘probably’ and ‘possibly’. In Chapter 2, I sketched a semantic analysis of ‘Yes’. That analysis is repeated here as (19):

(19)  $p$ , where  $p?$  is maximal in QUD.

I can now flesh out this analysis with the benefit of the theory of questions (Chapter 3), theory of dialogue context (Chapter 4), and basic grammar (Chapter 5). What we concluded in those chapters of relevance is the following:

- Polar questions are 0-ary propositional abstracts. Applying such an abstract  $([ ]).p$  to any record  $r$ , in particular the empty record  $[ ]$ , yields as result  $p$ .
- A polar question  $p?$  becomes QUD-maximal as a consequence of either a query  $p?$  or an assertion  $p$ .

The remaining set of issues are grammatical and derive in the first instance from the question: what is the syntactic category of ‘yes’? In English ‘yes’ (and equally ‘no’) is a rather solitary beast, modifying neither sentences nor VPs and resisting embedding, as illustrated in (20a, c).<sup>12</sup> These possibilities *are* realized in other languages, as e.g. Modern Hebrew (20b,d). The sole compositional process ‘yes’ participates in is being modified by other propositional modifiers:

(20) a. \*She yes left

<sup>12</sup> Nonetheless, it can apparently occur as a complement in an equative construction, as illustrated in the following attested example:

- (i) Will Des Browne survive as defence secretary in the wake of the fuss over the 15 Iranian detainees, sailors and marines, who were allowed to sell their stories to the media wolf pack, albeit briefly, during a quiet Easter news cycle? It all depends on how he is seen to have handled himself before MPs in the Commons. But my hunch is yes. (Michael White, *The Guardian*, 12 April 2007.)

- b. hi ken azva.  
     she yes left-3rd-sg.  
     She did indeed leave.
- c. \*I believe that yes.
- d. ani xošev šeken.  
     I think-sg that=yes.  
     I think so.
- e. A: Did Bo leave?  
     B: Actually yes. / Unfortunately no. / Possibly yes.

Consequently, a simple proposal one can adopt is to treat ‘yes’ as an intransitive, IC[+] adverb, where IC is a feature from Ginzburg and Sag (2000), discussed in section 5.5.2 for signs that cannot be complements of an embedding predicate. The rationale behind this is that it would make ‘yes’ resemble its counterparts in other languages, as well as other adverbials in English (‘probably’, ‘possibly’) which have a use as a propositional lexeme. Semantically the content of ‘yes’ is the proposition whose 0-ary abstract is MaxQUD. Recall from (13) in Chapter 5 that we assimilated MaxQUD to DGB-PARAMS, the field specifying the meaning elements that need to be instantiated within the DGB. Given that we can specify a lexical entry for ‘yes’ as in (21):

- (21) 
$$\left[ \begin{array}{l} \text{phon : yes} \\ \text{cat.head = adv[+ic] : syncat} \\ \text{dgb-params.max-qud : PolQuestion} \\ \text{cont = max-qud([ ]): Prop} \end{array} \right]$$

This strategy does entail two syntactic wrinkles. It requires us to assume that adverbs can modify proposition-denoting adverbs, in order to deal with cases such as (20e). The argument structure associated with such adverbs cannot be merely specified semantically, given contrasts such as (22) between a proposition-denoting NP and ‘yes’:

- (22) a. A: The claim is that Bo left early. B: \*Usually that claim.  
     b. A: The claim is that Bo left early. B: Usually yes.

Similarly we will need to modify our earlier assumption that the head daughter of a root clause needs to be verbal. This latter assumption is motivated by data analogous to (22), examples (42) from Chapter 5. One can remain within a more orthodox position and enforce the ‘verbality’ of bare ‘yes’, but the price is complications that seem to yield very little in return.<sup>13</sup>

<sup>13</sup> One could, for instance, postulate a non-branching categorially verbal construction type which builds propositions from intransitive (propositional) adverbs and MaxQUD.

(23) exemplifies (21). In the current chapter, I will for the most part simplify question and propositional notation, using ? and  $\lambda$  notation for polar and *wh*-questions respectively and omitting situational components of propositions.

- (23) a. A: Did Jo visit? B: Yes.  
 b. As a result of A's utterance:  
     MaxQUD = ?Visit(j) : PolQuestion;  
 c. B's utterance:
- $$\left[ \begin{array}{l} \text{cat} = \text{adv}[+ic] : \text{syncat} \\ \text{dgb-params.max-qud} = ?\text{Visit}(j) : \text{PolQuestion} \\ \text{cont} = \text{max-qud}([ ]) \mapsto \text{Visit}(j) : \text{Prop} \end{array} \right]$$

### Exercise 7.5.1 Sketch a treatment of 'Yes?'

In a strictly bipolar world, 'no' would be the direct negative correlate of 'yes', identical in all respects save for a negation applied to the content. However, for better or worse, 'no' resists being a direct correlate: when its antecedent is negative, 'no' absorbs one of the negations:<sup>14</sup>

- (24) a. A: Jo didn't leave. B: No (= Jo did not leave.).  
 b. A: Did Jo not leave? B: No (= Jo did not leave.).  
 c. A: Didn't Bo phone? B: No (= Bo did not phone.).  
 d. A: Did no one help Bo? B: No (= No one helped Bo.)

We can capture this by specifying that 'no' resolves to a negative proposition, which is a simple answer to MaxQUD.<sup>15,16</sup>

- (25)  $\left[ \begin{array}{l} \text{phon} : \text{no} \\ \text{cat.head} = \text{adv}[+ic] : \text{syncat} \\ \text{dgb-params.max-qud} : \text{PolQuestion} \\ \text{cont} : \text{Prop} \\ \text{cl} : \text{NegProp}(\text{cont}) \wedge \text{SimpleAns}(\text{cont}, \text{max-qud}) \end{array} \right]$

<sup>14</sup> Ginzburg and Sag (2000) claimed that this is simply a preference; that there is potential for ambiguity. They developed an account thereof using polyadic quantification. It seems to me though that the non-affirmative reading requires a distinct tune, a rise-fall, whereas the affirmative reading is most naturally associated with a fall. Moreover, I think that the double-negation reading is possible only for cataphoric 'no' as in (i); if the follow-up sentence is omitted, a reading with a single negation ensues, regardless of intonation.

(i) A: Did no one help Bo? B: No, someone did help him get up.

<sup>15</sup> Recall from Chapter 3 that for a polar question  $p?$ , the simple answers are  $\{p, \neg p\}$ .

<sup>16</sup> I do not offer here an explicit definition of the type NegProp. For such a definition, see Cooper and Ginzburg (2011).



This sort of analysis of ‘yes’ and ‘no’ extends to the intransitive use of propositional adverbials like ‘possibly’ and ‘probably’, the difference is that these words do have a use where they occur in construction with a proposition-denoting phrase:

- (26) a. A: Did Bo leave? B: Probably/Possibly.  
 b. A: Did Bo leave? B: She probably/possibly did.

Although I won’t bother formulating it here, the implication is that these two uses need to be related by lexical rule or similar device.

**Exercise 7.5.2** Provide lexical entries for French ‘Oui’ and ‘Si’, assuming that they presuppose respectively positive and negative polar questions. Is English ‘yes’ neutral in this respect?

**Exercise 7.5.3** Characterize the class of Propositional Modifiers.

## 7.6 Focus-Establishing Constituents

### 7.6.1 Basic Motivation

In all the cases we have considered so far, the NSU can be described completely on the basis of the fragment’s own grammatical characteristics and MaxQUD (MaxPending in the case of acknowledgements.). In Chapter 6 I motivated the need to preserve in context (in PENDING and in MOVES) the sign information associated with an utterance: the question is how much of this information, if any, is used in NSU resolution, particularly in cases that do not involve MCI?

It turns out that in order to analyse +ParPar NSUs context needs to be supplemented with one additional parameter, an antecedent sub-utterance which specifies semantic, syntactic and/or phonological parallelism conditions for the fragment utterance. Intuitively, this parameter provides a partial specification of the focal (sub-)utterance, and hence I dub it the *focus-establishing constituent* (FEC).<sup>17</sup> I use this term given the connection we have drawn between QUD and information structure. As mentioned earlier, the FEC plays a role somewhat akin to the role played by the *parallel element* in higher-order unification-based approaches to ellipsis resolution. Different NSU types make use of varying combinations of this information: semantic and syntactic (direct sluicing), syntactic (short answers), phonological (CE), semantic (exclams). We illustrate these possibilities in more detail.

**7.6.1.1 Semantic and syntactic parallelism** Consider the assertion/direct sluice pair in (27a). Direct sluicing involves in essence building a question whose domain derives from the fragment whP and whose range derives from MaxQUD. More concretely here we need to abstract over the index associated with a *wh*-phrase, in (27c) the index

<sup>17</sup> In Ginzburg and Sag (2000), we used the terminology SAL-UTT, but that is somewhat anodyne and does not convey the essence of this parameter.

$z$ . As Fernández (2006) points out, this abstraction needs to be linked to a role indexed by an antecedent NP, in this case  $x$  or  $y$ ; however, if no identification of  $z$  with  $x$  or  $y$  happens, the resultant content will be as in (27d), whereas, as Fernández (2006) suggests, what we desire is (27e):

(27) a. A: A student complained about one of our teachers.

B: Who?

b. Content of MaxQuD after A's utterance:<sup>18</sup>

$$(\left[ \begin{array}{l} x : \text{IND} \\ c1 : \text{student}(x) \\ y : \text{IND} \\ c2 : \text{teacher}(y) \wedge \text{possess}(w, y) \\ c0 : \text{complain}(x, y) \end{array} \right])$$

c. quest-dom of the whP:  $\left[ \begin{array}{l} z : \text{IND} \\ c1 : \text{person}(z) \end{array} \right]$

d.  $(r : \left[ \begin{array}{l} z : \text{Ind} \\ c1 : \text{person}(z) \end{array} \right]) \left[ \begin{array}{l} x : \text{IND} \\ c1 : \text{student}(x) \\ y : \text{IND} \\ c2 : \text{teacher}(y) \wedge \text{possess}(w, y) \\ c0 : \text{complain}(x, y) \end{array} \right]$

e.  $(r : \left[ \begin{array}{l} z : \text{Ind} \\ c1 : \text{person}(z) \end{array} \right]) \left[ \begin{array}{l} x = r.z : \text{IND} \\ c1 : \text{student}(x) \\ y : \text{IND} \\ c2 : \text{teacher}(y) \wedge \text{possess}(w, y) \\ c0 : \text{complain}(x, y) \end{array} \right]$

For sluicing, in parallel with this semantic dependency comes a syntactic dependency, which I mentioned when discussing *sententialism*: Ross (1969) pointed out, with reference to German, that the fragment must concord to the case requirements of the antecedent NP. Similar facts hold in various other languages where case is overtly expressed, as documented in detail in Merchant (2002).

**7.6.1.2 Syntactic parallelism without semantic parallelism** There are a number of NSU types where a syntactic dependency exists between an antecedent and the fragment, without there being a semantic dependency above and beyond what MaxQUD encodes already. In our discussion of *sententialism*, we saw an illustration of case concord in Hebrew short answers (example (3) above).

<sup>18</sup> This assumes a DRT-like analysis of indefinites, as in (9) of Chapter 3. A somewhat more refined analysis of non-referential NPs is provided in Chapter 8, but it does not affect the argument here.

Possessive pronouns yield perhaps the strongest parallelism argument in English. (28a) and (28b) are close to synonymous, but involve distinctly different case conditions on the short answer:

- (28) a. A: Whose book did you borrow? B: Jo's. / #Jo  
 b. A: Who owns the book you borrowed? B: #Jo's. / Jo. / It's Jo's.

Locative and temporal arguments provide another set of interesting data, cross-linguistically. For short answers, as Schlangen (2003) observes, the categorial specification imposed in English is very weak, including what seem to be PPs, adverbials, and NPs:

- (29) a. A: When did Bo arrive? B: At 2./Recently./The day after Mo left.  
 b. A: Where is Bo living? B: At 29 Moring Road./Locally./?The street next to ours.

This seems like a feature of English morphosyntax, wherein spatiotemporal expressions inhabit a neighbourhood that in terms of external syntax is cross-categorial. In Hebrew both 'when' and 'where' questions require a PP short answer, hence omitting 'be' ('in/at') or 'le' ('to') results in ill-formedness:

- (30) a. A: matay higia'a                      moti? B: be 2/la'axrona (\*)be yom šliši  
           when arrived-3rd-sg Moti?    at 2/to-lately (\*)in Tuesday  
           A: 'When did Moti arrive?' B: 'At 2/recently/on Tuesday'  
 b. A: eyfo mitgorer    moti? B: (\*)be rexov haxabašim/(\*)ba'ezor.  
           where live-3rd-sg Moti?    (\*)In Ethiopia Street 2/(\*)In the area  
           A: 'Where does Moti live?' B: 'In Ethiopia Street/In the area.'

As we noted in Chapter 6, *RF<sub>clausal confirmation</sub>* is another case where categorial parallelism with the source is exhibited, as was illustrated in (9) of that chapter, repeated here as (31):

- (31) a. A: I phoned him. B: Him? / #He?  
       b. A: Did he phone you? B: He? / #Him?  
       c. A: Did he adore the book? B: Adore? / #Adored?  
       d. A: Were you cycling yesterday? B: Cycling?/Biking?/#Biked?

**7.6.1.3 Phonological parallelism** *RF<sub>intended-content</sub>* constitutes perhaps the most extreme case of parallelism, since it involves, as noted in Chapter 6, segmental phonological parallelism with the source:

- (32) A: Did Bo leave? B: Max? (cannot mean: intended content reading: 'Who are you referring to?' or 'Who do you mean?')

**7.6.1.4 Semantic without syntactic parallelism** Exclams constitute an NSU type in which there is a purely semantic dependency between an antecedent sub-utterance

and the NSU; the antecedent supplies the referent of which the property encoded in the bare *whP* predicates:

- (33) a. A: Hercules cleaned the Augean stables.  
       B1: What a hero (Hercules).  
       B2: What a terrible task (to clean the Augean stables).  
       B3: What a horrible place (the Augean stables were).

Even in languages where case is clearly marked, exclams do not exhibit categorial concord with their antecedents. This is illustrated in the Hebrew example in (34), where the noun 'haoto' bears accusative case in the antecedent but not in the exclam:

- (34) A: natati        et-haoto    le-moti.  
       A: gave-1st-sg the-car(acc) to Moti.  
       'I gave the car to Moti.'  
       B: eyzo                gruta'a!  
       B: What-a(nom) wreck!  
       'What a wreck!'

### 7.6.2 Contextual dynamics of FECs

The data we have just seen illustrates the varying roles played by the FEC: in some cases it is crucial for the semantic composition, in others it plays a disambiguating role via morphosyntactic or phonological parallelism. This is why as a general rule, its type needs to be *LocProp*: it must have access to phonological, categorial, and semantic type and token information.<sup>19</sup>

As with any dynamic feature of context we need to specify how a given FEC becomes contextually available and what its lifetime is. So the first question to ask is: can the current FEC be predicted *a priori* given knowledge of the latest *utterance*? In general the answer is: not quite—the previous utterance *partially* specifies a range of FECs in line with various potential QUDs. Prototypical cases for the main NSU types discussed here are the following:<sup>20</sup>

- The FEC associated with a *wh*-interrogative utterance is the utterance associated with the *wh*-phrase.<sup>21</sup>

- (35) A: **Who** complained about one of our teachers?  
       B: Jo.

<sup>19</sup> Strictly speaking, if we were very picky we could define a new data type that would abstract away from certain aspects of a sign to which FECs do not make access. But I will not do so here.

<sup>20</sup> Some trickier cases involving mismatches between syntax and semantics, primarily with regard to sluicing, are discussed in section 7.8.2.

<sup>21</sup> I am restricting attention to unary *wh*-interrogatives, given that I am ignoring short answers to multiple *wh*-phrases (see footnote 25 for discussion). I am also restricting attention to matrix *wh*-interrogatives. See Fernández (2006) for an account that avoids this simplifying assumption.

- The FEC associated with a polar interrogative or declarative utterance can be any of the QNP utterances:

(36) A: A student complained about one of our teachers.

B: Who?

- The FEC associated with an exclamation can be any nominal, as well as major predicative constituent:

(37) A: Hercules cleaned the Augean stables.

B1: What a hero (Hercules was).

B2: What a terrible task (to clean the Augean stables).

B3: What a horrible place (the Augean stables were).

- The FEC associated with a clarification request is the to-be-clarified sub-utterance, which can be any sub-utterance:

(38) A: Was Natalie Portman smart? B: Was?/ Natalie Portman? / Smart?

How does an FEC become contextually available? There are essentially two strategies to go about this: one is forward-looking—akin to singular pronoun specification in DRT, the other backward-looking—akin to plural pronoun specification in DRT. The forward-looking strategy, employed in implementations such as Fernández et al. (2006) and Purver (2004b), for short answers and sluicing, involves ensuring that a side effect of a query (assertion) is that the FEC of the post-query (assertion) context is the *wh*-phrase utterance (the QNP utterances). The backward-looking strategy involves finding the FEC on the basis of information specified by the NSU. Adopting such a strategy makes sense in cases such as RFs or exclams, where potentially any constituent is clarifiable (exclaimable about). In such a case, a forward-looking strategy would require generating multiple potential MaxQUDs, each with their FEC, linear in the number of constituents, involving a horrific combinatorial pile-up. There is, I think, no knock-down argument against adopting a backwards-looking strategy for all NSUs. In fact, if we were to do so, the context is already set up for this to account for the various types of NSUs. But the benefits of a forward-looking strategy are in efficient processing, particularly in long-distance resolution cases, which, as discussed in Chapter 8, are common for short answers in multilogue. In practice, in two person dialogue, adjacency is pretty overwhelming, so a backwards-looking strategy is viable.

The emergence of FECs in context varies somewhat across distinct clausal types (interrogatives, declaratives, and, in the case of exclamation or clarification via accommodation)—I will consider the different cases shortly. What of the disappearance of FECs? Here the simplest hypothesis to make is that **the FEC needs to be accessible exactly like the associated QUD**. On the whole,<sup>22</sup> this seems like an accu-

<sup>22</sup> There are some tricky cases discussed below in the section on direct sluicing at least one of which suggests the possibility that an FEC can have influence beyond the QUD with which that FEC is associated.

rate generalization. Given this, we can pair QUDs and FECs as part of contextual specification: in this way downdating of the compound entity (as a consequence of the fact update/qud downdate rule, (44), discussed in Chapter 4) ensures an identical lifetime. Concretely, this amounts to changing the type of QUD from *list(Question)* to *list(Info-struct)*, where Info-Struct is the type in (39): we use a set-valued field given that in certain cases there will be multiple candidates and in others none. However, in practice, since the typical value is a singleton, I will usually be sloppy and avoid enclosing this entity within curly brackets

$$(39) \quad \text{InfoStruc} = \begin{bmatrix} q : \text{Question} \\ \text{fec} : \text{set}(\text{LocProp}) \end{bmatrix}$$

## 7.7 Declarative Fragments: Short Answers

The majority propositional NSU is the class of short answers. Our first task is to specify a construction by means of which its basic grammatical properties can be explicated. Given our treatment of questions as propositional abstracts, the semantic aspect of things is straightforward: I assume, following much previous work going back at least to Hull (1975), Hausser (1983), and Scha (1983), that the content arises by predicating (the question component of) MaxQUD of the fragment. Assuming we wish this construction to

- be categorially sentential (viz. verbal)
- embody categorial concord between the FEC and the fragment (here the HD-DTR), as discussed earlier,

then we posit (40) as the specification, exemplified in (41).<sup>23</sup>

$$(40) \quad decl\text{-}frag\text{-}cl =_{def} \left[ \begin{array}{l} cat = v : syncat \\ dgb\text{-}params.max\text{-}qud : \left[ \begin{array}{l} q : UnaryWhQuestion \\ fec : LocProp \end{array} \right] \\ cont = max\text{-}qud.q(hd\text{-}dtr.cont.x) : Prop \end{array} \right]$$
  

|

$$hd\text{-}dtr : \left[ \begin{array}{l} cat = max\text{-}qud.fec.cat : Syncat \\ cont : [x : IND] \end{array} \right]$$

As we will see, such cases seem to involve inheritance of the FEC specification by the new context for which the FEC is underdefined.

<sup>23</sup> I suppress grammatically predictable information from NSU constructions, e.g. *dgb*-params inheritance.

- (41) a. A: Who did Jo visit? B: Bo.  
 b. As a result of A's utterance:

$$\text{MaxQUD} = \left[ \begin{array}{l} q = \lambda x \text{ Visit}(j, x), \\ \left[ \begin{array}{l} \text{sit} = u_{\text{who}} \\ \text{fec} = \left[ \begin{array}{l} \text{phon} : \text{who} \\ \text{cat.head} = N : \text{POS} \\ \text{sit-type} = \left[ \begin{array}{l} \text{cont} : [x : \text{IND}] \\ \text{quest-dom} = \left\langle \left[ \begin{array}{l} y = \text{cont}.x : \text{IND} \\ \text{restr} : \text{person}(y) \end{array} \right] \right\rangle : \text{list}(\text{RecType}) \end{array} \right] \end{array} \right] \end{array} \right] \end{array} \right]$$

- c. Short-answer analysis using *decl-frag-cl*:<sup>24</sup>

$$\left[ \begin{array}{l} \text{dgb-params.max-qud} = \left[ \begin{array}{l} q = \lambda x \text{ Visit}(j, x) \\ \text{fec} = \text{who} : \text{LocProp} \end{array} \right] : \text{InfoStruc} \\ \text{cont} = \lambda x \text{ Visit}(j, x)(\text{hd-dtr.cont}.x) \\ \mapsto \text{Visit}(j, \text{hd-dtr.cont}.x) : \text{Prop} \end{array} \right]$$

$$\downarrow$$

$$\text{hd-dtr} : \left[ \begin{array}{l} \text{phon} : \text{bo} \\ \text{cat.head} = N : \text{POS} \\ \text{dgb-params} : \left[ \begin{array}{l} x : \text{IND} \\ \text{restr}^{\text{facts}} : \text{Named}(x, \text{bo}) \end{array} \right] \\ \text{cont} : [x = \text{c-params}.x : \text{IND}] \end{array} \right]$$

The remaining task is to specify how the FEC utilized in the above construction actually becomes available. As I mentioned in discussing FECs in the previous section, one strategy for accounting for their emergence is forward-looking. Such a strategy is particularly feasible for this case, as there is typically a single candidate for the FEC.<sup>25</sup> A simple way to implement this strategy is via a minor modification to the

<sup>24</sup> I use phon values as shorthand notation for FECs, where convenient.

<sup>25</sup> Cases with a multiple *wh*-question have been reported in the literature, e.g. (i) from Ginzburg (1999b):

- (i) A: Who was interacting with whom at the party?  
 B: Bill with Mary./ Some of my friends with each of her friends./ One thing's for sure: None of my friends with none of my sister's friends.  
 (ii) A: Who arrived when?  
 B: Bill at 5, Mustafa a couple of hours later./

The proposal Ginzburg (1999b) makes for such cases could, in principle, be adopted here.

QUD update caused by asking (see (30), Chapter 4).<sup>26</sup> The modification, depicted in (42) in boldface, involves selecting as the FEC the *wh*-utterance. Concretely, that constituent of LatestMove which is a *wh*-utterance, gets paired with the asked question *q* as MaxQUD:<sup>27</sup>

(42) Ask *wh*-question QUD-incrementation =<sub>def</sub>

$$\left[ \begin{array}{l} \text{pre} \\ \text{effects} \end{array} : \left[ \begin{array}{l} \text{spkr: IND} \\ \text{addr: IND} \\ \text{q : WhQuestion} \\ \text{LatestMove.cont} = \text{Ask}(\text{spkr}, \text{addr}, \text{q}) : \text{IllocProp} \\ \text{new-fec} = \left[ \begin{array}{l} \text{sit} = \text{wh1} \\ \text{sit-type} = T_{\text{wh1}} \end{array} \right] : \text{LocProp} \\ \text{cl1 : member}(\text{wh1}, \text{LatestMove.sit.constits}) \\ \wedge \text{WhIntUtt}(\text{new-fec}) \end{array} \right] \right]$$

$$\text{effects} : \left[ \text{qud} = \langle \text{q}, \text{pre.new-fec} \rangle \oplus \text{pre.qud} : \text{list}(\text{info-struct}) \right]$$

As an illustration of this slightly revised contextual development, consider (43):

- (43) A: Hi.  
 B: Hi.  
 A(1): Who's coming tomorrow?  
 B(2): Jo.  
 A(3): I see.  
 B(4): She's back from Mauritania.  
 A(5): Ah.  
 B(6): Mike.  
 A(7): Moroney?  
 B(8): Yeah.  
 (9) A bunch of others too.

A trace through this dialogue focussing primarily on QUD's evolution is the following:

<sup>26</sup> See Fernández (2006) for a more general solution, which allows for QUDs and FECs to emerge also from embedded constituents.

<sup>27</sup> Here *WhIntUtt(u)* amounts to the fact that *u* has a non-null value for QUEST-DOM and is categorially nominal.



(44)

| Utt.    | Move Update                           | QUD                                    |
|---------|---------------------------------------|----------------------------------------|
| initial | MOVES = ⟨⟩<br>QUD = ⟨⟩<br>FACTS = cgl |                                        |
| 1       | Ask(A, B, q0)                         | ⟨q0, Who⟩                              |
| 2       | Assert(B, A, p1)                      | ⟨p1?, fec: null⟩, ⟨q: q0, fec: Who⟩    |
| 3       | LatestMove := Accept(A, B, p1)        | ⟨q: q0, fec: Who⟩                      |
| 4       | LatestMove := Assert(B, A, p2)        | ⟨q: p2?, fec: null⟩, ⟨q: q0, fec: Who⟩ |
| 5       | LatestMove := Accept(A, B, p2)        | ⟨q: q0, fec: Who⟩                      |
| 6       | LatestMove := Assert(B, A, p3)        | ⟨q: p3?, fec: null⟩, ⟨q: q0, fec: Who⟩ |
| 7       | LatestMove := Ask(A, B, m?)           | ⟨q: m?, fec: null⟩, ⟨q: q0, fec: Who⟩  |
| 8       | LatestMove := Confirm(A, B, m)        | ⟨q: q0, fec: Who⟩                      |

**Exercise 7.7.1** Provide a detailed trace through the dialogue in (43).

7.8 Sluicing

7.8.1 *The distribution of sluices*

Sluices are a very common means of asking *wh*-questions. In fact, as Tables 7.6 and 7.7 indicate, they make up about 33 per cent of all *wh*-interrogative queries in the BNC.<sup>28</sup>

**TABLE 7.6** Total of sluices in the BNC

|             |                |            |              |
|-------------|----------------|------------|--------------|
| <i>what</i> | <i>why</i>     | <i>who</i> | <i>where</i> |
| 3045        | 1125           | 491        | 350          |
| <i>when</i> | <i>which N</i> | <i>how</i> | <i>which</i> |
| 107         | 160            | 50         | 15           |
| Total: 5343 |                |            |              |

Although there is a large generative literature on sluicing, it has almost without exception ignored the fact that bare *wh*-phrases are systematically ambiguous between at least three uses:<sup>29</sup>

<sup>28</sup> Some care is due when considering these numbers, given that they derive from a search engine, Matt Purver’s SCoRE, and the results depend on a particular search specification. The sluices arose from a search for the given bare *wh*-phrase, whereas the sentential *wh*-interrogatives were found by searching for strings with the given *wh*-phrase followed by a verb and then other optional material.

<sup>29</sup> There is a fourth use, dubbed by Fernández, Ginzburg, and Lappin (2004) the *wh*-anaphoric use, in which the antecedent of the sluice is a *wh*-phrase:

- (i) Larna: We’re gonna find poison apple and I know where that one is.  
Charlotte: Where? (BNC, KD1)

**TABLE 7.7 Non-elliptical *wh*-interrogative clauses in the BNC**

|             |              |                |              |
|-------------|--------------|----------------|--------------|
| <i>what</i> | <i>where</i> | <i>who</i>     | <i>how</i>   |
| 7965        | 1877         | 1696           | 1640         |
| <i>why</i>  | <i>when</i>  | <i>which N</i> | <b>Total</b> |
| 1328        | 420          | 225            | <b>15151</b> |

### Direct

The utterer of the sluice understands the antecedent of the sluice without difficulty. The sluice queries for additional information that was explicitly or implicitly quantified away in the previous utterance.

- (45) a. A: Can I have some toast please?  
       B: Which sort? (BNC, KCH)  
       b. Caroline: I'm leaving this school.  
       Lyne: When? (BNC, KP3)

### Reprise

The utterer of the sluice cannot understand some aspect of the previous utterance which the previous (or possibly not directly previous) speaker assumed as presupposed:

- (46) a. Geoffrey: What a useless fairy he was.  
       Susan: Who? (BNC, KCT)  
       b. Pat: You might find something in there actually.  
       Carole: Where? (BNC, KBH)

This use is rather rare in the BNC, but is quite prevalent in parent-child interaction (see Ginzburg and Kolliakou 2010). For a treatment, see section 7.9.2.

### Repetition

The sluice is used to ask for repetition of the previous utterance as a whole.

- (47) June: Only wanted a couple weeks.  
       Ada: What?  
       June: Only wanted a couple weeks. (BNC, KBI)

Beyond these basic functional distinctions, there are a number of linguistic diagnostics from Ginzburg and Sag (2000). Example (48a) shows that a non-quantified proposition constitutes a possible antecedent only for a reprise sluice, despite the felicity of a non-elliptical form that seems close in meaning to what the direct sluice might be expected to mean. Conversely, (48d) illustrates a context where, assuming A has no problems hearing what B says, only the direct sluice is appropriate:

- (48) a. A: Jo phoned.  
      B: WHO?/#who?/Oh, who is Jo?  
      b. A: I ate the ragaqat I found in the fridge.  
          B: the WHAT?/#what?  
      c. A: I met several of your graduate students at the party.  
          B: who?/WHO?  
      d. A: Did anyone phone for me?  
          B: Yes.  
          A: Aha. Who?/#WHO?

Fernández, Ginzburg, and Lappin (2007) showed that the direct/reprise/repetition ambiguity could be reliably distinguished by human annotators and developed an automatic classifier for the ambiguity, using machine-learning techniques. One basic fact that emerges from that study is that different *wh*-words have very different direct/reprise distributions. Table 7.8, from the paper cited, provides an estimate for distributions of the different readings, based on combining averaged percentages of the readings obtained for a sample of fifty tokens of each word in Table 7.6.

Explaining a distribution like this is a task any sufficiently rich theory of interaction should be able to do. Here I summarize the basic proposal Fernández, Ginzburg, and Lappin (2004) make regarding two striking patterns exhibited in Table 7.8.

One such pattern is the low frequency of *when* sluices, particularly by comparison with what one might expect to be its close cousin, *where*; indeed the direct/reprise splits are almost mirror images for *when* vs *where*. Another very notable pattern, alluded to above, is the high frequency of *why* sluices.

The *when* vs *where* contrast provides one argument against (49), which is probably the null hypothesis for the distribution of reprise sluices:

(49) **Frequency of antecedent hypothesis:**

The frequency of a class of reprise sluices is directly correlated with the frequency of the class of its possible antecedents.

**TABLE 7.8 Sluice Class Frequency—  
Estimated Tokens**

|                            |      |                             |     |
|----------------------------|------|-----------------------------|-----|
| <i>what<sub>rep</sub></i>  | 2040 | <i>whichN<sub>rep</sub></i> | 135 |
| <i>why<sub>dir</sub></i>   | 775  | <i>when<sub>dir</sub></i>   | 90  |
| <i>what<sub>rep</sub></i>  | 670  | <i>who<sub>dir</sub></i>    | 70  |
| <i>who<sub>rep</sub></i>   | 410  | <i>where<sub>dir</sub></i>  | 70  |
| <i>why<sub>rep</sub></i>   | 345  | <i>how<sub>dir</sub></i>    | 45  |
| <i>where<sub>rep</sub></i> | 250  | <i>when<sub>rep</sub></i>   | 35  |
| <i>what<sub>dir</sub></i>  | 240  | <i>whichN<sub>dir</sub></i> | 24  |

Clearly locative expressions do not outnumber temporal ones and certainly not by the ratio that the data in Table 7.8 would require in order to sustain (49).<sup>30</sup> Also relevant here is the data of Purver (2004b), discussed in Chapter 6, which shows that clarification requests of all types in the BNC that pertain to nominal antecedents outnumber CRs that relate to verbal antecedents by 40:1. This does not correlate with the relative frequency of nominal vs verbal antecedents (about 1.3:1).

A more refined hypothesis, then, which I will state here only informally, is (50):

(50) **Ease of grounding of antecedent hypothesis:**

The frequency of a class of reprise sluices is directly correlated with the ease with which the class of its possible antecedents can be grounded.

This latter hypothesis offers a route towards explaining the *when* vs *where* contrast. There are at least two factors that make grounding a temporal parameter, in general, significantly easier than grounding a locative parameter. The first is that conversation-*alists* typically share a temporal ontology based on a clock and/or calendar. Although well-structured locative ontologies do exist (e.g. grid points on a map), they are far less likely to be common currency. The natural ordering of clock/calendar-based ontologies reflected in grammatical devices such as *sequence of tenses* is a second factor that contributes to the more pervasive and systematic grounding of temporal parameters than locatives.

Seen in this light, the high frequency of *why* reprises is not surprising. Such reprises query either the justification for an antecedent assertion or the goal of an antecedent query. Neither of these is usually specified explicitly by the speaker. An obvious question is why these reprises do not occur even more frequently than they actually do—I discuss this issue in section 8.3.6.

I turn to the rather different task of offering a grammatico-contextual description of sluicing.

## 7.8.2 *Direct Sluicing*

I start by offering a characterization of direct argument-filling sluices, which require a somewhat more complex treatment than short answers, which they resemble in certain respects. Reprise sluices are discussed in section 7.9, whereas ‘Why’ sluices, which differ from other sluices in their direct and reprise uses, are treated in Chapter 8.<sup>31</sup>

<sup>30</sup> A rough estimate concerning the BNC can be extracted by counting the words that occur more than a thousand times. Of these, about 35,000 tokens are locative in nature and could serve as antecedents of *where*; the corresponding count for temporal expressions and *when* yields about 80,000 tokens. These numbers are derived from a frequency list (Kilgariff 1998) of the Demographic portion of the BNC.

<sup>31</sup> The account described below improves on that provided in Ginzburg and Sag (2000) in a number of respects:

An account of direct sluicing needs to characterize the content of such uses and their contextual background. Let us start with the latter: what is the context for a direct sluice? In broad terms, it is a non-negatively quantified proposition. Presupposed? Probably not: it seems quite clear that a direct sluice can be felicitous in a post-assertoric context where B is dubious about the assertion's truth:

- (51) a. A: Someone finished the milk.  
       B: No way.  
       A: Oh yes.  
       B: Who?  
       b. A: Someone will solve the problem. B: Really? Who?

So the antecedent need not be in FACTS. Does the antecedent need to be explicitly asserted<sup>32</sup> or as with 'yes' is there commonality with queried polar questions? It is clear that sluices *can* follow a polar query, as examples such as (52) show:

- (52) a. A: Could the Pope appoint an Israeli as his deputy? B: Gosh, who?  
       b. A: Has Bo given you one of my books? Go on, tell me which one.  
       c. A: Did anyone call? B: Such as who?/Who, for instance?  
       d. A: Did you see anything suspicious? B: What? A: You tell me.

The question is whether such cases are 'canonical' direct sluices or involve some sort of accommodation of a positive answer, or perhaps are reprise sluices—requesting clarification of a specific indefinite.<sup>33</sup> I will assume that such cases are direct sluices. This will simplify the treatment of one thorny class of cases, while somewhat complicating the dynamics of FECs.

The second issue we need to address is the 'fine structure' of the propositional component of the context for a direct sluice. As we saw earlier, the propositional component needs to be quantified. The prototypical antecedent is existentially quantified.<sup>34</sup> Negative universals ('no miners', 'no one', etc.) are not viable antecedents:

- It allows for multiple antecedents.
- It is explicit about the calculation of FECs.
- It is semantically more streamlined and transparent: the existence of a semantic dependency enabled via the FEC was not evident on that account. Whereas on the account in Ginzburg and Sag (2000) short answers and direct sluicing are very similar, in the current proposal sluicing is analysed as more complex semantically, in a sense made explicit in section 8.4, where this complexity is correlated with the relative order of acquisition of short answers and sluicing.

<sup>32</sup> Or at least embedded under a positively veridical predicate.

<sup>33</sup> See Chapter 8 for discussion of the fluidity of the referential/quantificational distinction.

<sup>34</sup> Can other quantifiers work? There is a clear preference for existentials because they are typically non-referential, in contrast to various other QNP uses. But, as shown in Ginzburg and Sag (2000) and Merchant (2002) (in response to accounts such as Chung, Ladusaw, and McCloskey 1995 and Reinhart 1997, whose accounts presupposed existential antecedents), sluices can certainly occur with other quantificational antecedents.

- (53) A: No miners will support the proposal. B: #Who? (Cf. Who will?)

An explanation for this will emerge in terms of our constraint QIAC against introducing a resolved question into QUD. The content I will be assigning (53) can be paraphrased as the infelicitous (54):

- (54) B: #Which miners will support the proposal?

What about the content of a direct sluice? Here it is not easy to give a definite answer, given the difficulty of providing a truly synonymous sentential paraphrase. Let me, nonetheless, mention two options, which if they are to work are ultimately not very distinct. One option is to think of the sluice as a question whose domain emanates from the bare *wh*-phrase, while its range emanates from the VP and the FEC. Call this the *conservative view*, given the analogy with the notion of *conservativity* in generalized quantifier theory. This leads to a paraphrase as in (55a) or (55b)—the former is a more euphonious paraphrase, but the latter is more pedantically correct since it avoids the uniqueness presuppositions associated with ‘which’ and actually conjoins the VP with restrictor associated with the FEC:

- (55) a. A: Some N V’ed. B: Who? (= Which N V’ed?)  
 b. A: Some N V’ed. B: Who? (= Who are Ns that V’ed?)

The second option involves simply querying a witness for a contextually given property. This is more general of course, but requires us to delimit context severely in order to ensure that the resolution possibilities coincide with those exhibited by sluicing. An obvious tactic would be to assume that ‘givenness’ coincides with anaphoric possibilities. But this is problematic, as illustrated in (56c, d)—the difficulty is that the pronoun-based paraphrase is felicitous with various antecedents which do not support direct sluicing:

- (56) a. A: A friend of Jo’s is coming for tea. B: Who?  
 b. Who is s/he?  
 c. A: Jo phoned. I don’t know who he is. # I don’t know who.  
 d. A: The doctor phoned. I don’t know who he is. # I don’t know who.

Note that the conservative view-oriented paraphrase is not susceptible to this problem:

- (57) a. A: Jo phoned. # I don’t know which Jo phoned. # I don’t know who.  
 b. A: The doctor phoned. # I don’t know which doctor phoned. # I don’t know who.

Given this, I will assume the conservative view, despite its more restricted generality. I turn now to formalize this characterization, in terms of a phrasal construction and an update rule that underpins the context. For the former we posit the specification in (58). The domain of the question arises from the head daughter, the bare *wh*-phrase,<sup>35</sup> whereas the range of the question derives from MaxQUD by (a) propositionalizing it (*p?* applied to an empty record) and (b) substituting the fragment's content for that of the FEC, which is none other than the semantic dependency I discussed earlier in (27). Note also that MaxQUD will already include in it the restriction associated with the FEC:

$$(58) \quad slu\text{-}frag\text{-}cl =_{def} \left[ \begin{array}{l} cat = v : syncat \\ dgb\text{-}params.max\text{-}qud : \left[ \begin{array}{l} p : QuantProp \\ q = ([ ] )p : PolarQuestion \\ fec : set(LocProp) \end{array} \right] \\ fec : LocProp \\ cl : member(fec, max\text{-}qud.fec) \\ cont = (r : G)max\text{-}qud.q([ ]) (fec.cont.x \rightsquigarrow r.head\text{-}dtr.cont.x) : Question \end{array} \right]$$
  
$$\qquad \qquad \qquad |$$
  
$$hd\text{-}dtr : \left[ \begin{array}{l} cat = max\text{-}qud.fec.cat : Syncat \\ cont : [x : IND] \\ F : RecType \\ G = [x : IND] \wedge_{merge} F : RecType \\ quest\text{-}dom = \langle G \rangle list(RType) \end{array} \right]$$

How do the antecedents for sluicing emerge? If we assumed sluicing arose only from assertions, we could directly follow the forward-looking strategy specified above for *wh*-interrogatives in the modified Ask QUD-incrementation rule *mutatis mutandis*. What would mutate in the modified Assert QUD-incrementation rule is the need to consider cases where the clause contains more than one potential FEC, in which case there is a potential for ambiguity, as in (59):

<sup>35</sup> The requirement ‘ $G = [x : \text{Ind}] \wedge_{\text{merge}} F$ ’ ensures that the quest-dom of the fragment is non-trivial, hence that the head daughter is indeed a *wh*-phrase. How to elucidate the type *QuantProp* will become clear in Chapter 8, when I sketch a treatment of quantification. An explicit definition is provided in Appendix A.

- (59) a. **A suspect** was arrested by **one of our policemen**.  
       B: Who (is the suspect/the policeman)?  
       b. **A student** complained about **one of our teachers**.  
       B: Who (is the student/the teacher)?

This means that we need to gather all the quantified sub-utterances of a given declarative (polar interrogative) clause as members of a set of potential FECs. Given the need to accommodate polar queries as well we need to generalize things slightly. This results in the specification in (60)—given a context in which LatestMove is either an assertion of  $p$  or a query of  $p?$ , MaxQUD will now consist of  $p?$  paired with all the quantified sub-utterances of a given declarative (polar interrogative) clause as members of a set of potential FECs:<sup>36</sup>

$$(60) \quad p? \text{ QUD-incrementation} =_{def} \left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{spkr: IND} \\ \text{addr: IND} \\ p : \text{Prop} \\ \text{LatestMove.cont} = \text{Assert}(\text{spkr}, \text{addr}, p) \vee \\ \text{Ask}(\text{spkr}, \text{addr}, p?) : \text{IllocProp} \\ \text{new-fec} : \text{set}(\text{LocProp}) \\ \text{cl} : \forall x [x = \left[ \begin{array}{l} \text{sit} = \text{quant1} \\ \text{sit-type} = T_{\text{quant1}} \end{array} \right] \wedge \text{member}(x, \text{new-fec})] \\ \rightarrow \text{Quantutt}(x) \wedge \text{member}(\text{quant1}, \text{LatestMove.sit.constits}) \end{array} \right] \\ \text{effects} : [ \text{qud} = \langle p?, \text{pre.new-fec} \rangle \oplus \text{pre.qud} : \text{list}(\text{info-struct}) ] \end{array} \right]$$

As an example consider (61a): (61b) exemplifies MaxQUD after A's utterance, the steps in calculating the contents of B's utterance are in (61c–e):

- (61) a. A: A student complained about one of our teachers.  
       B: Who?

<sup>36</sup> As explained in Chapter 8, a quantified utterance, picked out here via the property *Quantutt*, is a (sub)-utterance whose Q-PARAMS is a non-trivial type.



b. MaxQUOD after A's utterance

$$\left[ \begin{array}{l} q = ([ ]) \left[ \begin{array}{l} x : \text{Ind} \\ c1 : \text{student}(x) \\ y : \text{IND} \\ c2 : \text{teacher}(y) \wedge \text{possess}(w, y) \\ c0 : \text{complain}(x, y) \end{array} \right] : \text{Question}, \\ \\ \text{fecset} = \left\{ \left[ \begin{array}{l} \text{cat} = \text{NP}[+\text{nom}] : \text{syncat} \\ \text{cont} : [x : \text{IND}] \\ \text{q-params} : \left[ \begin{array}{l} y = \text{cont}.x : \text{IND} \\ r : \text{student}(\text{cont}.x) \end{array} \right] \end{array} \right], \right. \\ \left. \left[ \begin{array}{l} \text{cat} = \text{NP}[+\text{acc}] : \text{syncat} \\ \text{cont} : [x : \text{Ind}] \\ \text{q-params} : \left[ \begin{array}{l} y = \text{cont}.x : \text{IND} \\ r : \text{teacher}(\text{cont}.x) \end{array} \right] \end{array} \right] \right\} : \text{set}(\text{LocProp}) \end{array} \right]$$

c. hd-dtr of B's utterance :

$$\left[ \begin{array}{l} \text{cat} = \text{NP}[+\text{nom}] : \text{syncat} \\ \text{cont} : [x : \text{IND}] \\ \text{quest-dom} = \left\langle \left[ \begin{array}{l} z = \text{cont}.x : \text{IND} \\ r : \text{person}(z) \end{array} \right] \right\rangle : \text{list}(\text{RType}) \end{array} \right]$$

d. Assume 'a student' selected as FEC.

$$\begin{aligned} \text{max-qud}.q([ ])(\text{fec}.cont.x \rightsquigarrow r.\text{hd-dtr}.cont.x) &= \\ \left[ \begin{array}{l} x : \text{IND} \\ c1 : \text{student}(x) \\ y : \text{IND} \\ c2 : \text{teacher}(y) \wedge \text{possess}(w, y) \\ c0 : \text{complain}(x, y) \end{array} \right] & \quad (x \rightsquigarrow r.\text{head-dtr}.cont.x) \\ &= \\ \left[ \begin{array}{l} r.\text{head-dtr}.cont.x : \text{IND} \\ c1 : \text{student}(r.\text{head-dtr}.cont.x) \\ y : \text{IND} \\ c2 : \text{teacher}(y) \wedge \text{possess}(w, y) \\ c0 : \text{complain}(r.\text{head-dtr}.cont.x, y) \end{array} \right] \end{aligned}$$

e. Content of B's utterance =

$$(r : \left[ \begin{array}{ll} \text{head-dtr.cont.x} & : \text{IND} \\ r & : \text{person}(\text{head-dtr.cont.x}) \end{array} \right] \Bigg) \left[ \begin{array}{ll} r.\text{head-dtr.cont.x} & : \text{IND} \\ c1 & : \text{student}(r.\text{head-dtr.cont.x}) \\ y & : \text{IND} \\ c2 & : \text{teacher}(y) \wedge \text{possess}(w, y) \\ c0 & : \text{complain}(r.\text{head-dtr.cont.x}, y) \end{array} \right]$$

There are a variety of issues that a full treatment of direct sluicing brings with it, and many of these revolve around the dynamics of the FEC. Let me mention a few such issues and the basic strategy we can take towards resolving them.

Consider for a start the examples in (62a), where a direct sluice occurs following an utterance of 'yes' answering a polar question. Crucially, in languages with explicit case marking (e.g. the Greek exemplified in (62b)), there is case agreement between the existential in the initial utterance (62b(1)) and the sluice:<sup>37</sup>

- (62) a. A: Did you call anyone while I was out?  
       B: Yes.  
       A: Who?  
       b. A: pires kapon [acc] otan imouna ekso?  
       B: nai.  
       A: pion? [acc]/ #pios [nom]

The issue is how to ensure connectivity between the FEC emanating from the initial utterance and the sluice. One way would be to assume that 'Yes.' is a fully spelled-out sentence at some syntactic level, but given the usual divergences between NSU and reconstruction correlate I will not adopt this view.<sup>38</sup> There is a simple alternative, which involves us assuming that 'yes' is the identity not merely on content, but also on QUD. This requires us to formulate the lexical entry of 'yes' dynamically, see Fernández (2006) or Exercise 7.8.1.

<sup>37</sup> Thanks to Dimitra Kolliakou (p.c.) for help with the Greek data in this section.

<sup>38</sup> One problem for a sententialist approach to 'Yes.' is that it presupposes some sort of transformational mechanism that generates well-formed declarative sentences from a variety of inputs, three of which are illustrated in (i, ii, iii) below. In the case of (ii) a consequence of the transformation is the emergence of positive polarity form corresponding to 'anyone'; in the case of (iii) a first-person subject needs to be inserted.

- (i) A: Did Jo leave B: Yes  $\mapsto$  Jo left  
 (ii) A: Did anyone leave? B: Yes  $\mapsto$  \*Anyone left  $\mapsto$  Someone left  
 (iii) A: Hungry? B: Yes  $\mapsto$  \*Hungry  $\mapsto$  I am/I'm hungry

**Exercise 7.8.1** Formulate ‘yes’ dynamically, in a similar style to the conversational rules provided throughout.

A related case is in (63)—where, as the Greek analogue (63b) shows, an implicit argument imposes case connectivity.

- (63) a. A: Good, Belinda won tonight. B: What competition?  
 b. A: Poli kala. I bella nikise apopse. B: Pion diagonismo<sub>acc</sub>?

The FEC can, nonetheless, be introduced into the context via a minor modification of (60), which collects non-referential arguments from whichever part of argument structure we choose to represent such implicit arguments. Note that one must not conflate the representation in which the argument is explicit from the one in which it is implicit, as *inter alia* they have distinct clarificational potentials:

- (64) a. A: Good, Belinda ate something tonight. B: Something?  
 b. A: Good, Belinda ate tonight. B: # Something?

As we noted earlier, after ‘what,’ ‘when’ sluices are the most frequent ones exhibiting a direct reading. And yet these occur typically without an overt temporal expression above and beyond tense-marking on the verb, as illustrated in (65). Hence, the account of sluices to implicit arguments should extend here, though there is more of a question as to the category of the FEC in this case (in a language like English), as we discussed earlier:

- (65) a. Anon 6: Who did you interview then? Be ... Simone: Benjamin. Anon 6: When? Simone: Last night. (KE0, 138)  
 b. Caroline: I’m leaving this school. Lyne: When? Caroline: The end of summer. (KP3, 537)  
 c. Dbillon: Well, apparently they’ve erm Dave was saying they’ve got to er vacate the premises Danny: When? Dbillon: I’m not sure when (KPA, 1379)  
 d. Unknown: Trina knows that he’s got a big schlong. Kath: Alistair [last or full name], the deaf bloke, yeah. Unknown: (laughing) I’ve seen it. Kath: Have you? When? Unknown: (laugh) I was lifeguarding at the swimming pool, I turned around and the silly shit was getting out of (laugh) his school uniform, strip naked, [last or full name] was standing at the other end of the pool didn’t say anything, Al Alistair [last or full name], deaf guy (KPH, 690)

**Exercise 7.8.2** Develop an analysis of sluices of the form ‘When?’ and offer a detailed trace through the dialogue in (66):

- (66) A: Sorry I’m late. A friend of mine arrived yesterday.  
 B: Who?

A: Isolde Tristram.

B: When?

## 7.9 Reprise Fragments and Reprise Sluices

Let us turn now to deal with the common MCI-oriented NSUs such as RF and reprise sluices, based on the theory of metacommunicative interaction, developed in Chapter 6. That theory will require one minor modification, in boldface in (67)—identifying the to-be-clarified sub-utterance as the FEC:

(67) a. Parameter Identification=*def*

$$\left[ \begin{array}{l} \text{pre} \\ \text{effects} \end{array} : \left[ \begin{array}{l} \text{MaxPending} = \left[ \begin{array}{l} \text{sit} = v_0 \\ \text{sit-type} = T_{v_0} \end{array} \right] : \text{LocProp} \\ \text{p1} = \left[ \begin{array}{l} \text{sit} = u_1 \\ \text{sit-type} = T_{u_1} \end{array} \right] : \text{LocProp} \\ \text{cl: member}(u_1, v_0.\text{constits}) \\ \text{spkr} = \text{pre.addr} : \text{IND} \\ \text{addr} = \text{pre.spkr} : \text{IND} \\ \text{MaxQUD} = \left[ \begin{array}{l} q = \lambda x \text{ Mean}(\text{pre.spkr}, \text{pre.u1}, x) \\ \text{fec} = \text{pre.p1} \end{array} \right] : \text{InfoStruc} \end{array} \right] \right]$$

b. Parameter Focussing=*def*

$$\left[ \begin{array}{l} \text{pre} \\ \text{effects} \end{array} : \left[ \begin{array}{l} \text{MaxPending} = \left[ \begin{array}{l} \text{sit} = v_0 \\ \text{sit-type} = T_{v_0} \end{array} \right] : \text{LocProp} \\ \text{p1} = \left[ \begin{array}{l} \text{sit} = u_1 \\ \text{sit-type} = T_{u_1} \end{array} \right] : \text{LocProp} \\ \text{cl: member}(u_1, v_0.\text{constits}) \\ \text{spkr} = \text{pre.addr} : \text{IND} \\ \text{addr} = \text{pre.spkr} : \text{IND} \\ \text{MaxQUD} = \left[ \begin{array}{l} q = \lambda x v_0.\text{cont}(u_1.\text{cont} \rightarrow x) \\ \text{fec} = \text{p1} \end{array} \right] : \text{InfoStruc} \end{array} \right] \right]$$

Clausal confirmation readings can be explicated on the basis of the existing grammatical apparatus. Intended content readings are a tad trickier.

7.9.1 *Clausal Confirmation Readings*

Recall that Parameter Focussing is a coercion operation that given an FEC, intuitively an unclear sub-utterance of MaxPending whose contextual parameter is  $i$ , specifies the context as having the question  $\lambda i$  [MaxPending.cont] as MaxQUD.

This yields a straightforward analysis of clausal RFs/reprise sluices such as (68):

- (68) a. A: Is George here? B: George?  
 b. Are you asking if George (of all people) is here?  
 c. B is unsure about the referent he has for ‘George’; uses Parameter Focussing to update MAX-QUD accordingly (see (85), Chapter 6 for a more detailed DGB):

$$\left[ \begin{array}{l} \text{spkr} = B \\ \text{addr} = A \\ \text{pending} = \left\langle \left[ \begin{array}{l} \text{sit} = w0' \\ \text{sit-type} = \text{IGH} \end{array} \right] \right\rangle \\ \text{maxqud} = \left[ \begin{array}{l} q = \lambda x \text{ Ask}(A, \text{In}(\text{lo}, x)) : \text{Question} \\ \text{fec} = p2 : \text{LocProp} \end{array} \right] : \text{InfoStruc} \end{array} \right]$$

The phrasal type *decl-frag-cl* builds a propositional content by predicating MaxQuD of the fragment; given this, *polarization* (recall (29), Chapter 5) builds the required question:

$$\begin{array}{c} (69) \qquad \qquad \qquad S \\ \left[ \begin{array}{l} \text{polarization} \\ \text{cont} = ?\text{hd-dtr.cont} = ?\text{Ask}(A, B, ?\text{In}(\text{lo}, g)) : \text{Question} \end{array} \right] \\ | \\ S \\ \left[ \begin{array}{l} \text{decl-frag-cl} \\ \text{hd-dtr} : \left[ \begin{array}{l} \text{cont} = [x = g] : [x : \text{IND}] \\ \text{cat} = \text{fec.cat} : \text{syncat} \end{array} \right] \\ \text{dgb-params.maxqud} = \left[ \begin{array}{l} q = \lambda x \text{ Ask}(A, B, \text{In}(\text{lo}, x)) : \text{Question} \\ \text{fec} = p2 : \text{LocProp} \end{array} \right] : \text{InfoStruc} \\ \text{cont} = \text{maxqud.q}(\text{hd-dtr.cont.x}) \\ \mapsto \lambda x \text{ Ask}(A, B, \text{In}(\text{lo}, x)) ([x = g]) \\ \mapsto \text{Ask}(A, \text{In}(\text{lo}, g)) : \text{Prop} \end{array} \right] \\ \\ \text{NP} \\ \text{george} \end{array}$$

In entirely similar fashion we obtain the corresponding reprise sluice:

- (70) a. A: Did Bo leave? B: WHO?  
b. Who<sub>x</sub> you asking if s/he<sub>x</sub> left?

- (71) S
- $$\left[ \begin{array}{l} \text{unary-dir-is-cl} \\ \text{cont} = (G).\text{hd-dtr}.\text{cont} = \lambda x.\text{Ask}(A, ?\text{leave}(x)) : \text{Question} \end{array} \right]$$
- |
- S
- $$\left[ \begin{array}{l} \text{decl-frag-cl} \\ \text{maxqud} = \left[ \begin{array}{l} q = \lambda x \text{ Ask}(A, \text{In}(\text{lo}, x)) : \text{Question} \\ \text{fec} = p2 : \text{LocProp} \end{array} \right] : \text{InfoStruc} \\ \text{hd-dtr} : \left[ \begin{array}{l} \text{cont} : [x : \text{IND}] \\ \text{quest-dom} = \langle G \rangle : \text{list}(\text{RecType}) \\ \text{cat} = \text{fec}.\text{cat} : \text{syncat} \end{array} \right] \\ \text{cont} = \text{maxqud}.q(\text{hd-dtr}.\text{cont}.x) \end{array} \right]$$
- NP
- who

### 7.9.2 Intended Content Readings for Reprise Fragments and Sluicing

‘Polar’ intended content readings along with the corresponding reprise sluices are exemplified in (72):

- (72) a. A: Did Bo call? B: My cousin? (= Are you referring to my cousin?)  
b. A: Do you speak Chinese? B: Mandarin? (= Do you mean to say Mandarin?)

These can be handled in precisely the same way that clausal confirmation readings are handled in section 7.9.1 above. The sole difference pertains to the fact that Max-QUd is now the MaxQUd that emerges from the CCUR parameter identification.

**Exercise 7.9.1** Provide a detailed account of polar constituent readings, as exemplified in (72).

However, intended content RFs of the kind exemplified in (73) are tricky:

- (73) a. A: Did Bo kowtow? B: Bo?  
b. Who do you mean ‘Bo’?

The content we wish to assign to such utterances is in fact identical to the question which constitutes the value of MAX-QUD. How to get such a content to arise? The difficulty here is exacerbated because the constituent 'Bo' which should be the basis of the semantic construction is being used in a somewhat 'non-standard' way. 'Bo' is not being used to refer to an individual—the point of such an utterance is after all to highlight the inability of the speaker to do so. In some sense these uses of 'Bo' involve mention and not use, to use a familiar distinction. However, dubbing these uses as *mention* is not quite right either, at least if by *mention* one means a use in which reference is made simply to the form itself, as in (74):

- (74) a. 'Bo' has two letters.  
b. 'Bo' is a noun.

The point is that uses such as (73), as well as the non-elliptical (75a–c), involve an intrinsic anaphoricity. This extends to (75d), where the 4-year-old D malapropically uses 'accident' for 'accent' and J comments on that utterance:

- (75) a. A: Did Bo leave? B: Who is Bo?  
b. A: We're fed up. B: Who is we?  
c. A: Bo is coming. B: Who do you mean 'Bo'?  
d. D: I have a Geordie accident. J: 'accident' that's funny. [attested example]

A decision about how to analyse the quasi-quotative cases in (75) is a precondition of an analysis of the intended content readings RFs. I will assume, following Ginzburg and Cooper (2004) and Purver (2004a), that such cases involve reference to an utterance event under the constraint of segmental phonological parallelism. Purver (2004a), generalizing the treatment of Ginzburg and Cooper (2004), proposed rules that would mean in effect that any phrase has two analyses, one with its 'standard' semantics, the other as an *utt(erance)-anaph(oric) ph(rase)*.<sup>39</sup> A rule that creates an *utt-anaph-ph* from a word is given in (76): it allows a word whose phonology is type-identical with the FEC to refer to the FEC utterance event:<sup>40</sup>

- (76) *utt-anaph-ph* = *def*
- |                                         |
|-----------------------------------------|
| tune = max-qud.fec.sit-type.phon : Type |
| phon : <i>tune</i>                      |
| cat.head = V : PoS                      |
| dgb-params.max-qud : InfoStruc          |
| hd-dtr : <i>word</i>                    |
| cont = max-qud.fec.sit : Rec            |

<sup>39</sup> Postulating such an ambiguity goes back to Frege and Quine, who suggested phrases within quotative operators denoted the string itself. In other words, this is the context-independent mention-oriented use akin to (74).

<sup>40</sup> There is here, as with many NSUs which do not live a life as complements, an issue as to what syntactic category to associate with this construction. I will assume it to be *v*, though the positive arguments in favour of this option are not strong.

With this in hand, we turn back to consider the issue of how intended content RFs arise grammatically. It is worth emphasizing that there is no way to bring about the desired content using *decl-frag-cl*, the short-answer/reprise sluice phrasal type we have been appealing to extensively, regardless of whether we analyse the NP fragment as denoting its standard conventional content or alternatively as denoting an anaphoric element to the phonologically identical to-be-clarified sub-utterance. This is a prototypical instance of appeal to constructional meaning—a complex content that cannot be plausibly constructed using ‘standard combinatorial operations’ (function application, unification, etc.) from its constituents. Given this, we posit a new phrasal type, *qud-anaph-int-cl*, which also does service in explicating *gap* CRs (Purver 2004a) (see Exercise 7.9.2). This encapsulates the two idiosyncratic facets of such utterances: their content can be identified with that of *MaxQUD* and the *HD-DTR* is an *utt-anaph-ph*:

$$(77) \quad \text{qud-anaph-int-cl} =_{\text{def}} \left[ \begin{array}{l} \text{dgb-params.max-qud : InfoStruc} \\ \text{cont=max-qud.q : Question} \end{array} \right]$$

|

hd-dtr: *utt-anaph-ph*

Given this, we can offer the following analysis of (78):

- (78) a. A: Is George here? B: George?  
 b. B lacks referent for ‘George’; uses parameter identification to update *MAX-QUD* accordingly (see (75), Chapter 6 for a more detailed DGB):

$$\left[ \begin{array}{l} \text{spkr} = \text{B} \\ \text{addr} = \text{A} \\ \text{pending} = \left\langle \left[ \begin{array}{l} \text{sit} = \text{w0'} \\ \text{sit-type} = \text{IGH} \end{array} \right] \right\rangle \\ \text{maxqud} = \left[ \begin{array}{l} \text{q} = \lambda x \text{ Mean}(\text{A}, \text{u2}, x) : \text{Question} \\ \text{fec} = \left[ \begin{array}{l} \text{sit} = \text{u2} \\ \text{sit-type} = \text{T2} \end{array} \right] : \text{LocProp} \end{array} \right] : \text{InfoStruc} \end{array} \right]$$

Using *qud-anaph-int-cl* yields:



$$\begin{array}{c}
 (79) \qquad \qquad \qquad S \\
 \left[ \begin{array}{l}
 \text{qud-anaph-int-cl} \\
 \text{maxqud} = \left[ \begin{array}{l}
 q = \lambda x \text{ Mean}(A, u2, x) : \text{Question} \\
 \text{fec} = \left[ \begin{array}{l}
 \text{sit} = u2 \\
 \text{sit-type} = T2
 \end{array} \right] : \text{LocProp}
 \end{array} \right] : \text{InfoStruc} \\
 \text{cont} = \text{maxqud.q}
 \end{array} \right] \\
 | \\
 S \\
 \left[ \begin{array}{l}
 \text{utt-anaph-ph} \\
 \text{george} = \text{max-qud.fec.sit-type.phon} : \text{Type} \\
 \text{phon} : \text{george}
 \end{array} \right] \\
 \\
 NP \\
 \text{george}
 \end{array}$$

Having developed this constructional account of intended content readings of RF, it is worth noting that we could develop an analogous treatment of reprise sluicing, where, as we discussed in Chapter 6, the intended content reading vs clausal confirmation reading distinction is difficult to tease apart. In section 7.9.1, I showed how clausal confirmation reading RFs and sluicing can be handled without adding new rules to the grammar. The motivation for considering an alternative analysis comes from developmental data, discussed further in section 8.4, which indicates that reprise sluicing is invariably acquired before RF. This suggests the need for grammatically teasing apart reprise sluicing from RF, assuming, as is justified below, that this needs to be accounted for in the grammar.<sup>41</sup> A simple analysis of reprise sluicing we can offer is in (80): it allows a bare *wh*-phrase to denote MaxQUD given that the domain of the *wh*-phrase is the same as the domain of MaxQUD:

$$\begin{array}{c}
 (80) \quad \text{wh-anaph-int-cl} =_{\text{def}} \\
 \left[ \begin{array}{l}
 \text{dgb-params.max-qud} : \text{InfoStruc} \\
 G : \text{RecType} \\
 \text{cont} = \text{max-qud.q} : (G)\text{Prop}
 \end{array} \right] \\
 | \\
 \text{hd-dtr: } \left[ \text{quest-dom} = \langle G \rangle : \text{list}(\text{Rtype}) \right]
 \end{array}$$

<sup>41</sup> To be more precise, it requires a grammar where distinct mechanisms underlie reprise sluicing and RF. In the ultimate adult grammar a unified mechanism *could* be in operation.

Formula (80), in addition to allowing reprise sluices to be analysed, also allows certain *wh-anaphor* uses exemplified in (81a, b), which are not CRs. The *wh-anaphor*, on this analysis, reposes the question. It doesn't, as it stands, deal with cases like (81c), where the interrogative is embedded. Nonetheless this latter deficiency, noted earlier in other contexts, would be remedied once QUDs inherit from such clauses.

- (81) a. A: Who came yesterday? B: Who?  
 b. A: When are we going to conclude this discussion? B: mmh, when?  
 c. A: You know who called me up yesterday? B: Who?

**Exercise 7.9.2** Develop an analysis of *gap* CRs, exemplified in (82):

- (82) A: Is Mary an abouliac? B: An? (= *what did you say/mean in the utterance after 'an'?*)

### 7.9.3 Acknowledgements

The content of a repetition acknowledgement such as (83(2)) is, in certain respects, even harder to pin down than that of a clarification request.

- (83) Ariadne(1): Did Bo kowtow?  
 Bora(2): Bo, hmm. (I'm not sure.)

This is, in part, because such an utterance is intended primarily to acknowledge the entire previous utterance, not merely its antecedent sub-utterance, and sometimes is uttered almost as a reflex follow-up. Thus, any of the three following contents could be argued to be the content of (83(2)):

- (84) a. I notice your utterance, the one that contained the word pronounced *Bo*.  
 b. (I notice) You're asking if BO (of all people) kowtowed.  
 c. (I notice) You're referring to Bo.

It is arguable that (83(2)) does not have a univocal paraphrase—a speaker could be intending any of these three contents in this context; (84a) is most plausibly associated with utterances by children, as exemplified in section 6.3 in connection with the game *ack-huh*. (84b, c) seem convincing in contexts where the acknowledger is being reflective about the utterance:<sup>42</sup>

- (85) A: Did Bo kowtow?  
 B: Bo, hmm, good question. (I'm not sure)

<sup>42</sup> There are, however, certainly contexts in which an acknowledgement arises via repetition of a constituent but this need not be understood as involving an intention to highlight that constituent:

(i) (Context: B is a waitress in an Edinburgh diner) J: I'll be having chips and beans and a capuccino.  
 B: and a capuccino, OK. (attested example)

Both these readings arise straightforwardly: given contexts in which MaxQUD is either ‘Who do you mean “Bo” ’ or ‘Who are you asking if s/he kowtowed’ (as occur respectively by application of parameter identification and parameter focussing, with ‘Bo’ as FEC), the requisite readings arise via *decl-frag-cl*.

## 7.10 Initiating NSUs

In all previous sections the cases of NSU resolution we discussed were reactive moves, hence based on MaxQUDs which are contents of prior utterances. As I noted earlier, this is the contextual set-up which is, statistically speaking, overwhelmingly the most common. Nonetheless, there are certain conversational genres, in the terminology of Chapter 4, in which NSUs can occur as initiating utterances.

Consider the following examples, all of which are based on conversational genre types discussed in Chapter 4:

- (86) a. Buying a train ticket:  
       Client: A return to Newcastle please. (= I want a return ..., please give me a return ..., ...)
- b. Buying in a bakery:  
       C: Four croissants and one of these please. (=I want four croissants ..., please give me four croissants ...)
- c. Buying goods in a minimarket stationed in a petrol station:  
       C: [puts basket on till]  
       S: What number pump? (= What number pump were you using, what number pump did you fill up with...)

One characteristic of such utterances, noted by Clark (1996) and exemplified above, is that their content does not have a univocal representation. It depends on how conversational participants characterize a conversational genre type—distinct characterizations can still result in successful communication as long as they do not materially affect the underlying activity.

Our analysis of these cases, inspired by that of Larsson (2002), is a consequence of our rule for initiating utterances from Chapter 4. We repeat this here slightly modified in two respects, given developments in Chapter 6 and the current chapter: the initiating move is now an utterance—so of type LocProp—whereas the accommodated element from QUD is a question, sub-utterance pair, the latter for a reason I explain shortly. Thus, (87) indicates the following: a conversational participant can make an initiating move  $m_0$  if (a) QUD is empty and (b) given that the current genre is  $G_0$ , A believes that  $m_0$ 's content uttered relative to  $q_0.q$ —an accommodated question that is constitutive of the genre  $G_0$ —is relevant to  $G_0$  in  $dgb_0$ :

(87) Initiating Move = *def*

$$\left[ \begin{array}{l}
 \text{dgb} : \left[ \text{qud} = \langle \rangle : \text{eset}(\text{info-struc}) \right] \wedge \text{DGBType} \\
 \\
 \text{pre} : \left[ \begin{array}{l}
 \text{genre} : \text{GenreType} \\
 \text{beliefs} : \text{Prop} \\
 \text{agenda} : \text{list}(\text{IllocProp}) \\
 \text{private} = \left[ \begin{array}{l}
 m0 : \text{locProp} \\
 q0 : \text{info-struc} \\
 c1 : \rightarrow (\text{beliefs}, \\
 \quad \text{GenreRelevant}^{\text{qudpresupp}}(m0^{\text{cont}}, q0.q, \text{dgb}, \text{genre}) )
 \end{array} \right] : \text{PRTType}
 \end{array} \right] \\
 \\
 \text{effects} : \text{Turnholder-Underspec} \wedge_{\text{merge}} \\
 \left[ \begin{array}{l}
 \text{LatestMove} = \text{pre.private.m0} : \text{LocProp} \\
 \text{qud} = \langle q0 \rangle : \text{list}(\text{info-struc}) \\
 c3 : \text{Co-propositional}(\text{qud-contrib}(m0.\text{cont}), q0.q)
 \end{array} \right]
 \end{array} \right]$$

(87) allows for initiating moves *m0* relative to an incrementation of QUD by a question which is co-propositional with the content of *m0*. In particular, this allows for an analysis of (86a, b) as short answers and (86c) as a direct sluice.

Where the formulation given here differs from that of Chapter 4 is that the notion of QUD accommodation employed here is not purely semantic, but also requires a specification of the categorial aspects of the FEC. This allows us to capture linguistic restrictions on such uses such as the German and Hebrew examples in (88), where asking for a cup of coffee or loaf of bread is naturally done with an NSU bearing accusative case.<sup>43</sup>

- (88) a. A: et                      haxalla      hazoti    bevakasha.  
              defobj-marker the-challah the-that please.  
              A: That challah (Sabbath loaf) please
- b. A: Einen              normalen              Kaffee bitte.  
              A-masc-acc normal-masc-acc coffee please.  
              A: A regular coffee please

These data can be accounted for by stipulating the category of the FEC for the issue that makes up the corresponding genre as accusative. The empirical situation, however, is quite complex since conversations of this type can also involve NSUs with distinct case requirements. Although the definite accusative marker is obligatory for a definite object in a non-elliptical setting (as in (89a)),<sup>44</sup> it could be omitted in (89b).<sup>45</sup> Example (89c), a variant of (88a) with nominative case, is also apparently possible:

<sup>43</sup> I am indebted to David Schlangen for his advice on the German data, here and below.

<sup>44</sup> The sentence is acceptable in some archaic dialects.

<sup>45</sup> A related class of examples comes from Morgan (1989). He points out the existence of caseless NPs in Korean whose only use is as NSUs, case being obligatory in 'canonical' constructions.

(89) a. A: \*ten li haxalla hazoti bevakasha.

Give-imp to-me the-challah the-that please.

A: Give me please that challah.

b. A: haxalla hazoti bevakasha.

the-challah the-that please.

A: That challah please

c. A: Ein normaler Kaffee bitte.

A-masc-nom normal-masc-nom coffee please.

A: A regular coffee please

Whether the right approach to such cases is underspecification of the case requirements of the FEC is unclear. This might be correct for the Hebrew examples (88b) and (89b). However, an alternative possibility, put forward to me by David Schlangen (p.c.), with respect to (88a) and (89c), is that they relate to distinct interaction scenarios; the former corresponding to a typical coffee-ordering scenario at a café, the latter more appropriate, e.g. as an affirmative response to an offer of coffee (communicating *A regular coffee would be nice now*). A more definite resolution of this issue requires a far more extensive empirical basis.

## 7.11 Summary

In this chapter I have provided an empirical characterization of the NSU types that occur in a general corpus, the BNC. Taking this characterization as the basic quarry, I developed an account of the grammar of a wide range of NSU types, building on KoS's theory of context developed in earlier chapters. The approach is one I have dubbed *dialogue-oriented constructionism*. This embodies the claim that the context-oriented aspect of NSU words and constructions is as conventional as, for instance, 'traditional' indexicals ('I', 'here', 'now'), there being no reasonable criterion which will leave the former outside the purview of grammar and leave the latter inside. The theoretical cost has been as follows:

- **Context:** enriching QUD, update rules with an additional field for tracking potential focus-establishing constituents, antecedent sub-utterances that encodes various parallelism conditions.
- **Grammar:** positing a small number of constructions, lexical types.

In general terms I claimed in the introduction to this chapter that *given* our explicit account of context, the grammar of NSUs is relatively straightforward. Tables 7.9, 7.10, and 7.11 summarize the analyses provided for various NSU types discussed in this chapter.

One rather obvious observation about these tables is that the NSU types in Table 7.9 are, intuitively, simpler than those in Tables 7.10 and 7.11. Is there any evidence for this

TABLE 7.9 NSUs: analysis of context and content (part 1)

| NSU class                                                                                                                                                                                                                                                                        | Example                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Context                                                                                                                                                                                                                                                                          | Content                                                                          |
| Plain Acknowledgement                                                                                                                                                                                                                                                            | <i>A: ... B: mmh</i><br>Acknowledge(u, spkr)                                     |
| $\left[ \begin{array}{l} \text{pending.sit} = u : \text{Rec} \\ \text{spkr} : \text{IND} \end{array} \right]$                                                                                                                                                                    |                                                                                  |
| Repeated Acknowledgement                                                                                                                                                                                                                                                         | <i>A: Did Bo leave? B: Bo, hmm.</i><br>Acknowledge(u, spkr)                      |
| $\left[ \begin{array}{l} \text{pending.sit} = u : \text{Rec} \\ \text{spkr} : \text{IND} \\ \text{const1} : \text{sign} \\ \text{c} : \text{Constit}(\text{const1}, u) \end{array} \right]$                                                                                      |                                                                                  |
| Check Question                                                                                                                                                                                                                                                                   | <i>A: Bo isn't here. Okay?</i><br>max-qud.q                                      |
| $\left[ \begin{array}{l} \text{spkr} : \text{IND} \\ \text{addr} : \text{IND} \\ \text{p} : \text{Prop} \\ \text{LatestMove} = \\ \text{Assert}(\text{spkr}, \text{addr}, \text{p}) : \text{IllocProp} \\ \text{max-qud.q} = \text{p?} : \text{PolQuestion} \end{array} \right]$ |                                                                                  |
| Affirmative Answer                                                                                                                                                                                                                                                               | <i>A: Did Bo leave? B: Yes.</i><br>max-qud.q([])                                 |
| $\left[ \text{max-qud.q} : \text{PolQuestion} \right]$                                                                                                                                                                                                                           |                                                                                  |
| Rejection                                                                                                                                                                                                                                                                        | <i>A: Did Bo leave? B: No.</i><br>NegProp(p) $\wedge$<br>SimpleAns(p, max-qud.q) |
| $\left[ \text{max-qud.q} : \text{PolQuestion} \right]$                                                                                                                                                                                                                           |                                                                                  |
| Short Answer                                                                                                                                                                                                                                                                     | <i>A: Who left? B: Bo.</i><br>max-qud.q(frag.cont)                               |
| $\left[ \text{max-qud.q} : \text{whQuestion} \right]$                                                                                                                                                                                                                            |                                                                                  |

**TABLE 7.10** NSUs: analysis of context and content (part 2)

| NSU class<br>Context                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Example<br>Content                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Direct Sluice</p> $\left[ \begin{array}{l} \text{max-qud} : \left[ \begin{array}{l} p : \text{QuantProp} \\ q = ([\ ] )p : \text{PolarQuestion} \\ \text{fec} : \text{set}(\text{LocProp}) \end{array} \right] \\ \text{fec} : \text{LocProp} \\ \text{cl} : \text{member}(\text{fec}, \text{max-qud.fec}) \\ \text{hd-dtr} : \left[ \begin{array}{l} F : \text{RecType} \\ G = [x : \text{IND}] \wedge_{\text{merge}} F : \text{RecType} \\ \text{cont} : [x : \text{IND}] \\ \text{quest-dom} = \langle G \rangle : \text{list}(\text{RType}) \end{array} \right] \end{array} \right]$ | <p><i>A: Someone left. B: Who?</i><br/> <math>(r : G)\text{max-qud.q}([\ ]) (\text{fec.cont.x} \sqcap r.\text{hd-dtr.cont.x})</math></p> |
| <p>Reprise Sluice</p> $\left[ \begin{array}{l} G : \text{RecType} \\ \text{MAX-QUD} : \left[ \begin{array}{l} q : (G)\text{Prop} \\ \text{fec} : \text{LocProp} \end{array} \right] \\ \text{hd-dtr} : [\text{quest-dom} = \langle G \rangle : \text{list}(\text{RecType})] \end{array} \right]$                                                                                                                                                                                                                                                                                            | <p><i>A: Did Bo leave? B: Who?</i><br/> <math>\text{max-qud.q}</math></p>                                                                |

**TABLE 7.11** NSUs: analysis of context and content (part 3)

| NSU class<br>Context                                                                                                                                                                                                                                                                                                             | Example<br>Content                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <p><math>\text{RF}_{\text{int-cont}}</math></p> $\left[ \begin{array}{l} \text{pending.sit} = u : \text{Rec} \\ \text{spkr} : \text{IND} \\ \text{u1} : \text{sign} \\ \text{c} : \text{Constit}(\text{u1}, u) \\ \text{max-qud.q} = \lambda x \text{Intend}(\text{spkr}, \text{u1}, x) : \text{whQuestion} \end{array} \right]$ | <p><i>A: Did Bo leave? B: Bo?</i><br/>       (= Who do you mean, 'Bo?')<br/> <math>\text{max-qud.q}</math></p>                           |
| <p><math>\text{RF}_{\text{cl-conf}}</math></p> $\left[ \begin{array}{l} \text{pending.sit} = u : \text{Rec} \\ \text{u1} : \text{sign} \\ \text{c} : \text{Constit}(\text{u1}, u) \\ \text{i} = \text{u1.cont} \\ \text{max-qud.q} = \lambda i \text{pending.cont} : \text{whQuestion} \end{array} \right]$                      | <p><i>A: Did Bo leave? B: Bo?</i><br/>       (= Are you asking if BO left?)<br/> <math>? \text{max-qud.q}(\text{hd-dtr.cont})</math></p> |

complexity or is it simply a theoretical artifact? In fact, as I discuss in section 8.4, it turns out that for the majority classes the predicted complexity is consistent with the order of acquisition among children. Moreover, there is further evidence that this ordering does not merely reflect the order of pragmatic acquisition of conversational move types or parental input, which suggests that the explanation for the ordering has a grammatical component.



## Extensions

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In this chapter I pursue somewhat tentatively various extensions to KoS:

- From duologue to multilogue
- Unifying CRs and self-repair
- A deeper view of grounding and CRification (including irrelevance implicatures and the metacommunicative ‘Why?’)
- Developmental evidence on NSUs
- Anaphora and Quantified NPs in dialogue

### 8.1 Duologue and Multilogue

#### 8.1.1 Introduction

Duologue—two-person conversation—is by now a topic with an ever increasing theoretical, corpus-based, and implementational literature. In contrast, the study of *multilogue*—conversation which can include three or more participants—is still in its early stages.<sup>1,2</sup> The importance of multilogue should be clear: it constitutes the general case of a conversational setting, including familial interaction, tribal gatherings, and—in a more modern perspective—genres such as meetings, pedagogy, courtroom proceedings, and many types of games (bridge, poker, etc.).

The fundamental issue in tackling multilogue is: to what extent do mechanisms motivated for duologue (e.g. information states, protocols, conversational rules, etc.) scale up directly to multilogue?

There are of course various plausible views of the relation between duologue and multilogue. One possible approach to take is to view multilogue as a sequence of duologues. Something like this approach seems to be adopted in the literature on communication between autonomous software agents. However, even though many situations considered in multi-agent systems do involve more than two agents, most

<sup>1</sup> The work described in this section slightly reworks and extends joint work with Raquel Fernández (Ginzburg and Fernández 2005b,a).

<sup>2</sup> Various terms have been introduced to cover such interaction: *multilogue*, *multi-party dialogue*, *polylogue*. When I wish to contrast two-person conversation with more-than-two-person conversation I will use the terms *duologue* and *multilogue* respectively.

interaction protocols are designed only for two participants at a time, perhaps in parallel; see e.g. the protocol specifications provided by FIPA (FIPA 2003). Modelling of obligations and grounding becomes more complex when considering multilogue situations. The model of grounding implemented in the Mission Rehearsal Exercise (MRE) Project (Traum and Rickel 2002), one of the largest multilogue systems developed hitherto, derives from the one designed by Matheson, Poesio, and Traum (2000) for duologue and can only be used in cases where there is a single initiator and responder. It is not clear what the model should be for multiple addressees: should the contents be considered grounded when any of the addressees has acknowledged them? Should evidence of understanding be required from every addressee?

I proceed as follows. In section 8.1.2 I present data from the British National Corpus (BNC) comparing the resolution of NSUs in duologue and multilogue. The main focus of these data is on the distance between antecedent and fragment. These data is used to extract certain benchmarks concerning multilogue interaction. In section 8.1.4 I consider two possible transformations on duologue protocols into multilogue protocols. These transformations are entirely general in nature and could be applied to protocols stated in whatever specification language. The protocols that are generated by these transformations are evaluated with reference to the benchmarks extracted in section 8.1.2. Finally, in section 8.1.5 I discuss how these protocols can be implemented in terms of conversational update rules: in fact, our existing rules scale up smoothly, with only minor modifications required to enable them to apply to multilogue.

### 8.1.2 *Long-Distance Resolution of NSUs in Duologue and Multilogue: the Facts*

The corpus employed here is the sub-portion of the BNC conversational transcripts mentioned in Chapter 7. The corpus, consisting of 14,315 sentences, was created by randomly excerpting a 200-speaker-turn section from each of 54 BNC files. Of these files, 29 are transcripts of conversations between two duologue participants, and 25 files are multilogue transcripts.

Table 8.1 shows the raw counts of NSUs found in the duologue and multilogue transcripts, respectively.

**TABLE 8.1 Total of NSUs in Duologue and Multilogue**

|            | NSUs | BNC files |
|------------|------|-----------|
| Duologue   | 710  | 29        |
| Multilogue | 573  | 25        |
| Total      | 1285 | 54        |

TABLE 8.2 NSUs sorted by class and distance

| NSU Class          | Example                 | Total | Distance |     |    |     |     |     |     |
|--------------------|-------------------------|-------|----------|-----|----|-----|-----|-----|-----|
|                    |                         |       | 1        | 2   | 3  | 4   | 5   | 6   | >6  |
| Acknowledgment     | <i>Mm mm.</i>           | 595   | 578      | 15  | 2  |     |     |     |     |
| Short Answer       | <i>Ballet shoes.</i>    | 188   | 104      | 21  | 17 | 5   | 5   | 8   | 28  |
| Affirmative Answer | <i>Yes.</i>             | 109   | 104      | 4   |    |     | 1   |     |     |
| Reprise Fragment   | <i>John?</i>            | 92    | 76       | 13  | 2  | 1   |     |     |     |
| Repeated Ack.      | <i>His boss, right.</i> | 86    | 81       | 2   | 3  |     |     |     |     |
| Rejection          | <i>No.</i>              | 50    | 49       | 1   |    |     |     |     |     |
| Factual Modifier   | <i>Brilliant!</i>       | 27    | 23       | 2   | 1  | 1   |     |     |     |
| Repeated Aff. Ans. | <i>Very far, yes.</i>   | 26    | 25       | 1   |    |     |     |     |     |
| Helpful Rejection  | <i>No, my aunt.</i>     | 24    | 18       | 5   |    | 1   |     |     |     |
| Check Question     | <i>Okay?</i>            | 22    | 15       | 7   |    |     |     |     |     |
| Filler             | <i>... a cough.</i>     | 18    | 16       | 1   |    | 1   |     |     |     |
| Bare Mod. Phrase   | <i>On the desk.</i>     | 16    | 11       | 4   |    |     | 1   |     |     |
| Sluice             | <i>When?</i>            | 11    | 10       | 1   |    |     |     |     |     |
| Prop. Modifier     | <i>Probably.</i>        | 11    | 10       | 1   |    |     |     |     |     |
| Conjunction Phrase | <i>Or a mirror.</i>     | 10    | 5        | 4   | 1  |     |     |     |     |
|                    | <b>Total</b>            | 1285  | 1125     | 82  | 26 | 9   | 7   | 8   | 28  |
|                    | <b>Percentage</b>       | 100   | 87.6     | 6.3 | 2  | 0.6 | 0.5 | 0.6 | 2.1 |

All NSUs encountered within the corpus were classified according to the NSU typology presented in Chapter 7. Additionally, the distance from their antecedent was measured. Table 8.2 shows the distribution of NSU categories and their antecedent separation distance.<sup>3</sup> The last row in Table 8.2 shows the distribution of NSU-antecedent separation distances as percentages of the total of NSUs found. This shows that about 87 per cent of NSUs have a distance of one sentence (i.e. the antecedent was the immediately preceding sentence), and that the vast majority (about 96 per cent) have a distance of three sentences or less.

<sup>3</sup> The BNC annotation includes tagging of units approximating to sentences, as identified by the CLAWS segmentation scheme (Garside 1987). Each sentence unit is assigned an identifier number. By default it is assumed that sentences are non-overlapping and that their numeration indicates temporal sequence. When this is not the case because speakers overlap, the tagging scheme encodes synchronous speech by means of an alignment map used to synchronize points within the transcription. However, even though information about simultaneous speech is available, overlapping sentences are annotated with different sentence numbers.

In order to be able to measure the distance between the NSUs encountered and their antecedents, all instances were tagged with the sentence number of their antecedent utterance. The distance reported here is therefore measured in terms of sentence numbers. It should however be noted that taking into account synchronous speech would not change the data reported in Table 8.2 in any significant way, as manual examination of all NSUs at more than distance 3 reveals that the transcription portion between antecedent and NSU does not contain any completely synchronous sentences in such cases. In examples throughout the chapter, italics are used to indicate speech overlap. When italics are not used, utterances occur sequentially.

TABLE 8.3 NSUs in duologue and multilogue sorted by distance

| Distance   | 1         | 2        | 3        | 4       | 5       | 6       | >6        |
|------------|-----------|----------|----------|---------|---------|---------|-----------|
| Duologue   | 658 (59%) | 37 (45%) | 11 (45%) | 1 (12%) | 1 (14%) | 1 (13%) | 0 (0%)    |
| Multilogue | 467 (41%) | 45 (55%) | 15 (55%) | 8 (88%) | 6 (86%) | 7 (87%) | 28 (100%) |

Table 8.2 gives a hint of the main duologue/multilogue differences:

- Most long-distance NSUs occur in multilogue.
- Most long-distance NSUs are short answers.
- The other major NSU classes, used in grounding/CRification and in acceptance/rejection, overwhelmingly involve adjacent antecedents.

Let me spell this out in more detail. Although the proportion of NSUs found in duologue and multilogue is roughly the same (see Table 8.1 above), when taking into account the distance of NSUs from their antecedent, the proportion of long-distance NSUs in multilogue increases radically: the longer the distance, the higher the proportion of NSUs that were found in multilogue. In fact, as Table 8.3 shows, NSUs that have a distance of seven sentences or more appear exclusively in multilogue transcripts. These differences are significant ( $\chi^2 = 62.24$ ,  $p \leq 0.001$ ).

One striking result exhibited in Table 8.2 is the uneven distribution of long-distance NSUs across categories. With a few exceptions, NSUs that have a distance of three sentences or more are exclusively short answers. The data in Table 8.4 highlights a fundamental characteristic of the remaining majority classes of NSUs, Ack(nowledgements), Affirmative Answer, RF (Reprise Fragments), Repeated Ack(nowledgements), and Rejection. These are used either in grounding interaction, CRification, or to affirm/reject propositions. The overwhelming adjacency to their antecedent underlines the locality of these interactions.

Not only is the long-distance phenomenon almost exclusively restricted to short answers, but the frequency of long-distance short answers stands in strong contrast to the other NSU classes; indeed, over 44 per cent of short answers have more than

TABLE 8.4 NSUs in duologue and multilogue: majority classes ( $\geq 4\%$ )

| NSU Class          | Example                 | Total | Distance |    |    |   |   |   |    |
|--------------------|-------------------------|-------|----------|----|----|---|---|---|----|
|                    |                         |       | 1        | 2  | 3  | 4 | 5 | 6 | >6 |
| Acknowledgment     | <i>Mm mm.</i>           | 595   | 578      | 15 | 2  |   |   |   |    |
| Short Answer       | <i>Ballet shoes.</i>    | 188   | 104      | 21 | 17 | 5 | 5 | 8 | 28 |
| Affirmative Answer | <i>Yes.</i>             | 109   | 104      | 4  |    |   | 1 |   |    |
| Reprise Fragments  | <i>John?</i>            | 92    | 76       | 13 | 2  | 1 |   |   |    |
| Repeated Ack.      | <i>His boss, right.</i> | 86    | 81       | 2  | 3  |   |   |   |    |
| Rejection          | <i>No.</i>              | 50    | 49       | 1  |    |   |   |   |    |

distance one, and over 24 per cent have distance four or more, like the last answer in the following example:

- (1) a. Allan: How much do you think?  
 Cynthia: Three hundred pounds.  
 Sue: More.  
 Cynthia: A thousand pounds.  
 Allan: More.  
 Unknown: <unclear>  
 Allan: Eleven hundred quid apparently. (BNC, G4X)
- b. Liz: Who's produced them?  
 Sue: It's...  
 Allan: Erm the people who who our training consultant's just gone to I I P L?  
 Cynthia: Golly.  
 Allan: Somebody anyway.  
 Cynthia: Oh Andrea Andrea [last or full name]. (BNC, G4X)

Table 8.5 shows the total number of short answers found in duologue and multilogue respectively, and the proportions sorted by distance over those totals.

From this it emerges that short answers are more common in multilogue than in duologue—134 (71 per cent) vs 54 (29 per cent). Also, the distance pattern exhibited by these two groups is strikingly different: only 18 per cent of short answers found in duologue have a distance of more than one sentence, with all of them having a distance of no more than three, like the short answer in (2).

- (2) Malcolm: [...] cos what's three hundred and sixty divided by seven?  
 Anon 1: I don't know.  
 Malcolm: Yes I don't know either!  
 Anon 1: Fifty four point fifty one point four. (BNC, KND)

In contrast, long-distance short answers are common in multilogue: 56 per cent of those found in multilogue have more than distance one, while 37 per cent have more than distance three (see (1) above for examples).

As Table 8.5 shows, all short answers at more than distance three appear in multilogues. Following Fay, Garrod, and Carletta (2000), one could distinguish between

**TABLE 8.5 % over the totals found in duologue and multilogue**

| Short Answers | Total | 1  | 2  | 3 | > 3 |
|---------------|-------|----|----|---|-----|
| Duologue      | 54    | 82 | 9  | 9 | 0   |
| Multilogue    | 134   | 44 | 11 | 8 | 37  |

**TABLE 8.6 Long-distance short answers in small and large groups**

| Group Size | d > 3         | d ≤ 3         | Total |
|------------|---------------|---------------|-------|
| ≤ 5        | 20<br>(21.5%) | 73<br>(78.5%) | 93    |
| > 5        | 26<br>(63%)   | 15<br>(37%)   | 41    |

small groups (those with three to five participants) and large groups (those with more than five participants). The size of the group is determined by the number of participants who are active when a particular short answer is uttered. *Active participants* are those that have made a contribution within a window of thirty turns back from the turn where the short answer was uttered. Table 8.6 shows the distribution of long-distance short answers (distance > 3) in small and large groups respectively. This indicates that long-distance short answers are significantly more frequent in large groups ( $\chi^2 = 22.17$ ,  $p \leq 0.001$ ), though still reasonably common in small groups.

Large-group multilogues in the corpus are all transcripts of tutorials, training sessions, or seminars, which exhibit a rather particular structure. The general pattern involves a question being asked by the tutor or session leader, the other participants then taking turns to answer that question. The tutor or leader acts as turn manager. She assigns the turn explicitly, usually by addressing the participants by their name without need to repeat the question under discussion. An example is shown in (3):

- (3) Anon1: How important is those three components and what value would you put on them [...]  
 Anon3: Tone forty five. Body language thirty.  
 Anon1: Thank you.  
 Anon4: Oh.  
 Anon1: Melanie.  
 Anon5: twenty five.  
 Anon1: Yes.  
 Anon5: Tone of voice twenty five. (BNC, JYM)

Small group multilogues on the other hand have a more unconstrained structure: after a question is asked, the participants tend to answer freely. Answers by different participants can follow one after the other without explicit acknowledgements or turn management, like in (4):

- (4) Anon 1: How about finance then? <pause>  
 Unknown 1: Corruption  
 Unknown 2: Risk <pause dur=30>  
 Unknown 3: Wage claims <pause dur=18> (BNC, JJH)

### 8.1.3 *Duologue vs Multilogue: Two Benchmarks*

What operative conclusions can we draw from the data just presented? Theory-independently, one can summarize the data using the following two benchmarks for protocols for querying, assertion, grounding, and CRification interaction in multilogue:

- (5) a. **Multilogue long-distance short answers (MLDSA)**: querying protocols for multilogue must license short answers an unbounded number of turns from the original query.
- b. **Multilogue adjacency of grounding/acceptance (MAG)**: assertion and grounding protocols for multilogue should license grounding/clarification/acceptance moves only adjacently to their antecedent utterance.

MLDSA and MAG have a somewhat different status: whereas MLDSA is a direct generalization from the data, MAG is a negative constraint, posited given the paucity of positive instances. As such MAG is more open to doubt and I will treat it as such below, indicating ways which would allow it to be bypassed. Although MAG reflects identical NSU distribution in duologue and multilogue of non-short-answer NSUs, in fact it reflects a significant difference between duologue and multilogue: it indicates that, on the whole, only one acknowledgement/acceptance is needed per utterance/assertion, from the audience. But whereas in duologue the audience comprises the addressee, in multilogue this is not the case.

MLDSA, or rather the underlying empirical duologue/multilogue asymmetry, argues against reductive views of multilogue as sequential duologue. Nonetheless, in the current set-up the main work we have to do, as will become clear shortly, is actually **suppressing long-distance short answers in duologue**—if one assumes QUD to be (at least) a stack, as I have hitherto, this actually affords the potential for non-adjacent short answers in duologue (see, for instance, example (43), in Chapter 7). Two commonly observed *duologue* conditions will jointly enforce adjacency between short answers and their interrogative antecedents: (a) Questions have a simple, one-phrase answer; (b) Questions can be answered immediately, without preparatory or subsequent discussion. For multilogue (or at least certain genres thereof), both these conditions are less likely to be maintained: different conversational participants can supply different answers, even assuming that relative to each conversational participant there is a simple, one-phrase answer. The more participants there are in a conversation, the less their common ground and the more likely the need for clarificatory interaction. A pragmatic account of this type of the frequency of adjacency in duologue short answers seems clearly preferable to any actual mechanism that would *rule out* long-distance short answers. These can be perfectly felicitous—see e.g. example (1) above, which would work fine if the turn uttered by Sue had been uttered by Allan instead. Moreover such a pragmatic account leads to the expectation that the frequency of

long-distance antecedents is correlated with group size, as indeed is indicated by the data in Table 8.6.

### 8.1.4 Scaling-up Protocols

Goffman (1981) introduced the distinction between *ratified participants* and *overhearers* in a conversation. Within the former are located the speaker and participants whom she takes into account in her utterance design—the intended addressee(s) of a given utterance, as well as *side participants*. This work was extended significantly by Clark and Carlson (1986), who offer a semi-formal analysis of a variety of speech acts in a multilogue context. Their main tool is a multilayered illocutionary proposition of the form.  $\text{Inform}(\text{spkr}, \text{audience}, \dots, I_1) \rightarrow I_1 = \text{Illocforce}(\text{spkr}, \text{addressees}, p)$ , where the addressees are part of the audience and  $I_1$  is an *addressee-directed* illocutionary act (a request, a query, a command etc.), That is an illocutionary act which the speaker indicates to the audience he is directing to the addressees. In this section I consider two general principles of protocol extension, which can be viewed as adding roles for the *direct addressees* and the *side participants* respectively; dealing with *overhearers* is discussed in an exercise. The protocols that result are evaluated relative to the benchmarks introduced in section 8.1.3. Seen in this light, the principle Add Side Participants (ASP), arguably, yields the better results. I state the principles informally and framework independently as transformations on operational construals of the protocols.

The first principle I consider is Duplicate Responders (DR):

#### (6) Duplicate Responders

Given a dialogue protocol  $\pi$ , add roles  $C_1, \dots, C_n$  which duplicate the responder role.

Applying DR to the querying protocol from Chapter 4 yields the following protocol:

#### (7) Querying with multiple responders

Latest Move = Ask(A, q):

|             |                  |                                 |                                 |                                 |
|-------------|------------------|---------------------------------|---------------------------------|---------------------------------|
| A           | Push q<br>on QUD |                                 |                                 |                                 |
| Responder1: | Push q<br>on QUD | Make<br>q-specific<br>utterance |                                 |                                 |
| Responder2: | Push q<br>on QUD | WAIT                            | Make<br>q-specific<br>utterance |                                 |
| Responder3: | Push q<br>on QUD | WAIT                            |                                 | Make<br>q-specific<br>utterance |



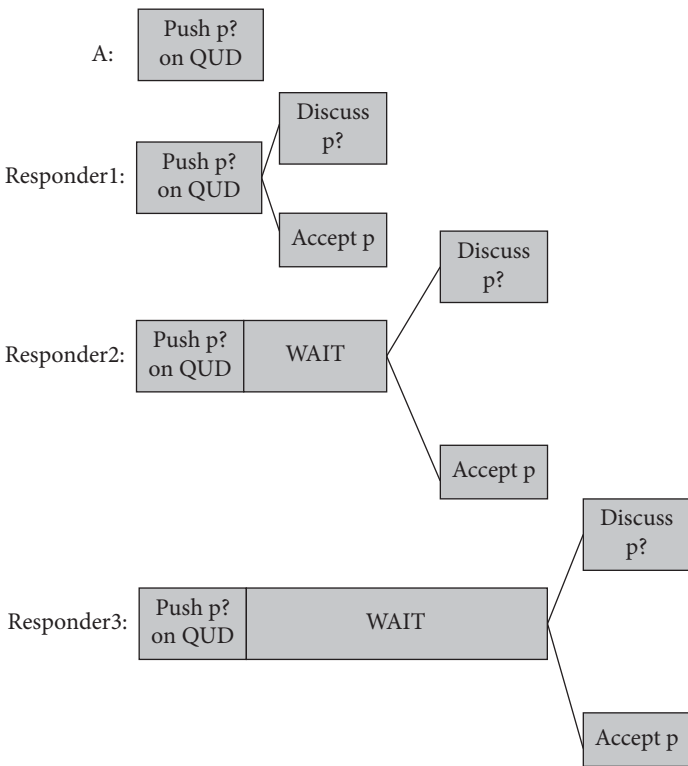
This yields interactions such as (4) above. The querying protocol in (7) licenses long-distance short answers, so satisfies the MLDSA benchmark. On the other hand, the contextual updates it enforces will not enable it to deal with the following (constructed) variant on (4)—in other words it does not allow responders to comment on previous responders, as opposed to the original querier:

- (8)    A: Who should we invite for the conference?  
         B: Svetlanov.  
         C: No (= not Svetlanov), Zhdanov.  
         D: No (= not Zhdanov,  $\neq$  not Svetlanov), Gergev.

Applying DR to the assertion protocol will yield the following protocol:

- (9)    Assertion with multiple responders

Latest Move = Assert(A, p):



One arguable problem with this protocol—equally applicable to the corresponding DRed grounding protocol—is that it licenses long-distance acceptance and is, thus, inconsistent with the MAG benchmark. On the other hand, it is potentially useful for interactions where there is explicitly more than one direct addressee.

The second principle I consider is Add Side Participants (ASP):

(10) Add Side Participants (ASP)

Given a duologue protocol  $\pi$ , add roles  $C_1, \dots, C_n$ , which effect the same contextual update as the interaction initiator.

Applying ASP to the duologue assertion protocol yields the following protocol:

(11) Assertion/Acceptance for a conversation involving

$\{A, B, C_1, \dots, C_n\}$

Latest Move = Assert(A, p):

A: 

|                   |
|-------------------|
| Push p?<br>on QUD |
|-------------------|

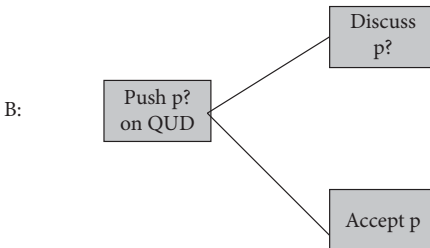
SideParticipant1: 

|                   |
|-------------------|
| Push p?<br>on QUD |
|-------------------|

SideParticipant2: 

|                   |
|-------------------|
| Push p?<br>on QUD |
|-------------------|

...



Latest Move = Accept(B, p):

B: 

|                                    |
|------------------------------------|
| Add p to Facts; pop p?<br>from QUD |
|------------------------------------|

SideParticipant1: 

|                                    |
|------------------------------------|
| Add p to Facts; pop p?<br>from QUD |
|------------------------------------|

SideParticipant2: 

|                                    |
|------------------------------------|
| Add p to Facts; pop p?<br>from QUD |
|------------------------------------|

...

A: 

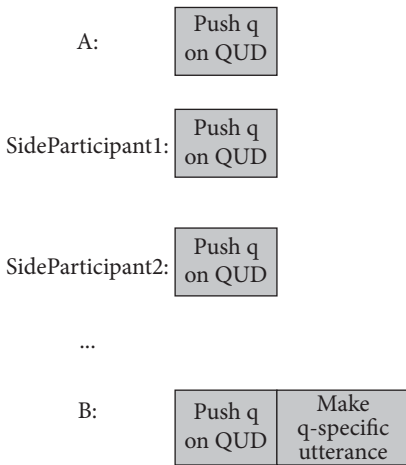
|                                    |
|------------------------------------|
| Add p to Facts; pop p?<br>from QUD |
|------------------------------------|

This protocol satisfies the MAG benchmark in that acceptance is strictly local. This is because it enforces *communal acceptance*—acceptance by one conversational participant can count as acceptance by all other addressees of an assertion. There is an obvious rational motivation for this, given the difficulty for a conversational participant in constantly monitoring an entire audience (when this consists of more than one addressee) for acceptance signals—it is well known that the effect of visual access on turn-taking is highly significant (Dabbs and Ruback 1987). It also enforces quick reaction to an assertion—anyone wishing to dissent from  $p$  must get their reaction in early, i.e. immediately following the assertion, since further discussion of  $p$ ? is not countenanced if acceptance takes place. The latter can happen of course as a consequence of a dissenter not being quick on their feet; on this protocol to accommodate such cases would require some type of backtracking.

Applying ASP to the duologue-querying protocol yields the following protocol:

- (12) Querying for a conversation involving  
 $\{A, B, C_1, \dots, C_n\}$

Latest Move = Ask( $A, q$ ):



This improves on the DR-generated protocol because it does allow responders to comment on previous responders—the context is modified as in the duologue protocol. Nonetheless, as it stands, this protocol won't fully deal with examples such as (4)—the issue introduced by each successive participant takes precedence, given that so far, apart from cases where the turn does not change (see the rule QCoord in section 4.5), QUD behaves like a stack. This can be remedied by a minor modification of the ordering mechanism: I will assume that when a question  $q$  is pushed onto QUD it doesn't subsume *all* existing questions in QUD, but rather only those on which  $q$  does not *depend*:

- (13)  $q$  is  $\text{QUD}^{\text{mod}(\text{dependence})}$  maximal iff for any  $q_0$  in QUD such that  $\neg \text{Depend}(q, q_0): q \succ q_0$ .

This is conceptually attractive because it reinforces the assumption that the order in QUD has an intuitive semantic basis. One effect this has is to ensure that any polar question  $p?$  introduced into QUD, whether by an assertion or by a query, subsequent to a *wh*-question  $q$  on which  $p?$  depends does not subsume  $q$ . Hence,  $q$  will remain accessible as an antecedent for NSUs, as long as no new unrelated topic has been introduced. This is illustrated in (14), where the question introduced by A remains MaxQUD throughout; on the other hand, the polar questions introduced into QUD by each successive assertion are qud-maximal only locally:

- (14) **ASP: A simple example**  
 A: Who should we invite for the conference ( $q_1$ )?  
 MaxQUD:=  $\{q_1\}$   
 B: Svetlanov. ( $p_1?$ )  
 MaxQUD:=  $\{q_1, p_1?\}$   
 C: No (= not Svetlanov), Zhdanov. ( $\neg p_1, p_3$ )  
 MaxQUD:=  $\{q_1, p_3?\}$   
 D: No (= not Zhdanov,  $\neq$  not Svetlanov), Gergev. ( $\neg p_3, p_4$ )  
 MaxQUD:=  $\{q_1, p_4?\}$

If we adopt the modification to the ordering on QUD given in (13), then the above ASP-generated protocols are such that both MLDSA and MAG benchmarks are fulfilled.

**Exercise 8.1.1** How should silent participants be integrated into the ontology of conversational roles? Offer a transformation on protocols that can capture the requisite effects. Does this necessitate any new conversational rules?

### 8.1.5 Conversational Rules for Multilogue

In this section I consider how the protocols scaled up according to the principles ASP and DR discussed in section 8.1.4 can be compositionally decomposed from conversational rules akin to those in section (4.4.3). One minor modification I need to make concerns an expanded set of participants. This applies in two areas: the *spkr*, *addr* roles in conversational rules, and the illocutionary propositions that constitute the values of  $\text{LatestMove}^{\text{content}}$ . Turning first to the latter and using an informal specification, we get rules of the following form:

- (15) a. Ask QUD-incrementation (plural audience): given  
 $\text{LatestMove}^{\text{content}} = \text{Ask}(A, B, \{C_1, \dots, C_n\}, q)$ ,  $q$  becomes QUD maximal  
 for  $\{A, C_1, \dots, C_n\}$

- b. UpdateFacts + DowndateQUD (plural audience):

Given  $\text{LatestMove}^{\text{content}} = \text{Accept}(B, \{A, C_1, \dots, C_n\}, p), \{B, A, C_1, \dots, C_n\}$   
 conjoin  $p$  with FACTS, downdate  $p?$  and all other  $qs$  from QUD resolved by  
 FACTS.

Recall that the *spkr*, *addr* roles in conversational rules are our means for specifying who holds the turn. Some rules involve no turn change (e.g. Ask QUD-incrementation), some involve obligatory turn change (e.g. Accept Move), whereas some are underspecified for this (e.g. QSPEC). The first two cases hitherto have been handled by using ‘role anaphora’<sup>4</sup>, as in (16a, b), whereas the latter exploited type dependence, as in (16c):

- (16) a. *spkr* = *pre.spkr*: IND; *addr* = *pre.addr*: IND  
 b. *spkr* = *pre.addr*: IND; *addr* = *pre.spkr*: IND  
 c.  $\text{PrevAud} = \{\text{pre.spkr}, \text{pre.addr}\} : \text{Set}(\text{IND})$ ; *spkr* : Ind; *c1* : member(*spkr*,  
*PrevAud*); *addr* : IND; *c2*: member(*addr*, *PrevAud*)  $\wedge$  *addr*  $\neq$  *spkr*

In a multilogue setting, these specifications become the more general (17a–c) respectively:

- (17) a. *spkr* = *pre.spkr*: IND; *addr* = *pre.addr*: Set(IND)  
 b. *spkr* : IND; *c1* : member(*spkr*, *pre.addr*); *addr* =  $\text{pre.addr} \cup \{\text{pre.spkr}\} \setminus \{\text{spkr}\} : \text{Set}(\text{IND})$ ;  
 c.  $\text{PrevAud} = \{\text{pre.spkr}, \text{pre.addr}\} : \text{Set}(\text{IND})$ ; *spkr* : Ind; *c1* : member(*spkr*,  
*PrevAud*); *addr* =  $\text{PrevAud} \setminus \{\text{spkr}\} : \text{Set}(\text{IND})$ ;

(17b, c) are underspecified, but could be integrated with a theory of turn-taking, for instance one integrating the normative approach of Sacks, Schegloff, and Jefferson (1974) with the cue-based approach initiated in Duncan (1972), as envisioned in Kronlid (2008).<sup>5</sup> In this setting, it is straightforward to specify a turn-assigning move, as exemplified in (18a(3)):

- (18) a. Anon1 (1): How important is those three components  
 and what value would you put on them?  
 Anon3 (2): Tone forty five. Body language thirty.  
 Anon1(3): Melanie.  
 Anon5 (4): twenty five.

<sup>4</sup> Technically, by means of manifest fields.

<sup>5</sup> Kronlid implements a version of the Schegloff–Sacks–Jefferson theory in a state-chart-based dialogue manager. De Ruiter, Mitterer, and Enfield (2006) offer psycholinguistic evidence against Duncan’s theory.

b. Turn assignment =  $_{def}$

$$\left[ \begin{array}{l} \text{pre} \\ \text{effects} \end{array} : \left[ \begin{array}{l} \text{spkr: IND} \\ \text{addr: Set(IND)} \\ j : \text{IND} \\ \text{rest}^{facts_1} : \text{member}(j, \text{addr}) \\ \text{LatestMove} = \text{TurnAssign}(\text{spkr}, j) : \text{IllocProp} \\ \text{spkr} = \text{pre.j} : \text{IND} \\ \text{addr} = \text{pre.addr} \cup \{ \text{pre.spkr} \} \setminus \{ \text{spkr} \} : \text{Set(Ind)} \end{array} \right] \right]$$

**Exercise 8.1.2** Provide a type description for a construction in which proper names get used as turn assignment moves.

With these minor modifications of the rules in place, let us consider what new conversational rules, if any, need to be posited. For ASP-modified protocols, no new conversational rules are needed.

Given the modification to QUD proposed in section 8.1.4, an implementation of DRed querying follows: if a querier addresses a group with a question  $q$ , the structure of QUD ensures that  $q$  will remain maximal and hence accessible as one of the main issues for each of the participants. Concretely, following a query  $q$  by A, Ask QUD-Incrementation enables the next speaker to provide a  $q$ -specific answer. By the ordering in QUD,  $q$  will remain maximal for any subsequent speaker who has not downdated it. Hence, we obtain the effect of a distributed access to answer  $q$ .

The main additional modification concerns distributed acceptance. Consider first the preconditions for an acceptance move—the difference from the duologue case is that they no longer involve adjacency of the assertion in question. They now involve the combination of the existence of a prior assertion of  $p$  and the maximality of  $p$ ? in QUD:

- (19) **Distributed Accept:** given Moves =  $\langle \dots \text{Assert}(A, \{C_1, \dots, C_n\}, p) \dots \rangle$ , and Max-QUD =  $p?$ ,  $C_i$  can make an utterance such that LatestMove = Accept( $C_i, \{A, C_1, \dots, C_n\}, p$ ).

Fact incrementation/QUD-downdate needs to be divided into two subcases: one that concerns the addressees, the other that concerns the original asserter. To take these in order: for the addressees, given the distributed nature of acceptance here, the precondition for fact incrementation/QUD-downdate has to be an acceptance by that particular individual. For the original asserter the precondition for fact incrementation/QUD-downdate is the existence of acceptance acts of  $p$  by all addressees:

- (20) a. **Distributed UpdateFacts + DowndateQUD (audience version):** Given  $\text{LatestMove}^{\text{content}} = \text{Accept}(C_i, \{A, C_1, \dots, C_n\}, p)$ ,  $C_i$  conjoin  $p$  with FACTS, downdate  $p?$  and all other  $qs$  from QUD resolved by FACTS
- b. **Distributed UpdateFacts + DowndateQUD (asserter version):**  
 Given Moves =  

$$\left\langle \dots \text{Assert}(A, \{C_1, \dots, C_n\}, p) \dots, \text{Accept}(C_1, \{A, \dots, C_n\}, p), \dots \right\rangle, A$$
  

$$\left\langle \text{Accept}(C_n, \{A, C_1, \dots\}, p) \right\rangle$$
  
 conjoin  $p$  with FACTS, downdate  $p?$  and all other  $qs$  from QUD resolved by FACTS

**Exercise 8.1.3** Reformulate the various conversational rules to take account of plural audiences.

### 8.1.6 Two Examples

I now illustrate the analysis developed here for multilogue, using two examples from the BNC. The first example is (4) repeated here as (21). Anon1 poses the question  $q_0$ . This thereby becomes MaxQUD for whoever seizes the turn first, in this case Unknown1. She then provides an answer to  $q_0$ . Call it  $p_1$ . At this stage the other two have two options, either believe that  $p_1$  resolves  $q_0$ , in which case they could accept  $p_1$  and downdate both  $p_1$  and  $q_0$ . Alternatively, as happens here, Unknown2 pursues an alternative route. Her QUD consists of  $q_0$  and  $p_1?$ ; both issues are unordered with respect to each other at that point. So she can address either  $p_1$  (agreeing or disagreeing with it) and/or  $q_0$ . She asserts  $p_2$  about  $q_0$ . Now we reach Unknown3's turn—in her QUD the maximal elements are  $q_0$  and  $p_2?$ , so she can address either (and she could also address  $p_1?$  but only using a sentential answer).

- (21) Anon 1 (1):           How about finance then? <pause>  
 Unknown 1 (2):   Corruption  
 Unknown 2 (3):   Risk <pause dur=30>  
 Unknown 3 (4):   Wage claims <pause dur=18>

| Utt. | DGB Update<br>(Conditions)                                                                                                                                                                    | Rule                                  |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| 1    | $\text{LatestMove} := \text{Ask}(\text{Anon1}, \{\text{unk1}, \text{unk2}, \text{unk3}\}, q_0)$<br>$\text{Anon1.QUD} := \langle q_0 \rangle$                                                  | Free Speech<br>Ask QUD-incrementation |
| 2    | $\text{LatestMove} := \text{Assert}(\text{unk1}, \{\text{Anon1}, \text{unk2}, \text{unk3}\}, p_1)$<br>$\text{Unk1.QUD} := \langle p_1?, q_0 \rangle$<br>$\text{About}(p_1, q_0)$              | QSPEC<br>Assert QUD-incrementation    |
| 3    | $\text{LatestMove} := \text{Assert}(\text{unk2}, \{\text{Anon1}, \text{unk1}, \text{unk3}\}, p_2)$<br>$\text{Unk2.QUD} := \langle p_2?, q_0 > p_1? \rangle$<br>$\text{About}(p_2, q_0)$       | QSPEC<br>Assert QUD-incrementation    |
| 4    | $\text{LatestMove} := \text{Assert}(\text{unk3}, \{\text{Anon1}, \text{unk1}, \text{unk2}\}, p_3)$<br>$\text{Unk2.QUD} := \langle p_3?, q_0 > p_2?, p_1? \rangle$<br>$\text{About}(p_3, q_0)$ | QSPEC<br>Assert QUD-incrementation    |

The second example is a slightly cleaned-up version—disfluencies have been omitted—of (2) from Chapter 1: John asks a question and answers it; Sarah then poses a clarification question about John's answer, which Sue effectively resolves (hence my assumption that her answer downdates the clarification question and her answer to it from QUD); Sue and John disagree about the answer to John's original question. Once again a crucial element in the explanation here is the fact that questions only take precedence over questions on which they do not depend:

- (22) a. John: (1a) Okay which one do you think it is?  
 (1b) Maybe F1 F1.  
 Sarah: (2) Shift and F1?  
 Sue: (3a) No. (3b) It's F7.  
 John: (4a) No, (4b) just F1 F1.

| Utt. | DGB Update<br>(Conditions)                                                                      | Rule                                           |
|------|-------------------------------------------------------------------------------------------------|------------------------------------------------|
| 1a   | LatestMove := Ask(J, {Sa, Su}, q0)<br>J.QUD := ⟨q0⟩                                             | Free Speech<br>Ask QUD-incrementation          |
| 1b   | LatestMove := Assert(J, {Sa, Su}, p1)<br>J.QUD := ⟨p1?, q0⟩<br>About(p1, q0)                    | QSPEC<br>Assert QUD-incrementation             |
| 2    | LatestMove := Ask(Sa, {J, Su}, p2?)<br><br>Sa.QUD := ⟨cq1? > q0⟩<br>Su.QUD := ⟨cq1? > p1?, q0⟩  | Parameter focussing,<br>Ask QUD-incrementation |
| 3a   | LatestMove := Assert(Su, {J, Sa}, p3)<br>QUD := ⟨¬cq1? > cq1? > p1?, q0⟩<br>QUD := ⟨p1?, q0⟩    | QSPEC                                          |
| 3b   | LatestMove := Assert(Su, {J, Sa}, p4)<br>Su.QUD := ⟨p4?, q0 > p1?⟩                              | Fact update/QUD-downdate<br>QSPEC              |
| 4a   | LatestMove := Assert(J, {Sa, Su}, ¬p4)<br>J.QUD := ⟨¬p4?, q0, > p4? > p1?⟩                      | QSPEC                                          |
| 4b   | LatestMove := Assert(J, {Sa, Su}, p5)<br>J.QUD := ⟨p5?, q0 > ¬p4? > p4? > p1?⟩<br>About(p5, q0) | QSPEC                                          |

### 8.1.7 Summary

In this section, I considered how to scale up dialogue protocols to multilogue, settings with multiple conversationalists. Two benchmarks, MLDSA and MAG, have been extracted to evaluate scaled-up protocols based on the long-distance resolution possibilities of NSUs in dialogue and multilogue in the BNC. MLDSA, the requirement that multilogue protocols license long-distance short answers, derives from the statistically significant increase in frequency of long-distance short answers in multilogue as opposed to dialogue. MAG, the requirement that multilogue protocols enforce



adjacency of acceptance and grounding interaction, derives from the overwhelming locality of acceptance/grounding interaction in multilogue, as in duologue. In light of these benchmarks, I then consider two possible transformations to duologue protocols formulated within an issue-based approach to dialogue management. Each transformation can be intuited as adding roles that correspond to distinct categories of an audience originally suggested by Goffman. The two transformations would appear to be complementary—it seems reasonable to assume that application of both (in some formulation) will be needed for coverage of multilogue in a variety of genres. MLDSA and MAG can be fulfilled within an approach that combines the Add Side Participants transformation on protocols with an independently motivated modification of the structure of QUD from a canonical stack to a stack where maximality is conditioned by issue dependence. As we have seen, the conversational rules posited in earlier chapters scale up quite straightforwardly to multilogue.

With respect to long-distance short answers the account provided here licenses their occurrence in duologue, as in multilogue. I offer a pragmatic account for their low frequency in duologue, which indeed generalizes to explain a statistically significant correlation between their increased incidence and increasing active participant size.

## 8.2 Unifying Self- and Other-Correction

The ever-acerbic physicist Wolfgang Pauli wrote to Einstein in 1929: “All that is left ... is to congratulate you (or had I better say ‘express ... condolences?’) on your having gone over to the pure mathematicians.” (Lee Smolin, ‘The Other Einstein’, *New York Review of Books*, 14 June 2007)

### 8.2.1 Introduction

In Chapter 6 I discussed the CA notion of *repair*, a cover term for recovery from a wide variety of phenomena, ranging from misarticulation via mishearing to incorrect understandings by recipients. Nonetheless, in that chapter I restricted attention to cross-utterance CRs, which in CA terms are a subclass of *other-initiated repair*. In their seminal paper Schegloff, Jefferson, and Sacks (1977) emphasized that, while clearly distinct, there are strong similarities between self-repair and other-initiated repair, which make an approach that captures these similarities desirable.

In contrast to this long-existing, but unformalized view, standard in CA, most recent computational work on detecting and repairing disfluent speech seems to take a view of self-repair as a *pre-semantic* ‘editing operation’, which need not be interleaved with higher-level contextual processes. Such a view is expressed e.g. in the following quote from Heeman and Allen (1999): ‘we propose that these tasks [among others detecting and correcting speech repairs—J.G.] can be done using local context and early in the processing stream.’

---

|                      |                   |              |               |  |                           |              |
|----------------------|-------------------|--------------|---------------|--|---------------------------|--------------|
| <i>until you're </i> | <i>at the le-</i> |              | <i>I mean</i> |  | <i>at the right-hand </i> | <i>edge</i>  |
| start                | reparandum        | moment of    | editing terms |  | alteration                | continuation |
|                      |                   | interruption |               |  |                           |              |

---

FIGURE 8.1: General pattern of self-repair

In this section I will present an approach, developed in an ongoing collaboration with Raquel Fernández and David Schlangen,<sup>6</sup> in which mid-utterance self-repair is treated as akin to but distinct from cross-utterance CRification—the basic commonality being the existence of a problem containing utterance which gets fixed via a follow-up utterance, leaving the problematic material with a special status, but within the discourse context. This approach ties in with recent psycholinguistic work, summarized below, which has shown that self-corrected material has a processing effect.

Before that, it's worth briefly mentioning some basic concepts relating to self-repair. As has often been noted (see e.g. Levelt (1983), and references therein for earlier work), speech disfluencies follow a fairly predictable pattern. The example in Figure 8.1 is annotated with the labels introduced by Shriberg (1994), building on Levelt (1983), for the different elements that can occur in a self-repair.

Of these elements, all but the moment of interruption and the continuation are optional. The relation between reparandum and alteration can be used as the basis of a further classification:<sup>7</sup> if the alteration differs strongly from the reparandum and does not form a coherent unit together with the start, or if alteration and continuation are not present at all, the disfluency can be classified as an *aborted utterance* / *fresh start*. Other classes are *repair* (alteration 'replaces' reparandum) and *reformulation* (alteration elaborates on reparandum). The following gives examples for all three classes:<sup>8</sup>

- (23) a. { I mean } [ I, + I, ] -/ [ there are a lot, + there are so many ] different songs,  
 b. [ We were + I was ] lucky too that I only have one brother.  
 c. at that point, [ it, + the warehouse ] was over across the road

Ginzburg, Fernández, and Schlangen (2011) propose an additional distinction among disfluencies such as the above, which, with a nod to the similarly named distinction

<sup>6</sup> Initially published as Ginzburg, Fernández, and Schlangen (2007), subsequently expanded and revised as Ginzburg, Fernández, and Schlangen (2011). This section is based, almost entirely, on these two papers, with non-insubstantial modifications. I thank Raquel Fernández and David Schlangen for their permission to use this material, though any errors here are entirely due to me.

<sup>7</sup> This classification is based on McKelvie (1998), Heeman and Allen (1999).

<sup>8</sup> The examples in this section are all taken from the Switchboard corpus (Godfrey, Holliman and McDaniel 1992), with disfluencies annotated according to conventions established in that corpus: '+' marks the moment of interruption and separates reparandum from alteration, '{}' brackets editing terms and filled pauses.

in the DAMSL annotation scheme (Core and Allen 1997), they label *backward-looking* disfluencies—the moment of interruption is followed by an alteration that refers back to an already uttered reparandum. These can be distinguished from types of those disfluencies where the moment of interruption is followed not by an alteration, but just by a completion of the utterance which is delayed by a filled or unfilled pause (hesitation) or a repetition of a previously uttered part of the utterance (repetition). This kind of disfluency can be dubbed *forward-looking*.<sup>9</sup> In what follows the account I develop will focus exclusively on *backward-looking* disfluencies; a treatment of *forward-looking* disfluencies can be found in Ginzburg, Fernández, and Schlangen (2011).

### 8.2.2 CRification and Self-Repair

Consider (24), where the (a, c) examples are CRs and the (b, d) examples are mid-utterance self-repairs:

- (24) a. A: Did Bo leave? B: Bo? A: Your cousin.  
 b. A: [ Did Bo + {I mean} my cousin ] leave?  
 c. A: Did Bo B: leave? A: yeah.  
 d. [ A: Did Bo +{I um} leave ]?

The analogy between the CRs and the self-repairs should be clear: a problem with some of the material is detected and signalled (= there is a ‘moment of interruption’); the problem is addressed and repaired.

On the analysis of CRs developed in Chapter 6, CRification involves a query-answer sequence no different from that involved in ‘regular’ illocutionary interaction. Like the latter, this interaction gets recorded in the DGB and remains there for a certain period. What of self-repair? Is there any justification for thinking that this process gets recorded in the DGB? One argument to this effect is that disfluencies can also bring about linguistic effects in whose interpretation they play a significant role, for instance anaphora, as in (25a) from Heeman and Allen (1999). In fact, disfluencies yield information: (25a) entails (25b) and defeasibly (25c), which in certain settings (e.g. legal), given sufficient data, can be useful.

- (25) a. Andy: [Peter was + {well} he was] fired.  
 b. Andy was unsure about what he should say, after uttering ‘was’.  
 c. Andy was unsure about how to describe what happened to Peter.

Indeed there is increasing psycholinguistic evidence to indicate that self-corrected material has a processing effect. Brennan and Schober (2001) demonstrate that in a situation with two possible referents, the fact that a description was self-corrected enabled listeners to draw the conclusion that the respective other referent was the

<sup>9</sup> Levelt (1983) refers to such disfluencies as ‘covert repair’.

correct one, before the correction was fully executed. Lau and Ferreira (2005) show that garden-path sentences which included verb replacement were more likely to be judged grammatical if the reparandum verb was structurally unambiguous. Conversely, ambiguous non-garden-path sentences were more likely to be judged ungrammatical if the structurally unambiguous disfluency verb was inconsistent with the final reading.

The structural similarities between CRs and SC established, it is important to note, in line with the observations of Schegloff, Jefferson, and Sacks (1977) and our earlier discussion of the TTP, that the range of utterances that occur within utterance by a single speaker are distinct though not disjoint from those that occur by a distinct speaker at a transition relevance point, as illustrated in (26) and (27):

- (26) A: Who likes Bo? Bo? (= Does Bo like Bo? or Did I say/mean 'Bo?')  
 A: Who likes Bo?  
 B: Bo? (= Does Bo like Bo? or Who do you mean 'Bo'? or Are you asking who likes BO?)
- (27) a. Jo ... {wait/no/hang on/...} Jack is arriving tomorrow (= I meant to say Jack, not Jo is arriving tomorrow)  
 b. Jo ... {wait/no/hang on/...} yeah Jo is arriving tomorrow (= I did mean to say Jo is arriving tomorrow)  
 c. Jo... {wait/no/hang on/...} Jo? (= Did I say/mean 'Jo?')... {wait/no/hang on/...} Jack is arriving tomorrow (= I meant to say Jack, not Jo is arriving tomorrow)

The task, then, is to show how KoS can capture the similarities exhibited by self-initiated within-utterance repair and other-initiated cross-utterance repair, while ensuring that their contextual effects remain differentiated.

### 8.2.3 *From CRs to Self-Initiated Mid-Utterance Repair*

As things stand, there are two things that prevent KoS from accounting for self-initiated mid-utterance repair: (1) all CR/corrections are forced to occur after complete utterances, and (2) CR/corrections can only be posed by others. This, provided that the speaker downdates PENDING in consequence of the move update rules (see section 6.5.2). Let us take up each of these issues in turn.

The first move we can make is to extend PENDING to incorporate utterances that are *in progress*, and hence, incompletely specified semantically and phonologically. Conceptually this is a natural step to make. Formally and methodologically this is a major step, as it presupposes the existence of a system of types that are applicable to utterance events in a more or less word-by-word fashion (rather than top/down or bottom/up, as is the case for the grammar introduced in Chapter 5).<sup>10</sup> One strategy

<sup>10</sup> This raises of course various issues: is this strategy used routinely, invariably, under pressure. For some empirical evidence on this see Lau and Ferreira (2005).

in this respect derives from Categorical Grammar (Steedman 1996, Morrill 2000), Dependency Grammar (Milward 1994), and Dynamic Syntax (Kempson, Meyer-Viol, and Gabbay 2000), where the grammar is explicitly designed to accommodate such incrementality.

TTR provides resources which could enable types appropriate for the task at hand to be defined on the basis of the grammar described in Chapters 5–7, though beyond some simple examples I will not attempt to prove this conjecture in the current work.<sup>11</sup> For current purposes, the decisions we need to make can be stated independently of the specific grammatical formalism used. The main assumptions we are forced to make concern where pending instantiation and contextual instantiation occurs, and more generally, the testing of the fit between the speech events and the types assigned to them. We assume that this takes place incrementally, say word by word.<sup>12</sup>

The incrementalization of PENDING has two good consequences: first, it allows me to follow in the pioneering footsteps of work in the PTT framework (e.g. Poesio and Traum 1997), one of whose original contributions is an account of acknowledgements to incomplete utterances. A word like ‘mmh’ is commonly used to acknowledge a partial sub-utterance, as in (28):

- (28) A: Move the train ...  
 B: Aha  
 A: ... from Avon ...  
 B: Right  
 A: ... to Danville. (Adapted from the Trains corpus)

A minor modification of our treatment from Chapter 6 (see (25), there) will allow such uses to be accommodated. In (29), ‘mmh’ is posited to enable A to acknowledge B’s most recently ungrounded utterance, regardless of whether it is ‘complete’ (in which case its content would be an IllocProp) or not:

- (29) 
$$\left[ \begin{array}{l} \text{phon : mmh} \\ \text{cat = interjection : syncat} \\ \\ \text{dgb-params : } \left[ \begin{array}{l} \text{spkr : IND} \\ \text{addr : IND} \\ \text{MaxPending : LocProp} \\ \text{rest}^{facts} 2 : \text{address(addr, spkr, MaxPending)} \end{array} \right] \\ \text{cont = [c1 : Understand(spkr, addr, MaxPending) ] : RecType} \end{array} \right]$$

<sup>11</sup> How Dynamic Syntax could be integrated with TTR is explored in Purver et al. (2010).

<sup>12</sup> This is of course a simplification over what happens in real-time processing, as Susan Brennan (p.c.) has reminded me.

Second, and relatedly, we can now account also for CRs/other corrections that occur mid-utterance, dispreferred as Schegloff, Jefferson, and Sacks (1977) suggest that they might be. One such corpus example is (30a). The constructed (30b) shows that in such contexts the same ambiguities are maintained as in cross-utterance cases exemplified above:

- (30) a. A: There are subsistence farmers that ...  
       B: There are what? (example from the Potsdam Pentomino Corpus)  
       b. A: Did Bo... (no pause) B: Bo? (= Who do you mean 'Bo'? or Are you asking something about BO?) A: I mean Mo/Yeah, Mo's partner.

Our CCURs hitherto are explicitly specified to enforce turn change: the speaker of the CR is specified as the addressee of the CRified utterance. How then to explicate the potential for mid-utterance self-correction? I assume that backward-looking disfluencies are possible essentially at any point where there is 'correctable material'. Technically this amounts to PENDING not being empty. I assume that editing phrases are, at least in many cases, contentful constituents of the repair.<sup>13</sup> Given that  $u0$  is a constituent of A's utterance in MaxPending, it is possible for A to accommodate as MaxQUD the issue 'what did A mean by  $u0$ ' and provide an utterance which, as usual in our CCURs, is constrained to be co-propositional with MaxQUD.

(31) Backwards-looking appropriateness repair = *def*

$$\left[ \begin{array}{l} \text{pre} \\ \text{effects} \end{array} : \left[ \begin{array}{l} \text{spkr} : \text{IND} \\ \text{addr} : \text{IND} \\ \text{pending} = \langle p0, \text{rest} \rangle : \text{list}(\text{LocProp}) \\ \text{cl} : \text{member}(u0, p0.\text{sit.constits}) \\ \text{spkr} = \text{pre.spkr} : \text{IND} \\ \text{addr} = \text{pre.addr} : \text{IND} \\ \text{MaxQUD} = \\ \left[ \begin{array}{l} q = \lambda x \text{ Mean}(\text{pre.spkr}, \text{pre}.u0, x) \\ \text{fec} = u0 \end{array} \right] : \text{InfoStruc} \\ \text{LatestMove} : \text{LocProp} \\ \text{c2} : \text{Co-propositional}(\text{LatestMove}^{\text{content}}, \text{MaxQUD}) \end{array} \right] \right]$$

With this machinery in hand, I consider two examples of self-correction. The first, (32), involves self-correction *mid-utterance*. The basic analysis is this: A utters

<sup>13</sup> For *forward-looking* disfluencies an editing phrase is mandatory and, following Clark and FoxTree (2002) and Horne (2008), it is a *conventional* signal for the imminent arrival of a soon-to-be-uttered word.

‘Peter’. This makes PENDING non-empty, which allows for a potential use of (31). A accommodates ‘What did you mean by uttering ‘Peter’?’ as MaxQUD and utters ‘I mean Paul’, which is co-propositional with this question and, thereby licensed as a follow-up. Then by the normal DGB update operations this answer is integrated and updates the utterance content.

(32) A: Peter, I mean Paul quit.

In more detail: after the utterance of ‘Peter’, A’s FACTS will include the presuppositions that the most recent speech event is  $u0$  (‘Peter’), classified by a type  $T_{u0}$  (‘Utterance whose first word is Peter; involves reference to  $p \dots$ ’); the DGB is essentially the following:<sup>14</sup>

$$\begin{aligned}
 (33) \quad a. \quad A.dgb1 = & \left[ \begin{array}{l} \text{spkr} = A \\ \text{addr} = B \\ \text{pending} = p0 = \left\langle \left[ \begin{array}{l} \text{sit} = u0 \\ \text{sit-type} = T_{\text{Peter} \dots} \end{array} \right] \right\rangle \\ \text{qud} = \langle \rangle \\ \text{facts} = \left\{ \begin{array}{l} \text{Named}(\text{'Peter'}, \text{'peter'}), \\ \text{MostRecentSpeechEvent}(u0), \\ \text{Classify}(T_{\text{Peter} \dots}, u0) \end{array} \right\} \\ \text{moves} = \langle \rangle \end{array} \right] \\
 b. \quad T_{\text{Peter} \dots} = & \left[ \begin{array}{l} \text{phon} : \text{peter} \\ \text{cat} = N : \text{syncat} \\ \text{constits} = \{\text{Peter}\} : \text{set}(\text{sign}) \\ \text{dgb-params} : \left[ \begin{array}{l} \text{spkr} : \text{IND} \\ \text{addr} : \text{IND} \\ \text{c1} : \text{address}(s, a) \\ \text{p} : \text{IND} \\ \text{c3} : \text{Named}(p, \text{'peter'}) \end{array} \right] \\ \text{cont} = p \end{array} \right]
 \end{aligned}$$

<sup>14</sup>  $T_{\text{Peter} \dots}$  could be further ‘incrementalized’ by, for instance, having its content as a sit-type with an existentially quantified predicate emanating from an existentially quantified constituent. But such decisions are not easy to make without further empirical evidence concerning speakers’ intuitions about incremental content. This, given our wish to remain faithful to the Reprise Content Hypothesis.

This allows for Backwards-looking appropriateness repair to be used—the issue ‘What did A mean by *u0*’ becomes MaxQUD which licenses as LatestMove ‘I mean Paul’:

$$(34) \quad A.dgb2 = \left[ \begin{array}{l} \text{spkr} = A \\ \text{addr} = B \\ \text{pending} = \left\langle \left[ \begin{array}{l} \text{sit} = u0 \\ \text{sit-type} = T_{\text{Peter}...} \end{array} \right] \right\rangle \\ \text{qud} = \langle ?\text{Mean}(A, u0, \text{paul'}), \lambda x \text{Mean}(A, u0, x) \rangle \\ \text{facts} = \left\{ \begin{array}{l} \text{Named}(\text{'Peter'}, \text{'peter'}), \text{Named}(\text{'Paul'}, \text{'paul'}), \\ 2\text{ndMostRecentSpeechEvent}(u0), \\ \text{Classify}(T_{\text{Peter}...}, u0) \\ \text{MostRecentSpeechEvent}(u1), \\ \text{Classify}(T_{\text{I mean paul}}, u1) \end{array} \right\} \\ \text{moves} = \langle \text{Assert}(A, \text{Mean}(A, u0, \text{'paul'}) \rangle \end{array} \right]$$

Accepting this gives rise to an application of pending extension, which modifies the original locutionary proposition: *u0* is modified to a record *v0* with the referent *paul* replacing *peter* and the utterance type is now  $T_{\text{Paul}...}$  whose phon includes the form Paul; MAXPENDING is modified accordingly:

$$(35) \quad A.dgb3 = \left[ \begin{array}{l} \text{spkr} = A \\ \text{addr} = B \\ \text{pending} = \left\langle \left[ \begin{array}{l} \text{sit} = v0 \\ \text{sit-type} = T_{\text{Paul}...} \end{array} \right] \right\rangle \\ \text{qud} = \langle \rangle \\ \text{facts} = \left\{ \begin{array}{l} \text{Named}(\text{'Peter'}, \text{'peter'}), \\ 2\text{ndMostRecentSpeechEvent}(u0), \\ \text{Classify}(T_{\text{Peter}...}, u0) \\ \text{MostRecentSpeechEvent}(u1), \\ \text{Classify}(T_{\text{I mean Paul}}, u1) \text{Named}(\text{'Paul'}, \text{'paul'}), \end{array} \right\} \\ \text{moves} = \langle \text{Assert}(A, \text{Mean}(A, u0, \text{paul})) \rangle \end{array} \right]$$



As can be readily observed, the utterance *u0* is still a component of facts in FACTS, though not a component of PENDING. Hence, referential possibilities to this utterance—and to the referent of ‘Peter’—are not eliminated.

The second example I consider briefly is (36)—it involves self-correction *post-utterance*. The analysis proposed for (32) applies here, with the main difference that the starting point of the sequence of updates leading to correction is initiated with the *completed utterance* as MaxPending. ‘Not Peter’ is an initial utterance co-propositional with the issue ‘Who did A mean by uttering *u0*’, and by the modified ordering on QUD (13), this latter question remains MaxQUD after the utterance ‘Not Peter’ allowing for a further utterance about this issue.

(36) A: Peter quit. Not Peter, I meant Paul.

### 8.2.4 Summary

In this section we have seen how self- and other-initiated repair can be related. The basic claim made here, following a long but unformalized tradition in Conversation Analysis, is that the two processes bear significant similarities: a problem is detected with an utterance, this is signalled, and then the problem is addressed and repaired, leaving the incriminated material with a special status, but within the discourse context. A unified account is provided: a single repository, PENDING carries CR/correctable material within and across utterances. Consequently, a single set of rules regulate the up- and downdating of PENDING, as well as the modification of its elements by answers to CRs or corrections, regardless of whether the utterances are in progress or completed. Different rules trigger within- and cross-utterance CRs/corrections, but that is as should be, as the form and content of these differ, as we have seen.

## 8.3 A Deeper View of Grounding and CRification

The issue for N.B.A. officials is how to interpret Thomas’ words, and his intent. Was Thomas merely advising a young star that he should not embarrass the Knicks by running up the score? Or was he issuing a warning? (*The New York Times*, 18 December 2006)

Alan Pardew has launched a thinly veiled attack on the divisive atmosphere undermining his former club West Ham and believes Charlton’s unified spirit will prove crucial in securing their Premiership safety. Charlton’s manager had steadfastly refused to discuss issues concerning Upton Park since leaving in December but,

although he failed to mention West Ham by name yesterday, his intentions were clear when he noted the improvement in Charlton's recent performances. (*The Guardian*, 10 February 2007)

The view of grounding and CRification I developed in Chapter 6 was oriented to utterances whose content is pretty directly *grammatically governed*, though of course by a grammar that is strongly driven by context. In so doing there are two aspects of meaning that have not figured, ones commonly referred to as *conversational implicature* and *indirect speech acts*. I will consider only restricted aspects of each, giving a sense of how to scale up the account within KoS to tackle these phenomena. The interest in considering the diverse class of moves that, following Searle, came to be known as *indirect speech acts* (ISAs) relates primarily to the issue of the extent of *non-public intention recognition* that one needs to add to our previous picture. Although there is a considerable literature on these (for an extensive review and detailed recent proposal see Asher and Lascarides 2001), particularly under the rubric of recognizing the *plan* underlying an utterance, there has been very little attempt to consider the frequency of cases where such recognition fails and some sort of repairing interaction takes place, let alone the mechanisms involved. I will sketch an account of several classes of ISAs, relating it to the basic issue of whether utterance plan recognition is a routine feature of utterance grounding. I will suggest that it need not be, though a somewhat easier task is—*genre recognition*. Related to this is the issue of the coherence of a very common metacommunicative device—the question ‘Why?’, which can be used to ask about non-public intentions. I will argue that this use of ‘Why?’ differs in a number of significant ways from CRs, subsuming instead to a class of explanation-like follow-up utterances.

### 8.3.1 Gricean Reasoning and Grounding

My focus here will be on implicatures that revolve around the Gricean maxim of relevance. A prototypical example in this respect is given in (37a). Two further examples from literary texts convey a similar import:

- (37) a. A: Horrible talk by Rozzo. B: It's very hot in here. **Implicates:** B does not wish to discuss A's utterance.
- b. Rumpole: Do you think Prof Clayton killed your husband? Mercy Charles: Do you think you'll get him off? (John Mortimer 'Rumpole and the Right to Silence', in *Rumpole à la Carte*, London: Viking (1990), 100)
- c. Harry: Is that you James? Stella: What? No, it isn't. Who is it? Harry: Where's James? Stella: He's out. Harry: Out? Oh, well, all right. I'll be straight round. Stella: What are you talking about? Who are you? (Harold Pinter 'The Collection', in: *Plays Two* (London: Faber & Faber, 1996), 133)

In current terms we could formulate the inference informally in (38a). Slightly more precisely, replacing current context with DGB and ‘previous utterance’ with DGB’s LatestMove,<sup>15</sup> we obtain (38b):

- (38) a. If an utterance does not cohere with the current context, the speaker conveys that she does not want to discuss the previous utterance.  
 b. **Relevance:**  $\text{IrRelevant}(u^{\text{content}}, \text{dgb}) \mapsto$   
 A does not wish to address  $\text{dgb.LatestMove}$ .

Questions that arise are: What does *IrRelevance* amount to? What is the status of such an inference? Where do we wish to represent such inferences? To the first of these we can offer a very explicit answer; the remaining questions are less clear-cut and we return to them shortly. Our earlier definition of U-Coherence in Chapter 6 offered us a notion of relevance and exemplified it by the case where the input context is a query. Identifying *Relevance* with U-Coherence, then, means that an utterance  $u$  is *relevant* to an information state  $I$  just in case there is a way to successfully update  $I$  with  $u$ . So *Irrelevance* means that there is *no* way to successfully update  $I$  with  $u$ . Given our earlier example, relative to a conversationalist C, being irrelevant in reacting to a query means *not* being able to find a sequence of updates such that the locutionary proposition  $p_u = \text{prop}(u, T_u)$  becomes the value of LatestMove (condition 39b); and the final element of the sequence of updates  $I_{k+1}$  is such that one of the conditions in (39c–e) is satisfied— $u$  is neither  $q$ -specific, nor an appropriate CR, nor is it genre-relevant:

- (39) a.  $I_0.\text{DGB.LatestMove} = v; v.\text{content} = \text{Ask}(A, q)$ ,  
 b.  $I_{k+1}.\text{DGB.LatestMove} = p_u$   
 c.  $p_u.\text{content}$  is  $q$ -specific relative to  $I.\text{DGB}$ , Or  
 d.  $p_u.\text{content}$  is Co-propositional with some question  $q_0$  that satisfies  $q_0 = \text{CCUR1.effects.maxqud}(I_0.\text{DGB.MaxPending})$  for some CCUR CCUR1, Or  
 e. One of C’s beliefs in  $I_0$  is that: for some  $G_0$  there exists  $\text{dgb1}$  such that  $(I_0.\text{DGB} \oplus p_u) \sqsubset \text{dgb1}$ , and such that  $\text{dgb1} : G_0$

Now it is worth emphasizing that Gricean reasoning has, with the significant exception of Relevance Theory (Sperber and Wilson 1986), been tied to cooperative interaction—for Grice it was important to demonstrate that conversational implicatures are a consequence of reasoning performed by rational agents on the communicative process, and there have been derivations of various implicatures in these

<sup>15</sup> LatestMove is more precise than ‘previous utterance’, in view of data such as (i):

(i) A: Is Bill coming? B: Bill? A: Your cousin. B: mmh. It is very late you know.

terms.<sup>16</sup> KoS's perspective on interaction is more general, in not being restricted by a cooperativity assumption. Moreover, Gricean accounts, and Relevance-Theoretic ones, by and large, ignore the possibility of interaction to address the possibly sub-optimal aspect of the utterance, a possibility which cancels the generation of the requisite implicatures. In (40a) the fact that A poses a CR indicates that she resists drawing an irrelevance implicature from B's utterance and indeed B clears up the confusion. This extends to other types of implicatures as well—(40b, c) exemplify potential quantity implicatures ('B does not know where in the South of France Hugo lives', 'Emma does not know how long the affair has been going on') that are not drawn:

- (40) a. A: Horrible talk by Rozzo. B: It's very hot in here. A: What do you mean?  
           B: Hard to sit still, let alone discuss Bourdieu.  
       b. A: Where is Hugo living these days? B: Somewhere in the south of France.  
           A: Yeah but where?  
       c. Robert: It's quite well established then, your . . . uh . . . affair? Emma: Yes.  
           Robert: How long? Emma: Some time. Robert: But how long exactly?  
           Emma: Five years.

Now the potential for CRs concerning the relevance of an utterance is already, with one potentially significant caveat, accommodated by the rule parameter identification from Chapter 6: the latter allows B to query what A meant by *u*, specifically by posing a CR about *u* or any sub-utterance of *u*. The one significant difference between these and relevance CRs is that whereas the (non-repetition) CRs discussed previously revolved around partially instantiated DGB-PARAMS, here the trigger is typically the irrelevance of a fully instantiated utterance. In other words, failure here occurs when there is an attempt to integrate a fully instantiated utterance in the DGB. The answer to such a CR will not in general be represented in the DGB, in contrast to other CRs where it could be found in DGB-PARAMS or PHON of the responder. This means that, arguably, we need to offer an alternative definition for the Mean predicate to the one we offered in (68), which will be applicable to speaker meaning associated with entire utterances. What we would need would be a definition that makes reference to the private part of the information state of the utterance's speaker, discussed in Chapter 4—specifically, identifying the speaker meaning with the maximal element of the agenda of the utterance's speaker. We have a way of singling out the entire utterance from its parts, since only the former has as its content an illocutionary proposition. In light of this, we can offer (41) as a definition of the value of the function Mean for the entire utterance:

<sup>16</sup> For a particularly explicit account, which interfaces with semantic input, see Asher and Lascarides (2003).

- (41) Given  $u^{content} : \text{IllocProp}$ ,  $\text{Mean}(A, u, c)$  iff  $u.\text{dgb-param.spkr} = A$  and  $A.\text{private.maxagenda} = c$

We could restore uniformity by associating a contextual parameter with the entire utterance, one which by default might be identified with the ‘semantic content’, but which could in principle be distinct.<sup>17</sup> Whether this would blur the distinction between sentential speaker meaning, questions about which are often difficult to answer, and e.g. NP speaker meaning, which is usually far easier to answer (‘Bo? yeah, my cousin’), is not easy to resolve without significantly more empirical investigation.

Irrelevance *implicatures* are an instance of metadiscursive interaction—interaction about what should or should not be discussed at a given point in a conversation:

- (42) a. I don’t know.  
 b. Do we need to talk about this now?  
 c. I don’t wish to discuss this now.  
 d. Whatever. Millie called yesterday.

A natural way to analyse such utterances is along the lines of QSPEC discussed in Chapter 4:  $q$  being MaxQUD gives B the right to follow up with an utterance specific to the issue which we could paraphrase informally as  $?WishDiscuss(q)$ . Such a rule is sketched in (43), where the notation ‘ $\text{qud} = \langle \text{max} = \{?WishDiscuss(q0), q0\}, Q \rangle$ ’ indicates that both  $?WishDiscuss(q0)$  and  $q0$  are maximal in QUD, and unordered with respect to each other:

- (43) Discussing  $u? =_{def}$

$$\left[ \begin{array}{ll} \text{pre} & : \left[ \text{qud} = \langle q0, Q \rangle : \text{poset}(\text{Question}) \right] \\ \text{effects} & : \left[ \begin{array}{l} \text{spkr} = \text{pre.addr} : \text{Ind} \\ \text{addr} = \text{pre.spkr} : \text{Ind} \\ r : \text{AbSemObj} \\ R : \text{IllocRel} \\ \text{Moves} = \langle R(\text{spkr}, \text{addr}, r) \rangle \oplus \text{pre.Moves} : \text{list}(\text{IllocProp}) \\ \text{cl} : \text{Qspecific}(R(\text{spkr}, \text{addr}, r), ?WishDiscuss(\text{pre.maxqud})) \\ \text{qud} = \langle \text{max} = \{?WishDiscuss(q0), q0\}, Q \rangle : \text{poset}(\text{Question}) \end{array} \right] \end{array} \right]$$

<sup>17</sup> This could be implemented, for instance, as a contextual parameter which conjoins on to content at the root clause level:

- (i)  $\left[ \begin{array}{l} \text{dgb-params} : \left[ p : \text{IllocProp} \right] \\ \text{content} = \text{hd-dtr.cont} \wedge \text{dgb-param.p} : \text{IllocProp} \end{array} \right]$

When the two are identical, then the contextual parameter has no effect on the sentential content.

Retaining the original MaxQUD,  $q_0$ , in QUD, neither subsumed nor subsuming ?WishDiscuss( $q_0$ ) is surprising at first. One might think that a consequence of a responder's failure to accept  $q$  for discussion is that  $q$  will only resurface if explicitly reposed. There is evidence, however, that actually  $q$  remains in a conversational participant's QUD even when not initially adopted, its very posing makes it temporarily DGB-available:

(44) A: Who are you meeting next week?

B(2): No comment.

A: Why?

B: Personal reasons.

A: I see.

B(6): Oh, OK, Jill.

Here the original question has definitely *not* been reposed and yet B still has the option to address it, which he should be unable to do if it is not added to his gameboard before (44(2)). DOWNDATING a question in case B does not wish to discuss it can be captured by a minor enhancement to the fact-update/qud-downdate rule from Chapter 4.<sup>18</sup>

Let us consider how we can analyse two variants of (44), given in (45a, b): in both cases initially A poses  $q_1$ ; B does not wish to discuss  $q_1$ , hence he accommodates ?WishDiscuss( $q_1$ ) into QUD and offers an utterance concerning this issue. A accepts B's assertion, so using the new version of fact-update/qud-downdate  $q_1$  could be downdated and either conversationalist could introduce a new topic, as in (45a). We can also consider the alternative conversation in which B changes tack, as in (45b), and at utterance (4') decides to address  $q_1$  after all. What is crucial is that A's acceptance of B's 'No comment' does not lead to ?WishDiscuss( $q_1$ ) getting negatively resolved—which would trigger  $q_1$ 's downdate. There are a number of mechanisms for achieving this, either by appealing to some fast self-correction mechanism along the lines discussed in section 8.2, signalled by B's concessive 'Oh, OK', or alternatively—what is ultimately not very different—assuming that A's acceptance leads to downdating  $p_1$ ?, but that B's change of tack means that ?WishDiscuss( $q_1$ ) is not negatively resolved. This being the case since the inference from  $p_1$  being accepted to ?WishDiscuss( $q_1$ ) being negatively resolved is fixed with reference to B's agenda and his changing tack means that that has now changed. Whichever path we adopt, if we block ?WishDiscuss( $q_1$ ) getting negatively resolved, then at this point both  $q_1$  and ?WishDiscuss( $q_1$ ) are MaxQUD and B can address  $q_1$ , using a short answer, whose resolution crucially depends on  $q_1$  being MaxQUD.

(45) a. A(1): Who are you meeting next week?

B(2): No comment.

<sup>18</sup> This modification, implemented in Appendix B on p. 370 simply involves a modification of the function NonResolve which fixes the value of QUD after the fact update to take into account questions  $q$  for whom ?WishDiscuss( $q$ ) is negatively resolved.

- A(3): I see.  
B(4): What are you doing tomorrow?  
b. A(1): Who are you meeting next week?  
B(2): No comment.  
A(3'): I see.  
B(4'): Oh, OK, Jill.

| Utt.    | DGB Update<br>(Conditions)                                                                                                                                                                | Rule                                       |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| initial | MOVES = $\langle \rangle$<br>QUD = $\langle \rangle$<br>FACTS = $\text{cg1}$                                                                                                              |                                            |
| 1       | LatestMove := Ask(A, B, $q1$ )<br>QUD := $\langle q1 \rangle$                                                                                                                             | Ask QUD-incrementation<br>Discussing $u$ ? |
| c. 2    | LatestMove := $\langle \text{Assert}(B, A, p1) \rangle$<br>QUD := $\langle p1? > ?\text{WishDiscuss}(q1), q1 \rangle$                                                                     | Assertion QUD-incrementation               |
| 3       | LatestMove := $\langle \text{Assert}(B, A, p1) \rangle$<br>QUD := $\langle \rangle$                                                                                                       | Accept<br>Fact update/QUD downdate         |
| 3'      | FACTS := $\text{cg1} \cup \{p1\}$<br>LatestMove := $\langle \text{Assert}(B, A, p1) \rangle$<br>QUD := $\langle ?\text{WishDiscuss}(q1), q1 \rangle$<br>FACTS := $\text{cg1} \cup \{p1\}$ | Accept<br>Fact update/QUD downdate         |

Formula (43) is, then, is a way of underwriting the possibility of making an utterance that *directly* addresses B’s willingness to discuss (the issue raised by)  $u$ . However, this issue can also be addressed indirectly—I return to the classical Gricean irrelevance case, (37), repeated here as (46):

(46)    A: Horrible talk by Rozzo. B: It’s very hot in here.

A dialogical reformulation of the classical Gricean account is the following: A makes the initial utterance  $u0$ , which B does not wish to address—maybe Rozzo is hovering in the background or one of his friends. On the level of the DGB, B pretends that  $u0$  did not take place—a behavioural strategy available for various events, not necessarily linguistic ones (e.g. A burped or spat). In order to avoid giving the impression that she is grounding  $u0$ , B now produces an utterance  $u1$  which is irrelevant to  $u0$ . How does A interpret  $u1$ ? He has a number of options: the most obvious option is to pose a CR. The more complex option is to assume that  $u0$  has been ignored and then reflect on the reason for that: B might have not been paying attention, in which case repetition of  $u0$  is on the cards. The evidence might not support this interpretation, in which case, A can reason that  $u0$  has been ignored for a reason and use the modified Fact update/QUD-downdate to downdate the issue relating to his assertion. What I am not formalizing here, though it’s not particularly difficult to do so, are the two not specifically linguistic mechanisms—ignoring an event and the effect of such an act.

What I do offer here is a *short-circuited* account (in the sense discussed e.g. in Sadock 1974) of irrelevant response interaction. This involves positing a conversational rule along the lines of (47)—given that MaxPending is *irrelevant* to the DGB, one can make MaxPending into LatestMove while updating Facts with the fact that the speaker of MaxPending does not wish to discuss MaxQUD:

(47) Irrelevance inference = *def*

$$\left[ \begin{array}{lcl} \text{pre} & : & \left[ \begin{array}{l} \text{dgb} : \text{DGBType} \\ \text{c} : \text{IrRelevant}(\text{maxpending}^{\text{content}}, \text{dgb}) \end{array} \right] \\ \text{effects} & : & \left[ \begin{array}{l} \text{LatestMove} = \text{pre.pending} : \text{LocProp} \\ \text{Facts} = \text{pre.Facts} \cup \\ \{ \neg \text{WishDiscuss}(\text{pre.spkr}, \text{pre.maxqud}) \} \end{array} \right] \end{array} \right]$$

Note that this does not make the *unwillingness to discuss* be the *content* of the offending utterance; it is merely an inference. Still this inference will allow MaxQUD to be downdated, via fact update/question downdate.

Alex Lascarides (p.c.) has pointed (48) out to me, which illustrates the dangers of making the ‘unwillingness to discuss’ the *content* of B’s utterance in (48a).<sup>19</sup> Exchange (48a) is not as felicitous an antecedent to ‘Why?’ as (48b) is:

- (48) a. A: Did you like Rozzo’s talk? B: It’s very hot in here. A: Why?  
 b. A: Did you like Rozzo’s talk? B: I don’t want to answer. A: Why?

Note that in my update-rule-based account B’s *unwillingness to discuss* merely gets introduced to FACTS, whereas LatestMove is the conventional content of the utterance. Indeed, as we shall see in section 8.3.6, where I discuss ‘Why?’, the update-rule-based account is not vulnerable to this datum: being in FACTS is not, as we shall see, a sufficient condition to being an antecedent of Why?—there are lots of facts that get presupposed without being possible antecedents for ‘Why?’ I shall suggest that what makes a fact a possible antecedent for Why? is, essentially, that it concerns some question currently or recently in QUD; in this case, B’s unwillingness to discuss Rozzo’s talk does not satisfy this condition. This contrasts with the situation where B explicitly says ‘I don’t want to answer’, where that issue *is* in QUD.

In a Gricean account developed by Lascarides and Asher (2009), B’s utterance is not rhetorically connected to A’s utterance as far as the logical form of dialogue goes (in current terms—B’s utterance does not get incorporated into the DGB). This triggers inferences in a separate *cognitive logic* (in current terms—these inferences occur only in the private component of the information state). These implicatures, not being a

<sup>19</sup> I would like to thank Alex Lascarides (p.c.) for this example and for much valuable discussion, which made me rethink key aspects of the account sketched here.



part of the logical form of dialogue, are not antecedents to surface anaphors in the dialogue. This strategy is not without its risks, as illustrated in (49), where A and B are, say, a bickering couple. Here the anaphor 'this' seems to resolve to 'Your refusing to discuss our problems' (or some less specific variant on that):

- (49) A: Jason, we're in a bad situation!  
B: Oh gosh it's pretty hot tonight.  
A: This is what annoys me so much about you.

Pinning down the meaning of 'this' is notoriously difficult, but presumably should be done with reference to the DGB.

### 8.3.2 *Indirect Content in Grounding/CRification: Basic Data*

The data in (50) consists of cases where in one way or the other the grammatically governed content does not fully explain the effect achieved by the utterance: in (50a)–a direct answer is given to the query in (50a(3)), however the utterance (50a(6)) illustrates that more was deduced from (50a(3)) than merely a query about degrees of desperation to use the toilet; the infamous (50b) is most naturally understood as a request for salt not a question about B's salt-passing technique; in (50c) B's answer is understood as an affirmative answer to A's query, not as a (trivial) question:

- (50) a. [Context: outside a toilet on an aeroplane]  
Anon woman(1): Is this the queue?  
Me(2): Yes  
Anon woman(3): How desperate are you?  
Me(4): (shrugs)  
(5) (Person leaves toilet cubicle)  
Me(6): Go ahead.  
b. A: Can you (please) pass the salt? B: Sure (passes salt over).  
c. A: Is Millie going to be annoyed? B: Is the Pope Catholic? A: Oh, right.

On the interactive side of things, one can ask: what triggers the asking of queries such as (51(1)), where the question explicitly addresses the underlying motivation of the previous query. (For future reference, we refer to this reading of 'Why?' as *Why?meta* .)

- (51) Cherrilyn (1): You got a pound?  
Jessica (2): Why? (=Why does Cherrilyn ask if Jessica has got a pound?)  
Cherrilyn (3): ch I mean in change  
Jessica (4): no. (BNC, KBL)

Work on computational dialogue in the 1980s and 1990s took as its lead an assumption about plan recognition articulated by Allen and Perrault:

We assume that agents attempt to recognize the plans of other agents and, then, use this plan when deciding what response to make (from the abstract of Allen and Perrault 1980)

Do data such as (50) and (51) justify this assumption? Should all examples above be analysed uniformly somehow, using an ‘intentionalized’ grounding protocol? This could be implemented straightforwardly in the current set-up by adding the speaker’s utterance purpose as a contextual parameter. Instantiating this parameter would be incorporated as a process interleaved with the ‘regular’ contextual instantiation.

**Exercise 8.3.1** Offer a concrete proposal for a *purpose* contextual parameter in *generic-root-cl*.

Such an account is certainly technically viable. The question is whether there is evidence to support this move, which raises the bar of utterance understanding a few notches higher.

There are a number of considerations that seem to militate against this:

- The evidence from corpus studies of CRs is that intention-based CRs are rare, despite the much higher uncertainty potential than with referential CRs. Two recent studies discussed in Chapter 6, Rodriguez and Schlangen (2004) and Rieser and Moore (2005), were explicitly searching for CRs that target the ‘proposal’/‘consideration’ level (in the sense of Allwood and Clark, see section 6.2.2). They did find some examples which, in my view, are convincing instances of such CRs. In these examples, each of the queries can be paraphrased as starting *So you mean ...*:<sup>20</sup>

- (52) a. Constructor: (noise, sil: 48) hm, irgendwie geht das nicht so toll fest. (hum: lachen) (*Hmm, somehow it’s not so tight.*)  
 Instructor: (noise, sil: 1) hält das nicht? (*It’s not sticking?*) (Bielefeld Baufix Corpus, turns 01K022–01I024)
- b. Instructor: dass du ähm das Plättchen mit den drei Löchern wieder abmachen kannst, und einen nach vorne setzen kannst (*That, um, you need to remove the flat one with the three holes again and place one in front*)  
 Constructor: also ich montier jetzt die beiden Würfel erstmal ab. (*So, I first dismount the two cubes?*)

<sup>20</sup> Though in the case of the ATIS dialogues, the CRs might be analysed as part of the specified protocol for telephone-based travel agents.

- (53) a. X: You know what the conference might be downtown Seattle so I may have to call you back on that.  
 PT: Okay. Did you want me to wait for the hotel then?  
 (ATIS corpus)
- b. X: Okay now I don't have to purchase this right now.  
 X: Okay that sounds fine. (\*PAUSE\*)  
 PT: Would you like me to go ahead and book that? (ATIS corpus)

In Rodriguez and Schlangen (2004) the percentage of CRs classified as *int-eval* is 22 per cent, whereas in Rieser and Moore (2005) it is 8 per cent. My own recoding of the cases coded with this tag yielded much lower figures of approx 4 per cent for Rodriguez and Schlangen (2004) and 1.5 per cent for Rieser and Moore (2005). This, in view of the fact that many of the CRs could be understood perfectly well as *intended content* CRs:<sup>21</sup>

- (54) a. Constructor: Ja, aber dies habe ich jetzt an einer Ecke nur festgemacht. (Yes, but this I've just tightened on a corner.)  
 I: wie an einer Ecke? (How so on a corner?)
- b. PT: Okay /um/ Can you go over your credit card with me one more time? It doesn't seem to be taking it for the hotel. X: /MM/ The number?  
 PT: Mm um

Even assuming the higher figures, they are substantially lower than the corresponding figures for referential CRs cited in these studies—50 per cent and 40 per cent, respectively. As discussed in Chapter 6, CRs constitute about 4–6 per cent of all moves, from which we could assume 20 per cent as an upper bound of *int-eval* cases and 4 per cent as an alternative recoded figure. This means we have between 1 and 0.2 per cent of all dialogue moves involving *int-eval* CRs, a very low figure for what is, in principle, a very tricky task. David Schlangen (p.c.) has suggested to me that this dearth can be explained, in part, by the fact that *int-eval* CRs are potentially intrusive, and hence highly dispreferred. This seems like a plausible explanation, though if *int-eval* CRs are continually arising without being explicitly uttered, this would require significant self-repression resources.

- An additional consideration relates to *Why<sub>meta</sub>*, a device which explicitly allows for *int-eval* CRs. In contrast to other CRs, which are overwhelmingly local (distance = 1 in 85 per cent of the CRs in Purver, Ginzburg, and Healey (2001), 95 per cent in Rodriguez and Schlangen (2004); Rieser and Moore (2005), 100 per cent within  $n \leq 4$  in all three studies), examples of long-distance uses of *Why<sub>meta</sub>* can be readily found in the BNC, as illustrated in (55). This belies (a strong interpretation of) the quote above from Allen and Perrault that '[agents] attempt to recognize the plans of other agents and, *then* (italics added—J.G.), use this plan ...'.

<sup>21</sup> I would like to thank David Schlangen and Verena Rieser for allowing me to examine their data and for discussion on these issues.

- (55) a. Norrine: When is the barbecue, the twentieth? (pause) Something of June.  
 Chris: Thirtieth.  
 Norrine: A Sunday.  
 Chris: Sunday.  
 Norrine: Mm.  
 Chris: Why? (= *Why do you ask when the barbecue is*)  
 Norrine: Becau Because I forgot. (pause) That was the day I was thinking of having a proper lunch party but I won't do it if you're going out. (BNC, KBK)
- b. Cherrilyn: Are you still (pause) erm (pause) going to Bristol (pause) on Monday?  
 Fiona: Dunno.  
 Cherrilyn: No?  
 Fiona: I dunno. Doubt it, why? (= *Why do you ask if I'm going to Bristol?*)  
 Cherrilyn: I just wondered.  
 Fiona: Why?  
 Cherrilyn: I just wondered! (BNC, KBL)

The main analogy I wish to put forward is to relate plan recognition, or more precisely *genre* recognition to *language* recognition: in most conversational settings, in many countries it is clear what language the conversation will be in. There is, nonetheless, some varying uncertainty and the possibility of being approached by a foreigner, in which case one can adapt (minimally by recognizing that a non-local language is being spoken). In more multilingual societies the linguistic uncertainty is larger and recognizing the language in which the interaction will be conducted or at least initiated is a routine task, reaching an extreme in places like Catalonia or Hong Kong where the language is liable to change in mid-conversation. By the same token, in many conversational settings one recognizes the genre one is in. But not always, in which case clarification is called for. To be sure, some conversationalists can on occasion be continually monitoring their interlocutors' purposes (just as they might be monitoring them for idiosyncratic performance errors)—e.g. psychoanalysts are (reputedly) trained to do this. Whether we do so continually, despite the potential for this capability, is an open question. But we should prefer accounts that, while recognizing the potential, do not enforce this.

### 8.3.3 ISAs: an Analysis and Some Data

Asher and Lascarides (2003) propose an analysis of indirect speech acts (ISAs) which involves assigning to utterances multiple, conflicting illocutionary forces. These are encoded in dot-product semantic types. These can arise either via conventionalized means, as in (56a, b), or via Gricean reasoning, as in (56c):

- (56) a. Can you pass the salt  $\mapsto_{\text{grammar}}$  question · request  
 b. Is the Pope Catholic  $\mapsto_{\text{grammar}}$  question · assertion  
 c. I have a ticket for tonight's performance  $\mapsto_{\text{griceanreasoning}}$  assertion · offer

For some semantic counterarguments to the strategy of semantic supertypes see Beyssade and Marandin (2006). In addition to this, one might express an intuitive unease about Asher and Lascarides's proposal: can an utterance simultaneously bear conflicting IFs in terms of comprehension/understanding? Data grounding this intuition are in (57):

- (57) a. A: Can you pass the salt? B: Is that a question or a request? A: # Both .../  
 B: # It's a simple request but a silly question.  
 b. A: I have a ticket for tonight's performance. B: # That's a surprising statement and a tempting offer.

CR potential distinguishes conventionalized from non-conventionalized ISAs. In (58a) B's CR can certainly be understood as attempting to confirm the request, whereas (58b) can be understood as attempting to confirm B's reassertion:

- (58) a. A: Can you pass the salt? B: The salt? (= Are you requesting me to pass the salt? (Schlangen, 2004))  
 b. A: ... B: Is the Pope Catholic? A: That weird German ... You're saying it's definite? B: Yeah, yeah.

In contrast, in (59a) B's 'Why?' cannot pick up on the non-conventionalized indirect offer. Similarly, (59c) contrasts minimally with (59b): in the former the offer is direct and CRable, in the latter it is indirect and non-CRable:

- (59) a. [A is a ticket tout, standing outside a theatre] A: It's the best show in London. I have a ticket for tonight's performance. B: (But) Why? ( $\neq$  Why do you want to sell your ticket).  
 b. A: I'm offering to sell you a ticket for this show. B: Offering?/to sell?  
 c. A: I've got a ticket for tonight's show. ( $\rightarrow$  I'm offering to sell you a ticket for tonight's show.) B: #Offering?/A ticket? (= Are you saying you have a ticket?  $\neq$  Are you offering to sell me a ticket?)

### 8.3.4 *Conceptual Analysis*

We have, then, two notions to explicate: conventionalized indirect moves and unconventionalized indirect moves. The former, to which I turn in detail in the next subsection, I shall analyse as grammatically internalized. What of non-conventionalized indirect moves? One cannot make sweeping claims about a rather loosely defined class of cases. I sketch a proposal that covers several of the examples considered here, based on the notion of genre relevance developed in Chapter 4.

My explication of unconventionalized ISAs draws an analogy with the issue of genre recognition. More concretely, the ‘indirect content’ emerges by searching for a dependent issue within the current genre. Recall that in Chapter 4, I introduced the notion of a move  $m_0$  being relevant to a genre type  $G_0$ . This amounted to it being the case that making move  $m_0$  given what has happened so far (represented in  $dgb$ ) can be *anticipated* to conclude as final state  $dgb1$  which is a conversation of type  $G_0$ . We could go two ways at this point: either broaden the range of conversational types by allowing in extra moves that are ‘tightly connected’ to existing moves in a given genre, or broaden the range of genre relevance to allow in such moves. We follow the latter strategy, as it will allow us to capture the intuition that understanding such moves involves inference and also distinguish the NSU possibilities of direct vs indirect moves. An illustration of this is in (60): whereas moves that are directly relevant to a genre license QUD accommodation, it seems that this is not the case with indirect genre relevance:

- (60) a. [Dialogue initially, in a bakery:] A: Three croissants.  
 b. [Dialogue initially, in a queue outside a toilet:] A: ?Very desperate.

In light of this, we define a notion of *Indirect Genre Relevance* in (61). A move  $m_1$  is indirectly relevant to a genre  $G_0$  if its contribution to QUD influences (the contribution to QUD of) a move  $m_0$  that is directly relevant to  $G_0$ :

- (61) **Indirect Genre Relevance:** A move  $m_1$  is *indirectly relevant* to a genre type  $G_0$  iff  $\text{qud-contrib}(m_1)$  influences  $\text{qud-contrib}(m_0)$  for some  $\text{IllocProp } m_0$  that is relevant to a genre type  $G_0$ .  
 (i)  $\text{qud-contrib}(m_0^{\text{content}})$  is  $m_0^{\text{content}}$  if  $m_0^{\text{content}}$  : Question  
 (ii)  $\text{qud-contrib}(m_0^{\text{content}})$  is ? $m_0^{\text{content}}$  if  $m_0^{\text{content}}$  : Prop

With this definition in hand, we introduce a variant on the rule of Initiating Move from Chapter 4. This is identical to the latter except for the fact that it uses **IndirectGenreRelevance** and does not allow for QUD accommodation. This difference is motivated by data such as (60).

- (62) Indirect Move:

$$\left[ \begin{array}{l} \text{dgb : DGBType} \\ \text{pre : } \left[ \begin{array}{l} \text{genre : GenreType} \\ \text{beliefs : Prop} \\ \text{agenda : list(IllocProp)} \\ \text{ip0 : IllocProp} \\ \text{c1 : } \rightarrow (\text{beliefs, IndirectGenreRelevant(ip0, dgb, genre)}) \end{array} \right] \text{ : PRTYPE} \\ \text{effects : TurnUnderspec} \wedge_{\text{merge}} \\ \left[ \text{LatestMove0}^{\text{content}} = \text{pre.private.ip0 : IllocProp} \right] \end{array} \right]$$

I exemplify how indirect content arises on the basis of (62). Consider first the toilet queue example (50a), repeated here as (63):

- (63) Anon woman(1): Is this the queue?  
 Me(2): Yes  
 Anon woman(3): How desperate are you?  
 Me(4): (shrugs)  
 (Person leaves toilet cubicle)  
 Me(5): Go ahead.

Let us assume that this genre has the QNUD specification informally rendered in (64):

- (64) QNUD = *where does the queue end, who goes next.*

Given this, the felicity of (63(3)) requires it to be indirectly relevant to the genre of toilet chat. For this to be the case, (65) needs to be a presupposition, if possibly an accommodated one. This encapsulates much if not all of the indirect meaning which (63(3)) expresses:

- (65) How desperate one is influences who should enter the toilet next.

As for (59a), repeated here as (66):

- (66) [A is a ticket tout, standing outside a theatre] A: It's the best show in London.  
 I have a ticket for tonight's performance.

Assuming that the ticket-selling genre has a QNUD specification informally rendered in (67a), then (67b) represents a presupposition that needs to be accommodated. Reflecting on the dependent issue conveys the indirect content of the utterance.

- (67) a. *who could sell a ticket, how much is the ticket*  
 b. Whether A has a ticket influences who could sell a ticket.

Of course this is a very genre-dependent implication, which does not go through in the context in (68). Indeed in this context, A implies that she wants to keep her ticket and B recognizes that. This fact is directly captured by the current account:

- (68) a. [Context, A and B are standing outside KCL Union.]  
 A: I have a ticket for tonight's show.  
 B: Lucky you.

A related case is Sperber and Wilson's celebrated (69):

- (69) A: Do you want some coffee?  
 B: Coffee would keep me awake.

(69) is related to (66) because it can also give rise to contradictory implications, the default being 'No', but 'Yes' is conveyed in a situation where the need for B's being

alert is at issue. Technically, what is different about it is that it is an indirect *response*, not an initiating move. This links us to what is ultimately one of the main issues with indirect moves: what triggers the search for *additional* content? In what way are we avoiding potential intractability akin to requiring the resolution of a pervasive non-public goal parameter or similar (e.g. each utterance raises the issue of why it was performed)?

Although I will not propose an explicit processing strategy here (see e.g. Purver 2006 for one example), the approach here, whereby Indirect Move is reified as a distinct option, allows for coherence to be checked via an ordering among moves. For instance, direct move updates (e.g. QSPEC) can be preferred, ensuring that if integration is successful, there will be no attempt at the more costly indirect route. Only if the direct route fails need the indirect move option be considered. In addition, if processing is difficult ('times out'), a CRification option is always available.

### 8.3.5 Conventionalized ISAs

I now proceed with some grammatical proposals concerning conventionalized ISAs. In Chapter 5, I introduced the assumption, following Ginzburg and Sag (2000), that illocutionary force is integrated at level of root clause. 'Canonical' root clauses satisfy (70):

$$(70) \left[ \begin{array}{l} R : \text{IllocReln} \\ p = \text{hd-dtr.cont} : \text{SemObj} \\ \text{cont} = R(\text{spkr}, \text{addr}, p) : \text{IllocProp} \end{array} \right]$$

For conventionalized ISAs the relationship between illocutionary force and head daughter content is more complex. The general strategy I sketch here is that conventionalized ISAs involve an ambiguity of illocutionary force, in line with CR data such as (57a) and (58a) above. In other words, the illocutionary force in such cases is a contextual parameter.

Let us start with reassertions: in Chapter 4 I introduced the Question Introduction Appropriateness Condition (QIAC), repeated here as (71), as a defining condition on the DGB as a means of preventing *inter alia* the posing of questions, answers to which exist in the DGB:

#### (71) Question Introduction Appropriateness Condition (QIAC):

A question  $q_1$  can be introduced into QUD by A only if there does not exist a fact  $\tau$  such that  $\tau \in \text{FACTS}$  and  $\tau$  resolves  $q_1$

Nonetheless, the fate of an interrogative used in a context where the question expressed is *resolved* is not brute infelicity but, as pointed out in Ginzburg (1992), construal as a so-called *rhetorical question* or *reassertion*. The move evokes the resolving answer to the question.



- (72) a. A: Well let's put Max next to Cinderella.  
B: Oh, but who did I say Max likes to sit next to?!
- A: You're quite right. Then we should change the sitting arrangements.
- b. A: I can't see the boss here tonight.  
B: Where do you think he is?!
- A: Geeze, that's incredible.

In neither of these cases does an answer-conveying response get elicited. In (72a) the utterance by B gets affirmed by A, indicating that B has succeeded to highlight some propositional content. In (72b), a propositional content deriving from B's utterance is embedded by the factive predicate *incredible*. In order for these discourses to work, however, both participants must assume they have common knowledge of the answer. Otherwise, obvious breakdowns will ensue. Respectively:

- (73) a. (Frown). No, I don't remember you mentioning Max's neighbourly preferences. OK, remind me...
- b. Sorry pal, what the devil are you insinuating about our beloved boss?

Formally, we can capture this by postulating an additional root clause rule: in Chapter 5, we had postulated a root clause construction that assigned a ‘default’ conversational move type to a clause based on their semantic type (see section 5.5). Here we provide a distinct construction.<sup>22</sup> The construction involves the force of an interrogative denoting  $q$ , uttered in a context where  $q$  is resolved, to be a reassertion of a resolving answer to  $q$ :

$$(74) \quad \text{reassert-cl} =_{\text{def}} \left[ \begin{array}{l} \text{dgb-params:} \left[ \begin{array}{l} \text{u0 : Rec} \\ \text{spkr: IND} \\ \text{addr: IND} \\ \text{utt-time: TIME} \\ \text{c1 : addressing(spkr, addr, utt-time)} \\ \text{p : Prop} \\ \text{rest}^{\text{facts}} \text{2: resolve(p, hd-dtr.cont)} \\ \text{c3: member(p, FACTS)} \end{array} \right] \\ \\ \text{cont} = \left[ \begin{array}{l} \text{sit} = \text{u0} \\ \text{sit-TYPE} = \left[ \begin{array}{l} \text{R = Reassert : IllocReIn} \\ \text{c1 : R(spkr, addr, utt-time, p)} \end{array} \right] \end{array} \right] : \text{Prop} \end{array} \right]$$

$\text{hd-dtr: } \left[ \text{cont : Question} \right]$

<sup>22</sup> This construction could be unified with the default root clause construction by complicating slightly the mapping *IllocRel* introduced in section 5.5.2.

**Exercise 8.3.2** Formulate a protocol for reassertion. What update rules does it require?

Moving on to a related, second type of ISA, exemplified in (75b, c, d)—the use of trivially resolved issues:

- (75) a. A: Will Bo flatter Merrill?  
b. B: Is the Pope Catholic?  
c. B: Does a bear shit in the woods?  
d. B: Do lions eat bananas?

These cases all seem to involve polar questions—the polar question  $q_1$  functions like ‘yes’ or ‘no’.<sup>23</sup> Hence, what emerges is an assertion involving not the resolving answer to  $q_1$ , but to the issue that is the MaxQUD. I propose to capture this by means of the construction in (76): the content of the clause is a reassertion of a proposition that resolves MaxQUD, so long as the triviality of the question constituting the head daughter is presupposed.

$$(76) \quad \textit{idiom-assert-cl} = \textit{def} \left[ \begin{array}{l} \text{dgb-params:} \left[ \begin{array}{l} \text{u0 : Rec} \\ \text{spkr: IND} \\ \text{addr: IND} \\ \text{utt-time: TIME} \\ \text{c1 : addressing(spkr, addr, utt-time)} \\ \text{p : Prop} \\ \text{rest}^{\textit{facts}}\text{2: resolve(p, max-qud)} \\ \text{c3: member(Trivial(head-dtr.cont), FACTS)} \end{array} \right] \\ \\ \text{cont} = \left[ \begin{array}{l} \text{sit = u0} \\ \text{SIT-TYPE} = \left[ \begin{array}{l} \text{R = Reassert : IllocReIn} \\ \text{c1 : R(spkr, addr, utt-time, p)} \end{array} \right] \end{array} \right] : \text{Prop} \end{array} \right]$$

$\downarrow$   
 head-dtr: [ cont : PolarQuestion ]

Apart from yielding the desired results as far as interpreting answers such as (75) in concerned, this analysis also explains (58b)—the conventional content associated with ‘Is the Pope Catholic’ (with its reference in 2011 to Joseph Ratzinger) is the content of the head daughter. This is what licenses that referential confirmation CR.

The final case we consider is what one might call *addressee ability questions* relating to cases like (50c) and (58a). We postulate a third construction, whose head daughter

<sup>23</sup> On which, see section 7.5.

denotes a question  $q$ , where  $q$  concerns the addressee's ability to bring about outcome  $o$ . The content of the root clause in such a case is a request for  $o$ :

$$(77) \text{ addressee-ability-cl} =_{\text{def}} \left[ \begin{array}{c} \text{dgb-params:} \left[ \begin{array}{l} \text{u0 : Rec} \\ \text{spkr: IND} \\ \text{addr: IND} \\ \text{utt-time: TIME} \\ \text{cl : addressing(spkr, addr, utt-time)} \\ \text{o : Outcome} \end{array} \right] \\ \text{cont} = \left[ \begin{array}{l} \text{sit = u0} \\ \text{SIT-TYPE} = \left[ \begin{array}{l} \text{R = Request : IllocReIn} \\ \text{cl : R(spkr, addr, utt-time, o)} \end{array} \right] \end{array} \right] \end{array} \right] : \text{Prop}$$

|

head-dtr: [ cont = ?Able(a, o) : PolarQuestion]

Given (77), an utterance like (78a) is *ambiguous* between a request and a query reading. Nonetheless, even the request reading does have a question (NB: not a *query*) as a constituent, which is what licenses (78c) as a reprise even under the request reading, as well as (78b):

- (78) a. Can you pass the salt?  
       b. You want me to pass the salt?  
       c. Can I pass the salt?

### 8.3.6 *Why?*

Jerry: How's Robert?

Emma: When did you last see him?

Jerry: I haven't seen him for months. Don't know why. Why?

Emma: Why what?

Jerry: Why are you asking when I last saw him? (*Betrayal*, Pinter).

Last but by no means least I turn to queries of the form 'Why?', in particular uses like we saw in examples (55). An obvious starting proposal would be to assimilate such uses to CRs and use the mechanisms we have in place since Chapter 6 to do the rest. However, such a proposal runs into two significant problems.

The first problem is constituted by long-distance CRs like (55) and Pinter's example above—the to-be-clarified utterance is no longer pending—a conventional response has after all been provided—and hence the standard triggering context for a CR is missing.

A related problem derives from a consideration of the value sought in a *Why?*<sub>meta</sub> query: *when*, *where*, and indeed virtually all other *wh*-phrases used in CRs query a constituent of what one might call the nucleus of a given content (boldface in (79b)), whereas *Why?*<sub>meta</sub> outscopes this nucleus to query at the illocutionary level (boldface in (79d)):<sup>24</sup>

- (79) a. A: Did Bo leave? B: Who?/When?/Where?  
 b.  $\lambda b/t/l \text{ Ask}(\mathbf{A}, \text{?Leave}(\mathbf{b}, \mathbf{t}, \mathbf{l}))$   
 c. A: Did Bo leave? B: Why?  
 d.  $\lambda r. \text{Cause}(r, \text{Ask}(\mathbf{A}, \text{?Leave}(\mathbf{b}, \mathbf{t}, \mathbf{l})))$

(80a) and (80b) are close synonyms. However, the contrast between the behaviour of ‘Why?’ and ‘For what reason?’ is particularly striking. The latter only allows for the reading in (80b), not for a *Why?*<sub>meta</sub> reading (though the reading in (80b) is, as it happens, pragmatically not unrelated to *Why?*<sub>meta</sub>).

- (80) a. Why is Bo leaving?  
 b. For what reason is Bo leaving?  
 c. A: Can Bo be granted leave? B: For what reason?  
 d.  $\lambda r. \text{Ask}(\mathbf{A}, \text{?Cause}(r, \text{GrantLeave}(\mathbf{b}, \mathbf{t}, \mathbf{l})))$

This is another illustration that the information sought in a *Why?*<sub>meta</sub> query, like that in sentential ‘What do you mean’ CRs, is not a contextual parameter of the utterance, so the answer provided would not be integrated as a consequence of contextual instantiation, as with standard contextual parameters.

The conclusion is that *Why?*<sub>meta</sub> cannot be treated as a CR (in the sense of CR developed in Chapter 6; pre-theoretically it is a CR in that it queries an aspect of the communicative process) and requires a distinct treatment.

The obvious alternative strategy to consider is to extract an account of *Why?*<sub>meta</sub> from a general treatment of non-sentential ‘Why?’, direct or metacommunicative. A partial unification is possible, based on a single conversational rule underwriting the coherence of ‘why’-queries as follow-ups to utterances. A completely unified treatment, however, tempting though it is,<sup>25</sup> is quite tricky and ultimately, I will suggest, not clearly worthwhile. To see this it is necessary first to sketch an analysis of direct

<sup>24</sup> There are some other contrasts between ‘Why’ and the others. These include the ability of ‘why’ to take fragments as a complement and to take *not* as a complement:

- (i) A: Bo is leaving this week. B: Why Bo/this week?  
 A: Bo is leaving this week. B: \*Where Bo/this week?  
 (ii) A: Bo isn’t leaving this week.  
 B: Why not?  
 A: Bo isn’t leaving this week.  
 B: \*Where/\*when/\*for what reason not?

<sup>25</sup> For accounts of this type see Ginzburg (1997) and Ginzburg (2003).



LatestMove. It cannot be the case that just any fact licenses such an utterance, but the relevant class of facts is quite a bit wider than Topical, let us call these Broadly-Topical facts. They seem to include all the potential utterance-related presuppositions discussed in Chapters 2 and 6. Note that (83a–f) are not available with a bare ‘Why?’:

- (83) a. A: Terrible weather recently. B: But it’s nice and sunny today. A: Yes. Why have we had such terrible weather recently?  
 b. A: Bo is resigning. B: Why is he resigning at his advanced age?  
 c. A: Bo is resigning. B: Why do you talk about Bo so much?  
 d. A: Bo is resigning. B: Why do you talk about his resigning in the present tense? The resignation came into effect last month.  
 e. A: Bo is resigning. B: Why do you refer to Professor Magnusson with his first name?  
 f. A: Bo is resigning. B: Why do you make these curt three-word announcements?  
 g. A: Bo is resigning. B: # Why was your grandmother born in Lvov?

A conversational rule that underwrites the coherence of (83) would be roughly as in (84)—this says that a broadly topical fact enables the issue ‘Why *f*’ to become MaxQuD, in which case the LatestMove needs to be co-propositional with MaxQuD:

- (84) Explanation rule = *def*
- $$\left[ \begin{array}{l} \text{pre} \\ \text{effects} \end{array} : \left[ \begin{array}{l} f : \text{BroadlyTopical} \\ \text{MaxQuD} = \text{Why}(\text{pre}.f) : \text{Question} \\ \text{co-prop}(\text{LatestMove}^{\text{content}}, \text{MaxQuD}) \end{array} \right] \right]$$

In addition to capturing the coherence of (85(a)(2)), we can also explain cases like (85(b)(2)):

- (85) a. A: Are you in cahoots with Tony? B: Why are you asking this?  
 b. A: Do you find it cold here? A/B: Should I close the window? / I should probably close the window. (= The reason A is asking if it’s cold here because the window should be closed.)

Now the rule in (82) only has substance in so far as we are explicit about what it means to be TOPICAL. A first approximation is to say is that a fact is TOPICAL in case it is **About** some question currently or recently in QUD. This can be transparently related to work on anaphora in discourse (Polanyi 1987, Webber 1991, Asher 1993, Asher and Lascarides 2003). One central constraint proposed in these works is a constraint known as the RIGHT-FRONTIER CONSTRAINT (RFC), which, assuming a configurational view of discourse structure, stipulates that antecedents can be found only on the right frontier of a discourse (structure).

Here we introduce TOPICAL by means of minor modifications of FACTS update and Ask/Assert QUD-incrementation:<sup>26</sup> in the former case, acceptance or confirmation of  $p$  add  $p$  to the TOPICAL facts, whereas the incrementation of QUD by  $q_0$  after a query updates TOPICAL by removing those facts which pertain to questions that are not in QUD. This is where there is an analogy to the RFC.

(86) Fact update/QUD-downdate (topicalized) = *def*

$$\left[ \begin{array}{l} \text{pre} \\ \text{effects} \end{array} : \left[ \begin{array}{l} p : \text{Prop} \\ \text{LatestMove} = \text{Accept}(\text{spkr}, \text{addr}, p) \vee \text{Confirm}(\text{spkr}, \text{addr}, p) : \text{IllocProp} \\ \text{qud} = \langle p?, \text{pre.qud} \rangle : \text{poset}(\text{Question}) \\ \text{topical} : \text{Set}(\text{Prop}) \\ \text{facts} = \text{commonground1} : \text{Set}(\text{Prop}) \\ \text{cl} : \text{topical} \sqsubseteq \text{Facts} \\ \text{topical} = \text{pre.topical} \cup \{p\} : \text{Set}(\text{Prop}) \\ \text{qud} = \text{NonResolve}(\text{pre.qud}, \text{facts}) : \text{poset}(\text{Question}) \end{array} \right] \right]$$

(87) Ask QUD incrementation (topicalized) = *def*

$$\left[ \begin{array}{l} \text{pre} \\ \text{effects} \end{array} : \left[ \begin{array}{l} q_0 : \text{Infostruc} \\ \text{LatestMove}^{\text{content}} = \text{Ask}(\text{spkr}, \text{addr}, q_0.q) : \text{IllocProp} \\ \text{qud} : \text{poset}(\text{InfoStruc}) \\ \text{Topical} : \text{set}(\text{Facts}) \\ \text{facts} = \text{commonground1} : \text{Set}(\text{Prop}) \\ \text{cl} : \text{topical} \sqsubseteq \text{Facts} \\ \text{Topical} = \text{Pre.Topical} \setminus \{f \mid \text{No } q \in \text{QUD AND About}(f, q)\} : \text{Set}(\text{Prop}) \\ \text{qud} = \langle q_0, \text{pre.qud} \rangle : \text{poset}(\text{Question}) \end{array} \right] \right]$$

Table (88c) exemplifies the dynamics of QUD and Topical for (88a); ‘WRY’ is an abbreviation for the question denoted by the use here of ‘Who resigned yesterday?’, similarly with the rest of the notation here. As far as (88a) goes, after utterance (7) ‘Why’ could be resolved as either the preferred ‘Why René Matrasso resigned’ or ‘Why Pierre Lapin and René Matrasso resigned’. Conversation (88b) is a variant in which after utterance (5) ‘Why’ would be resolved as ‘Why Pierre Lapin resigned’. These resolutions are available to either A or B.

- (88) a. A(1): Who resigned yesterday? B(2): Pierre Lapin. A(3): Pierre Lapin? B(4): He is the restaurateur. A(5): OK. B(6): René Matrasso. A(7): Aha. (8) Why? / B(8’): I have no idea why.

<sup>26</sup> I illustrate here only the modification of Ask QUD-incrementation, since the Assert case is entirely analogous.

- b. A(1): Who resigned yesterday? B(2): Pierre Lapin. A(3): Pierre Lapin? B(4): He is the restaurateur. A(5): OK. A(6): Why? / B(6'): I have no idea why.

| Utt. | QUD                        | Topical  |
|------|----------------------------|----------|
| 1    | QUD := ⟨WRY⟩               | {}       |
| 2    | QUD := ⟨PL?, WRY⟩          | {}       |
| 3    | QUD := ⟨WPL, WRY⟩          | {}       |
| c. 4 | QUD := ⟨TR, WPL, PL?, WRY⟩ | {}       |
| 5    | QUD := ⟨WRY⟩               | {TR, PL} |
| 6    | QUD := ⟨RM?, WRY⟩          | {PL}     |
| 7    | QUD := ⟨WRY⟩               | {PL, RM} |

Now what of Why?<sub>meta</sub>? It differs from the ‘canonical’ use of ‘Why?’ in a number of significant respects:

- Addressee specificity:** in contrast to direct ‘Why?’ (see example (81a)), (89a, b) only get a Why?<sub>meta</sub> reading if A cedes the turn to B; if A keeps the turn such a reading is unavailable:
 

(89) a. A: Are you in cahoots with Tony? \*A/B: Why? (= Why are you asking?)  
 b. A: Who is going to win the race? \*A/B: Why? (= Why are you asking?)
  - Lexical idiosyncrasy:** in English the Why?<sub>meta</sub> reading is unavailable with other cause-querying *wh*-collocations such as ‘How come?’ and ‘For what reason?’. Similarly in Hebrew Why?<sub>meta</sub> is available only with ‘lama’ but impossible with ‘madua’ and ‘eyx ze’; in German Why?<sub>meta</sub> is possible with ‘warum?’ but impossible with ‘weshalb’:<sup>27</sup>
- (90) a. A: Are you in cahoots with Tony? B: # How come?/For what reason? (=What might be the reason due to which I’m in cahoots with Tony?)  
 b. A: ha'im bar tavo maxar B: lama? / madua?  
 Is-it-the-case Bar come-fut-3rd-sg tomorrow? Why?/Why?  
 A: Will Bar be coming tomorrow? B: Why (do you ask)?/Why would she come?
- Intonational difference:** Why?<sub>meta</sub> typically involves a simple rise, whereas the ‘canonical’ ‘Why?’ involves a rise-fall tune.

These differences indicate the difficulty of unifying Why?<sub>meta</sub> with direct ‘Why?’ The fundamental difficulty is this: given that Why?<sub>meta</sub> is never available to the original speaker, we would need to somehow ensure that Move<sup>content</sup> is Topical for a certain period—for instance in so far as the issue it introduces is in QUD—but *solely for B*. One can of course stipulate this in a number of ways—one needs both to introduce it and to eliminate it—but there does not seem to be a way to do this in a way that naturally generalizes over (86) and (87).

<sup>27</sup> Thanks to David Schlangen (p.c.) for this latter datum.



One might suppose that the reason that  $\text{Why?}_{meta}$  is unavailable to the original speaker is coherence—it is a rather odd, almost self-defeating question to ask. Nonetheless, the same content expressed non-elliptically is not, it seems, incoherent, for instance when the initial query is unusual.

- (91) a. A: Are you in cahoots with Tony? Why am I asking this?  
 b. A: Where was your grandmother born? Why am I asking this?

This suggests we probably need to bite the bullet and assume that a distinct mechanism underlies  $\text{Why?}_{meta}$ . The lexical and intonational differences noted above provide linguistic support for this. Indeed, all we will be doing to account for  $\text{Why?}_{meta}$  is to posit a construction akin to other obligatorily second-part adjacency pair lexemes such as ‘Aha’, ..., ‘sabax elnur’ (see footnote 21, section 4.4.2). (92) is a description of the  $\text{Why?}_{meta}$  construction: the content of this construction is a query by A of the cause of an utterance  $m$  by B, an utterance whose QUD contribution is still in QUD.

- (92)  $\text{why-mc-cl} =_{def}$
- $$\left[ \begin{array}{l} \text{phon} : \left[ \begin{array}{l} \text{seg} : \text{wai} \\ \text{tune} : \text{H} \end{array} \right] \\ \text{cat.head} = \text{V} : \text{PoS} \\ \text{dgb-params} : \left[ \begin{array}{l} \text{A} : \text{IND} \\ \text{B} : \text{IND} \\ \text{r} : \text{Semobj} \\ \text{q} : \text{Question} \\ \text{c0} : \text{QUDContrib}(\text{r}, \text{q}) \\ \text{c1} : \text{In}(\text{q}, \text{QUD}) \\ \text{R} : \text{IllocRel} \\ \text{m} : \left[ \text{cont} = \text{R}(\text{B}, \text{A}, \text{r}) : \text{IllocProp} \right] \wedge \text{sign} \\ \text{c2} : \text{In}(\text{m}, \text{MOVES}) \end{array} \right] \\ \text{cont} = \text{Ask}(\text{A}, \text{B}, \text{hd-dtr}(\text{m}^{content})) : \text{IllocProp} \end{array} \right]$$
- $$\left[ \begin{array}{l} \text{hd-dtr} : \left[ \begin{array}{l} \text{cat} = \text{adv} : \text{PoS} \\ \text{cont} = (\text{f} : \text{Prop}) \lambda \text{r}. \text{Cause}(\text{r}, \text{f}) : (\text{Prop})\text{Question} \end{array} \right] \end{array} \right]$$

(93) exemplifies this rule. In (1), A asks a question which triggers a CR from B. After the clarification interaction is resolved, B provides an answer to A’s question, but then poses a  $\text{Why?}_{meta}$  about the reason for that query.

- (93) a. A(1): Is Bo leaving tomorrow?  
 B(2): Bo?  
 A(3): Bo Hendriksson.  
 B(4): Maybe. (5) Why?

| Utt.    | DGB Update<br>(Rule)                                                                                                                     |
|---------|------------------------------------------------------------------------------------------------------------------------------------------|
| initial | MOVES = {}<br>QUD = {}<br>FACTS = cgl                                                                                                    |
| 1       | Moves := ⟨Ask(A, B, q1)⟩<br>QUD := ⟨q1⟩<br>Ask QUD-incrementation                                                                        |
| 2       | Moves := ⟨Ask(B, A, q2), Ask(A, B, q1)⟩<br>parameter identification<br>QUD := ⟨q2, q1⟩<br>Ask QUD-incrementation                         |
| 3       | Moves := ⟨Assert(A, B, p1), Ask(B, A, q2), Ask(A, B, q1)⟩<br>QSPEC<br>QUD := ⟨p1?, q2, q1⟩<br>Assertion QUD-incrementation               |
| 4       | Moves := ⟨Assert(B, A, p2), Assert(A, B, p1), Ask(B, A, q2), Ask(A, B, q1)⟩<br>QSPEC<br>QUD := ⟨p2?, q1⟩<br>Assertion QUD-incrementation |

- c. 
$$\left[ \begin{array}{l} \text{phon.seg : wai} \\ \text{phon.tune : H} \\ \text{cat.head = V :syncat} \\ \text{cont = hd-dtr.cont(m}^{content}\text{)} = \\ (f : \text{Prop}) \lambda r. \text{Cause}(r, f)(\text{Ask}(A, B, q1)) = \\ \lambda r. \text{Cause}(r, \text{Ask}(A, B, q1)) : \text{IllocProp} \end{array} \right]$$

hd-dtr:  $[ \text{cont} = (f : \text{Prop}) \lambda r. \text{Cause}(r, f) : (\text{Prop})\text{Question} ]$

### 8.3.7 Summary

In this section I have sketched extensions of KoS to deal with various phenomena that are typically viewed as *pragmatic*—certain kinds of implicatures and indirect speech acts. My main concern has been to situate the requisite reasoning within the picture of grounding and CRification developed in earlier chapters: the picture that emerges is one in which reasoning at the DGB level can retain a certain primacy, allowing

as it does the possibility for indirect moves and CRification concerning relevance if integration of direct moves fails. I have also suggested that the potential for CRs concerning private intentions can be explicated as an instance of the coherence of explanation moves.

## 8.4 Developmental Evidence on NSUs

In this section I present a summary of some recent work on the acquisition of NSUs. The main interest of such work in the current context is the potential it offers to shed light on the comparative complexity of NSU types: this work provides evidence about the order of acquisition among some of the main classes of NSUs, a characterization which somewhat surprisingly perhaps has not been provided in previous acquisition literature. I then show that there is a good fit between this and notions of complexity that emerge here.

In Ginzburg and Kolliakou (2009) and (2010), we present evidence that the order of acquisition of the main classes of NSUs is as in (94a): this generalization is based on two subparts, (94b, c):

- (94) a. **NSU order of emergence:** short answers, yes/no < sluice < Reprise Fragments  
 b. **The Late Short Query Effect:** Non-Sentential Answers (NSAs) (short answers, yes/no) < Non-Sentential Queries (NSQs) (sluices, Reprise Fragments)  
 c. **The Repetition Avoidance Effect:** sluices < Reprise Fragments

This generalization is based primarily on evidence from English, on which I will concentrate exclusively here, but is supported by evidence from Greek.<sup>28</sup>

### 8.4.1 *The Late Short Query Effect*

The principal finding of Ginzburg and Kolliakou (2009) is the Late Short Query Effect: at a stage where the child has productive use of sentential (primarily *wh*- and polar) questions and has mastered short answers and the polar lexemes 'yes' and 'no', RFs and sluices are virtually absent. The Late Short Query Effect emerges from a study of a sub-corpus of the Manchester corpus in CHILDES (Theakston et al. 2001, Rowland et al. 2003) which involved manual coding of a random selection of the transcripts of four children.<sup>29</sup> The following categories were tagged and their numbers tallied: sentential answers (declaratives used to answer questions) and sentential

<sup>28</sup> Ginzburg and Kolliakou (2009) provide data from the Stephany corpus of child/adult interaction in Greek (Stephany 1995) supporting the Late Short Query Effect.

<sup>29</sup> The corpus studied consisted of 25,060 turns (111,969 words), including at least six files per child (average size of file: 830 turns) and covering age span 2;3 to 2;10 approximately.

**TABLE 8.7** Distribution of sentential/non-sentential answers for four children aged 2;1–2;10 in the Manchester Corpus

| subject | period   | decl.      | yes + no | short-ans. | non-sent. total | ans. total |
|---------|----------|------------|----------|------------|-----------------|------------|
| Aran    | 2;3–2;6  | 125 (29%)  | 207      | 100        | 307 (71%)       | 432        |
| Becky   | 2;3–2;7  | 88 (25%)   | 210      | 55         | 265 (75%)       | 353        |
| Carl    | 2;1–2;8  | 99 (50%)   | 41       | 58         | 99 (50%)        | 198        |
| Dominic | 2;4–2;10 | 85 (25.5%) | 130      | 108        | 238 (74.5%)     | 323        |

**TABLE 8.8** Distribution of sentential/non-sentential questions for four children in the Manchester Corpus (same period as Table 8.7)

| Subject | wh         | Polar | Total     | Sluicing       | RF | Total   | Total     |
|---------|------------|-------|-----------|----------------|----|---------|-----------|
|         | Sentential |       |           | Non-sentential |    |         | Questions |
| Aran    | 45         | 4     | 49 (96%)  | 2              | 0  | 2 (4%)  | 51        |
| Becky   | 106        | 98    | 204 (92%) | 14             | 4  | 18 (8%) | 222       |
| Carl    | 47         | 5     | 52 (96%)  | 2              | 0  | 2 (4%)  | 54        |
| Dominic | 15         | 2     | 17 (85%)  | 1              | 2  | 3 (15%) | 20        |

(WH-, polar and intonation) interrogatives, as well as four NSU types (short answers, 'yes'/'no', RF, and sluices).

Table 8.7 above summarizes the distribution of sentential versus non-sentential answers of the four subjects. Table 8.7 can be compared with Table 8.8, which summarizes the distribution of the children's sentential versus non-sentential queries. At a point when all other categories have emerged, for three of the four subjects studied sluicing and RFs are almost entirely absent. The different sentential/non-sentential distributions for answers versus questions is of high statistical significance (Aran:  $p \leq .001$ ,  $\chi^2 = 86$ ; Becky:  $p \leq .001$ ,  $\chi^2 = 241$ ).

Ginzburg and Kolliakou (2009) further demonstrate that the Late Short Query Effect does not reflect the input of NSUs available to the children. The mothers use NSQs productively, whereas these are entirely absent from three of the children's productions and rare for the other. Evidence is also provided that the stage when non-sentential queries *are* acquired does not correlate with a significant change in the distribution of non-sentential answers. This indicates that that the increasing frequency of NSQs with time cannot be simply attributed to a newly developed predilection for phonologically reduced forms. Finally, Ginzburg and Kolliakou (2009) argue in some detail against a pragmatic explanation for the Late Short Query Effect, by showing (a) the existence of contexts in which the child could have produced an NSQ (but did not) and that the rate of this omission differs in a statistically significant way from the

**TABLE 8.9** RF and sluicing figures for four children from the Belfast Corpus

|              | Barbara        | David         | Michelle        | Rachel          |
|--------------|----------------|---------------|-----------------|-----------------|
|              | (2;4.9–4;1.18) | (2;0.3–4;2.3) | (2;4.28–4;4.19) | (2;5.25–3;2.03) |
| Sluice Mean  | 2.5            | 2.5           | 2               | 0.5             |
| Sluice Total | 34             | 37            | 33              | 3               |
| RF Total     | 0              | 2             | 0               | 0               |

behaviour of their carers; (b) that the contents communicated by NSQs are ones that the children have the ability to express using alternative, *sentential* means.

#### 8.4.2 *The Repetition Avoidance Effect*

The Repetition Avoidance Effect, reported in Ginzburg and Kolliakou (2010), is that Reprise Fragments (RF), the majority clarification request construction among adults, emerges with some delay in comparison with sluicing. In a study of all transcripts of the Belfast corpus (Henry 1995, Wilson and Henry 1998) as well as the transcripts of eight children of the Manchester corpus, it is shown that (a) on the whole children go through long periods where sluicing is productive and RF is *absent*, whereas (b) adults' usage of RF in interaction with children (and elsewhere) significantly exceeds their usage of sluicing and RFs are *prima facie* optimally accessible in context, given their priming by the prior utterance.

The distribution of RFs and sluicing in the data of the younger ( $\leq 4;6$ ) Belfast children are given in Table 8.9 above. The mean and totals are over fourteen conversations (nine conversations for Rachel).

Finally, using the taxonomy of sluicing from section 7.8, Ginzburg and Kolliakou (2010) show that the early emergence of sluicing cannot be attributed to high frequency, rote-learned instances.

#### 8.4.3 *Basic Implications*

The findings surveyed in sections 8.4.1 and 8.4.2 establish some significant patterns concerning the order of acquisition of NSUs. I consider now how these findings are to be explained and some implications they have for the grammatical analysis of NSUs.

The delay in emergence of a construction  $C_1$  where construction  $C_0$  has already emerged can be explained in two ways, pragmatically or mechanistically. A pragmatic explanation sees the absence of  $C_1$  as a consequence of two possible factors: (a) the *contextual unavailability* of antecedents which make  $C_1$  felicitous, and/or (b) the *semantic undesirability* of the contents  $C_1$  expresses. A mechanistic explanation ties the absence of  $C_1$  to an insufficiently developed linguistic/conversational competence, one which is however sufficient to underpin use of  $C_0$ . We consider first with respect to the order of emergence of NSUs to what extent a pragmatic explanation is plausible:

1. NSAs < NSQs: a pragmatic account is on the cards since the NSAs and NSQs have distinct functions and distinct triggering contexts. Nonetheless, as mentioned above, Ginzburg and Kolliakou (2009) provide evidence that in fact (at a stage where NSAs are produced and NSQs absent), children *do* get appropriate contexts for NSQs and avoid producing them in a manner that differs in a statistically significant way from their carers.
2. Sluicing < CE: in this case a pragmatic account is not very plausible. Both constructions are used, for the most part, for the same function<sup>30</sup>—to pose clarification requests—and indeed they can be used to express contents that seem to be equally expressive and appropriate in similar contexts (e.g. A: ...there. B: There?/ Where?).

This discussion suggests that an explanation of the order of acquisition of NSUs requires, at the very least, a mechanistic component. What mechanistic account is to be offered depends, to a significant extent, on the linguistic description one associates with NSUs. I believe that the acquisition data presented here rule out one rather influential strategy by means of which certain NSUs have been analysed, and these same considerations have a more general implication.

The commonest strategy for dealing with NSUs in generative linguistics is the strategy I dubbed earlier *Underlying NSU sententialism*, wherein NSUs are analysed as underlyingly *sentential* constructions such that a non-grammatical module serves—under appropriate conditions—to recover material that is phonologically redundant and remains unpronounced.

However, it appears that *Underlying NSU sententialism* is not a strategy which could be used to develop an account of the order of acquisition of NSUs. Consider explaining the delay of NSQs relative to NSAs. As discussed earlier, the subjects whose data on this we considered lacked NSQs at a stage where NSAs were present as well as *wh*-interrogative sentences. This means that both the input and the mechanism that, according to this strategy, should give rise to sluicing are present. This should entail the presence of sluicing, contrary to what is desired. The problem we diagnose here for *Underlying NSU sententialism* can be generalized to other *unitary* strategies for NSUs, theories in which NSU resolution is associated with a single, typically extra-grammatical mechanism. Once the child has acquired one type of NSU, s/he has by necessity acquired the mechanism for generating NSUs. In order to explain the absence of a delayed NSU type, one will have to ensure that the input to this mechanism which gives rise to the delayed type is absent. Particularly, when an NSU type with a similar pragmatic background to the delayed one is already present (e.g. sluicing in the case of CE), this condition will be unlikely to occur.

A case in point is higher-order unification (HOU) (Dalrymple, Pereira, and Shieber 1991, Pulman 1997) and the delayed appearance of RFs relative to sluicing. As

<sup>30</sup> The frequency of direct sluices in the data is small compared to reprise and repetition sluices.

explained in Ginzburg and Cooper (2004), HOU can explicate the clausal confirmation reading of a RF and similar readings evinced by sluicing. Once the HOU mechanism is available to the child, assuming it to be the mechanism underlying NSU meaning, then the availability of *wh*-phrase denotations, will ensure that sluicing is available. But this in turn will also ensure that the clausal confirmation reading of an RF is available, which goes against the observed order of emergence.

#### 8.4.4 *Explaining the order of NSU acquisition*

In what follows, I provide an account of the Late Short Query Effect using a dialogue-oriented construction grammar. Such an account comprises two components, given in (95).

- (95) a. **Propositional NSUs and sentential interrogatives without NSQs; sluicing without RF:** The specification of grammars and contextual systems in which (a) short answers and sentential interrogatives coexist without NSQs; (b) sluices exist without RFs.
- b. **The order of NSU acquisition:** An explanation for the emergence of NSAs before NSQs and sluicing before RF.

With respect to (95a), I will restrict attention to NSAs vs NSQs; the independence of sluicing from RF was discussed earlier in section 7.9. Ginzburg and Sag (2000) specifies a grammar which accommodates *inter alia* sentential declaratives, sentential interrogatives, and the various NSU constructions discussed above—polar lexemes, short answers, direct and reprise sluices, and RF. In this grammar, as in other work in the Construction Grammar tradition, constructions are organized into a type hierarchy. This enables higher-level types to be posited which allows relevant cross-cutting generalizations to be captured. For our current purposes it is worth mentioning two relevant pieces of the construction hierarchy, that which concerns interrogatives and that which concerns NSUs.

The hierarchy for interrogatives, most of whose constitutive types have been reconstructed in TTR in Chapters 5, 6, and 7, is displayed in Figure 8.2 opposite.

The crucial point about a grammar organized in terms of a hierarchy like Figure 8.2 is that its acquisition can be modelled in a straightforward monotonic way, as pointed out originally by Green (2000) (see also Goldberg 1995, Tomasello 2003). In our case a grammar without sluicing involves an interrogative hierarchy with only the encircled branches. Acquiring the additional constructions involves a monotonic refinement of the hierarchy. This applies equally to the hierarchy of NSUs, shown in Figure 8.3: a general type here is *hd-frag-ph* which specifies the aforementioned fragment-FEC parallelism. A grammar for NSUs without direct sluicing involves merely the encircled branch in Figure 8.3.

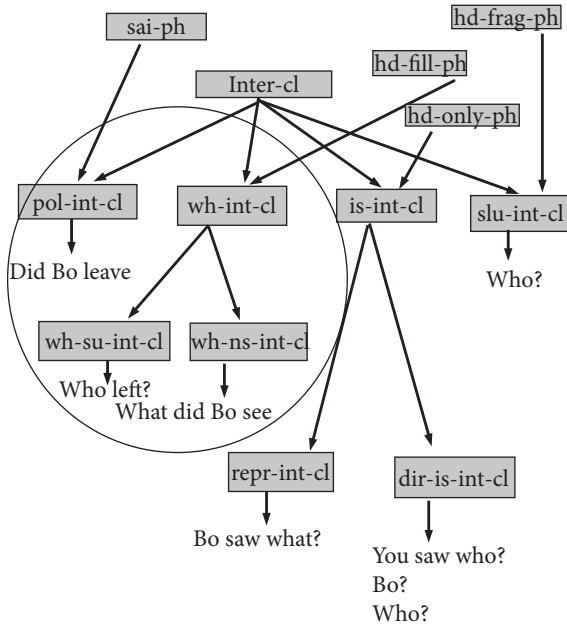


FIGURE 8.2: Sketch of a type hierarchy for interrogatives

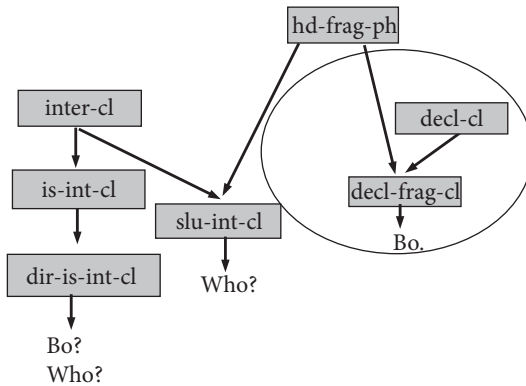


FIGURE 8.3: Sketch of a type hierarchy for NSUs

This concludes the first part of our explanation of the Late Short Query Effect: the specification of a grammar and contextual system in which short answers and sentential interrogatives coexist without NSQs.

We can now consider how semantic and contextual processing might play a role in the observed order of acquisition of NSUs, most specifically the fact that NSAs emerge



before NSQs and sluices before RF. We pursue an account of this in terms of a notion that combines semantic and contextual complexity, as explained shortly. This type of account is in line with many accounts of the order of acquisition of lexical items and constructions that have appealed to some notion of complexity (see e.g. Brown and Hanlon 1970, Bloom, Merkin, and Wooten 1982).

When considering the semantic complexity of NSUs, there is a rather obvious notion that is worth considering, namely the complexity of the semantic operations involved in constructing the content. However, to the extent that this is a relevant factor, it clearly tells us only half the story since access to the contextual parameters plays such a crucial role in the semantic processing. Hence, our notion of complexity should make reference also to contextual accessibility. There are a number of ways by means of which such notions could be combined. I assume a notion of complexity that gives contextual complexity priority, ties to be broken by combinatorial complexity:

- (96)  $C1 \leq C2$  iff either C1 has a simpler contextual background than C2, or C1 and C2 have equivalent contextual backgrounds but C1 is combinatorially simpler than C2.

Let us consider first a notion of *complexity of content construction*. In our case, the operations involved are: *function application*,  *$\lambda$ -abstraction*, *proposition extraction*, *substitution*, *negation*, and *polar questioning*. In so listing the operations, I am actually taking a somewhat theory-neutral view of semantic combinatorics: proposition extraction, substitution, negation, and polar questioning can all be explained in TTR terms, as we have seen in Chapter 3.

Using this expanded set of operations, we can evaluate the *complexity of content construction* in terms of the operations involved in each construction type. Let us consider some examples: the content of ‘yes’ is specified in Table 8.10 as arising by

**TABLE 8.10** Content construction complexity for NSUs in a DOC grammar

| NSU type                   | Content Spec                                                          | Content Construction Operations                            |
|----------------------------|-----------------------------------------------------------------------|------------------------------------------------------------|
| yes                        | max-qud.prop                                                          | 1 proposition extraction                                   |
| no                         | NegProp $\in \{(1), \neg(1)\}$                                        | 1 proposition extraction<br>+ 1 negation + 1 choice-from-2 |
| short-answer               | max-qud(frag.cont)                                                    | 1 function application                                     |
| direct sluice              | $\lambda$ frag.dom<br>(max-qud.prop(FEC.cont $\mapsto$<br>frag.cont)) | 1 $\lambda$ -abstraction +<br>1 substitution +             |
| RF <sub>confirmation</sub> | ?max-qud(frag.cont)                                                   | 1 proposition extraction                                   |
| reprise sluice             | max-qud                                                               | ?-op + function application                                |

TABLE 8.11 Contextual parameters involved in NSU contents

| NSU                | contextual params involved                              |
|--------------------|---------------------------------------------------------|
| yes, no, short ans | max-qud                                                 |
| direct sluice      | max-qud, FEC                                            |
| RF, reprise sluice | max-qud, FEC via CCUR on partially understood utterance |

extracting the proposition from max-qud. The complexity of constructing ‘yes’ therefore involves one proposition extraction operation. Similarly, given the specification of short answer content in Table 8.10, its complexity involves one function application operation. A direct sluice involves a propositional extraction, a  $\lambda$ -abstraction, and a substitution. Its complexity involves, therefore, 1 proposition extraction + 1 substitution + 1 abstraction. The analysis of reprise sluicing we discuss here is the second of the accounts we develop for it, in section 7.9.2.

Table 8.11 summarizes the contextual parameters involved in the content construction of the main NSU constructions.

Given the data in Tables 8.10 and 8.11 some clear conclusions emerge:

- (97) a.  $\text{NSA} \leq \text{NSQ}$ : short answers and yes/no have fewer contextual parameters than direct sluices and are also simpler combinatorially; short answers and yes/no have simpler contextual backgrounds than reprise sluices and RF.
- b.  $\text{Reprise sluice} \leq \text{RF}$ : reprise sluices and RFs have similar contextual backgrounds, but RFs are more complex combinatorially.

## 8.5 Quantification and Anaphora in Dialogue

In Chapter 5, I introduced the *Reprise Content Hypothesis*, repeated here in both weak (98a) and strong (98b) versions:

- (98) a. *A fragment reprise question queries a part of the standard semantic content of the fragment being reprised.*
- b. *A fragment reprise question queries exactly the standard semantic content of the fragment being reprised.*

This was used to argue against a unification-based approach to semantic composition. In section 8.5.1, following the lead of Purver and Ginzburg (2004), I apply the criteria of the RCH to the analysis of nominal meaning, arguing that it provides data which, among other things, goes against a treatment of NPs as generalized quantifiers.

The quantificational approach goes back to Montague and remains one of the most popular analytical strategies.<sup>31</sup>

I then set out to provide a new formulation of Purver and Ginzburg's (HPSG-based) approach using TTR, with the aim of developing an account of the dynamics of grounding, clarification, and anaphora for discourse involving *both* referential and non-referential NPs. Integrating non-referential meaning into an account of grounding and CRification is important since otherwise one works under a distorted picture where every content is a contextual parameter that needs to be instantiated in context.

### 8.5.1 *Q-params and dgb-params*

Contrast (99a) with (99b) as uttered by Aaron to Belinda: the former is typically uttered with the strong expectation that Belinda can recognize who Jo is, whereas the latter involves use of the indefinite precisely because Aaron himself has at best only a relatively weak idea of the identity of the thief, and presumably does not expect Belinda to be in any more knowledgeable a position:

- (99) a. A: Jo arrived yesterday.  
b. A: A thief broke in here last night.

These contrasting expectations are reflected in the differing clarificational potentials of the two utterances, as we see when we examine possible clarification requests and responses thereto. In (100a), Belinda's question can only be understood as concerning Jo's identity; while in (100b), the corresponding question is very hard to interpret as asking about the thief's identity, but rather seems to ask about the property predicated of them (that of being a thief):

- (100) a. A: Jo arrived yesterday.  
B: Jo?  
A: Yes, that's right. / Oh, you know, my friend Jo from school.  
b. A: A thief broke in here last night.  
B: A thief?  
A: Yes, thief. / Well, burglar then, but certainly someone up to no good.

These differences can be captured by assuming that an utterance of the referential 'Jo' contributes to the DGB-PARAMS of the utterance, as we saw in previous chapters, whereas non-referential NPs such as 'a thief' do not. But, if they do not contribute to DGB-PARAMS, what do they contribute to semantically? In broad terms the answer is clear: non-referential NPs need to contribute their descriptive conditions to the sentential content, while any associated individuals are existentially quantified within the sentence. I introduce a Q-PARAMS field to indicate this existential quantification,

<sup>31</sup> This section is a significantly revised version of the main body of Ginzburg and Purver (2008). I would like to thank Matt Purver for his permission to use this material here, though all errors here are my own.

and so the representation of (100b) becomes (roughly, as we are as usual ignoring temporal arguments and tense) as in (101a).<sup>32</sup> A witness for this content is given in (101b): it requires finding a value for the contextual parameters *s0* and *l* (corresponding to the location ‘here’), **but not for the label *q-params.x*, which is in effect existentially quantified**:

$$\begin{array}{ll}
 (101) & \text{a. } \left[ \begin{array}{l} \text{dgb-params} : \left[ \begin{array}{l} s0 : \text{SIT} \\ l : \text{loc} \end{array} \right] \\ \\ \text{cont} = \left[ \begin{array}{l} \text{sit} = \text{sit0} \\ \text{sit-type} = \left[ \begin{array}{l} \text{q-params} : \left[ \begin{array}{l} x : \text{IND} \\ r2 : \text{thief}(x) \end{array} \right] \\ \text{nucl} : \text{break\_in}(\text{q-params}.x, l) \end{array} \right] \end{array} \right] : \text{Prop} \end{array} \right] \\
 & \text{b. } \left[ \begin{array}{l} \text{dgb-params} = \left[ \begin{array}{l} s0 = \text{sit0} \\ l = l0 \end{array} \right] \\ \\ \text{cont} = \left[ \begin{array}{l} \text{sit} = s0 \\ \text{sit-type} = \left[ \begin{array}{l} \text{q-params} : \left[ \begin{array}{l} x : \text{IND} \\ r2 : \text{thief}(x) \end{array} \right] \\ \text{nucl} : \text{break\_in}(\text{q-params}.x, l) \end{array} \right] \end{array} \right] \end{array} \right]
 \end{array}$$

Generalized Quantifier Theory (GQT) will allow us to arrive at a denotation like (101a), but only in a somewhat longwinded way—in a TTR version of the standard account, the denotation (102a) is posited for the NP ‘a thief’, which via  $\beta$ -reduction given a V denotation (102b) becomes (102c) (see Cooper 2004: 12).

$$\begin{array}{ll}
 (102) & \text{a. } \lambda R : \left[ \begin{array}{l} x : \text{IND} \end{array} \right] \text{RecType} \left[ \begin{array}{l} \text{par} : [x : \text{IND}] \\ \text{restr} : \text{thief}(x) \\ \text{scope} : R(\text{par}) \end{array} \right] \\
 & \text{b. } \lambda r : \left[ \begin{array}{l} x : \text{IND} \end{array} \right] \left[ \begin{array}{l} c0 : \text{break\_in}(r.x) \end{array} \right] \\
 & \text{c. } \left[ \begin{array}{l} \text{par} : [x : \text{IND}] \\ \text{restr} : \text{thief}(x) \\ \text{scope} : [c0 : \text{break\_in}(\text{par}.x)] \end{array} \right]
 \end{array}$$

<sup>32</sup> In fact, as we should account for the fact that the subconstituent common noun ‘thief’ must be grounded, and can be CRified as in (100b) above, we should make the associated noun predicate a member of DGB-PARAMS, but I will ignore this complication here.

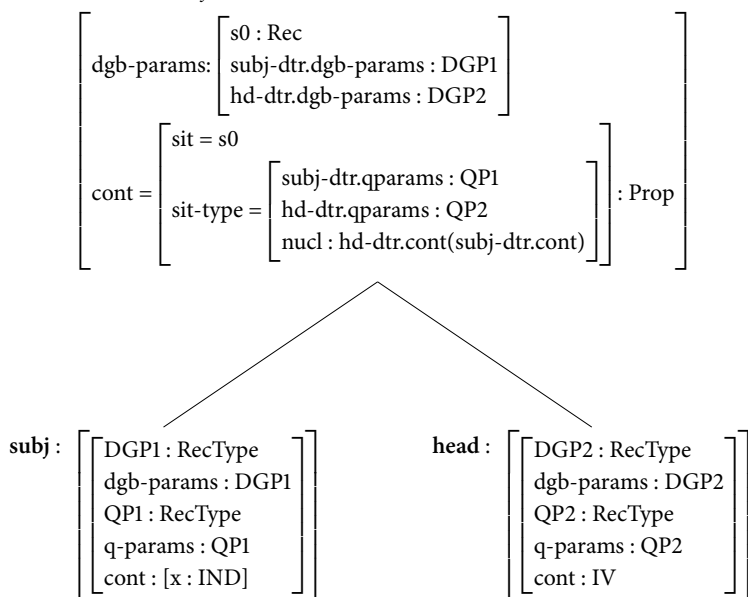
However, we could get to that route rather more directly by postulating (103) as the NP's contribution to content. Note also that we specify DGB-PARAMS as typed by the empty record type, i.e. the record type that involves no constraints.

$$(103) \left[ \begin{array}{ll} \text{dgb-params} & : [ ] \\ \text{q-params} & : \left[ \begin{array}{ll} x & : \text{IND} \\ r & : \text{thief}(x) \end{array} \right] \\ \text{cont} & : [x = \text{q-params}.x : \text{IND}] \end{array} \right]$$

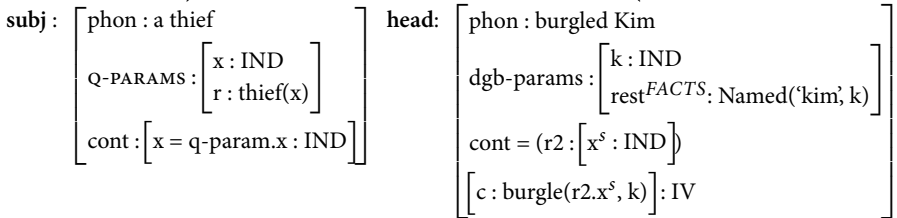
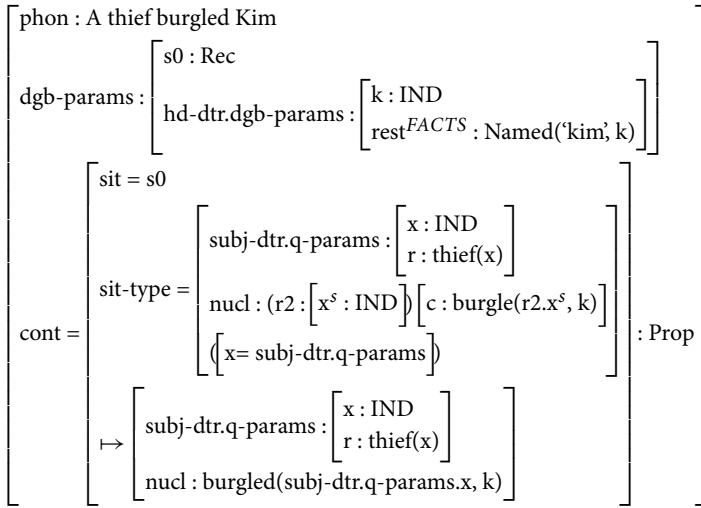
This allows us not only to arrive at a suitable overall representation, but to directly express the distinction between two types of NP via the distinction between inclusion in DGB-PARAMS or in Q-PARAMS. 'Referential' NPs (the reason for the scare quotes will become obvious shortly) help build up the contextual DGB-PARAMS component; 'non-referential' NPs help build the sentential content via Q-PARAMS; while both have contents of the same semantic type (individuals) and fill argument roles of a predicate uniformly.

A revised version of our rule for *decl-hd-subj-ph* from Chapter 5 is given in (104a). This incorporates amalgamation of both DGB-PARAMS and Q-PARAMS. A simple example is provided in (104b):

(104) a. *decl-hd-subj-cl* =<sub>def</sub>



b. A thief burgled Kim



In fact, this set up also allows for a fairly straightforward analysis of intensional verbs such as ‘seek’. (Cooper 2005a) proposes to treat sentences such as (105a) as relating an agent with a record type, as in (105b). This is because the witnessing conditions of the record type that fills the object argument role seem to describe well the success conditions of a search, as in Montague’s account:

- (105) a. Jill seeks a unicorn.  
 b.  $\text{seek}(j, \left[ \begin{array}{l} \text{x : IND} \\ \text{r : unicorn(x)} \end{array} \right])$

Cooper shows how to implement this analysis via a GQ analysis of NPs. On the current analysis, this would follow straightforwardly by having an intensional verb select for the Q-PARAMS type of its object.

In comparing a GQT-based approach and the non-GQ approach sketched so far simplicity is not a good guide: GQT has, arguably, a somewhat simpler synsem interface, but its denotations are more complex. If we take a dialogic perspective, however, things are not so equal. For a start, the simplicity GQT provides is conditional: it is maintained only in so far as indubitably<sup>33</sup> referential NPs are type-raised to the type of GQs, a move that falls foul of the RCH. However, if a GQ-oriented theory buys into distinctness of types among NPs, other problems come up. The first is that the simplicity of the synsem interface is lost—composing NPs and verbs is no longer possible by a single rule.<sup>34</sup>

The second is that reprise questions show no more evidence that GQ-like readings are available from non-referential NPs than they are from referential ones. Purver and Ginzburg (2004) present reprise data for indefinites and other quantified NPs, extracts of which are repeated in (106), none of which seem to permit anything other than the NP-referential or subconstituent (corresponding to the common noun phrase) readings we have already seen:

- (106) a. Unknown: What are you making?  
           Anon 1: Erm, it's a do– it's a log.  
           Unknown: A log?  
           Anon 1: Yeah a book, log book.
- b. Anon 2: Was it nice there?  
           Anon 1: Oh yes, lovely.  
           Anon 2: Mm.  
           Anon 1: It had twenty rooms in it.  
           Anon 2: Twenty rooms?  
                     [↘ *Is it **twenty** rooms you're saying it had?*]  
                     [↘ *Is it **twenty rooms** you're saying it had?*]  
                     [↘ *??Which **twenty rooms** are you saying it had?*]  
           Anon 1: Yes.
- c. Richard: No I'll commute every day.  
           Anon 6: Every day? [↘ *Is it **every** day you'll commute?*]  
                     [↘ *Is it **every day** you'll commute?*]  
                     [↘ *?Which **days** do you mean by **every day**?*]  
           Richard: as if, er Saturday and Sunday.  
           Anon 6: And all holidays?  
           Richard: Yeah [pause]

<sup>33</sup> In fact, there *are* reasons for doubt, as we will see shortly, if we take the addressee's perspective.

<sup>34</sup> Of course there are elegant ways to streamline the interface in type-driven frameworks, via type shifting etc.

So far, I follow Purver and Ginzburg (2004) closely apart from the difference in framework. However, the advantage of the type-theoretic approach developed here becomes clear when considering examples like (107). Here, while Aaron knows that the arriver was Jo, he uses an indefinite because he assumes Belinda might not have a name for Jo:

- (107) A: A friend of mine arrived yesterday.  
 B: A friend of yours?  
 A: Right – Jo, in fact.

Note that in such cases, it is perfectly easy to understand Belinda's question as asking about the friend being referred to—the reference of the NP 'a friend'. We can also imagine it as querying the appropriateness of the noun property 'friend', of course (in which case Aaron's response might be more along the lines of 'yes, I think I'd count her as a friend—certainly more than an acquaintance'). But while Purver and Ginzburg (2004) offer an information-structure-based account of how both the NP query and the noun property query might be available, they must assume that indefinites are ambiguous (between referential and non-referential versions) to explain the difference between (100b) and (107).

However, here one needs to make no such assumption. Cooper and Ginzburg (2002) (and Exercise 6.5.1 in Chapter 6) show how a type-theoretic approach lends itself to modelling the process of *accommodation* of a nominal reference—the process by which Belinda can process an example like (99a) even without knowing who Jo is, essentially by existentially quantifying Jo away. In current terms, what is needed is a simple coercion operation which allows (108b) to be produced from (108a):

- (108) a. 
$$\left[ \begin{array}{l} \text{dgb-params} : \left[ \begin{array}{l} x : \text{IND} \\ r : \text{named}('Jo', x) \end{array} \right] \\ \text{q-params} : [] \\ \text{cont.sit-type} : \text{arrive}(\text{dgb-params}.x) \end{array} \right]$$
- b. 
$$\left[ \begin{array}{l} \text{dgb-params} : [] \\ \text{q-params} : \left[ \begin{array}{l} x : \text{IND} \\ r : \text{named}('Jo', x) \end{array} \right] \\ \text{cont.sit-type} : \text{arrive}(\text{q-params}.x) \end{array} \right]$$

In the case of example (107), we merely require the opposite move: Belinda and Aaron, inferring or knowing that a particular friend is in fact being referred to, can coerce (108b) to (108a), allowing a referential CR to be asked and successfully interpreted.



Note that this move is available to us because the analysis maintains the same type (individuals) for both referential and non-referential versions.

**Exercise 8.5.1** Generalize the NSU constructions such as *decl-frag-cl* to accommodate QNPs.

**Exercise 8.5.2** Develop an analysis of Exclams (see section 7.6.1.4). Your analysis will require a lexical entry for an exclamative word (see e.g. Ginzburg and Sag 2000 for one proposal) and a means of accessing the FEC, which plays an important role in the semantics.

### 8.5.2 *Anaphora*

A related argument can be made from what has always been an important testing ground for any theory of NP meaning, namely anaphora. We consider for now *discourse anaphora*—anaphora that occur across sentential boundaries—returning to intrasentential anaphora, in particular its ‘bound variable’ variant, in section 8.5.4. The vast literature on quantification and anaphora has, with very few if notable exceptions (e.g. Groenendijk 1998, Asher and Lascarides 2003, Poesio and Traum 1997), been designed for monologue. Dialogue brings new challenges on this score: querying, disagreement, partial understanding.<sup>35</sup> Both referential and non-referential NPs give rise to discourse anaphora—the challenge is to provide a uniform theory. An account like the one we develop here, where referential and non-referential NPs are of the same semantic type—the level of individuals which directly provide the referents for anaphora—is at an advantage compared to a theory where witnesses need to be ‘extracted’ independently from GQ denotations.<sup>36</sup>

The main challenge for a theory of meaning for pronouns is of course how to characterize their antecedency conditions. Dialogue takes us away quite quickly from certain received ideas on this score. Example (109) indicates straight off that antecedents are not going to be located in the contextual repository of shared assumptions, namely FACTS, given the possibility of anaphora out of questions:

(109) A: Did John phone? B: He’s out of contact in Daghestan.

Another reason why FACTS is inappropriate relates to a more general issue: in theories like DRT and DPL (though not SDRT)—once an antecedent (for discourse anaphora), always an antecedent. This strategy of pooling all antecedents together once they become available is problematic in light of the fact, well noted in the AI literature on anaphora resolution (see e.g. Grosz and Sidner 1986), that changing the topic of conversation drastically changes anaphoric possibilities. Thus, in (110(5)) ‘he’ cannot

<sup>35</sup> Groenendijk (1998) and Asher and Lascarides (2003) abstract away from metacommunication and, for the most part, disagreement.

<sup>36</sup> See e.g. Cooper (2004: 26).

felicitously refer to Jake, despite the fact that the equivalent utterance using a directly referential expression is perfectly coherent:

- (110) A: Jake hit Bill. / B: No, he patted him on the back. / A: Ah. Is Bill going to the party tomorrow? / B: No. / A(5): Is #he/Jake?

This leaves QUD, Moves, or Pending. QUD we can dismiss since its elements lack sufficient structure. So we are left with Moves and/or Pending. (111a) is an example of anaphora from an ungrounded utterance, whereas (111b), repeated from section 8.2, is an example of anaphora from a disfluent utterance:

- (111) a. A: Did John phone? B: Is he someone with a booming bass voice?  
b. Peter was, well he was fired.

This suggests, then, that antecedents are to be located in both MOVES and PENDING. Where within the representation of an utterance? The obvious candidate is CONSTITS, which has fields for all sub-utterances, given that that yields a straightforward account of agreement between antecedent and anaphor. Such an account is unavailable to standard dynamic theories (including SDRT), where the dynamics is defined on a purely semantic level. This is particularly important in grammatical gender languages, where gender is not well correlated with the intrinsic characteristics of a referent, as illustrated in (112): (112a) shows grammatical gender agreement in German across turns. Similarly, in (112b, c), we see examples from Hebrew, which lacks a neuter gender and correspondingly has no neuter pronouns. There exist two words corresponding to English 'car', one is masculine, the other feminine. Subsequent pronominal reference must agree with the gender of the antecedent:

- (112) a. A: kommt jetzt ein Zug? B: Er kommt von Gleis 2.  
coming now a-sg-masc train-sg-masc? He comes from platform 2.  
A: Is there a train coming now? B: Yeah, it's coming from platform 2.  
b. A: ledani yeš mexonit yafa B: eyfo hu kana ota?  
to-Dani exists car-sg-fem nice-sg-fem Where he bought her-sg-fem?  
A: Dani has a nice car. B: Where did he buy it?  
c. A: ledani yeš oto yafe B: eyfo hu kana oto?  
to-Dani exists car-sg-masc nice-sg-masc Where he bought him-sg-masc?  
A: Dani has a nice car. B: Where did he buy it?

Given that all NPs are here of individual type, grounding both referential NP and non-referential NP utterances involves finding witnesses for their content. This is illustrated in (113): an utterance type including a QNP sub-utterance type is given in (113a), while a token for this type is given in (113b). The pertinent semantic fields are boldfaced:

$$\begin{array}{l}
(113) \quad a. \quad \left[ \begin{array}{l}
\text{phon : a thief fleeced jo} \\
\text{cat.head} = V[+\text{fin}, +\text{root}] : \text{PoS} \\
\text{constits} = \left\{ \begin{array}{l}
\text{u}_{\text{a thief}} : \left[ \begin{array}{l}
\text{phon : a thief} \\
\text{cat} = N : \text{PoS} \\
\text{q-params} : \left[ \begin{array}{l}
x : \text{IND} \\
r1 : \text{thief}(x)
\end{array} \right] \\
\text{cont} : [x = \text{q-params.x} : \text{IND}]
\end{array} \right], \\
\text{u}_{\text{fleeced}} : \left[ \begin{array}{l}
\text{phon : fleeced} \\
\text{cat} = V[+\text{fin}] : \text{PoS} \\
\text{cont} = (r1 : [x^c : \text{Ind}]) (r2 : [x^s : \text{IND}]) \\
[c : \text{fleece}(r2.x^s, r1.x^c)] : \text{TV}
\end{array} \right], \\
\text{u}_{\text{jo}} : \left[ \begin{array}{l}
\text{phon : jo} \\
\text{cat} = N : \text{PoS} \\
\text{dgb-params} : \left[ \begin{array}{l}
x : \text{Ind} \\
r1 : \text{named}(\text{jo})(x)
\end{array} \right] \\
\text{cont} : [x = \text{dgb-params.x} : \text{IND}]
\end{array} \right]
\end{array} \right\} \\
\text{dgb-params} : \left[ \begin{array}{l}
\text{spkr} : \text{Ind} \\
\text{addr} : \text{Ind} \\
\text{u}_{\text{jo}}.\text{cont.x} : \text{IND} \\
\text{s0} : \text{Sit}
\end{array} \right] : \text{RecType} \\
\text{cont} = \text{Assert}(\text{spkr}, \text{addr}, \\
\left[ \begin{array}{l}
\text{sit} = \text{s0} \\
\text{sit-type} = \left[ \begin{array}{l}
\text{q-params} : \left[ \begin{array}{l}
x : \text{IND} \\
r2 : \text{thief}(x)
\end{array} \right] \\
\text{nucl} : \text{fleece}(\text{q-params.x}, \text{u}_{\text{jo}}.\text{cont.x})
\end{array} \right] \end{array} \right] : \text{IllocProp}
\end{array} \right]
\end{array}
\end{array}$$

$$\begin{array}{l}
 \text{b. } \left[ \begin{array}{l}
 \text{phon} = \text{athieffleecedjo} \\
 \text{cat} = \text{V}[+\text{fin}, +\text{root}] \\
 \\
 \text{constits} = \left\{ \begin{array}{l}
 \text{u}_{\text{a thief}} = \left[ \begin{array}{l}
 \text{phon} = \text{athief} \\
 \text{cat} = \text{N} \\
 \text{q-params} = \left[ \begin{array}{l}
 \text{x} = \text{th1} \\
 \text{r1} = s_{\text{thief}(\text{th1})}
 \end{array} \right] \\
 \text{cont} = [\text{x} = \text{th1}]
 \end{array} \right], \\
 \\
 \text{u}_{\text{fleeced}} = \left[ \begin{array}{l}
 \text{phon} = \text{fleeced} \\
 \text{cat} = \text{V} \\
 \text{cont} = (\text{r1} : [\text{x}^c : \text{IND}]) (\text{r2} : [\text{x}^s : \text{IND}]) \\
 \left[ \text{c} : \text{fleece}(\text{r2.x}^s, \text{r1.x}^c) \right]
 \end{array} \right], \\
 \\
 \text{u}_{\text{jo}} = \left[ \begin{array}{l}
 \text{phon} = \text{jo} \\
 \text{cat} = \text{N} \\
 \text{dgb-params} = \left[ \begin{array}{l}
 \text{x} = \text{j1} \\
 \text{r1} = s_{\text{named}(\text{jo})(\text{j1})}
 \end{array} \right] \\
 \text{cont} = [\text{x} = \text{j1}]
 \end{array} \right]
 \end{array} \right\} \\
 \\
 \text{dgb-params} = \left[ \begin{array}{l}
 \text{spkr} = \text{A} \\
 \text{addr} = \text{B} \\
 \text{u}_{\text{jo}}.\text{cont.x} = \text{j1} \\
 \text{s0} = \text{sit0}
 \end{array} \right] \\
 \\
 \text{cont} = \text{Assert}(\text{spkr}, \text{addr}, \left[ \begin{array}{l}
 \text{sit} = \text{s0} \\
 \text{sit-type} = \left[ \begin{array}{l}
 \text{q-params} : \left[ \begin{array}{l}
 \text{x} : \text{IND} \\
 \text{r2} : \text{thief}(\text{x})
 \end{array} \right] \\
 \text{nucl} : \text{fleece}(\text{q-params.x}, \text{u}_{\text{jo}}.\text{cont.x})
 \end{array} \right]
 \end{array} \right] )
 \end{array} \right]
 \end{array}
 \right]
 \end{array}$$

Nonetheless the referential witnesses and the non-referential witnesses have a distinct status in a number of respects. In contrast to the referential utterances, non-referential utterances do not contribute referents to sentential content, as we have seen and is illustrated in (113). Moreover, non-referential witnessing has a distinct status from finding witnesses for DGB-PARAMS, as reflected in the different CRIfication status of referential and non-referential utterances: witnessing the former carries with it a condition that the witnesses introduced are those intended by the speaker. The witnesses for non-referential utterances can be viewed as ‘arbitrary

individuals'. In some cases, this remains their status, in line with the fact that for non-referential NPs anaphora is not generally possible from within a query (polar or *wh*), as originally pointed out in Groenendijk (1998), or from an assertion that has been rejected:<sup>37</sup>

- (114) a. A: Do you own an apartment? B: Yes. A: Where is it located?  
 b. A: Do you own an apartment? B: No. A: #Where might you buy it?

This means, naturally enough that quasi-referential witnesses to QNPs can only emerge in a context where the corresponding assertion has been accepted. A natural move to make in light of this is to postulate a process in which these witnesses get incorporated into the DGB-PARAMS of that utterance as a side effect of assertion acceptance, a consequence of which will be the emergence of referents for discourse anaphora to QNPs. This means that DGB-PARAMS serves uniformly as the locus for antecedents of discourse anaphora. The witnesses to genuinely non-referential utterances such as *wh*-phrases still have utility in intrasentential anaphora, discussed below, and in subordination phenomena, which I do not discuss here.

The rule of incorporating non-referential witnesses in DGB-PARAMS is actually simply a minor add-on to Accept move (see (42), Chapter 4), the rule underpinning the utterance of acceptances—it can be viewed as providing for a witness for situation/event anaphora since this is what gets directly introduced into *dgb-params*. In (115) the preconditions involve the fact that the speaker's latest move is an assertion of a proposition whose type is T1. The effects change the speaker/addressee roles (since the asserted to becomes the acceptor) and adds a record *w*, including a witness for T1,

<sup>37</sup> The data here are quite subtle. Consider the following clearly felicitous anaphor:

- (i) A: Did a man with a limp pass by here a few minutes ago?  
 B: Was he squat with a red beard?

This would seem to be a *specific* indefinite, whose existence is not at issue, and would constitute a contextual parameter, as discussed above with respect to example (107).

Similarly, in (ii), based on examples due to Dekker (1997), anaphora is possible despite the rejection of the assertion because B has referential access to the fake priest:

- (ii) A: A priest was looking for you. B: He's not a priest, just an actor, and I doubt he wanted to see me.

Compare this with the following somewhat similar cases: (iii) seems infelicitous, though (iii') seems acceptable, but involves modal subordination—(iv) is felicitous only to the extent that it seems as though B has perhaps grudgingly changed his tune after his initial utterance.

- (iii) A: Mike, we are getting reports of an attempted escape at Wandsworth prison. Did you see anyone wearing a striped jump suit running near the park around 9:30? Mike: # Was he wearing a hat on as well?  
 (iii') Would he have been wearing a hat?  
 (iv) A: A priest was looking for you. B: No way—priests don't come looking for a grubby atheist like me.  
 # What did he want?

to the contextual parameters. As with any grounding process, one needs to include an additional condition—not just any witnesses are acceptable, the desired ones are those intended by the speaker (see Chapter 6):

(115) Accept move =<sub>def</sub>

$$\left[ \begin{array}{l} \text{pre} \\ \text{effects} \end{array} : \left[ \begin{array}{l} \text{spkr} : \text{IND} \\ \text{addr} : \text{IND} \\ p = \left[ \begin{array}{l} \text{sit} = \text{sitl} \\ \text{sit-type} = \text{T1} \end{array} \right] : \text{Prop} \\ \text{LatestMove}^{\text{content}} = \text{Assert}(\text{spkr}, \text{addr}, p) : \text{IllocProp} \\ \text{spkr} = \text{pre.addr} : \text{IND} \\ \text{addr} = \text{pre.spkr} : \text{IND} \\ t : \text{T1} \\ w = \text{pre.LatestMove.dgb-params} \cup [\text{sit} = t] : \text{Rec} \\ p1 = \text{IntendClassify}(\text{spkr}, w, m0.\text{dgb-params}) : \text{Prop} \\ c2 : \text{member}(p1, \text{private.beliefs}) \\ \text{Moves} = m1 \oplus \text{pre.Moves} : \text{list}(\text{LocProp}) \\ m1^{\text{content}} = \text{Accept}(\text{spkr}, \text{addr}, p) : \text{IllocProp} \\ m0.\text{dgb-param} = w : \text{Rec} \end{array} \right] \right]$$

An antecedent for a pronoun is to be sought in an *active* move. What is an active move? Clearly this is an intricate notion, but the data we saw here suggests a rough approximation drawing on our dialogue-oriented conceptualization of the right frontier constraint (see section 8.3): the ‘right frontier’ is constituted by elements of QUD, answers to such (TOPICAL facts), and utterances under grounding/correction:

- (116) For a given DGB dgb0, an ActiveMove is an element of dgb0.Moves or dgb0.Pending such that either (a) *qud-contribution*( $m^{\text{content}}$ ) is in dgb0.QUD or  
 or  
 (b)  $m^{\text{content}}$  is TOPICAL or  
 (c)  $m$  is in dgb0.Pending

We can now state the meaning of a singular pronoun as follows: an antecedent is to be sought in the CONSTITITS of an *active move*; the pronoun is identical in reference to this antecedent and agrees with it.<sup>38</sup> More technically, the DGB-PARAMS of a singular

<sup>38</sup> Agreement is a more general property of antecedents, which should be factored out. I include it here in the mix to ensure that the right kind of information is available for this task. Here agreement ranges over {mascsg, femsg, neutsg}, with a variety of cross-linguistic differences. The modelling of agreement here follows the approach of Cooper (2008).

pronoun specify an active move *m* and a constituent of that move *a* (the antecedent). The antecedent *a* is then constrained to be identical in reference with the pronoun and to have an identical agreement value:

$$(117) \left[ \begin{array}{l} \text{phon : she} \\ \\ \text{dgb-params : } \left[ \begin{array}{l} m : \text{LocProp} \\ a : \text{LocProp} \\ c1 : \text{member}(a, m.\text{constits}) \\ c2 : \text{ActiveMove}(m) \\ m.\text{sit.dgb-params} : [a.\text{dgb-params}.x = a.\text{cont}.x : \text{IND}] \end{array} \right] \\ \\ \text{cat = } \left[ \begin{array}{l} \text{head} = N : \text{PoS} \\ \text{ana} : + \\ \text{agr} = \text{dgb-params}.a.\text{sit-type.cat.agr} : \left[ \begin{array}{l} \text{num} = \text{sg} : \text{Number} \\ \text{gen} = \text{fem} : \text{Gender} \\ \text{pers} = \text{third} : \text{Person} \end{array} \right] \end{array} \right] \\ \\ \text{cont : } [x = a.\text{sit.cont}.x : \text{IND}] \end{array} \right]$$

For notational simplicity, we abbreviate the DGB-PARAMS type in (117) as (118a)—here *m* is the move and *a* the antecedent utterance. We can thereby rewrite (117), omitting the agreement specification, as (118b):

$$(118) \quad \begin{array}{l} \text{a. } \left[ \begin{array}{l} m : \text{LocProp} \\ a : \text{LocProp} \\ c1 : \text{member}(a, m.\text{constits}) \\ c1 : \text{DiscourseAnt}(m, a) \end{array} \right] \\ \\ \text{b. } \left[ \begin{array}{l} \text{dgb-params : } \left[ \begin{array}{l} m : \text{LocProp} \\ a : \text{LocProp} \\ c1 : \text{member}(a, m.\text{constits}) \\ c2 : \text{DiscourseAnt}(m, a) \end{array} \right] \\ \\ \text{cont : } [x = a.\text{cont}.x : \text{IND}] \end{array} \right] \end{array}$$

How will this meaning, aligned with the acceptance move rule, enable us to explicate the cases we have seen? Consider (114a): accepting ‘yes’ will result in augmenting

the DGB-PARAMS of 'yes' with a witness for  $\left[ \begin{array}{l} a : \text{IND} \\ c1 : \text{addressee}(a) \\ x : \text{IND} \\ c2 : \text{apartment}(x) \\ c3 : \text{have}(a, x) \end{array} \right]$ , which in particular

provides an entity constrained to be an apartment which can serve as antecedent for

'it'. In contrast, for (114b), a witness for the type  $\rightarrow \left[ \begin{array}{l} a : \text{IND} \\ c1 : \text{addressee}(a) \\ x : \text{IND} \\ c2 : \text{apartment}(x) \\ c3 : \text{have}(b, x) \end{array} \right]$  is some type

that is incompatible with  $\left[ \begin{array}{l} a : \text{Ind} \\ c1 : \text{addressee}(a) \\ x : \text{Ind} \\ c2 : \text{apartment}(x) \\ c3 : \text{have}(b, x) \end{array} \right]$ .

This leaves things open-ended: the addressee could be renting or homeless or living in a hotel room, but without saying more there is no concrete witness. As far as (109) goes, a referent for 'John' is in dgb-params once the query is grounded, and this is available as an antecedent since the query is at that point an active move. In (111) we assume that this is a case where CRification is performed not to *find* a missing contextual value but to *confirm* it. Hence, PENDING has a value for 'John'. Given the presence of utterance antecedents in CONSTITS, we note also an explanation for examples like (119), where a pronoun is CRed and the answers mentioned are the possible antecedents:

- (119) A: A teacher gave a parent a book from the school library. He liked it a lot. B:  
The teacher or the parent?

Finally, to illustrate the workings of the notion *active move*, I offer a trace through (120) by considering the DGB-PARAMS of active moves. After (1), (1) constitutes the active move, with two potential pronominal antecedents, Jake and Bill. After (2) is accepted the question posed in (1) is no longer in QUD, so the move is not active. (2) is deactivated once (4) gets uttered: at that point Bill is the sole pronominal antecedent. After (5) is accepted 'a party' also becomes an antecedent:

- (120) A(1): Jake hit Bill. / B(2): No, he patted him on the back. / A(3): Ah. (4) Is Bill going to a party tomorrow? / B(5): Yes. / A(6): I see. (7) Is #he/Jake?



|       |      |                                        |                                                                                                                                                                                                                                                                                                     |
|-------|------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (121) | Utt. | active moves                           | DGB-PARAMS TOKEN                                                                                                                                                                                                                                                                                    |
|       | (1)  | (1)                                    | $\left[ \begin{array}{l} \text{spkr} = A \\ \text{addr} = B \\ c1 = s_{\text{address}}(s, a) \\ j = j0 \\ \text{rest}^{\text{facts}}_1 = s_{\text{named}}(j0, \text{'jake'}) \\ b = b0 \\ \text{rest}^{\text{facts}}_2 = s_{\text{named}}(b0, \text{'bill'}) \end{array} \right]$                   |
|       | (2)  | (1),(2)                                | (2).dgb-param = $\left[ \begin{array}{l} \text{spkr} = A \\ \text{addr} = B \\ c1 = s_{\text{address}}(s, a) \\ h1 = j0 \\ \text{rest}^{\text{facts}}_1 = s_{\text{named}}(j0, \text{'jake'}) \\ h2 = b0 \\ \text{rest}^{\text{facts}}_2 = s_{\text{named}}(b0, \text{'bill'}) \end{array} \right]$ |
|       | (3)  | (2)                                    |                                                                                                                                                                                                                                                                                                     |
|       | (4)  | (4)                                    | $\left[ \begin{array}{l} \text{spkr} = A \\ \text{addr} = B \\ c1 = s_{\text{address}}(s, a) \\ b = b0 \\ \text{rest}^{\text{facts}}_2 = s_{\text{named}}(b0, \text{'bill'}) \end{array} \right]$                                                                                                   |
|       | (5)  | (4), (5)                               |                                                                                                                                                                                                                                                                                                     |
|       | (6)  | (5) (with<br>witness for 'a<br>party') | $\left[ \begin{array}{l} \text{spkr} = A \\ \text{addr} = B \\ c1 = s_{\text{address}}(s, a) \\ b = b0 \\ \text{rest}^{\text{facts}}_2 = s_{\text{named}}(b0, \text{'bill'}) \\ p = p0 \\ c4 = s_{\text{party}}(p0) \end{array} \right]$                                                            |

8.5.3 *Plural NPs*

So far, the discussion has all involved singular NPs, both referential and non-referential. How does this approach carry over to plurals? With referential plurals,

the transition seems clear: we simply need plural NPs to denote sets of individuals rather than individuals,<sup>39</sup> and this seems perfectly in keeping with the RCH as long as this set is a member of DGB-PARAMS (see (122a)). Predicating the VP content directly of the set must result in a collective reading, as in (122b) where the set of students left together; distributive readings can equally be handled, as I discuss in section 8.5.4.

- (122) a. John: they'll be working on the, they'll be working on the kidnapper's instructions though wouldn't they? They would be working on the kidnapper's instructions, the police?

Sid: The police? [ $\rightsquigarrow$  *Who do you mean by 'the police'?*]

[ $\rightsquigarrow$  *Is it the police who you are saying would be working ...?*]

John: Aye

Sid: On

Unknown: [unclear]

Sid: aye the, the senior detectives

- b. 'The students left'

$$\left[ \begin{array}{l} \text{dgb-params} : \left[ \begin{array}{l} s : \text{Set}(\text{IND}) \\ r : \text{student}'(s) \end{array} \right] \\ \text{cont} = \left[ \begin{array}{l} \text{sit} = \text{so} \\ \text{sit-type} = \text{left}(\text{dgb-params}.s) \end{array} \right] \end{array} \right]$$

Turning to non-referential NPs, so far the discussion has concentrated on indefinites, capturing their quantificational force implicitly by the witnessing relation between records and types. It seems clear that the same approach will work here: examples such as (106b) above can be expressed via a suitable set member of Q-PARAMS, and a witness for the record type will require the existence of a suitable set (here, of twenty rooms). And as before, a simple coercion operation between inclusion in DGB-PARAMS and Q-PARAMS can then account for any shift between truly non-referential and specific uses.

And one can apply this approach to other quantified plurals as well. Barwise and Cooper (1981) provide us with the notion of a *witness set* for a GQ, and show an equivalence (for monotone-increasing quantifiers) between a GQ holding of a verbal predicate and that predicate holding of the witness set. Examples such as (123a) can thus be analysed—on its collective understanding—by taking the NP to denote its witness set directly (and for this to be a member of Q-PARAMS or DGB-PARAMS as the

<sup>39</sup> Or sums, groups, etc. see e.g. Lönning (1997).

(non-)referentiality of its use demands).<sup>40</sup> An analysis of the former case is provided in (123b).

(123) a. Most students left

$$b. \left[ \begin{array}{l} \text{sit} = s1 \\ \text{sit-type} = \left[ \begin{array}{l} \text{q-params} : \left[ \begin{array}{l} x : \text{Set(IND)} \\ r : \text{most}(x, \text{student}) \end{array} \right] \\ \text{nucl} : \text{left}(\text{q-params}.x) \end{array} \right] \end{array} \right]$$

here  $\text{most}(x, y)$  holds if  $x \subset y$  and  $x$  contains a majority of the  $y$ s

This approach is not sufficient, of course, for monotone-decreasing (e.g. ‘few students’) or non-monotone (‘exactly two students’) quantifiers: as Barwise and Cooper (1981) point out, we must show not only that the verb predicate holds of the witness set, but that it holds of no other members of the restriction set. However, Purver and Ginzburg (2004) propose an analysis of such QNPs as denoting *pairs* of reference set (the few students who left) and complement set (the students who didn’t), noting that this might also help explain the phenomenon of complement set anaphora by providing suitable antecedents – and such an approach could equally be developed in the current approach, as exemplified in (124):

(124) a. Few students left.

$$b. \left[ \begin{array}{l} \text{sit} = s1 \\ \text{sit-type} = \left[ \begin{array}{l} \text{q-params} : \left[ \begin{array}{l} \text{refset} : \text{Set(IND)} \\ \text{compset} : \text{Set(IND)} \\ \text{c1} : \text{partition}(\text{refset}, \text{compset}, \text{student}) \\ \text{c2} : \text{few}(\text{refset}, \text{student}) \end{array} \right] \\ \text{nucl} : \text{left}(\text{q-params}.\text{refset}) \wedge \neg \text{left}(\text{q-params}.\text{compset}) \end{array} \right] \end{array} \right]$$

here  $\text{few}(x, y)$  holds if  $x \subset y$  and the cardinality of  $x$  is small

#### 8.5.4 Scope Ambiguity

Scope ambiguities are among the most studied semantic indeterminacies. GQT provides a particularly simple analysis of these: the ambiguity is explained in terms of the distinct order of application of GQs.

I have already argued against one of the components of this account in section 8.5.1, pointing out that there is dialogue evidence against the higher-order denotations

<sup>40</sup> See, for instance, the possible referential reading given for (106c) above, or examples like:

(i) A: Most folks were upset by the decision.

B: Most folks?

A: Yeah well like Bianca, Amanda, Luigi, and me. That’s what? About two thirds of the company?

GQT postulates for QNPs. As I pointed out in Chapter 6, a significant finding that has emerged from recent corpus studies of CRs in dialogue (see Purver, Ginzburg, and Healey 2001, Rodriguez and Schlangen 2004, Rieser and Moore 2005) is the complete absence of CRs that somehow relate to uncertainty regarding scope. One might draw various conclusions from this, but perhaps as significant is what emerges when one *constructs* CRs that relate to scopal uncertainty: the ambiguity appears to be localized in an NP:

- (125) a. A: The boys kept a cat.  
           B: One cat for all the boys or different ones?  
           A: They each kept a cat.  
       b. A: I'm going to give all you kids a present.  
           B: Will we need to share or do we each get something for ourselves?

Note that contrary to what underspecification theories (such as Minimal Recursion Semantics (Copestake, Flickinger, Pollard, and Sag 2005) or Hole Semantics (Bos 1995)) would suggest, this data indicates that the various scope possibilities *are* computed and can be localized with a single NP; see Fox and Lappin (2009) for an approach to underspecification consistent with the former of these data.<sup>41, 42</sup> Disambiguation similarly proceeds by using distributive adverbs and adjectives ('each', 'the same/different cat' etc.). GQT provides no help in localizing the ambiguity since the meanings of QNPs are constant across distinct readings.

The alternative sketched here is to distinguish between an independent use of an QNP and a *dependent* use. In the latter case, the NP contributes a function to Q-PARAMS, whose value on an argument fills the role associated with the NP.<sup>43</sup> Such uses have been argued to exist on a number of independent grounds so eliminating scope alternation as a mechanism has added attractions:

- **Functional readings in questions/answers:** ever since Engdahl first pointed out that interrogatives such as (126a) give rise to answers such as (126b) (see Engdahl 1986), it has been recognized that questions can specify *dependencies*. More controversially, some have argued that pair-list answers such as (126c) should be analysed in similar terms (see Ginzburg and Sag 2000):

- (126) a. Which problem did each scientist solve?  
       b. The one his supervisor assigned him.  
       c. Dennett solved consciousness, Chomsky solved Plato's problem, Dawkins solved religion ...

<sup>41</sup> Fox and Lappin, who presuppose a GQ treatment of NPs, develop an approach which enumerates all possible readings for a sentence by specifying a family of functions  $PermsScope_k$  (where  $k > 1$ ) that generate all  $k!$  indexed permutation products of a  $k$ -ary indexed product term (the  $k$  QNPs in the sentence).

<sup>42</sup> Admittedly, there are only two scopings possible here. But there again, it is a defensible hypothesis that in spoken language at least, there are rarely more possibilities than this, at least as far as QNPs.

<sup>43</sup> Since this is merely a sketch, I do not address scope ambiguities due to sentence embedding, implicated e.g. in *de dicto* / *de re* bifurcations.

- **Dependent adjectives:** certain adjectives are intrinsically relational, requiring either a discourse antecedent or a QNP (see e.g. Keenan and Westerståhl 1997):

- (127) a. # (Requires a context establishing the existence of a prior book) Jo gave the girls a different book.  
 b. (OK in neutral context) Each teacher gave the girls a different book.

- **Intrinsically narrow scope QNPs:** Certain QNPs like ‘at least N’ seem never to take ‘wide scope’, as exemplified in (128), which lacks a reading asserting that all linguists speak, say, English and German. Steedman (1999a) observes that this can be captured by assuming that such NPs are obligatorily dependent or at least can never serve as antecedents for dependent uses.

- (128) Every linguist speaks at least two languages.

Let us start with a simple example, as in (129a). We associate with this the content in (129b): Q-PARAMS specifies a function  $f$  from individual-providing records to (types of) cat-including situations and the nucleus provides for a mapping linking each boy (from the contextually provided sets of boys  $B$ ) to the cat he keeps (each such cat is the witness for the  $z$  field in the cat-including situation from Q-PARAMS):

- (129) a. The boys each kept a cat.

$$\begin{array}{l}
 \text{b.} \left[ \begin{array}{l} \text{dgb-params : } \left[ \begin{array}{l} s : \text{Rec} \\ B : \text{Set}(\text{Ind}) \\ r : \text{boy}(s) \end{array} \right] \\ \\ \text{cont} = \left[ \begin{array}{l} \text{sit} = s \\ \text{sit-type} = \left[ \begin{array}{l} \text{q-params : } \left[ \begin{array}{l} f : \left[ \begin{array}{l} x : \text{IND} \end{array} \right] \left[ \begin{array}{l} z : \text{IND} \\ c1 : \text{cat}(z) \end{array} \right] \end{array} \right] \\ \text{nucl : } r : \left[ \begin{array}{l} x : \text{Ind} \\ c1 : \text{In}(\text{dgb-params}.B, x) \end{array} \right] [c0 : \text{kept}(r.x, f(r).z)] \end{array} \right] \end{array} \right] \end{array} \right]
 \end{array}$$

The content in (129b), arises in a straightforward fashion, using rules like those postulated in section 8.5.1: ‘kept’ predicates of the content of ‘a cat’, ‘each’ acts as a distributive adverb on the VP, which then predicates of the set-valued subject.<sup>44</sup> The Q-PARAMS values of the two NPs get merged into a joint record type, which is a constituent of the content.

- (130) a. kept a cat  $\mapsto$
- $$\left[ \begin{array}{l} x^s : \text{Ind} \end{array} \right] \left[ \begin{array}{l} \text{q-params : } \left[ \begin{array}{l} f : \left[ \begin{array}{l} x : \text{IND} \end{array} \right] \left[ \begin{array}{l} z = f(x) : \text{IND} \\ c1 : \text{cat}(z) \end{array} \right] \end{array} \right] \\ \text{cont : } [c0 : \text{kept}(x^s, f(x^s))] \end{array} \right]$$

<sup>44</sup> This treatment of ‘each’ derives from Link (1983), via Winter (1999).

$$\text{b. each} \mapsto (T \left( \left[ \begin{array}{c} x^s : \text{Ind} \end{array} \right]_{RType} \right) (z : \text{Set}(\text{Ind})) (r : \left( \left[ \begin{array}{c} x : \text{Ind} \\ \text{cl} : \text{In}(z, x) \end{array} \right] \right) T(r.x^s))$$

From this, it becomes clear in all but one respect how to specify the meaning of a dependent use of ‘a cat’. The one remaining issue, one which is typically abstracted away from, but which we cannot do in TTR<sup>45</sup>—is how to specify  $x$ , the independent variable which appears in the content of the dependent NP? One way to do so is to specify the independent variable to be the content of a contextually provided (but intrasentential) sub-utterance with a non-null Q-PARAMS value. What is right about this is that it leads such dependence to be fixed by the speaker’s intentions, since the grammar does not do this. The primary challenge this strategy faces is how to specify *intrasententiality* (without introducing within the dependent utterance reference to itself). This can be accomplished by exploiting the proposal in section 8.2 to extend Pending to ongoing utterances. Given that, a sub-utterance intrasentential with the dependent utterance is simply a constituent of MaxPending.<sup>46</sup>

In light of this, we can specify a dependent use as in (131):

(131) Dependent use of ‘a cat’:

$$\left[ \begin{array}{ll} \text{MaxPending} & : \text{LocProp} \\ \text{dgb-params} & : \left[ \begin{array}{ll} \text{ant} & : \left[ \text{q-params} : [x : \text{IND}] \right] \\ \text{c} & : \text{member}(a, \text{MaxPending.constits.sit}) \end{array} \right] \\ \text{q-params} & : \left[ \begin{array}{c} f : \left( \left[ \begin{array}{c} x : \text{Ind} \end{array} \right] \right) \left[ \begin{array}{c} z = f(x) : \text{IND} \\ \text{cl} : \text{cat}(z) \end{array} \right] \end{array} \right] \\ \text{cont} & : \left[ x = \text{q-params}.f(\text{ants.q-param}.x) : \text{IND} \right] \end{array} \right]$$

A similar analysis will work for (132a): the sole difference is that in such a case the distributive operator is implicit—(132b) suggests we do not want to build it into the NP meaning. A number of ways have been proposed for incorporating distributivity into the combinatorial process, an issue we cannot resolve here. For discussion and a type-logical account see Winter (1999).

<sup>45</sup> There being no tolerance for free variables.

<sup>46</sup> One alternative which avoids integrating incrementality into the grammar would be to keep track of Q-PARAMS-bearing NPs within a sentence by means of a ‘long-distance dependency’, equating intrasentential non-referential antecedent with membership of this set. An option used in SDRT (Asher and Lascarides 2003) and within TTR (Cooper 2004) for specification of pronoun antecedents is to use a metavariable for the antecedent and to invoke some principle of resolution of DRSs or types. This is a viable possibility, which is combinatorially simple. Such a move is methodologically questionable: for a start, it is preferable to remain on the object level in so far as possible. Moreover, it is not clear why pronouns should require such a move, as opposed to various other context-dependent elements. This criticism applies *mutatis mutandis* to their use in the current context.

- (132) a. Most villagers owned a cat.  
 b. Most villagers own a cat and gathered in the square to listen to the visiting ghattologist.

How then to account for clarificational interaction as in (125a), repeated here as (133)?

- (133) A(1): The boys kept a cat.  
 B(2): A cat? One cat for all the boys or different ones?  
 A(3): They each kept a cat.

In Chapter 6 I discussed the need to underspecify grammatical types in the context of repetition-oriented CRification. In such cases, categorial or phonemic uncertainty accompanies referential uncertainty. In this case, the ambiguity between an independent and dependent use leads to referential uncertainty at the sentential level, viz. which content is the intended one—there is also localized referential ambiguity given the need to resolve the sub-utterance from which the independent variable emanates if a dependent reading is intended. We can accommodate such cases using either of the two possibilities I entertained in section 8.3, namely by identifying sentential utterance meaning with the intended private meaning (the maximal element of the agent's agenda) or by positing a contextual parameter for the sentential content.

Adopting for now the latter, this yields (134a) as a possible locutionary proposition for the utterance, a proposition which fails to be true given that the sentential content parameter is not instantiated. This triggers CRification. An answer that is provided can then be used, via pending extension (section 6.5.1), to obtain a locutionary proposition that is stronger than (134a) and which is true, e.g. (134d), if the independent use of 'a cat' was intended:

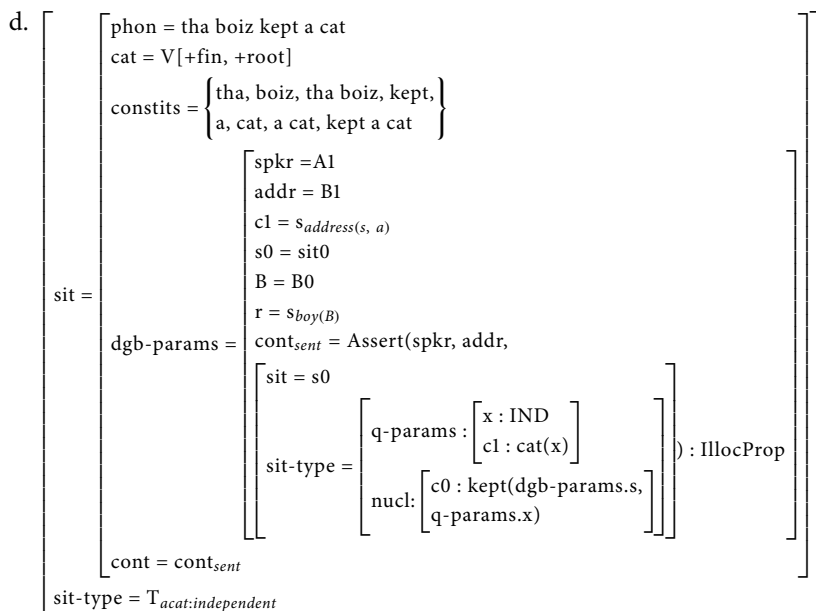
- (134) a. Locutionary proposition for utterance:

$$\left[ \begin{array}{l} \text{sit} = \left[ \begin{array}{l} \text{phon} = \text{tha boiz kept a cat} \\ \text{cat} = V[+\text{fin}, +\text{root}] \\ \text{constits} = \left\{ \begin{array}{l} \text{tha, boiz, tha boiz, kept,} \\ \text{a, cat, a cat, kept a cat} \end{array} \right\} \\ \text{dgb-params} = \left[ \begin{array}{l} \text{spkr} = A1 \\ \text{addr} = B1 \\ \text{cl} = s_{\text{address}(s, a)} \\ \text{s0} = \text{sit0} \\ B = B0 \\ r = s_{\text{boy}(B)} \end{array} \right] \end{array} \right] \\ \text{sit-type} = T_{\text{acat:dependent}} \vee T_{\text{acat:independent}} \end{array} \right]$$

$$\begin{aligned}
 \text{b. } T_{\text{acat:dependent}} = & \left[ \begin{array}{l} \text{phon : the boys kept a cat} \\ \text{cat} = V[+\text{fin}, +\text{root}] : \text{syncat} \\ \text{constits} = \left\{ \begin{array}{l} \text{the, boys, the boys, kept,} \\ \text{a, cat, a cat, kept a cat} \end{array} \right\} : \text{set}(\text{sign}) \\ \\ \text{dgb-params : } \left[ \begin{array}{l} \text{spkr: IND} \\ \text{addr: IND} \\ \text{c1 : address(s, a)} \\ \text{s0: SIT} \\ \text{B : Set(IND)} \\ \text{r : boy(B)} \\ \text{ant : } \left[ \begin{array}{l} \text{q-params : } [x : \text{IND}] \\ \text{c : IntraSent(ant)} \end{array} \right] \\ \text{cont}_{\text{sent}} : \text{IllocProp} \end{array} \right] \\ \text{cont} = \text{Assert}(\text{spkr, addr,} \\ \left[ \begin{array}{l} \text{sit = s0} \\ \\ \text{sit-type} = \left[ \begin{array}{l} \text{q-params : } \left[ \begin{array}{l} f : ([x : \text{Ind}]) \left[ \begin{array}{l} z = f(x) : \text{Ind} \\ \text{c1 : cat(z)} \end{array} \right] \end{array} \right] \\ \text{nuc1 : r : } \left( \begin{array}{l} x : \text{IND} \\ \text{c1 : In(dgb-params.B, x)} \end{array} \right) \\ \text{[c0 : kept(r.x, q-params.f(r.x))]} \end{array} \right] \end{array} \right] \end{array} \right] : \text{IllocProp} \end{array} \right]
 \end{aligned}$$

$$\begin{aligned}
 \text{c. } T_{\text{acat:independent}} = & \left[ \begin{array}{l} \text{phon : the boys kept a cat} \\ \text{cat} = V[+\text{fin}, +\text{root}] : \text{syncat} \\ \text{constits} = \left\{ \begin{array}{l} \text{the, boys, the boys, kept,} \\ \text{a, cat, a cat, kept a cat} \end{array} \right\} : \text{set}(\text{sign}) \\ \\ \text{dgb-params : } \left[ \begin{array}{l} \text{spkr: IND} \\ \text{addr: IND} \\ \text{c1 : address(s, a)} \\ \text{s0: SIT} \\ \text{B : Set(Ind)} \\ \text{r : boy(B)} \\ \text{cont}_{\text{sent}} : \text{IllocProp} \end{array} \right] \\ \text{cont} = \text{Assert}(\text{spkr, addr,} \\ \left[ \begin{array}{l} \text{sit = s0} \\ \\ \text{sit-type} = \left[ \begin{array}{l} \text{q-params : } \left[ \begin{array}{l} x : \text{IND} \\ \text{c1 : cat(x)} \end{array} \right] \\ \text{nuc1 : [c0 : kept(dgb-params.B, q-params.x)]} \end{array} \right] \end{array} \right] : \text{IllocProp} \end{array} \right]
 \end{aligned}$$





### 8.5.5 Non-Referential Anaphora

Finally, we return to pronominal anaphora, which we believe can be analysed by means of the mechanism of dependence sketched above. One cannot evade the fact that pronouns are invariably contextually indeterminate, whatever the intentions of the speaker producing them. They are hence subject to CRification, as illustrated by the examples in (135):

- (135) a. A: Everybody likes his sister. B: Whose sister? A: Everyone's.  
 b. A: No woman thinks she is a genius. B: that who is a genius? A: Herself.  
 c. A: Teachers tell students they cannot participate in the demonstration. B: Who cannot participate? A: The teachers.

We need to ensure that pronouns bound by non-referential NPs are non-referential, while ensuring at the same time that such uses are CRifiable. This can be achieved with a minimal modification of our lexical entry for a 'discourse-bound' pronoun (see the entry in (117)), merely by ensuring that a pronoun can also find antecedents intrasententially. Here though antecedents need not be in DGB-PARAMS, it is sufficient that their content is of type IND. I assume an explication of intrasententiality identical to the one discussed in the previous section in the context of scope dependence.

- (136) a. 
$$\left[ \begin{array}{l} \text{phon : she} \\ \text{dgb-params : } \left[ \begin{array}{l} m : \text{LocProp} \\ a : \text{LocProp} \\ c2 : \text{member}(a, m.\text{constits}) \\ c : \text{DiscourseAnt}(m, a) \vee \text{IntraSentAnt}(m, a) \end{array} \right] \\ \text{cat = } \left[ \begin{array}{l} \text{head = N : PoS} \\ \text{ana : +} \\ \text{agr = dgb-params.m.cat.agr : } \left[ \begin{array}{l} \text{num = sg : Number} \\ \text{gen = fem : Gender} \\ \text{pers = third : Person} \end{array} \right] \end{array} \right] : \text{syncat} \\ \text{cont : } [x = a.\text{cont.x} : \text{IND}] \end{array} \right]$$
- b. 
$$\text{IntraSentAnt}(m, a) = \left[ \begin{array}{l} c1 : \text{Maxpending}(m) \\ a : [\text{cont} : [x : \text{IND}]] \end{array} \right]$$

Given this, a use of (137a), where *a* is resolved intransententially to ‘a thief’, will yield the content in (137b), where ‘his’ is indeed non-referential—its content, *dgb-params.a.cont.x*, is identified with *subj-dtr.q-params.x*:

- (137) a. A thief abandoned his jemmy.
- b. 
$$\left[ \begin{array}{l} \text{dgb-params = } \left[ \begin{array}{l} a = \text{'a thief'} \\ c1 = s_{\text{IntrasentAnt}(a)} \\ s0 = \text{sit0} \end{array} \right] \\ \text{cont = } \left[ \begin{array}{l} \text{sit = dgb-params.s0} \\ \text{sit-type = } \left[ \begin{array}{l} \text{subj-dtr.q-params : } \left[ \begin{array}{l} x : \text{IND} \\ r : \text{thief}(x) \end{array} \right] \\ \text{comp-dtr.q-params : } \left[ \begin{array}{l} x : \text{IND} \\ r : \text{jemmy-of}(\text{dgb-params.a.cont.x}, x) \end{array} \right] \\ \text{nucl : Abandon}(\text{subj-dtr.q-params.x}, \text{obj-dtr.q-params.x}) \end{array} \right] \end{array} \right] \end{array} \right]$$

One can, then, if one so wishes, formulate a theory within the current framework that underspecifies intra- and cross-turn pronominal singular anaphora into a single lexical entry, where resolution takes place at the utterance token level. This formulation exploits the fact that we can explicitly specify utterance antecedents as the tokens realizing constituents, thereby allowing us to account for cross-turn agreement. This also yields an explicit account of the clarification potential for pronouns—an answer specifies the antecedent in one way or another.

Of course one is not obliged to adopt this uniform approach. One could introduce an ambiguity among pronouns between discourse-bound and intrasententially bound, following approaches that make a sharp distinction between bound variable and discourse anaphora (e.g. Reinhart 1983, Chierchia 1995). The intrasentential mechanism would then be handled by the grammar, which in any case is known to constrain such resolution (via a binding theory). Nonetheless, integrating binding theory constraints into the theory sketched here is straightforward in principle, given that moves are locutionary propositions and include both utterance types and tokens simultaneously.

#### 8.5.6 *Summary*

Generalized quantifier theory in differing doses is the strategy of choice for analysing QNPs. Since its inception by Montague, it has been a highly insightful approach but one whose cognitive status is in some doubt. The Reprise Content Hypothesis, which originates in Purver and Ginzburg (2004), offers a concrete means of establishing this doubt: whereas data from CRification supports the existence of NPs denoting individuals and sets of individuals, of predicate-denoting verbs and common nouns, and even of function-denoting NPs, there is no evidence whatever of talk about generalized quantifiers. Although GQ theory provides an elegant solution to the problem of scope ambiguity, once again what evidence dialogue provides for this phenomenon does not support an analysis in terms of scope alternation. In this respect underspecification theories are also unsupported since constructed examples of available CRs concerning scope ambiguities support an approach where ambiguities are resolved and disjoined.

I have sketched here an analysis of NP meaning in dialogue using TTR. On this analysis, NPs, be they referential or non-referential, denote individuals, but contribute either to the repository of *contextual* or *quantificational* parameters. Scope is handled via dependency. Anaphora is handled by specifying the antecedents as contextual parameters; discourse anaphora involves searching among the *active* moves for resolution, whereas intersentential anaphora involves a local search. The antecedent utterance is, thus, always a contextual parameter and subject to CRification, though if the antecedent's content is non-referential the same will hold of the pronoun. This analysis yields a relatively simple picture of grounding and CRification within a more general view of meaning, given that it allows also for non-referential content; it also allows for an account of intensional verbs. Scaling up the account is a task for future work.

## An Interaction-Oriented Theory of Meaning

In this final chapter I pull together certain of the threads opened throughout earlier chapters, in particular linking up with the various issues raised in Chapter 1. This also provides a guide to reading the book from the end.

### 9.1 The Interactive Stance

Anyone who wants to contend with spoken human language must recognize that grammar and interaction are intrinsically bound. That is, unless we wish to exclude from analysis a large body of words and constructions, then the right way to construe *grammar* is as **a system that characterizes types of talk in interaction**. This ultimately rather obvious conclusion is a direct implication of the phenomena described and formally analysed in this book and summarized below. One can only view this claim as surprising in light of the recent intellectual hegemony of paradigms that for various reasons sought to avoid contact with actually occurring spoken language. Anyone who wishes to reject such an interactionist view of grammar needs to provide counter-analyses of the phenomena described here of corresponding explicitness on both grammatical and contextual fronts.

Moreover, I have given in this work some brief indications—see in particular section 6.3—that an interactionist view of grammar reflects necessary and constitutive features of human language—for instance, evidence from computer simulations that a language without clarification requests and correction has no long-term viability; see the papers in Cooper and Kempson (2008) for additional illustrations of this general point.

In this book I have been engaged in developing KoS, a theory of meaning for spoken language. I have argued for the need to replace the communitarian approach to semantics—rooted in characterizing communication via its successful episodes, thereby abstracting away from individual differences and from the communicative process—with the far more general interactive stance. The interactive stance—a variant of which used to be dominant among medieval logicians (Hamblin 1970)—

involves taking seriously the fact that communication involves multiple agents with distinct beliefs and desires and places importance on explicating the potential for misunderstanding, rejection, and correction, as well as success. A concrete desideratum for semantics that I have pursued here is **the ability to characterize for any utterance type the update that emerges in the aftermath of successful mutual understanding and the full range of possible clarification requests otherwise**. This manifests itself in the methodological principle (from Purver and Ginzburg 2004) that drives the grammar here, the Reprise Content Hypothesis, a significant strengthening of compositionality.

In the following three sections I summarize respectively (a) the tools with which I developed one version of the interactive stance, (b) some of the main empirical phenomena this has enabled me to tackle, and (c) some of the disciplines whose concerns can thereby be linked.

## 9.2 Linking up the World, Grammar, and Interaction

The work in this book links the external world, grammar, and interaction. Using the logical formalism of Type Theory with Records, I construct models of these three domains. The external domain, building particularly on earlier work in situation semantics, provides us with concrete entities such as individuals and events, as well as abstract entities such as questions and propositions. The grammatical domain, building on earlier work on sign-based grammars, provides us with utterance events and utterance types, the latter akin to Saussurian *signs*. The content component of utterances and their types derive from the external domain. The interactional domain, the main novel contribution of this work, provides us with information states for agents who can engage in conversation. Our main emphasis in this domain has been on explicating two main classes of entities: (a) the *dialogue gameboard* (DGB), an entity associated with each conversing agent, corresponding in essence to that agent's record of the publicized aspects of interaction, and (b) *conversational rules*, the regularities that describe how conversational interaction changes dialogue gameboards. Dialogue gameboards (and by extension conversational rules) are structured in terms of entities from the external and grammatical domains, as exemplified in Figure 9.1. Dialogue gameboards feed back into the grammatical domain to provide its contextual component.

## 9.3 Results: an Overview

In this section I summarize some of the main results that emerge in this work. I partition the results in three, depending on the nature of the contextual component that is exploited—initially results that depend on the DGB as it is introduced in Chapter

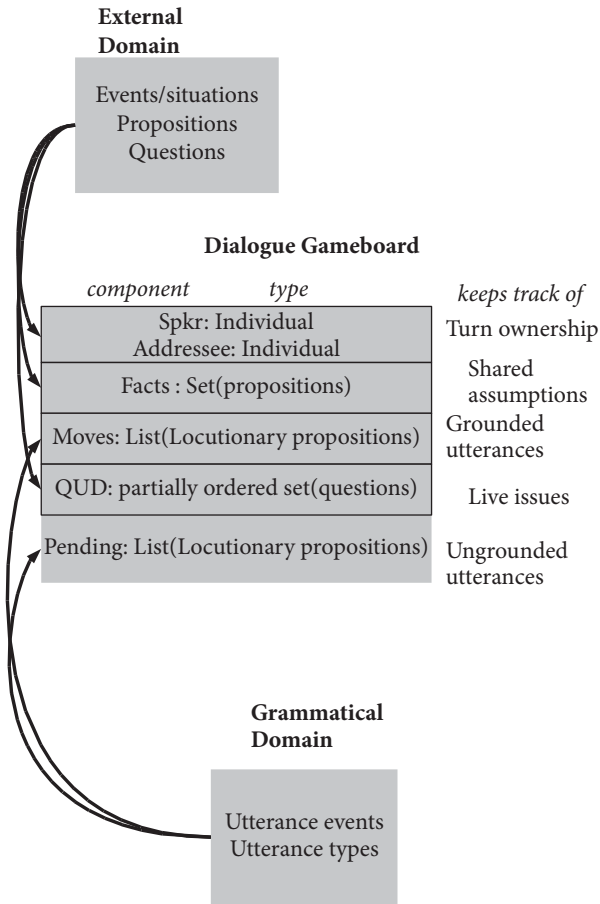


FIGURE 9.1: The structure and function of dialogue gameboards

4, then on the modified DGB introduced in Chapter 6; finally, results that depend on reference to the total information state.

### 9.3.1 QUD-Based Results

In Chapter 4, I introduced the DGB as a data structure that goes beyond standard architectures of public context in two main respects: first, it involves QUD, a field that keeps track of unresolved issues; second, each conversationalist is associated with their own DGB, which allows for mismatches to occur. Figure 9.2 summarizes the main results which make reference to this variant of the DGB:

- **Greeting/Parting interaction:** greeting interaction, as befits its relatively lowly status (occurring as it does in various non-human species), can be described

within any architecture that can encode ‘initial move of interaction.’ Parting, as we saw, is quite a bit more complex and involves non-trivial presuppositions.

- **Non-resolving answers, follow-up queries, disagreement:** these are all consequences of QSPEC, a principle I formulate in Chapter 4. This principle licenses a *q*-specific utterance—an utterance that is a partial answer to or sub-question of *q*, whenever *q* is the maximal element of QUD. Disagreement is accommodated since asserting *p* makes *p?* MaxQUD, the maximal element in QUD, and *p?*-specific utterances include disagreements.
- **Rhetorical questions:** these are analysed as *reassertions*: utterances of interrogatives for which a resolving answer exists in FACTS.
- **Self-answering, scaling up to multilogue:** self-answering is directly accommodated by QSPEC since it licenses MaxQUD-specific utterances regardless of who the speaker of the most recent Move is. The accounts of querying and assertion scale up to multilogue, conversations involving more than two participants. Given A’s query *q*, QSPEC and the ordering on QUD ensures that *q*-specific utterances can be given by multiple participants as long as *q* remains under discussion. As far as assertion goes, the default possibility that emerges is *communal acceptance*—acceptance by one conversationalist can count as acceptance by all other addressees of an assertion, a possibility whose robustness is supported by corpus evidence.
- **Meta-discursive interaction:** the possibility of making an utterance concerning whether to address a prior utterance emerges from a principle similar to QSPEC.
- **Contexts, grammar for ‘yes’, ‘no’, and other propositional lexemes:** asserting *p* and posing *p?* both make *p?* maximal in QUD. This constitutes the basic contextual condition licensing ‘yes’, ‘no’, and other propositional lexemes; the content of such NSUs is a straightforward combinatorial operation on *p?*.
- **Contexts, grammar for short answers and direct sluicing:** the semantic and syntactic parallelism that characterizes both NSU types necessitates a slight type modification of QUD, which consists of a pairs of questions and sub-utterance antecedents. Given this, the context for these NSU types can be specified as emerging from queries whose content is a unary question or assertions of a quantified proposition, respectively.
- **Distinct conversational genres:** a technique for classifying conversations into genres (e.g. shopping in a bakery, casual chat, etc.) is provided by means of a description of the DGB of a conversationalist who has *successfully* completed such a conversation.

### 9.3.2 QUD and Pending-Based Results

One of the most direct consequences of adopting the interactive stance is the need to accommodate Semantic non-determinism—**interpretation can lead to distinct**

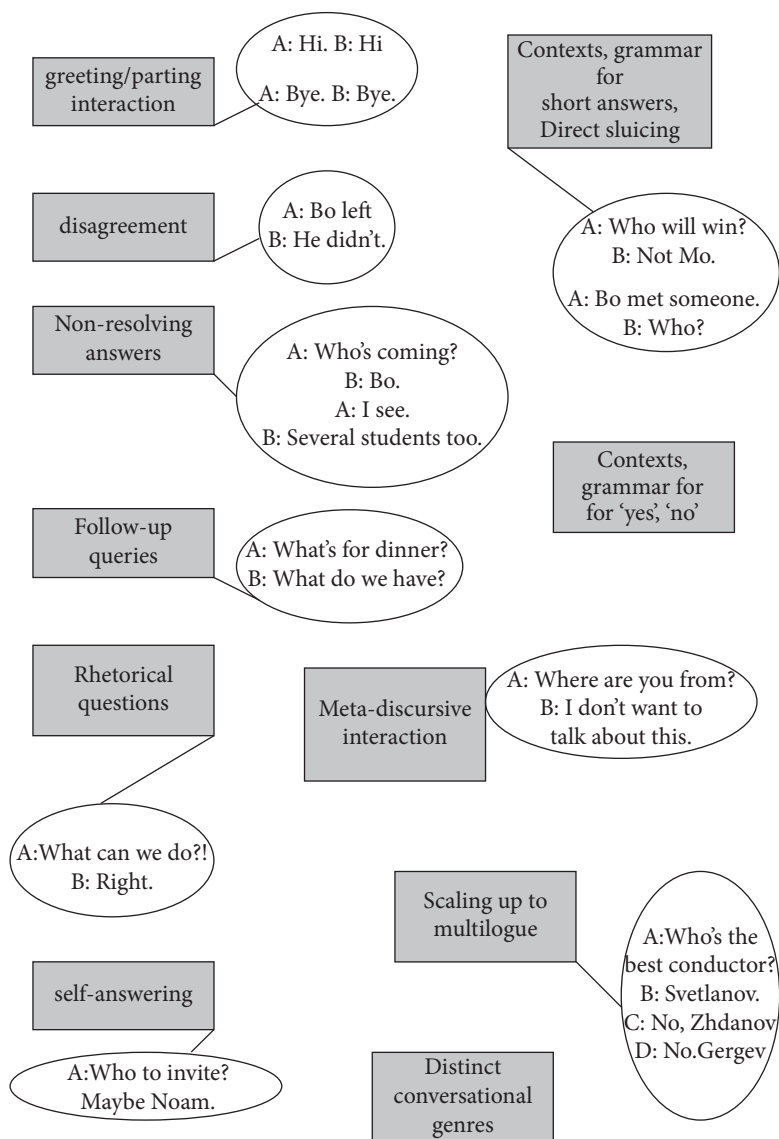


FIGURE 9.2: Results dependent on FACTS/MOVES/QUD architecture

**updates across conversational participants.** Just one modification is required to the structure of the DGB, the postulation of a field **Pending**, keeping track of as yet ungrounded utterances, whose type is a list of *locutionary propositions*, propositions consisting of an utterance and a (grammatical) type which classifies it. Figure 9.3 summarizes the main results which make reference to this variant of the DGB:



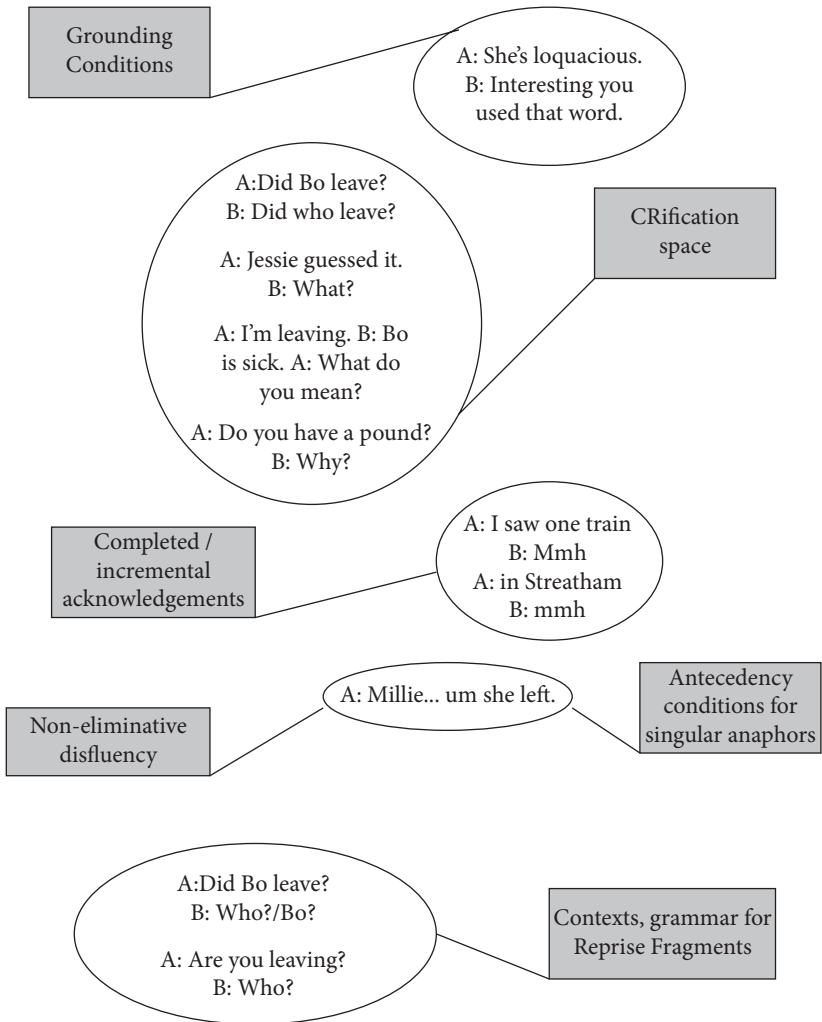


FIGURE 9.3: Results dependent on FACTS/MOVES/QUD/Pending architecture

- **Grounding conditions:** are characterized as *the utterance type fully classifies the utterance token*. This allows for updates effected by utterances to include a variety of information about the utterance, going far beyond its content, including syntactic and phonological information.
- **CRification space:** the potential for four classes of clarification interaction is characterized—those concerned with literal content, repetition requests, relevance clarifications, and requests for underlying motivation. The first three

classes are specified using conversational rules that accommodate into QUD an issue (e.g. 'What did you mean/say in sub-utterance *u*<sub>1</sub> of the target utterance') which follow-up responses must address. Underlying motivation CRs are argued to belong to a quite distinct genus of utterances, instances of utterances that provide *explanations* for a previous utterance.

- **Contexts, grammar for Reprise Fragments:** these are handled identically to other NSUs—the content of such NSUs emerges from straightforward combinatorial operations on MaxQUD. Distinct parallelism conditions characterize distinct NSU types and these make reference to phonological and/or syntactic information down to the word level.
- **Completed/incremental acknowledgements:** given the direct interface between grammar and DGBs, we can formulate lexical entries for words like 'mmh'—the contextual parameters include the maximal utterance in Pending, the content being that the speaker understands that utterance. Accommodating incremental acknowledgements, as with **Non-eliminative disfluency**, presupposes that Pending contains incomplete sentential increments.
- **Non-eliminative disfluency:** disfluencies are analysed using the same tools as clarification requests. The analysis provided ensures that disfluent material, although no longer active in content construction, still remains in context.
- **Antecedency conditions for singular anaphors:** the antecedents for discourse anaphors are analysed as contextual parameters of an *active move*, a move which relates to an issue in QUD or is in Pending. Intrasentential antecedents are characterized analogously, without the requirement that they be contextual parameters.

### 9.3.3 Results dependent on the total information state

Most of the results in this book make reference solely to the DGB, in other words to a level of public information. One cannot avoid some phenomena making reference to the total information state, which encodes hidden variables such as the conversationalists' private beliefs and goals. Figure 9.4 summarizes the main results which make reference to the total information state:

- **Contexts for initiating NSUs:** the context for such an NSU is determined by the conversational genre which the participants believe they are in.
- **Indirect Moves:** the potential for *non-conventionalized* indirect moves is characterized using the theory of conversational genres. Conventionalized indirect moves are handled as grammaticized root construction types.
- **Construing Irrelevant responses:** a response is irrelevant if its coherence cannot be established using conversational rules. This notion of irrelevance can be used to formulate an inference that the author of the irrelevant response does not wish to address the original utterance.

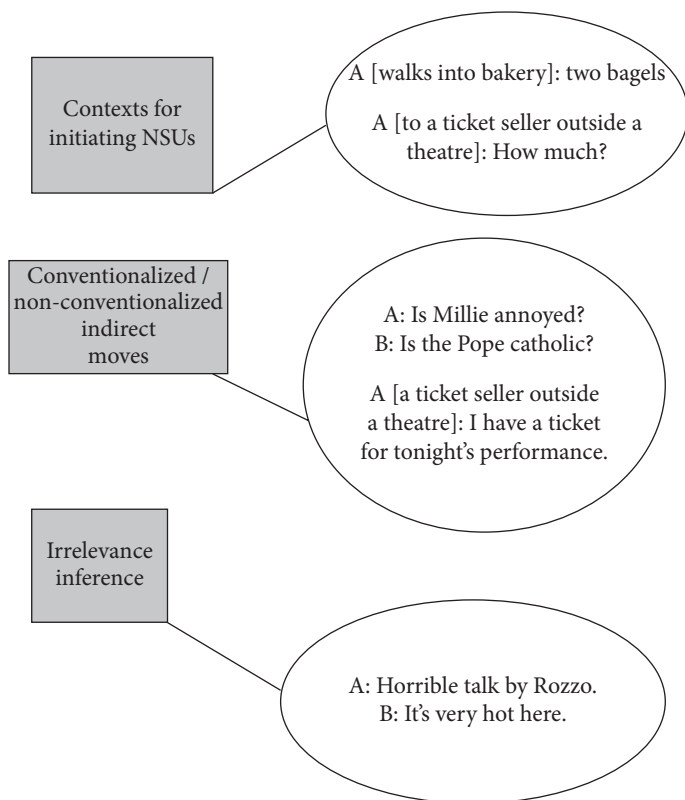


FIGURE 9.4: Results dependent on Total Information State architecture

## 9.4 Embedding Grammar in Interaction

In this book I have tried to show how adopting the interactive stance enables us to provide a unified account of various phenomena characteristic of interaction at the macro and micro levels. I would like to highlight some phenomena analysed here which offer the prospects of a merging of perspectives for a number of the language sciences:

- **Formal Grammar:** formal descriptions of words/phrases in genuine spoken context.
- **Conversation Analysis:** formal theory unifying other-repair (CRification) with self-repair (disfluency).
- **Psycholinguistics:** grounding at various levels of utterance representation.
- **Gricean pragmatics:** Irrelevance inferences tied with relevance CRification. *Non-conventionalized* indirect moves explicated as relevance inferences within a conversational genre.

- **Child Language Acquisition:** matching semantic/contextual complexity with the order of acquisition of NSUs.
- **Dialogue systems:** a declarative theory of conversational genres.

## 9.5 What Every Theory of Meaning for Spoken Language Must Embrace

KoS is a framework for describing meaning in spoken interaction. In some cases I have pointed out that a given phenomenon could be analysed in more than one way; obviously this applies to most, if not all other phenomena discussed here. More generally, there are of course a variety of theoretical approaches that could be adopted. Still I think that, above and beyond a view which places interaction at the centre of grammar, we can take away from this work a small number of fundamental requirements that any viable theory needs to satisfy. All of these are at present quite a distance out in left field.

1. **No single public context:** this seems inevitable if one wishes to accommodate semantic non-determinism.
2. **Rich grammar/context interface:** such an interface needs to make available to the grammar *inter alia* notions that correspond to KoS's MaxQUD, MaxPending. This seems inevitable for any grammar wishing to deal with NSUs.
3. **Type/token egalitarianism**—utterance types and tokens are simultaneously first-class citizens. This seems inevitable for any semantics/grammar wishing to deal with repair.
4. **Hybrid contexts**—the theory must enable non-semantic information to be filtered 'long-term'—(surviving across several turns)—into contexts. This seems inevitable for any theory wishing to deal with cross-turn parallelism and priming phenomena.

## APPENDIX A

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### *The External Domain*

The aim of this appendix is to collate the various notions that make up the external world ontology that underpins this book. This constitutes a sketch of a type-theoretic universe—a far more detailed and precise picture is to be found in Cooper (2005a, 2011).

#### A.1 The Type-Theoretic Universe

- (1) **Type-Theoretic Universe** (Cooper 2011, here simplified version)  
 $\text{TYPEDWORLD} =_{\text{def}} \langle \text{Type}^n, \text{BasicType}, \text{ComplexType}^n, \langle \text{Pred}^n, \text{Arity}^n \rangle, \text{RecType}^n, \langle A^n, F^n \rangle \rangle$
- a.  $\text{Type}^n$  is the set of types of order  $n$ , built up recursively using type construction operations:
    - **function types**: if  $T_1$  and  $T_2$  are types, then so is  $(T_1)T_2$  the type of functions from elements of type  $T_1$  to elements of type  $T_2$ . The set  $f : (T_1)T_2$  iff  $\{a|a : T_1\}$  is the domain of  $f$  and the range of  $f$  is a subset of  $\{a|a : T_2\}$
    - **List/finite set types**: if  $T$  is a type, then  $\text{List}(T)$  ( $\text{FiniteSet}(T)$ ), the type of lists (finite sets) each of whose members is of type  $T$ , is also a type. The list  $[a_1, \dots, a_n] : \text{List}(T)$  (finite set  $\{a_1, \dots, a_n\}$ ) iff for all  $i$   $a_i : T$
    - **Meet/Join types**: if  $T_1$  and  $T_2$  are types, then so are  $T_1 \wedge T_2$  and  $T_1 \vee T_2$ :  $a : T_1 \wedge T_2$  ( $T_1 \vee T_2$ ) iff  $a : T_1$  and (or)  $a : T_2$ .
    - **Dual types**: if  $T$  is a type, then so is  $\neg T$ .  $a : \neg T$  iff for some type  $S$ , where  $S$  and  $T$  are incompatible,  $a : S$
    - **Singleton types**: if  $T$  is a type and  $x : T$ , then  $T_x$  is a type.  $a : T_x$  iff  $a = x$ .
  - b.  $\text{BasicType}^n$  : IND, TIME, LOC, ...
  - c.  $\text{ComplexType}^n$ : if  $P$  is a member of  $\text{Pred}^n$ ,  $\text{Arity}^n(P) = \langle T_1, \dots, T_m \rangle$  and  $a_1, \dots, a_m$  are such that  $a_1 : T_1, \dots, a_m : T_m$ , then  $P(a_1, \dots, a_m) \in \text{ComplexType}^n$
  - d.  $\text{RecType}^n$ : set of records, record types defined with respect to a set of objects used as labels:
    - The record:
 
$$\left[ \begin{array}{lcl} l_1 & = & a_1 \\ l_2 & = & a_2 \\ \dots & & \\ l_n & = & a_n \end{array} \right] \text{ is of type: } \left[ \begin{array}{lcl} l_1 & : & T_1 \\ l_2 & : & T_2 \\ \dots & & \\ l_n & : & T_n \end{array} \right]$$

iff  $a_1 : T_1, a_2 : T_2, \dots, a_n : T_n$
  - e.  $\langle A^n, F^n \rangle$  is a model:  $A^n$  assigning entities to  $\text{BasicType}$ , and  $F^n$  assigns tuples to elements of  $\text{ComplexType}^n$ .

## A.2 Propositions, Questions, Outcomes

### A.2.1 Basic Definitions

- (2) a.  $\text{Prop} =_{\text{def}} \begin{bmatrix} \text{sit} & : & \text{Record} \\ \text{sit-type} & : & \text{RecType} \end{bmatrix}$
- b. A proposition  $p = \begin{bmatrix} \text{sit} & = & s_0 \\ \text{sit-type} & = & ST_0 \end{bmatrix}$  is true iff  $s_0 : ST_0$
- c.  $\text{TrueProp} =_{\text{def}} \begin{bmatrix} \text{sit-type} & : & \text{RecType} \\ \text{sit} & : & \text{Record} \wedge \text{sit-type} \end{bmatrix}$
- d.  $\text{QuantProp} =_{\text{def}} \begin{bmatrix} \text{sit} : \text{Record} \\ \text{sit-type} : [\text{q-params} : [\text{x} : \text{Ind}]] \end{bmatrix}$
- (3) a. If  $p = \begin{bmatrix} \text{sit} = s \\ \text{sit-type} = T \end{bmatrix}$ , then  $\neg_{\text{prop}} p =_{\text{def}} \begin{bmatrix} \text{sit} = s \\ \text{sit-type} = \neg T \end{bmatrix}$
- b. Given  $s_1, s_2 : \text{Rec}$  and  $T_1, T_2 : \text{RecType}$ ,  
 $\text{Disjrec}((s_1, T_1), (s_2, T_2)) =_{\text{def}} s_1$ , if  $s_1 : T_1$   
 $s_2$ , otherwise, if  $s_2 : T_2$   
 $[\ ]$  (the empty record), otherwise
- c.  $\begin{bmatrix} \text{sit} = s_1 \\ \text{sit-type} = T_1 \end{bmatrix} \vee_{\text{prop}} \begin{bmatrix} \text{sit} = s_2 \\ \text{sit-type} = T_2 \end{bmatrix} =_{\text{def}} \begin{bmatrix} \text{sit} = \text{Disjrec}((s_1, T_1), (s_2, T_2)) \\ \text{sit-type} = T_1 \vee T_2 \end{bmatrix}$
- d.  $\langle s_1, s_2 \rangle : T_1 \wedge_{\text{ctt}} T_2$  if  $s_1 : T_1$  and  $s_2 : T_2$ .
- e.  $\begin{bmatrix} \text{sit} = s_1 \\ \text{sit-type} = T_1 \end{bmatrix} \wedge_{\text{prop}} \begin{bmatrix} \text{sit} = s_2 \\ \text{sit-type} = T_2 \end{bmatrix} =_{\text{def}} \begin{bmatrix} \text{sit} = \langle s_1, s_2 \rangle \\ \text{sit-type} = T_1 \wedge_{\text{ctt}} T_2 \end{bmatrix}$
- (4) a.  $\text{Irrealis} =_{\text{def}} ([t : \text{TIME}])\text{RType}$
- b.  $\text{Outcome} =_{\text{def}} \begin{bmatrix} \text{sit} & : & \text{Record} \\ \text{irr-sit-type} & : & \text{Irrealis} \end{bmatrix}$
- c. For an outcome  $\begin{bmatrix} \text{sit} = s_0 \\ \text{irr-sit-type} = p_0 \end{bmatrix}$   
 $\text{Fulfillers} =_{\text{def}} \begin{bmatrix} s_1 : \text{Record} \\ \text{fulfil-time} : \text{TIME} \\ c_1 : \text{anterior}(s_0, s_1) \\ p = \begin{bmatrix} \text{sit} = s_1 \\ \text{sit-type} = p_0(\text{fulfil-time}) \end{bmatrix} : \text{True} \end{bmatrix}$
- (5) a.  $\text{PolarQuestion}_0 =_{\text{def}} \begin{bmatrix} p : \text{Prop} \\ q : ([\ ])\text{Prop}_p \end{bmatrix}$

- b.  $\text{UnaryWhQuestion}_0 =_{\text{def}} \left[ \begin{array}{l} A : \text{Type} \\ s : \text{Rec} \\ R(\text{arg1}_A) : \text{ComplexType} \\ q = \left( [x : A] \right) \left[ \begin{array}{l} \text{sit} = s \\ \text{sit-type} = R(\text{arg1}_A : q.x) \end{array} \right] : [x : A] \text{Prop} \end{array} \right]$
- c.  $\text{BinaryWhQuestion}_0 =_{\text{def}} \left[ \begin{array}{l} A_1 : \text{Type} \\ A_2 : \text{Type} \\ s : \text{Rec} \\ R(\text{arg1}_{A_1}, \text{arg2}_{A_2}) : \text{ComplexType} \\ q = \left( \left[ \begin{array}{l} x_1 : A_1 \\ x_2 : A_2 \end{array} \right] \right) \left[ \begin{array}{l} \text{sit} = s \\ \text{sit-type} = R(\text{arg1}_{A_1} : q.x_1, \text{arg2}_{A_2} : q.x_2) \end{array} \right] : \left( \left[ \begin{array}{l} x_1 : A_1 \\ x_2 : A_2 \end{array} \right] \right) \text{Prop} \end{array} \right]$
- d.  $\text{N-aryWhQuestion}_0 =_{\text{def}} \left[ \begin{array}{l} A_1 : \text{Type} \\ A_2 : \text{Type} \\ \dots \\ A_n : \text{Type} \\ s : \text{Rec} \\ R(\text{arg1}_{A_1}, \text{arg2}_{A_2}, \dots, \text{argn}_{A_n}) : \text{ComplexType} \\ q = \left( \left[ \begin{array}{l} x_1 : A_1 \\ x_2 : A_2 \\ \dots \\ x_n : A_n \end{array} \right] \right) \left[ \begin{array}{l} \text{sit} = s \\ \text{sit-type} = R(\text{arg1}_{A_1} : q.x_1, \text{arg2}_{A_2} : q.x_2, \dots, \text{argn}_{A_n} : q.x_n) \end{array} \right] : \left( \left[ \begin{array}{l} x_1 : A_1 \\ x_2 : A_2 \\ \dots \\ x_n : A_n \end{array} \right] \right) \text{Prop} \end{array} \right]$
- e.  $\text{Question}_0^n =_{\text{def}} \text{PolarQuestion}_0 \vee \text{UnaryWhQuestion}_0 \vee \dots \vee \text{N-aryWhQuestion}_0$
- f.  $x : \text{Question}_0^n =_{\text{def}} \text{There exists } r : \text{Question}_0^n \text{ and } x = r.q$
- g.  $x : \text{PolarQuestion} =_{\text{def}} \text{There exists } r : \text{PolarQuestion}_0 \text{ and } x = r.q$
- h.  $x : \text{UnaryWhQuestion} =_{\text{def}} \text{There exists } r : \text{UnaryWhQuestion} \text{ and } x = r.q$

### A.2.2 Answerhood Notions

(6) Given a question  $q : (A)B$ :

- a.  $\text{AtomAns}(q) =_{\text{def}} \left[ \begin{array}{l} \text{shortans} : A \\ \text{propans} = q(\text{shortans}) : \text{Prop} \end{array} \right]$
- b.  $\text{NegAtomAns}(q) =_{\text{def}} \left[ \begin{array}{l} \text{shortans} : A \\ \text{propans} = \neg q(\text{shortans}) : \text{Prop} \end{array} \right]$
- c.  $\text{SimpleAns}(q) =_{\text{def}} \text{AtomAns}(q) \vee \text{NegAtomAns}(q)$

- d.  $\text{Aboutness}(q) =_{\text{def}} \left[ \begin{array}{l} r1 : \text{SimpleAns}(q) \\ p1 = r1.\text{propans} : \text{Prop} \\ r2 : \text{SimpleAns}(q) \\ p2 = r2.\text{propans} : \text{Prop} \\ \text{propans} : \text{Prop} \\ c : \text{Entails}(\text{propans}, p1 \vee_{\text{prop}} p2) \end{array} \right]$
- e.  $\text{StrongExhAns}(q) =_{\text{def}} \left[ \begin{array}{l} \text{exhans} : \text{Prop} \\ c0: \left( \left[ \begin{array}{l} r : \text{SimpleAns}(q) \\ p = r.\text{propans} : \text{TrueProp} \end{array} \right] \right) \left[ c1 : \text{Entails}(\text{exhans}, p) \right] \end{array} \right]$
- f.  $\text{PotResolvAns}(q) =_{\text{def}} \text{AtomAns}(q) \vee \left[ \text{propans} = \forall x : A \neg q(x) : \text{Prop} \right]$   
 $\text{ResolvingAnswers}(q) =_{\text{def}} (p \text{ resolves } q \text{ relative to } B\text{'s desired outcome } o)$
- $\left[ \begin{array}{l} p : \text{TrueProp} \\ B : \text{IND} \\ o : \text{outcome} \\ p1 = \text{PotResolveAns}(q).\text{propans} : \text{Prop} \\ c2 : \text{Entails}(p, p1) \\ c3 : \text{want}(B, o) \\ c4 : \text{ep-sufficient}(p, B, o) \end{array} \right]$

### A.2.3 Question Relations

- (7) a.  $q_2$  influences  $q_1$  iff any proposition  $p$ , such that  $p$  Resolves  $q_2$ , also satisfies  $p$  entails  $r$  such that  $r$  is About  $q_1$ .  
 b. Two questions  $q_0$  and  $q_1$  are co-propositional if there exists a record  $r$  such that  $q_0(r) = q_1(r)$ .

### A.2.4 Dialogue and Grammatical Types

- (8) a.  $\text{AbstSemObj} =_{\text{def}} \text{Prop} \vee \text{Question} \vee \text{Outcome}$   
 b.  $\text{SemObj} =_{\text{def}} \text{AbstSemObj} \vee \left[ x : \text{IND} \right] \vee \left( \left[ y : \text{Ind} \right] \left( \left[ x : \text{Ind} \right] \text{Rectype} \vee \left( \left[ x : \text{Ind} \right] \text{RecType} \vee \dots \right) \right) \right)$   
 c.  $\text{IllocProp} =_{\text{def}} \left[ \begin{array}{l} \text{sit} : \text{Record} \\ x : \text{IND} \\ y : \text{IND} \\ a : \text{AbstSemObj} \\ R : \text{IllocRel} \\ \text{sit-type} = \left[ c1 : R(x, y, a) \right] : \text{Rectype} \end{array} \right],$



d.  $\text{LocProp} =_{\text{def}} \left[ \begin{array}{l} \text{sit} : \text{Record} \\ \text{ph1} : \text{list}(\text{phonform}) \\ \text{pos1} : \text{PoS} \\ \text{cont-obj} : \text{SemObj} \\ \text{G} : \text{RecType} \\ \text{pos2} : \text{PoS} \\ \text{qd1} : \text{list}(\text{RecType}) \\ \\ \text{sit-type} = \left[ \begin{array}{l} \text{PHON} : \text{ph1} \\ \text{CAT} : [\text{head} : \text{pos1}] \\ \text{CONT} = \text{cont-obj} : \text{SemObj} \\ \text{C-PARAMS} : \text{G} \\ \text{GAPS} : \text{list} \left( \left[ \begin{array}{l} \text{CAT} : [\text{head} : \text{pos2}] \\ \text{CONT} : [\text{x} : \text{IND}] \end{array} \right] \right) \\ \text{QUEST-DOM} : \text{qd1} \end{array} \right] : \text{RecType} \end{array} \right]$

(9)  $\text{Info-struc} =_{\text{def}} \left[ \begin{array}{l} \text{q} : \text{Question} \\ \text{fec} : \text{set}(\text{LocProp}) \end{array} \right]$

## APPENDIX B

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### *The Interaction Domain*

One of the main explanatory strategies of this book has been to posit certain types to represent agent information states and different kinds of interactional patterns. An initial system is formulated in Chapter 4, then slightly modified in Chapters 6 and 7, with additional modifications in Chapter 8. The aim of this appendix is to present this system as a unified whole, indicating what parts are strictly necessary and which are more tied to explaining particular phenomena.

#### B.1 Information States and Dialogue Gameboards

At any given point an agent is described in terms of her information state, an entity of type *TotalInformationState* (TIS). This type has two fields *dgb*, *private*, representing public and non-public information respectively. Most, though by no means all, of my explanations relate to the public component whose type is the type *DialogueGameBoard* (DGBType). This type is initially introduced in Chapter 4, with the additional field **Pending** being introduced in Chapter 6. Chapter 6 also argued for a type change for *Moves*, from *IllocProp* (the content of an illocutionary act) to *LocProp* (the representation of an utterance, including non-semantic information):

- (1) a. *TotalInformationState* (TIS) =<sub>def</sub>

$$\left[ \begin{array}{l} \text{dgb} : \text{DGBType} \\ \text{private} : \text{PRTType} \end{array} \right]$$

- b. *PRTType* =<sub>def</sub>

$$\left[ \begin{array}{l} \text{genre} : \text{GenreType} \\ \text{beliefs} : \text{Prop} \\ \text{agenda} : \text{list}(\text{IllocProp}) \end{array} \right]$$

*DGBType* =<sub>def</sub>

$$\left[ \begin{array}{l} \text{spkr} : \text{IND} \\ \text{addr} : \text{IND} \\ \text{utt-time} : \text{TIME} \\ \text{c-utt} : \text{addressing}(\text{spkr}, \text{addr}, \text{utt-time}) \\ \text{Facts} : \text{Set}(\text{Prop}) \\ \text{Pending} : \text{list}(\text{LocProp}) \\ \text{Moves} : \text{list}(\text{LocProp}) \\ \text{QUD} : \text{poset}(\text{Question}) \\ \text{non-resolve-cond} : \forall q \in \text{QUD} [\neg \text{Resolve}(\text{FACTS}, q)] \end{array} \right]$$

Certain additional fields and type modifications were posited in Chapters 4, 6, and 8: QNUD (plays an important part in characterizing the global structure of conversations); Topical (a subclass of ‘salient’ Facts); QUD is posited to be of the type *InfoStruc* (to keep track of *focus-*

establishing constituents); an additional ordering condition is imposed on QUD. The type  $DGBType_{rich}$  incorporates all these modifications; one can substitute that for  $DGBType$  or introduce types that incorporate only a subset of these:

- (2)  $DGBType_{rich} =_{def}$
- $$\left[ \begin{array}{l} \text{spkr: IND} \\ \text{addr: IND} \\ \text{utt-time} \\ \text{c-utt : addressing(spkr, addr, utt-time)} \\ \text{Facts : Set(Prop)} \\ \text{Topical : Set(Prop)} \\ \text{c-top : Topical} \subset \text{Facts} \\ \text{Pending : list(LocProp)} \\ \text{Moves : list(LocProp)} \\ \text{QUD : poset(InfoStruc)} \\ \text{QNUD : list(Question)} \\ \text{non-resolve-cond : } \forall q \in \text{QUD} [\neg \text{Resolve(FACTS, q)}] \\ \text{ordering-cond : } \forall q, q' \in \text{QUD}, q > q' [\neg \text{Depend}(q, q')] \end{array} \right]$$

The type  $GenreType$ , used to characterize  $PRTType$ , by describing the state of an agent that completes a certain kind of conversation is given in (3):

- (3)  $GenreType =_{def}$
- $$\left[ \begin{array}{l} \text{facts : Set(Prop)} \\ \text{qnud : set(question)} \\ \text{moves : list(IllocProp)} \end{array} \right]$$

## B.2 Conversational Rules

### B.2.1 Preliminaries

Conversational rules are mappings relating information states, a *preconditions* state to an *effects* state. In most cases, we focus solely on the change at the DGB level, though rules that make reference to the total information state are indispensable. Both preconditions and effects are subtypes of  $DGBType$  (TIS), so can be written as in (4a, b): *PreCondSpec* is a type that includes information specific to the preconditions of this interaction type, whereas *ChangePreconSpec* represents those aspects of the preconditions that have changed. In order to maximize perspicuity I have adopted a notational convention of including in the specification of a conversational rule solely the non-predictable components, viz *PreCondSpec* and *ChangePreconSpec*. Hence, a conversational rule is generally notated as in (4d), though this is merely an abbreviation:

- (4) a.  $\left[ \begin{array}{l} \text{preconds: Precondtype} \\ \text{effects : Effectstype} \end{array} \right]$   
 b.  $\text{Precondtype} =_{def} DGBType^- \wedge_{merge} \text{PreCondSpec}$   
 c.  $\text{Effectstype} =_{def} DGBType^0 \wedge_{merge} \text{ChangePreconSpec}$   
 d.  $\left[ \begin{array}{l} \text{preconds: PreCondSpec} \\ \text{effects : ChangePreconSpec} \end{array} \right]$

With respect to the labels associated with speaker and addressee there are three basic possibilities in two-person interaction: the turn remains the same (5a), the turn changes (5b), or the turn is underspecified—it is up for grabs for either conversationalist, as in (5c), which picks the speaker from among the conversationalists and the addressee as the distinct other element in this set. No turn change gets abbreviated away in the notational convention introduced in (4). The other two cases get notated explicitly. Turn change by specifying the effects as in (5b), whereas turn underspecification by means of a merge-type conjunction, as in (5d):

- (5) a. No-Turn-Change =<sub>def</sub> 
$$\left[ \begin{array}{l} \text{pre : } \left[ \begin{array}{l} \text{spkr : IND} \\ \text{addr : IND} \end{array} \right] \\ \text{effects : } \left[ \begin{array}{l} \text{spkr = pre.spkr : IND} \\ \text{addr = pre.addr : IND} \end{array} \right] \end{array} \right]$$
- b. Turn-Change =<sub>def</sub> 
$$\left[ \begin{array}{l} \text{pre : } \left[ \begin{array}{l} \text{spkr : IND} \\ \text{addr : IND} \end{array} \right] \\ \text{effects : } \left[ \begin{array}{l} \text{spkr = pre.addr : IND} \\ \text{addr = pre.spkr : IND} \end{array} \right] \end{array} \right]$$
- c. Turnholder-Underspecified =<sub>def</sub> 
$$\left[ \begin{array}{l} \text{pre : } \left[ \begin{array}{l} \text{spkr : IND} \\ \text{addr : IND} \end{array} \right] \\ \text{effects : } \left[ \begin{array}{l} \text{PrevAud} = \{\text{pre.spkr}, \text{pre.addr}\} : \text{Set(IND)} \\ \text{spkr : IND} \\ \text{c1 : member(spkr, PrevAud)} \\ \text{addr : IND} \\ \text{c2 : member(addr, PrevAud)} \\ \quad \wedge \text{addr} \neq \text{spkr} \end{array} \right] \end{array} \right]$$
- d. effects : TurnUnderspec  $\wedge_{\text{merge}}$  
$$\left[ \dots \right]$$

A number of the rules I formulated appear in two guises, an initial formulation relative to the DGB as it was postulated in Chapter 4 and a subsequent formulation in Chapter 5, where metacommunicative and illocutionary interaction is interleaved. The main effect this has is that the content of utterances ‘originates’ in PENDING and updates MOVES only if MAX-PENDING is *true*. Given that this pendification can be applied straightforwardly to each rule I will *as far as notation goes* abstract away from it, though it is implicit in each rule that updates MOVES.

I note some additional general rule modifications proposed at various stages. They are all independent of each other, so any subset of them could be adopted including of course the entire set:

- Change in type of QUD from *poset(Question)* to *poset(InfoStruc)* affects all rules that update QUD.
- Introduction of QNUD affect the dynamics of Facts.
- Introduction of Topical: affects the dynamics of QUD and Facts.
- Expand the audience for multilogue: affects all rules.

In this appendix I take into account only the change in type of QUD.

### B.2.2 Utterance Coherence

Conversational rules combine with grammars to yield the following coherence notion:

- (6) a. **U(tterance)–Coherence:** Given a set of conversational rules  $\mathcal{C}$ , a grammar  $\mathcal{G}$ , and an information state  $I_0 : TIS$ , an utterance  $u_1$  is **U(tterance) $_{\mathcal{C}, \mathcal{G}}$  $^{I_0}$ -coherent** iff there exist  $c_1, \dots, c_{k+1} \in \mathcal{C}, T_{u_1} \in \mathcal{G}, k \geq 0$  such that  $c_1(I_0) = I_1, \dots, c_{k+1}(I_k) = I_{k+1}$  such that  $I_{k+1}.DGB.LatestMove = prop(w_1, T_{u_1})$ , where  $prop(w_1, T_{u_1})$  contextually extends  $prop(u_1, T_{u_1})$ , and for every  $0 < i \leq k$   $I_i.DGB.MaxPending = prop(w'_i, T_{u_1})$ , where  $prop(w'_i, T_{u_1})$  contextually extends  $prop(u_i, T_{u_1})$ .<sup>1</sup>
- b. **Pairwise U–Coherence:** Given a set of conversational rules  $\mathcal{C}$  and a grammar  $\mathcal{G}$ , two utterances  $u_1, u_2$  are **U $_{\mathcal{C}, \mathcal{G}}$ -pairwise-coherent** iff there exists  $I_0 : TIS$  such that  $u_1$  is **U(tterance) $_{\mathcal{C}, \mathcal{G}}$  $^{I_0}$ -coherent** and there exist  $c_1, \dots, c_{k+1} \in \mathcal{C}, T_{u_1}, T_{u_2} \in \mathcal{G}, j, k \geq 0$  such that  $c_1(I_0) = I_1, \dots, c_{k+1}(I_k) = I_{k+1}$  such that  $I_j.DGB.LatestMove = prop(w_1, T_{u_1}), I_{k+1}.DGB.LatestMove = prop(w_2, T_{u_2})$ , where  $prop(w_1, T_{u_1})$  contextually extends  $prop(u_1, T_{u_1})$ , and  $prop(w_2, T_{u_2})$  contextually extends  $prop(u_2, T_{u_2})$ .
- c. **Sequential Utterance Coherence:** Given a set of conversational rules  $\mathcal{C}$  and a grammar  $\mathcal{G}$ , a sequence of utterances  $u_1, \dots, u_n$  is **U(tterance) $_{\mathcal{C}, \mathcal{G}}$  $^{I_0}$ -sequentially coherent** iff, for any  $1 \leq i, u_i, u_{i+1}$  are **U $_{\mathcal{C}, \mathcal{G}}$ -pairwise-coherent**.

## B.3 Move Update Rules

### B.3.1 Conversation Initialization and Termination

- (7) a.  $Greeting =_{def} \left[ \begin{array}{l} \text{preconds : } \left[ \begin{array}{l} \text{moves} = \langle \rangle : \text{list}(\text{LocProp}) \\ \text{qud} = \{ \} : \text{poset}(\text{InfoStruc}) \end{array} \right] \\ \text{effects : } \left[ \text{LatestMove}^{Content} = \text{Greet}(\text{spkr}, \text{addr}) : \text{IllocProp} \right] \end{array} \right]$
- b.  $CounterGreeting =_{def} \left[ \begin{array}{l} \text{preconds : } \left[ \begin{array}{l} \text{LatestMove}^{Content} = \text{Greet}(\text{spkr}, \text{addr}) : \text{IllocProp} \\ \text{qud} = \{ \} : \text{poset}(\text{InfoStruc}) \end{array} \right] \\ \text{effects : } \left[ \begin{array}{l} \text{spkr} = \text{pre.addr} : \text{IND} \\ \text{addr} = \text{pre.spkr} : \text{IND} \\ \text{LatestMove}^{Content} = \text{CounterGreet}(\text{spkr}, \text{addr}) : \text{IllocProp} \end{array} \right] \end{array} \right]$
- c.  $Parting =_{def} \left[ \begin{array}{l} \text{preconds : } \left[ \begin{array}{l} \text{qud} = \{ \} : \text{poset}(\text{InfoStruc}) \\ \text{facts} : \text{Set}(\text{Prop}) \\ f = \text{MinInteraction}(\{\text{spkr}, \text{addr}\}) : \text{Prop} \\ \text{cl} : \text{member}(f, \text{facts}) \end{array} \right] \\ \text{effects : } \left[ \text{LatestMove}^{Content} = \text{Part}(\text{spkr}, \text{addr}) : \text{IllocProp} \right] \end{array} \right]$

<sup>1</sup> ‘Every  $0 < i \leq k \dots$ ’ is to be read logically, so that in case  $k = 0$ , the rest of the clause is not required to be true.

$$\begin{aligned}
 \text{d. CounterParting} &=_{\text{def}} \left[ \begin{array}{l} \text{preconds} : \left[ \begin{array}{l} \text{LatestMove}^{\text{Content}} = \text{Part}(\text{spkr}, \text{addr}) : \text{IllocProp} \\ \text{qud} = \{ \} : \text{poset}(\text{InfoStruc}) \end{array} \right] \\ \text{effects} : \left[ \begin{array}{l} \text{spkr} = \text{preconds.addr} : \text{IND} \\ \text{addr} = \text{preconds.spkr} : \text{IND} \\ \text{LatestMove}^{\text{Content}} = \text{CounterPart}(\text{spkr}, \text{addr}) : \text{IllocProp} \end{array} \right] \end{array} \right] \\
 \text{e. Disengagement} &=_{\text{def}} \left[ \begin{array}{l} \text{preconds} : \left[ \begin{array}{l} \text{LatestMove}^{\text{Content}} = \text{CounterPart}(\text{spkr}, \text{addr}) : \text{IllocProp} \\ \text{qud} = \{ \} : \text{poset}(\text{InfoStruc}) \end{array} \right] \\ \text{effects} : \left[ \text{LatestMove}^{\text{Content}} = \text{Disengage}(\{ \text{spkr}, \text{addr} \}) : \text{IllocProp} \right] \end{array} \right]
 \end{aligned}$$

### B.3.2 Reactive Moves

$$(8) \text{ QSPEC} =_{\text{def}} \left[ \begin{array}{l} \text{pre} : \left[ \text{qud} = \langle i, I \rangle : \text{poset}(\text{InfoStruc}) \right] \\ \text{effects} : \text{TurnUnderspec} \wedge_{\text{merge}} \left[ \begin{array}{l} r : \text{AbSemObj} \\ R : \text{IllocRel} \\ \text{LatestMove} = R(\text{spkr}, \text{addr}, r) : \text{IllocProp} \\ c1 : \text{Qspecific}(r, i.q) \end{array} \right] \end{array} \right]$$

$$(9) \text{ Explanation rule} =_{\text{def}} \left[ \begin{array}{l} \text{pre} : \left[ f : \text{BroadlyTopical} \right] \\ \text{effects} : \left[ \begin{array}{l} \text{MaxQud.q} = \text{Why}(\text{pre.f}) : \text{Question} \\ \text{co-prop}(\text{LatestMove}^{\text{content}}, \text{MaxQud.q}) \end{array} \right] \end{array} \right]$$

$$(10) \text{ Discussing } u? =_{\text{def}} \left[ \begin{array}{l} \text{pre} : \left[ \text{qud} = \langle i0, I \rangle : \text{poset}(\text{InfoStruc}) \right] \\ \text{effects} : \left[ \begin{array}{l} \text{spkr} = \text{pre.addr} : \text{IND} \\ \text{addr} = \text{pre.spkr} : \text{IND} \\ r : \text{AbSemObj} \\ R : \text{IllocRel} \\ \text{Moves} = \langle R(\text{spkr}, \text{addr}, r) \rangle \oplus \text{pre.Moves} : \text{list}(\text{IllocProp}) \\ c1 : \text{Qspecific}(R(\text{spkr}, \text{addr}, r), ?\text{WishDiscuss}(\text{pre.maxqud})) \\ \text{qud} = \left\langle \text{max} = \left\{ \left[ \begin{array}{l} q = ?\text{WishDiscuss}(q0) \\ \text{FEC} = \{ \} \end{array} \right], i0 \right\}, I \right\rangle : \text{poset}(\text{InfoStruc}) \end{array} \right] \end{array} \right]$$

(11) Irrelevance Inference = *def*

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{dgb} : \text{DGBType} \\ \text{c} : \text{IrRelevant}(\text{maxpending}^{\text{content}}, \text{dgb}) \end{array} \right] \\ \text{effects} : \left[ \begin{array}{l} \text{LatestMove} = \text{pre.pending} : \text{LocProp} \\ \text{Facts} = \text{pre.Facts} \cup \\ \{\neg \text{WishDiscuss}(\text{pre.spkr}, \text{pre.maxqud})\} \end{array} \right] \end{array} \right]$$

(12) Assertion checking = *def*

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{p} : \text{Prop} \\ \text{LatestMove}^{\text{content}} = \text{Assert}(\text{spkr}, \text{addr}, \text{p}) : \text{IllocProp} \\ \text{qud} = \left\langle \left[ \begin{array}{l} \text{q} = \text{p?} \\ \text{FEC} = \text{p0} \end{array} \right], \text{I} \right\rangle : \text{poset}(\text{InfoStruc}) \end{array} \right] \\ \text{effects} : \text{TurnUnderspec} \wedge_{\text{merge}} \left[ \text{LatestMove}^{\text{content}} = \text{Check}(\text{spkr}, \text{addr}, \text{p?}) : \text{IllocProp} \right] \end{array} \right]$$

This version of acceptance, refined in Chapter 8 from an initial proposal in Chapter 4, is more intricate than most other reactive moves here because it also has the side effect of updating DGB-PARAMS.

(13) Accept move = *def*

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{spkr} : \text{IND} \\ \text{addr} : \text{IND} \\ \text{p} = \left[ \begin{array}{l} \text{sit} = \text{sit1} \\ \text{sit-type} = \text{T1} \end{array} \right] : \text{Prop} \\ \text{LatestMove}^{\text{content}} = \text{Assert}(\text{spkr}, \text{addr}, \text{p}) : \text{IllocProp} \end{array} \right] \\ \text{effects} : \left[ \begin{array}{l} \text{spkr} = \text{pre.addr} : \text{IND} \\ \text{addr} = \text{pre.spkr} : \text{IND} \\ \text{t} : \text{T1} \\ \text{w} = \text{pre.LatestMove.dgb-params} \cup \left[ \text{sit} = \text{t} \right] : \text{Rec} \\ \text{p1} = \text{IntendClassify}(\text{spkr}, \text{w}, \text{m0.dgb-params}) : \text{Prop} \\ \text{c2} : \text{member}(\text{p1}, \text{private.beliefs}) \\ \text{Moves} = \text{m1} \oplus \text{pre.Moves} : \text{list}(\text{LocProp}) \\ \text{m1}^{\text{content}} = \text{Accept}(\text{spkr}, \text{addr}, \text{p}) : \text{IllocProp} \\ \text{m0.dgb-param} = \text{w} : \text{Rec} \end{array} \right] \end{array} \right]$$

(14) Confirm move = *def*

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} p : \text{Prop} \\ \text{LatestMove}^{\text{content}} = \text{Check}(\text{spkr}, \text{addr}, p?) : \text{IllocProp} \\ \text{qud} = \left\langle \left[ \begin{array}{l} q=p? \\ \text{FEC}=p0 \end{array} \right], I \right\rangle : \text{poset}(\text{InfoStruc}) \end{array} \right] \\ \text{effects} : \left[ \begin{array}{l} \text{spkr} = \text{pre.addr} : \text{Ind} \\ \text{addr} = \text{pre.spkr} : \text{Ind} \\ \text{LatestMove} = \text{Confirm}(\text{spkr}, \text{addr}, p) : \text{IllocProp} \end{array} \right] \end{array} \right]$$

#### B.4 QUD/FACT Up/Downdates

The rules for QUD-incrementation as a consequence of asking and assertion have simple formulations in Chapter 4. They become more intricate in Chapter 7, where they also involve the dynamics of FECs.

(15) a. QUD-incrementation

b. Ask *wh*-question QUD-incrementation = *def*

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{spkr} : \text{IND} \\ \text{addr} : \text{IND} \\ q : \text{WhQuestion} \\ \text{LatestMove.cont} = \text{Ask}(\text{spkr}, \text{addr}, q) : \text{IllocProp} \\ \text{new-fec} = \left[ \begin{array}{l} \text{sit} = \text{wh1} \\ \text{sit-type} = T_{\text{wh1}} \end{array} \right] : \text{LocProp} \\ \text{c1} : \text{member}(\text{wh1}, \text{LatestMove.sit.constits}) \\ \quad \wedge \text{WhIntUtt}(\text{new-fec}) \end{array} \right] \\ \text{effects} : \left[ \text{qud} = \langle q, \text{pre.new-fec} \rangle \oplus \text{pre.qud} : \text{list}(\text{info-struc}) \right] \end{array} \right]$$

c. *p?* QUD-incrementation = *def*

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{spkr} : \text{IND} \\ \text{addr} : \text{IND} \\ p : \text{Prop} \\ \text{LatestMove.cont} = \text{Assert}(\text{spkr}, \text{addr}, p) \vee \\ \text{Ask}(\text{spkr}, \text{addr}, p?) : \text{IllocProp} \\ \text{new-fec} : \text{set}(\text{LocProp}) \\ \text{c1} : \forall x[x = \left[ \begin{array}{l} \text{sit} = \text{quant1} \\ \text{sit-type} = T_{\text{quant1}} \end{array} \right] \wedge \text{member}(x, \text{new-fec})] \\ \rightarrow \text{Quantutt}(x) \wedge \text{member}(\text{quant1}, \text{LatestMove.sit.constits}) \end{array} \right] \\ \text{effects} : \left[ \text{qud} = \langle p?, \text{pre.new-fec} \rangle \oplus \text{pre.qud} : \text{list}(\text{info-struc}) \right] \end{array} \right]$$

The formulation provided here of Fact Update/ QUD Downdate ignores the topicality dimension among FACTS. A formulation that takes this into account is discussed in Chapter 8. It is formulated for QUDs that consist of InfoStrucs. The function NonResolve removes from



QUD all elements whose question component  $q$  is such that an element of FACTS resolves  $q$  or the lack of desire to discuss  $q$ .

- (16) a. Fact Update/ QUD DOWNDATE:

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} p : \text{Prop} \\ \text{LatestMove} = \text{Accept}(\text{spkr}, \text{addr}, p) \vee \text{Confirm}(\text{spkr}, \text{addr}, p) \\ \text{qud} = \left\langle \left[ \begin{array}{l} q = p? \\ \text{FEC} = f0 \end{array} \right], \text{pre.qud} \right\rangle : \text{poset}(\text{InfoStruc}) \end{array} \right] \\ \text{effects} : \left[ \begin{array}{l} \text{facts} = \text{pre.facts} \cup \{p\} : \text{Set}(\text{Prop}) \\ \text{qud} = \text{NonResolve}(\text{pre.qud}, \text{facts}).Q' : \text{Poset}(\text{InfoStruc}) \end{array} \right] \end{array} \right]$$

b.  $\text{NonResolve} =_{\text{def}}$

$$r : \left( \left[ \begin{array}{l} B : \text{IND} \\ F : \text{set}(\text{Prop}) \\ Q : \text{poset}(\text{InfoStruc}) \end{array} \right] \right) \left[ \begin{array}{l} Q' : \text{poset}(\text{InfoStruc}) \\ c1 : Q' \subset r.Q \\ c2 : \forall q_0 \in Q' \neg \exists f \in F \\ \text{Resolve}(f, q_0.q) \vee \text{Resolve}(f, ?\text{WishDiscuss}(r.B, q_0.q)) \end{array} \right]$$

I retain here the formulation of QCoord from Chapter 4, where QUD consists solely of questions, as the InfoStruc formulation unnecessarily obscures the main point here.

- (17) QCoord:

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} q : \text{Question} \\ \text{LatestMove} = \text{Ask}(\text{spkr}, \text{addr}, q) : \text{IllocProp} \\ \text{qud} = \langle q, Q \rangle : \text{poset}(\text{Question}) \end{array} \right] \\ \text{effects} : \left[ \begin{array}{l} q1 : \text{Question} \\ c1 : \neg \text{Influence}(q1, \text{pre}.q) \\ \text{LatestMove} = \text{Ask}(\text{spkr}, \text{addr}, q1) : \text{IllocProp} \\ \text{qud} = \langle \text{pre}.q, q1, \text{pre}.Q \rangle : \text{poset}(\text{Question}) \end{array} \right] \end{array} \right]$$

## B.5 Total Information State Rules

- (18) a.  $m_0$  is relevant to  $G_0$  in  $\text{dgb}_0$  for A (**GenreRelevant**( $m, G_0, \text{dgb}_0$ )) iff A believes that  $\text{outcome}(\text{dgb}_0 \oplus_{\text{moves}} m_0, G_0)$  will be fulfilled. That is, iff there exists  $\text{dgb}_1$  such that  $\text{dgb} \sqsubset \text{dgb}_1$  and such that  $\text{dgb}_1 : G_0$
- b.  $m_0$  is relevant to  $G_0$  in  $\text{dgb}_0$  QUD-presupposing  $q(m_0)$  (**GenreRelevant**<sup>*qudpresupp*</sup>( $m, q(m_0), \text{dgb}_0, G_0$ )) iff A believes that the outcome  $\text{outcome}(\text{dgb} \cup \left[ \begin{array}{l} \text{dgb.moves} := \langle m_0, \text{dgb.moves} \rangle \\ \text{dgb.qud} := \langle q_0, \text{dgb.qud} \rangle \\ c1 : \text{Co-propositional}(\text{qud-contrib}(m_0.\text{cont}), q(m_0)) \end{array} \right], G_0)$  will be fulfilled.

(19) Initiating Move = *def*

$$\left[ \begin{array}{l} \text{dgb} : [\text{qud} = \langle \rangle : \text{eset}(\text{info-struc})] \wedge \text{DGBType} \\ \\ \text{pre} : \left[ \begin{array}{l} \text{genre} : \text{GenreType} \\ \text{beliefs} : \text{Prop} \\ \text{agenda} : \text{list}(\text{IllocProp}) \\ \text{private} = \left[ \begin{array}{l} \text{m0} : \text{locProp} \\ \text{q0} : \text{info-struc} \\ \text{c1} : \rightarrow (\text{beliefs}, \\ \quad \text{GenreRelevant}^{\text{qudpresupp}}(\text{m0}^{\text{cont}}, \text{q0.q}, \text{dgb}, \text{genre}) ) \end{array} \right] \end{array} \right] : \text{PRTType} \\ \\ \text{effects} : \text{Turnholder-Underspec} \wedge_{\text{merge}} \\ \left[ \begin{array}{l} \text{LatestMove} = \text{pre.private.m0} : \text{LocProp} \\ \text{qud} = \langle \text{q0} \rangle : \text{poset}(\text{InfoStruc}) \\ \text{c3} : \text{Co-propositional}(\text{qud-contrib}(\text{m0.cont}), \text{q0.q}) \end{array} \right] \end{array} \right]$$

(20) Utterance interpolation (with question accommodation) = *def*

$$\left[ \begin{array}{l} \text{pre} : \text{DGBType} \\ \left[ \begin{array}{l} \text{T}_u : \text{GramType} \\ \text{u} : \text{Rec} \\ \text{p1} = \left[ \begin{array}{ll} \text{sit} & = \text{u} \\ \text{sit-type} & = \text{T}_u \end{array} \right] : \text{Prop} \end{array} \right] \\ \\ \text{effects} : \left[ \begin{array}{l} \text{c2} : \text{True}(\text{p1}) \\ \text{il} : \text{InfoStruc} \\ \text{qud} = \langle \text{il}, \text{pre.qud} \rangle : \text{list}(\text{question}) \\ \text{c3} : \text{Co-propositional}(\text{p1}^{\text{content}}, \text{il.q}) \\ \text{facts} = \text{pre.facts} \cup \{\text{MostRecentSpeechEvent}(\text{u})\} \\ \text{moves} = \langle \text{p1}, \text{pre.moves} \rangle \end{array} \right] \end{array} \right]$$

- (21) a. Indirect Genre Relevance: A move  $m_1$  is *indirectly relevant* to a genre type  $G_0$  iff  $\text{qud} - \text{contrib}(m_1)$  influences  $\text{qud} - \text{contrib}(m_0)$  for some  $\text{IllocProp } m_0$  that is relevant to a genre type  $G_0$ .
- (i)  $\text{qud-contrib}(m_0^{\text{content}})$  is  $m_0^{\text{content}}$  if  $m_0^{\text{content}} : \text{Question}$
  - (ii)  $\text{qud-contrib}(m_0^{\text{content}})$  is  $?m_0^{\text{content}}$  if  $m_0^{\text{content}} : \text{Prop}$

(22) Indirect Move = *def*

$$\left[ \begin{array}{l} \text{dgb : DGBType} \\ \text{pre : } \left[ \begin{array}{l} \text{genre : GenreType} \\ \text{beliefs : Prop} \\ \text{private = } \left[ \begin{array}{l} \text{agenda : list(IllocProp)} \\ \text{ip0 : IllocProp} \\ \text{c1 : } \rightarrow (\text{beliefs, IndirectGenreRelevant(ip0, dgb, genre)}) \end{array} \right] : \text{PRTYPE} \end{array} \right] \\ \text{effects : TurnUnderspec } \wedge_{\text{merge}} [\text{LatestMove}^{\text{content}} = \text{pre.private.ip0 : IllocProp}] \end{array} \right]$$

## B.6 Pending-Based Rules

### B.6.1 Utterance Integration

(23) Pending Update = *def*

$$\left[ \begin{array}{l} \text{T}_u : \text{GramType} \\ \text{u = } \left[ \begin{array}{l} \text{phon = p} \\ \text{cat = syncatl} \\ \text{constits = conl} \\ \text{dgb-params = } \left[ \begin{array}{l} \text{spkr = s} \\ \text{addr = a0} \\ \text{utt-time = t0} \end{array} \right] : \text{sign} \\ \text{cont = conl} \end{array} \right] \\ \text{pre : } \left[ \begin{array}{l} \text{audio : Rec} \\ \text{parse : Rec} \\ \text{d : dgb} \\ \text{p1 = } \left[ \begin{array}{l} \text{sit} = \text{audio} \\ \text{sit-type} = \text{MostRecentSpeechEvent(u)} \end{array} \right] : \text{Prop} \\ \text{p2 = } \left[ \begin{array}{l} \text{sit} = \text{parse} \\ \text{sit-type} = \text{Classifies}(\text{T}_u, \text{u}) \end{array} \right] : \text{Prop} \\ \text{c1 : member(p1, FACTS)} \\ \text{c2 : member(p2, FACTS)} \end{array} \right] \\ \text{effects : } \left[ \begin{array}{l} \text{spkr = u.dgb-params.spkr : Ind} \\ \text{addr = u.dgb-params.addr : Ind} \\ \text{pending = } \left\langle \left[ \begin{array}{l} \text{sit = u} \\ \text{sit-type = T}_u \end{array} \right], \text{pre.pending} \right\rangle : \text{list(locProp)} \end{array} \right] \end{array} \right]$$

(24) Pending Extension =  $_{def}$

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} T_u : \text{GramType} \\ u : \text{Rec} \\ \text{spkr} : \text{IND} \\ \text{MaxPending} = \left[ \begin{array}{l} \text{sit} = u \\ \text{sit-type} = T_u \end{array} \right] : \text{LocProp} \\ p1 : \text{LocProp} \\ c1 : \text{Extension}(p, p1) \\ p1 = \text{IntendClassify}(\text{spkr}, u, p1.\text{sit-type}) : \text{Prop} \\ c2 : \text{member}(p1, \text{private.beliefs}) \end{array} \right] \\ \text{effects} : \left[ \text{MaxPending} = p1 : \text{LocProp} \right] \end{array} \right]$$

(25)  $\text{LocPropExtension}(p1 = \left[ \begin{array}{l} \text{sit} = u \\ \text{sit-type} = T_u \end{array} \right], p2 = \left[ \begin{array}{l} \text{sit} = v \\ \text{sit-type} = T_v \end{array} \right])$  iff  $p1, p2 : \text{LocProp}$   
and (a) for all sign fields either  $u.f = v.f$  or  $u.f \sqsubset v.f$  and (b)  $T_u \sqsubset T_v$

### B.6.2 Repair

The general form of a Clarification Context Update Rule is given in (26a). Rules differ with respect to the identity of the question that gets accommodated as MAX-QUD (26b, c, d) indicate the questions associated with distinct CCURs. Combining a CCUR with the rule of utterance interpolation yields (26e):

(26) a. Clarification Context Update Rule =  $_{def}$

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{MaxPending} = \left[ \begin{array}{l} \text{sit} = v0 \\ \text{sit-type} = T_{v0} \end{array} \right] : \text{LocProp} \\ p1 = \left[ \begin{array}{l} \text{sit} = u1 \\ \text{sit-type} = T_{u1} \end{array} \right] : \text{LocProp} \\ c1 : \text{member}(u1, v0.\text{constits}) \end{array} \right] \\ \text{effects} : \left[ \begin{array}{l} \text{spkr} = \text{pre.addr} : \text{IND} \\ \text{addr} = \text{pre.spkr} : \text{IND} \\ \text{MaxQud} = \left[ \begin{array}{l} q = q1 : \text{Question} \\ \text{fec} = \text{pre.p1} : \text{LocProp} \end{array} \right] : \text{InfoStruc} \end{array} \right] \end{array} \right]$$

b. parameter identification:  $q1 = \lambda x \text{ Mean}(\text{pre.spkr}, \text{pre.u1}, x)$

c. parameter focussing:  $q1 = \lambda x v0.\text{cont}(\text{pre.u1.c-param} \rightsquigarrow x)$

d. repetition:  $q1 = \lambda x \text{ Utter}(\text{pre.spkr}, \text{pre.u1}, x)$

e. Utterance Interpolation  $\wedge_{merge}$  CCUR = *def*

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{MaxPending} = \left[ \begin{array}{l} \text{sit} = v0 \\ \text{sit-type} = T_{v0} \end{array} \right] : \text{LocProp} \\ p0 = \left[ \begin{array}{l} \text{sit} = u1 \\ \text{sit-type} = T_{u1} \end{array} \right] : \text{LocProp} \\ c1: \text{member}(u1, v0.\text{constits}) \end{array} \right] \\ \\ \text{effects} : \left[ \begin{array}{l} \text{spkr} = \text{pre.addr} : \text{IND} \\ \text{addr} = \text{pre.spkr} : \text{IND} \\ u : \text{sign} \\ T_u : \text{GramType} \\ q1 : \text{Question} \\ p1 = \left[ \begin{array}{l} \text{sit} = u \\ \text{sit-type} = T_u \end{array} \right] : \text{Prop} \\ c2: \text{True}(p1) \\ \text{MaxQud} = \left[ \begin{array}{l} q = q1 \\ \text{fec} = \text{pre.p0} \end{array} \right] : \text{InfoStruc} \\ c3: \text{Co-propositional}(p1^{\text{content}}, \text{MaxQuD}, q) \\ \text{facts} = \text{pre.facts} \cup \{\text{MostRecentSpeechEvent}(u)\} \\ \text{moves} = \langle p1, \text{pre.moves} \rangle \end{array} \right] \end{array} \right]$$

- (27) a. Given an utterance  $u : \text{sign}$ :  
b.  $\text{Mean}(A, u, c)$  iff  $u.c\text{-param.spkr} = A$  and  $u.c\text{-param.x} = c$   
c.  $\text{Utter}(A, u, T)$  iff  $u.c\text{-param.spkr} = A$  and  $u.\text{phon} : T$

(28) CR Accommodation = *def*

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{MaxPending} = \left[ \begin{array}{l} \text{sit} = v1 \\ \text{sit-type} = T_{v1} \end{array} \right] : \text{LocProp} \\ \text{LatestMove} = \left[ \begin{array}{l} \text{sit} = v0 \\ \text{sit-type} = T_{v0} \end{array} \right] : \text{LocProp} \\ \text{addr} = v0.\text{dgb-params.spkr} : \text{Ind} \\ u1 : \text{Sign} \\ c1: \text{member}(u1, v0.\text{constits}) \\ q1 = \text{CCUR.qud}(u1) : \text{InfoStruc} \\ c2 : \text{Co-propositional}(q1.q, \text{MaxPending}) \end{array} \right] \\ \\ \text{effects} : \left[ \begin{array}{l} \text{LatestMove} = \text{Pre.MaxPending} : \text{LocProp} \\ \text{MaxQud} = \text{pre.q1} : \text{InfoStruc} \end{array} \right] \end{array} \right]$$

(29) Backwards-looking appropriateness repair = *def*

$$\left[ \begin{array}{l} \text{pre} : \left[ \begin{array}{l} \text{spkr} : \text{IND} \\ \text{addr} : \text{IND} \\ \text{pending} = \langle \text{p0}, \text{rest} \rangle : \text{list}(\text{LocProp}) \\ \text{cl} : \text{member}(\text{u0}, \text{p0.sit.constits}) \end{array} \right] \\ \\ \text{effects} : \left[ \begin{array}{l} \text{spkr} = \text{pre.spkr} : \text{Ind} \\ \text{addr} = \text{pre.addr} : \text{Ind} \\ \text{MaxQUD} = \\ \left[ \begin{array}{l} \text{q} = \lambda x \text{ Mean}(\text{pre.spkr}, \text{pre.u0}, x) \\ \text{FEC} = \text{p0} \end{array} \right] : \text{InfoStruc} \\ \text{LatestMove} : \text{LocProp} \\ \text{c2} : \text{Co-propositional}(\text{LatestMove}^{\text{content}}, \text{MaxQUD}) \end{array} \right] \end{array} \right]$$

## B.7 Multilogue-Oriented rules

Scaling up to multilogue involves modifying the audience set of the rules in the earlier parts of the appendix as follows:

(30) a. Duologue roles:

$$\left[ \begin{array}{l} \text{spkr} = \text{pre.spkr} : \text{IND} \\ \text{addr} = \text{pre.addr} : \text{IND} \end{array} \right] \mapsto \left[ \begin{array}{l} \text{spkr} = \text{pre.spkr} : \text{IND} \\ \text{addr} = \text{pre.addr} : \text{Set}(\text{IND}) \end{array} \right]$$

b. Duologue roles:

$$\left[ \begin{array}{l} \text{spkr} = \text{pre.addr} : \text{IND} \\ \text{addr} = \text{pre.spkr} : \text{IND} \end{array} \right] \mapsto \left[ \begin{array}{l} \text{spkr} : \text{IND} \\ \text{cl} : \text{member}(\text{spkr}, \text{pre.addr}) \\ \text{addr} = \text{pre.addr} \cup \{\text{pre.spkr}\} \setminus \{\text{spkr}\} : \text{Set}(\text{IND}) \end{array} \right]$$

c. Duologue roles:

$$\left[ \begin{array}{l} \text{PrevAud} = \{\text{pre.spkr}, \text{pre.addr}\} : \text{Set}(\text{IND}) \\ \text{spkr} : \text{IND} \\ \text{cl} : \text{member}(\text{spkr}, \text{PrevAud}) \\ \text{addr} : \text{IND} \\ \text{c2} : \text{member}(\text{addr}, \text{PrevAud}) \wedge \text{addr} \neq \text{spkr} \end{array} \right] \mapsto \left[ \begin{array}{l} \text{PrevAud} = \{\text{pre.spkr}, \text{pre.addr}\} : \text{Set}(\text{IND}) \\ \text{spkr} : \text{IND} \\ \text{cl} : \text{member}(\text{spkr}, \text{PrevAud}) \\ \text{addr} = \text{PrevAud} \setminus \{\text{spkr}\} : \text{Set}(\text{IND}) \end{array} \right]$$

(31) Turn assignment = *def*

$$\left[ \begin{array}{l} \text{pre : } \left[ \begin{array}{l} \text{spkr : IND} \\ \text{addr : Set(IND)} \\ \text{j : IND} \\ \text{cl : member(j, addr)} \\ \text{LatestMove = TurnAssign(spkr, j) : IllocProp} \end{array} \right] \\ \text{effects : } \left[ \begin{array}{l} \text{spkr = pre.j : IND} \\ \text{addr = pre.addr} \cup \{ \text{pre.spkr} \} \setminus \{ \text{spkr} \} : \text{Set(IND)} \end{array} \right] \end{array} \right]$$

## APPENDIX C

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### *The Grammatical Domain*

The aim of this appendix is to sketch a grammar fragment by collating the various lexical and constructional entries posited in this book. It is no more than a sketch, making no attempt at compact description in terms of type hierarchies and constraint inheritance. The grammar is a version of the grammar in Ginzburg and Sag (2000), with some simplifications—for expository reasons—on the syntactic front and some significant differences in the semantic and contextual components. The TTR formalism, however, allows utterance types to include dependencies involving phonological types, unavailable in standard HPSG. This possibility is exploited on a couple of occasions.

#### C.1 General Notions

$$(1) \quad \text{Sign} =_{\text{def}} \left[ \begin{array}{l} \text{phon} : \text{list}(\text{phonform}) \\ \text{cat} : \left[ \text{head} : \text{PoS} \right] \\ \text{gaps} : \text{list} \left( \left[ \begin{array}{l} \text{cat} : \left[ \text{head} : \text{PoS} \right] \\ \text{cont} : \left[ x : \text{IND} \right] \end{array} \right] \right) \\ \text{dgb-params} : \text{RecType} \\ \text{q-params} : \text{RecType} \\ \text{cont} : \text{SemObj} \\ \text{quest-dom} : \text{list}(\text{RecType}) \end{array} \right]$$

#### C.2 Lexicon

$$(2) \quad \left[ \begin{array}{l} \text{jo} : \text{Phontype} \\ \text{phon} : \text{jo} \\ \text{cat.head} = N : \text{POS} \\ \text{cont} : \left[ x : \text{IND} \right] \\ \text{dgb-params} : \left[ \begin{array}{l} y = \text{cont.x} : \text{IND} \\ \text{rest}^{\text{facts}} : \text{Named}(y, \text{jo}) \end{array} \right] \end{array} \right]$$

$$(3) \quad \left[ \begin{array}{l} \text{phon} : \text{who} \\ \text{cat.head} = N : \text{POS} \\ \text{cont} : \left[ x : \text{IND} \right] \\ \text{quest-dom} = \left\langle \left[ \begin{array}{l} y = \text{cont.x} : \text{IND} \\ \text{rest1} : \text{person}(\text{index}) \end{array} \right] \right\rangle : \text{list}(\text{RecType}) \end{array} \right]$$



- (4) a. 'a cat', independent use:

$$\left[ \begin{array}{l} \text{phon} : \text{a cat} \\ \text{cat.head} = \text{N} : \text{POS} \\ \text{dgb-params} : [] \\ \text{q-params} : \left[ \begin{array}{l} x : \text{IND} \\ r : \text{thief}(x) \end{array} \right] \\ \text{cont} : [x = \text{q-params}.x : \text{IND}] \end{array} \right]$$

- b. 'a cat', dependent use:

$$\left[ \begin{array}{l} \text{phon} : \text{a cat} \\ \text{cat.head} = \text{N} : \text{POS} \\ \text{MaxPending} : \text{LocProp} \\ \text{dgb-params} : \left[ \begin{array}{l} \text{ant} : [\text{q-params} : [x : \text{IND}]] \\ c : \text{member}(a, \text{MaxPending.constits.sit}) \end{array} \right] \\ \text{q-params} : \left[ f : ([x : \text{Ind}]) \left[ \begin{array}{l} z = f(x) : \text{IND} \\ c1 : \text{cat}(z) \end{array} \right] \right] \\ \text{cont} : [x = \text{q-params}.f(\text{ants.q-param}.x) : \text{IND}] \end{array} \right]$$

- (5) a.  $\text{IV} =_{\text{def}} ([\text{subj.cont}.x : \text{Ind}])\text{RecType}$   
b.  $\text{TV} =_{\text{def}} ([\text{comp.cont}.x : \text{IND}])([\text{subj.cont}.x : \text{IND}])\text{RecType}$   
c.  $x^s =_{\text{def}} \text{subj.cont}.x$   
d.  $x^c =_{\text{def}} \text{comp.cont}.x$

(6) 
$$\left[ \begin{array}{l} \text{phon} : \text{runs} \\ \text{cat} : \left[ \begin{array}{l} \text{head} = \text{V} : \text{PoS} \\ \text{subj} : <\text{synsem}> \end{array} \right] \\ \text{cont} = (r : [x^s : \text{Ind}]) [c : \text{run}(r.x^s)] : \text{IV} \end{array} \right]$$

(7) a. 
$$\left[ \begin{array}{l} \text{phon} : \text{visit} \\ \text{cat} : \left[ \begin{array}{l} \text{head} = \text{V} : \text{PoS} \\ \text{subj} : <\text{synsem}> \\ \text{comps} : <\text{synsem}> \end{array} \right] \\ \text{cont} = (r1 : [x^c : \text{Ind}]) (r2 : [x^s : \text{IND}]) \\ [c : \text{visit}(r2.x^s, r1.x^c)] : \text{TV} \end{array} \right]$$

b. The gaps version to be used in object extraction cases:

$$\left[ \begin{array}{l} \text{phon : visit} \\ \text{cat : [head = V : PoS]} \\ \text{gaps : nelist} \left[ \begin{array}{l} \text{cat : [head : PoS]} \\ \text{cont : [x : IND]} \end{array} \right] \\ \text{cont = (r : [x^S:IND])} \left[ \begin{array}{l} \text{c : visit(r.x,gaps.cont.x)} \end{array} \right] : \text{IV} \end{array} \right]$$

$$(8) \left[ \begin{array}{ll} \text{phon} & : \text{did} \\ \text{cat} & : \left[ \begin{array}{ll} \text{head = V} & : \text{PoS} \\ \text{subj} & : \text{<synsem>} \end{array} \right] \\ \text{cont = ID}_{(\text{RecType})\text{RecType}} & : (\text{RecType}) \text{ RecType} \end{array} \right]$$

$$(9) \left[ \begin{array}{l} \text{phon : huh} \\ \text{cat.head = interjection : PoS} \\ \text{dgb-params :} \left[ \begin{array}{l} \text{spkr : IND} \\ \text{addr : IND} \\ \text{pending : utt} \\ \text{c2 : address(addr, spkr, pending)} \end{array} \right] \\ \text{cont = Ask(spkr, addr,} \\ \quad \lambda x \text{ Mean(addr, pending, x)) : IllocProp} \end{array} \right]$$

$$(10) \left[ \begin{array}{l} \text{phon : yes} \\ \text{cat.head = adv[+ic] : syncat} \\ \text{dgb-params.max-qud : PolQuestion} \\ \text{cont = max-qud} \left[ \begin{array}{l} \end{array} \right] : \text{Prop} \end{array} \right]$$

$$(11) \left[ \begin{array}{l} \text{phon : no} \\ \text{cat.head = adv[+ic] : syncat} \\ \text{dgb-params.max-qud : PolQuestion} \\ \text{cont : Prop} \\ \text{c1 : NegProp(cont) } \wedge \text{ SimpleAns(cont, max-qud)} \end{array} \right]$$

$$(12) \left[ \begin{array}{l} \text{phon : mmh} \\ \text{cat.head = interjection : syncat} \\ \text{dgb-params :} \left[ \begin{array}{l} \text{spkr : IND} \\ \text{addr : IND} \\ \text{MaxPending : LocProp} \\ \text{c2 : address(addr, spkr, MaxPending)} \end{array} \right] \\ \text{cont =} \left[ \text{c1 : Understand(spkr, addr, MaxPending)} \right] : \text{RecType} \end{array} \right]$$

$$(13) \left[ \begin{array}{l} \text{phon : she} \\ \text{dgb-params : } \left[ \begin{array}{l} \text{m : LocProp} \\ \text{a : LocProp} \\ \text{c2 : member(a, m.constits)} \\ \text{c : DiscourseAnt(m, a) } \vee \text{ IntraSentAnt(m, a)} \end{array} \right] \\ \text{cat = } \left[ \begin{array}{l} \text{head = N : PoS} \\ \text{ana : +} \\ \text{agr = dgb-params.m.cat.agr : } \left[ \begin{array}{l} \text{num = sg : Number} \\ \text{gen = fem : Gender} \\ \text{pers = third : Person} \end{array} \right] \end{array} \right] : \text{syncat} \\ \text{cont : [x = a.cont.x : IND]} \end{array} \right]$$

- (14) a. For a given DGB dgb0, an ActiveMove is an element of dgb0.Moves or dgb0.Pending such that either  
 (a) *qud-update-contribution*( $m^{\text{content}}$ ) is in dgb0.QUD or  
 (b)  $m^{\text{content}}$  is TOPICAL or  
 (c) m is in dgb0.Pending  
 b.  $\text{DiscourseAnt(m, a)} = \left[ \begin{array}{l} \text{cl : ActiveMove(m)} \\ \text{m.sit.dgb-params : [a.dgb-params.x = a.cont.x : IND]} \end{array} \right]$   
 c.  $\text{IntraSentAnt(m,a)} = \left[ \begin{array}{l} \text{cl : Maxpending(m)} \\ \text{a : [cont : [x : Ind]]} \end{array} \right]$

### C.3 Phrasal Types

#### C.3.1 General types

$$(15) \text{hd-comp-ph} =_{\text{def}} \left[ \begin{array}{l} \text{dgb-params : } \left[ \begin{array}{l} \text{comp-dtr.dgb-params : DGP1} \\ \text{hd-dtr.dgb-params : DGP2} \end{array} \right] \\ \text{cont = dtrs.hd-dtr.cont(dtrs.comp-dtr.cont) : IV} \end{array} \right]$$

**head:**

$$\left[ \begin{array}{l} \text{DGP2 : RecType} \\ \text{dgb-params : DGP2} \\ \text{cont : (CT1)IV} \end{array} \right]$$

**comp:**

$$\left[ \begin{array}{l} \text{DGP1 : RecType} \\ \text{dgb-params : DGP1} \\ \text{CT1 : Type} \\ \text{cont : CT1} \end{array} \right]$$

The version of the declarative ‘NP/VP’ rule given here includes the integration of q-params in sentential content. Although I haven’t spelled out similar effects in other phrasal rules, it is straightforward to do so.

$$(16) \left[ \begin{array}{l} \text{dgb-params:} \left[ \begin{array}{l} s0 : \text{Rec} \\ \text{subj-dtr.dgb-params} : \text{DGP1} \\ \text{hd-dtr.dgb-params} : \text{DGP2} \end{array} \right] \\ \text{cont} = \left[ \begin{array}{l} \text{sit} = s0 \\ \text{sit-type} = \left[ \begin{array}{l} \text{subj-dtr.qparams} : \text{QP1} \\ \text{hd-dtr.qparams} : \text{QP2} \\ \text{nucl} : \text{hd-dtr.cont}(\text{subj-dtr.cont}) \end{array} \right] \end{array} \right] \end{array} \right] : \text{Prop}$$

$$\begin{array}{cc} \text{subj:} \left[ \begin{array}{l} \text{DGP1 : RecType} \\ \text{dgb-params : DGP1} \\ \text{QP1 : RecType} \\ \text{q-params : QP1} \\ \text{cont : [x : IND]} \end{array} \right] & \text{head:} \left[ \begin{array}{l} \text{DGP2 : RecType} \\ \text{dgb-params : DGP2} \\ \text{QP2 : RecType} \\ \text{q-params : QP2} \\ \text{cont : IV} \end{array} \right] \end{array}$$

$$(17) \text{decl-inv-cl} =_{\text{def}} \left[ \begin{array}{l} \text{gaps : nelist} \left( \left[ \begin{array}{l} \text{cat :} \left[ \begin{array}{l} \text{head : PoS} \\ \text{cont : [x : IND]} \end{array} \right] \end{array} \right] \right) \\ \text{dgb-params:} \left[ \begin{array}{l} s0 : \text{Rec} \\ \text{subj-dtr.dgb-params} : \text{DGP1} \\ \text{comp-dtr.dgb-params} : \text{DGP2} \end{array} \right] \\ \text{cont} = \left[ \begin{array}{l} \text{sit} = s0 \\ \text{sit-type} = \text{hd-dtr.cont}(\text{comp-dtr.cont}(\text{subj-dtr.cont})) \end{array} \right] \end{array} \right] : \text{Prop}$$

$$\begin{array}{ccc} \text{head : inv-aux-word} & \text{subj:} \left[ \begin{array}{l} \text{DGP1 : RecType} \\ \text{dgb-params : DGP1} \\ \text{cont : [x}^s \text{ : IND]} \end{array} \right] & \text{comps:} \left[ \begin{array}{l} \text{DGP2 : RecType} \\ \text{dgb-params : DGP2} \\ \text{cont : IV} \end{array} \right] \end{array}$$

(18) CONSTITITS amalgamation:

$$\left[ \begin{array}{l} \text{dtrs} = \text{d0} : \text{list}(\text{sign}) \\ \text{constits} : \text{set}(\text{sign}) \\ \text{cl} : \forall r [\text{member}(r, \text{constits}) \leftrightarrow \text{member}(r, \text{d0}) \vee \text{Constitsmember}(r, \text{d0})] \end{array} \right]$$

Constitsmember( $r$ ,  $d0$ ) holds iff  $r$  is a member of one of  $d0$ 's constits' sets. In other words: if there exist  $c$ ,  $d$  such that  $\text{member}(d, d0)$ ,  $c = d.\text{constits}$ , and  $\text{member}(r, c)$ .

### C.3.2 Interrogative Constructions

(19) *pol-int-cl* =<sub>def</sub>

$$\left[ \begin{array}{l} \text{dgb-params} : \left[ \begin{array}{l} \text{s0} : \text{Rec} \\ \text{subj-dtr.dgb-params} : \text{DGP1} \\ \text{comp-dtr.dgb-params} : \text{DGP2} \end{array} \right] \\ \text{cont} = ?. \left[ \begin{array}{l} \text{sit} = \text{s0} \\ \text{sit-type} = \text{hd-dtr.cont}(\text{comp-dtr.cont}(\text{subj-dtr.cont})) \end{array} \right] : \text{Question} \end{array} \right]$$

**head** : *inv-aux-word*    **subj**:  $\left[ \begin{array}{l} \text{DGP1} : \text{RecType} \\ \text{dgb-params} : \text{DGP1} \\ \text{cont} : [x^s : \text{IND}] \end{array} \right]$     **comps** :  $\left[ \begin{array}{l} \text{DGP2} : \text{RecType} \\ \text{dgb-params} : \text{DGP2} \\ \text{cont} : \text{IV} \end{array} \right]$

(20) a. *wh-ns-int-cl* =<sub>def</sub>

$$\left[ \begin{array}{l} \text{dgb-params} : \left[ \begin{array}{l} \text{filler-dtr.dgb-params} : \text{DGP1} \\ \text{head-dtr.dgb-params} : \text{DGP2} \end{array} \right] \\ \text{quest-dom} = \langle \rangle : \text{list}(\text{RType}) \\ \text{cont} = (\langle r : F \rangle) \text{hd-dtr.cont}(\text{gaps.cont.x} \sqcap r.\text{filler.cont.x}) : \text{Question} \end{array} \right]$$

**filler** :  $\left[ \begin{array}{l} \text{DGP1} : \text{RecType} \\ \text{dgb-params} : \text{DGP1} \\ F : \text{type} \\ \text{quest-dom} = \langle F \rangle : \text{list}(\text{RType}) \end{array} \right]$     **head** :  $\left[ \begin{array}{l} \text{DGP2} : \text{RecType} \\ \text{dgb-params} : \text{DGP2} \\ \text{cont} : \text{Prop} \\ \text{gaps} : \left[ \begin{array}{l} \text{cat} : [\text{head} : \text{PoS}] \\ \text{cont} : [x : \text{IND}] \end{array} \right] \end{array} \right]$

(21) a. *polarization* =<sub>def</sub>

$$\left[ \begin{array}{l} \text{dgb-params} : \text{DGP1} \\ \text{cont} = ?[\text{hd-dtr.cont}] : \text{Question} \end{array} \right]$$

$$\text{hd-dtr: } \left[ \begin{array}{l} \text{DGP1} : \text{RecType} \\ \text{dgb-params} : \text{DGP1} \\ \text{cont} : \text{Prop} \\ \text{quest-dom} = \langle \rangle : \text{RType} \end{array} \right]$$

 b. *unary-dir-is-cl* =<sub>def</sub>

$$\left[ \begin{array}{l} \text{dgb-params} : \text{DGP1} \\ \text{F} : \text{RecType} \\ \text{G} = [\text{x} : \text{IND}] \wedge_{\text{merge}} \text{F} \\ \text{cont} = (\text{r} : \text{G})\text{hd-dtr.cont}(\text{quest-dom.l.x} \rightsquigarrow \text{r.x}) : \text{Question} \end{array} \right]$$

$$\text{hd-dtr: } \left[ \left[ \begin{array}{l} \text{DGP1} : \text{RecType} \\ \text{dgb-params} : \text{DGP1} \\ \text{cont} : \text{Prop} \\ \text{quest-dom} : \langle \text{G} \rangle \end{array} \right] \right]$$

 (22) *repr-int-cl* =<sub>def</sub>

$$\left[ \begin{array}{l} \text{cat.head} = \text{V}[\text{+fin}] : \text{PoS} \\ \text{dgb-params} : \left[ \begin{array}{l} \text{spkr} : \text{IND} \\ \text{addr} : \text{IND} \\ \text{cl} : \text{Addr}(\text{spkr}, \text{addr}) \\ \text{P} : \text{IllocRel} \\ \text{r} : \text{AbstSemObj} \\ \text{maxpending}^{\text{content}} = \text{P}(\text{addr}, \text{spkr}, \text{p}) : \text{IllocProp} \end{array} \right] \\ \text{hd-dtr} : \left[ \begin{array}{l} \text{F} : \text{Type} \\ \text{quest-dom} = \langle \text{F} \rangle : \text{list}(\text{RType}) \\ \text{cont} = \text{p} : \text{IllocProp} \end{array} \right] \\ \text{cont} = ([\text{r} : \text{F}])\text{MaxPending}^{\text{content}}(\langle \text{l}_i \rightsquigarrow \text{r.l}_i \rangle_{i \in \text{labels}(\text{F})}) : \text{Question} \\ \text{quest-dom} = \langle \rangle : \text{list}(\text{RType}) \end{array} \right]$$

### C.3.3 Non-Sentential Utterance Types

(23) exemplifies the *dgb-params* inheritance assumed here for all NSU types. I avoid specifying it in subsequent NSU-type specifications.

(23) *decl-frag-cl* (quantifier-free version) = *def*

$$\left[ \begin{array}{l} \text{dgb-params} : \left[ \begin{array}{l} \text{max-qud} : \left[ \begin{array}{l} q : \text{UnaryWhQuestion} \\ \text{fec} : \text{LocProp} \end{array} \right] \\ \text{hd-dtr.dgb-params} : \text{DGP1} \end{array} \right] \\ \text{cat.head} = v : \text{PoS} \\ \text{cont} = \text{max-qud.q}(\text{hd-dtr.cont.x}) : \text{Prop} \end{array} \right]$$

$$\text{hd-dtr} : \left[ \begin{array}{l} \text{cat} = \text{max-qud.fec.cat} : \text{Syncat} \\ \text{DGP1} : \text{RecType} \\ \text{dgb-params} : \text{DGP1} \\ \text{cont} : [x : \text{IND}] \end{array} \right]$$

(24) *slu-frag-cl* = *def*

$$\left[ \begin{array}{l} \text{cat.head} = v : \text{PoS} \\ \text{dgb-params.max-qud} : \left[ \begin{array}{l} p : \text{QuantProp} \\ q = ([ ]) p : \text{PolarQuestion} \\ \text{fec} : \text{set}(\text{LocProp}) \end{array} \right] \\ \text{fec} : \text{LocProp} \\ \text{cl} : \text{member}(\text{fec}, \text{max-qud.fec}) \\ \text{cont} = (r : G) \text{max-qud.q}([ ])(\text{fec.cont.x} \rightsquigarrow r.\text{hd-dtr.cont.x}) : \text{Question} \end{array} \right]$$

$$\text{hd-dtr} : \left[ \begin{array}{l} \text{cat} = \text{max-qud.fec.cat} : \text{Syncat} \\ \text{cont} : [x : \text{IND}] \\ F : \text{RecType} \\ G = [x : \text{IND}] \wedge_{\text{merge}} F : \text{RecType} \\ \text{quest-dom} = \langle G \rangle : \text{list}(\text{RType}) \end{array} \right]$$

(25) *utt-anaph-ph* =<sub>def</sub>

$$\left[ \begin{array}{l} \text{tune} = \text{max-qud.fec.sit-type.phon} : \text{Type} \\ \text{phon} : \text{tune} \\ \text{cat.head} = \text{V} : \text{PoS} \\ \text{dgb-params.max-qud} : \text{InfoStruc} \\ \text{hd-dtr} : \text{word} \\ \text{cont} = \text{max-qud.fec.sit} : \text{Rec} \end{array} \right]$$

(26) *qud-anaph-int-cl* =<sub>def</sub>

$$\left[ \begin{array}{l} \text{MAX-QUD} : \text{InfoStruc} \\ \text{cont} = \text{max-qud.q} : \text{Question} \end{array} \right]$$

hd-dtr: *utt-anaph-ph*

(27) *bare-why-cl* =<sub>def</sub>

$$\left[ \begin{array}{l} \text{cat.head} = \text{V} : \text{PoS} \\ \text{dgb-params} : \left[ \begin{array}{l} \text{f} : \text{Prop} \\ \text{Topical} : \text{set}(\text{Prop}) \\ \text{cl} : \text{member}(\text{f}, \text{Topical}) \end{array} \right] \\ \text{cont} = \text{hd-dtr.cont}(\text{f}) : \text{Question} \end{array} \right]$$

hd-dtr:  $\left[ \begin{array}{l} \text{cat} = \text{adv} : \text{syncat} \\ \text{cont} = (\text{f} : \text{Prop}) \lambda r. \text{Cause}(r, \text{f}) : (\text{Prop})\text{Question} \end{array} \right]$



(28) *why-mc-cl* =<sub>def</sub>

$$\left[ \begin{array}{l}
 \text{phon} : \left[ \begin{array}{l} \text{seg} : \text{wai} \\ \text{tune} : \text{H} \end{array} \right] \\
 \text{cat.head} = \text{V} : \text{PoS} \\
 \text{dgb-params} : \left[ \begin{array}{l}
 \text{A} : \text{IND} \\
 \text{B} : \text{IND} \\
 \text{r} : \text{Semobj} \\
 \text{q} : \text{Question} \\
 \text{c0} : \text{QUDContrib}(\text{r}, \text{q}) \\
 \text{c1} : \text{In}(\text{q}, \text{QUD}) \\
 \text{R} : \text{IllocRel} \\
 \text{m} : \left[ \text{cont} = \text{R}(\text{B}, \text{A}, \text{r}) : \text{IllocProp} \right] \wedge \text{sign} \\
 \text{c2} : \text{In}(\text{m}, \text{MOVES})
 \end{array} \right] \\
 \text{cont} = \text{Ask}(\text{A}, \text{B}, \text{hd-dtr}(\text{m}^{\text{content}})) : \text{IllocProp}
 \end{array} \right]$$
  

$$\text{hd-dtr} : \left[ \begin{array}{l}
 \text{cat.head} = \text{adv} : \text{PoS} \\
 \text{cont} = (\text{f} : \text{Prop}) \lambda \text{r}. \text{Cause}(\text{r}, \text{f}) : (\text{Prop})\text{Question}
 \end{array} \right]$$

### C.3.4 Root clauses

I assume that all root clauses are specified as (29) is syntactically. I do not repeat this specification for the other root clauses, which have only their semantics specified.

(29) *generic-root-cl* =<sub>def</sub>

$$\left[ \begin{array}{l}
 \text{cat} = \left[ \begin{array}{l} \text{head} = V[+\text{fin}] : \text{PoS} \\ \text{IC} = + : \text{Boolean} \\ \text{ROOT} = + : \text{Boolean} \end{array} \right] : \text{syncat} \\
 \\
 \text{dgb-params} : \left[ \begin{array}{l} \text{u0} : \text{Rec} \\ \text{s} : \text{IND} \\ \text{t} : \text{TIME} \\ \text{a} : \text{IND} \\ \text{cl} : \text{addressing}(\text{s}, \text{a}, \text{t}) \\ \text{head-dtr.dgb-params} : \text{DGPI} \end{array} \right] \\
 \\
 \text{cont} = \left[ \begin{array}{l} \text{sit} = \text{u0} \\ \text{SIT-TYPE} = \left[ \begin{array}{l} \text{R} = \text{IllocRel}(\text{hd-dtr.cont}) : \text{IllocReln} \\ \text{cl} : \text{R}(\text{s}, \text{a}, \text{t}, \text{hd-dtr.cont}) \end{array} \right] \end{array} \right] : \text{Prop}
 \end{array} \right]$$

|

$$\text{head-dtr:} \left[ \begin{array}{l}
 \text{cat} = \left[ \begin{array}{l} \text{head} = V[+\text{fin}] : \text{PoS} \\ \text{IC} = + : \text{Boolean} \\ \text{ROOT} = - : \text{Boolean} \end{array} \right] : \text{syncat} \\
 \\
 \text{DGPI} : \text{RecType} \\
 \text{dgb-params} : \text{DGPI} \\
 \text{cont} : \text{AbstSemObj} \\
 \text{quest-dom} : \langle \rangle \\
 \text{gaps} : \langle \rangle
 \end{array} \right]$$

(30) *interj-cl* =<sub>def</sub>

$$\left[ \begin{array}{l} \text{cat} = \left[ \begin{array}{l} \text{head} = V[+\text{fin}] : \text{PoS} \\ \text{IC} = + : \text{Boolean} \\ \text{ROOT} = + : \text{Boolean} \end{array} \right] : \text{syncat} \\ \text{dgb-params} : \left[ \begin{array}{l} \text{head-dtr.dgb-params} : \text{DGP1} \\ \text{u0} : \text{Rec} \end{array} \right] \\ \text{cont} = \left[ \begin{array}{l} \text{sit} = \text{u0} \\ \text{SIT-TYPE} = \text{CT1} \end{array} \right] : \text{Prop} \end{array} \right]$$

$$\text{head-dtr} : \left[ \begin{array}{l} \text{cat} = \left[ \begin{array}{l} \text{head} : \text{interj} \\ \text{IC} = + : \text{Boolean} \\ \text{ROOT} = - : \text{Boolean} \end{array} \right] : \text{syncat} \\ \text{DGP1} : \text{RecType} \\ \text{dgb-params} : \text{DGP1} \\ \text{cont} = \text{CT1} : \text{RecType} \end{array} \right]$$

(31) *reassert-cl* =<sub>def</sub>

$$\left[ \begin{array}{l} \text{dgb-params} : \left[ \begin{array}{l} \text{u0} : \text{Rec} \\ \text{s} : \text{IND} \\ \text{t} : \text{TIME} \\ \text{a} : \text{IND} \\ \text{c1} : \text{addressing}(\text{s}, \text{a}, \text{t}) \\ \text{p} : \text{Prop} \\ \text{c2} : \text{resolve}(\text{p}, \text{hd-dtr.cont}) \\ \text{c3} : \text{member}(\text{p}, \text{FACTS}) \end{array} \right] \\ \text{cont} = \left[ \begin{array}{l} \text{sit} = \text{u0} \\ \text{SIT-TYPE} = \left[ \begin{array}{l} \text{R} = \text{Reassert} : \text{IllocReIn} \\ \text{c1} : \text{R}(\text{s}, \text{a}, \text{t}, \text{p}) \end{array} \right] \end{array} \right] : \text{Prop} \end{array} \right]$$

$$\text{hd-dtr} : \left[ \text{cont} : \text{Question} \right]$$

(32) *idiom-assert-cl* =<sub>def</sub>

$$\left[ \begin{array}{l} \text{dgb-params:} \left[ \begin{array}{l} u0 : \text{Rec} \\ s : \text{IND} \\ t : \text{TIME} \\ a : \text{IND} \\ c1 : \text{addressing}(s, a, t) \\ p : \text{Prop} \\ c2 : \text{resolve}(p, \text{max-qud}) \\ c3 : \text{member}(\text{Trivial}(\text{head-dtr.cont}), \text{FACTS}) \end{array} \right] \\ \text{cont} = \left[ \begin{array}{l} \text{sit} = u0 \\ \text{SIT-TYPE} = \left[ \begin{array}{l} R = \text{Reassert} : \text{IllocReIn} \\ c1 : R(s, a, t, p) \end{array} \right] \end{array} \right] : \text{Prop} \end{array} \right]$$

head-dtr : [ cont : PolarQuestion]

(33) *addressee-ability-cl* =<sub>def</sub>

$$\left[ \begin{array}{l} \text{dgb-params:} \left[ \begin{array}{l} u0 : \text{Rec} \\ s : \text{IND} \\ t : \text{TIME} \\ a : \text{IND} \\ c1 : \text{addressing}(s, a, t) \\ o : \text{Outcome} \end{array} \right] \\ \text{cont} = \left[ \begin{array}{l} \text{sit} = u0 \\ \text{SIT-TYPE} = \left[ \begin{array}{l} R = \text{Request} : \text{IllocReIn} \\ c1 : R(s, a, t, o) \end{array} \right] \end{array} \right] : \text{Prop} \end{array} \right]$$

head-dtr : [ cont = ?Able(a, o) : PolarQuestion]

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